

**Norms, Natures and God: Metaphysics for a Complex
World**

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Acknowledgments

Central ideas for this paper were developed as part of the Wilde Lectures in Natural and Comparative Religion at Oxford University, Trinity Term, 2019.

CHAPTER I

Introduction

1. Introductory remarks

I have human nature or human form that governs my activity, both voluntary and not. Much as the government governs the activity of the people *both* by legislating norms and encouraging people to follow the norms, my nature's governance also has the dual role of setting norms for me and influencing my activity to follow these norms. This nature is something real and intrinsic to me, something that makes me be what I am, a human being. You have one, too, and yours is intrinsically just like mine.

When extended to other fundamental beings besides humans, the above is the center of Aristotle's metaphysics. I will show that this center is extremely fruitful, providing compelling solutions to problems in ethics, epistemology, the philosophy of mind, semantics, metaphysics and philosophy of science. Many of these are prominent problems that have been the subject of much discussion, such as the problem of priors in Bayesian epistemology or of vagueness in semantics, while others are problems that have not attracted much attention, such as the problem of seemingly arbitrary detail in moral rules. I shall discuss problems and solutions in Chapters II–VIII.

The metaphysical apparatus of what I have been calling the “Aristotelian center” will be gradually elaborated as we move through the problems and details of their solutions. At the same time, not every detail of the solutions needs to be adopted by the reader to find the general Aristotelian strategy compelling. However, in Chapter X a number of issues of detail will be discussed more specifically.

The ability to give unified solutions to an array of problems spread through many areas of philosophy gives one good reason to accept the central Aristotelian theses. However,

in Chapter XI, I will also argue that this center does not plausibly hold on its own, and the best way to be an intellectually satisfied Aristotelian, especially after Darwin, is to be a theist as well.

The form of argument for Aristotelianism is similar to David Lewis's argument for extreme modal realism (**LewisPlurality1986**): the variety of problems solved by the theory is evidence for the theory. At the same time, the sheer variety of these problems forces a limitation on this book. There is a vast philosophical literature on some of the problems, offering a large number of solutions. It is not practical to survey all these, though some notable ones will be discussed and criticized at times. But my excuse for limiting the engagement with the extant solutions is not merely practical. For the argument is not that Aristotelianism provides the best solution to each individual problem, but that it provides a unified solution to a broad variety of problems. As long as competing solutions for one problem do not solve the others, or solve them but not overall as well, the argument has force. And the judgment in the end is left to the reader, a judgment that will need to draw on a broader literature.

There are several lines of thought readers attracted to the unified Aristotelian solutions may follow. Some may deny that the problems facing the central Aristotelian theses are as serious as I contend. Some may agree that the problems are serious, and regretfully reject the Aristotelian apparatus, either because they take the cost of the theistic solution to be too great or are unconvinced that the theistic solution works on its own terms. Others may agree that the problems are serious but find some other solution than the theistic one. But some, I hope, will conclude that the Aristotelian solutions are so attractive, and the theistic solution to the problems is sufficiently plausible, that this book provides not only a good reason to accept the Aristotelian center but also to accept theism.

I will now mention a crucial assumption: this book is predicated on a robust realism about ethics, epistemology, semantics and metaphysics. This realism is both metaphysical, holding that there are objective facts about the states of affairs, and epistemological, claiming that we have epistemic access to this objective reality, and our best views about it tend

to be right. A number of the arguments on this book implicitly put pressure on this realist assumption, in that they show that there is a choice between a variety of non-realisms and a highly controversial metaphysics (see Section XI.2.5.6 for specific engagement with this point). The methodology here, however, will be that realism is taken for granted, and no matter how high the metaphysical price for solving problems in a realist way, that price is worth paying. In particular, I take normative facts to be as much a part of our objective data about the world as observable physical facts.

I do not have any particularly original argument for realism about norms. Like many others, I find persuasive an argument based on particular cases. For instance, it is as indisputable as my having two hands that torturing minority children would be wrong even if we had a racist society that thought we had no duties towards members of minority groups and even if we ourselves agreed with that society's abhorrent views. And it is as indisputable as there being other minds that one is epistemically defective if one comes to believe in fairies solely to thumb one's nose at the scientific establishment, and this is true even if society agrees that such nose-thumbing is an epistemic virtue. Likewise, to use **McInernyET's** (**McInernyET**, p. 2) vivid example, eating thumbtacks does not contribute to flourishing no matter how much one wants to eat them or how much one enjoys this due to a rewiring of one's pleasure system. A non-realist reader will say I am just thumping the table, of course. Indeed, I have thumped it, and for most of the rest of the book I will simply assume realism.

Another limitation of this book is that I will minimize engagement with the rich literature as to the details of Aristotelian metaphysics. Much of the book is predicated on a robust but not metaphysically detailed Aristotelianism, one that many Aristotelians should be able to agree on.

In the rest of this first chapter, two things will happen. First, I will sketch the central Aristotelian metaphysics in slightly greater detail. Second, I will discuss a neglected science-based argument from the 17th century polymath Marin Mersenne for the existence of God. This argument does not work very well, I will argue. However, a central thread

running through this book will be how “Mersenne problems” analogous to the problems in science raised by Mersenne arise in many areas of philosophy and provide a compelling case for the existence of Aristotelian natures or forms.

2. Aristotelian natures

2.1. A quick introduction. According to Aristotle, reality is fundamentally built out of substances, which are real mind-independent entities that bear properties and are not related to another thing in the way that a property is to its bearer. These substances are not limited to microphysical entities like quarks and photons. Aristotle takes biological organisms like oak trees and human beings to be paradigm cases of substances. In fact, not only are substances not limited to microphysical entities, but there is some reason to doubt that that microphysical entities are substances at all. For one can have a superposition of states with different numbers of particles of a given type, whereas there should be a well-defined number of substances.¹

Each material substance has a form or nature—I will use the terms interchangeably in this book. This form or nature performs a number of roles including unifying the matter of the substance into a single thing, setting norms for the structure and activity of the substance, and guiding the actual development and activity of the object. The nature of

¹One might hope that sometimes the wavefunction will be in an eigenstate of particle number for a particle type, and think that at such times at least some particles will be substances. However, one would expect a system to stay long in an exact eigenstate of particle number. Consider a system in an eigenstate of electron number. Real-world systems are not isolated. Thus pretty much always there is a non-zero probability of a positron quantum tunnelling and annihilating that electron to produce two photons, and then, unless the wave-function is immediately collapsed, we will presumably have a system that’s in a superposition of a state with two photons and a state with an electron and positron. While it is possible to think that particles are extremely short-lived substances, existing only when we have an appropriate eigenstate, we do tend to think of substances as moderately long-lived. At the same time, one might worry that this line of thought may push some thinkers to suppose a single universal substance—as in quantum holism ([IsmaelSchafferQHolism](#))—and that if one tries to avoid that conclusion, doing so will offer some way of restoring particle-based substances. (I am grateful to Robert Verrill for this objection.)

the oak tree is not merely an arrangement of its particles, since an arrangement lacks normative force. But it's also not just a norm, because a norm does not guide the activity of a mindless thing. Indeed, in living things, the form of a substance has traditionally been considered to be its life or soul: it makes the substance be alive.

Natures are innate to their substances. Nonetheless, this statement underdetermines an important question, namely whether substances of the same sort—say, red oaks—all numerically share one nature, or each individual substance has its own nature, albeit in relevant respects. They are all exactly alike in substances of the same kind. For two things could in principle share something innate to them. It could be that all people have the same soul, much as two conjoined twins can have the same stomach. Aristotle scholarship is divided on the question whether Aristotle believed in “individual forms”, one per substance. However, at least one of the advantages of an Aristotelian theory of form will be accentuated if we accept individual forms, as we shall see, and on the whole I will argue in that there is good philosophical reason to take natures to be individual. Nonetheless, if you like shared forms, *many* of the benefits I will draw out for a theory of forms will be ones you, too, can have.

2.2. Aristotelian optimism. Natures not only define how a thing should function, but also actively lead the thing to function in that way. This means there is an inherent bias in each substance towards acting well. This bias contributes to Aristotle's optimistic thought that natural states occur “for the most part” (**Aristotle**), which is quite useful for figuring out what is in fact natural, since the frequency of the occurrence of a state is evidence of its naturalness.

There is, however, a tension in Aristotle's own thought between the above optimism and his pessimistic thought that few are virtuous. Exploring the tension will help us understand Aristotelian optimism more clearly as it faces the problem of moral evil.

There are many substances with different natures in the world. The flourishing of some requires involves the languishing of others: the lion's feeding is the gazelle's death. Moreover, a substance's nature directs it to behavior that works well for the substance in

its natural niche. But things do not always stay in their niche. Because of this, Aristotle has many resources for explaining why there is a significant set of cases where substances find themselves in unnatural states.² But Aristotle nonetheless thinks that misfortune will only be a minority of the cases.

Let us return to the Aristotelian optimism that things function well “for the most part”. What is and is not “for the most part” depends on the reference class. Most humans have legs, but most living substances do not. If the reference class of the “for the most part” is all activities of all substances, then human moral behavior forms such a small portion of that class—it is so outnumbered by bacterial reproduction, say—that even if all human moral behavior were wicked it would be unlikely to make a difference with respect to the Aristotelian optimism. However, at the same time, with such a broad reference class, the optimism would be of little use to us in understanding normativity for humans, for humans could simply be an outlier in all respects.

A more optimistic reference class would be all the activities of a particular kind of substance. On this reading, Aristotle would lead us to expect that each kind of substance does well in most of its activities. But moral activity is only a small proportion of the activity of a human. We also breathe, circulate blood, repair cells, etc. Leibniz in his *Monadology* estimated that three quarters of our activity is at an animal level (**RescherMonadology**). Stalin was an utter moral failure, but still he maintained homeostasis until the age of 74. Human moral activity could, thus, be mostly bad even though most human activity is good. Again, the tension between Aristotle’s general optimism and his pessimism about human morals would be resolved.

A yet more optimistic reference class would be a particular major type of activity—say, moral activity or reproduction—of a particular kind of substance. Now we would have the prediction that most human moral activity will be good, and this seems to contradict Aristotle’s thesis about typical human moral badness. But even this is not clear.

²For further discussion of the harmony between substances, see ??forwardref.

In MacDonald's *The Princess and Curdie*, Curdie has just expressed to the princess's great-great-grandmother a pessimistic thesis that unavoidably most things humans do are bad.

"There you are much mistaken," said the old quavering voice. "How little you must have thought! Why, you don't seem even to know the good of the things you are constantly doing. Now don't mistake me. I don't mean you are good for doing them. It is a good thing to eat your breakfast, but you don't fancy it's very good of you to do it. The thing is good—not you." (MacDonaldPC)

The old woman makes two important points. First, we should not forget that we perform *many* morally significant actions each day. Curdie ate breakfast. He could have thrown it at his mother, or just ungratefully poured it out on the ground. His eating breakfast was morally good, an instance of nourishing a hungry person. And we perform many such morally good actions each day. Second, the fact that we perform these morally good actions does not do us much credit, the grandmother insists. I suspect that the reason for her pessimism here is Curdie's lack of the kinds of motivations that would render breakfast-eating positively creditable. But the mere motivation to nourish himself was already good, even if not particularly creditable.

There is a further point we may add. While on a mathematics exam, it might be enough to get 60% to pass, morally speaking it is not enough that 60% of one's actions be good. If in the morning I kick a neighbor's puppy, at lunch I charge my private meal to a research budget, in the afternoon I plagiarize something from a foreign language journal for inclusion in my book, and in the evening I cheat in order to beat my kid at chess, I am a bad person even if each of these actions is paired with two morally good actions of the eating-breakfast level of goodness. Having a majority of one's actions be good is not nearly enough to avoid being bad.

Thus with the reference class of "for the most part" restricted to moral activities, Aristotle's optimism and pessimism can be both maintained. And the above considerations

also show that Aristotle's optimism is quite compatible with realism or pessimism about human morality.

A further optimistic ingredient that we will at times draw on is the idea that the different ways of being well in an organism have a tendency to mutually support each other in a unified kind of way. There will be trade-offs, sometimes tragic ones, but by and large a healthy heart supports healthy lungs, a healthy mind supports a healthy body, courage supports justice, justice supports courtesy, and courtesy supports kindness, all of which tend to make one live a happier life even by hedonistic standards.

Aristotelian ethics is sometimes accused of inferring "ought" from "is". The Aristotelian optimism should plead guilty, but with a mitigating factor. Since things only typically act how they ought, the move from "is" to "ought" is defeasible. Nonetheless, that things act a certain way is some evidence that they ought to do so. This is, in fact, quite plausible apart from Aristotelian assumptions. For there are three possible views on the statistical correlation between "is" and "ought":

- (1) There is no correlation whatsoever between how things behave and how they ought to behave.
- (2) There is a negative correlation between how things behave and how they ought to behave.
- (3) There is a positive correlation between how things behave and how they ought to behave.

Option (1) is quite implausible. That a behavior obtains is surely *relevant* evidence for the question of how a thing should behave.

Further, take at random any two distinct binary features A and B of human beings, say, having green eyes and having won an egg-and-spoon race. Suppose we actually measure the correlation of these two features across all humanity, by calculating the covariance $\text{Cov}[A, B] = P(A \ \& \ B) - P(A)P(B)$, where $P(C)$ is the probability that a random human has C . We might find that the covariance is very small. But given that there are eight

billion people, how likely is it that $P(A \ \& \ B)$ should *exactly* equal $P(A)P(B)$ across the population? Surely very unlikely indeed. Thus, very likely, the actual distribution of any two distinct binary features are statistically correlated, positively or negatively.

So now the choice is between supposing a positive or a negative correlation between behavior and norm. The hypothesis of a negative correlation does not seem very plausible. Imagine that you are thrown into a situation about which you know nothing, but see a line of people one by one doing something—say, pressing a red button. And now it's your turn: you're at the head of the line. It is unjustified pessimism to think that your observation provides you with evidence for the hypothesis that you should *not* press the red button, as would be the case if the correlation between behavior and norm were negative. If you know nothing more, it's a better bet to act like others. And the same, surely, goes for other kinds of things than humans. That a behavior is exhibited is not evidence against the behavior being right. It is very unlikely to be evidentially irrelevant. So it is at least some evidence for the behavior being right.

This argument does not rule out the possibility that the correlation between behavior and norm is very weak, weaker than the Aristotelian optimist needs to get significant evidence for norms from behavior. But it is a start.

Nor need it be the case that “is” is always evidence for “ought”. In some cases we may have independent evidence that a particular entity is so defective that certain sorts of its behavior are of no positive evidential value. We shouldn't learn ethics from the depraved. That said, I happen to be sufficiently optimistic that I doubt that there are many people who are so depraved that they exhibit *no* positive correlation between behavior and norm, even if that correlation is not mediated by right decision-making but only by self-interest.

2.3. Who are the humans?

2.3.1. *Rational animals.* It is traditional in the Western philosophical tradition to describe the human being as a rational animal. One can take this further and *define* the human being as a rational animal, and indeed this is common from the medieval Aristotelian

tradition³ to recent times. It follows that if rational dolphins or octopuses evolved, they would also be human in the philosophical sense, having the same nature as we do. However, while Aristotle himself certainly held that we were rational and we were animals, he did apparently does not *define* us by our rational animality (**Kietzmann2019**). And such a move would fit poorly with Aristotle's hierarchical way of defining species that inspired the Porphyrian tree. For if some but not all possible primates were human and some but not all possible cephalopods were human, then *human* couldn't be a subdivision of vertebrate or of invetebate. Rather, it would be a species that cuts across multiple genera. This might not bother those of us who do not accept the hierarchical mode of definition⁴, but it would be highly problematic for Aristotle.

But there is another serious related problem. Human nature specifies what is normal for human beings. For a rational primate, having four limbs adapted to movement on land is normal; for a rational dolphin, having flippers and a tail would be normal; for a rational octopus, having eight tentacles would be normal. This suggests that there isn't a single human nature that would be shared by rational primates, cetaceans and cephalopods.

However, this argument is much too quick. After all, the variety-of-normalcy problem may well arise even within the biological species *Homo sapiens*, though particular examples are apt to be controversial. The majority of the human adult population worldwide, especially outside of Europe, is lactose intolerant. At the same time, sometimes lactose intolerance is described as an "impaired ability to digest lactose" (**CDHF**), which suggests that lactose intolerance is abnormal (we would not talk of an impaired ability to digest cellulose in humans). Yet this alleged impairment is found in the majority of the human adult population worldwide, with prevalence varying regionally from below 5% to nearly 100% (**Sahi1994**). We could say that speaking of lactose intolerance as an impairment is

³For instance, Aquinas grants that "seeing that man adds rational to animal, it follows that a rational animal is a man" (**AquinasPower3**).

⁴However, see **KoonsPorphyrean** (**KoonsPorphyrean**) for a promising contemporary account of the Porphyrian tree.

just a mistake. But there is another possibility: it may be that lactose intolerance is normal for some members of our species and abnormal for others, and the description of it as an impairment is correct when restricted to persons of certain ancestries but not others. In that case, assuming that all *Homo sapiens* do share the same nature, what makes it abnormal for an adult of one ancestry to be lactose intolerant but normal for an adult of another cannot just be our shared human nature.

Instead, it might be that our shared human nature encodes some conditional like “If you have the complex feature *F*, then you should be able to digest lactose even as an adult.” This conditional applies to humans worldwide, but only a minority of humans actually exhibit *F*. This feature might, for instance, be genetic coding for life-long lactose digestion, so that those who have this coding ought to be lactose tolerance and those who do not need not be. In that case, even among the Irish lactose intolerance need not always be an impairment: it would only be an impairment among those who have the genetic coding for lactose tolerance but are nonetheless for some other reason intolerant. Or this feature could specify some other aspect of the genome that correlates with, but does not cause, lactose digestion.

An even more controversial and politically charged example could be sex-linked traits: for some members of our species, having a uterus may be normal, and for others it may be abnormal. Again, our human nature could encode a conditional like “If you have feature *G*, then you should have a uterus.” Note that while it is controversial whether there is such a conditional, holding that there is does not by itself decide important questions about gender identity. For instance, someone who thinks that gender identity is determined by genetics could say that the conditional holds with *G* being a genetic feature (say, having two X chromosomes), while someone who thinks that gender identity is determined by personal self-identification could say that the conditional holds with *G* being self-identification as a woman. They could then both use the conditional to argue for opposite answers to the question of whether hysterectomy for sex-reassignment is a medical treatment that restores normalcy.

Much less controversial are more temporary conditions that affect what is normal. A high temperature is normal for a sick human but abnormal for a healthy one. Yet we surely don't want to say that getting sick changes our nature.

The problem of the variety in what is normal among logically possible biological species of rational animals could be handled analogously by a defender of the view that humans are defined as rational animals. Human nature could specify that if you are a primate you have four limbs, if you are a cetacean you have flippers and if you are an octopus you have eight tentacles. But while this is theoretically possible, it would mean that human nature would have to encode an infinite number of such conditionals. Indeed, taking this to its logical conclusion, we would have to include conditionals to fit not just rational animals in worlds with laws of nature like ours, but in worlds with a physics radically different from ours. It seems very implausible to suppose that there is such infinite normative complexity in our nature.

Moreover, the expansive view of humanity results in a very implausible restriction of the space of possible natures. For while Aristotelian harmony may not allow for every combination of normative features—such as an animal that ought to live all its life deep underwater but also ought to breathe atmospheric oxygen—we would expect there to be a broad selection of logically possible natures harmoniously combining different normative features. Thus, just as it is possible to have an animal that is supposed to both echolocate and fly (the bat), and animal that is supposed to both echolocate and be snake-like (perhaps this is unexemplified, but possible), it should be possible to have an animal that is supposed to be both snake-like and rational. But such an animal would not be human, because everyone agrees that it is false that humans are supposed to be snake-like (on the expansive view, they can be non-defectively snake-like, while on narrower view, being snake-like would be a defect). So it should be possible to have non-human rational animals.

Finally, note that even if dolphin and octopus persons are not human, it is very plausible that our human nature would require us to treat them with the respect that persons

deserve. After all, even on the broad definition of humanity, disembodied beings like deities or angels would not count as human, and yet many people who have believed in such beings have held that we should show them *at least* (and sometimes more than) the kind of respect we do for fellow humans, for instance by keeping promises made to them.

2.3.2. *The narrower view.* Even within the narrower view of humanity that would exclude dolphin and octopus persons from being human, we still have the difficult question of where exactly the boundaries of humanity are to be drawn. We can make use of our best ethical intuitions to give us good reason to draw them in a way that includes at least all members of *Homo sapiens*. For instance, according to one study, rescuers of Jews from the Holocaust were more likely than non-rescuers (39% vs. 15%) to emphasize that “ethical values were to be applied universally, that they extended to all human beings” (Oliner1988). The intuition of many rescuers, as well as of typical anti-racists and anti-sexists in our time, supports taking all members of *Homo sapiens* to be human.

But there are two further questions. One is about other historical species closely related to us, such as Neanderthals and Denisovans. While we could identify humanity with *Homo sapiens*, we need not. Probably, we do not know enough about Neanderthal and Denisovan life to see whether it is plausible that one form encompasses the norms for them as well as for us. And fortunately for us this question is of merely theoretical importance, though it might have been of practical importance for our ancestors.⁵

The second question is this. Should we accept as human all biological humans who do not satisfy the criteria for developed personhood of philosophers like **WarrenAbortion** (**WarrenAbortion**), such as language or generalized problem-solving skills? In the case of adult *Homo sapiens*, there is a simple Aristotelian argument for doing so. Adult biological humans who do not fulfill such criteria suffer from an abnormality. The evidence for this is that if we could treat them medically in a way that leads to the fulfillment of the criteria,

⁵For instance, we might think that we have special duties towards members of our kind, or that we are obligated to procreate only with members of our kind (if so, and if our species does not include Neanderthals, many of our ancestors violated this).

we would have good reason to do so. Indeed, if these adult biological humans were not of the same kind as us, then such medical treatment by transforming them into beings like us would constitute what Aristotelians call a substantial change, a change from one kind of thing to another. But objects do not survive substantial change. Thus, such treatment would be a killing. That is implausible.

The remaining subquestion deals with immature members of *Homo sapiens*, such as zygotes, embryos, fetuses and infants, prior to the development of the features that meet these philosophers' criteria for developed personhood? Here, the questions become more controversial. On a view that excludes such immature members, we would say that they are a member of a different kind from us, and their ordinary course of life is that they mature and die *in utero*, giving rise to a being like us, a human in the metaphysical sense. Yet this does not seem an entirely plausible interpretation of maturation processes. Death seems to be a catastrophic event for a substance, and in the ordinary growth of from zygote to embryo to fetus to infant to child to adult there does not appear to be such a catastrophe. This suggests that these members of our biological species are also human in the metaphysical sense, but more needs to be said (**PrussAbortion; ColgroveZygotes**).

3. Mersenne questions

3.1. Mersenne's argument. Marin Mersenne was a monk, philosopher, theologian, and the 17th century equivalent of the arXiv preprint archive—a crucial line of communication between a broad variety of thinkers and scientists. He drew on his broad knowledge of the science of the time to offer an argument that begins with many pages of questions, of which the following are representative:

Why is the moon 56 earth-radii away from the earth? Why is the sun 1182 [earth-radii] away from us at its apogee? ... and why is its distance at perigee not other than 1101 [earth-radii]? ... I could equally ask you about Saturn, and Jupiter, and Mars ... Who gave more strength to the lion than to the ant? Who made it be that earth is not in the moon's place,

and that the planets aren't larger or smaller, closer or further? Who has ordered all the parts of the world as we see them? (**MersenneImpiety**)⁶

These "Mersenne questions" go on and on, with a mind-numbing number of examples. And Mersenne has one answer to all these questions, posed in a rhetorical question: "Was it not God?" (**MersenneImpiety**).

The argument sounds similar to fine-tuning arguments for theism which became popular in the late 20th century. These arguments, too, list a variety of physical parameters and offer God as an explanation of them all.⁷ Now, a crucial argument not given in the above quote but explicit in fine-tuning arguments is an argument that the parameters needed for intelligent life as we know it, or for some other valuable trait of the universe, like its amenability to scientific investigation (**CollinsDiscoverability**). The basic idea behind fine-tuning arguments is, very roughly, that nature is indifferent to value but God cares about value, so the fact that the parameters are valuable provides evidence for theism over naturalism.

It is, thus, natural to look in Mersenne for an argument that it is somehow particularly valuable for the moon to be 56 earth-radii from the earth, but at least in this work, Mersenne does not even hint at such an argument. Nor is there any argument that it is better that lions are stronger than ants, or for the moon to orbit the earth rather than the other way around. If Mersenne is giving a fine-tuning argument, the argument is oddly incomplete. And Mersenne's penchant for adumbrating detail at great length makes the incompleteness reading unlikely.

Rather, it appears that Mersenne is simply looking for an explanation of the scientific details he cites, sees no prospect of a scientific explanation, and offers theism as the alternative. And indeed it is only in the 20th century with computer models of solar system

⁶The moon-earth distance is approximately correct. The earth-sun distances are an order of magnitude off.

⁷For one of many defenses, see **LewisBarnes** (**LewisBarnes**), and for one of many criticisms, see **MMV2001** (**MMV2001**).

formation that we have much in the way of plausible answers to Mersenne's questions about the distances between solar system bodies. For instance, the leading theory of lunar formation involves the earth being hit by another body and a large chunk being pushed into orbit. Given assumptions about the impact, one can then sketch an explanation of the evolution of the size of the moon's orbit (**Canup2014**), and hence of the current distance between the earth and the moon.

Still it is worth noticing that such an explanation only gives an answers to the Mersenne question about the earth-moon distance at the cost of raising similar Mersenne questions about the parameters of the impact such as the mass distribution of the pre-impact earth, the angle and location of impact, the mass distribution of the impacting body, etc. So it may seem that Mersenne's argument still has some force.

But Mersenne's argument has a fatal flaw. Even if we grant that it is very unlikely that a future science will predict these exact numbers, there is always the possibility of a stochastic explanation, one that does not predict exact values, but supposes a random natural process that generates a set of values at random. Now, if Mersenne had an argument showing that the values of the parameters are suspiciously valuable—say, necessary for intelligent life—then a stochastic explanation might not be as good as a theistic one. From a Bayesian point of view, we might be able to argue that it is extremely unlikely that a random selection of parameters would have such value, but not nearly so unlikely that God would choose such parameters. But given that Mersenne makes no case that the parameters have anything to recommend them to God for creation, we have no reason to think that the probability of God choosing is these parameters is any higher than the probability of them arising randomly, and hence we have no support for theism.

Suppose, however, that we had a Mersenne-type case where randomness was not a satisfactory explanation. Then the argument would have hope.

It is, however, worth mentioning one more potential weakness of the argument. If one is willing to deny the Principle of Sufficient Reason, one could simply say that the parameters are what they are and there is no reason why they are like this—that they are

a *brute* fact. This, however, is less satisfying than the stochastic answer, for adverting to brute fact should be a last resort, to be chosen when no explanation is available. But here there is an explanation available, namely theism. So in cases where randomness is not an available recourse, a Mersenne-style argument has some force, but is weakened by the brute fact objection.

3.2. Appearance of contingency. Now, Mersenne gives a dizzying number of examples, and he seems to relish the sheer appearance of arbitrariness of the numbers like “56” and “1182”. While this has some rhetorical force, it also has argumentative force. The more arbitrary-looking parameters the parameters are, the less epistemically likely it is that they are what they hold of necessity or that good scientific theories will predict their exact values. And the greater the number of parameters, the less likely it is that science can provide an explanation of them all.

The appearance of arbitrariness is evidence of contingency, and contingency calls out for explanation.⁸ But at the same time, we have to be careful here. For instance, it might seem arbitrary that protons have (approximately) 1836 times the mass of electrons, but the masses of protons and electrons could well be essential properties of them, so that a pair of particles whose mass ratio were different from 1836 could not be a proton-electron pair. A similar point can be made about Mersenne’s own example of lions being stronger than ants. So in some cases, the arbitrary-seeming parameter does in fact hold of necessity. But that does not mean that the Mersenne question disappears. For while the parameter itself is not contingent in these cases, there is nearby contingency. Even if the masses of protons and electrons are essential properties, it is possible to have other particles with similar behavior but different masses, and so it is still contingent that the world contains a pair of opposite-charge particles with mass ratio (approximately) 1836 that form atom-like entities similar to the atoms of our world.

⁸In **PrussPSR (PrussPSR)**, I have argued for a Principle of Sufficient Reason (PSR) that holds that all contingent facts have an explanation. But even if one rejects the PSR, one should hold that explaining relevant contingencies is a good feature of a theory, one that provides evidence for the theory.

The point generalizes: Sometimes the apparently arbitrary parameters can be explained by the necessary features in the essences of things, but in those cases it will often be the case that it is contingent that these essences, rather than other similar ones, are exemplified. In those cases, the appearance of arbitrariness yields an appearance of contingency, and the true contingency is nearby.

There is, however, a further worry here. Consider the apparent arbitrariness of the fact that the ratio of the circumference of a circle to its diameter in decimal notation has 1 and 4 as its second and third digits, respectively. Yet this fact can be wholly mathematically explained by necessary mathematical truths such as that $\pi = 4 - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \dots$.⁹ Thus the appearance of arbitrariness of a parameter is merely *defeasible* evidence of contingency in the parameter or even nearby.

We thus have to be cautious: moving from apparent arbitrariness to contingency, whether of the parameter itself or of something nearby, is always going to be a defeasible and non-deductive move. This is why there is a value in Mersenne's giving as many examples as he does, since non-deductive arguments tend to stack up. And many of Mersenne's particular examples, such as the astronomical distance examples, are ones where it would be difficult to believe in a necessity-based explanation without any contingency involved.

In the rest of the book we will find that if we turn our attention away from science and towards philosophy, we will find a myriad of cases like Mersenne's where there are seemingly arbitrary parameters. But these will be cases where a randomness explanation is implausible, bruteness is not satisfactory and the appearance of contingency is undefeated. However, unlike in Mersenne's cases, I won't be arguing—at least not in the first instance—that theism provides the solution. Rather, the solution will be Aristotelian metaphysics of form. But then I will argue that, especially after Darwin, theism is needed to make Aristotelian metaphysics intellectually palatable.

⁹This point is very similar to an argument in **Hume1779** (**Hume1779**, Part IX).

CHAPTER II

Mersenne questions in ethics

1. Motivating examples

1.1. The rule of preferential treatment. Let us begin with a more detailed discussion of an example from Thomas Aquinas's discussion of the order of charity. Aquinas thinks, along with common sense, that those who are more closely related¹ to us have a greater moral call on us. Thus, if it is a question of bestowing the same good on one of two people, if one is more closely related, we should benefit the closer one. But Aquinas writes: "The case may occur, however, that one ought rather to invite strangers [to eat with us], on account of their greater want." But then he immediately raises the question of what one should do "if of two, one be more closely connected, and the other in greater want" (**AquinasST**).

We might hope that here Aquinas would give us some clever rule for weighing connection against need. But instead he writes something that is very sensible and yet disappointing: "it is not possible to decide, by any general rule, which of them we ought to help rather than the other, since there are various degrees of want as well as of connection" (**AquinasST**).

It is tempting at this point to throw up one's hands and simply say that in these in-between cases there is no fact of the matter as to what should be done, or both options are permissible, or else relativism applies to the case. But that would not do justice to the way we agonize when we find ourselves in such a difficult situation, trying to discover the truth of the matter. (It is interesting to note that the most common real-life moral dilemmas tend to be like these kinds of cases, rather than highly controversial questions about trolleys,

¹In some relevant way, that is. A torturer is closely related to the victim, but that doesn't produce a special duty on the part of the victim!

strategic bombing or bioethics much discussed by philosophers.) And indeed Aquinas maintains a realist attitude to the question while simply offering this advice for how to figure out the answer in a particular case: “the matter requires the judgment of a prudent [person]” (**AquinasST**).

We can formalize this as the problem of specifying a function $f(r_1, b_1, r_2, b_2)$ of four variables, two of them, r_1 and r_2 , being degrees of relation and the other two, b_1 and b_2 , being degrees of benefit, where the function yields one of three values corresponding to whether it is (i) obligatory, (ii) permissible but not obligatory or (iii) impermissible to bestow a benefit of degree b_1 on a person with relation of degree r_1 to the agent in place of bestowing a benefit of degree b_2 on someone related to degree r_2 .

In fact, the problem of a rule of preferential treatment is much more complicated than the above indicates. First, the *kinds* of benefit and relation also matter: “we ought in preference to bestow on each one such benefits as pertain to the matter in which, speaking simply, he is most closely connected with us” (**AquinasST**). So the function will depend not merely on quantitative features but qualitative ones. Second, although Aquinas does not mention it here, the evaluation will no doubt depend on various features of the circumstances. And, third, in practice instead of choosing between two certain benefits, we are choosing between two probability distributions over the space of possible benefits.

Now, as Aquinas admits, we do not know what the moral evaluation function for choices between benefits to different people is. But given moral realism, once enough relevant features are satisfied, there has to be some such function, even if we do not know what it is, just as there is a function that assigns to each person alive now the number of hairs they now have, even though we cannot specify any of the values of the function. And we have good reason to expect the moral evaluation function to be very complicated. Indeed, probably the only serious proposal for a relatively simple function f here is the utilitarian suggestion that $f(r_1, b_1, r_2, b_2)$ yields obligation when $b_1 > b_2$, mere permission when $b_1 = b_2$ and prohibition when $b_1 < b_2$. But this simple utilitarian solution would

betray the intuition that the degrees of relation are relevant to the moral evaluation. (For further discussion of utilitarianism, see [forwardref](#).)

Indeed, whatever the function is, it is likely to look arbitrary. Fix the degrees of relationship to be those of one's parent and of a total stranger, and fix a specific and certain financial benefit of \$1000 to one's parent, and fix the circumstances. Then as we increase the financial benefit to the stranger from zero to infinity, we will presumably initially have a requirement of benefiting the parent (it would be wrong to give \$1 to a stranger instead of \$1000 to a parent in ordinary circumstances), then a permission either way, and then a requirement to benefit the second party. There will be boundaries between these regions of logical space, and these boundaries will look as arbitrary and contingent as the boundaries between different tax brackets. Like the tax brackets, some proposals for boundaries will be *clearly* unreasonable, but there will be many proposals that appear reasonable. And whatever the actual boundaries will look arbitrary.

Of course, seemingly arbitrary numbers can come out of an elegant and simple rule: it seems arbitrary that the fifth and sixth digits of π are 5 and 9 respectively, but there is an elegant mathematical explanation. However apart from the utilitarian proposal, we do not have any at all plausible simple proposal for f . And when we realize that there is a large number of other factors besides degrees of relationship and degrees of benefit—for instance, surely children owe more to those who have put more effort and genuine love into parenting²—the idea of a simple proposal for a determining rule becomes quite untenable.

These seemingly arbitrary boundaries in the order of charity call out for an explanation at least as much as the exact distance between the earth and the moon does. Just as it seems implausible that the distance between the earth and the moon *must* be exactly what it is, it seems implausible to think that the boundaries must be exactly where they are—unless the utilitarian is right about f being very simple.

²For a fictional exploration of some of these complexities, see **DaviesST's (DaviesST)** *Salterton Trilogy*.

In fact, the ethics case calls out for an explanation more than Mersenne's scientific examples did. For we might be able to swallow the earth-moon distance being a contingent and brute unexplained fact. But a brute fact is intuitively unfitting for a moral rule. A claim that it just so happens, with no explanation at all, that you should ϕ undercuts the moral force of the alleged moral obligation. We expect anything seemingly arbitrary in our moral norms to have an explanatory ground.

To further argue for this point, consider a version of Divine Command Theory on which obligations are divine commands, and God rolled indeterministic dice to decide which actions to command, and by chance God's commands coincided with our common-sense morality, though they could just as well have commanded cruelty and dishonesty. A Divine Command Theory on which it is mere chance that cruelty is forbidden rather than commanded provides an unacceptable answer to the Euthyphro problem of explaining why God prohibits what he prohibits. Intuitively, a set of injunctions that is as arbitrary as that cannot constitute morality. But this point generalizes beyond divine command theory. Suppose that that we have some preferential treatment rules that are brute and contingent, rules that could have been so different from what they are that they could have enjoined on us the anti-utilitarian rule that we should always prefer the lesser benefit. Then whatever these brute rules are, they do not constitute morality, but at best happen to agree with morality in content.

Thus, even if there is some bruteness in the rules of preferential treatment, the rules in our world must be generated in a way that makes rules such as the anti-utilitarian rules not be among the possible outcomes. But this makes it very unlikely that the rules would be brute. For what force would limit the brute rules to avoid unacceptable options? Such a view of limited bruteness would be akin to a metaphysical optimism on which banana peels can come into existence *ex nihilo*, but not where we might trip over them.

It is important to remember that the Mersenne question here is a metaphysical question: What explanatory grounds are there for why this rule, rather than some competitor, holds? The *epistemic* question about preferential treatment may well have a virtue-theoretic

answer like Aquinas's: if we acquire the requisite virtues, we will be able to judge particular cases fairly reliably, and until then our best bet is to ask the advice of virtuous others.

But before I continue the discussion of the possible explanation for the above ethical Mersenne question, let me follow Mersenne's lead and multiply the examples, in order to defend against potential answers that only work in some cases, and to make clear how widespread the problem is. In doing so, I take the risk of boring the reader. But it is also my hope that, apart from the benefit of identifying a broad pattern of questions in furtherance of my projects, I will also bring to the reader's attention a number of questions that have some intrinsic interest.

1.2. Risk and uncertainty. Some people—perhaps you—would accept a 92% chance of winning a thousand dollars at the cost of an 8% chance of losing ten thousand. I wouldn't. I say that both I and they are reasonable. On the other hand, someone who (in ordinary circumstances) rejects a 99.9999% chance of winning a thousand dollars at the cost of a 0.0001% chance of losing ten thousand and someone someone who accepts a 10% chance of winning a thousand dollars at the cost of a 90% chance of losing ten thousand are unreasonable. It is well known that attitudes to risk vary between people, and while there are clearly unreasonable attitudes, it is very plausible that there is a broad range of reasonable attitudes. So, as we vary the probabilities of wins and losses, we move between cases where accepting the risk is unreasonable, to cases where both accepting and rejecting are reasonable, to cases where rejecting is unreasonable.

This, once again, raises the Mersenne problem of why the transitions between the various evaluative categories fall right where they do. And of course things are more complicated than described above. Plausibly, the rational evaluation function will depend not just on the probabilities involved but also on the values of the potential gains and losses, and perhaps even on the *types* of values (lives, money, etc.) that are involved.³

³*There are technically two ways of taking risk into account in decision theory. The classical approach is to apply a non-affine monotonic non-decreasing transformation function f to the original utility or value U assignment on the epistemic probability space and then simply maximizing expected rescaled utility

While in the case of preferential treatment, utilitarianism provided a neat but implausible solution, so too in this case, expected utility maximization provides a neat but implausible solution. On expected utility maximization, you are rationally required to accept a chance p of a good of degree α despite a chance q of a bad of degree β against a status quo of value zero just in case the expected utility $p\alpha + q\beta$ is strictly positive; when it is zero, you are permitted but not required; and when it is negative, you are not permitted. The solution is neat and simple.

One problem with this solution is it requires all goods to be neatly quantifiable (cf. the next example for difficulties related to that). But the more serious problem is that it requires an implausibly negatively judgmental attitude towards ordinary people's attitudes to risk. Indeed, here is a plausible trio of theses about risk that are incompatible with expected utility maximization:

- (1) There is no upper bound on possible finite utilities.
- (2) A decade of the worst tortures the KGB could think of has a finite negative utility.
- (3) There is no possible good G of finite utility such that one would be rationally required in accepting a certainty of a decade of the worst tortures the KGB could think in exchange for a one in billion chance of G .

For as long as $10^{-9}\alpha + \beta > 0$, where α is the value of G and β is the (highly negative) value of the tortures, one would rationally required to accept the deal on expected utility maximization, and by (1) and (2) there exists a possible G that makes $10^{-9}\alpha + \beta$ strictly

$E[f(U)]$ (**WilkinsonRisk**). On the other hand, **BuchakRisk** (**BuchakRisk**) offers a novel way of calculating expected utilities without rescaling them. In the special case where the utility function U is non-negative, this consists in applying a monotonic transformation g and replacing $E[U]$ with $\int_0^\infty g(P(U > \lambda)) d\lambda$ (in the case where g is the identity, this is $E[U]$). In both formulations, there will be clear cases where the transformation function is unreasonable and, if risk aversion is rational, cases where it is reasonable. And one can combine the two formulations for an even more complex attitude to risk, by maximizing $\int_0^\infty g(P(f(U) > \lambda)) d\lambda$ in the case of a non-negative f .

positive. Hence, we should reject expected utility maximization, and absent expected utility maximization, it is unlikely that the rationality evaluation function for risk is anything other than messy and arbitrary-looking.

The most plausible objection to my argument against expected utility maximization is to reject the no-upper-bound thesis (1). Here is one way an argument for such a rejection might go. First, there is a maximum intensity of goods that our brain can handle. Second, goods may become significantly less valuable as they are repeated, decreasing in such a way that the sum of the values of any goods you could have over an arbitrarily long life has an upper bound.

But the repetition thesis is most plausible when boredom and other memory-based phenomena are in play. Suppose you have lived for a very long time. Then you suffer from partial amnesia: you have lost all episodic memory of your past meals and of your past pinpricks. You are offered what you are reliably informed is the most delicious and wholesome dessert ever prepared by the best chef on earth, a dessert which you are told you've eaten some large number n times in the past, and you may eat the dessert at the cost of a one in ten chance of a small pinprick. It's clearly worth it, regardless of what n is. So now suppose this happens to you every day of a very long life. The marginal value of each such dessert (i.e., the amount it contributes to total lifelong utility), absent memories of past desserts, must be at least one tenth of the marginal disvalue of the pinprick, at least given expected utility maximization. But the disvalue of the pinpricks clearly does not tend to zero with forgotten repetition. Hence, the value of the desserts does not tend to zero. And hence for any finite utility bound, enough such desserts will exceed the bound.

For a different example, not involving radically large utilities, imagine this Star Trek plot. Captain Kirk visits a planet inhabited by two intelligent alien species, all on par with respect to moral value: there are 1,000,000 oligons and 2,000,000,000 pollakons. Unfortunately, an asteroid is heading for the planet. If nothing is done, it will certainly hit the planet in such a way as to wipe out all the pollakons but leave the oligons untouched. The only thing Kirk can do is to fire phasers at the asteroid. Spock has calculated that if

this is done, the asteroid's track will be redirected in such a way that it will certainly wipe out all the oligons. Kirk asks whether that will help the pollakons? Spock's answer is that probably not: there is a 999/1000 chance that all the pollakons will still die, but there is a 1/1000 chance that they will all survive.

It seems plausible that Kirk ought not fire phasers. And it is even more plausible that Kirk is not required to fire phasers. He should not sacrifice the million oligons for a small chance of saving the two billion pollakons. But the expected utility of firing phasers is $-1,000,000 + (1/1000)(2,000,000,000) = +1,000,000$ lives. On the other hand, if the chance of saving the pollakons were high enough, say 99%, the story would turn into a plausible trolley case, and Kirk would be permitted and likely required to fire phasers. Where the switchover occurs is a Mersenne question.

Perhaps the most intuitive justification of expected utility maximization is that, under some technical assumptions, the Strong Law of Large Numbers guarantees that if in a sufficiently long sequence of independent wagers, you keep on choosing a wager whose expected value is better by (at least) some fixed increment $\delta > 0$ than the expected value of an alternate, with probability one your total revenue will eventually be higher than if you were always to take the lower expected-value wager.⁴

But the devil is in the details, and the technical assumptions are not always satisfied. One can construct an infinite sequence of wagers where going for the lower expected-value wager of each pair will, with probability one, eventually result in a higher total result. For

^{4*}Let $X_1, X_2, \dots$ be a sequence of independent random variables. The Strong Law of Large Numbers says that, under some technical assumptions, if $X_1, X_2, \dots$ are independent random variables, then with probability one we have $\lim_{n \rightarrow \infty} n^{-1}(X_1 + \dots + X_n) = \lim_{n \rightarrow \infty} n^{-1}(E[X_1] + \dots + E[X_n])$, with both limits finite, where $E[X]$ is the expected value of X . Thus, if $X_1, X_2, \dots$ are independent random variables satisfying the technical assumptions, and so are $Y_1, Y_2, \dots$, and the X_i represent one set of wagers while the Y_i represent the other set of wagers, so that $E[X_i] \geq \delta + E[Y_i]$, then the average revenue from selecting the X_i will converge to at least δ plus the average revenue from selecting the Y_i . One set of technical assumptions that suffices is that $n^{-1}(E[X_1] + \dots + E[X_n])$ converges to a finite value and $\sum_{n=1}^{\infty} n^{-2} \text{Var}[X_n]$, where $\text{Var}[X] = E[(X - E[X])^2]$ is the variance of X (**ChungProbability**).

suppose that on every day n of eternity, with $n \geq 1$, you are offered the opportunity to pay half a unit of utility in exchange for playing a game with a $1/2^n$ chance of winning 2^n units of utility. By expected utility maximization, you would value the n th game at $(1/2^n) \cdot (2^n) = 1$ units of utility, and at a price of $1/2$ units, it would be worth playing.

But consider what will almost surely happen if you adopt the policy of following expected utility maximization and playing the game. The sum of the probabilities of winning on the different days is finite: $1/2 + 1/4 + 1/8 + \dots = 1 < \infty$. The Borel-Cantelli Lemma (**ChungProbability**) then says that almost surely you will win only a finite number of times (“almost sure” is the technical term probabilists use to describe an event that happens with probability one, such as getting heads at least once if you toss a fair coin infinitely many times).⁵ In other words, almost surely, there will come a day after which you will win no more. At that point, you may well be ahead, having won more than you paid. But the sum of what you won up to that day is finite, and from then on you will just lose half a unit of utility every day. Eventually, there will come a day when your losses will overtake your winnings, and from then on, you will just fall further and further behind every day.⁶

The very unhappy situation of playing infinitely many times and eventually starting to lose every time is the almost sure result of following expected utility maximization on each day. We can compare this to the neutral situation of refusing ever to play, and getting

⁵We can give an elementary proof of this fact in the case at hand (the proof generalizes to the general case). Let W_n be the event that you will win at least once after day n . Then $P(W_n) \leq 2^{-(n+1)} + 2^{-(n+2)} + \dots = 2^{-n}$. Let W be the event that you will win infinitely many times. Then $P(W) \leq P(W_n)$ for every n , since if you win infinitely many times, you must win on infinitely many days after day n , and so you must win on at least one day after day n . Since $P(W_n) \leq 2^{-n}$, we have $P(W) \leq 2^{-n}$ for every n . But probabilities cannot be negative, and the only non-negative real number x such that $x \leq 2^{-n}$ for every n is zero. So $P(W) = 0$, and hence almost surely W does not happen, so that almost surely you win only finitely many times.

⁶The example above uses exponential growth. More moderate growth will work, as long as the sum of the probabilities is finite. Thus, we could say that on day n the prize has value $n(\log(n+2))^2$ and the probability of winning the prize is the reciprocal of that, since $\sum_{n=1}^{\infty} 1/(n(\log(n+2))^2) < \infty$.

zero each day, or the situation of accepting the expected utility maximizing gamble for a number of days, until the probability of winning becomes really small, and refusing from then on.

It is worth noting that this is not just a paradox involving the aggregation of infinitely many utilities. For, almost surely, after a finite number of days, the expected utility maximizer falls behind the consistent refuser, and every day after that, the expected utility maximizer is further and further behind, like someone who got a subscription to a streaming service and forgot to either use or cancel it. And all these amounts are finite, and occur a finite, albeit unknown, distance into the future.

We can also consider an interpersonal version of the story. Suppose we have (countably) infinitely many people, numbered $1, 2, \dots$, and person n is offered the chance to pay half a unit of utility in exchange for a chance $1/2^n$ of winning 2^n units. As before, by expected utility considerations it's worth it. So, if everyone is an expected utility maximizer, everyone will pay. But by the Borel-Cantelli Lemma, almost surely, only finitely many people will win. Thus, almost surely, we will have infinitely many people pay a cost of half a unit each, and finitely many people win some finite amount. This is a disastrous situation, with a negative infinite overall utility. Almost surely, it would be much better if everyone refused to play, or only those who had a "non-negligible" chance at winning played.⁷

Granted, in more realistic cases of long runs of independent lotteries, we can expect expected utility maximizers to do better.⁸ However the utilitarian claim isn't just that we should *typically* maximize utility, but that we should *always* do so. If we are to typically do so, then we will have a Mersenne question as to the grounds of the distinction between the cases where we should do so and the cases where we should not do so, and it is implausible to think any neat formula captures the distinction.

⁷It is worth noting that exponential growth is not necessary for the examples to work. All we need is that there is a chance p_n of winning a prize of $1/p_n$, and that $\sum_{n=1}^{\infty} p_n < \infty$. While for ease of calculation above I let $p_n = 1/2^n$, one can have much more moderate shrinkage, such as $p_n = 1/n^2$ or even $p_n = 1/(n(\log(n+2)))^2$.

⁸See, e.g., **ZhaoRisk** (**ZhaoRisk**) and critical discussion of this as an argument for expected utility maximization in general by **WilkinsonRisk** (**WilkinsonRisk**).

In any case, it appears that expected utility maximization is not rationally required. But it is the only clearly non-arbitrary solution to the problem of deciding under uncertainty. While some of the examples involved prudential rationality, the Kirk example was essentially moral. And the prudential rationality examples can all be switched to moral rationality by supposing that the victim or beneficiary of the action is someone that one has responsibility for, such as one's child. Or just consider the difficult questions involving balancing the risk that the innocent are punished with the risk that the guilty go free. There will also be Mersenne questions about involving morality and balancing risks and benefits between persons. For instance, what risks we may morally impose on others in exchange for a good to ourselves depends in a complex way on one's relationship to these others, the probability of the risk, the degree to which these others accept the risk, the benefit to self, and so on.

Whenever I drive, I risk killing other drivers, their passengers, pedestrians by the side road, and so on. But the probability of these awful outcomes is very small, and typically other people on or by the road have accepted reasonable risks (or have had them accepted by proxies, in the case of children), so these dire but unlikely outcomes typically do not render it impermissible for me to go to the grocery store to pick up ice cream.⁹ But when the risk is higher, say because I am tired and sleepy after a long day and hence less likely to be a safe driver, the matter becomes less clear. At some point, as the risk increases, it becomes impermissible to go to the grocery store for ice cream.

In all these cases, we have the Mersenne question of grounding the fact that the switchovers happen where they do. Expected utility utilitarians¹⁰ will have a nice answer

⁹I leave open the question whether concerns about global warming render it impermissible.

¹⁰As opposed to actual-outcome utilitarians who evaluate actions morally based on the actual utilities that would result from an action. **RailtonConsequentialism (RailtonConsequentialism)** calls this "objective consequentialism", as opposed to the "subjective consequentialism" of which expected utility consequentialism is a special case.

to this problem. But utilitarianism, as already noted^{??backref}, has many highly counterintuitive implications.

1.3. Orderings between goods. Under ordinary circumstances, it would not be reasonable to choose to be a mediocre mathematician rather than a superb musician. But suppose one's choice is whether to be a superb musician or a superb mathematician? Here we are dealing with incommensurable goods and either choice is reasonable.

So now let's ask this general question: Is it reasonable to choose to be a mathematician of quality α rather than a musician of quality β ? Again, we have a function that takes a number of variables, including α and β and the circumstances, and tells us whether (a) it is reasonable to opt to become a mathematician but not reasonable to opt for music, or (b) both are reasonable, or (c) opting for music is reasonable but opting for mathematics is not. And, just as before, it is very plausible that the function is quite complex.

The problem obviously generalizes to all the many kinds of pairings of incommensurable goods there are. In each case, there will be some function of many variables encoding the correct rational evaluation of the situation, and we will have the Mersenne question of what grounds the fact that this function, rather than one of the infinitely many others, encodes the correct rational evaluation.

We also have Mersenne questions here that involve qualitative rather than quantitative comparisons. Other things being equal, social pleasures are better than solitary ones. This seems rather arbitrary. What makes it be so?

In the preferential treatment and moral risk examples, utilitarianism offered a nice solution. But incommensurable goods are a problem for any plausible utilitarianism. Utilitarianism comes in two varieties, depending on whether the good is pleasure or the good is satisfaction of desire. As Mill famously noted, it is essential to the plausibility of utilitarianism that one be able to make a distinction between lower and higher pleasures, so as to get the common-sense conclusion that it is better to be Socrates unsatisfied than to be a satisfied pig (**MillUtilitarianism**).

But once one makes the distinction between lower and higher pleasures, or lower and higher desires, incommensurability quickly shows up, since different kinds of pleasures and desires do not simply come in a linear ranking. Let's suppose that you get more enjoyment and satisfaction of the desire for truth from mathematics and more enjoyment and satisfaction of aesthetic desire from music, and let us suppose (contrary to typical situations) that your choice of life will not affect anyone else. Then it seems right to say that the mathematical and musical lives are incommensurable even on utilitarianism. But even if they are not incommensurable, but equal or one is better than the other, we still have a Mersenne problem as to what level of quality of mathematical life exceeds, equals or falls below what level of quality of musical life. And in fact it will be more complex than that, in that the quality of a mathematical or musical life is clearly multidimensional.

One might try to get out of this by hoping for some precise definition of the degree of pleasure or the strength of a desire. Perhaps there is a perfect neural correlate of the degrees of pleasure or the strengths of desire that can be quantified in a single number. But such an approach is likely to lead to the swinish utilitarianism that Mill wisely rejects. For presumably the neural correlate can be manipulated directly, and the pig could be given pleasures which, in terms of neural intensity, exceed the highest of Socrates' refined joys, and could be made to have a degree of intensity of desire for its swill far exceeding Socrates' desire for virtue.

Moreover, any neural approach is likely to fall prey to questions of cross-species comparison. While pig and human brains are similar, they are not the same, and neural states of pleasure and desire are likely to be merely analogical between them (and if that's not right, then replace the pig by octopus with its decentralized brain). Clearly at least some comparisons between human and porcine goods are possible: a tiny human pleasure is worth less than a great porcine one. As one increases the human pleasure and/or decreases the porcine one, there will come cases where neither of the two is to be preferred, and then eventually cases where the human pleasure is to be preferred over the porcine one. But where exactly the cross-over points are is not something we can just read off the

neural correlates. And things get even messier when we compare humans to possible beings that have no brains, such as intelligent robots (if these are possible) or aliens with very different biochemistry.

And even if one could give some such precise neural formulation, we would still have the Mersenne problem of why *this* formulation corresponds with true value rather than some other.

1.4. Intersubject aggregation of value and population ethics. Whether or not consequentialism is true, there are some questions which need to be settled in a consequentialist way, for instance questions where the stakeholders are strangers one has no special obligations towards and where there are no relevant deontic considerations beyond the consequences. If we aim for the consequentialistically best outcome for more than one person, we will need a way of aggregating value between these subjects. A particularly difficult set of cases comes up when the number of subjects in existence varies between the options.

Perhaps the most straightforward option is to *add up* utilities across the affected population. This faces several problems. The most famous is Parfit's repugnant conclusion (**ParfitRP**). Any finite scenario full of highly fulfilled people can be beat by a scenario with a much larger number of people whose level of fulfillment is minimal. Yet it is implausible to think that we should aim at vastly multiplying human population at a minimal level of fulfillment.

A technical problem here is the following. In decision theory, utilities are normally considered to be defined up to positive affine invariance: they can rescaled and shifted without changing anything. More precisely, for any positive number α and any number β , if we consistently replace every utility x with $\alpha x + \beta$, then we have not changed anything. But if we add utilities across a variable number of individuals, although multiplying all utilities by a positive factor α makes no difference to our aggregate decisions, the addition of a constant β to every utility can make a difference. For instance, suppose we have a choice between these options:

- (i) there are ten individuals each enjoying a utility of 15 each
- (ii) there are twelve individuals enjoying a utility of 10 each.

As it stands, on the additive model of aggregation, (i) has a total utility of $10 \cdot 15 = 150$ and (ii) has a total utility of $12 \cdot 10 = 120$, thereby yielding a preference for the ten individual option. But if we apply the affine transformation with $\beta = 25$ and $\alpha = 1$, then the utilities enjoyed by the individuals in the two scenarios become $15 + 25 = 40$ and $10 + 25 = 35$, yielding respective totals of $10 \cdot 40 = 400$ and $12 \cdot 35 = 420$, flipping the preference in favor of the larger population option.

Essentially, the technical problem is that we need a “zero point” for utilities. If we have such a point, then increasing the number of people with utility above zero will always improve the outcome, while increasing the number of people below it will make things worse. Now, while there may be clear cases—someone might say that Einstein’s life was above the zero point, but a life of constant torture would be below—precisely defining a zero point in terms of the vast multitude of various good- and bad-making features of a human life involves a large number of Mersenne questions.

The most natural alternative to adding utilities is averaging them. As Parfit has noted, this leads to the another unpleasant conclusion: if the average of some nation’s utility is even slightly below the average utility for all human beings, then things would be better if the people in that nation didn’t exist.¹¹

Furthermore, averaging only escapes the repugnant conclusion for humans given assumptions about what the rest of reality is like. Suppose that it turns out that humans are outnumbered by non-human persons by a factor of ten, and the overall average utility in

¹¹The problem becomes perhaps even more vivid if we include non-human animals as subjects. For it may well be that no non-human animal enjoys a utility greater than that of an average human. One way to this conclusion is the intuition that no non-human animal is such that we should save its life instead of an average human’s. Another way is to reflect on the fact that an average human enjoys goods—say, moral or cultural ones—that are qualitatively higher than those of any non-human animal. On any averaging view, then, it seems it would be good to allow all non-human animals on earth die out. Some anti-natalists may disagree about the intuitions regarding average human life (**Benatar1997**).

reality is terrible, as most persons live lives of horrific misery, so that the average utility of a non-human person is -100 while that of a human is 10 . This makes the average utility overall be $(-100 \cdot 10 + 10)/11 = -90$. Now suppose we have a choice between two options:

- (i) make every human enjoy a utility of 100 (a life of deep fulfillment) without changing the number of humans
- (ii) multiply human population by a factor of 10 , but make each person's utility be a miserable -50 .

On (i), overall average utility becomes $(-100 \cdot 10 + 100)/11 = -81.8$. On (ii), it becomes $(-100 + (-50))/2 = -75.0$. In other words, multiplication of human misery, with sufficient increase of population, may be better than making all humans happy. This is even more repugnant than the original repugnant conclusion, since we are "improving" things by making people not merely slightly well-off but by making them actually miserable, just not as miserable as the average.

The last point vividly illustrates a well-known problem with averaging: what decision is right depends on epistemically inaccessible facts about intelligent life outside earth. One can, of course, solve this by restricting the averaging to our own species and hoping that we won't meet aliens. But we can still get some version of this repugnant conclusion simply within the human species. We can imagine a situation where three quarters of humans are found in a repressive nuclear state, on average have an utterly miserable flourishing level of -100 , and there is no way for those of us outside that state to remedy the situation. Suppose instead we have a choice between two policies for the rest of the world: keep the population of that part of the world constant and bring everyone to enjoying approximately a utility of 20 or triple the population and bring everyone down to a pretty miserable -35 . The resulting averages will be $(-100 \cdot 3/4) + 20 \cdot 1/4 = -70.0$ and $(-100 + (-35))/2 = -67.5$. As in the alien case, we have a nasty version of the repugnant conclusion: lots of people at a low level of happiness—indeed, a level of significant misery—outweigh a moderate amount of people at a high level of happiness.

It is worth noting that in the case of averaging, there is a decision point about aggregating across time. Averaging the utility across all subjects at all times is mathematically straightforward if the the number of subjects across time is finite. If it is infinite, there are many additional complications.¹² However, averaging across all subjects at all times could seriously exacerbate some of the repugnant conclusion worries. For instance, if non-rational conscious animals count, then the vast majority of conscious animals on earth across all time are relatively primitive, say on the level of lizards and squirrels, and hence not capable of much flourishing. As a result, the average level of happiness may easily end up being even lower than if we just average at the present time, thereby making it more practical to increase average flourishing by vastly multiplying subjects (human or not) with low levels of flourishing.

The other intuitively natural option is to average simultaneous utilities at each time, and then integrate them across time. This runs into at least three special problems. First, we have a difficulty of how to account for a multiverse whose universes do not share a common time-line. Second, given Relativity Theory, we need a privileged sequence of reference frames—a privileged foliation—to define the simultaneity of utilities. Third, and perhaps most problematically, many aspects of the flourishing or languishing of humans are difficult to localize in time. Examples include having a *diversity* of cultural experiences over a lifetime, having one's goals be posthumously fulfilled or frustrated, or failing to ever find a good friend.

It is possible that some simple function without free parameters can be used to aggregate utility across people in a way that avoids paradoxes. But it seems unlikely. Intuitively,

¹²*If the infinity is countable, then any averaging will have to involving a limiting procedure. For instance, if u_i is the utility of the i th subject, then we can aggregate by computing $\lim_{n \rightarrow \infty} \sum_{i=1}^n u_i$. However, the value of that limit is likely depend on the order of the subjects if infinitely many of the u_i are positive and infinitely many are negative (unless $\sum_{i=1}^n |u_i|$ is finite). If all the subjects are in the same spacetime, we might try to order the subjects temporally. That itself carries some decision points. Do we order subjects temporally by the beginnings of their existence, the middles, or the ends?

we have some sort of diminishing returns as we increase the number of people with a minimal level of happiness, but the diminishment is not the simple “one over the population count” multiplier of the averaging solution. It seems very plausible that an aggregation function that does not generate some repugnant conclusion will be rather complex and will have multiple free parameters—or even infinitely many, as when having to choose a concave transformation function as in Ng1989’s (Ng1989) Theory X’. And then we can ask what grounds these parameters’ having the values they do.

The above assumed, for ease of modeling, that we could reduce the utility enjoyed by each individual to a number. For reasons discussed in Section??backref, this is itself unlikely. Our flourishing seems to consist of a number of incommensurable goods such as friendship, understanding, play, etc. Moreover, even if we reduce flourishing to pleasure or desire satisfaction, it is likely that we have incommensurable pleasures and incommensurable desires. All this greatly complicates any aggregation procedure, multiplying the number of its free parameters.

Finally, even simple aggregation of pleasure or desire-satisfaction over a lifetime raises analogues of the repugnant conclusion. Replacing eighty years of a deeply fulfilled life by a million of barely positive utility does not seem appropriate. Averaging, on the other hand, is also wrong: my life would not improve by being reduced to its best day.

1.5. Double Effect. Consider these two cases, where all the people other than the dictator are assumed to be innocent.

TROLLEY: A runaway trolley is heading for a fork in the tracks. If nothing is done, it will turn left and kill five people who are on the track. You can redirect the trolley onto the right track, where there is one person on the track, who will be killed by the trolley.

DICTATOR: A dictator tells you to kill one person. If you fail to do so, the dictator will kill five others.

Assume there are no further relevant consequences. Deontologists tend to have the intuition that redirecting the trolley is permissible but obeying the dictator's orders is murder, while consequentialists assume that both are permissible, and indeed obligatory. Let us assume that deontologists are right.

The usual explanation of the difference is that in TROLLEY one merely foresees, without intending, the harm to the person on the right track if one redirects, while in DICTATOR one intends the death of the person if one obeys the order, even if one does so merely as a means to saving the five.

Often, this difference is formalized into a Principle of Double Effect, which says that an action with a foreseen evil effect and an intended good effect is not ruled out simply on account of the evil effect precisely when the evil effect is intended neither as an end nor as a means, and the bad consequences are not disproportionate to the good ones.

But now consider the following variant:

GUNSHOT: If you do not cause the death of one person, the dictator will kill five others.

The one person is tied up, and a loaded gun is permanently affixed pointing at that person. The dictator does not care about your intention. Never heard a gunshot before, you are curious what it sounds like, so you consider pulling the trigger.

In GUNSHOT, if you pull the trigger out of curiosity, you don't intend to kill the one. Moreover, the consequences on balance are slightly better than in TROLLEY: five are saved and one dies, but also your curiosity about gunshots is satisfied. But if we allow GUNSHOT, then with a bit of cleverness one can come up with a way of non-intentionally killing the one person in DICTATOR. Perhaps one is curious what size of hole a bullet makes in a shirt pocket.

How can one rule out the non-intentional killing in GUNSHOT without allowing one to accede to the dictator's demand?

Here are four lines of response. The first is that proportionality needs to hold not just between the good consequences and the bad ones, but between the *intended* good consequences and the foreseen bad consequences. In GUNSHOT, while the foreseen goods are

proportionate to the death of the person at the other end of the gun, the intended good of the satisfaction of curiosity is not proportionate to that evil. Effectively, this approach forbids one from counting goods that are caused by the foreseen evil towards proportionality, since if one were to intend these goods, one would be intending the evil as a means to them.

But there are other cases where counting goods causally downstream of evils towards proportionality seems exactly right.

BEAR: You have a leg caught in a trap. If you open the trap now, you can save the leg.

If you wait to open the trap later, your leg will become infected and will need to be amputated. However, nearby two other people are trapped in a way that they cannot escape (even with your help) until rescue comes tomorrow. A hungry bear is about to eat one of them, Alice. You foresee that if you open your leg-trap now, the noise of the opening will distract the bear from Alice so that it will eat Bob instead. Once the bear has eaten one person, it won't eat again until after rescue comes.

The problem here is that a foreseeable consequence of your opening your trap is Bob's getting eaten, and saving a leg is not proportionate to Bob's death. Now, it is true that Bob's getting eaten keeps Alice from getting eaten. But if we do not count goods causally downstream of evils towards proportionality, then we cannot count Alice's being saved among the goods in our proportionality calculation, and all we get to compare is your leg and Bob's life, and hence there is no proportionality. But, intuitively, it is permissible to escape the trap even if this leads to a different person getting eaten.

A second line of response is that intentionally firing a gun pointed in someone's direction is very "close" to intentionally killing them, and we should forbid not only intentionally killing innocent people, but also actions that are very close to such killing (**FootAbortion**; **DelaneyCloseness**). But it is difficult to see why intentionally firing a gun that happens to be pointed in someone's direction should count as too close to

killing, but redirecting a trolley at someone does not. After all, nothing of moral significance should hang on the means by which one redirects the trolley. Now imagine that the trolley is heading by inertia towards the left track, but a carefully placed barrel of gunpowder near the junction can push the trolley in the direction of the right track. And why should it matter whether it is a trolley or a bullet that is being powered by the gunpowder? (We can imagine the trolley is shaped like a bullet train!)

Another motivation for closeness theories is cases like the following variant of an example of Mark Johnston (**DelaneyCloseness**):

ARSON: A terrorist has hidden a bomb that, if not found, will kill a dozen people. You drive the terrorist to the house where he correctly believes his elderly bed-bound grandmother is sleeping, and you burn it down in order that the terrorist might think his grandmother was burned to death. Being afraid for his other grandmother's life, the terrorist reveals where the bomb was hidden.

Notice that nothing in your plan depends on the terrorist's grandmother getting burned to death, or even being in the house—if she died of natural causes five minutes before you arrived, your plan would have worked just as well. You just need the terrorist to *think* she was burned to death. And the means for getting the terrorist to think this is not the grandmother's death, but the house being on fire. If the grandmother was miraculously transported by angels out through the back window, that wouldn't harm your plan. Yet surely you did something very close to intentionally killing the terrorist's grandmother, so close that it is about as wrong as intentionally killing her.

There is indeed something intuitive about closeness theories. However, on such theories we have a seemingly arbitrary parameter defining the forbidden degrees of closeness to intentional killing, and we have a Mersenne question about what grounds the parameter's value. It is worth noting, too, that at least one of the parameters is actually numerical. Suppose one knows, but the terrorist does not, that the grandmother had serious health

problems and hence that there is a probability p that she was already dead. If p is sufficiently close to 1, there is no significant closeness between one's action and killing (assuming there is no one else in the house). If p is close to 0, the action is too close to killing. Where is the cut-off? One may suppose $1/2$ is a natural cut-off, but that does not seem right. For instance, playing Russian roulette with the terrorist's grandmother in order to terrorize the terrorist seems impermissible, too.¹³

OBKoons (OBKoons) have, however, offered a view of Double Effect on which it *does* matter whether we are dealing with a trolley or a bullet, and where Mersenne questions are less apparent. The reason is that trolleys and bullets are artifacts with different socially-assigned ends, and when one employs an artifact according to its usual operating instructions, the usual ends count morally. The end of a bullet is killing, and when one propels a bullet in that direction, we might say that one's action counts as close enough to intentional killing.¹⁴ One difficulty with this view is that it claims a perhaps implausible moral difference between firing a *gun* and detonating explosives that happen to be packed in a steel *pipe* sealed at one end with a ball-bearing inserted on the other side of the explosive, when both the gun and the pipe are pointed at the same innocent person.

¹³**LiaoCloseness (LiaoCloseness)** suggests a way out of the closeness problem by means of a pluralistic theory on which other considerations, such as intentionally involving someone in a situation that is in fact harmful or using someone as a mere means, are employed to argue that intuitively impermissible actions that are close to killing are in fact wrong. The ARSON case probably escapes Liao's suggestions. There is no intention to involve the grandmother, and *she* is not being used, since all that matters is that the terrorist *thinks* she is in the house. But it is likely that Liao's plural considerations will have seemingly arbitrary parameters, such as how harmful a situation has to be to rule out the involvement of the innocent person, or how intimately involved a person has to be involved for them to count as being used as a mere means.

¹⁴**OBKoons (OBKoons)** actually claim that one *intends* killing in such a case. This seems wrong. After all, after the bullet lands, one can rationally, and with no change of one's ends, try to save the person hit by it. But it is not rational to act against one's former ends if one has not repudiated them. However one does not need the implausible claim about intention: all one needs is that the action is close enough to intentional killing for moral purposes.

Additionally, the O'Brien–Koons account does not entirely escape Mersenne questions. An artifact has a particular end and particular set of operating instructions in virtue of patterns of social behavior. However, there are infinitely many functions from social behavior to ends and operating instructions that roughly match our intuitions but disagree on edge cases, such as exactly how often must forks be used to scratch one's back for them to acquire the end of scratching the back. Again, we need an explanation of why this function rather than another defines the artifact for moral purposes.

A third line of response depends on contingencies of psychology. It seems unlikely that one could fire the gun pointed at Bob's heart without intending Bob's death, unless one has done something to protect Bob (say, putting a shield between the gun and Bob). But note that a really callous person who does not care about human life probably *could* fire a gun that is pointed at Bob simply in order to hear the gunshot. Our actions have plenty of consequences we know about but that we are completely indifferent to, and hence do not intend. I know that moving my fingers to type this sentence disturbs air molecules. But I have no intention to disturb air molecules. For if it turned out that due to some random quantum oddity the air molecules were unmoved by my fingers, my prediction that I will disturb the air would be falsified, but my action would not be unsuccessful in any way. But an action that fails to achieve one of its ends is at least partly unsuccessful.

We might, however, make an Aristotelian move here. Perhaps it is abnormal to fire a gun pointed at Bob's heart without intending death, and we are morally responsible for the *normal* ends of an action in addition to its actual ends. And perhaps our human nature defines what ends an action *should* have, given the agent's knowledge of the circumstances. This is a version of the closeness view, perhaps similar to the **OBKoons (OBKoons)** version.

A fourth response notes that there is something callous about aiming at a minor end when there are evident great evils at stake. Consider:

CANDY: A runaway trolley is heading for a fork in the tracks. If nothing is done, it will turn left and kill five people who are on the track. You can redirect the trolley onto

the right track, where there are five persons on the track, who will be killed by the trolley, but turning the knob to redirect the trolley will cause a candy to pop out for you to eat.

It is callous to redirect the trolley for the sake of the candy. Similarly, it is callous to fire a shot in the direction of a person in order to see the hole the bullet makes in their clothing. The intended end of the action is absurdly small in comparison to the foreseen harm.

To expand on this response, imagine that Carl is the agent in TROLLEY, but Carl is someone who does not have much in the way of moral feelings, though he does intellectually desire to the right thing. Carl also enjoys redirecting trolleys, though like most people he rarely has a chance to do so. So when he finds himself in TROLLEY, he rejoices, and he redirects the trolley solely for the fun of it, while reasoning that by Double Effect he is acting permissibly, because the death of the one person on the left track is not a means to the pleasure of redirection, and proportionality holds, because on balance the effects are good.

What is wrong with Carl's action? It seems that he is being callous: the one person on the left track dies for Carl's pleasure. If Carl were intending to save the five on the right track, in the way that we normally expect someone in a trolley case to do, there would be no callousness. This suggests that in checking proportionality, we need to compare the *intended* good against the *foreseen* evils.

We could simply specify that the intended goods are proportionate to the foreseen evils. However, BEAR points away from that. In BEAR, your intended good, the saving of the leg, is not proportionate to the foreseen evil of Bob's death. Moreover, let's modify the case of Carl and the trolley.

LEG: Dave is strapped down near the trolley track in such a way that if the trolley turns left, it will destroy his leg. Dave terrified of his leg being crushed, and he is about to redirect the trolley, but he notices that there is a person on the right track. He regretfully decides that he must sacrifice his leg and do nothing. But then in turn he notices that on the left track there will be five people who will be saved by

redirecting the trolley, and concludes that redirecting is permissible because now the consequences are positive.

The overall consequences function in Dave's reasoning as a defeater to the observation that redirecting the trolley will result in the death of the person on the right track, rather than as an end.

While Carl is callously and acting wrongly in having a person die as a side-effect of a trivial end, Dave is acting for a quite serious end. It would be better if Dave adopted the lives of the five people on the track as an end as well, but given his terror at the amputation, it is not reasonable to *require* that (though it would be wrong for him to redirect the trolley if the only other relevant effect he saw was the death of the person on the right track).

Our fourth response thus suggests a conjunctive proportionality condition in the Principle of Double Effect:

- (1) First, the intended good is not trivial in comparison to the foreseen evil, and, second, the foreseen goods are proportionate to the foreseen evils.

Now, the triviality condition clearly involves an apparently arbitrary parameter measuring relative triviality, and raises serious Mersenne questions.

We have seen that of the available solutions to the problem presented by GUNSHOT, the two most tenable ones involve parameters calibrating closeness or triviality, and hence raise Mersenne questions.

Additionally, we should think a little about the condition that the goods—whether specified as foreseen or as intended—are proportionate to the foreseen evils. To a first approximation, one might opt for a utilitarian analysis: the overall utility is positive. We have already seen serious problems with this in ??backref in regard to uncertainty, expected value and risk, as well as incommensurability. Further, there is probably a significant overlap between deontology and a friendliness to partiality in ethics. Imagine a trolley problem where on the right track there is a stranger and a cat and on the left track there is one's child. The utilitarian consequences of redirection are negative, but the fact

that the person on the left track is one's child seems to make it permissible (some will even say required) to redirect.

But we have also seen in ??backref that taking relationships into account has many apparently arbitrary free parameters. So the complexity only multiplies.

1.6. A miscellany of other Mersenne questions.

1.6.1. *Introduction.* There are many other cases which involve thresholds or transitions that appear to be arbitrary. We will discuss them briefly. Some readers will doubtless disagree with a number of my examples. But a sufficient number of the remaining examples should still compellingly raise Mersenne problems. And the list is not exhaustive: the reader should be able to generate more items, as the following are largely given as examples.

1.6.2. *Ethics: Gratitude.* If x receives a benefit from y 's ϕ ing, the degree of gratitude that it is appropriate for x to show y covaries with:

- (i) benefit to x ,
- (ii) cost to y , and
- (iii) deontic status of ϕ ing (from wrong, to merely permissible, to obligatory, to supererogatory).

If the benefit is b , cost is c and deontic status is d (e.g., negative for wrong, zero for merely permissible, between zero and one for obligatory, and one or more for supererogatory), we will then have some function $g(b, c, d)$ that specifies how much gratitude is needed. This function has to include at least one parameter we can raise a Mersenne question about, because the function has to convert from the units of cost and benefit that b and c are in to some sort of units of gratitude. As a first attempt, we might try the product rule:

$$g_1(b, c, d) = \alpha b^+ c^+ d^+,$$

where α is a constant of proportionality, and for any real number x , we use x^+ to denote the positive part of x , i.e., $x^+ = x$ if $x > 0$ and $x^+ = 0$ if $x \leq 0$. This captures the idea that gratitude is only apt when there is a benefit from y 's ϕ ing to x (if $b < 0$, then $b^+ = 0$). It

also captures the idea that wrong and merely permissible actions, not being praiseworthy, deserve no gratitude, since for wrong and merely permissible actions we have $d^+ = 0$.

But the g_1 gratitude function is not quite right, because it implies that when the cost is zero, the appropriate gratitude is zero, despite the broadly held intuition that if there is a God, we owe him gratitude for our existence, even though creation has no cost to an omnipotent being.

We might try to combine b^+ with c^+ by addition rather than multiplication. This captures the idea that gratitude can be appropriate even when there is no cost to benefactor, but it misses the fact that where there is no benefit, no gratitude is appropriate. Perhaps a formula such as:

$$g_2(b, c, d) = \alpha b^+ (\beta + c^+) d^+,$$

is a better match to our intuitions, where β is some positive value (in the same units as the cost). This does imply that where benefit or deontic status is zero or negative, no gratitude is appropriate, but gratitude for costless things is appropriate. But now we have Mersenne questions about *two* parameters. Moreover, we have a Mersenne question about what grounds the gratitude formula having the exact form it does, with three multiplications and two additions.

But in fact things are even more complicated. We need to distinguish actual benefits received by x from the benefits that y intended x to receive. Both are relevant in subtly different ways to the degree of gratitude. Gratitude for trying to benefit need not be as large as for actually benefiting. And in the case of costs, while expected costs are probably more relevant than actual costs, the actual costs incurred by the benefactor likely enter into the rule of gratitude. We can thus expect a complex formula, with Mersenne questions about the form of the formula as well as about a number of constants in the formula.

1.6.3. *Ethics: Private property.* There is something quite compelling about Locke's idea that if, in the state of nature, you mix your labor with an object (including an area of land), leaving as much and as good for others, you gain exclusive rights over the object.

A number of Mersenne questions come up. What exactly counts as labor? Suppose I come across a stick in the grass, absentmindedly pick it up, and lay it back down. As a result, the stick is now a little easier to find, as some grass that was over it is now under it. It doesn't seem right to say that I now own the stick: my input was too minimal.

What is the exact extent of the exclusive rights? Suppose that I have erected a large building, and a science teacher uses it to demonstrate how to measure the speed of sound by recording the sound of a loud clap and the ensuing echo. In doing so, the teacher has deliberately bounced sound waves off my building, thereby vibrating it. This is surely too minimal a "use" to fall under the scope of exclusive rights. The example suggests that perhaps use that does not impede the owner's use is outside the scope of the owner's exclusive rights. But that principle is generally denied: I don't get to use books in my absent colleague's office unless past practice makes it clear that my colleague consents. We have some idea of a minimal use that falls outside the scope of exclusivity, but where the line lies is unclear.

And how do we abandon property rights? I pick up a rock, and hit it against another rock a couple of times in the way I would to knap it into the head of an axe. I then decide that perhaps this rock isn't the best candidate for an axe, and I put it down. I do have in the back of my mind the thought that if one day I want to make an axe and I can't find a better rock, I will come back to this one. My memory of the rock fades gradually. Early in the process, I own this rock with which I mixed my labor. Eventually, I have abandoned the rock. When exactly? Just how weak a thought of "I might come back for this" is enough for me to continue owning it?

These are nitpicky questions. But by the law of excluded middle, at any point either it is or is not permissible for another party to make a particular use of a perhaps owned object. What grounds exactly where, in the foggy area between the clear and unclear cases, the use is permissible and where it is impermissible?

1.6.4. *Ethics: Torture.* On strict deontological views, one shouldn't torture one innocent person to save any number of lives. But of course it would be permissible to gently prick

someone with a pin to save a single life. Somewhere between the pinprick and the torture is a transition. What makes the transition be where it is?

On threshold deontological views, it is wrong to torture one innocent to save a small number (say, one or two) of lives, but it is permissible to do so to save a very large number (say, a billion). Again, we have a transition to be explained.¹⁵ And note that even if one is a strict deontologist about torturing the innocent, likely one is a threshold deontologist about some other things. Thus, one may think it's permissible to save an innocent life but not permissible to lie to get a deserved salary raise, and hence there needs to be an explanation of the grounds of the transition from permissibility to impermissibility as the gravity of the case changes. Or one may think it is permissible to trespass on a neighbor's property to save a cat's life but not to save a grasshopper's. I expect that every moral realist who isn't a full-blown consequentialist is a threshold deontologist about some things. And each threshold raises a Mersenne question.

1.6.5. *Ethics: Self-defense.* It is widely held that lethal self-defense an aggressor is permissible. There are at least two questions that come up here. The first is akin to proportionality in double effect: what degree of aggression licenses lethal self-defense. For instance, if an assaulting party clearly threatens only to cut off your hair, lethal self-defense is unjustified, even though such an assault would constitute actual bodily harm according to the British case of *DPP v Smith* (McCombe2017). On the other hand, if the assaulting party is likely to kill you, then lethal self-defense is surely permissible. Lethality of an attack is a natural line to draw, but it is not the right line. If an evil surgeon is going to cut off your healthy legs without your consent, lethal self-defense appears permissible even if you are confident of survival. Furthermore, the probability of the aggressor's succeeding in inflicting the threatened damage matters. Lethal force is not permitted against a malicious child with a foam sword, even though there is a tiny chance that the sword will trip you up, resulting in your hitting your head on a rock and dying. At the same time, certainty of the threatened damage is more than is required. Finally, what I called "lethal self-defense"

¹⁵I am grateful to Philip Swenson for this example.

itself comes in probabilistic degrees: shooting an attacker in the head is nearly certain to kill; hitting them with a rock is less so. So there is now a Mersenne question of what explains what combinations of probability and damage justify self-defense with what probability lethality.¹⁶

The second, and more conceptually complex, question is what constitutes aggression. Again, there are clear cases of what does not and what does. If a terrorist hands Alice a weapon and informs her that she will be killed if Bob is still alive in fifteen minutes, and it is not physically possible for Alice to use the weapon against the terrorist, she cannot engage in lethal self-defense against *Bob* on the grounds that by being alive he too is an aggressor. On the other hand, the terrorist's credible threat makes the terrorist an aggressor, and it is very plausible that Alice could legitimately engage in lethal self-defense against terrorist if it were to become possible.

In between the extremes of threatening by simply being alive and threatening by culpable voluntary attack is a range of cases, such as threatening by breathing (e.g., there is only enough air in spaceship for one of two people), threatening by involuntary activity similar to voluntary activity (e.g., a muscle spasm is about to happen resulting in a detonator being activated), threatening in non-culpable non-moral ignorance (e.g., thinking one is controlling a video game when one is controlling an armed real-life robot), threatening by voluntary and right activity (e.g., a police officer using justified lethal force), wrongly threatening in non-culpable moral ignorance (e.g., innocently not knowing that the complex medical situation at hand is one where in fact disconnecting the ventilator is impermissible), and threatening in culpable moral ignorance (e.g., engaging in genocide because it is more convenient to come to agree with the leader than to investigate the moral questions for oneself).

Moreover, the lines between these options are themselves more continuous than may initially strike the eye. It may seem, for instance, that it is very natural to draw the

¹⁶Some will say that one's intentions are also relevant. **AquinasST** (**AquinasST**, II-II Q64 A7), for instance, held that a private individual is not permitted to intend the death of an assailant.

line between cases where the threatening party is acting rightly and where they are acting wrongly. But now consider cases where the threatening party is acting only *slightly* wrongly, for instance because they are a police officer engaging in a use of lethal force that is only slightly disproportionate.¹⁷ Likewise, one might think the difference between inculpable and culpable ignorance is sharp, but there are cases where the ignorance is only slightly culpable. For instance, consider the case of the doctor disconnecting someone from a respirator incorrectly thinking that the disconnection is permissible, where the reason the doctor was mistaken about the morality of the situation was because she was five minutes late for an ethics lecture many years ago in medical school, and their lateness was only very slightly culpably wrong (they should have put slightly more effort in running to catch the bus).

It is thus unlikely that there is a natural place where we draw the line as to the nature of the aggression. But given realism there are truths of the matter, and a Mersenne question as to why they are what they are.

1.6.6. *Ethics: Respect.* We need to show respect for rational beings. This respect includes such things as not killing them when they are innocent and non-aggressive, not eating them (except perhaps in extreme circumstances), not acting as if they were fungible, treating them as ends rather than as mere means, and so on. But what is a rational being? First, we have a distinction between an individual- and a kind-based concept of rationality: on the former, a being is rational to the extent that it currently has certain rational powers; on the latter, a being is rational to the extent that it is of a kind that should have certain rational powers. But whichever we choose, and plausibly there are principled reasons to

¹⁷There may also be cases where the wrongfulness is not of the relevant sort to justify lethal self-defense. For instance, suppose that for the sake of personal witness to the importance of solving problems non-violently, Alice had vowed non-violence for life. Bob now threatens Alice's life in a paradigmatically murderous way, and Alice pulls out a gun and is about to shoot Bob in self-defense. Assuming that a vow of non-violence is morally binding, it is wrong for Alice to shoot Bob. But the wrongness of Alice's shooting Bob does not justify Bob's switching to "self-defense mode" and pulling the trigger before Alice can pull hers.

choose one rather than the other¹⁸, we still have a Mersenne question as to the degree of rational power—whether actual or proper to the kind—that is needed for us to have duties of respect. Rational powers, after all, clearly come in degrees, and if at some point respect is called for, we need an explanation of why that point shows up where it does.

The question of what degree of rational powers is needed for respect is one that we actually face with regard to our treatment of higher mammals on earth, and merely hypothetically face with regard to extraterrestrial life. It is an important question. But, as usual, the Mersenne puzzle isn't that of determining what the fact is, but of what makes an answer be an answer, especially in light of the appearance that any threshold will be arbitrary.

1.6.7. *Ethics: Punishment.* Punishment should not be disproportionate to a crime. *Lex talionis* would provide a neat and elegant account of this: we do to criminals what they did to the innocent. But it is morally abhorrent to torture torturers or rape rapists. And even if we accept such abhorrent extremes, there are cases where punishment simply has to switch types of harm. If a thief does not have enough honest property of their own to make possible a deprivation equal to what they stole, imprisonment is a plausible alternate currency, but there is no simple and elegant formula providing an exchange rate between the value of stolen property (perhaps corrected for the economic state of the victim) and length of imprisonment, unless it turns out—as is quite unlikely^{back/forwardref}—that there is a precise way of quantifying personal utilities. Maybe there is no simple and elegant formula, but at least a complex one. If so, we have our Mersenne question about what grounds it. And even if there is no specific formula, there are bounds of moral acceptability: a day's imprisonment for stealing and destroying a car is insufficient; a lifetime's imprisonment for stealing a book is excessive. And what grounds the transitions between acceptability and unacceptability?

¹⁸Though they will be highly controversial, since a significant part of the fraught debate about the moral status of human zygotes, embryos and fetuses turns on this.

1.6.8. *Ethics: Consent.* Standards of consent necessary to permit one's being treated a certain way vary widely depending on the treatment. There are multiple dimensions in which we can measure the "strength" of a consent requirement: how well informed the consenting party needs to be, what age or level of intellectual development the party needs to have, what proxies if any can legitimately offer consent on the party's behalf, how unpressured the consent needs to be, how clearly formulated the consent needs to be, whether the consent must be specific to the case or whether prior blanket consent suffices, etc. Under ordinary circumstances, only a clear lack of refusal is needed for a pat on the shoulder. The permissibility of major surgery, however, has a consent requirement of significant "strength" along many of the above axes. And the permissibility of sex has a consent requirement of even greater "strength" along some of the above axes, since while proxy consent and prior blanket consent can suffice for major surgery, they do not suffice for sex.¹⁹ The mapping between the form of treatment and the multidimensional "strength" of consent is of great complexity, and has an appearance of significant arbitrariness. What grounds it?²⁰

Now, at least some of the norms regarding consent are constituted by social expectations. For instance, it is likely that many of the expectations concerning consent to companies' use of our personal information are such²¹, and a society where these expectations are more stringent or less stringent need not be corrupt. However, first, not all norms regarding consent are like that. A society that does not require sexual consent just *is* corrupt, no matter the context, and much if not all of our increased focus on patient autonomy in medical ethics, together with consequent emphasis on consent, is genuine moral progress. And

¹⁹It is tempting to explain this in terms of the fact that surgery—or at least the sort of surgery for which proxy consent suffices—benefits the patient regardless of the patient's consent, while sex is only beneficial when consented to. But this is arguably false. Parents can validly consent to an organ transplant between their children, even if the donor is not expected to benefit on balance (though generally there is a benefit from having one's sibling alive!).

²⁰This argument is further developed in **DurhamDissertation (DurhamDissertation)**.

²¹This example is due to Anne Jeffrey.

even in the case of privacy, it is plausible that some expectations are not merely socially constituted—it would plausibly be wrong, no matter what society said, to follow non-consenting people 24-hours a day with wall-penetrating sensor technology and broadcast all their activity to all willing to pay a small fee. Thus we need a ground for the distinction between the cases where the need for consent comes from social expectations and where it is independent of social expectations.

And even in the case where consent is constituted by social expectations, the mapping between social expectations and moral norms is likely a quite complex function involving apparently arbitrary parameters that raise Mersenne questions. After all, it's not always wrong to violate social expectations. Presumably, there is some kind of a proportionality between the strength of a social expectation and the strength of the resulting moral norm (how easily it can be overridden by other considerations, how much of a risk of violation it is worth taking, etc.), while the "strength" of a social expectation is a complex thing constituted by a distribution of individual expectations. Both the method of aggregating individual expectations into a social expectation and the proportionality constant between social expectation and normative force raise Mersenne questions.

1.6.9. *Ethics: Normative powers.* Consent is one special case of a normative power. We have a number of normative powers that we exercise through communicative acts with specific illocutionary force. With promises, we create obligations for ourselves, and with commands, we create obligations for others. With requests, we create reasons for others, and with permissions, we remove reasons for others.

The scope of our normative powers has limitations, though where exactly the limitations lie can be a matter of controversy. Perhaps the clearest case is requests, where as a society we have developed a broad spectrum of levels of insistence, signaled by linguistic and extra-linguistic cues. Normally, more insistent requests create stronger reasons and less insistent ones create weaker ones. We do not appear to have any lower limit on the strength of reasons we can create solely by requests. We can always add yet another "But, please, don't let me impose on you" to weaken the request. However, there is a

contextually-variable upper limit on the strength of reasons created by a request. One can roughly measure the strength of these reasons, say, by the cost to the requestee at which fulfillment becomes pretty unreasonable. The strength of the reasons to fulfill a request is then a function of variables including the intrinsic reason provided by the requester's needs (if someone is starving, one has reason to offer them food even if they don't ask for it), the degree of insistence, and the relationship between the two parties.²² Normally, then, the more insistent the request, the more it skews the requestee's reasons in favor of the requested action, but there is a limit to how far one can skew the strength of reasons away from the no-request *status quo*. If I am not actually in poverty, requesting money from strangers for my personal pleasure cannot create a very strong request-based reason, no matter how much I ask for it.²³

We have similar limits on the strength of reasons coming from the exercise of other normative powers. The example of promises is particularly interesting here in that we might think that our normative powers are strongest here since we use them to bind our own wills. But notice that while I can probably make a promise to defend my friend's life where the promise creates a reason whose strength is such that I am obligated to seriously risk my own life, no promise to come to my friend's party can create a reason strong enough to stand against serious risk to my life, unless there is something very special about that party.²⁴ Even in promises, thus, there are serious limits to how far we can affect the reasonableness of our decisions.

²²Note that the relationship may itself have been modified by the fact of the request. A friendship might be damaged by an unreasonable or rude request, and strengthened by the vulnerability revealed in a disclosure of need.

²³Though of course I might create a prudential reason to give me the money to shut me up, or out of fear that I am a mugger, but that's not a request-based reason in the sense I am talking about. There are also unusual cases where a request on balance creates a reason against doing something, such as when it is appropriate to punish a child by withholding from them something that they specifically want, but are not request-based reasons.

²⁴Perhaps the party is the best hope for reconciling with someone who is dying of cancer, and my friend cares deeply about the relationship.

Social convention sets certain aspects of the mapping from communicative acts exercising normative powers to the normative effects. It determines which normative powers are exercised in which words or gestures, and how various communicative and extra-communicative features affect the strength of reasons. But social convention works within the limits discussed above. Social convention can *say* that if I spit on my hand and shake hands after promising to come to your party, then I am obligated to come to the party even if it costs me my life, but in fact this action would not create any such reason, since skewing the intrinsic reasons thus far is just as beyond our normative power as running a one minute mile is beyond our locomotive power. But while we can find a physical explanation for the locomotive limits on an individual human, the normative limits raise Mersenne questions about all of their free parameters—and they have many, since there are multiple normative powers and in each one the limit will depend on multiple contextual factors.

1.6.10. *Political: Constitution problem.* Political philosophy also provides a number of examples of seemingly arbitrary parameters. Consider the constitution problem. A state has a written or unwritten constitution specifying what must happen for legislation to be valid and hence authoritatively binding on the citizens. But how is a constitution instituted? One theory is that it happens by the consent of the people, which can be understood as either the people at the time of the original passage of the constitution or the people currently governed by it. But obviously for any state of sizeable size it will be false that all the people have consented: some have not made their opinions heard and some have been overruled.

Requiring “consensus” or a supermajority raises the question of exactly how many dissenters can be tolerated, and once that question is answered we have a Mersenne question as to what grounds that cutoff being where it is. Requiring a simple majority or plurality involves one less free parameter: the cutoffs in having more than half of the voters or having more voters than any alternatives seem non-arbitrary. However, even a majority or plurality based system leaves questions about other parameters. Does one need a quorum of the governed? It would not seem right, for instance, to have a vote on a constitution

on a day where the bulk of the population is unable to get the polls due to a hurricane or a war, and as a result only a small number of unrepresentative citizens can express their opinion. But if a quorum is required, then of course we have a Mersenne question as to exactly what grounds the facts about what counts as quorum.

Voting cutoffs and quorums are relatively easy to quantify compared to the question of who the people giving their consent are? Presumably, small children should not be eligible. But where do we draw the line between small children and paradigmatic adult deciders? Any age-based line raises several Mersenne question—one about the numerical age cutoff and multiple questions about how age is measured (from fertilization, implantation, brain development, beginning of the birth process, completion of the birth process, etc.)? A cutoff based on mental capacity, on the other hand, involves many parameters that need to be set, because there is no single measure of mental capacity, and so one needs to have multiple measures with their respective weights. Moreover, we have a decision point on whether those who do not get a vote have proxies voting for them (e.g., parents) and, if so, who counts as whose proxy.

Or consider such details as how well-publicized the constitutional consultation needs to be, how clearly spelled-out the constitution needs to be for people's vote on it to be valid, how voting with more than two options is done (for, surely, there are more than two options for constitutional questions), and who gets to decide which options are presented for voting?

Some of the above questions only make sense in the case of a formal consultation process of a sort that has occurred rather rarely over the course of human history. If we are not to think the vast majority of polities to be illegitimate, the account needs to allow for implicit consent or custom.²⁵ But there things become even less clear. There will almost always be some citizens who regard the state as illegitimate—indeed, some regard any state as illegitimate. For many people, acceptance of a political system's legitimacy is not

²⁵Famously, **AquinasST** (**AquinasST**, I-II Q97 A3) holds that custom can create and abrogate law.

an simple binary question, but something that comes in degrees and has contexts. Imagine that two thirds of the population has a credence of two thirds that the political system is legitimate, and the remaining third of the population has a credence of ten percent in the system's legitimacy, and they express these credences in their actions. Should we then look at the average credence of the relevant citizens (e.g., those of age)—0.48 in my example above—and see if it meets some cutoff? If so, the cutoff will raise Mersenne questions. Moreover, people often do not just have a single credence in the all-or-nothing legitimacy of the political system, but rather have different credences regarding different aspects of legitimacy: is this a state that has the right to use violence against its citizens to enforce laws, is it a state that has a right to levy taxes, to draft citizens to defend it or do other work for it, etc.

Similar issues come up for other accounts of the constitution problem, such as **Hart's (Hart)** ultimate rule of recognition embodied in the customs of the officials. Who are the officials? What fraction of the officials can act differently before their dissent counts as undermining the custom?

If our interest is ethics, we can avoid these problems by adopting philosophical anarchism, and denying that law has any genuine authority.

1.6.11. *Political: Dissolution problem.* A set of Mersenne questions similar to those raised by the constitution problem is raised by the dissolution problem: the question of when it is that a political regime becomes illegitimate, and the respects in which it may be illegitimate (perhaps the street traffic regulations of the Third Reich were legitimate, but little else was). One might think of dissolution as resulting from the state's failure to keep its side of the social bargain. But there has probably never been a state that kept its side in every respect. It seems to be only gross failure that implies illegitimacy, but that raises a Mersenne question about what grounds the relevant degree of grossness.

1.6.12. *Political: Scope of authority.* We have a set of Mersenne questions regarding who lies within the scope of the state's authority. Typical human states have authority largely

but not entirely defined geographically—for an exception, consider ships under a country's flag that move on the high seas. Geography itself raises Mersenne questions. Suppose we say that a state in the future has authority over the same territory as it now occupies. But what grounds something counting as “the same territory”? We live on a planet that is constantly changing its shape, whether at a large scale due to movements of tectonic plates or a small scale due to children digging holes in beaches or a medium scale due to states building new off-shore islands. When a tectonic plate shifts, how does territory shift? There are precise ways one might try to answer these questions. We define latitude and longitude in terms of coordinates on a mathematically idealized oblate spheroid approximating the earth. We might then define “same territory” in terms of these coordinates. But there are infinitely many ways of mathematically modeling the earth at any given time and infinitely many ways of matching that model to the physical soil of the earth over time. And it seems unlikely that the geography of a planet is in the end what defines a state. It may well be the case that in the future a significant proportion of the earth's population will live on space stations or in the asteroid belt, and we will have ship of Theseus kinds of questions about identifying the sameness of a space station over time, and difficult questions about defining segments of the asteroid belt.

One might say that a state, or its people in their constitution, gets to define its own understanding of who counts as among the governed, and so a state can opt to define the governed as those occupying a certain portion of a certain specific diachronic oblate spheroid model or to define the governed as those within a certain specific distance of a particular landmark. But there must be limits to a state's normative power to define these boundaries, since otherwise a state could simply swallow up territory by mere fiat. Imagine, for instance, if France defined its territory in terms of all land within five hundred kilometers of the Eiffel tower, and then French agents conquered the world by secretly taking small bits of the tower and distributing them worldwide so that no place was more than 500 km away from one or more of them. It is very plausible that a state has some freedom in how it defines its boundaries, but that freedom is limited. And the range of ways of

defining the scope of authority seems like the sort of thing that would have many different parameters without privileged values, and hence raises many Mersenne questions.

One may wonder why questions about the legitimacy and scope of a political system are being raised as part of a discussion of ethical questions. There are two reasons. First, we have a moral duty to obey the commands of a legitimate state. Second, only those acting on behalf of a legitimate state are morally permitted to make certain onerous demands on the population, especially ones backed up by threats of violence.

1.6.13. *Just war*. Traditionally, at least the following three conditions have been required for a state to enter into a war justly:

- (i) the cause must be just,
- (ii) the value of the cause must be proportionate to the expected harms, and
- (iii) the chance of success must be sufficiently high.

Let's focus on (ii) and (iii). Note first that the conditions need to be combined rather than having independent cutoffs. If the harms threatened by the enemy are sufficiently large—say, a threat of large-scale genocide—a small chance of success is sufficient to justify fighting back, while if the threatened harms are small, then in light of the great harms of waging war, a larger chance of success is needed.²⁶

As in many questions like this, an expected-value utilitarian has a simple account of (ii) and (iii): you just compute the expected utility of waging war and not waging war. I will discuss utilitarian solutions in general in more detail in ??forwardref. But here is an intuition that suggests a problem with such solutions here. Suppose Country A intends to invade solely in order to murder the inhabitants of a small village with 1500 inhabitants in Country B, and the best military analysis shows that the attack can be easily repelled by using drones which will kill 2000 invaders with no friendly casualties. Clearly fighting back is right, even though on a short-term utilitarian analysis 2000 lives are lost by fighting back, and 1500 by not fighting back. Of course, there are standard things utilitarians will

²⁶For more on the interaction of the conditions, see Hurka2004 (Hurka2004) and AloyoChance (AloyoChance).

say: not fighting back will encourage more slaughter in the future, and so on. But it is easy to imagine cases where there will be no such encouragement. (E.g., this war might occur during a short malfunction in force-fields between countries, which force-fields when functional are perfectly effective at keeping peace, and a new engineering innovation can make the force-fields practically a hundred percent functional.)

This intuition is a part of the general conviction that in just war-time decisions when fighting on the side of justice, one prioritizes friendly non-combatants and friendly combatants to enemy combatants, while where exactly enemy non-combatants fit in is more controversial as is the question of how one prioritizes between friendly non-combatants and friendly combatants (**KasherYadlinMilitary**; **KasherYadlinAssassination**; **MargalitWalzer**; **McMahanInBello**). But the prioritization had better not be lexicographic: it is wrong to kill a million enemy combatants to save one friendly combatant. Thus, as many other cases, we have weights in proportionality computations—both in the *ad bellum* considerations of proportionality as well as in *in bello* decisions—and there do not appear to be any “natural and non-arbitrary” values for these weights. Once again, then, whatever these weights are, we have Mersenne questions as to what makes them be what they are.

Hurka writes:

Judgments about proportionality in war cannot be made simply or mechanically. Even if we have specified the types of good and evil relevant to these judgments, we must identify the specific effects of a given war or act, which requires comparing that war or act with alternatives that are merely hypothetical and can only be estimated given our available evidence. In addition, once the relevant effects are identified they must be weighed against each other, which often involves some indeterminacy because they are of irreducibly different types. (**Hurka2004**)

Given that Hurka is evidently right about the complexities involved, it is very implausible that there is a simple formula that gives the answer. Unless we want to embrace the

“indeterminacy” that he advocates for, it is very plausible that there are many parameters determining whether a war is just, parameters whose grounding needs to be explained. And even if there is indeterminacy, we need parameters determining when the matter is determinate and when it is on, on pain of a regress of indeterminacies—see the discussion of vagueness in ??forwardref.

2. Arbitrariness and continuity

Whatever the values of the parameters in the ethical Mersenne questions are, these values appear likely to be such that if we knew their exact values, we would find them arbitrary, like the 1102 earth-radii that Mersenne says is the distance between the earth and the sun at apogee, or the actual number of 23872. In physics, only a few seem to hold out a hope that the fundamental constants in the fundamental laws of nature may be “nice numbers” like 2, π , $\sqrt{2}$ or e . It seems intuitively even less plausible that things would so turn out in ethics.

And even if the parameters turned out to be such “nice numbers” in ethics, that would itself be a very surprising fact, because while, say, π seems very natural in physics, it seems rather less natural in ethics. Imagine that you should benefit your parent over a sibling just in case the ratio of benefits to parent vs. to the sibling is no lower than $1 : \pi$. That would itself seem arbitrary. It seems that whatever the numbers turn out to be, they will have an appearance of arbitrariness and of contingency.

Many of my examples involve thresholds, such as the amount of rational capacity needed for respect or the degree to which a government needs to care for the common good to have authority. It is plausible to reject the idea that there are discrete thresholds, and instead hold that there are continuous functions, say a function $r(x)$ specifying the degree of respect required to be shown to a being with rationality of degree x .

But then instead of explaining one threshold, one needs to explain the whole complex shape of the “respect function”. On the most naive version of this, rational capacity will be graphed along one axis and respect on another, which will raise Mersenne questions about

the slope of the graph at various points, the positions of any inflection points, and so on. And in reality, both rationality and respect have many dimensions, so what we have is a complex function of many arguments and whose values are multidimensional. In general, moving from thresholds to continuous functions only multiplies the degrees of freedom among the ethical parameters, and hence multiplies what is needed for explanation. Continuity does not solve the problem.

3. The human nature solution

3.1. The solution. On our Aristotelian picture, the nature of an organism grounds norms about what the organism's structure and behavior should be. In particular, the nature of the organism will ground many arbitrary-seeming norms, such as those governing the range of appropriate sizes of Indian elephants, the migratory behaviors of monarch butterflies, and the lengths of human femurs. Possession of this nature makes the organism be the kind of organism it is, and imposes on it the associated norms.

In the case of humans, behaviors include voluntary ones, and so it is unsurprising that there are norms governing these as well. And just as there are many parameters governing bodily structure and sub-voluntary behavior, there are many parameters governing moral behavior, all grounded in the form.

At the same time, Aristotelian optimism provides us with evidence as to what the parameters approximately are. The actual bodily structures of humans give defeasible evidence as to what normative human bodily structure is and the actual behaviors of humans give defeasible evidence of moral norms. And in both cases, we have ways of identifying healthier or more virtuous paradigms, using the optimistic idea that the various ways of doing well tend to hang together with some degree of unity, and the structure and behavior of such paradigms gives us further evidence as to the norms.

Admittedly, there appears to be a disanalogy between health and virtue. We might use a moral exemplar like Mahatma Gandhi or Mother Teresa to figure out moral norms, but we wouldn't use an athletic exemplar Usain Bolt or a Serena Williams to figure out

physical norms. One explanation of the difference is that Bolt and Williams have highly-developed traits that are specialized to a forms of life quite different from that of the typical human—namely, the life of a professional athlete—while Ghandi and Teresa’s excellences in justice, fortitude and mercy are as important in circumstances as in theirs.

The Aristotelian response to the Mersenne questions also raises the question of why the form includes these norms and not others. But here there is an easy answer available. The form is at least partly defined by the norms it includes. Thus, Mersenne’s question about the lion and the ant when reformulated into normative terms, as the question of why the lion’s strength *ought to* be greater than the ant’s, is easily answered: this follows from defining features of what make lions be lions and ants be ants.

The appearance of arbitrariness and of contingency in the ethical Mersenne problems is somewhat misleading: it is like the appearance of arbitrariness and contingency in the facts that water is H₂O and that carbon atoms have six protons. Water couldn’t have a different chemical structure and carbon atoms couldn’t have a different number of protons. But it is also an important truth here that there could be other substances that could have a different chemical structure or a different number of protons and similar behavior (KripkeNamNec1980). Similarly, *we* couldn’t have norms of preferential treatment other than the ones written on our nature, but there could be—and perhaps in this vast universe are—other rational animals with other such norms.

3.2. The incredulous stare. At this point, an objector may offer an incredulous stare. How can our human nature contain such a vast amount of moral information as the solutions to all the Mersenne questions require?

However, as **DurhamDissertation** (DurhamDissertation, Section 4.4) notes, it is plausible that a substance’s nature already grounds many complex dispositional facts about the substance. Furthermore, it is also plausible that it grounds many non-moral normative facts, such as about what size of heart is normal for a given body size, what force a bone ought to be able to withstand, and so on. Given that it is already this informationally rich,

why can't we also have great informational richness in the moral regime? On robust Aristotelianism, the form is not a mere summing up of the arrangement of the parts, so it can be extremely rich informationally.

In addition to this, there is also a somewhat technical line of thought. Let's try to quantify how much information needs to be specified for the Mersenne questions. In some cases, we have a finite number of alternative options, and the question is which of them is right (e.g., which hierarchical ordering on some finite set of incommensurable goods is the right one). Here, the information content is finite, and hence is that of specifying a single integer in a finite range. In other cases, we can encapsulate the answer to a question in a finite number of real numbers, say ones specifying the relative moral weight of some set of considerations. In these cases, the information content is that of a finite sequence of real numbers, which can be proved to have at most the information content of a single real number.²⁷ In other cases, perhaps we can encapsulate the information needed for a Mersenne question as a finite collection of continuous real-valued functions each of a finite number of real-valued arguments. But any such function can be encoded as a single real number.²⁸ In other cases, perhaps, the information for a Mersenne question may be

²⁷A single real number can encode two real numbers. One way to see this is to note that a real number x strictly between 0 and 1 can encode any real number y , say by the formula $x = (1/2) + (1/\pi)(\arctan y)$. And a single real number x in that range can also encode any *pair* of real numbers y and z in that range. If $y = 0.y_1y_2y_3\dots$ and $z = 0.z_1z_2z_3\dots$ in decimal (with the convention that we prefer an infinite string of 9s at the end to an infinite string of 0s), we can just write $x = 0.y_1z_1y_2z_2y_3z_3\dots$. Iterating, any finite sequence of reals can be encoded as a single real.

²⁸To specify a continuous function of real valued arguments, we only need to specify its values—which are real numbers—at points that have rational number coordinates, since at irrational ones the value can be determined by a limiting procedure. There are only countably many rational numbers. But a countable sequence of real numbers can be encoded as a single real number. First as in note 27, we can assume that each of the real numbers is between 0 and 1, and encode each as a decimal (with the same convention about trailing 9s). If the k th number is $x_k = 0.x_{k1}x_{k2}x_{k3}\dots$, then encode all the numbers as $0.x_{11}x_{12}x_{21}x_{13}x_{22}x_{31}x_{14}x_{23}x_{32}x_{41}\dots$. In other words, we put after the decimal point all the digits x_{ij} where $i + j = 2$ in lexicographic order of indices (that will be just x_{11}), then all the ones where $i + j = 3$ (that will be x_{12}, x_{21}), and so on.

specified as an open subset of some finite-dimensional Euclidean space, and that can be encoded as a single real number, too.²⁹ As a result, it may well be the case that all our ethical Mersenne distinctions can be specified to giving a finite sequence of real numbers, and, as before, a finite sequence of real numbers can be encoded as a single real number.

Thus, as far as our arguments go, the total informational content in human nature need only be equal to a single real number. And it is plausible that even a very simple substance can have a nature whose informational content contains a single real number. For instance, an electron's natural causal powers might encode the fine-structure constant that measures the strength of electromagnetic interactions, and for all we know, the fine-structure constant has the informational content of a real number. Indeed, this observation about electrons shows that there can well be great complexity in the nature of a simple thing.

Given Aristotelianism and if we live in a continuous world, it is plausible that apart from ethics our human nature does contain the informational content of at least one real number—say, a number specifying what breaking strength a human femur is supposed to withstand, or maybe the fine-structure constant since the human body engages in electromagnetic interaction. Since any two real numbers can be encoded as one³⁰, it follows that our solution to ethical Mersenne problems does not require human nature to have any more information than it may have independently of these.

And it might be that we don't need to go all the way to a real number, that in the end the amount of information in human nature is finite, since maybe we live in a discrete rather than continuous world (cf. **PrussInfinity**, **PrussInfinity**, Chapter 8), and all the parameters we need to specify can be finitely specified. If so, then maybe the fine-structure constant

²⁹* An open subset can be specified by specifying all the points with coordinates that are rational numbers. The set of points whose coordinates are rational numbers is countable. Thus, an open subset can be specified by giving a subset of the natural numbers, and any subset S of the naturals can be encoded as a single real number—say, by putting a one in the n th place after the decimal point if and only n is in S and putting a zero there otherwise.

³⁰See note 27, above.

is also capable of finite specification—but it is still epistemically possible that it has a finite but extremely long specification, and hence that the answers to the Mersenne questions in us do not exceed the informational content of an electron.

4. Other solutions

We have many Mersenne questions pointing to arbitrary-seeming parameters in ethical rules. I will now argue that a broad spectrum of non-Aristotelian ethical theories and solutions are unlikely to yield good answers to the Mersenne questions or else raise new Mersenne questions of their own.

4.1. Kantianism. Kantianism derives moral rules from the very concept of objective rationality. Famously, this leads to difficulties in accounting for the substantive content of rules. For instance, from the point of view of objective rationality, it is difficult to generate a presumption in favor of causing pleasure and against causing pain. The more tightly connected a moral rule is to the specifics of the human condition and of the circumstances, the more difficult it will be for the Kantian to account for it. But the Mersenne questions above thrive precisely on such detail. Consider, for instance, the improbability of a good Kantian account of how much we should, other things being equal, favor siblings over cousins, or of why proxy consent is sufficient for surgery but insufficient for sex. The “logical distance” between the high level principles, like the categorical imperative to treat others as ends or to act according to universalizable rules, and such specific moral content appears unlikely to be bridgeable in a pure Kantian account. Thus, precisely those cases that we have seen to raise compelling Mersenne problems make pure Kantianism an implausible ethical theory.³¹

³¹There is some hope that combining Kantianism with Aristotelianism would help (**MaglioDissertation**), but then we will need to have Aristotelianism, which is what I am arguing for.

Of course, such appearances can be deceiving. One might well have antecedently thought that the relatively simple axioms of set theory are unlikely to generate the richness of mathematical theorems that we have seen to come from them. So it would be good to go beyond an intuition of “distance”.

There are at least four ways to do that. First, proceed by intuitions regarding a specific example. Consider two different moral rules regarding to the relative treatment of siblings and cousins:

- (i) Benefits to siblings are to be slightly preferred to benefits to first cousins
- (ii) First cousins and siblings are to be treated on par.

Neither rule requires us to treat anyone as a mere means or takes away from treating people as ends. Both rules are universalizable. So we are not going to be able to derive one rule rather than the other from pure Kantianism.

Second, we can make use of a heuristic that I employ in checking whether a numbered argument given by undergraduate students is valid, i.e., whether its conclusion logically follows from its premises. I look to see if the conclusion of the argument contains any substantive terms that do not appear in any of the premises. If it contains such terms, it is unlikely that the argument is valid, though of course there are possible exceptions. If the premises are contradictory, then the logical rule of explosion makes every conclusion a valid consequence; alternately, the conclusion might be disjunctive and the substantive term that did not occur in the premises occurs in one disjunct while another disjunct follows from the premises (though I have yet to see this happen in a student paper). Now an argument from premises about the nature of rationality as such with a conclusion about specific familial relationships or about specific human activities such as sex or surgery fails the heuristic, and hence is unlikely to be valid. And the cases do not seem to be like the most obvious exceptions—the premises are not contradictory and the conclusion is not disjunctive.

Third, all or most of the examples that raised Mersenne questions have an appearance of contingency to them, in a way that does not fit with the hypothesis that they derive from

necessary principles about the nature of rationality. One way to formulate this contingency is to note that many of the rules are ones that we would not expect to apply to other rational species. If we came across an alien species that regarded familial ties as somewhat more or somewhat less important than we think permissible for humans, we should not judge them immoral. It would not surprise us if other rational animals—perhaps ones occupying other niches—were rationally or morally required to take greater or smaller risks than we.³²

Finally, we have an epistemological argument. While clearly we do not know the exact values of the parameters in the Mersenne questions, we have some approximate knowledge, as already indicated above in a number of the cases. We clearly did not come to this approximate knowledge by logically deriving it from Kantian first principles. Nor did we even do so by means of an intuition that they follow from these principles. For I take it that we do not in fact have an intuition that, say, the preference for siblings over cousins follows from Kantian principles. If anything, we have an intuition that it does not. So, it seems that if these rules in fact follow from Kantian principles, it's just a coincidence that our beliefs about the parameters are correct, a coincidence that makes the beliefs be mere justified true belief rather than knowledge. But the beliefs are knowledge. So, the Kantian explanation does not work.

The epistemological argument has some force, but less than the other arguments. First, the argument is related to the highly controverted literature on evolutionary debunking arguments (**PlantingaWarrant**; **KahaneDebunk**; **WielenbergDebunk**). Second, a theistic reader has an easy way out of the argument: God knows what values of parameters in fact logically follow from Kantian principles and could either directly instil in us correct beliefs about them or ensure that we evolve in a way that yields such true beliefs. Nonetheless, we have a strong cumulative case against there being a Kantian solution to the problem.

4.2. Act utilitarianism. The main problem with act utilitarianism is that it generates incorrect moral claims. It says that a healthy patient whose organs can save three others

³²One thinks, for instance, of the Klingons and Kelpians from the Star Trek universe, respectively.

can be killed when doing so doesn't have any other countervailing consequences such as making others more callous. It says that if you and I are loners who make no contribution to society, but I own a dog and you don't have any pets, then you have a duty to sacrifice your life for mine, to save my dog from being ownerless; and if neither of us has a pet, but you enjoy chocolate a little more than I do while everything else is equal, then I have a duty to sacrifice my life for you, since your life would include slightly more utility.

The act utilitarian, of course, can counter that such idealized cases are rare to non-existent and our intuitions regarding rare cases are unreliable. A perhaps more serious worry is that while utilitarianism resolves some Mersenne questions, it shifts the bump under the run to others. For instance, we already saw in ??back that for utilitarianism to be plausible and not swinish requires a hierarchy of goods, and there will be Mersenne questions regarding the hierarchy.

But now note that even hard-nosed desire-fulfillment or hedonistic utilitarianism that does not posit a hierarchy of goods will fail to escape Mersenne questions. For there are multiple mental state concepts that are good candidates for corresponding to the words "desire" and "pleasure", with none of them morally privileged in a clear way.

When the psychotherapist tells Jones that she always unconsciously wanted to kill her mother, is that a "desire" in the sense of desire-fulfillment utilitarianism or not? Reflection the complexity of human life, e.g., as seen in literature (**AllenGeometry**), shows that there are likely to be many "desire"-type concepts. We have desire as a felt attraction, inchoate or clear, as an overt behavioral tendency, as a hidden pattern among one's behaviors, as a belief in the worthiness of a goal, or as a personally stipulated explicit utility assignment. We have an orthogonal distinction between first, second, and higher order desires. If only one precisely specified option—say, second-order felt attraction—counts as defining our well-being, we have a Mersenne question as to which is the one that is the true goal of moral pursuit. But if, as is more plausible, more than one such option is worth pursuing, the proper "combination function", or method of generating a weighted overall value, raises Mersenne question.

Next, consider the different ways of quantifying the strength of a desire. The decision theorist has perhaps the only clean option. An ideal agent's strength of a desire is identified with the utility assignment which can be computed from binary preference relations between different probabilities of outcomes: if you would choose a 30% chance of *A* over a 60% chance of *B*, then you want *A* more than twice as much as you want *B*. But deriving a utility assignment from such binary comparisons only works if your binary preference relations satisfy an appropriate set of axioms, such as the ones for the von Neumann–Morgenstern Representation Theorem. However, likely no human being's binary preferences satisfy the axioms, and so we need a function from the binary preferences of non-ideal human beings to utility assignments. There does not appear to be a privileged way of choosing such a function, and so whichever function in fact defines the human good raises the Mersenne question of why this function and not some other that fits about as well with our intuitions.

And even binary preferences are rather problematic to define. Quite likely if you slightly change the wording of your question as to whether I choose a wager with a probability *p* of a philosophical achievement *A* or a wager with a probability *q* of a certain large sum of money, I will give different answers. What way of presenting the wagers defines my actual preference between them raises another Mersenne question.

And as for pleasure and pain, we will again have a broad variety of concepts and a multiplicity of ways of quantifying them. This can perhaps best be seen if we think about the mental life of possible and actual non-human sentients. Does a particular state of an earthworm count as a pleasure? It is unlikely to be exactly like a state of ours. There will likely be many ways of classifying mental states across species, and on some the worm's state will be a pleasure and on others it won't. So we have a degree of freedom in our act utilitarianism as to what we count as pleasure or pain in non-humans. And even within humans there are complex questions. Consider for instance masochism or the subtle morose "satisfaction" of the pessimist who sees everything going downhill. There are likely to be different ways of classifying states as pleasures or pains, and the hedonistic utilitarian

will have a Mersenne question as to why one rather than another classification is the one that defines ethics.

Finally, if our utilitarian is also a physicalist about the mental, the problems increase further: there are presumably infinitely many functions from brain-like states (*brain-like*, since aliens may not literally have *brains*) to desire-like or pleasure-like states. The functions are apt to have a multiplicity of parameters (??forwardref), and thus we will have a number of additional Mersenne questions.

4.3. Rule utilitarianism. Instead of requiring that each action optimize total utility, rule utilitarians require that each action follow rules that are themselves optimized for total utility. The main advantage of this is that we have a hope of escape from the counterintuitive consequences of act utilitarianism. The rule not to kill the innocent may well be the optimal rule for us, even if in a lifeboat situation it would maximize utility for the two stronger people to kill and eat the weaker third. For a rule that allows such occasional killings of the innocent would lead to a lower sensitivity to human life, and humans possessed of such a rule would likely misjudge the applicability of the rule in a way that would reduce total utility.

Rule utilitarianism can explain the apparently arbitrary specifics of the moral rules. The optimal rule of preferential treatment is likely to be messy and complicated, because optimizing utility for humans with their messy lives is complicated. We thus have a neat response to Mersenne questions. Furthermore, we have a good explanation of the appearance of arbitrariness and contingency. For the optimization procedure that would define the moral rules would be a vast and complex one, taking into account the impact of the actions falling under the rules both in the short and the long run, both on humans and on non-humans. It is unsurprising if a complex optimization procedure produces results that seem arbitrary but are in fact carefully chosen to their end. A computer-optimized airplane wing will have precise angles and bends that cannot really be explained without running through the whole computation. Furthermore, the precise values are likely to be

contingent, in that they will depend on details of the laws of nature, and the laws of nature appear to be contingent (and if laws of nature *merely* appear to be contingent, then we still have an explanation of the *appearance* of contingency).

Moreover, rule utilitarianism is less prone than Kantianism to make our limited but true beliefs about the moral rules be merely coincidental. For we have evolved biologically and mimetically in the service of survival and reproduction, and because of the perhaps contingent connections between these goods and other aspects of utility, evolution put pressures on us that directed our moral beliefs in a truthful direction. There are deep and difficult questions whether this is enough to make the connection between our beliefs and the truth be sufficient for knowledge, but there is more hope here than on the Kantian side.

However, famously, rule utilitarianism divides into two varieties, depending on exactly what the rules are optimized for. On ideal rule utilitarianism, the rules are such that everyone's successfully following them would be optimal, even if in fact they are too difficult for us to follow. Ideal rule utilitarianism, however, is widely held to reduce to act utilitarianism, since if everyone were to actually follow the rule of maximizing utility, that would be optimal with respect to maximizing utility. But act utilitarianism has already been put aside.??backref

Non-ideal rule utilitarianisms, on the other hand, inject a note of realism into the optimization procedures. For instance, what might render a set of rules correct is that if everyone were to *try* to follow them, optimal results would result. This already raises a Mersenne question. For trying is something that comes in degrees, and it is very likely that different rules will be generated when we optimize for the utility resulting from everyone's trying hard to follow them than if we optimize for the utility resulting from everyone's trying with minimal effort. And there will be a vast number of intermediate cases, so there will be a Mersenne question of what grounds the fact that α , say, is the right degree of effort for defining the optimization procedure that generates the moral rules.

Furthermore, specifying the degree to which the hypothetical agents try to follow the moral rules is not enough to specify the optimization procedure. For instance, one has to

specify the level of intelligence of the hypothetical agents, their non-moral interests and the non-human environment, which yields multiple Mersenne questions as to what the requisite levels of these for the hypothetical optimization procedure are.

Perhaps we can try to avoid such questions by requiring the counterfactual world to match our world in respects of the level of intelligence of the agents, their non-moral interests and the non-human environment, but this runs into two problems. First, when we try to imagine a world where all agents try to follow the moral rules, the world we imagine will be one whose agents have different non-moral interests and higher levels of intelligence than we do, since such a world would have a much more just educational system than ours and hence would nurture children into greater intelligence, and a rather different non-human environment. If we keep the three factors fixed while having the hypothetical agents try to follow the moral rules, we are likely to get some very unlikely counterfactual results, just as keeping too much of our world fixed in a counterfactual situation results in the odd claim that if Oswald did not kill Kennedy, Kennedy would have been buried alive. Second, we have to say that if our history had gone slightly differently, so that (say) the distribution of intelligence in the general population were slightly different, the optimization procedure would have generated different rules, and hence different moral rules would have been true. Indeed, on this view we would get the very strange idea that what we morally do can affect morality itself.

Besides this, there are other non-ideal aspects that we should probably introduce. Some of our important moral rules discuss how we should deal with culpable malefactors. But in a world where everyone tries to do the right thing, depending on the strength of trying, there might well be *no* culpable malefactors, or at least very few. And it is unlikely that moral rules optimized for such a very different situation would be likely to be the right ones for us. So we probably need to optimize the rules with respect to a hypothetical situation where not everyone tries to follow them. And that raises Mersenne questions as to the grounding of the specification of how many people in the hypothetical case follow these rules and how the non-followers behave.

In short, ideal rule utilitarianism collapses to act utilitarianism, while developing the non-ideal rule utilitarian project raises many Mersenne questions as to the details of the counterfactual situation.

4.4. Social contract. Social contract theories ground ethical rules in agreement between agents. We can divide this based on whether the agreement is actual or hypothetical. Actual agreement theories face obvious problems. First, it is highly implausible to think of the members of society as having *actually* agreed to live by moral rules, apart from special cases such as a religious person vowing to God to sin no more. Second, actual agents can agree to live by unjust rules, even rules unjust to themselves, and such rules would not constitute morality.

Contemporary social contract theories tend instead to be based on duties grounded in hypothetical agreement between agents in situations of ignorance (**RawlsJustice**). Anyone who has been in a long committee meeting knows that actual agreement between agents can result in complex rules with much apparent arbitrariness, and it would be unsurprising if hypothetical agreements were similar. Thus far, social contract fits our Mersenne-style data well.

But the hypothetical agreement condition involves multiple parameters such as how rational and smart the hypothetical agreeers are (and there are multiple dimensions of intelligence), what exactly are they ignorant of, how many of them are there, what are their attitudes towards risk and uncertainty, etc. We have here an explanation of the Mersenne parameters in terms of other Mersenne parameters, and the problem remains fully entrenched.

The risk and uncertainty point is worth emphasizing. Rawls would say that rational agents would only agree to rules that do not treat anyone inhumanly. But a rational agent who is more accepting of risk and does not have prior moral constraints will be willing to tolerate rules that create a minority group that is treated inhumanly if the risk of being a member of that group is sufficiently small—i.e., if the group is a small enough fraction of the general population—and the benefits to the majority are sufficiently large. There will

be types of inhuman treatment and levels of risk that it would not be rational to accept for the sake of a high probability of a large benefit, but the lines between these and the ones that it would be rational to accept do not seem derivable from any plausible set of basic principles of rationality.

Granted, typical Kantian constructivists will insist that certain kinds of inhuman treatment would never be rationally acceptable. But now consider the Mersenne questions about these kinds of treatment. For instance, destroying the autonomy of another person might be taken never to be rationally acceptable. But a minor limitation on another's autonomy clearly is acceptable for a sufficiently great good: if the only way to save a country from nuclear destruction by evil enemy would be to acquiesce in the enemy's demand that everyone wear jeans on Friday, then this limitation on sartorial autonomy should be enforced. Somewhere there is a line between minor limitations of autonomy and such deep destruction of autonomy that could not be tolerated no matter the price. The only "natural" place to draw this line would be at *complete* destruction of autonomy. But if it is only complete destruction of autonomy that is prohibited by the Kantian, then this does not place a sufficient constraint on the rules that could be accepted. For instance, the enslavement of persons would not be prohibited, as long as the enslaved persons were still capable of some minimal autonomous agency, say, a choice of color of clothing.

Furthermore, even prohibiting treatment that completely annuls someone's autonomy will not avoid Mersenne questions in the vicinity. For we will have probabilistic questions. Is it permissible to perform an action that has a 99.9% chance of destroying someone's autonomy, a 99% chance, a 90% chance, a 51% chance, a 10% chance, etc.? The only natural lines to draw seem to be at 100%, 50%,³³ or 0% but none of these will do. A prohibition on an action that has a 100% chance of destroying autonomy prohibits nothing: any action we perform can fail. A prohibition on an action that has more than a 0% chance of destroying autonomy prohibits everything: I scratch my head, and there is a tiny chance that due to some weird sequence of events this causes an earthquake that leads to you getting hit on

³³There are actually two options here: more than 50% and at least 50%.

the head by a beam that results in your life being reduced to a vegetative level. And while 50% is much more reasonable, in some difficult cases it is excessively restrictive. If a child is certain to die within a day, and is suffering from horrific pain that can only be relieved by a drug that has a 50% chance of reducing the child to a vegetative state for the rest of the day, administering the drug can be permissible.

4.5. Virtue ethics. Aquinas himself invoked the virtuous agent as providing at least the epistemic path to an answer to the preferential treatment question. We could also take virtue ethics to provide an answer to the Mersenne question: What makes these parameters, rather than others, hold is that the virtuous agent's patterns of behavior are thus and so parameterized.

But this of course simply shifts the Mersenne problem to that of why the virtuous agent's patterns of behavior are parameterized as they are. Virtue ethics helps with the epistemic aspects of the Mersenne question, but the metaphysical aspects remain. The answer to these most consonant with the virtue ethics tradition appears to be the one given in the Aristotelian tradition which grounds the parameters in the agent's nature. Thus, the parameters can be grounded directly or via the intermediary of virtues.

4.6. Divine command. On divine command ethics, the right is what is commanded by God. Divine command ethics, like social contract and rule utilitarianism, carries with it significant hope for explaining the apparent arbitrariness in ethical parameters. We would not be surprised if laws governing complex embodied beings like us and coming from an infinitely intelligent and good legislator had significant complexity that to us would look like arbitrariness.

It may initially seem the divine command ethics runs into the same problem of pushing the Mersenne questions back to the question of why God legislated these parameters and not others. But notice that the Mersenne problems I have been discussing are *grounding* questions. Even if God's legislation were completely arbitrary in a way that ultimately violated the Principle of Sufficient Reason, on divine command ethics we would have a

ground for the parameters in preferential treatment and other ethical rules being what they are. To say that we should prefer siblings over first cousins in a ratio of 17 : 10 because God commanded it is to give a ground for the obligation, even if that ground itself needs an explanation. Compare the moral prohibition on adding cyanide to friends' drinks. There would be something absurd if that prohibition were ungrounded. But it has a ground, or at least a partial ground: cyanide is fatal to humans. Imagine now that, contrary to fact, it was an unexplained brute fact that cyanide is fatal to humans. Nonetheless, the grounding problem for the moral prohibition would have been solved by citing the danger of cyanide, even though we would be relying on something unexplained.

In this way, our ethical grounding Mersenne problem is quite different from Mersenne's merely explanatory problem. In Mersenne's case to explain why the distance between the earth and the moon is what it is in terms of other parameters of earlier states of the solar system does not make significant progress, since the earlier states need an explanation. But when we have given a plausible *ground* to the moral obligation, we have indeed made progress. Mersenne's original argument depends for its plausibility on a fairly general Principle of Sufficient Reason. Here we just use a heuristic principle that moral truths with an appearance of arbitrariness need a deeper ground.

Moreover, the divine command theorist has nice answers available to the question of why God chose these rules. For instance, God could be an act consequentialist and could have optimized the rules to produce the best consequences, including perhaps such consequences as the value of following and disvalue of breaking moral rules in addition to first order values and disvalues like pleasure and pain. We would expect a complex optimization for a precise end to produce results with an appearance of arbitrariness. A sailboat hull computer-optimized to minimize drag is likely to have many parameters that look arbitrary to those who do not know how it was generated.

At the same time, we still have some serious Mersenne grounding problems given divine command theory. The plausibility of divine command ethics rests in the idea that God is a legitimate authority and legitimate authorities need to be obeyed. This suggests

that logically prior to divine command ethics there is some sort of a proto-ethical general rule about obedience to legitimate authority. That rule itself will have to have parameters specifying which authorities are legitimate and what the scope of their authority is. And we will have the Mersenne problem of grounding these parameters.

Moreover, even if we do not have such a general rule about all authority, but a specific rule about divine authority, this will still raise some Mersenne problems. For, as **AquinasST** (**AquinasST**, I-II Q90 A4) argues, legislation only has a claim on our obedience when it is “notified to [those ruled by it] by promulgation”. And promulgation is a complex concept involving thresholds and parameters. It is not necessary for promulgation that all those subject to the legislation have actually heard of it. But likewise it is not enough for the legislators to meet secretly, and write the legislation on a stone buried on public land. Intuitively, we need the legislation to be reasonably accessible to those governed by it, but there are many parameters hidden behind the word “reasonably”, and we need grounds for them all.

The divine command theorist needs a promulgation condition that intuitively has normative force, and to escape the Mersenne problems needs to have no apparently arbitrary parameters. But, further, to have moral realism, it needs to be a promulgation condition that it is plausible to think has actually been met.

A strict promulgation condition like that everyone governed knows of the rules has not plausibly been met, , since there is no agreement on what God’s commands are—or even on there being a God. What is worse, when we focus on the Mersenne cases in ethics that were enumerated earlier in this chapter, it unclear that divine commands instituting the parameters would even satisfy a fairly modest promulgation that requires those who try really hard to be able to find what the legislation is when it is relevant to life. There surely are cases where we have tried really hard to figure out what is the right thing to do and we didn’t succeed. Perhaps it could be argued that we didn’t try “hard enough”, but now we are the true Scotsman territory. And of course there would be a Mersenne question about the parameters defining the “hard enough”.

Divine command theory, like other major theories, fails to offer a plausible solution to the Mersenne problems.

4.7. Particularism. By pointing out the complexity and context-sensitivity of moral problems, the Mersenne questions may seem to be evidence for particularism. On particularism, while there may be general *ceteris paribus* rules like “Other things being equal, don’t torture people”, the application of these general rules to specific situations is not rule-governed. Hence, there won’t be a rule specifying when one, say, favors a sibling over a cousin. Instead, there are particular facts about what to do in particular situations.

However, particularism only multiplies the Mersenne questions. For whereas on rule-based systems we had Mersenne questions about why the parameters in the rules had the values they do, now we will have Mersenne questions about why in particular actual circumstances C_1 we should act one way while in slightly different particular actual circumstances C_2 we should act a different way.

Furthermore, plausibly, there will still abstractly speaking be a function that assigns to each maximally specific set of nomically possible circumstances a hypothetical determination of how one would be obligated to act in that circumstance. There may, of course, be no formula specifying the function, but that does not affect the Mersenne question of why this function rather than another, perhaps similar one, is the correct one.

4.8. Brute necessity. Perhaps we could say that it is a brute, unexplained but necessary truth that the answers to the ethical Mersenne questions are as they are. The boundaries lie where they do, but there is no special ontology behind them: it’s just a necessary truth that we should prefer parents to cousins, that an armed up-rising up against a regime responsible for Nazi-style atrocities is permissible while only non-violent protest against the faults of modern-day Canada is permitted, and so on.

Brute necessities should never be a first resort in theorizing, but sometimes they might be acceptable as a final resort. Consider Mersenne-type questions one could ask about set theory. If the Zermelo-Fraenkel with Choice (ZFC) axioms for set theory are consistent,

then for every natural number n they are compatible with the hypothesis CH_n that there are exactly n cardinalities strictly between the cardinality of the natural numbers and the cardinality of the real numbers (**SolovayCanBe**). Suppose it turns out that in fact CH_{15} is true. We would have an excellent Mersenne question as to why it is CH_{15} that is true rather than say CH_0 or CH_8 , but the mind boggles as to what could be a satisfactory answer to that question, much as it does in the ethical questions. Perhaps the truth of CH_{15} could be a brute fact, albeit a necessary one since it is generally held that mathematical truths are necessary (though see ??Pruss for a controversial Aristotelian metaphysical story on which some mathematical truths—perhaps even including things like the Continuum Hypothesis—are contingent).

Some brute necessities can perhaps be admitted in ethics. For instance, if CH_{15} is necessarily true, then it is necessarily impermissible for us to punish someone for falsely informing us that CH_{15} is true. This impermissibility would derive from the impossibility of CH_{15} being false (and hence the impossibility of falsely informing someone of that it's true) and the impermissibility of punishing people for actions that they did not do. (It is possible, of course, to insincerely inform someone of a necessary truth. But that's a different wrong action, even if equally bad.)

But truly ethical brute necessities are deeply implausible. Here is one way to see this. Suppose there is a sequence s of one or more English sentences expressing your favorite set of fundamental and necessarily true ethical norms. For instance s might be a single injunction like "Love your neighbor as yourself" or "Maximize total pleasure minus pain of all sentients", or it might be a longer list. Encode s into a sequence of decimal numbers in some natural way, for instance by encoding each symbol in s into a three digit ASCII number from 032 to 126. It is widely believed—though it has not been proved—that π is a "normal number", i.e., every possible sequence of digits occurs in π . If so, then the decimal encoding of s occurs somewhere inside π —and even if not, it may well still do so. Suppose that the decimal encoding of s occurs in π as the n th through $(n + m)$ th digits. Now consider this metaethical theory:

- (1) To do the right thing is to follow the English injunctions in three decimal-digit ASCII encoding between the n th and $(n + m)$ th digits of π .

Call this (π, n, m) -metaethics. On the hypothesis that the fundamental ethical injunctions are necessary and can be expressed in English, there are n and m for which (π, n, m) -metaethics necessarily yields all and only the correct ethical injunctions. But, nonetheless, no (π, n, m) -metaethics has any plausibility. For there is no plausible normative connection between an injunction occurring at some location in π and its being binding on us.

Admittedly, if we in fact found a sequence of English injunctions “near” the beginning of π in ASCII, say from the tenth to the 300th digits³⁴, we would have some reason to follow them. But the reason would be something like this: The best explanation for why these injunctions are found in π is found in a powerful being or beings that can somehow control mathematical truths or, more plausibly, the evolution of our Indo-European languages and ASCII code, and there is good pragmatic reason to follow the commands of such powerful beings. Perhaps they have our good in mind, perhaps they will get mad if we don’t follow their commands, or perhaps they are trying to inform us of the true ethics. But nonetheless $(\pi, 10, 300)$ -metaethics would be false. The reason these injunctions would have force wouldn’t be that they are found in π , but something else, such as that a powerful being wants us to follow them.

One might object that the above argument against (π, n, m) -metaethics begs the question. After all, if the metaethics is true, then to be obligated to ϕ just is for ϕ ing to be commanded between the n th and $(n + m)$ th digits of π , and so that’s what ethical normativity *is*—the normatively significant connection is *identity*. But here is another way to see the point at issue. Moral obligation is not the only normativity we have. We also have the normativity of emotions: different emotional attitudes are apt for different situations. Moreover, emotional normativity is hyperintensional. We can have propositions p and q such that, necessarily, p is true if and only if q is true, but nonetheless we should have different emotions towards them. For instance, an emotion of great wonder and delight is

³⁴Alas, in fact, the three digits starting with the tenth digit of π are 358, which is out of range for ASCII.

appropriate towards fascinating mathematical theorems like Fermat's Last Theorem, but not towards the fact that $48848 \times 849 = 41471952$, even though, necessarily, Fermat's Last Theorem is true if and only the multiplication fact is true (since both are necessarily true). Now, it seems that the emotions appropriate towards the fact that Alice has performed a wrongful action are quite different from those that are appropriate towards the fact that Alice has performed an action against which an injunction is encoded in π between the n th and $(n + m)$ th digits—indeed, quite possibly there is no emotion appropriate towards the latter fact—even if the two facts are such that necessarily each obtains if the other does.

We can use the normativity of our emotions to probe and criticize metaethical theories: a metaethical theory like (π, n, m) -metaethics would force an implausible revision in our emotional life. Similarly, an action's violating a brute injunction in the Platonic heaven feels like something towards which we should have a lesser kind of negative attitude than towards the action's being morally wrong. And that gives us reason to think being so brutally forbidden is not what moral prohibition really is.

In other words, a metaethics where the ethical claims are grounded in something intuitively of no relevance to our moral activity, such as the content of the digits of π , is not plausible. To be a candidate for a grounds of ethical claims, a thing needs to be ethically compelling—in a way that we should be able to *feel*. For a more controversial illustration of this point, consider **MacIntyreDivineCommand's** (**MacIntyreDivineCommand**) argument that no collection of the traditional attributes of God is such as to make it plausible that the commands of a being with those attributes are what ethics is, and this is a strong reason to doubt divine command metaethics, even if it turns out that God exists and has necessarily commanded all and only the fundamental moral truths. And while this claim about divine command metaethics is controversial, given that the traditional attributes of God include that God has created us and is perfectly good which might be held to ground obligations³⁵, if one were to replace God with a God* who has all the attributes other than

³⁵Cf. **EvansGodMoral's** (**EvansGodMoral**) gratitude-based account.

goodness and lovingness³⁶, the point would be even more plausible. Divine* command metaethics would clearly be unsatisfactory even if it turned out that necessarily God commanded all and only the fundamental moral truths.

But now take some theory T , like (π, n, m) -metaethics, that grounds arbitrary-seeming ethical principles on a non-compelling but necessary ground, say the digits of π . But now modify the theory by keeping all the same ethical principles, but removing all claims of grounding. Removal of the ground surely does not make the story any better. Someone who said that what explained why we should favor siblings over cousins by a margin of twenty percent by saying that it is thus written starting with the n th digit of π would be ethically ridiculous (though if n is small, finding the injunction might be some evidence for its correctness). But suppose we drop the spurious π -based ground: surely the ungrounded ethical claim is no better off than the spuriously grounded one.

There may be fundamental ethical truths that are not themselves grounded. But these truths need to be compelling ethically—perhaps the Golden Rule is like that—and not have an appearance of arbitrariness. And there may be arbitrary-seeming truths in ethics, but they are not fundamental. The arbitrary-seeming truths need an ethically compelling ground.

A variant of a brute necessity approach is a Platonic approach on which ethics is grounded in the brute features of the universal humanity. Since this universal can be thought of as just another theory about what a robust human nature is, we can think of this for now as a variant on the Aristotelian story—a variant that will be more carefully evaluated in ??ch10:forwardref.

4.9. Informationally complex natural universals. The property or universal *being a natural number* appears quite natural, in the LewisNewWork’s (LewisNewWork) sense that is opposed to gerrymandered properties like *being a dog or a book*. Yet we learn from Gödel’s First Incompleteness Theorem (BoolosJeffrey1995) that there is no way to capture

³⁶This is a *per impossibile* conditional if the other attributes entail goodness or lovingness, as SwinburneGod (SwinburneGod, p. 13), for instance, has argued in the case of goodness.

the truths about natural numbers in a finite (at least first-order) way. Thus a universal may be quite informationally complex despite not being gerrymandered.

Perhaps, similarly, there is a natural universal that is nonetheless highly informally complex and that captures the ethical phenomena. This could be a thin property like *obligation* or *virtuousness*, or a thick property like *love*.³⁷ Then ethics could perhaps be captured by a simple rule, like “Do what’s obligatory” or “Love everyone”, which has a vast richness of entailments due to the complexity in the referenced universal.

As noted earlier, many of our Mersenne questions involve an appearance of contingency as well as pointing to kind relativity. It seems very plausible, for instance, that the parameters in our initial example of preferential treatment are highly dependent on the kind of rational animal we are talking about. Rational animals that are more dependent on their parents in childhood would likely be such that they ought to have a greater preference for their parents over strangers than rational animals that have little contact with parents after procreation. And what one says about rational animals that reproduce in ways radically different from us—asexually or with more than two individuals involved in reproduction—is likely going to be quite different.

On this Platonic view, there is a vast number of apparently arbitrary parameters, such as that dragons ought to have a preference of degree d_D for parents while humans one of degree d_H . These parameters have an appearance of contingency, and yet there is no contingency here at all—the necessary Platonic facts about universals fix them. But we can still try to explain the appearance of contingency as follows. There are infinitely many possible humanoid kinds, and within some range appropriate to humanoids any possible degree of preference for parents will be the right degree for one or more humanoid kinds. And it is contingent that the specific kind of humanoids that is actually exhibited on earth is, say, H_{1249} , so it is contingent that we have humanoids with a degree of parental preference $d_{H_{1249}}$, though it is not contingent humans—i.e., H_{1249} s—have that degree of parental preference.

³⁷I am grateful to Charity Anderson for the last suggestion.

Indeed, this is very much what the Aristotelian will say. It is necessary that we humans have the exact degree of parental preference we do, but earth could easily have had a subtly different kind of humanoid instead.

There is, however, a relevant difference between our Platonic and Aristotelian views. On the Platonic view, the kinds of rational beings will not be defined in terms of the ethical norms applicable to them, since otherwise we have essentially the Aristotelian view again. When members of a kind K are obligated to have a preference of degree d_K for parents, the kind K is not defined with respect to the degree of preference, nor presumably with respect to any ethical normativity (and maybe any normativity at all). In particular, we have supervenience of ethical properties on non-ethical properties. And while this supervenience has been a part of the orthodoxy in philosophical ethics in the past century, the vast amount of detailed Mersenne questions and apparently arbitrary parameters makes such supervenience implausible. For it is implausible to think there is a rule that maps the non-ethical features of a rational animal to a unique degree of preference for aiding parents and that does not involve apparently arbitrary parameters of a sort that produce an appearance of contingency.

On the Aristotelian view, on the other hand, the kind K is defined by the form, which intrinsically encodes ethical and other norms. Thus, we can explain the appearance of contingency in the parameters ethically governing our lives by noting that there are other humanoid kinds—likely including ones that are biologically just like we are—whose ethical norms are different. This is akin to the Kripkean explanation of the appearance of contingency in water being H_2O : something else with the same “watery” behavior could fail to be H_2O . We need not suppose a single necessary mapping from the non-ethical features of a kind to the ethical features.

And while rules like “Love everyone” appear initially simple, if we realize that there is a vast collection of necessary truths about love that involve apparently arbitrary parameters or rules that cannot be simply derived from a few elegant principles, the appearance of simplicity becomes less of an advantage over an Aristotelian rule of “Follow your nature”,

where in fact your nature is in fact very complex. The Aristotelian rule is intrinsically simpler because the concept of a nature or form as such is less informationally complex than that of love—though, of course, your actual nature is highly informationally complex. (The concept of a book as such is not very informationally complex, but a given book may be highly informationally complex.) The simplicity in the Aristotelian rule is like the simplicity in a silly consequentialist rule like “Maximize the universe’s average temperature”—all the concepts in the rule are fairly simple, though of course facts about what actions fall under the rule depend on very complex features of the actual universe, just as facts about what actions fall under “Follow your nature” depend on very complex features of your actual nature.

Additionally, it’s worth noting that some of the attractiveness of the Platonic approaches disappears on closer examination. It is appealing to think that one can explain all of ethics in terms of one universal, or perhaps universal per kind, and a simple rule like “Do what what’s obligatory” or “Love everyone.” However, such proposals leave out a significant portion of ethical discourse. Granted, figuring out what we are obligated to do is indeed ethically central. But obligations come in degrees of weightiness. I don’t mean by this that weightier obligations override less weighty ones. Rather, some obligations are worse to break than others. Under ordinary social circumstances, it is equally true that I am obligated not to slap you in the face and that I am obligated not to shoot you in the kneecap. But it is worse if I shoot you in the kneecap—that violates a weightier obligation. One can try to ground comparisons between obligations in terms of obligation *simpliciter* by noting that some obligations override others. Thus, if an evildoer tells me that they will kill you unless I either slap you in the face or shoot you in the kneecap, I should slap you in the face. However, the question was what grounds the fact that I am more weightily obligated to refrain from shooting than from slapping under ordinary circumstances, and it does not seem right to ground this in my obligations in extraordinary circumstances. And, in addition to obligations, there are moral reasons of varying force, including ones generating cases of supererogation.

Thus, the theory will likely require some universals other than the binary relation of obligation between an agent and an action, such as a ternary comparative obligation relation between an agent and two actions, or a ternary moral-reason relation between an agent, an action and a force. Similarly, in the case of love, there are actions that are more loving than others. These will include loving actions one of which is more loving than the other as well as unloving actions one of which is, nonetheless, more loving than the other. It does not seem possible to define comparisons between degrees of love simply in terms of the predicate “loves”. We will need universals corresponding to the quaternary relation expressed by the sentence “ x ϕ ing is more loving towards y than x ψ ing”.

4.10. A two-step vagueness strategy. It is very tempting to dismiss the Mersenne questions in ethics with a two-step strategy. *Step one:* In each of the Mersenne-type problems, give non-arbitrary grounds for an approximate and vague determination of the parameters involved. Thus, while it is implausible to think that, say, social contract theory will generate a precise answer to the preferential treatment question, it is reasonable to think it will generate claims like: “Benefits to siblings are to be *somewhat* preferred to benefits to cousins.” *Step two:* Insist that the Mersenne question as to the grounds of the exact dividing line has the false presupposition that there is an exact dividing line—instead, we have insuperable vagueness.

An initial concern with the two-step strategy is to worry whether other ethical theories can actually generate sufficient non-arbitrary grounds that have the degree of precision that we think really is there. This concern has two variants. One involves cases where we know what is the right moral judgment. Kantianism, for instance, is unlikely to generate even a vague morally-relevant distinction favoring siblings over cousins, and yet we know that we should favor siblings over cousins. The problem of ranking types of goods generates difficult Mersenne questions as to what grounds comparisons for utilitarianism, yet we know that fundamental philosophical truths are more valuable than the pleasures of chocolate. The second variant of the concern involves cases where we agonize over what

to do. The agonizing is a sign of intuition that there is an answer to a moral problem, albeit one we cannot discern. While we may not be seeking absolute precision, and may be willing to accept some level of vagueness, in a number of cases we seek for more precision than the various alternatives to the form-based theory can ground.

Suppose the initial concern can be allayed in both of its forms, perhaps by clever development of a theory that does generate the known vague moral claims and by biting the bullet and admitting that moral agonizing is out of place in the other cases. There is still another question: how do we account for the vagueness at all. There are three main contemporary accounts of vagueness: (i) non-classical logic, (ii) supervaluationism and (iii) epistemicism.³⁸

It is difficult to make sense of what it is for a proposition to be not settled by the world and yet true, and

On non-classical logic approaches to vagueness, one typically increases the number of truth values beyond two. Consider an ethical Sorites series, where we fix some circumstances *C* and then say:

(A₀) Giving \$1000 to a stranger is better than giving \$0 to one's parent.

³⁸Elizabeth Barnes' classical-logic version of ontic vagueness **BarnesOntic** is difficult to classify. On this view, every proposition is either true or false, and indeed it is determinate (i.e., non-vaguely true) that it is true or false, and logic is classical. It is also true that we do not know whether a vague proposition is true or false. So far this sounds like epistemicism. After all, it follows that in a Sorites sequence starting with a hairless person and moving through more and more hirsute people until we get a person with a paradigmatic full head of hair there *is* some first non-bald person. However, Barnes also insists that the world does not "settle" whether a vague proposition is true or false. This seems problematic. Suppose Jim determinately exists but is vaguely bald. Then we can say that either it is true that Jim is vaguely bald or it is false that Jim is vaguely bald. It follows by classical logic that it is true-and-not-settled that Jim is vaguely bald or it is false-and-not-settled that Jim is vaguely bald. Since a proposition is settled if and only if its negation is settled, it follows that one of these propositions is true-and-not-settled: that Jim is bald or that it is not the case that Jim is bald. In either case, there is a true-and-not-settled proposition. But it is difficult to see how a proposition could be true, and hence true at the actual world, and yet not settled by the actual world.

Now for each positive integer n , the following material conditional is very plausible:

(A_n) If giving \$1000 to a stranger is better than giving $\$(n - 1)$ to one's parent, then giving \$1000 to a stranger is better than giving $\$n$ to one's parent.

From A_0 and A_1 , one concludes by *modus ponens* that giving \$1000 to a stranger is better than giving \$1 to one's parent. From this and A_2 , by *modus ponens* one concludes that this is true even if what one gives one's parent is \$2. Continuing onward, once we get to A_{1001} , we conclude that it's better to give \$1000 to a stranger than \$1001 to one's parent, which is clearly false. Thus, we need to reject one of the premises A_n . Presumably it's one with $n > 0$, since A_0 is clearly true. But a material conditional $p \rightarrow q$ is false just in case p is true and q is false. Thus if A_n is false and $n > 0$, we have:

(2) Giving \$1000 to a stranger is better than giving $\$(n - 1)$ to one's parent and giving \$1000 to a stranger is not better than giving $\$n$ to one's parent.

And that is exactly the kind of sharp transition that the vagueness theorist wishes to deny.

The non-classical approach to vagueness (**Williamson Vagueness**) will involve a logic with one or more additional truth values inserted between *false* and *true*. On a three-valued option we insert a single such truth value, say *indeterminate*, but it is probably more natural to insert a continuum of truth values and have a truth value for every number between 0 (fully false) and 1 (fully true). Then the statement:

(B_n) Giving \$1000 to a stranger is better than giving $\$n$ to one's parent

is true for $n = 0$ (note that B_0 is just A_0), but will become less true as n increases (where in the three-valued case, I take indeterminate to be less true than true, and false to be less true than indeterminate).

But B_n is fully true not just for $n = 0$, but is also fully true for $n = 1$ and $n = 2$ and at least a few other small values of n . On the other hand, not only is B_n fully false for $n = 1001$, but it's also fully false for $n = 1000$, for $n = 999$, and at least for a few other values that are a bit less than 999. On the other hand, somewhere for $2 < n < 999$ on our non-classical view we have B_n taking on one or more intermediate truth values, or else the

view is useless. And hence whereas in the classical logic reading we had one transition to be explained, from true to false, now we have at least two: from full truth to the first n where B_n has an intermediate truth value, and then from the last n where B_n has an intermediate truth value to where it has full falsity. Thus we have doubled the number of Mersenne questions.

We might try to say that where the transitions from full truth/falsity and intermediate truth values occur is itself an instance of second-order vagueness, and hence there is no point where it's fully true that we have such a transition. But if we use our non-classical multivalent logic to account for this second-order vagueness again, then we have a further multiplication of Mersenne questions. For where we started with with a case where grounding the transition from false to true among the B_n was our only Mersenne question, now we have a such transitions as fully truly fully false to intermediately fully false, from fully truly intermediately truly to intermediately truly intermediately truly, and so on. Going second-order doesn't just shift the Mersenne bump under the rug, but makes it grow.

Perhaps, though, one can stay on the first-order level, go for a continuum of truth values, and dig in one's heels by insisting on strict decrease of truth value for all n : thus B_1 is slightly truer than B_0 , and so on. But the precise assignment of intermediate truth values—say, B_{505} getting a truth value of $T_{0.51}$ —also calls for an explanation. Thus it seems we have a vast multiplication of Mersenne questions. But there is a response to this argument: some hold that exact truth values are a mere feature of the logical model of vagueness and what matters is their ordering (**Williamson Vagueness**). And the ordering of the truth values is, perhaps, quite non-arbitrary in that B_m is truer than B_n precisely when $m < n$. But the insistence that the ordinal properties of truth values is what has reality still does not escape the multiplication of Mersenne questions. For consider a different set of ethical questions involving a threshold. For instance, let C_x say that one has a duty to obey the orders of a government that cares to degree x about the common good, for some method of quantifying care about the common good, where, say, $x = -100$ corresponds to the Nazi German

state and $x = 5$ corresponds to modern Finland. Then C_{-100} is very much false while C_5 is very much true. But even if all we insist on is the ordering of truth values, then we will still have a vast, perhaps infinite, number of Mersenne questions about comparing the truth values of the C_x to the B_n like:

(3) What grounds the point N at which B_n becomes less true than $C_{0.24}$?

For clearly B_0 is truer than $C_{0.24}$ while B_{1001} is falsier.

However, the most common response to vagueness these days is supervaluation. The terms of a sentence can have multiple precisifications, with “precisifications” or “interpretations” (**Williamson Vagueness**) corresponding to different classical truth values. “Bob is bald” may be true if we precisify “bald” as having less than half a cubic centimeter of scalp hair and false if we precisify it as having fewer than a meter of hair. Then we have vagueness. When, on the other hand, a sentence is true (respectively, false) under all precisifications, we say it is super-true (super-false).

In the ethical examples, such as whether it is better to give \$1000 to a stranger or \$500 to a parent in circumstances C , presumably the supervaluationist escape from Mersenne questions will be that no matter how far we precisify C , the statement will be vague due to a vagueness in ethical terms such as “better” or “right” or “wrong” which have multiple precisifications yielding different truth values for the ethical claim. For instance, it may be better₁₇ to give the double amount to the stranger but not better₄₀. Indeed, on a view like this, we will have precisely specified cases where for some precisifications of “better” it will be better to give to the parent and for others it will be better to give to the stranger.

Just as in the multivalued logic case, this multiplies Mersenne questions. For where previously it looked like we have a transition from its being true that it’s better to favor the stranger to its being not true, now we have two transitions: from its being super-true that it’s better to favor the stranger (say, when the amount of benefit to the stranger is extremely large) to its being vague whether it’s better to favor the stranger to its being

super-false that it's better to favor the stranger. And supervaluating at the next level up—say, supervaluating “super-true”—only multiplies the Mersenne questions, as we saw in the non-classical logic case.

And there are some additional problems for the supervaluationist response. A standard objection to supervaluationism in general is that it implies that it is super-true that there is a sharp boundary of “bald”: for, given any precisification “bald_{*i*}”, there is a sharp boundary for it. In doing this, supervaluationism explicitly forces the denial of its governing intuition that there are no sharp boundaries.

Finally, the application of supervaluationism to ethics is itself deeply problematic. It is truism that we have reason to do what is better. Truisms had better be super-true. This implies two possibilities with regard to the truism. Either for every precisification “better_{*i*}” we have reason to do what is better_{*i*}, and we don't need to precisify “reason”, or else we need to precisify “reason” and “better” in lockstep when we precisify the truism, so that for every *i* it will be true that we have reason_{*i*} to do what is better_{*i*}. Neither option is satisfactory.

If for every *i* we have reason to do what is better_{*i*}, given the existence of infinitely many precisifications here, it seems that the choice whether to favor the stranger and the parent is governed by infinitely many reasons on both sides. This infinite multiplication of reasons is implausible. Moreover, the supervaluationist responder to the Mersenne questions has to say there is no overall winner here—no on-balance reason all things considered—for if there were, then we could raise our Mersenne question with regard to the overall winner, and we would be no further ahead. But saying that there is no on-balance reason here denies the intuition that cases near the boundary are hard cases, that it is a difficult question to figure out whether to favor the parent or the stranger, since as soon as one can see that one is in the vague region, one could just conclude that neither action is on balance required by one's reasons.

But, on the other hand, if there are infinitely many ways to precisify “reason”, none of them privileged, then this undercuts the very idea of our life being governed in a non-arbitrary way by rationality. It seems entirely arbitrary whether we follow reasons₁₇ or reasons₄₀ in our lives. Many questions of rationality turn into purely verbal questions as to how “reason” is to be precisified. And the same goes for related terms like “morality” and “virtue”. This does not seem to do justice to the non-arbitrariness that is central to a realist conception of reason, morality and virtue, or to the agonizing that is appropriate in hard cases. The point here is similar to the one raised in ??backref regarding (π, m, n) -metaethics: it would be arbitrary to require obedience to the commands that are found starting with the m th digit of π , rather than the commands found in some other location.

Furthermore, there are moral reasons that a virtuous agent would rather die than transgress. We can, however, find Mersenne cases involving such cases—for instance, one should rather die than torture the innocent, but we saw in ??backref that what kind of infliction of pain is absolutely wrong raises Mersenne questions. But in the supervenient cases, it will then be a merely verbal or “non-substantive” (SiderWriting) question whether one has a certain “reason”—it depends on the precisification. But to ask that people die for what is “purely verbal” is obscene, and to die oneself for such a “reason” is a failure to respect one’s own humanity (to put it in Kantian terms).

Finally, consider a standard epistemicist theory of vagueness according to which there is a true semantic theory that assigns to each term the precise meaning it has in the light of the patterns of our use of that term, but neither that semantic theory nor the empirical data on the patterns of language use are available to us in sufficient detail to settle the meaning of vague terms **WilliamsonVagueness**. Thus, there is a precise fact as to how much hair one can have and yet have “bald” apply to us, a fact grounded in the patterns of our use of the word “bald”, but it is a fact that is not accessible to us. Similarly, there is a precise meaning of “right”, “better” and similar ethical terms, a fact grounded in the patterns of our linguistic usage. If the transition between bald and non-bald occurs between 98 and 99 hairs, there is nothing mysterious about the fact that someone with 98 hairs is bald and

someone with 99 is not, just as there is nothing mysterious about the fact that a backless chair is a stool.

But the problem here is exactly the same as the last problem with supervenience. Ethical questions are turned into purely verbal questions. Just as on supervenience, there is a multiplicity of concepts closely corresponding to our words “right”, “better” and “reason”. On supervenience, none of these concepts was privileged, which turned ethical questions into purely verbal questions, undercutting our understanding of the way our lives are to be governed by reasons and morals. On epistemicism, there *are* privileged concepts that exactly correspond to the words, but they are privileged purely linguistically—it just so happens that these privileged concepts better fit with our usage under the correct semantic theory. We get an unacceptable arbitrariness on which if our linguistic practices were somewhat different, we would be using the word “better” differently, and there would be nothing less natural about that usage. If so, then our actions’ being governed by the better or the right, rather than by some variant property, would be entirely arbitrary. As **SchoenfieldVagueness** (**SchoenfieldVagueness**, p. 265–266) observes, it is absurd to think that linguistic usage settles the question of the permissibility of abortion, and adding knowledge of the correct semantic theory to the facts of linguistic usage should not help either.³⁹

In summary, non-classical logic violates classical logic, which should only be a last resort, and further multiplies rather than resolving Mersenne questions. Supervenience likewise multiplies Mersenne questions. And, perhaps most seriously, both supervenience and epistemicism as applied to ethics turn ethical questions into purely verbal ones, undercutting a robust moral realism.

HawthorneMoralVagueness (**HawthorneMoralVagueness**, p. 218) notes that one can imagine a continuum of patterns of linguistic usage, across possible worlds, on which

³⁹**HawthorneMoralVagueness** (**HawthorneMoralVagueness**) criticizes Schoenfield by observing that it is not enough on epistemicism to know the patterns of linguistic usage—one also needs to know the way that the semantic facts depend on the patterns of usage.

“morally good” shades between the meaning it now has and its meaning *made of gorgonzola cheese*. Plausibly there is a possible world along that continuum where “morally good” refers to the morally good but does so in a way that is indeed epistemically vague in the Williamsonian way, because it depends on fine details of usage and semantic theory whether in that world “morally good” refers to the morally good (as it does) or to the gorgonzolan. Thus I should not overstate my case by claiming that a term that refers to the morally good could not be epistemically vague. However, such a world would not be one where the concept expressed by the term “morally good” would legitimately play the kind of central normative role that the concept of the morally good plays in our lives (unless the denizens had some other and non-vague way of expressing the concept). Indeed, it would not be right for us to say that the denizens of that world know that what falls under their concept of the good is indeed supremely worth pursuing (since one does not know borderline cases on epistemicism), even though in fact it *is* supremely worth pursuing. Interestingly, it might even be the case that in their variant of English “The morally good is always worth pursuing” is non-vaguely true. For “worth pursuing” might march semantically in lockstep in the continuum of worlds along with the “morally good” in such a way that in worlds where “morally good” means *made of gorgonzola*, “worth pursuing” means *made of cheese*. This thought experiment suggests that it may not be enough to characterize our robust moral realism by saying we *know* central moral truths, such as that the morally good is always worth pursuing, but we need to be committed to there not being an undermining outside view, like the one we have with regard to the world where “morally good” is semantically next door to meaning *made of gorgonzola*. How exactly to state this constraint on moral realism is a difficult question, but it does seem that something has gone seriously wrong in a world where people sacrifice their lives for the sake of a property that for aught they know is identical to *made of gorgonzola*—that is just not a world where robust moral realism is true.

In this critique, I was following the standard understanding of epistemicism as grounding the vagueness in unknown patterns of use and an unknown mapping from patterns

of use to meanings (**Williamson Vagueness**), so that if moral properties are vague there are no privileged moral properties. There is, however, a more general understanding of epistemicism as any theory on which there are genuine facts of the matter about vague transitions, but these facts are epistemically unavailable to us. Saying this, however, does nothing to solve the Mersenne problems, but has this much going for it: it is compatible with the basic metaphysics of the Aristotelian human-nature account that I am defending, and hence is compatible with there being a solution.

The Aristotelian human-nature account can but does not have to be taken to be a species of this more general epistemicism. The crucial question is whether all specific moral truths—say, that one should in circumstances *C* give \$*n* to one's parent rather than \$*m* to a stranger—are in fact knowable by us. One could have a more optimistic theory on which they are, but some require more time to figure out than is *practically* available to us. Or one could hold that there are some that are utterly beyond us. The second may seem absurd: how could a consideration beyond our ken count as a reason? But if we understand morality in terms of teleology as we will in ??forward, and if we grant that we could have aspects of our physical nature whose teleology is not knowable by us (why not?), then we might be open to this less optimistic option—as long as such cases are rare enough that most of the time one can make the right decisions. I incline towards the more optimistic view, but will not predicate anything on either option.

4.11. Anti-realism. Retreating from realism in ethics to error theory does, of course, remove all the Mersenne problems in ethics. But the cost is high: it is incorrect to say that genocide is wrong. Moreover, since some of the Mersenne problems involve not just morality but also prudential reasoning, this requires one to deny the correctness of standard prudential reasoning. But perhaps the most serious problem with the error theoretic solution is that we will have parallel Mersenne problems in other normative areas, such as epistemology (??forward) and semantics (??forward), and the cost of error theory there is very high indeed: indeed one will no longer be able to correctly say that one epistemically *ought to* accept error theory.

A more moderate solution is to opt for a form of ethical relativism. Relativism, of course, suffers from serious and standard objections. Perhaps the most obvious is that it justifies an ultra-conservative approach: for if what I (in the case of individual relativism) or my society (in the social variant) thinks is guaranteed to be true, then I or society has no reason to take variant views into account, since if you disagree with me or my society, you're guaranteed to be wrong (from my or my society's point of view).

Moreover, relativism is itself prone to Mersenne questions. First consider individual relativism, on which a moral claim is true just in case one believes it. The Mersenne question here will be most obvious if one opts for a view on which belief reduces to having a credence above some probabilistic threshold, say 0.95. For then the relativist view comes down to the thesis that a moral claim is true just in case one assigns it a credence of at least 0.95. But that seems arbitrary. Why should one be obligated to do what one has credence of 0.95 in, but not obligated what one has credence of 0.93 in? So we have a threshold problem.

Many, however, resist the reduction of belief to a credence threshold. But if we do not so reduce belief, we should then see belief as just one positive doxastic state among many, such as surmising, being inclined to think, believing, being confident that, and being sure that. Moreover, a little reflection shows that such classifications are too coarse grained to do justice to the richness of our mental life. So we have a Mersenne question: Why are moral claims made true by my believing them rather than my surmising them or being sure of them?

And thinking that the problem here just involves degrees of confidence is probably neglecting much complexity in the human mind. There is likely a continuum between fully believing and merely acting as if one believes. Why does moral truth show up in the continuum where it does? Or think of the case when the psychotherapist diagnoses one with a subconscious belief. Either such states define moral truths or they do not, and whichever option is right, we have a Mersenne question as to why it is. And consciousness itself may come in degrees.

Furthermore, a narrow relativism restricted to those moral claims that we actually believe is very implausible. Suppose I believe that it is wrong to eat animals, and I know that cows are animals, but I do not actually draw the conclusion that it is wrong to eat cows. On such a narrow relativism, it would be wrong for me to eat animals but it would not be wrong for me to eat cows, even though I know them to be animals. This is incredible. So we want to extend moral truth at least to things that clearly follow from my moral beliefs. But probably we do not want to extend it to things that follow in ways that are far beyond our ability to know. For, first, if we extend it thus far, then a Kantian might end up agreeing with a relativist, since the Kantian thinks that moral truths are necessary truths, and might well hold that necessary truths follow from everything. And, second, such an approach loses the internalist motivations of relativism. But if we restrict moral truth to things that follow *sufficiently easily* from our beliefs, we will have a Mersenne question of grounding where the line of sufficient easiness lies.

If our relativism is of the social sort, we will have analogues to the above Mersenne questions raised by belief and consequence at the social level. And we will have more Mersenne questions. There is a complex and difficult literature on how to attribute doxastic states to a community (Quinton1976; Hakli2006; deRidder2022; Lackey2020; Tuomela1992; Gilbert1987; Gilbert2006). A reasonable reading of that literature is that there is a multiplicity of concepts that can be expressed with a phrase like “The committee believes that *p*.” For instance, belief by the vast majority of the committee members is enough on the more reductive concepts, while on more procedural versions of the concepts the committee’s belief requires some sort of a joint procedure, such as a vote. There will be many answers here, corresponding to a broad spectrum of takes on what a community’s beliefs is. And a social relativist will then have a Mersenne question as to why moral truth is defined by the particular take in question.

The second set of Mersenne question arises from the question of identifying what counts as one’s community. I am a citizen of two countries and a permanent resident of a third. Are the moral beliefs of all of these communities—no doubt, mutually contradictory

in various ways—true for me, or only the moral beliefs of one apply to me? Do moral beliefs come to be true as applied to me because I am legally a member of the community, or because I identify with it emotionally, or because I would like to identify with it emotionally? Or is it, perhaps, that every community's beliefs are true for the community, but there is no such thing as being true for the members of the community? How large does a community have to be to define moral truths? Is a chess club a community that defines moral truths? What if the chess club goes down to one member? Are a pair of friends a community? It is clear that there are many degrees of freedom in a social relativistic theory, and we would have a Mersenne question corresponding to each of them.

5. Hume's objection: Complexity, instinct and nature

Hume saw the complexity of property, inheritance, contract and jurisdiction, and used this complexity to argue *against* a Natural Law account:

For when a definition of *property* is required, that relation is found to resolve itself into any possession acquired by occupation, by industry, by prescription, by inheritance, by contract, &c. Can we think that nature, by an original instinct, instructs us in all these methods of acquisition?
(??Enquiry::Morals)

To further expand on the complexity, Hume notes the vast variety between these rules in different societies, and analogizes them to the variety of architectures found in housing across societies. A better view, Hume insists, is that we simply engineer rules for the sake of social utility, just as we engineer houses for various ends, and in different environments this results in different solutions, albeit ones with a lot of commonality.⁴⁰

Three responses are possible.

First, we do actually have good empirical reason to think that our moral intuitions and instincts vary quite a bit from case to case, and hence can take into account the moral

⁴⁰It is natural to try to solve the problem by adverting to positive law. But Hume notes that there is much complexity with respect to the question of which institutions get to make the positive laws.

complexity that Hume points to. In some cases (etiquette is the most uncontroversial), different rules are appropriate in different societies, but the imperfectly intuitively grasped function from society to rules is complex. An account of our actual moral instincts is likely to have enormous complexity.

Second, the account I am defending is one on which our forms set norms. These norms can be arbitrarily complex. Granted, there will be a harmony between these forms and our instincts and intuitions. But this harmony is not a one-to-one mapping. There is such a thing as normal and abnormal food for a type of organism, and we expect the organism to have instincts that tend to direct them to normal consumption and away from abnormal consumption. But just as correctly functioning sight can still err, so too a correctly functioning feeding instinct can lead organisms to ingest what is abnormal and to refrain from what is normal, especially in environments that have abundances different from the ones present where the organisms evolved. Thus, our natural instinctive preference for high-calorie foods leads humans in affluent Western countries to unhealthy consumption. Reflection can yield additional information as to *normative* consumption on the basis of high rates of obesity. By reflecting on our nutritive instincts and *other data*—such as the teleology of nutrition and medical facts about us—we can get additional information as to what is normative consumption for an organism, including a human one. This additional information is still fallible, and may fall short of the complexity of the norms involved. And what goes for our nutritive instincts is even more strongly applicable to our moral ones.

Third, recall Hume's own solution to the problem that the complexity of the rules is a result of our social engineering for social utility. Hume's solution is subject to complexity problems of its own. For instance, what groups count as societies and how do we aggregate the benefits to individuals to get a social utility, etc.? But something similar to Hume's solution can be appropriated for the natural law account. We can suppose norms in our nature establishing the goals for certain social institutions, such as property or state authority, and perhaps establishing some constraints on how these goals are to be pursued,

and at the same time requiring us to engineer institutions that satisfactorily pursue these goals within the constraints. These norms will be complex, but will be less complex than the vast complexity in our social institutions. Thus, we have a bootstrapping, from fairly complex norms setting the ends and constraints for social institutions, to the institutions themselves.

6. Concluding remarks

We have a vast panoply of Mersenne problems about the grounding of the complex norms that we find ourselves with. Positing a robust view of a human nature encoding and grounding these norms solves the problems as far as they go, though it leads to further questions that will be examined in ??forwardrefs. The major alternatives all have Mersenne questions of their own, and many have serious additional problems.

CHAPTER III

Natural Law's advantages for metaethics and ethics

1. Metaethics

1.1. Some desiderata. Metaethics is an account of why the most fundamental ethical truths are true. If we were to make a wish-list for metaethics, it would arguably include the following desiderata for what should follow from the theory:

- (4) Ethical truths are objective
- (5) Ethical truths are knowable
- (6) The explanation of fundamental ethical truths makes them morally compelling to us
- (7) The normative implications are plausible.

Recall our (π, m, n) -metaethics on which what made ethical claims true is that they were encoded at some specific position in the digits of π . This gave us objectivity and knowability (at least given the specific position and encoding system), and perhaps plausibility of normative implications, but not compellingness.

An individual relativism that says that the right is what agrees with one's belief as to what is right, on the other hand, gives us knowability, and insofar as we find morally compelling the idea that we should obey our conscience it gives us some compellingness, but it lacks objectivity. Furthermore, its normative implications as to what I ought to do are very plausible to me, since obviously I find my own moral views plausible, but the theory's implications for what people with odious moral beliefs should do—namely, those actions they believe are right—are highly implausible.

Utilitarianism, on the other hand, considered as a metaethical theory about the nature of the right, yields objectivity, knowability and moral compellingness (the idea that what

we should do is maximize the good is among the *prima facie* most plausible of moral ideas), but it famously has implausible normative consequences.

A Natural Law metaethics on which for an action to be right is for it to be a proper exercise of the will according to our nature yields a mildly circumscribed objectivity: it makes the right be relative to our kind. But as we saw in Chapter II, this degree of relativity is highly plausible: it is plausible that ethical requirements do vary between different kinds of intelligent beings.

Natural Law metaethics yields knowability when we accept the optimistic Aristotelian harmony theses that things generally function correctly and that the various norms for a thing tend not to conflict. For instance, given such a thesis, the norms for our emotions—including emotions such as moral repugnance or moral admiration or the feeling of obligation—are likely to cohere with our norms for our actions, and by and large our emotions and actions are apt to be correct. This enables us to evaluate normative ethical theories according to the constraint of whether their requirements fit sufficiently with our emotions and require actions that are not too distant from those typically performed by people whose lives appear to be harmoniously flourishing. We thus have a rational equilibrium epistemology for our ethics.

The basic in Natural Law metaethics idea here is that we ought exercise our will correctly. This is so compelling that it smacks of triviality. Nonetheless, the claim is not trivial, since it provides an analysis of the moral ought in terms of the functional correctness of our wills. We find compelling the idea that we should be true to ourselves. But to be true to ourselves is not just being true to our changing beliefs and values, but to be true to that which makes us be human: our human nature.

The metaethics of right action as the proper functioning of the will is *prima facie* compatible with quite a broad variety of normative theories. We can imagine a being whose will's proper function is to will maximal total utility¹, or to will in accordance with God's

¹Thus, Aristotelian metaethics is compatible with utilitarian normative ethics, but not with metaethical utilitarianism on which the right is *defined* as what maximizes utility.

commands, or to will what is universalizable, or to will one's flourishing, or even to cause maximal harm to self. Some of these views will, however, be less plausible given other Aristotelian commitments, such as harmony theses. The harmony theses make it unlikely, for instance, that the right thing be maximal self-harm. Indeed, Aristotelian optimism pushes us to an ethical theory whose normative consequences are, by and large, fairly intuitive. At the same time, there is a real possibility of error, and of correction of that error.

Natural Law metaethics does justice to the idea that the source of our obligations is in us, rather than in some external fact—such as a divine command—whose moral relevance is questionable. The human form or nature is a constituent part of us, and thus we compellingly have the source of morality in ourselves. In a sense, then, we are our own moral legislators. But because our nature is metaphysically not up to us, we do not have a choice as to what we legislate and we can be wrong about what we have in fact legislated. Natural Law metaethics will thus accept with modifications both the relativist's and the Kantian's insistence on autonomy, but without the ultra-conservative consequences of relativism on which we are always guaranteed to be right and hence never have reason to change our views, and while avoiding the merely formal character of Kantianism which makes it unlikely to yield sufficient normative consequences to guide our lives.

Other metaethical theories may satisfy the four desiderata as well to varying degrees, though if I am right, Natural Law does it overall best.

1.2. Internality of morality. An attractive feature of a metaethics is thus that it grounds moral truths in features of the moral subjects themselves. It seems compelling to ask "So what?" about an external ground of moral truth, such as divine or social commands, or Platonic moral facts.

Plato argued that that virtue is the center of our flourishing on the plausible grounds that moral virtue is the health of the soul (**PlatoRepublic**). What makes this argument an appealing invitation to moral virtue is that health is an internal good of the person. Socrates' analogy opens us to seeing moral virtue as a good grounded internally to the person. But since virtue is a disposition to living by the fundamental moral truths,

making these truths be internally grounded makes virtue more fully internally internally grounded.²

Additionally, an internalist intuition is probably a part of the explanation of the attractiveness of subjectivist metaethics: it is harder to say “So what?” to what is at the center of one’s self. Aristotelian ethics grounds moral truths in the agent’s form, which is a central metaphysical constituent of the agent, indeed literally the very soul on classical Aristotelianism. It is not exactly the same kind of ground as a subjectivist offers, but it is still an intimately *internal* ground.

In fact, in an important way the agent’s form provides a *more* internal ground of moral facts than the agent’s own beliefs. Typically, we do not choose our beliefs, but catch them, much as we catch flus and colds from those around us, as Richard Feldman is said to have quipped. It is largely an accident of our environment that they are what they are, especially if there is no metaphysical truth for them to reflect. One of the motivations for subjectivist theories is the intuition that morality is internally grounded. But when the internal ground is something that is itself produced by the accidents of surrounding culture, as is clearly the case for some of our moral beliefs, then we have a betrayal of the internalist intuition. On the other hand, on an Aristotelian theory, the ground of moral truth is our form, which is a central essential metaphysical constituent of the human individual. Note that a Kantian has a similar advantage: the ground of morality is our rationality, and our rationality is central and, very plausibly, essential to us.

Furthermore, a plausible subjectivism needs to take into account potential disagreements within the agent’s beliefs and values. A reasonable solution is to insist that more deep-seated aspects of one’s psyche take precedence over more accidental features. If you believe that people should get the benefit of the doubt, but in the heat of the moment you believe that the person you are arguing with should not get the benefit, the subjectivist

²Some non-fundamental moral truths, such as that I ought not sell the laptop that I am writing this on, will not be fully internally grounded, since they depend on circumstances and facts outside of us, such as that this laptop belongs to my employer.

should probably opt for defining your obligation in terms of the general belief, as it is more likely to be deeply planted in you.³ We might say that our nature as human beings, understood in the optimistic Aristotelian way as both instituting norms and impelling us to believe and follow them, has a kind of depth that trumps the deepest of our contingent beliefs.

That Natural Law makes the source of morality be literally a part of each of human being also makes it more plausible than contemporary non-naturalist Platonist-style theories (e.g. **ShaferLandau**, **ShaferLandau**), and divine command theory, on which the source of morality is found in an objective reality outside of us—in a heaven, Platonic or theistic—that we merely participate in rather than have as a part of us.⁴

An ethics grounded in our natures has something in common with both internalism and externalism. Like in internalism, fundamental ethical truths are grounded in facts wholly about us. But these facts about us are, like in externalism, not directly available

³An alternative is the idea that higher-order values and ideas should take precedence over lower-order ones (cf. **FrankfurtHigherOrder**, **FrankfurtHigherOrder**). Thus, if Bob believes that Alice should not get the benefit of the doubt, but also believes that he should believe that she should, then it is the second-order belief that trumps. This does not settle all the questions of disagreement between first-order beliefs—there need not be a second-order belief that decides between them. Moreover, although it is psychologically uncommon for a higher-order belief or value to be a mere whim, there is nothing to rule that possibility out. Suppose that I think to myself: “To illustrate the point I am making in this footnote, it would be good if I had the really wacky belief that it’s required to insult blue-eyed friends on Friday the 13th of January in a prime-numbered year.” Nobody should think that such a whimsical second-order value or belief yields a first-order obligation.

⁴Some Platonists, especially bundle-theorists, hold that properties are Platonic abstract objects that are nonetheless a part of the entities that exemplify them—they are immanent universals (**ArmstrongWorld**). Such Platonists can also say that the source of morality is truly a part of us. As **OckhamBoehner** (**OckhamBoehner**, p. 39) vividly argued by noting that then Christ in heaven has a component—namely, *humanity*—in common with Judas in hell, it is somewhat implausible to think that all human beings literally overlap, but philosophers are used to biting such bullets. In ??forward, we will discuss the Aristotelian version of this theory on which the human form is common to all humans rather than an individual constituent of each. Here it is perhaps worth noting that an *individual and unshared* form perhaps does a little bit more justice to the self-legislation intuition.

to consciousness. They need to be teased in Aristotelian methodology out by looking for an elegant moral theory that reconciles the most plausible insights of virtuous agents and ourselves and that fits well with a picture of us as an animal whose form of life is what it is.

1.3. What are moral or rational norms? The species-relativity of norms makes it plausible that in the space of possibilities—and perhaps in extraterrestrial regions of actuality as well—there will be species of beings that are intelligent enough to have advanced science and technology but whose natural behavior will be quite different from us. This raises a difficult question as to what makes a particular set of natural norms count as a set of moral or even rational norms, and hence makes a species that possesses them a species of moral or rational agents.

Not all natural norms are moral or rational norms. The natural norm behind a properly functioning horse shedding in the spring is neither moral nor rational. Perhaps a necessary condition for a moral norm is that it govern voluntary behavior or a disposition to voluntary behavior. But the question of what behavior counts as voluntary is difficult. It is tempting to say that the behavior of an entity is voluntary if it is subject to reasons. But reasons live in a space made possible by rational norms, and so it seems we need an account of rational norms, at least, to make sense of what behavior is voluntary.

Here is one speculative Aristotelian functionalist way to answer the questions in three steps. First, we connect reasons and norms with goods considered as such: an (internal) reason for a behavior is an apprehension of the behavior as *good*. A mouse may apprehend cheese as yummy to eat, or maybe even as nutritious, but not as *good*. Being yummy or being nutritious is a way of being good to eat, but to apprehend as yummy or nutritious is not to apprehend either the cheese or its consumption as good.

Second, we say that a necessary condition for a behavior to be voluntary is that it comes from such a good-regarding reason. This, of course, raises the infamous problem of in-the-right-way causation. A behavior can be deviantly caused by a reason without being voluntary. Davidson's famous example is the climber who wishes to lose the burden

of holding up another person on a rope, intends to drop the other, and the desire to do so “might so unnerve him” that the rope slips (**DavidsonEssays**). Whatever reason the climber had for the dropping is the cause of the dropping, but the dropping is involuntary. Aristotelian metaphysics does, however, seem to have a tool for solving this problem. Causation can be seen to be teleological in nature, and we might say that it is a primitive fact that sometimes the effect *is* a fulfillment the teleology of the cause, in which case we can say that the effect is caused in the right way. A voluntary behavior is then one which is a fulfillment of the teleology of the cause, where the cause is a good-regarding reason.

Third, the will is then functionally defined as the system by which reasons lead in the right way to behavior that promotes the goods apprehended by these good-regarding reasons. A rational norm is a norm of the will’s behavior favoring some or all reasons. We can then define a moral norm as identical to a rational norm, if we like. Or, if we do not want to say that prudential reasons are all moral, we might say that a moral norm is a norm of behavior of the will favoring some or all reasons that themselves are focused on a good not considered primarily as a good to self.

This is not, of course, the only way to define which norms are moral or rational. And it is quite possible that the question of which norms are moral or rational is largely a verbal question. Go back to the characterization of reasons in terms of goods. The mouse takes the cheese to be yummy, and that is not taking the cheese to be good. But humans often represent good things in thicker ways: as beautiful, courageous, or even divine. Couldn’t we imagine a continuum of animals where at one end the cheese is represented as yummy and on the other as having gustatory beauty? Somewhere in the continuum we have moved to representing the cheese as having a thick form of goodness. But where this happens is unclear.

We have a certain set of norms for the will. We can call these rational, and a subset of them moral. What hangs on whether a different set of norms governing the behavior of a different class of beings counts as rational or moral? Couldn’t this be like the merely verbal question of which animals’ hard cephalic projections count as “horns”?

But perhaps the question of which things are moral agents matters for first-order moral questions about interspecies relations, such as which organisms we are permitted to eat, whose lives we should save, etc. However, it is not clear that the answers to these questions will neatly line up with determinations of the boundaries of moral agency. Consider intelligent whales that are fine philosophers and scientists in their epistemic life, and that even enjoy contemplating the good, but do so purely non-practically, without the good being any motivator for their actions, all of which are instinctive. It would seem wrong to eat such beings, even if they turn out not to be moral agents. On the other hand, imagine a shrimp that has the normal behavioral complement of a shrimp, with one exception: it represents the ingestion of algae as good, and voluntarily pursues this good as such. And its intellectual abilities are the minimum needed for an ordinary shrimp's life as combined with the most minimal possible concept of the good. It may well be wrong to eat such shrimp, but it is difficult to say whether one should save the life of one of these minimally moral shrimp or the life of one of the intelligent but amoral whales (of course, we should not be biased here by size!) given a choice.

1.4. Supervenience. It is widely held that moral facts supervene on non-moral facts: if two possible worlds differ with respect to moral facts, they must differ with respect to at least one non-moral fact. Similarly, it is held that normative facts in general supervene on non-normative facts. The difficulty is then to explain the supervenience relations.

The nature-first theorist has a complex relationship to both supervenience claims. Begin with the supervenience of the normative on the non-normative, and first consider general normative claims such as that every sheep should have four legs and every human should refrain from torturing the innocent. Such normative claims are necessary truths, since their truth is a part of what makes a sheep a sheep and a human and a human. Necessary truths vacuously supervene on any basis we might choose, since there are no possible worlds that differ with respect to necessary truths.

But what about *particular* normative claims, such as that Sally ought to have four legs or that Trump ought to discharge the duties of the President of the United States? If we are

interested in the supervenience of the normative on the non-normative, we face a serious problem on Aristotelian metaphysics: there are very few non-normative facts. It seems that every natural kind is defined in part by normative properties. That Sally is a sheep is itself a normative thesis, since a part of what it is to be a sheep is to be such that one ought to have four legs. That Sally is an animal is also normative. And Sally is *essentially* a sheep, and hence being a sheep is central to Sally's identity in such a way that it may even be the case that the very claim that Sally exists counts as a normative claim. Aristotelian metaphysics likely reaches even down to the fundamental particles. Electrons not only do but should repel other electrons. A non-normative fact, it seems, cannot make reference to any natural kinds, and cannot even make reference to particulars that fall under natural kinds. But on Aristotelianism, every particular substance, with the exception of God if there is a God, falls under a natural kind. And facts about God are through-and-through normative, since God is essentially perfectly good. Thus, on Aristotelianism, all facts about the existence of particular substances seem to be normative.

Nor can *any* existential claims escape normativity, since to be a substance or to be appropriately related to a substance (say, by being its accident). If so, then every existentially quantified claim is normative. The denial of a normative claim is normative as well, and since a universally quantified claim is the denial of an existentially quantified claim (everything is *F* if and only if there does not exist an object that is not *F*), universally quantified claims will be normative as well.

But if all facts about particulars and all quantified facts are normative, it seems that *all* facts are normative on Aristotelianism. If this is right, then every pair of worlds agrees with respect to all non-normative facts, because there are no such facts. And if all facts are normative, then any two worlds that are the same in normative terms are altogether the same. Thus, the thesis of the supervenience of the normative on the non-normative can only hold given the modal fatalist thesis that there is only one possible world. And of course modal fatalism is not only highly counterintuitive but also does not allow one to do any justice to the appearance of contingency in all the parameters we have been

identifying in this book—or in Mersenne’s original observations about astronomical objects (**MersenneImpiety**).

But while this goes against mainstream views of normativity, it is perhaps an advantage of Natural Law metaethics. For by eliminating non-normative facts, we no longer have any puzzling phenomenon of the relationship between the normative and the non-normative to be explained. And at the same time, this elimination of non-normative facts is not a form of reductionism or eliminativism, since the ordinary facts that are usually taken to be non-normative, such as that the sky is blue, are still there, but turn out to be richer, having normative aspects to them.

Next, what about specifically moral normativity supervening on the non-moral? Moral facts are normative facts about the will. Again, general moral facts, such as that humans should refrain from torturing the innocent or should discharge the duties of the President of the United States if they have voluntarily sworn the relevant oath of office, are necessary truths and hence trivially supervene on whatever facts we want, including non-moral or even non-normative ones. What about particular normative facts, such as that Trump should discharge presidential duties? Since having a human nature is an essential feature of Trump, and the human nature essentially has moral norms, it does not seem that even such facts can supervene on facts about humans that aren’t morally normative.

But now imagine a world w_1 that *looks like* our world w_0 . It has bipeds that look just like us. The history of these bipeds is empirically indistinguishable from our history. There is, for instance, a biped that is empirically indistinguishable from Napoleon and one empirically indistinguishable from Marie Curie. Even apart from formal statements of supervenience, the moral supervenientist is apt to have the intuition that in w_1 there would have to be the same moral facts as in w_0 . The Aristotelian, however, appears to be committed to holding that there is a world w_1 as above where the bipeds’ moral norms are different from ours, although their bodies and behavior looks just like ours. Perhaps denying a theoretical intuition, namely that such a w_1 is impossible, is not problematic. But there is something more problematic in the vicinity: moral skepticism appears to follow for the

Aristotelian. For these beings will have the same collective moral beliefs and intuitions as we do, but presumably some of these beliefs and intuitions will be wrong. If so, then how can we be confident in our collective beliefs and intuitions?

First, however, we should note very likely a number of our collective moral beliefs and intuitions *are* wrong. Certainly this was true for many of our ancestors—take, for instance, the acceptance of slavery across large swathes of the world in times past—and it would be very implausible to think the same is not true of us. It was suggested that the Aristotelian is committed to the claim that there could be a world w_1 where *some* of the moral norms of the bipeds differ from our norms. Assuming they have the same moral beliefs as we do, it follows that either some of their collective moral beliefs or some of our collective moral beliefs are false (or both!). But that does not yield moral skepticism, since we already had good reason to think that some of our collective moral beliefs are false, and surely that is no more a justification for moral skepticism than our occasional errors are justification for skepticism about the physical world, *pace* **DescartesMeditations** (**DescartesMeditations**).

What if we strengthen the story by supposing that *for the most part* the moral norms of the bipeds in w_1 fail to match their collective beliefs and intuitions? First, however, it is not completely clear that this strengthened story is possible. If a perfectly good God necessarily exists, perhaps that goodness would logically preclude the creation of a world with a wholesale mismatch between moral norms and moral beliefs. But even if it is logically possible to have a world with such a large-scale mismatch, the mere logical possibility of a world with a skeptical scenario does not undercut our beliefs or even knowledge. It is logically possible—and even has a non-zero probability—that next time you pour cream into coffee it will spell out, in tiny swirls, Hamlet's most famous speech, but surely we know that this won't happen—or at least justifiedly believe it won't. To get skepticism, one would need the skeptical scenario to be fairly probable, and supervenience-style arguments do not yield any probability.

1.5. Agent-centrism.

1.5.1. *The egoism objection.* According to Natural Law metaethics, an action is right provided that its performance constitutes the will's flourishing, and is wrong provided that its performance constitutes the will's languishing. This seems objectionably egoistic. Paradigm cases of moral wrongness involve harm to others, to the *patient*, and are wrong because of that harm. The thought that the action makes the *agent* languish is a thought too many (cf. **WilliamsPCM**, **WilliamsPCM**). Therefore, the argument goes, we should opt for an other-centered metaethics.

My response will be two-pronged. First, I will argue that the very features criticized in Natural Law provide a significant advantage in a number of cases. Second, however, I will argue that the argument against Natural Law's agent-centric character only works against some normative developments of the Natural Law metaethics, rather than against the metaethics.

1.5.2. *The normative advantages of agent-centrism.* Other-centered theories nicely account for what is wrong with murder: it gravely harms the victim. They account slightly less well for what is wrong with typical cases of attempted murder: it is an attempt to harm to harm the victim. But they do not account for atypical cases of attempted murder where the victim simply does not exist. Suppose Alice thinks that she has an identical twin living somewhere in Toronto, and sets out to kill her, to avoid the twin's claiming an inheritance. Alice has an extremely rare genetic disorder which an identical twin would share, but which is very unlikely to be otherwise exhibited even in a city as large as Toronto. She adds a poison to Toronto's water supply that targets only people with this genetic disorder, and then takes care to avoid drinking Toronto's tap water. But in fact Alice never had a twin.

There is thus no one that Alice is attempting to kill. Yet morally speaking, she is just as guilty as in an attempted murder case where her twin exists, and depending on one's views on moral luck maybe even as guilty as in the case where she succeeds in killing her twin. It is worth noting that in Anglo American jurisprudence, Alice might get away under the doctrine of impossible attempts, on which an attempt has to have some feasibility, and

trying to kill a non-existent person has none. (Of course, Alice is likely to be convicted for pollution and for reckless endangerment of people with this disorder, but these miss out on her murderousness.) However, it is clear that notwithstanding the law of impossible attempts, it makes no difference to Alice's guilt whether in fact she has a twin or not.

We may, of course, try to save the doctrine that wrongdoing always wrongs another by trying to identify other victims, such as society or God. We can try to tweak the Alice case to exclude the society solution. Perhaps Alice is trying to kill her twin sister in a world where she thinks they are the only survivors of a disaster, but in fact Alice is the only survivor and she's never had a twin. That won't help with the God case, at least not within classical theism, since God is traditionally and plausibly thought of as a necessary being.⁵ Furthermore, the intuition that I am responding to is that there is something particularly centered on the ordinary direct human victim of a wrongdoing that contributes much of the wrong. And if God or society is what we count as the victim in the Alice case, then it seems that we have to say that in the Alice case there is less wrong than in the more ordinary case of attempted murder where the victim actually exists. And yet it seems that how wrong Alice's action is does not depend on whether she has a twin.

The above focused on patient-centric wrongs. We can also think about patient-centric duties. Again, it seems that our duties go beyond these. Plausibly, we have a duty not to deliberately produce a human being who is so genetically constituted as to be practically guaranteed to have a life of unrelenting suffering. Now suppose that Bob and Carl both know that they and their spouses have genes such that if they reproduce, the child will have a life of unrelenting suffering. In light of this knowledge, Bob refrains from reproduction, while Carl's sadistic tendencies impel him to reproduce in light of this fact. Bob and Carl both have a duty. In Carl's case, we can identify the individual to whom he has

⁵Findlay1948 (Findlay1948) gives a convincing case that God would have to be a necessary being, but because he does not have a distinction between narrowly logical necessity and the broadly logical necessity of post-Kripkean metaphysics, he thinks this refutes the existence of God.

this duty, an individual whom he has wronged. But in Bob's case, the analogous individual does not exist—precisely because Bob has fulfilled his duty. Again, we can try to identify others to whom Bob owes not having a child—society, God and likely Bob's wife. But since there is one less individual here than in Carl's case, on a patient-centric theory Carl has a somewhat more stringent duty, since the unfortunate child exists. However, it does not seem that Carl has a more stringent duty than Bob.

Finally, moving away from cases, it is obvious that some degree of agent-centrism is needed for any plausible story about wrongs and duties. It is *agents* that do wrongs and have duties. We have a duty not to eat humans. Lions, pigs and horses have no such duty. Therefore, an account of what makes it wrong for me to eat other humans has to involve some facts about *me*. It cannot be wholly other-based.

One might respond that on the Natural Law account, what makes it be wrong for me to eat other humans involves *only* facts about me, while it should also involve facts about the prospective victims. But how we understand this objection depends on how we understand the question of "what makes it wrong for me to eat other humans".

First, we can understand it as a question about the grounds of the general moral rule that it is wrong for humans to eat other humans. On Natural Law the grounds of that general moral rule are entirely within the agent. However, that is how it should be. For the general moral rule would also hold even if there were no other human beings in the world. And in fact the general moral rule would hold *non-trivially* even if there were no other humans, since even if there were in fact no other human beings, I should avoid actions that are likely to constitute the eating of a fellow human being (e.g., shooting and eating an animal that has a significant epistemic probability of being human). An account of the wrongness of eating other human beings that requires other humans to exist is unsatisfactory.

Second, we can understand the question as asking about particular cases: What is it that makes it wrong for me to eat, say, Carl? But then the Natural Law story is going to

include a fact about Carl: I am the sort of thing that shouldn't eat other humans and Carl is another human.

On neither reading do we have an argument against Natural Law metaethics.

1.5.3. *Avoiding agent-centrism in normative Natural Law ethics.* Let me start with a personal confession. For many years I objected to the eudaimonism I took to be at the heart of Natural Law, a eudaimonism one might take to consist of the twin theses:

(8) What makes an action right is its promotion of the agent's flourishing.

(9) An agent's right actions are aimed at the agent's flourishing.

And these theses seemed objectionably egoistic.

However, while there are ways of pairing Natural Law metaethics with a normative ethics that embraces (8) and (9), both theses are dispensable for a Natural Lawyer.

Indeed, (8) is so clearly wrong that it is unlikely that many Natural Law theorists accept it, given the well-known anti-consequentialism of the Natural Law community. Let us grant the Socratic thesis that moral flourishing is central to our flourishing. This thesis has historically been taken to support deontology. However, it is wrong to rob a bank in order to pay for the tuition of an ethics class even if there is strong empirical evidence that this class will be so morally transformative that on the whole one's flourishing will be promoted, even if one takes into account the temporary harm done to one's moral self by the robbery. Similarly, one may have a duty to continue working in a job that is just barely moral and empirically likely to be destructive of one's flourishing as a moral agent in order to pay for one's child's medical treatment.

Now, on the metaethics that I am defending, what makes an action right is that it *constitutes* the agent's flourishing with respect to the will. If we add to this the thesis that whatever constitutes the agent's flourishing with respect to the will constitutes the agent's flourishing as a whole, then we get a version of (8) with "promotion" replaced by "constitution". However we should not think that what constitutes the agent's flourishing with

respect to the will constitutes the agent's flourishing as a whole. If an agent is flourishing with respect to the will, but is full of ignorance, in great pain, and lying in a bed of vomit??Vlastos-ref, the agent is not flourishing on the whole.

And the thesis that what makes an action right is its constitution of the agent as flourishing in respect of the will seems to be simply a thesis about proper function, and does not imply any selfishness. Consider, for instance, that a bee's defending the hive at the expense of its life fulfills the bee with respect to whatever we might call the driver of the bee's activity, a driver analogous to our will. But this does not make the bee in any real way selfish: the bee, presumably, does not think about itself as it does not have a self-concept. And certainly a guided missile is not selfish just because it fulfills its nature by exploding.

It is tempting to think that in the case of an agent who is driven by a rational will, if what makes the action right is its constituting the agent as flourishing (in one respect), then by willing the action under the description "right action", the agent aims at flourishing, in a way in which neither the bee nor the guided missile's actions are aimed at flourishing. If this line of thought is correct, then the characterization of rightness in terms of flourishing implies (9), and that seems objectionably egoistic.

However, a rational agent's intentions are hyperintensional. It is possible to aim at heating up a room without aiming at increasing the kinetic energy of the molecules in the room, even though what makes there be heat in a room is the kinetic energy of molecules, and necessarily one is present if and only if the other is. Indeed, during the millenia before the relationship between heat and kinetic energy was known, no one aimed at increasing the kinetic energy of molecules while heating a room, and even now when the relationship is well-known, few people's intentions in turning a thermostat dial make reference to molecular motion. Similarly, even if the rightness of an action is grounded in, constituted by or even identical with the action's being an instance of the agent's flourishing as a willer, rightness can be aimed at without aiming at flourishing.

Furthermore, virtuous agents rarely aim at rightness as such; indeed the literature on moral fetishism insists, I believe excessively, that they never do so (**SmithMoralProblem**).

Instead, virtuous agents aim at thicker right-making features of an action, such as its being an expression of loyalty to a friend, its fulfilling a stranger's need, or its having been promised. It is because the action has such thick features that its performance is an instance of volitional flourishing.

There are, of course, times when a human agent aims at rightness as such. One set of cases is provided by agents who cannot figure out on their own what is right and have to take the rightness of an action on the authority of another, without understanding what makes the action right. Such agents include small children, but also sometimes ordinary well-functioning adults who find themselves in such situations of such moral complexity that they turn to a professional ethicist or a trustworthy friend for advice. Furthermore, in cases where an agent is not sure which action is right but decides on probabilistic grounds, it seems plausible that they are acting for the sake of rightness as such.⁶

Is this objectionably egoistic, assuming the rightness is constituted by the agent's volitional flourishing? There are at least four reasons to doubt this. The first was already mentioned: the hyperintensionality in intentions.

To see the second reason, observe that we actually have a *three layer* story:

(10) the rightness of the action

(11) the action's being such as to constitute volitional flourishing

(12) the thick features of the action because of which the action constitutes volitional flourishing.

The egoism objection in the case of agents aiming at right as such insists that the willing of (10) inherits an egoistic character from the agent-centrism of (11). But note that (11) is not the end of the story. Just as (10) is grounded in (11), so likewise (11) is grounded in (12). And in paradigmatic cases, the thick features in (12) are other-centric features. If we

⁶In fact, **Weatherson2014** (**Weatherson2014**) has used this to argue against avoiding moral risks—avoiding performing actions which have a non-negligible probability of being wrong. But that one should avoid moral risks once the degree of wrongness is sufficient is so plausible that it seems better to use this kind of a case to argue that there is no objectionable moral fetishism here.

think that willing the rightness of the action inherits egocentrism from the flourishing, we should even more think that it inherits other-centrism from the thick features, since the thick features are a yet more ultimate ground of rightness than the flourishing is.

Third, recall that we already noticed that an action can be right and constitute volitional flourishing but hamper one's flourishing as a whole, as in the case of working a soul-destroying job to pay family medical bills or refusing to rob a bank in order to pay for a morally transformative class. In such cases, it is absurd to say that by aiming at volitional flourishing one is being selfish, since the action does not, in fact, contribute to one's good overall.

Indeed, a metaethics that grounds rightness in flourishing as a willer is compatible with one's never being required to intentionally pursue one's own good. We can (perhaps with some difficulty) imagine an alien species whose members pair off in such a way that the proper functioning of each one's will is just to will the good of the other member of the pair. Perhaps this is a species so physically constituted that they are always more effective at benefiting others than at self. In such a species, the Natural Law metaethics would require utter unselfishness—and yet what would *ground* the rightness of an action would still be that the action is proper to one's will and hence constitutes the will as flourishing.

We could imagine two versions of such aliens. They might be less reflective than ourselves, and never act on higher-order reasons like rightness as such. Or they might be reflective, and might even come to a Natural Law metaethics on which an action is made right by its constituting the agent as flourishing. In such a case, they might aim at an action under the description "right", but only because they know that an action's rightness is ultimately grounded in their species in the action's benefiting the other member of the pair, though mediately in its constituting the agent's flourishing. In such a case, there is no more objectionable egoism than you exhibit in a case where an eccentric rich person says that they will donate lots of money to charity if you do something that is good for you and so you eat a healthy and delicious salad.

For our fourth and final response, consider somewhat odd cases of intentional activity. Suppose that I want to test a device that detects nerve signals between between the brain and the arm, and so I wiggle my fingers. My action aims at finger movement, but in a sense I don't actually care about finger movement. My end is to trigger the nerve signal detector. The movement of fingers not only is not my end, but it is not even a means to triggering the nerve signal detector. To make this point clear, we might suppose that the detector is triggered before the nerve signal reaches the muscles controlling the fingers, so that the finger movement comes after the triggering of the detector. Or, for a more common case, consider the practice of follow-through in racquet sports. Players are counseled to follow-through on their hits, i.e., to keep their racquet moving after it has made contact with the ball or shuttle. This movement makes no noticeable causal difference to the flight of the ball or shuttle, but aiming at a longer movement makes the racquet movement at the time of impact stronger. If one weren't aiming to continue the motion after the hit, one would likely begin slowing down the motion before the hit. In a game, the post-impact motion itself is not a useful end, nor need it be a means to anything one cares about in a game.

We might say in cases like this that one aims at "unnecessary event", the finger wiggling or the extra movement, in order to better calibrate one's action with regard to the ends that one cares about. We can call this "calibrational" aiming or intention. This calibrational aiming has something in common with the case of a hunter who knows how their gunsight is misaligned, and hence aims a meter to the right of the deer, though the difference is that in the finger wiggling and follow-through cases one really does intend the fingers to move and the follow-through to occur, while in the crooked sight case the hunter does not intend the bullet to go to the right of the deer. But in all three cases one merely uses the aim in order to calibrate the aspects of the action that one really cares about.

It is compatible with the Natural Law story that the agent aims at rightness, and hence at the flourishing of the will, only calibrationally. And merely calibrational aiming at one's

good is in no way selfish—it does not imply any care, not even instrumental, about getting one's good. This is something that our altruistic paired aliens could do.⁷

That said, we are not such aliens. The goods that our will naturally aims at include our own goods. The fact that an action results in or constitutes our own flourishing counts in favor of the action about as much as the fact that it results in or constitutes the flourishing of another individual. Thus in our own case we can have two sets of reasons for doing the right thing with regard to other people. First, we have the thick reasons that render the action right, reasons that in typical cases are other-concerning. In cases where we act on another's counsel, we may not be aware of these reasons, and we may be aiming calibrationally at rightness. But, second, acting rightly is good for us. One would be failing to have the proper attitude to oneself if one didn't take the fact that the action is good for one into account. Thus there is an egocentric reason available whenever we know we act rightly. However, this is as it should be. That acting well is good for us is one of the great philosophical discoveries of all time, going back to Socrates, Plato and Aristotle, and every moral theory should include this fact, whether the theory uses this fact in grounding the right, as on Natural Law, or whether this fact is just an additional observation independent of the grounding of the right.

2. Supererogation

It initially appears that one should perform an action best supported by moral reasons. But this principle leads to a very demanding ethics with no room for supererogation. Consider a heroic sacrifice of one's life for the sake of others—say, jumping on a grenade to save innocent lives. In typical cases, this is more praiseworthy than refusing to make the sacrifice. If it is more praiseworthy, it is surely better supported by moral reasons. But nonetheless, someone who refuses to make the sacrifice is typically not to be blamed. Thus there are “actions that go beyond duty” (Urmson), which we call “supererogatory”. These

⁷cf:Howard: <https://jesp.org/index.php/jesp/article/view/1249/332>

are actions that are morally better than their omission, but nonetheless their omission is not a bad thing.

It is very puzzling how to make sense of the permissibility of doing an action less well supported by moral reasons. But a Natural Law account has a way by making a distinction between flourishing and languishing that applies very broadly. A cheetah capable of reaching 120 km/h flourishes less than a cheetah capable of reaching 125 km/h, but neither languishes, while a cheetah only capable of 12 km/h does languish in respect of running. Many areas of functioning have norms allowing both a binary distinction between failing to meet the norm and meeting the norm, and a degreed distinction between levels of flourishing in respect of the norm.

We can say that an individual's state or activity *S* is supernormal in respect *R* provided that the individual flourishes in *S* with respect to *R* more than they would in the case of some alternative state or action *S'* that still would not be an instance of languishing with respect to *R*. And now we can say that an action is supererogatory provided that it is supernormal with respect to the exercise of the will.

The mystery of supererogation was how one can count as permissibly acting if on balance one has moral reason to do something better. But once we see that this is just a special case of a phenomenon that extends far beyond morality, and indeed beyond the life of rational beings, the mystery should be significantly decreased. And the general framework of the supernormal matches our intuitions that there is such a thing as falling short of what is normal, and such a thing as exceeding it.

3. Equality and human dignity

We have the intuition that all humans, or at least all mentally functioning adult humans, are in some important sense equal. It is difficult, however, to figure out what the relevant sense of equality is.

A utilitarian has a neat procedural account of equality. It is not that humans have equal value, but rather that when we maximize total utility, this total utility is the sum

of individual utilities with *equal* weights. But there are serious problems with utilitarianism. And indeed, the equal weight part of the view is itself implausible. Suppose that Alice has kidnapped Bob and taken him to a distant uninhabited planet. On her way back to earth, Alice had an accident and got marooned on a different uninhabited planet. Alice and Bob have enough supplies to survive a year. It is certain that neither Alice nor Bob will ever come in contact with other people. Carl knows the above, and is himself marooned on a third planet, but has enough supplies for two lifetimes. He can send a drone with half of his supplies to Alice or to Bob, but not to both. No one besides him and his recipient will know. It seems clear that Carl should send supplies to Bob rather than to Alice the kidnapper. But on an equal weight utility view, this is false.

Furthermore, a standard utilitarian view is that *all* loci of utility are equally weighted, including non-rational animals. Humans only count for more because they are capable of higher utilities—they have more sophisticated pleasures or higher order desires. If equal weighting of utilities is how the utilitarian counts as doing justice to the intuition that all humans are equal, then the utilitarian fails to do justice to the equally strong intuition that humans and non-rational animals are *not* equal.

Perhaps a deontologist can do better, saying that people have equal *rights*. Now, rights are correlative to duties, so presumably the equality of rights is correlative to equality of duties of non-infringement. But while it is equally true of everyone that they have the same basic rights, and it is equally true that it is wrong to infringe on these rights, this is not actually enough for equality of rights. For instance, if *E* is your right not to be deprived of your eyes and *T* is your right not to be deprived of your toes, it is equally true that you have *E* and you have *T*, and it is equally true that it is wrong to infringe on your *E* as that it is wrong to infringe on your *T*. But *E* and *T* are not equal rights, since *E* is more stringent than *T*. For instance, if one attacker is about to infringe your *E* and another is about to infringe your *T*, we would expect the police to prioritize your right to your eyes over your right to your toes. Moreover, while it is equally true that both attacks are morally wrong, they are not equally morally wrong (similarly, it is equally true that elephants and whales

are large compared to mice, but whales are larger, both absolutely and as compared to mice): both assaults deserve significant penalty, but the assault on eyes deserves a greater. Within a single individual there are many unequal rights but that are, nonetheless, equally truly possessed by one. Thus equal truth of possession of rights is insufficient for equality of rights.

Do we in fact have equal rights? Well if we compare rights by stringency, as we did *E* and *T*, this is not clear. We have stronger reason to stop the murder of a complete innocent than to stop the murder of a someone who just falls short of deserving the death penalty, and the murder of a complete innocent deserves a greater punishment—if not in law, then in public opinion (cf. Mill). Similarly, it seems clear that we should make a greater effort to stop the murder of a person with a decade of life ahead of them than the murder of someone who otherwise has five minutes of life remaining.

Or perhaps equality is a political matter. Each mature adult, perhaps, should have equal input in social decisions. But while this may be the best practical way of running things, it is not clear that it has an imperative beyond the practical. Suppose that we had an indisputably correct way of measuring wisdom and virtue. It does not seem clearly wrong for a society to weigh votes according to the wisdom and virtue of the voter. The main reason not to do so is the pragmatic one that any way of measuring wisdom and virtue will be contentious, unreliable and open to abuse. Besides this, it seems that our intuitions about equality go beyond the sphere of democratic politics, and extend to non-mature humans as well.

If we look at the range of actual human abilities, we see significant incommensurabilities and inequalities. But on a robust theory of human nature, we can say that possessing the human form is not a function of possessing multiple valuable properties that some have to a greater and some to a lesser degree. Each human has the *human form* to exactly the same degree. The possession of this form is what makes us human, and it is the valuable thing that grounds our dignity.

But does not everyone, Aristotelian or not, agree that there is such a thing as being human, and that we are all equally human? Yes, of course. But on non-Aristotelian views, we differ from one another in regard to the properties that make us human.

Let's say that we define humanity in terms of having DNA sufficiently close to some standard *H*. Then, first, surely we differ in how closely they hew to that standard. Second, since human DNA varies from person to person, even if you and I are equally close to the standard, we are close to it for different reasons: I may be close to *H* because I am very close to *H* in portion *A* of my DNA and somewhat close to *H* in portion *B*, while you are close to *H* because you are very close to *H* in portion *B* while being somewhat close in portion *A*. Third, if we take humanity to be the valuable thing that we are equal in, then it seems we are taking closeness to *H* to be valuable. But closeness to *H* seems valuable in virtue of the fact that *H* is a way of coding for various valuable properties, such as a variety of aspects of intelligence (insofar as these are heritable) and a variety of valuable physical features. It is indeed valuable to have DNA that codes for these good things, but nonetheless we differ in this valuable thing—for some of us may have DNA that better codes for, say, musicality, and others have DNA that codes less well for musicality (**WAGUM**). Thus our being human on a DNA standard is constituted by a variety of independently valuable genetic features, and we differ in these values. This does not seem to be a good account of human equality.

Or suppose that we define humanity in the way biologists define species, as groups with significantly more genetic interchange within the group than outside it. We are members of the human species, then, because of patterns of genetic interchange in our ancestry. Again, this is something that comes in degrees: there are human populations that are more or less isolated from the bulk of humanity. And, second, it is far from clear that being a part of this genetically interchanging population as such has the kind of immense value that our equality qua human beings needs to have in order to do justice to our intuitions about human equality. If a small group of people became isolated for several centuries they would no less be our equals.

The Aristotelian account is perhaps the unique account that defines humanity in terms of possession of an *exactly* similar constituent—the human nature—rather than the possession of non-exactly similar properties.

But how does having the human form ground a deep fundamentally equal value of human beings? We can say that we differ with regard to many valuable features we have, but a particularly valuable, perhaps the most valuable one, feature is our form—it is that which makes us human, which sets the norms that define our dignity, that gives us our high calling of pursuing intellectual and moral virtue, and that is the ground of the possibility of our flourishing. Items used in worship—holy books, vestments, thuribles, altars, and sacred locations—are considered sacred in significant part due to their ends. Humans have high ends in virtue of their form or nature, and there is thus a deep value we all have in virtue of these ends. Granted, there is a further value when these ends are fulfilled. But there is an equality in the baseline—in the having of the ends—and we can plausibly say that it is this equal possession of the same high ends that constitutes our dignity.

We can thus distinguish our valuable features into two classes: dignity, which we have in virtue of the teleological structure specified by our form, and what one might call accidental value, which consists in flourishing according to that teleological structure. We are all equal in dignity but vary in accidental value.

At the same time, we talk of “dignity” in the case of roles that one can accidentally have, such as the dignity of a monarch (which can be infringed with *lèse majesté*). Here we can follow Aristotle’s idea that philosophically important terms have a focal sense and non-focal derivative senses, so that the focal sense of “health” is the proper function of an organism’s body, while derivative senses apply to that which indicates focal health (e.g., healthy urine) or produces it (e.g., healthy food) or is in some other relevant way related to focal health. Focal dignity is the value we have in virtue of having the role *human being*. Non-focal dignity is value we have in virtue of having some accidental role, like monarch or parent. And just as we have a relationship between focal dignity and accidental value, where the accidental value is subordinate to focal dignity and fulfills the teleology in it,

likewise subordinate to non-focal dignity (which itself is a kind of accidental value) there will be accidental values of fulfilling the teleological structure in the non-focal dignity (e.g., being a just monarch or a loving parent). It may not be a coincidence that it is natural to talk of our humanity as imposing a high *calling* on us, and we call many accidental roles *callings*.

4. Outlandish paradoxes

It is easy to generate paradoxes in ethics and decision theory by invoking outlandish situations. Many, but not all, such situations involves infinities. I will give three representative examples.

First, we have the Satan's Apple paradox about infinite sequences of choices on which something further depends:

Satan has cut a delicious apple into infinitely many pieces, labeled by the natural numbers. Eve may take whichever pieces she chooses. If she takes merely finitely many of the pieces, then she suffers no penalty. But if she takes infinitely many of the pieces, then she is expelled from the Garden for her greed. Either way, she gets to eat whatever pieces she has taken. (AEH2004)

The puzzle is that for each piece, Eve has conclusive reason to take the piece, but if she acts on all these reasons, something terrible happens. As presented, this is a paradox about self-interest, but we can turn it into an ethical one by supposing that the rewards and penalties of Eve's choices devolve on someone else, say Adam. In that case, we can say that Eve should accept each piece and yet that's the worst option.

Another kind of paradox involves infinite numbers of beneficiaries. Imagine that there is an infinite number of complete strangers, numbered with the integers (negative, zero and positive), as well as two cats, all facing a deadly danger, and you have a choice between one of three equally convenient options:

- (13) Save the strangers numbered 0, 1, 2,

(14) Save the strangers numbered $-1, -2, -3, \dots$ and one cat.

(15) Save the strangers numbered $1, 2, 3, \dots$ and two cats.

Now, you have no reason to prefer the stranger numbered 0 over the stranger numbered -1 , the stranger numbered 1 over the stranger numbered -2 , and so on. So as far as the saving of people, (13) and (14) are a wash, but it's better to save a cat than not to, so (14) is morally preferable.⁸

But likewise there is no reason to prefer saving the people numbered with negative integers over the people numbered with positive integers, so as far as the saving of people goes, (14) and (15) are balanced. However, saving two cats is better than saving one, so (15) is better than (14).

But now, (13) is clearly better than (15): for in (13), you save the stranger numbered 0 instead of the two cats, and wonderful as cats are, it is much better to save that one human over two cats.

So we have a moral preferability circle, and whatever you do, there is something better you could have done at no greater cost. It seems plausible that you have a duty to do better if you can do so at no greater cost, and yet whatever you do, you violate that duty. And so it seems that you cannot act as you ought, thereby violating the plausible maxim that ought implies can.

Third, consider some particularly extreme apparent counterexample to deontology. We have the intuition that it is wrong to kill innocent people, but even deontologists find it difficult to maintain that intuition in the face of cases where killing an innocent person would save a vast number of lives. In those cases we are apt to be uncomfortable both with killing the innocent, and with letting the vast number of others die lest we get our hands dirty. Here there appears to be a conflict between the principle that innocent blood is not to be shed and the reason for that principle, the sacredness of life.

⁸If the reader thinks that cats do not fall in our moral purview, just replace the saving of a cat with saving a human from some harm much less than death.

There have been various realist attempts to defuse such paradoxes⁹, and a defender of human nature as the foundation of ethics can accept any of them as well as any other ethical realist. However, there is also a simple and highly intuitive alternative to these defusions. A horse's nature may ground facts about the appropriate gait when browsing on grass and the appropriate gait when fleeing a predator through water. But equine nature is simply silent on a horse's gait when fleeing aliens in a zero-gravity environment. Similarly, our human nature could be silent on how we should act in outlandish situations, and our principles just need not extend to such cases. This fits very well with the ordinary person's disdain for philosophers like me who spend a lot of time thinking about such cases.

There is a second, and similar, solution. It could be that our ordinary moral rules *do* extend to outlandish cases. Thus, the moral reasoning by which we generated the moral paradoxes in Satan's Apple and the infinite saving case may be correctly grounded in norms in our nature. It may well be that, say, (14) is morally better than (13), that (15) is morally better than (14), that (13) is morally better than (15), and that you ought to do the morally best (or one of the morally best, if there is a tie) between the three options. It's just that these specifications of our nature are impossible to fulfill under these circumstances. In other words, it is very plausible to say that ought implies can in situations that are a part of humans' natural environment, but there may be logically possible outlandish situations that go far beyond this environment where ought no longer implies can. Insofar as our nature gives us norms fitted to our human environment, we should not be surprised if these norms have counterintuitive implications, such as violating ought implies can, in situations far outside that environment.

We might think of these cases as ones where morality "glitches out". This kind of glitching could have a variety of forms. We might have genuine moral dilemmas where our moral requirements outright contradict each other. Or more mildly we might have cases where true moral requirements conflict with some of our intuitions, or with what one might think of as the reason for the moral requirements. For moral requirements often

⁹For an excellent critical survey, see **Carlsmith** (Carlsmith, Chapter 3).

come with a reason in terms of a good that is typically promoted by the requirement, even if all the details of the moral rule cannot be logically derived from that reason. Compare how in the American Constitution, copyright and patent law is expressly justified by the value of “the Progress of Science and useful Arts” (**USConst**). Thus a prohibition on killing the innocent promotes respect for the sacredness of life, but there may be other ways of respecting that sacredness as well. But in an extreme case where the human race would die out if we refuse to kill an innocent, a prohibition on killing is in tension with the reason for it. In these cases, we would expect to find ourselves pulled in different ways, as we indeed are. And it is unsurprising if norms for the governance of a particular species should work strangely or poorly in situations far from what one might think of as the natural environment of this species. Nor is this some sort of a defect in the norm, any more than it is a defect in a laptop that it does not function in a blast furnace.

Similar solutions might well be available on at least two other ethical theories where the laws may be customized to humanity: contractarianism and divine command theory. But, on the other hand, such solutions will be implausible on theories such as utilitarianism and Kantianism that purport to apply to any kind of rational being at all.

A similar point can be made about outlandish epistemological paradoxes. On the natural law epistemology that we will discuss in ??forwardref, we should not expect our nature to give us guidance, or at least satisfactory guidance, in situations too far out of the human environment.¹⁰ And while in ethics there are at least two common anthropocentric alternatives to natural law, contractarianism and divine command, in epistemology anthropocentric alternatives are harder to find.??Hawthorne? Thus, we have perhaps an even stronger consideration in favor of a normativity based on human nature on the epistemological side.

More will be said in ??forward about outlandish scenarios.

¹⁰This might give a different resolution to the epistemological paradoxes of infinity in **PrussInfinity** (**PrussInfinity**, Chapter 5).

5. Conclusions

Natural Law provides an attractive metaethics that allows one to ground objective ethics in something that is truly a constituent of ourselves, while avoiding subjectivism or formalism. While one can develop Natural Law along eudaimonist lines, the central thesis that rightness of actions is the proper functioning of the rational will can be detached from eudaimonism and avoids the objections to it. Moreover, the constituent, the form, that grounds our morality is something that we all have in common, and hence is an appropriate and valuable ground for our equal human dignity. Finally, Natural Law predicts that ethics is likely to have “glitches” in outlandish situations, since we do not expect the nature of a living thing to specify how it ought to function in environments far from those it is meant for, and this prediction is indeed borne out.

CHAPTER IV

Applied ethics

1. Introduction

Thinking that ethical duties are grounded in norms innate to human nature does not by itself logically entail answers to controversial questions of applied ethics. One can think that our nature requires us to kill those whose suffering we cannot stop, and hence that euthanasia is required, one can think that our nature prohibits the killing of the innocent even if that killing would be in their interest, and one can have an in-between view.

But the nature-based approach provides at least two benefits for applied ethics. First, because of the Aristotelian harmony principle, it allows facts about our natural behaviors and needs as the kinds of organisms we are to provide us with defeasible but often strong evidence about what we should do. Second, it makes it more plausible than it would be on a number of competing theories that the answers to applied ethics questions might be irreducibly intricate—not reducible to a small number of simple principles—and might include domain-specific ethical rules for the various areas of our natural lives, such as family relationships or sexuality or (if it's natural) property rights.

We will thus explore some things that we can say on nature-based ethics. The plausibility of what we will say will serve as indirect evidence for the underlying Aristotelian metaphysics.

2. Natural relationships

2.1. Siblings and cousins. Among many disconnects between academic ethics and ordinary life is that in ordinary life, many of our pressing ethical questions have to do with how we should deal with different kinds of relatives, but philosophical investigation

on this is sparse. I propose that a test case for an ethical theory is whether it can make good sense of our duties to our siblings and cousins.

Duties to friends and spouses plausibly arise from commitments we make. Duties to parents have traditionally been grounded in our obligation of gratitude for our life. Duties to children can typically be grounded in the decision to perform actions that have a non-negligible probability of producing a person dependent on us. Duties to strangers might be grounded in our shared rationality. But we owe more to our siblings and cousins than we do to strangers, even though typically we had no say in whether we are to have siblings and cousins, and even when we have no favors to return.

On utilitarianism, our duties to siblings and cousins come mainly from the contingent fact that we tend to be better positioned to do good to them, say because we know their needs better, are likely to be physically closer, and help from us is likely to be more welcome. But if such contingencies are all that is involved, then we also have to accept an error theory about our intuitions when they go beyond these contingencies. If a sibling or a stranger is drowning, other things being equal, one normally¹ should try to rescue the sibling, even if the stranger is slightly easier to pull out, or is likely to have a slightly better future life. If one finds out that a local homeless person is a cousin one has not seen since early childhood, it is more vicious to ignore their needs than to ignore similar need in a random stranger. Murder of a stranger is evil, but fratricide is worse.

In general, utilitarianism, contractarianism and most versions of Kantianism² focus on the agent's rationality, taking the details of the agent's humanity to provide no direct normative input into ethical decisions. The fact that most humans hate eating mud gives one

¹The "normally" is not just a restatement of "other things being equal." If you are a captain choosing which drowning person gets the last space in a lifeboat, with the candidates including your sibling and a stranger, plausibly you should not have a preference for your sibling even if other things are equal—instead, you should break the tie by a chance process like flipping a coin.

²But see **MaglioDissertation (MaglioDissertation)**.

reason not to feed mud to them, and the fact that humans are unable to instantly teleport ensures we do not have the same obligation to those on other continents as to those nearby. But these facts about humans are not relevantly normative³, and the normativity of the conclusions here comes from general normative considerations applicable to all rational beings on these theories. There is some *prima facie* plausibility to the idea that the non-normative facts about the relationships between parents and children, together with normative facts applicable to all rational beings, could explain distinctively filial and parental duties. But this is not plausible for the cases of siblings and cousins.

On the other hand, if we see ethics as based on the norms written into our *human* nature, given a harmony between the rational and animal aspects of this humanity, we should expect distinctive ethical norms tied to particular kinds of natural human relationships, including perhaps in the first instance familial ones. There is no need on our Natural Law ethics to derive the duties to cousins from non-normative facts about cousinhood and norms for all rational beings: such rules can be fundamental and irreducible. And the laws can, in principle, be at any level of precision, whether simply consider one's siblings at a higher weight in one's moral calculus than more distant relatives (we are all relatives, after all, as we learn from evolutionary theory) or to prefer one's siblings over one's cousins to such-and-such a specific degree (given the discussion in Chapter??backref, this option seems better). The laws could even have social construction built in: they could require us to respect our relatives in the ways that our society prescribes, and require us to establish societies that institute ways for us to respect our relatives.

Divine command theory has a similar advantage: God's commands can be at any level of generality or precision, be it to love one's neighbor or to telephone one's cousins at least twice a year if one can. In principle, rule utilitarianism can do this as well: it is plausible that having rules concerning special relationships like fraternal ones could maximize utility. But Natural Law arguably gives a better explanation of the duties tied to these special

³Though it might be that they are still normative in some way, since it may be that the concept of a human is normative as we saw in ??backref.

relationships. For the nature of these special relationships is very plausibly tied to our humanity, and hence it makes sense that the special obligations attached to them should flow from that humanity rather than the commands of a God or the results of an abstract hypothetical optimization procedure.

Indeed, on Aristotelian natural law, we can say that having these kinds of special obligations is an important aspect of what makes us human—for it is an important aspect of our form, which is precisely what makes us human. And that fits with how we think of it: someone who neglects a homeless cousin is acting *inhumanly*.

2.2. Less natural relationships. We have a broad variety of socially-instituted and culturally-variable relationships which are very unlikely to have norms encoded for them in human nature. In English-speaking countries the relationship to the parent of one's godchild or the godparent of one's child tends not to have sufficient importance to even have a name, while in other cultures it is important and specifically named. The relationship between an employer and employee varies so broadly with legal and social structures that it is probably best seen as an umbrella for a number of different relationships, none of which is likely to be encoded in human nature.

Admittedly, a relationship could fail to be culturally widespread and yet could have norms encoded for it in human nature, but there is a more elegant approach to analyzing such relationship: we can see them as cultural determinations of a more fundamental relationship type, with some of the norms coming from human nature's rules for the more fundamental relationship and others from the culture. Moreover, human nature may prescribe the scope for cultures to establish the rules. Relationships arising from such cultural determination can be thought of as "less natural". At the same time, the difference between these relationships and the "more natural" ones like siblinghood are likely to be largely of degree. For while there may be a fundamental normative relationship of siblinghood, it has further culturally-determined norms.

2.3. Marriage. A particularly interesting question, of significant relevance to controversies in our society over the past century, is where *marriage* lies on the naturalness spectrum. I shall argue that it is likely to be quite natural, with a number of fundamental norms grounded in our human nature, by arguing against two main alternative theories and combinations of them.

The first theory holds that marriage is an institution defined by many human societies. Like other such social institutions, such as judgeship, parliament membership, monarchical sovereignty, exchequer chancellorship, and presidency, it is defined by the rights and obligations conferred by society on those who enter into the institution. While we use the same words “judge” and “monarch” across societies, there is only a family resemblance between the institutions these terms refer to, since the actual rights and obligations defining the institutions are often very different indeed. The resemblance may be very weak: the rights and obligations of the monarch of England in the 13th century are about as different from the rights and obligations of the current monarch as the rights and obligations of modern day judge are from a modern day executioner. Nonetheless, for historical reasons we may use the same word “monarch”, sometimes clarifying with adjectives like “absolute” or “constitutional”.

The second theory has it that individual couples choose to undertake certain obligations with respect to each other, which obligations give rise to rights, and this complex of rights and obligations defines the marriage. In more traditional societies, a couple may not choose the obligations specifically but will rather opt for a package of “customary” obligations and their consequent rights. In modern Western society, many couples write their own wedding vows, specifying general obligations. But even in those cases, it is likely that these vows are not typically thought of as a precise and exhaustive legal contract, but rather as a way of customizing one of the prevalent packages of obligations. Again, on the individual theory, we use the same word “marriage” for all these different packages of rights of obligations due to some sort of vague family resemblance between them.

A more sophisticated theory may combine aspects of the social and individual theories, holding that not only do couples undertake obligations to each other and gain rights with respect to each other, they also undertake obligations to society and gain rights with respect to society.

But the individual and social theories are unsatisfactory for multiple reasons.

2.3.1. *Discovery.* People in good marriages come to discover new normative aspects to marriage as they go through life together. Some of these can be put into trite words like “communication is important”, but to this day not everyone knows these words when they get married, and the words often do not do justice to the nuance. We saw in ??backref that reasons of beneficence have different weight depending on the agent’s relationship to the patient. One of the things that is discovered in marriage is facts about the weight that the marriage relationship carries in moral deliberation, say in cases of beneficence, a weight that cannot be put into words. Some of the normative (and empirical) discoveries go into self-help books and marriage preparation courses, but some are subtle and hitherto we are unable to put them into explicit words, though we can recognize paradigmatic violations and fulfillments, both in reality and in fiction. And there are even occasions where instead of discovering a general norm of what to do when married, one discovers what x , with x ’s particular characteristics, is to do if married to y , with y ’s particular characteristics. This may well be a special case of a general rule that has different implications for different couples, but the general rule may yet be undiscovered.

If the norms of marriage were simply whatever it was that the parties to the marriage chose, there would be nothing to discover. And if the norms of marriage were simply set by society, it would be odd to think that it is particularly by living the married life that one discovers the norms. Rather, the norms would be discovered by study of the history of the social institution of marriage, the laws surrounding it, the intentions of the legislators, and so on.

We might, admittedly, in individual and social institution cases discover new normative facts by logical derivation from previously known ones, but that is not the main way

in which we learn about marriage: we learn about it by observing marriages, both from the inside and from the outside, over a lifetime.

Now, we discover new facts, including normative ones, about natural kinds of entities precisely by observing these entities. By observing water, we come to see that it is H₂O and by observing mammals, we come to see that their middle ear should have the malleus, incus and stapes bones??check. However, we are often best positioned to observe marriage at work in our own case, so it is unsurprising that such observation is particularly productive of knowledge of normative aspects of the relationship, though often we also learn from observation of others' marriages and their sharing of their discoveries. Often, too, these observations are mediated through literature. This is rather different from the case of natural science, but it is to be expected in the normative case, because normative laws are often easier to illustrate than to put into precise words. In any case, whether the discoveries come from one's own marriage or others' marriages, many of these discoveries come long after one has already entered into marriage, and hence the norms so discovered are not our creation at the time of marriage.

Central this growth in normative understanding is the Aristotelian harmony between different norms. Living according to the norms of marriage tends to fulfill us in other respects, while living contrary to the norms of marriage tends to be bad for us in other respects, and these are things we can often see. A happy marriage makes for happy spouses and an unhappy marriage for unhappy spouses. By observing how marriage works for oneself and others one learns what marriage should be like.

The discovery argument makes the individual commitment especially implausible, but also provides evidence against the social theory, since the discovery of norms can be made by a couple's self-observation rather and not just be observation of society.

2.3.2. *Travel*. Generally speaking, when a married couple emigrates to or visits another society, they are deemed married in their new place, unless there is some general reason that precludes them from counting as married in the new jurisdiction (e.g., the new jurisdiction does not recognize same-sex marriages).

The recognition of the couple as married is not just an honorific indexed to their country of origin. When the Queen of Denmark visited the United States in 1991, she was referred to as a “queen”??check, but obviously she did not have rights and obligations of a monarch with respect to the United States, and so “queen” here was indexed with respect to the Kingdom of Denmark, and similarly for the title “prince” held by her husband. However, if someone referred to Henrik as Margrethe’s *husband* or to Margreth as a *married woman* during the visit, these terms would not be merely indexed to Denmark. Rather, they would have the rights and obligations of an American married couple, as modified by their diplomatic immunity to persecution, and an ordinary non-diplomatic visitor from Denmark would not even have that modification.

Should we say that by the mere fact of entering a country, a couple that was married in their country of origin enters into a new marriage institution? On the social theory that is exactly what happens: the couple receives a new package of rights and obligations, definitive of marriage in a new society. But this would be quite surprising: it would mean that a couple going for a honeymoon in another country would have had two weddings (one might tongue-in-cheek wonder if they shouldn’t then be entitled to a second honeymoon?), and globetrotting couples would rack up marriage after marriage. Moreover, relinquishing one’s citizenship in a country one no longer lives in would be tantamount to a divorce.

Or perhaps instead of the new institution being entered into upon *entry* into a country, a couple by marrying becomes subject to the marriage institution of every jurisdiction that is willing to recognize them as married, but perhaps with the rights and obligations of these institutions are merely conditional on their being in those countries. While this would alleviate the problem of multiple weddings, such automatic entry into institutions in states that have no jurisdiction over one seems implausible. Moreover, the problem of multiple weddings is not solved. When Margrethe and Henrik married in 1967, there was no state of East Timor. Then in 1975 it declared independence. By that declaration, did they impose a new marriage institution on Margrethe and Henrik, a marriage institution that

disappeared next year when East Timor was annexed by Indonesia, and then reappeared in 2002?

On the purely individual theory, the travel problem disappears. Different states may add rights and obligations, but what defines the marriage is the complex of rights and obligations that the couple entered into on their own, and it counts as a “marriage” in their travel destination because of the family resemblance between these rights and obligations and those that members of that society take on when they enter into an analogous relationship.

Similarly, the travel problem disappears on the view that marriage is a natural norm, independent of society.

2.3.3. *Cross-cultural criticism.* Andronia is an especially sexist society, and Bob and Alice is an Andronian married couple. You haven’t previously interacted with Bob in a context that made his sexist views clear, but one day you find out that Alice is sick, and Bob is not showing any consideration for Alice besides the minimum needed to be shown to any human being. You call Bob out on this, and he tells you that in Andronia it is the wife’s job always to show consideration for her husband while the husband need only keep the wife alive and show her the kind of consideration one owes every human being when she is sick. He adds: “This is how my parents behaved, how Alice’s parents behaved, and Alice knew that this is what she was getting into when we got married.”

If marriage is a natural relationship, we can correctly say that Bob and the rest of Andronian society is just wrong about what marriage requires: “Bob, that may all be, but it’s not how husbands *should* behave!” We could then show Bob examples of virtuous, caring egalitarian couples in the hope that these examples would open his mind to what marriage really entails. Or we might say to Bob: “If that’s all you’ve committed to, then you’re not really married, and so you are reaping the benefits of marriage from Alice under false pretenses.”

But if the complex of obligations in marriage is either socially or individually defined, and if neither Andronian society nor the couple included any special obligation of husband

to wife in sickness beyond that which we owe any other human being, then Bob could well be simply right in his understanding of his marital duties. This is an unattractive position.

Granted, if Bob and Alice have now moved to a less sexist society, we could tell Bob that by moving to this society they have accepted the additional duties of husbands to wives. This is, however, dubious. It may be that by immigrating to a country we take on the *legal* obligations of that country, but it could well be the case that Bob is meeting these legal obligations, as these tend to be fairly minimal, and fall far short of what is morally expected from a spouse. What Bob is failing to do is to meet the customary obligations attached to marriage in less sexist societies, but it is implausible that by moving to a country one becomes obligated by the non-legal customs of the country. No moral criticism would necessarily attach to an American couple if after moving to Canada they failed to celebrate Thanksgiving in October. Furthermore, nothing of significance is changed in the above story if we specify that Bob and Alice are *still* living in Andronia. Be they in Andronia or elsewhere, a husband owes more to his wife than Bob thinks.

Perhaps we could tell Bob: "If that's all you committed to, then you aren't married in *our* sense of the word." But that isn't a criticism of Bob's behavior with regard to Alice. Bob could just say: "So what?" At most it is a criticism of his misuse of words if Bob claimed to be married. Moreover, even as a linguistic criticism it is unlikely to hold water. For we do in fact use "marriage" and related words for relationships in a vast array of historical and present societies, many of which are quite sexist indeed.

Admittedly, on both the social and the individual view, we could criticize Bob for the relationship that he is in. We could tell him: "If that's what marriage in your context is, it's a corrupt institution, and you shouldn't be married to Alice." But this is unacceptably weak tea. It implies the false claim that Bob really is married to Alice but does not owe her any consideration in sickness beyond that owed a stranger.

2.3.4. *Fulfillment of a natural desire.* Plausibly, apart from reasonable moral and practical restrictions, people should be able to marry those whom they wish to. A society that did not make this possible would be failing its members.

Now, society has no obligation to make possible the fulfillment of every desire people have. Rather, it is reasonable to make a distinction between natural desires and more contingent desires, and hold that society should support the fulfillment of natural desires, such as for food, drink, shelter, useful employment, and knowledge. Given the plausibility of the fact that marriage is one of those things society ought to make available to its members, we have evidence that marriage is an object of natural human desire. But if it is an object of a natural human desire, then it is likely that marriage itself is natural rather than constructed.

This is perhaps the weakest of the arguments for marriage being a natural relationship, however. First, not everyone shares the intuition that a society ought to make marriage possible. Second, it is not clear that we couldn't have a natural desire to construct—individually or socially—an institution of a certain type.

2.3.5. *Same-sex marriage.*

2.3.5.1. An argument for liberals. Let us assume that egalitarian justice requires one to advocate for same-sex marriage in jurisdictions where same-sex marriage is not available.

But suppose that marriage is socially constructed, and that we are in a locality in which one of the norms of marriage is that it be a relationship between a man and a woman. Then, if we understand “marriage” as the word is locally understood, it makes no more sense to advocate for same-sex marriage than to advocate for chess without pawns: these are simply contradictions in terms. Granted, we may choose to advocate for social recognition of another, more egalitarian institution than marriage. But that will be a different institution.

If we advocate for this different institution, we have two choices. Either, we propose to maintain the institution currently called marriage, whether for everyone who wishes to enter into it or just for those grandfathered into it, or not. If we propose to maintain the current non-egalitarian institution, then we are not really advocating for same-sex marriage. We are advocating for a two-institution model, closely akin to marriage plus civil unions compromises that have generally been seen as unacceptable by advocates of same-sex marriage in Western countries.

On the other hand, if it is proposed not to maintain the current institution of marriage, then the common and plausible arguments that extending marriage to same-sex couples does no harm to opposite-sex couples (**TwentyYears**) will ring hollow. For it is a part of the proposal that the institution they are a part of be annihilated. Furthermore, in practice, in jurisdictions where marriage has been extended to same-sex couples, generally those who were previously married still count as married. Therefore, on the assumption that marriage is socially constructed, not only is there annihilation of the institution that couples used to be a part of, but these couples are, without their express consent, inducted into the new institution. Such automatic induction into a new relationship does not seem consistent with the ideals of a free society, and yet generally defenders of same-sex marriage have not been bothered by this.

If, however, one holds that marriage is a natural human relationship, then one can argue for marriage equality without arguing for a two-institution model or for the annihilation of the existing institution. Instead, one can hold that marriage is a natural human relationship which can be non-defectively instantiated by couples of the same sex as well as couples of the opposite sex. Given that marriage, understood univocally, can be entered in on by both same-sex and opposite-sex couples, it is clear why it is discriminatory for a state to limit recognition of it to opposite-sex couples. And in advocating for the end of this inequality, one isn't advocating for an end of an existing institution, but simply for the state's recognition of the fact that this natural institution can equally well include same-sex and opposite-sex couples. The argument that no harm is done to opposite-sex married couples appears strong on this view.

It is worth noting that defenders of marriage equality who hold that marriage is constructed by individual couples can also avoid the above problematic consequences of social construction. On the individual construction view, in recognizing a marriage, the state is recognizing is a certain type of contract, where the type is defined by a kind of family resemblance. But recognition of opposite-sex contracts of a certain type without recognition of same-sex contracts of a relevantly similar type would be unreasonable. Imagine if

one could only sell a house to someone of the opposite sex, after all. On the individual construction view, the claim that no harm is done to opposite-sex couples by state recognition of same-sex marriage is also easily defensible. So, the above argument from same-sex marriage advocacy supports the natural relationship view and the individual construction view, but not the social construction view.

That said, as noted in Section 2.3.4, the natural relationship view may provide the liberal with a better defense of the claim that denial of same-sex marriage is a very thing, if the natural desire for marriage is of significant importance in human nature.

2.3.5.2. An argument for conservatives. Here is a plausible principle: If we limit access to an institution on the grounds of gender or sex, absent very strong reason we should strive to make an equivalent available. For instance, perhaps there is some reason for colleges to limit certain sports such as gridiron football to one gender, but if so then they should make other sports available to persons of other gender (**FrancisSports**). But many conservatives have not only objected to same-sex marriage but also to the availability of civil-union institutions for same-sex couples. I will argue that such conservatives should embrace a view of marriage as a natural relationship.

For if marriage is constructed, either individually or socially, then even if the norms of that construction in fact limit marriage to persons of the opposite sex, an equivalent institution without that limitation could be constructed, and by the principle at the top of this argument, it ought to be. In fact, it seems that the best way to resist this argument would be for the conservative to hold that marriage is a natural relationship, and that this relationship is only possible or only normatively appropriate for opposite-sex couples, while any superficially similar relationship between persons of the same sex is not a natural relationship. Because no merely social institution, such as a civil union, would be a natural relationship, it would not be an equivalent to marriage. Therefore, the conservative can respond to the original argument by saying that there is very strong reason not strive to make an equivalent available, namely that no equivalent is *possible*.

In response, as per our previous argument, the defender of same-sex marriage should say that marriage is a natural relationship that *can* legitimately hold between persons of the same sex. So this conservative response does not close the debate. But it provides the conservative with a way forward. Indeed, it seems that both sides on the same-sex marriage debate will be better served by moving from a social construction to a natural relationship view of marriage, and then discussing whether this natural relationship has norms that make it possible and permissible for persons of the same sex to instantiate it.

2.3.6. *Summing up.* We have three main views of marriage: natural relationship, social construction and construction by the individual couple. The discovery, cross-cultural criticism and natural desire arguments all support the natural relationship view, and there is good reason for those holding the conservative view on same-sex marriage to accept the view as well. The travel argument and the liberal view on same-sex marriage support either the individual couple construction or natural relationship views, though the importance attached to marriage inequality by those on the liberal side may fit a little better with the natural relationship view. On balance, we have a good case that marriage is a natural relationship. And our Aristotelian account then explains how the naturalness of such a relationship is to be grounded, namely in human nature.

3. The task of medicine

The realism about teleology and normalcy provided by the Aristotelian framework allows for an elegant solution to the problem of what the task of medicine is.

The medical professional is a *professional*. Of course, everyone should refuse to act immorally on behalf of a client. But a professional has norms and pursues goals that go beyond general morality, and has reason to refuse to further the client's aims even when there is nothing generally immoral about these aims but the aims nonetheless violate the professional goals. Thus, while it is not immoral to create kitsch, a professional artist nonetheless has good reason to refuse a commission that would be unavoidably kitschy. And it is

especially true of the medical professional that they are not mere providers of legally permissible services requested by a client (**CurlinTollefsen**). A medical professional should reject a patient's request to have a toe removed from each foot simply because the patient wants to use the number of toes on each foot express their disagreement with the ancient Pythagorean preference for odd numbers over even ones.

In the case of some professions, the goals are very much socially defined, and apart from legal minutiae, the delineation of these goals is of relatively small importance. For instance, we have at least three professions that deal with the directing of water: gutter installers, sewer maintainers and plumbers. All three professions are important, but the division of labor between them is not of great importance. It would do little harm to society if we had a single profession for all three tasks, or if we divided up the tasks in some other way, say in terms of dealing with potable and non-potable water, or incoming and outgoing water relative to a house.

However, the division of labor between the medical professions and other professions does seem to cut nature at its joints. The medical professions directly aim at the goods of bodily health, a very natural subdivision of the space of human goods.

Moreover, there is a special value in the medical professional having a very sharp focus on health. Medical considerations are of great importance to everyone's life. But in the end, the patient (or their representative, but let's simplify) needs to be able to make a prudent decision about the recommendations from a medical professional, weighing this recommendation against non-medical considerations such as ones of economics, interpersonal relationships, personal pleasure and convenience, and so on. The patient is typically not an expert in biological matters, but tends to have a good grasp of other relevant goods: for instance, they will know what effect giving up alcohol would have on their social life, or what goods their children would have to give up if a medical procedure is to be paid for. It is important, however, that a medical recommendation be primarily concerned with the good of the patient's health, so that the weighing between medical goals and other goods be delegated to the patient as much as possible, and that the non-medical goods not

be double-counted (once by the medical professional and again by the patient) in figuring out the prudent course of action.

At the same time, it is also important for guiding patients to prudent decisions that medical professionals understand health holistically, rather than narrowly thinking only of kidneys or feet. Thus, a focus on health in general is important for the medical professional, or at least for a medical professional who has an advising relationship with a patient.

Health cannot be defined purely statistically—it has a normative component **LennoxHealth**. One way to see this is that there is a directionality in health. Having muscles capable of deadlifting four hundred kilograms and having muscles incapable of deadlifting four kilograms are both significant deviation from the average for an adult human, but only the second is unhealthy. The ability to deadlift a great weight might be said to be supernormal, but its possessor not thereby abnormal, nor do is there a need of medical treatment to reduce their strength, as in **VonnegutHB**'s (**VonnegutHB**) dystopian "Harrison Bergeron" story.

One may try to respond to the deadlift example by focusing on the distinction between capabilities and incapacities, and saying that having a *capability* of at least statistically normal degree or an *incapability* of at most statistically normal degree is healthy. This is a neatly non-normative account, but it is unsatisfactory, since a significantly above average capability to involuntarily collapse from a small impact to the head is not healthy. Capabilities are not always good or healthy.⁴ Furthermore, the distinction between capabilities and incapacities appears arbitrary. We can describe a cell-membrane as having the capability to transmit some chemicals and the incapability of transmitting others (and both can be healthy), but we can also describe it as having the incapability to block some and the capability to block others.

⁴One might think that voluntary capabilities are always good as they contribute to the person's agency. However, voluntary capabilities are only a small subset of an organism's capabilities.

Or one might go for a *partially* statistical account of health as in **BourseHealth** (**BourseHealth**), where we have a non-statistical notion of the directionality of function a bodily system—muscles are for contracting and skulls are for resisting impact—and a statistical notion of what degree of function or readiness for function is normal. Health then involves function or readiness for function of *at least* statistically normal degree.

This still requires an account of the function of a bodily system. For Bourse, this is defined by the system's statistically typical contribution to individual survival and reproduction. This is implausible, however. If a crazed dictator of the world were to institute a policy of slaughtering all four-legged sheep, three-leggedness would not become intrinsically healthy (cf. **PlantingaWarrant**, **PlantingaWarrant**), though it would typically contribute to individual survival. Furthermore, such a non-normative account of health makes it unclear why the task of medicine is worth pursuing. For while survival and reproduction are very plausibly good for a human being, health on this definition is defined by the *statistically typical* contribution of a putative function to survival and reproduction. But the question of whether a condition *typically* contributes to survival and reproduction is only evidentially relevant to the question of whether the condition contributes to an individual's survival and reproduction, and it is surely the individual's survival and reproduction that we care about. The mere fact that a feature tends to contribute to survival and reproduction in the population does not constitute the feature as choiceworthy when it does not do so in the individual, while health does constitute a feature as choiceworthy. On this account, then, by pursuing health we end up pursuing a mere proxy for what we care about.

One might alternately try to account for the function here in evolutionary terms as a number of authors have done (**Wright1973**; **Millikan1984**; **Neander1991**; **Neander1995**; **Agar1993**; **Sterelny1990**; **Papineau1993**; **Dretske1995**). The basic idea of these accounts is that a bodily system's function is the behavior which contributed to the phenotype having been evolutionarily selected for. However, besides technical objections one might make to

such accounts (e.g. **KoonsPruss, KoonsPruss**), reducing health to conformity to evolutionary history faces exactly the same problem as Bourse's account of function: there is little or no value in being healthy, and hence what medicine pursues is only a proxy for what is valuable. For there is little if any value in having bodily systems that function in ways that millions of years ago helped our ancestors survive and reproduce.

Besides this, even if one can define healthy directionality in one of these non-normative non-Aristotelian ways, the question of statistically defining an abnormal deviation is highly problematic. Excessive sugar consumption tends to decrease heart function. Thus, on a statistical account, the soft-drink industry could bring about cardiac health in a number of people with marginal cardiac dysfunction by promoting sugary drinks and thereby bringing down the population average or median of cardiac function—yet it is absurd to think that anyone would become healthy as a result. Conversely, a government program to promote exercise and thereby improve cardiac function, if successful, makes some non-participants get heart disease by shifting the average or median upward—and indeed without the consent of these non-participants. This is absurd. Furthermore, the degree to which one must fall short of the statistical norm to count as unhealthy appears quite arbitrary. Does health require the 40th percentile for heart function, the 20th, or the 5th?

Instead, I take we should thus take health to be something through-and-through normative, a good of the body, such that its lack is bad for the body. At the same time, health is not simply *the* good of the body. There are goods of the body besides health, such as athletic achievement, beauty, and actual reproduction. These goods depend on health, but are not a part of health: for instance, a relatively healthy reproductive system is needed for reproduction, but one can have such a system without actually using it to obtain the good of reproduction.

Among our goods, it is perhaps health that is most clearly species-relative. We can define health as sufficient capacity to function bodily in accordance with our human teleology, and then an Aristotelian metaphysics of forms is perfectly fitted to grounding norms of health. And then, in turn, we can define the medical task as that of promoting health.

4. Our animal nature

4.1. The moral significance of our animal nature. On the theory being defended, we have a (rational) will whose proper function defines the right for us. At the same time, we are biological organisms, and our will plays a certain functional role in driving our behaviors (which we might distinguish from mere activities, which will include things like circulating the blood) and organizing our lives. Among those higher animals, such as cats, that are not rational agents, that functional role is still filled by *something*—some kind of behavior driver which we may or may not wish to call a will. It is plausible when we reflect on these higher animals that their behavior driver will be malfunctioning if they are not driven to live the kind of animal lives that are proper to their kind. By analogy, if we are not driven to live the kind of animal lives that are proper to our kind, our behavior driver is malfunctioning. But our behavior driver is the will, and on the Natural Law metaethics I am defending, right action is defined by the proper functioning of the will. Thus the argument from analogy combined with the metaethics yields a case that we *ought* to pursue lives with behaviors proper to the kinds of animals we are. Indeed, we expect a certain degree of harmony between our animal lives and our lives as agents.

This harmony has some plausible implications for environmental ethics and our relationship to other animals.

4.2. Living naturally.

4.2.1. Transhumanism. Various animal behaviors such as drinking, eating and walking are part of a flourishing life for the kind of mammal that we are. To the extent that we should be seeking our own flourishing, we have reason to pursue animal behaviors. However, this may not yield a very strong norm of pursuit of such behaviors for two reasons. First, our own flourishing does not appear to be the only thing the norms of our will call us to promote—the flourishing of others seems just as important. Second, flourishing as animals is likely only a small part of our flourishing—the bulk of our

flourishing consists in specifically human forms of flourishing such as understanding and love.

In addition to seeing animal behaviors as part of a flourishing life, there is another consideration in favor of living a life consistent with our animal nature. Aristotelian optimism gives us reason to think there is a harmony between the norms of the will and the norms of the rest of a rational animal's organism, and so a rational animal typically ought to act in accord with, and not contrary to, its broader animal nature.

This rules out more radical forms of transhumanism. It would not accord with our broader animal nature to upload ourselves to a computer, even if (which is dubious) we could survive such an upload. The virtual life would not include activities such as breathing, drinking, eating and walking that are a part of a natural human life.

Exactly how much modification of the human body is compatible with the norms of our will is not immediately clear. But the above lines of thought suggest a significant degree of moral caution. And, again, this is an area where we expect many ethical parameters specifying permissible and impermissible modifications.

4.2.2. *Ecology*. It is in the nature of any organism to interact in certain ways with its natural environment. While that interaction may involve out-competing some other organisms, the natural behavior of an organism tends to harmonize with its environment to a significant degree (from an evolutionary point of view, organisms and their environment tend to co-evolve). Given a harmony between the norms of the will and the norms of the rest of the rational animal, we would expect a rational animal to have norms that aim it at a certain degree of integration with a natural environment. Unless we choose to limit the word "moral" to reasons having to do with other persons^{??backref}, these norms will then constitute moral norms to harmonize with the environment, and even if we do choose to limit the word in this way, the reasons here will nonetheless have serious rational force.

It could be an intelligent organism has such norms because harm to the environment harms the organism. But even if so, it is likely that the norms are not simply derivative from the organism's need to do well in its own niche. Imagine that biological warfare

has wiped out all humans beyond you and a handful of friends, leaving the ecosystem intact but lethal to humans. You prepare to flee earth for another solar system, expecting humanity never to return. The spaceship has a good library, but the last novel in a series of fantasy novels that you like has not been released. Instead, it is stored in a computer system so set up by a crazy copyright owner that if you download the novel, all the nuclear weapons on earth will be launched. Your safety is assured if you wait to download the novel once your spaceship has left the earth's atmosphere, but the ecosystem will never recover from the damage. While under more normal circumstances, severe damage to the ecosystem has evident repercussions for humans, in this case it need not.

The evident wrongfulness of destroying the ecosystem for the sake of one enjoyable novel is not simply grounded in our duties to protect our species. It may, nonetheless, be *explained* by the need to protect our species. For it might be better for us to have coarser-grained duties to the environment that forbid large-scale destruction for the sake of minor goods than to have the finer-grained duties that would allow such destruction in extremely rare circumstances. After all, we know that it is all too easy to fool ourselves as to where the boundaries of the "rare" circumstances lie. The rule utilitarian has room for such an account, but so does an Aristotelian who thinks that there is an inner harmony in our nature such that the various aspects of our nature are good for us in other respects. This pattern of explanation will be discussed further in Section 6.1 of Chapter X.

5. The definition of life

Here is an intuition that until fairly recently would have been widely shared: There are deep metaphysical divides between non-living and living things, and between merely living things and persons, and these divides mark a hierarchy of value, a chain of being. If we could defend such a divide, it would dovetail with the idea that persons are in an important way *sacred*, having rights while other things have mere interests, if that.

I want to offer a highly speculative Aristotelian reconstruction of this intuition. To introduce the reconstruction, start with a puzzle for Aristotelian views. It seems that on such views:

(16) Each thing naturally strives for its own perfections.

(17) The natural activity of a thing is a perfection of it.

But this generates a regress. Let's say that reproduction is an oak tree's perfection. Then by (16), the oak tree naturally strives for reproduction. This natural activity of striving for reproduction, by (17), is then itself a perfection of the oak tree. Therefore, by (16), the oak tree must naturally strive for it: hence the oak tree naturally strives for striving for reproduction. And so on, *ad infinitum*. But surely an oak tree does not pursue infinitely many things. And even after a few level of meta-striving we exhaust plausibility.

I suggest that we deny (16). Some perfections of a thing are not actually naturally striven for by the thing.⁵ The oak tree does strive for reproduction with its reproductive organs. Moreover, it has a second-order striving: it strives to strive for reproduction, by growing the reproductive organs with which it strives for reproduction. There may be one or two more meta-levels, but at some level we can say: it just does this, without striving to do it.

Non-living substances, on an Aristotelian metaphysics, also have form and also strive for ends. But plausibly they don't strive to strive: they just strive. We thus have a hierarchical division between inorganic things which do not strive to strive and living things which have second order teleological strivings.

The problem of the definition of life is a thorny conceptual problem in biology or its philosophy. Infamously, different authors give quite different lists of empirically-focused

⁵An interesting theological example may be the idea in the Thomistic tradition that both the beatific vision and our striving for it are gifts of God's grace, rather than natural for us, even though the beatific vision perfects us.

features such as homeostasis, growth, reproduction, information encoding and metabolism as part of the definition of life **MalaterreChartier** (**MalaterreChartier**) use data-mining techniques to identify eight different categories (the first three are metabolism, proteins/catalysis, and cellular/structural features), each with many features under it. The multiplicity of features listed makes the concept of life seem arbitrary. Indeed, Malaterre and Chartier's conclusion is that life is a multidimensional degreed concept. But the Aristotelian approach may allow for a metaphysical approach that escapes the multiplicity of empirically-driven categories.

Indeed, it is philosophically problematic to tie the the concept of life too tightly to the physical forms of life around us. For if there are immaterial agents such as deities, spirits or angels, they should also count as alive. After all, those who believe in such beings often hold them to be *immortal*, and only a living thing can be immortal.⁶ Generally these beings are conceptualized as alive, even when they cannot engage in homeostasis, growth or reproduction. And yet while a particular claim about the actual existence of immortal immaterial agents might be false, such claims do not seem to be fundamentally conceptually confused. Thus, a good account of life should include the kind of life that is attributed to immaterial agents, and typical accounts in the philosophy of biology fail to do that.⁷

Furthermore, it is a merit of a definition that when applied to cases where we do not know how to classify a thing, the definition does not trivially decide the issue, but points

⁶We might say that a non-living thing cannot die because it was never alive in the first place, but that is not what we mean by "immortal".

⁷The closest in Malettere and Chartier's survey is information encoding. However, it is unlikely that information encoding *by itself* is enough to define life—an ordinary computer is constantly encoding information without being alive. Moreover, it is not clear that information encoding can capture the intellectual activity of immaterial agents. For instance, classical theism's doctrine of divine simplicity does not allow God's knowledge of contingent reality to be internal state encoding the external contingent reality—instead, perhaps, it is some sort of an extrinsic "gaze" (**GrantSimplicity**).

to the question we need to answer if we are to decide the issue. Consider two borderline cases: viruses and sophisticated robots, like Star Trek's Data. In neither case are we confident whether we have life. Viruses are famously a borderline case. And while intelligent robots like Data are described as "synthetic life forms" in the *Star Trek: Picard* series, and the Star Trek canon clearly favors their being actually alive, the question is not so philosophically clear. Data obviously fails typical biological definitions of life: while he engages in self-maintenance, he doesn't grow or reproduce in the biological sense of the word (though he does make other androids). Yet typical viewer intuition is to treat him as alive.⁸ And whether a virus qualifies as alive varies from definition to definition in a way that makes it sound like the question of viruses being alive is merely verbal. Yet given the strong intuition that there is significant value to life, even something sacred, the question of what is and is not alive should not be merely verbal.

On the other hand, an account on which what it is to be alive is to have a second order teleological striving—to strive to strive for a perfection—will nicely include immaterial agents. It will include any entity that prepares itself for future teleological activity, say by growth, and hence will include all the physical forms of life we know about. It will exclude elementary particles. And whether it includes viruses or sophisticated robots is unclear—as it should be. For it is unclear whether viruses and sophisticated robots have form at all. If viruses have form, then it is likely that their activity of attaching to hosts for purposes of future replication is a striving for replicative striving, and hence they are alive. If sophisticated robots have form, they also exhibit meta-striving, and hence are alive. But in both cases we do not know whether there is form, or whether we are dealing with a mere agglomeration of particles. Aristotle himself seems to have thought that artifacts only had form in the analogical sense of an art in the mind of the designer (**Aristotle**), but he could have been wrong in the case of artifacts like Data. (For more on the epistemic issues here, see Section [forwardref](#) in Chapter XI.)

⁸Though, admittedly, there may be some confusion in Star Trek between consciousness and life.

We thus have two levels in a chain of being: things that strive but don't meta-strive, and things that meta-strive. Now, among the things that meta-strive, we can describe a higher kind of thing: a thing that strives for all of its perfections. The premises of the regress argument with which we started this section apply to such a being. Thus, this is a being that strives for striving for ... for perfection, at any number of levels. While this is implausible for an oak tree or even a dog, we do actually know of one kind of being that does that: humans. Human beings not only conceptualize particular perfections, such as friendship or striving for striving for health, but they conceptualize perfection as such, and strive for it as such. If a trustworthy being offered to increase some perfection or other of yours, and assured you that you would in no way be harmed, it would be rational for you to accept the offer, because perfection as such is one of the things you and I pursue.

At the same time, in a minded being, the infinite chain that results from striving for all one's perfections need not be a chain of separate desires and hence does not require a being that is actually infinite. Rather, all that's needed is for the being to be such that it has or teleologically strives to have the concept of a perfection as such and a desire for perfection as such. This desire then can manifest in a striving to figure out what the perfections are—a striving that is central to the search for happiness (*eudaimonia*) that was so characteristic of Socratic and post-Socratic Greek philosophy—and a striving to be ready to pursue whatever one finds. In fact, it might be that for reasons having to do with the nature of infinity *only* a minded being can pursue an infinite number of ends—for any non-minded being that did that would need to have infinitely many distinct causal sources of its activity in a way that might well violate causal finitism, the thesis that it is impossible for an infinite number of causes to work together.⁹ And among minded beings, perhaps it is definitive of *persons* that they pursue all good.

We thus have a qualitative hierarchy of being between the mere strivers, the mere meta-strivers and the universal strivers. The first division in the hierarchy may well correspond

⁹For a defense, see **PrussInfinity** (**PrussInfinity**).

to that between the non-living and the living, and the second might—depending on speculative questions about infinity—align with the division between mere life and personhood. And it is very natural to see qualitative divisions of value here as well.

CHAPTER V

Epistemology

1. Balancing doxastic desiderata

I observe one raven, and it's black. I observe another and it's black, too. The story goes on. Every raven I observe is black. After a certain number of ravens, in a sufficiently broad number of settings, it becomes reasonable to believe that all ravens are black. But when?¹

William James famously identified two incommensurable doxastic desiderata: attainment of truth and avoidance of falsehood. The larger the number of black ravens that are needed for me to believe that all ravens are black, the more surely I avoid falsehood, but the more slowly I attain truth. Intuitively, there is room for differences between reasonable people: some tend to jump to conclusions more quickly, while others are more apt to suspend judgment. But on either extreme, eventually we reach unreasonableness. Both someone who concludes that all ravens are black based on one observation and someone who continues to suspend judgment after a million broadly spread observations are unreasonable.

There is, thus, a range of reasonable levels of evidence for an inductive belief. And, as in the myriad of ethical cases of Chapter II, this raises the Mersenne question: What grounds facts about the minimum amount of evidence required for an inductive inference and the maximum amount at which suspending judgment is still rational? Of course, the "minimum" and "maximum" may may depend on the subject matter, on higher-order evidence such as about how well previous inductive generalizations have fared, and even on

¹I am grateful to Sherif Girgis for raising the issue of incommensurable desiderata in connection with these issues.

pragmatic factors. But that added complexity does nothing to make the Mersenne question easier to answer. And, as we discussed in [backref](#), invoking vagueness does not solve the problem, but multiplies the complexity even further.

And, of course, my contention will be that conformity to the human form is what grounds the answers for us. The rational way to reason is the way that conforms to our form's specification of the proper functioning of our intellect.

It appears to be quite plausible that different answers to the rationality questions would be appropriate for species of rational animals adapted to different environments. First, some possible worlds as a whole have laws of nature implying a greater uniformity and simplicity than that found in other worlds, and hence make it appropriate to make inductive inferences more quickly. Second, the specific environments that the rational animals evolved in may have greater or lesser uniformity, despite the same laws of nature—an environment where natural kinds are found has greater uniformity, for instance. Third, the ecological niche occupied by the rational animals may punish falsehood more or may reward truth more, either in general or in specific cases: perhaps for prey animals an oversensitivity to danger is not a problem. Thus the Aristotelian species-relative answer to the Mersenne questions is particularly appealing.

2. Logics of induction

Attempts have been made to give precise answers to questions about the reasonableness of inductive inferences using a rigorously formulated logics of induction, most famously by **CarnapLFP2** (**CarnapLFP2**, **CarnapContinuum**). Let us suppose, first, that some such logic, call it L_{12} , does indeed embody the correct answers. Nonetheless, we will have a Mersenne question as to why L_{12} , rather than one of the many alternatives like L_{11} or L_{13} , is the logic by which we ought to reason inductively.

In the case of truth-functional deductive logic, there is a system that appears to be both particularly natural and matches our intuitions so well that it has gained a nearly universal following among philosophers, logicians, mathematicians and computer scientists:

two-valued boolean logic. No logic of induction has anything like this following, which is unsurprising given that no logic of induction has the kind of naturalness and fit with intuition that would privilege it over the others to a degree where it would seem non-arbitrary to say that it is *the* logic we should reason with.²

Further, observe that logics of induction can be divided into two categories: those with parameters (say, parameters controlling the speed of inductive inference) and those without. A logic of induction with parameters raises immediate Mersenne problems about what grounds the fact about which parameters, or ranges of parameters, are in fact rationally correct. A parameter-free logic of induction, however, is not likely to do justice to the fact that different ways of balancing rational goods are appropriate in different epistemic and pragmatic contexts. Moreover, it is unlikely to do justice to the intuition that the balancing should be different in different species of rational agents.

3. Goodman's new riddle of induction

All the emeralds we've observed are green, and it's reasonable to infer that all emeralds are green. But **GoodmanFFF's (GoodmanFFF)** famous riddle notes that all the emeralds we've observed are also grue, but it's not reasonable infer that all emeralds are grue. Let's say an emerald is grue if it is observed before the year 2100 and green, or if it is blue and not observed before 2100. According to Goodman (with year modified), the predicate "is green" is *projectible*, i.e., amenable to inductive inference, while the predicate "is grue" is not. But how do the two differ?

As Goodman notes, the fact that "grue" is defined in terms of "green" and "blue" does not help answer the question. For if we specify that something is bleen if it is observed before 2100 and blue, or it is blue and not observed then, then we can define something to be green provided it is observed before 2100 and grue or bleen and not observed in that time period, and similarly for "blue" with "grue" and "bleen" swapped.

²Once we get to quantification and modality, things become less clear.

Whatever the *justification* of inductive reasoning may be, it is clear that induction with “green” is reasonable, but not so with “grue”. Notwithstanding Goodman’s symmetry observations, “grue” is a gerrymandered predicate, as can be seen in accounting for it in terms of more fundamental physical vocabulary. But now observe that “green” is also gerrymandered. Giving an account of colors is a difficult philosophical project, but to a first approximation an object is green provided that the wavelength profile of its reflected, transmitted and/or emitted light is predominantly concentrated somewhere in the range of 500 to 570 nm. The actual boundaries of that region are messy and appear vague, the measure of predominant concentration is difficult to specify, and accounting for reflective, transmittive and emissive spectra is a challenge. The full account in terms of more fundamental scientific terms will be complex and rather messy, though not as badly as in the case of “grue”, which is more than twice as complex since it needs to account for blueness and the date of “2100”, which is a very messy date in more fundamental physics units (perhaps Planck times since the beginning of the universe?). Where the boundary between non-projectible and projectible lies—what counts as too gerrymandered for projectibility—is an excellent Mersenne question.

There is a very plausible way to measure the degree of gerrymandering of a predicate. We take a language the content of whose symbols are terms for perfectly natural concepts, ones corresponding to fundamental joints in reality (**LewisNewWork**), and we look for the shortest possible formula logically equivalent to the predicate, and say that the predicate is gerrymandered in proportion to the length of this formula. Now, to account for “is grue” in terms of fundamental terms, we need an account of greenness, an account of blueness and, worst of all, an account of observation, so “is grue” is much more gerrymandered than “is green”.

But this approach does not yield the correct judgments about inductive reasoning. Let t_1 abbreviate “ 10^{61} Planck times from the beginning of the universe”. Since Planck times are a rather natural unit, the specification of t_1 is pretty natural. Note that t_1 is about three billion years in the future. Say that something is “pogatively charged” provided that it

is positively charged and came into existence earlier than at t_1 or it is negatively charged and came into existence no earlier than at t_1 . All the protons we have seen are negatively charged. But we should not conclude that all protons across time are negatively charged, and hence that in about three billion years whenever new protons are formed, they will be negatively charged. However, notice that by the formula length account, "is green" is significantly more gerrymandered than "is negatively charged". Negative charge is much closer to the fundamental than specific human-named colors like green. It seems, thus, that our Mersenne question about the boundary between the non-projectible and projectible is not merely defined by a single number—a threshold such that predicates definable with a length below that number are projectible.

Perhaps, however, what is going on here is this. The hypothesis that all emeralds are grue is beaten by the hypothesis that all emeralds are green, even though both fit with observation. Similarly, the hypothesis that all protons are negatively charged is beaten by the hypothesis that all protons are positively charged. So perhaps rather than an absolute concept of projectibility, we have a relation of relative projectibility: "is green" is projectible relative to "is grue" and "is grue" is non-projectible relative to "is green".

We can once again try to account for this in terms of the complexity of formulae. But now we need to compare the complexity of two formulae. The most natural comparison is that P is projectible relative to Q provided that in perfectly natural terms Q is more complex than P . But a small difference in complexity does not make for a difference in whether a term is projectible or not. We shouldn't say that "is blackish" is projectible relative to "is brownish", just because "blackish" is slightly easier (I assume) to define in fundamental terms. Thus our account will say something like: P is projectible relative to Q provided that $F(\ell(P), \ell(Q)) < \lambda$ where $\ell(R)$ is the length of the shortest expression of R in perfectly natural terms, F is some sort of a comparison function, say $F(n, m) = n - m$ or $F(n, m) = n/m$, and λ is a threshold. Thus, while previously we had a single numerical threshold as our parameter of projectibility—predicates with complexity lower than the

threshold are projectible—we now have both the threshold λ and the function F to be specified.

Furthermore, while the idea of a language all of whose terms reflect fundamental joints in reality can be defended, the grammar of the language will make a difference to the precise complexity measurements. For instance, if we have the fundamental predicates Cx , Dx and Ex , then the complex formula expressing the predicate “is C as well as either D or E ” will be

$$Cx \ \& \ (Dx \vee Ex)$$

in infix notation, and hence five times longer than the formula Cx representing “is C ”, but in Polish notation will be

$$KCxADxEx$$

and hence only four times longer than Cx .

For a relative projectibility relation defined in terms of linguistic complexity, we thus have at least three free parameters, each a fit subject for a Mersenne question: a threshold, a comparison function, and a grammar for the basic language.

But in fact we probably should not think of a binary projectible / non-projectible distinction, whether relational or absolute. As **GoodmanFFF** (**GoodmanFFF**, p. 106) himself observed, what we have instead is a range of predicates that are more or less projectible. We have “is green” and “is grue”. But we can also say that x is grue* provided that x is green and observed by a French speaker before 2100 or by a non-speaker of French before 2107, or never observed, and “grue*” will be less projectible than “is grue”. On the basis of our observations, the probability that all emeralds is green is very high, and the probability that they are all grue or grue* is very low. But nonetheless, the probability that they are grue is somewhat higher than that they are grue*. After all, an alien conspiracy to re-color emeralds upon observation with a sharp cutoff in one year seems a little bit less unlikely than one where the cutoff depends on whether the observer speaks French. Similarly, it

makes sense to think of “is green” as less projectible than “is positively charged”, and of “is cute” as even less projectible.

Projectibility now becomes a matter of degree. An advantage of this is that perhaps we no longer need to make it relational. The reason for the superiority of the green-hypothesis to the grue-hypothesis and for the positive-charge-hypothesis to the negative-charge-hypothesis can be given in terms of the relationship between the degrees of projectibility. However, we will need a function from predicates to degrees of projectibility, and the choice of that function will have infinitely many degrees of freedom.

4. Epistemic value

4.1. Epistemic value on its own.

4.1.1. *Scoring rules.* Plausibly, the more sure you are of a truth, the better off epistemically you are, and similarly the more sure you are of a falsehood, the worse off you are.

But what exactly is the dependence of value on the degree of certainty? Fix some hypothesis H and let $T(p)$ be the epistemic value of having degree of belief or credence p (where $0 \leq p \leq 1$) in H if H is in fact true and let $F(p)$ be the value of credence p in H if H is in fact false. The pair T and F is called an accuracy scoring rule.³

We can put some plausible constraints on T and F . First, $T(p)$ cannot decrease if p increases, and $F(p)$ cannot increase if p decreases.⁴ But that still leaves infinitely many degrees of freedom for the selection of T and F .

We can, however, make some progress if we reflect on expected values. If your current credence in H is p , then by your lights there is a probability p of your having epistemic

³The literature tends to focus more on inaccuracy scoring rules, which are cost or penalty functions (PreddEtAl). An inaccuracy scoring rule is the negative of an accuracy scoring rule and *vice versa*, so one can freely go back and forth between the two options. Since plausibly the good is more fundamental than the bad, it makes sense to take accuracy scoring rules as more fundamental, however.

⁴We might more strongly specify that $T(p)$ always strictly increases with p , and $F(p)$ strictly decreases. That is plausible, but one might also have a view on which there is a finite number of discrete thresholds at which increase/decrease happens.

score $T(p)$ and a probability $1 - p$ of your epistemic score being $F(p)$, so your expected score is:

$$pT(p) + (1 - p)F(p).$$

Suppose now you consider doing something odd: without any evidence, you might brain-wash yourself to switch your credence from p to some other value p' . By your current lights, the expected epistemic value of this switch is:

$$pT(p') + (1 - p)F(p').$$

And this shouldn't be higher than the expected epistemic value of your actual credence p . For surely by the lights of your assignment of p to H , no other credence assignment should be expected to do better. Indeed, if another credence assignment p' were expected to do better by the lights of p , then p would be some kind of a "cursed probability", one such that if you assign it to H , then immediately expected value reasoning pushes you to replace it with p' . This is not rational. So, it is very plausible indeed that:

$$pT(p) + (1 - p)F(p) \geq pT(p') + (1 - p)F(p').$$

If T and F satisfy this inequality for all p and p' , we say that the pair T and F is a *proper* scoring rule. And if by the lights of the assignment of p to H , that assignment has better expectation than any other, i.e., if the inequality above is strict whenever $p \neq p'$, we say that the rule is *strictly proper*.

Propriety mutually constrains the choice of T and F . Given any non-decreasing function T , there is a function F that is unique up to an additive constant such that the pair T and F is a proper scoring rule, and conversely given any non-increasing function F , there is a T unique up to an additive constant such that T and F is a proper scoring rule.??? Hence, once we have one of the two functions, the other is almost determined. However, at the same time, this result shows what a profusion of proper scoring rules there is: for every non-decreasing function, there is a proper scoring rule that has that as its T component.

The question of epistemic value assignment may seem purely theoretical. However, it has real-world ramifications. Suppose a scientist has attained a credence p in a hypothesis H , and is considering which of two experiments to perform. One experiment will very likely have a minor but real effect on the credence in H (think here of a case where you've gathered 1000 data points, and you now have a chance of gathering 100 more). The other will most likely be turn out to be irrelevant to H , but there is a small chance that it will nearly conclusively establish H or its negation. For each experiment, the scientist can use their present credence assignments to estimate the probabilities of the various epistemic outcomes, and can then estimate expected epistemic values of the outcomes.

GreavesWallace (GreavesWallace) have proved that if the scoring rule is strictly proper, for each experiment that has potential relevance to H , the expected epistemic value of performing the experiment is higher than the expected epistemic value of the *status quo*. Thus if the experiments are cost-free, it is always worth performing more experiments, as long as the appropriate scoring rule is strictly proper, and it does not matter which strictly proper scoring rule we choose. But if in addition to deciding whether to perform another experiment, the decision to be made is *which* experiment to perform, then the choice of scoring rule will indeed be important, with different strictly proper scoring rules yielding different decisions (see Section 4.1.3 and the Appendix to this chapter for an example).

There are a number of mathematically elegant strictly proper scoring rules, such as the Brier quadratic score, the spherical score and the logarithmic score. Of these, the logarithmic score is the only that is a serious candidate for being *the* correct scoring rule, in the light of information-theoretic and other arguments **McCutcheonLog**. In our setting where we are evaluating the value of a credence in a single proposition H , the logarithmic score is $T(r) = \log r$ and $F(r) = \log(1 - r)$. However, there are also reasons to doubt that the logarithmic score is the One True Score.

4.1.2. *The logarithmic score is not the One True Score.* There is an immediate intuitive problem for the logarithmic score. If you are certain of a falsehood, your logarithmic score is $\log 0 = -\infty$, while if you are certain of a truth, your score is $\log 1 = 0$. Now, while there

is good reason to think that the disvalue of being sure of a falsehood exceeds the value of being sure of a truth, it is somewhat implausible that it infinitely exceeds it.

For the next two problems, note that logarithmic scores and the arguments for them only really come into their own when we are dealing with more than two propositions (in our above setting, we had H and $\sim H$ are the only relevant possibilities). Suppose we are dealing with n primitive possibilities or “cells”, $\omega_1, \dots, \omega_n$ (say, the sides of an n -sided die), and that our agent has assigned credence p_i to ω_i . If in fact ω_i eventuates, the logarithmic score yields epistemic value $\log p_i$.

One of the merits touted for the logarithmic score is that the epistemic value depends only on the credence assigned to the cell that eventuates. But this is also a serious demerit. Suppose that you and I are trying to figure out how many jelly beans there are in a jar. Let’s say that our range of possibilities is between 1 and 1000. I look very quickly and assign equal probability $1/1000$ to each number. You count very carefully and arrive at 390. But then you think that although you are really good at counting, you might be off by one. So you assign $998/1000$ to 390, and $1/1000$ to each of 389 and 391. It turns out that the number is 391. We both have the same logarithmic score, $\log(1/1000)$, since we both assigned the same probability $1/1000$ to cell 391. But intuitively your assignment is much better than mine: you are better off epistemically than I.

Finally, observe that in real life, credences are not consistent—do not satisfy the axioms of probability. And the logarithmic score allows one to have extremely inconsistent credences and still do well. If I assign credence 1 to *every* possible outcome, I am guaranteed to max out the logarithmic score no matter what. Thus one of the least rational credence assignments results in the best possible score.

4.1.3. *Two approaches.* We now have two different approaches to the Mersenne questions about epistemic value and scoring rules. First, we could suppose that there is such a thing as *the* One True Score. Since only the logarithmic score seems significantly mathematically privileged over all the other scores, and the logarithmic score is not the One True

Score, there will be a significant appearance of contingency about the One True Score even if there is one.

Second, and this seems the better approach, we might suppose that just as rational people can differ in prudential preferences, they can differ in epistemic preferences. Some may, for instance, have a strong sharpish preference for gaining near-certainty in truths, while being fairly indifferent whether their credence in a truth is 0.6 or 0.8, as neither is that close to certainty. Others, on the other hand, may value increased certainty in a gradual way, somewhat like the logarithmic rule does.

However, it is important to note that while there may be room for rational people to differ in epistemic preferences, there is reason to think that there are rational constraints on epistemic preferences that go beyond formal conditions such as strict propriety, continuity or symmetry—where the last is the condition that $T(p) = F(1 - p)$.

Let $T_0(x) = 1000$ if $x \geq 0.999$, $T_0(x) = -1000000$ if $x \leq 0.001$, and $T_0(x) = 0$ otherwise. Let $F_0(x) = T_0(1 - x)$. Then the pair T_0 and F_0 is a symmetric and proper scoring rule (see this chapter's Appendix).

Consider now a scientist who adopts this scoring rule for some hypothesis H of minor importance about some chemicals in her lab that she initially assigns credence $1/2$ to. She has a choice between two methods. She can use beat-up machine A that she has in her lab, which displays an answer to the question of whether H is true, but for either answer there is a 0.11% chance that the answer is wrong. Or she can use spiffy new machine B which has the slightly lower 0.09% chance of error either way. The only problem is that her lab doesn't own machine B and her grant can't afford the price. Her only hope for using machine B is to go and buy a scratch-off lottery ticket which has a one in a million chance of yielding a prize exactly sufficient to purchase machine B .

Suppose her only options are using machine A , and then learning whether H is true with a credence of 0.9989, or buying a lottery ticket, which gives her a one in a million chance of learning whether H is true with a credence of 0.9991 and a 999,999 out of a million chance of being no further ahead. Going for the second option seems irrational if

all that is at stake is epistemic value: the difference between 0.9989 and 0.9991 is just not worth it given that most likely the lottery route won't yield anything.⁵

But a scoring rule like the one described above prefers the lottery option. For the epistemic value of using machine *A* is guaranteed to be zero since after using machine *A*, the scientist will have credence 0.9989 or 0.0011, depending on whether the result favors *H* or not, and a credence of 0.9989 or of 0.0011 gets assigned a value of zero by both T_0 and F_0 .

On the lottery option, however, conditionally on getting a winning ticket, it is equally likely given the scientist's priors that the machine will return a verdict for or against *H*, which will result in a credence of 0.9991 or 0.0009, respectively, and in either case there will be a 0.0009 chance that the verdict is erroneous. Since $F_0(0.0009) = T_0(0.9991) = 1000$ and $T_0(0.0009) = F_0(0.9991) = -1000000$, it follows that the expected epistemic value, conditionally on winning the lottery, will be:

$$\frac{0.9991 \cdot 1000 + 0.0009 \cdot (-1000000)}{2} + \frac{0.9991 \cdot 1000 + 0.0009 \cdot (-1000000)}{2} = 99.1 > 0.$$

And if we multiply this by the 1/1000000 chance of winning the lottery, we still have something positive, so the expected epistemic value of playing the lottery with the plan of using machine *B* is positive, while that of using machine *A* is zero.

Thus, by considerations of epistemic value, the scientist with this scoring rule will prefer a 1/1000000 chance of gaining credence 0.9991 as to whether *H* is true to a certainty of gaining the slightly lower credence 0.9989. This is not rational.

Now, in the above example, our scoring rule while proper, symmetric and finite, was neither continuous nor strictly proper. However, we will show in the Appendix of this chapter that there is a sequence of continuous, strictly proper, finite and symmetric scoring rules T_n and F_n such that $T_0(x) = \lim_{n \rightarrow \infty} T_n(x)$ and $F_0(x) = \lim_{n \rightarrow \infty} F_n(x)$ for all x . If n

⁵If what was at stake wasn't epistemic value but something pragmatic, then things could be different. We could imagine a law where some life-saving medication can be administered to a patient only if we have confidence greater than 0.999 that it'll work, and then there will be no practical difference between 1/2 and 0.9989, but a big one between 0.9989 and 0.9991.

is large enough, then the pair T_n and F_n will require exactly the same decision from our scientist as T and F did, since the expected values of the expected (T_n, F_n) -scores of the two courses of action will converge to the expected values of the expected (T, F) -scores.

Hence not all epistemic valuations that satisfy the plausible formal axioms are rationally acceptable, and so we will have Mersenne questions about what grounds the further constraints on the epistemic valuations. These constraints are likely to include messy prohibitions, with multiple degrees of freedom, on the kinds of sharp jumps that our pathological scoring rule above exhibited as well as on the less sharp jumps that the approximating continuous scoring rules had.

Furthermore, things become more complicated when we consider that the epistemic value of a credence in a truth will differ depending on the importance of that truth. Getting right whether mathematical entities exist or whether humans are purely physical or how life on earth started has much more epistemic value than getting right Napoleon's shoe size at age 20. Epistemic value will thus not only be a function of credence and truth, but also of subject matter. Moreover, we will have further degrees of freedom concerning the operation of combining epistemic values for different propositions—addition may seem a plausible operation, but the logarithmic and spherical rules are not combined additively across propositions.

We thus have multiple indicators of a contingency about epistemic value assignments. And there is good reason to think that different forms of life are more suited to different epistemic value assignments. The most obvious aspect of this is that once we move away from the toy case of assigning a value to one's epistemic attitude to a single proposition and consider that attitudes to a large number of propositions need to be considered, it is obvious that the subject matter of the propositions will affect how great a weight we give credences about them in the overall evaluation. And the importance of subject matter obviously depends on the form of life. It is plausible that for intelligent agents whose natural environment is more hostile it would be more fitting to have a greater epistemic value assigned to practical matters, while agents that have fewer natural enemies and can

get food easily might more fittingly have a greater epistemic value assigned to theoretical matters. One imagines here that intelligent antelope might be properly expected to be less philosophical than intelligent elephants.

4.2. Connection of epistemic values with other values. Suppose a scientist is deciding whether to go home to be with family or perform an experiment at the end of a long day, an experiment that cannot be done on another date. The scientist weighs the epistemic goods arising from the experiment against the value of interacting with family. What is the right thing to do depends here on many unspecified factors. How much of a delay in returning home would the experiment create? What is the current state of the scientist's relationship with the family and what is their current level of need? What practical benefits, if any, could accrue to the scientist, the scientist's family, or humankind from the experiment? And, finally, how important is the hypothesis being tested and how much can the experiment be expected to contribute to confirming or disconfirming the hypothesis?

Only the last question adverts directly to epistemic value. It could be that the answers to the preceding questions suffice to determine whether scientist should go home. Perhaps the scientist's progress towards tenure is crucial to the family's livelihood. Or perhaps there is a sufficiently high chance that the experiment's answer will contribute to a cure for cancer that apart from any epistemic goods it is worth staying. Or, alternately, perhaps the scientist has a commitment to family to go home on time this evening, and regardless of the epistemic goods involved, that commitment needs to be honored.

But of particular present interest is what should be done if the non-epistemic goods do not answer the question of what the scientist should do. In that case, one needs to weigh the epistemic goods against the non-epistemic ones. There is no automatic priority of one over the other. Some purely epistemic goods are great enough that significant sacrifice of non-epistemic goods is worthwhile. Thus, much sacrifice would make sense if the outcome would be to know whether there is life in other galaxies, even if that knowledge has no practical value to us. On the other hand, very little sacrifice would be worthwhile to find out whether the lab's largest rat has an even or odd number of freckles.

If we could find the right scoring rule, that would let us compare different epistemic values. But we would still need a way to weigh the outputs of the scoring rule with non-epistemic values. And then we will have our old familiar appearance of contingency. For whatever is the right way of comparing these values, we can imagine beings slightly different from us for whom a different comparison is appropriate: beings that ought to be more contemplative or more practical. And we have the familiar Mersenne question about why it is that for us the weighing goes as it does, rather than in some other way.

5. Bayesianism

5.1. Introduction. Bayesianism is the best developed picture of what a precise and rigorous account of epistemic rationality would be like. It is thus worth looking carefully at what kind of answers the Bayesian could give to the questions we have been asking as a model of the kind of answers we can hope for in a well-developed theory.

On Bayesianism, as was already the case for scoring rules, the main interest is not in beliefs as such but in the agent's credences, which come in degrees. I will assume the most familiar model where credences are quantified by probabilities ranging from 0 to 1. Other models include qualitative probabilities where instead of a specific value being assigned one has fundamental comparisons between probabilities (**KLST**), interval-valued probabilities where a range of numerical probabilities is instead of a specific numerical probability (**Kyburg1961; Whitcomb2005**), and accounts on which conditional probabilities are fundamental (**HajekCond**). Most of the issues raised will apply *mutatis mutandis* to the other models.

On our model, if a perfectly rational agent assigns credence $C(p)$ to a proposition p , then these credences will satisfy plausible axioms for probabilities, such as:

- (i) Non-negativity: $0 \leq C(p)$
- (ii) Normalization: $C(p) = 1$ if p is necessarily true
- (iii) Finite Additivity: $C(p \vee q) = C(p) + C(q)$ if p and q are mutually exclusive.

Furthermore, perfectly rational agents update their credences on receipt of evidence by conditionalization. In other words, upon receipt of evidence E , the “prior” credence of p goes from $C(p)$ to the “posterior” conditional credence $C(p \mid E)$. If $C(E) > 0$, then we define $C(p \mid E) = C(p \ \& \ E)/C(E)$.⁶ This conditionalization is designed to ensure the plausible rule that if p and q each entail the evidence E , then the ratio of the credences of p and q does not change upon learning E .

5.2. Induction and priors. From a Bayesian point of view, how induction works is determined by the probabilities prior to all evidence, the ur-priors. Suppose, for instance, that I assign equal prior probability to every logically possible color sequence of observed ravens. For simplicity, suppose that there are only two colors, white and black. I find out that there are a million ravens, and I observe a thousand of them, and find them all black. I am about to observe another raven. The probability that the next raven will be black will be $1/2$. For the sequence $B, \dots, B, W$ (with 1000 Bs) is just as likely as the sequence $B, \dots, B, B$ (with 1001 Bs), and both sequences fit equally well with the observations.

On the other hand, suppose I assigned probability $1/3$ to the hypothesis that all ravens are white, $1/3$ to all black, and split the remaining $1/3$ equally among the $2^{1000000} - 2$ multicolor sequences. My observation of the first 1000 ravens then rules out the all-white hypothesis. And it rules out most of the multicolor sequences: there are $2^{999000} - 1$ remaining multicolor sequences that start with 1000 black ravens, which is a tiny fraction of the original $2^{1000000} - 2$. Since as a good Bayesian I keep the ratios between the probabilities unchanged when the events entail the observations, each of the remaining multicolor sequences has $1/(2^{1000000} - 2)$ of the probability of the all-black sequence, and since there are $2^{999000} - 1$ multicolor sequences remaining that yield the observation of the first thousand ravens being black, the ratio between the multicolor probability and the all-black probability is $(2^{999000} - 1)/(2^{1000000} - 2)$ to 1, or approximately 1 to 2^{1000} , while the all-white

⁶There are some technical issues with handling cases where $C(E) = 0$, say if one could find out that a spinner has landed in a precise location. However, such cases may not occur in practice.

hypothesis is completely ruled out. Thus, we have overwhelming confirmation of the all-black probability, and hence an even more overwhelming probability of the hypothesis that the next raven will be black.

Other ways of dividing the probabilities between the hypotheses yield other results. Carnap's c^* inductive logic (**CarnapLFP2**), for instance, had a division that worked as follows. For each number n between zero and a million we have the hypothesis H_n that there are exactly n black ravens, and Carnap proposed that the million-and-one such hypotheses should have equal probability, and then each hypothesis H_n is divided into equally likely subhypotheses specifying all the subhypotheses that make there be n ravens. Thus, H_0 and $H_{1000000}$ have only one subhypothesis: there is only one way to have no-black or all-black. But H_1 and H_{999999} have a million subhypotheses each: there are a million options for which raven has the outlying color.

Using the same constant-ratio Bayesian technique as before, after observing 1000 black ravens, the probability that the next one is black turns out to be approximately 0.999, but the probability that all million are black will only be about 0.001. More generally, if there are N ravens, and the first m of them have been observed to be black, and $n \geq m$, then the probability that the first n will be black will be $(1 + m)/(1 + n)$.⁷ Hence we have very good reason to think that the *next* raven is black, but unless we have observed the bulk of the ravens, we won't have reason to think that all the ravens are black.

Intuitively, while Carnapian c^* probabilities do support induction, they make induction too slow—it is only when we have observed the bulk of the cases being a certain way that we get to conclude that they are all like that. On the other hand, my 1/3–1/3–1/3 division is too fast. Even with 1000 black ravens having been observed, the probability

⁷*Let B_m be the claim that the first m ravens are black. Then for $n \geq m$ we have $P(B_m | H_n) = \binom{N-m}{n-m} / \binom{N}{n}$ since there are $\binom{N}{n}$ maximally specific subhypotheses of H_n , and there are $\binom{N-m}{n-m}$ of these that have the first m ravens be black. Thus $P(B_m) = (N+1)^{-1} \sum_{n=m}^N \binom{N-m}{n-m} / \binom{N}{n} = \frac{1}{1+m}$, where the sum was calculated with Mathematica software and is a sum of partial factorials. The probability that the first n are black given that the first m are black where $n \geq m$ will then be $P(B_n | B_m) = (1 + m)/(1 + n)$.

of a white raven shouldn't be *astronomically* small in the way that $1/2^{1000}$ is. Reasonable priors, thus, yield a speed of induction somewhere between these.

The Carnapian c^* measure has a kind of elegance and naturalness, but it makes induction too slow. We can gerrymander a formula for the priors which will fit with our intuitions of how fast induction should work. **CarnapContinuum** (**CarnapContinuum**) himself eventually explored continua of methods of induction which varied in speed of induction. But then we have parameters that govern the priors. If a specific assignment of priors were rationally required of us, then we would have the Mersenne question of why it is these and not some other very similar priors that are required. And if there is a range of priors rationally permitted to us, then we would have Mersenne questions about the boundaries of this range. And there would be such boundaries, because c^* is too slow, so some measures must be excluded.

Further, imagine beings other than us that inhabit a more c^* -friendly "Carnapian" world than we do. While in our world, we have a significant number of natural kinds that exhibit or fail to exhibit some basic property exceptionlessly—for instance, every electron is charged, and no photon has mass—in that world there are few such natural kinds. Instead, if we were to tabulate the frequencies of basic binary properties in various natural populations—say, tabulating the frequency of blackness among ravens, charge among electrons, mass among photons—we would find the frequencies to be spread out pretty uniformly between 0 and 1. In that world, Carnapian c^* priors would lead to the truth faster than the more induction-friendly priors that we have. And let us imagine that in that world we have intelligent beings who reason according to Carnapian priors. Even if we happily grant that c^* priors are irrational for us, it seems plausible to think that they could be rational for those beings. To insist that these Carnapians are irrational, because it would be irrational for us to have these priors, seems akin to saying that bigamy would be immoral for aliens who need three individuals to reproduce, or that there is something wrong with sharks because they lack lungs.

Consideration of the rationality of induction thus once again reveals an appearance of contingency in the normative realm, which once again yields an argument for an Aristotelian picture of human nature, where the rationally required priors or ranges of priors are those that we are impelled to by our human nature.

Finally, it is worth noticing the interesting way in which this Aristotelian account of original priors makes justification both internalist and externalist. It is like an internalist account in that what justifies having an original prior is internal facts about the agent: the norms grounded in the agent's form, which form constitutes the agent as a human being and hence as an agent. But like in externalist justifications, these normative internal facts are not directly available to the agent's consciousness. They can be teased out by generous application of Aristotelian optimistic methodology to our intuitions and to the patterns of inference exhibited by the movements between evidence and posteriors among those agents whom our intuitions hold up as exemplars of rationality (as in the maxim that as for priors, "by their posteriors ye shall know them"). Again, as in the ethical case^{??backref}, this lets us have *something* of internalism, while yet preserving some of the advantages of externalism.

5.3. Anti-skepticism. There are skeptical hypotheses that predict the same observations as our best non-skeptical theories about the world. Indeed, for any reasonable theory T of the world, there is a variety of skeptical hypotheses about how the world merely looks as if T , say due to an evil demon, or us being in a simulation, or random quantum fluctuations in a Boltzmann brain. If an as-if- T skeptical theory T' starts off with probabilities of the same order of magnitude as the probabilities of T , because all of the evidence fitting with T fits equally well with T' , no matter how much evidence we gather, the probabilities of T' and T will remain within an order of magnitude—indeed will have the same ratio. For instance, if T' starts off at half of T 's probability, then it will remain at half of T 's probability, and in particular, it follows that T can never have more than $2/3$ probability, no matter how much evidence we gather for it, since we will always have $1 \geq P(T') + P(T) = (1/2)P(T) + P(T) = (3/2)P(T)$.

A sane epistemology that lets us confirm reasonable theories to a degree bigger than, say, $2/3$, thus requires low priors for skeptical hypotheses. How is it going to achieve this? Perhaps the best initial candidate would be a strong preference for simplicity. We might then expect to assign a lower probability to the hypothesis T' on which an evil demon makes things seem as if theory T were true on the grounds that the evil demon complicates the story. But on reflection, an evil demon need not *greatly* complicate the story, and may even simplify it in some respects.

For instance, if T is the true philosophical and scientific story about the world (assuming that in fact the skeptical hypotheses are false), then T will need to have a complex account of the mind-body problem—either of how consciousness and intentionality can be grounded in a physical system or of how non-physical mental states and physical states interact. But a skeptical hypothesis can be entirely Berkeleian, supposing a non-physical evil “demon” to impose phenomenal states on a purely non-physical being, and supposing the non-physical evil “demon” to have precisely the causal powers needed for that. Likewise, a real physical world that fits with our best theories has vast amounts of complexity beyond our observational abilities: details of the behavior of the world too small or too far away for us to observe them. But an evil demon world need not include any of that complexity. Only what’s empirically relevant to us needs to be included in the evil demon’s deception. If the evil demon is just deceiving *me*, the evil demon’s deception at most needs to encompass a small pretend-universe of radius of about a hundred light-years, since anything outside that radius is irrelevant to my observations. Nor need the evil demon think about the exact positions of all the particles in that sphere: the evil demon need only work with an approximate physics good enough to fool me.

Similar points apply to a number of other skeptical hypotheses. A theory on which I am a Boltzmann brain that’s been floating for five minutes in a bubble of oxygen in an otherwise empty universe, about to die once the oxygen dissipates, involves much less in the way of informational complexity than the world of our best theories which posit

vastly many more degrees of freedom due to there being many orders of magnitude more particles.

One might think a simulation hypothesis—that I inhabit a computer simulation—would involve significant complexity, because not only would I need to be simulated, but we would have all the complexity of the physical cosmos in which the simulating computer lives. But that physical cosmos need not be nearly as complex as the universe of our best theories. It could, for instance, be a cosmos optimized for computing. Whereas in our world, the logic gates, memory cells, oscillators and wires of computers are built from many atoms each, and the atoms themselves are made of multiple particles with very significant informational complexity due to their apparently analogue (or at least very high resolution discrete) nature, we could suppose a cosmos whose basic particles *are* logic gates, memory cells, oscillators and connectors, and which have only the degrees of freedom relevant to their computational role. Such a cosmos could be vastly simpler than the cosmos of our best physics.

Complexity considerations thus do not appear to let us escape from skeptical hypotheses. What else can we do? Well, one option is to just go with common sense: assign low priors to skeptical hypotheses because they are “crazy”. But it seems like an anthropocentric cheat to build such an anti-skeptical bias into the conditions for our priors. It would be odd to think that our priors should be such as to please our intuitions.

But this is not at all odd on Aristotelian optimism. If our world is as Aristotelian optimism has it, we would expect the structure of our priors to match our world to some degree, and hence for the priors to reject skeptical hypotheses. Assuming that in fact skeptical hypotheses are false, and that without presupposing their falsity it is difficult to engage in any sort of reasoning, it is unsurprising that our human nature simply tells us to assign them low priors. It’s both epistemically and holistically good for us to do that, common-sense agrees with this, and on optimistic Aristotelianism that some practice is common and good for us is evidence that the practice is normative.

The above discussion was formulated in Bayesian terms. But similar points apply to any reasonable epistemological framework. We need some way to choose a non-skeptical over a skeptical hypothesis, even if the skeptical one fits our observations just as well. And an Aristotelian story on which our anthropocentric intuitions of what is and is not a “crazy” hypothesis are normative for us is a good account of why this kind of anthropocentric intuition is appropriate.

Outside of an Aristotelian story, however, it is difficult to see how to explain why skeptical hypotheses consistently should have very low priors. Moreover, this kind of anthropocentric approach allows for the possibility that beings of a different sort, whose natural niche is a different kind of environment—say, non-material beings whose proper environment is a Berkeleian world of spirits—would have different norms of reasoning, and what to us is a skeptical hypothesis might to them be the most common-sense thing.

But before we embrace the Aristotelian account, we should consider some Bayesian responses.

5.4. Subjective Bayesianism. Subjective Bayesians avoid all the difficulties of specifying permissible priors by merely requiring the (initial, ultimate or ur-) priors to satisfy some formal properties. These are taken to include the axioms of probability and, sometimes, the regularity constraint that all contingent propositions have non-zero probability. Rationality then constrains transitions from one set of probabilities to another: these must follow the Bayesian update rule that upon receiving evidence E , one’s probability in a hypothesis H goes from $P(H)$ to $P(H \mid E)$. But the initial choice of priors is up to the individual, subject to the formal constraints.

The resulting picture of rationality does not match common sense. Take the most ridiculous set of conspiracy theories that we would all agree is unsupported by our evidence, but where nonetheless the conjunction of the theories is logically consistent with the evidence. Then there is a possible assignment of priors such that updating in the Bayesian way on our actual evidence strongly confirms the conjunction of these theories, both in the

incremental sense of greatly increasing that probability and in the absolute sense of making that probability high. (All we need is that the prior probability of the conjunction of the theories be low, but that the conditional probability of that conjunction on the conjunction of our evidence be high.) Yet to reason that our evidence strongly supports these theories is paradigmatic of irrationality. It shouldn't be the case that a rational person could come to exactly the same conclusions on exactly the same evidence as a paradigm of irrationality would.

Or consider the implausible asymmetry between the freedom in choosing initial priors and the rigid constraint in updating. Suppose I don't like my current rather high evidentially-based posterior for the hypothesis that I will die or lose my mind by age 105, and I would feel better if I had a cheery high posterior for the hypothesis that I will live with full mental faculties past 105. Then there will be a set of priors such that, given the evidence E that I had received over my lifetime, would have yielded the cheery posterior.⁸ According to the subjective Bayesian there would have been nothing irrational in having adopted those priors in the first place, and thus having ended up at the cheerier posteriors. I didn't adopt those priors, but why should I be tied to the priors that I actually had?

The picture of priors here is like a rationally unbreakable vow to live one's life as an evolution of these priors under Bayesian application of evidence. But it is a vow made without any rational ground, indeed without any choice, and likely in childhood. We would think it unsupportable that someone be held committed for life to a promise made early in childhood. If we think with the subjective Bayesian that the initial priors are rationally arbitrary, why then should we be bound to them and hence to the posteriors coming from them? Yet if we are not bound to them, then subjective Bayesianism just turns into nearly total subjectivism (maybe subject to some consistency constraints).

Let us now consider some non-subjective responses.

⁸I just need to assume priors P where $P(E \& L_{105})$ is much higher than $P(E \& \sim L_{105})$ and $P(L_{105}) = 1/2$, where L_{105} is the hypothesis that I will live past 105 with full mental faculties.

5.5. Indifference. An intuitive way to solve the problem of priors is to place a constraint on one's priors that epistemically equivalent situations get equal credences. This is essentially a version of the Principle of Indifference: For each side of a perfect die, the proposition that the die will land with that side up is epistemically equivalent, so our prior credences should be indifferent between the sides. Assuming that the logical features of the situation necessitate that exactly one of the six sides eventuates, the finite additivity of probabilities ensures that each side has probability $1/6$.

There are many well-known problems with this approach. Some of these are technical. A classic example is Bertrand's Paradox: What is the probability that a random chord drawn in a circle will be shorter than the side of an equilateral triangle inscribed in the circle? There is more than one way of applying the the principle of indifference, yielding different results (**Shackel2007**). Or consider the Hausdorff Paradox: Given the Axiom of Choice, if S' is the surface of a sphere S with a certain countably infinite set removed, then there are three disjoint subsets A , B and C whose union is S such that the four sets A , B , C and $B \cup C$ (the union of B and C) are all congruent—i.e., each of the four sets can be rotated around the center of the sphere to give any of the others (**Hausdorff**). Since A , B and C are congruent, Indifference tells us that z is equally likely to be in A as in B and C . It follows that z is twice as likely to be in $B \cup C$ as in A . But likewise A and $B \cup C$ are congruent, so Indifference tells us that z is equally likely to be in A as in $B \cup C$, a contradiction.⁹

But there are also two non-technical problems. First, in many cases it is difficult to see how we should carve up the space of possibilities into equivalent options. Should we, for instance, divide grand metaphysical views into naturalism and non-naturalism, with the two getting equal probability $1/2$, or should we have a finer-grained division into

⁹*If $P(U)$ is the probability that z is in U , then $2P(A) = P(A) + P(A) = P(B) + P(C) = P(B \cup C)$ by the first Indifference argument and since B and C are disjoint, while $P(A) = P(B \cup C)$ by the second argument. The only way both equations can be true is if $P(A) = 2P(A)$, i.e., $P(A) = 0$. But then $P(S') = P(A \cup B \cup C) = 3P(A) = 0$. However, since S' omits countably infinitely many points of S , and the probability of a random point landing in a countably infinite set is zero or infinitesimal, $P(S')$ is either one or one minus an infinitesimal, so we cannot have $P(A) = 0$.

naturalism, anti-naturalism (nothing is natural) and dualism (there are both natural and non-natural things), with each getting probability $1/3$? Or should we have a longer list, with dualism being split into Platonism, monotheism, polytheism, etc., but naturalism left as one item, now of rather low prior probability?

Second, as we basically already saw in ??backref, if we are not careful, we will undercut induction. If every assignment of black and white coloration to a set of ravens has equal probability, then until they have observed *all* the ravens, the Bayesian's credence that all the ravens are black will be no more than $1/2$.¹⁰ Now, perhaps we should not consider every sequence of ravens on par with every other one. But now the question of which sequences are on par with which does not seem to have any canonical answer. As we discussed in ??backref, Carnap thought that lumping together sequences according to the *frequency* of blackness was the way to go. But if not all sequences are on par, why should we think all *sequences* are on par? Intuitively having exactly one out of N being black is equivalent to having exactly one out of N being white, but why should either of these be equivalent to having exactly half of them be black, or all of them be white, say? But if we have no good way to divide the raven color assignments into equivalent events, indifferentism does not help us.

5.6. Algorithmic priors. Suppose we have a language L of finite sequences of symbols chosen from some finite alphabet of basic symbols, with some of the sequences representing a member of some set S of situations.

For instance, S could be arrangements of chess pieces on a board¹¹, and L could be a declarative first-order language with no quantifiers, twelve piece predicates (specifying

¹⁰Specifically, if there are N ravens in total, and n have been observed to be black, there are 2^{N-n} sequences of raven coloration compatible with the observation, only one of which has all the ravens black, the probability that they are all black will be $1/2^{N-n}$, which is less than $1/2$ if $n < N$.

¹¹The arrangements contain less information than chess positions, since a chess position includes other information, such as whether a given king or rook has already moved, whose turn it is, as well as historical information needed for adjudicating draws.

both the color and piece type) and sixty-four names of squares. We could then say that a symbol sequence ϕ represents an arrangement s provided that ϕ is a syntactically valid sentence that is true of s and of no other arrangement. However, in general L need not be a declarative language. It could, for instance, be an imperative computer language for an abstract Turing machine or a physical computer, and the situations could be possible outputs of that machine. Then we might say that a symbol sequence ϕ represents a situation s just in case ϕ is a program that, when run, halts with the output being s . Or if we like we might add an additional layer of representation between the outputs of the machine and the situations—for instance, the outputs of an abstract Turing machine might represent the physical arrangement of particles in a universe, with the chess piece arrangements supervening on these. We can even chain languages. For instance, we could have a hardware computer language L_1 , with the outputs being sequences of symbols in some declarative language L_2 , whose sentences in turn represent members of a set S of situations.

Next, consider a natural way of choosing at random a finite sequence of symbols of L . Here is one. Add to L 's finite alphabet a new "end" symbol. Then randomly and independently, with each symbol being equally likely (i.e., having probability $1/(n+1)$, where n is the number of non-end symbols), choose a sequence of symbols until you hit the end symbol. The sequence preceding the "end" symbol will then count as the randomly selected sequence in L . Every sequence of length k has probability $1/(n+1)^k$, so the probabilities decrease exponentially with the length of the sequence. We repeat the random selection process until we get a sequence that is both syntactically correct and represents a situation in S .¹² We now stipulate that the prior probability of a situation s is equal to the probability that the above process will generate a sequence ϕ that represents s .

We can be a little bit more precise. If ϕ is a sequence of symbols of L and s is a situation in S , write $R(\phi, s)$ if ϕ is syntactically correct and represents s , and let $R(\phi)$ be shorthand for the claim that ϕ syntactically correctly represents something in S , i.e., $\exists x(x \in S \ \&$

¹²If at least one finite sequence is syntactically correct and represents a situation in S , then with probability one, we will eventually get to a sequence that syntactically correctly represents some sequence.

$R(\phi, x)$). Then if A is a randomly chosen sequence of symbols of L , we can define the prior probability $Q(s)$ of s as the conditional probability that A syntactically correctly represents s on the supposition it syntactically correctly represents something, i.e.,

$$Q(s) = P(R(A, s) \mid R(A)) = \frac{P(R(A, s))}{P(R(A))}.$$

We can call these priors L -Solomonoff priors (**PrussRegularityForward**, **PrussRegularityForward**; cf. **Solomonoff78**, **Solomonoff78**). Such priors favor situations that can be more briefly represented in L over ones whose representations are long. The effect of this choice of priors depends heavily on the choice of language L and how well it can compress some situations over others. For instance, in our chess case, if we have no quantifiers, it is easy to see that any two piece arrangements with the same number of pieces will have equal prior probability, because each square's contents have to be separately specified. Thus, if we know that every square contains a pawn, and we have observed the first 63 of these pawns and found them all to be black, the probability that the 64th square will be black is still 1/2. On the other hand, if quantifiers and identity are allowed into our language, then the all-black-pawn situation can be briefly represented by

$$(1) \forall x \text{ BlackPawn}(x)$$

(where the domain is squares on the board), while the situation where squares 1,...,63 have black pawns and square 64 has a white pawn is harder to represent. We might, for instance, use a sentence like:

$$(2) \forall x (\sim(x = 64) \rightarrow \text{BlackPawn}(x)) \ \& \ \text{WhitePawn}(63).$$

Since the probability of generating a given sequence of symbols decreases exponentially with the number of symbols, (2) is much less likely to be randomly generated than (1), and it is intuitively very likely (though proving this rigorously would likely be quite difficult) that in general the probability of generating a sentence representing 64 black pawns is higher than that of generating a sentence representing 63 black pawns followed by one white pawn. We can thus expect that the conditional probability of the 64th pawn being

black on the first 63 being black to be very high. (Maybe even too high? It is very difficult to do anything other than use intuition, because there are so many ways that a single situation can be represented linguistically.)

Just as in the case of using linguistic complexity to quantify projectibility, the choice of language here provides many Mersenne questions. If we opt for the algorithmic version of the theory, we need to choose some computer language for a real or abstract computer, and then we need to choose a representation map between outputs and situations in the external world, with infinitely many possible candidates. And on the more descriptive versions, we still need to choose a language, with many decision points as to syntax and vocabulary. It is very unlikely that there is a privileged language. And not every language will yield priors that fit with our intuitions about induction. For instance, we can easily create a language, whether algorithmic or descriptive, where 63 squares of black pawns followed by one square with a white pawn are much more briefly describable than 64 squares with black pawns. For instance, on the descriptive side, we might use a $\text{BlitePawn}(x)$ predicate, where something is a blite pawn provided it is a black pawn and on one of the first 63 squares or a white pawn and on the 64th, and an analogous $\text{WhackPawn}(x)$, and then we can say $\forall x \text{ BlitePawn}(x)$.

In the unlikely case that there is a single language L such that L -Solomonoff priors are rationally required for us, we will have a vast number of Mersenne questions about the various parameters of the language and its representation relation. In the more plausible case that there is a set of admissible languages such that we are required to have L -Solomonoff priors for some admissible language, we will have a vast number of Mersenne questions about the parameters that control the range of admissible languages. All of this gives rise to a significant degree of appearance of contingency.

Consider, too, the following observation. It is implausible that the languages defining rational priors for us should be ones that are completely beyond our ken. But, on the other hand, it is plausible that there are possible languages that to the smartest human are as incomprehensible as one of the less intuitive computer languages like Haskell or Verilog

or one of the creations of logicians further from natural language like lambda-calculus is to a typical six-year-old. Imagine now beings to whom such these languages beyond human ken are easy. It *could* be the case that the norms for rational priors for them are formulated in terms of L -Solomonoff priors for one of the “baby languages” that humans can understand, but this does not seem a particularly plausible thesis. It seems more likely that for those beings, the algorithmic rational priors would be different than for us.

The standard way to defend algorithmic measures of complexity from the problems presented by a plurality of languages is to observe that sufficiently sophisticated languages have translational resources. Thus, one can write a Haskell interpreter in Javascript, and so anything that can be expressed in Haskell can be expressed in Javascript by including the code for a Haskell interpreter, and then using a string constant that contains the Haskell code. The result is that the difference in the length of code needed to generate a given output in different computer language will typically be just an additive constant: if one can produce the output with Haskell code in n bytes, then one can produce it in approximately¹³ $n + k_{J,H}$ bytes in Javascript, where $k_{J,H}$ is the length of the Haskell interpreter in Javascript. For large enough n , the additive constant will be unimportant. If we are to measure the complexity of a one-hour broadcast-quality video by the length of code needed to compress the video, the addition of a constant like $k_{J,H}$ will likely be negligible: a Haskell interpreter is about half a megabyte, while an hour of video compressed losslessly can be reasonably expected to be in the gigabytes.

However, two points need to be made in our epistemological context. First, even if two different languages give very similar sets of priors, if they give even slightly different priors, we either have the Mersenne question of what makes one of these sets of priors be the objectively correct one, or we have the Mersenne question about the boundaries of the range of permissible languages. Second, unlike in the case where we are measuring the complexity of a large set of data, such as a video file, in the inductive cases we need to look

¹³The approximation is due to complications due to having to embed code in a string constant, which may involve various escape characters.

at ways of expressing relatively simple statements, such as “All electrons are negatively charged” or Schrödinger’s equation or “The first 63 squares have a black pawn and the last square has a white pawn.” But for such statements, a translation manual will dwarf the length of the translated text. To say “All electrons are negatively charged” in French by first describing how English works, and then saying that this description should be applied to the English sentence, will produce a French sentence that is many orders of magnitude longer than the English one, and hence not a sentence that is relevant to measuring the prior probability that all electrons are negatively charged.

Finally, while there is something elegant and natural about randomly choosing items in L by randomly choosing within the set of symbols with an end marker added, there are other ways to proceed. For instance, instead of making the end marker equally likely as each of the ordinary symbols, one could at each step of generation flip a fair coin. On heads, one is done generating. On tails, one then uniformly randomly chooses one of the n symbols.¹⁴ Or one might first randomly choose a positive integer specifying the length of the sequence of symbols according to some probability distribution on the positive integers, and then make all the sequences of that specified length be equally likely. Or one might randomly choose a positive integer, and then choose the n th sequence of symbols in some ordering (e.g., alphabetical). While the initial symbol-by-symbol method with an end-symbol may seem more elegant, it is hard to say that it is rationally privileged to the point that the priors generated with it are rationally required. But if it’s not thus privileged, then the range of random choice methods will provide more Mersenne questions. For not every random choice method yields priors that are plausible candidates for rational permissibility. There will be random choice methods where the sentence “Birds are a government-run drones” has a probability $1 - (1/10^{100})$ of being generated (with all the

¹⁴This will actually not make a difference in those languages where the syntax already determines where a syntactically valid sequence ends. This will be the case with some Polish notation languages, where a valid sequence ends when the main operator is filled out with arguments.

other sentences having probabilities adding up to $1/10^{100}$), and so a boundary would need to be posited between the admissible and inadmissible random choice methods.¹⁵

5.7. Basic probabilities, norms and explanationism. Recently, **ClimenhagaBasic** (**ClimenhagaBasic**) introduced the very helpful notion of *basic* epistemic probabilities, which determine other epistemic probabilities. For instance, the algorithmic approach tends to take unconditional probabilities as the basic ones, and then conditional ones are defined by the rule $P(A \mid B) = P(A \ \& \ B)/P(B)$. Or on an indifferentist approach to a die roll, the basic probabilities are the equal probability $1/6$ of each of the six sides, and the other probabilities are determined by these and by logical relations. Thus, the probability of the die showing an even number will be $P(2) + P(4) + P(6) = 3/6$, because the three options are logically mutually exclusive.

An Aristotelian meta-epistemology can accommodate a picture of the grounding of epistemic probabilities that takes some of these probabilities to be more fundamental than others. For it is open to the Aristotelian to have a distinction between basic and derived norms. Thus, the prohibition of torturing blue-eyed persons is derived from the prohibition of torturing persons. The latter norm might be basic, or might derive from more basic ones. The question of the grounding structure of norms is a substantive question worthy of investigation, and indeed comprises a significant amount of the work in normative ethics.

It is worth noting that the grounding structure of basic norms is not exhausted by norms of the form: “Assign x as the epistemic probability of A (or of A given B)”, nor does Climenhaga claim it is. There surely are basic norms like: “Make your epistemic probabilities satisfy the axioms of probability.” But there could also be substantive constraining norms, like: “Make your epistemic probabilities for A -type events independent of your

^{15*}One might think that there is a privileged way to randomly generate sequences: one just makes each sequence has equal probability, regardless of the sentence’s length. Since there are countably infinitely many finite sequences, however, this option leads to a countably infinite fair lottery which is known to have many paradoxes (**PrussInfinity**). A similar difficulty arises if we first choose a positive integer as the length, with all positive integers being equally likely, and then make all sequences of that length equally likely.

epistemic probabilities for *B*-type events.” Furthermore, it is quite possible that some of the basic norms will be range-based, such as “Assign *A* an epistemic probability between *x* and *y*”, or qualitatively or quantitatively comparative: “Assign *A* a higher epistemic probability than *B*” or “Assign *A* half the epistemic probability of *B*.”

Bracketing some important technical detail, Climenhaga’s own approach is that the basic probabilities are conditional ones like $P(A \mid H)$, where *H* is a hypothesis putatively immediately explanatory of *A*, and the basic probability values are determined by the explanatory relation between *H* and *A*. We could, for instance, suppose that *H* is a thorough description of the chemistry and physics of a match being struck and *A* is the event of the match igniting. The probability value then would be determined by the chemical and physical facts in *H*.

There is reason to be pessimistic about this as a complete solution, however. One set of difficulties comes from the thought that such a system of basic probabilities might work for an ideal agent, but humans are not ideal. The explanatory hypotheses we formulate, especially in the pre-scientific era or in ordinary life, are not formulated with the kind of precision that allows for precise probabilities to be read off. If Alice feels grave resentment against Bob, that could explain her insulting Bob, but the hypothesis of grave resentment does not yield a specific probability. We might think that the hypothesis of grave resentment is a disjunction of a large number of more scientifically precise hypotheses, ones that most people would not even be able to formulate. But if our basic probabilities only involve conditioning on these hypotheses, we will need norms for what probabilities a real human should assign in light of a very vague sense of what the range of the more precise hypotheses is, what their probabilities are, and so on. We will need norms, coming along with many seemingly arbitrary parameters and consequent Mersenne questions, on how a real human being needs to approximate the ideal agent.

A second set of difficulties comes from the difficulty of applying the explanationist approach to explanatorily fundamental hypotheses, such as that God exists, or that the beginning of the world is entirely a natural event satisfying some precise unexplained

physical theory. Here it seems we either need one fundamental hypothesis to be *a priori* certain or we need basic unconditional probabilities for the various fundamental hypotheses. The option of basic unconditional probabilities raises Mersenne questions. As for the possibility of an *a priori* certain fundamental explanatory theory, there is perhaps only one candidate for such a theory in the philosophical literature: theism, either as supported by classical deductive arguments from premises that are themselves certain (such as one's own existence, and various metaphysical principles), or supported as a basic belief required of all human beings by their nature.??cf.Plantinga

Such an *a priori* theism is worth considering, but it raises its own difficulties for the explanationist project. For while the hypothesis that God exists would yield immediate explanations for various claims about the world, it does not seem theologically plausible to think that God is bound by precise probabilities. Thus even with an *a priori* theism, we would have Mersenne questions about priors for conditional probabilities about what God would do if God existed.

5.8. Non-Bayesian update. We have thus seen that our Aristotelian metaepistemology provides a grounding for priors that seems superior to main Bayesian alternatives on offer. But there is at least one more area where Aristotelianism does better. Bayesianism is committed to updating by conditionalization being the only permissible way to update credences. If my current credence in a hypothesis H is $P(H)$, and I receive evidence E , my credence should move to $P(H \mid E)$, which is equal to $P(H \ \& \ E)/P(E)$ if $P(E) > 0$. No other changes of credence are permitted. There is an elegance and non-arbitrariness to this, of course modulo the above-discussed issues about the choice of the priors $P(H)$ and $P(H \mid E)$.

But as is the case for many other formally simple philosophical theories, this is too simple. Consider several cases.

PILL: A trustworthy oracle offers you a pill which will shift your credences closer to truth and does not introduce any incoherence.

MISTAKE: An hour ago, I made an arithmetical evidence when updating on evidence. I moved from credence 0.4 in H to credence 0.6, even though in fact $P(H \& E)/P(E)$ equalled 0.7. I have just discovered my mistake. It would be good for me to go back and correct my credences, essentially rewinding my epistemic life from the mistake and re-conditionalizing on all the subsequent evidence.

PREJUDICE: My original priors had extremely high probabilities for some conspiracy theory involving members of a certain minority group, so high that despite the fact that all my life I have been receiving strong evidence against the theory, nonetheless I am still quite convinced of the theory. I reflect on my original priors, and note that my priors of the conspiracy theory in question were quite out of proportion to my priors for similar theories involving other minority groups. I conclude that my original priors were stupid and racist. I go back and change them to treat the various groups more equally, and then as best I can I fix up my posteriors to match the evidence I recall having had.

In all three cases, my update of credences seems quite rational, but is not a case of conditionalization. We should thus suppose that while it may be a good general rule that we should update by conditionalization, we should at times depart from it. But the question of how we should depart from conditionalization now introduces complexity in the theory that raises significant Mersenne questions. We need more than just simple exceptions for each case: there will be parameters to set.

The case of PILL as given is straightforward. We can say that you should depart from Bayesian update when doing so would move some of your credences closer to truth and none further away from it, where a credence's "distance from truth" is the distance of the credence from 0 or 1 depending respectively on whether the proposition the credence is in is false or true. However, there are variants of PILL. Suppose that the pill has a 0.9 chance of moving you significantly closer to truth and a 0.1 chance of moving you slightly away from truth. It seems like it could be worth it. Or suppose that the pill moves your credences

in important propositions closer to truth at the expense of moving your credences in some unimportant propositions further from the truth.

This suggests that we will need to quantify epistemic value, much as we did in Section??backref, and then say something like this: You are epistemically required (respectively, permitted) to opt for a pragmatically costless change of credence when doing so increases (does not decrease) expected epistemic value. If we quantify epistemic value in terms of strictly proper scoring rules, a delightful result of this move is that we do not need to handle the ordinary case of Bayesian update any differently. For as we noted, **GreavesWallace (GreavesWallace)** proved that in the kinds of cases where one simply receives evidence and sets one's credences according to it, conditionalization is the unique best policy for maximizing expected epistemic value as measured by a strictly proper scoring rule.¹⁶

However, as we also saw in Section??backref, the choice of a scoring rule or measure of epistemic value involves infinitely many free parameters, and at the same time not every scoring rule that satisfies the formal constraints is rationally plausible. In our present context of evaluating non-conditionalizing updates the point is particularly clear. For when evaluating credence-changing pills, we need to take into account the *importance* of the affected credences, and that is not a formal criterion.

Note that PILL is not as outlandish as it initially seems. We do in fact modify people's thinking with psychiatric medication as well as with psychotherapy.

Next, consider MISTAKE. We don't in fact remember all our evidence: we update our credences on the more important things and forget the less important, which often includes some or all of the evidence. I know that Beijing is the capital of China, but I don't know where I learned it from. I know that there was a cat in the yard earlier this morning, but much of the rich sensory evidence is now gone from my mind. An attempt to go back and correct a past update error is bound to be a messy affair. Sometimes when the mistake is

¹⁶Cf. **IsaacsRussellUpdating (IsaacsRussellUpdating)** discuss how maximizing such epistemic value also helps in cases where you are unreliable in fulfilling your epistemic plans.

minor it might be better to let it go rather than miss out on some of the evidence gathered since. And the situation is often too messy and too complex for any sort of an expected epistemic utility calculation.

Yet the fact that a situation is messy does not get us off the hook. There will still be a distinction between right and wrong ways to proceed, even if the rule cannot be formalized. We have here something very similar to the kinds of messy everyday ethics cases that are grist for the situationist's mill.^{??backref} It is unlikely that the answer is given by any elegant principle without the kinds of parameters that raise Mersenne questions, but at the same time there *is* an answer, and we expect there to be something to ground it.

And what goes for the messiness in correcting a calculational mistake applies even more to the messiness involved in the kind of intellectual conversion that occurs when one realizes that all of one's thinking for years has been based on irrational prejudice, and one attempts to dig oneself out of the resulting heap of epistemic defects. Again, there are right and wrong ways to proceed, but the likelihood of a clear and elegant principle that solves the problem is nearly nil.

5.9. Bayesianism versus natural law metaepistemology. Bayesian update says that upon learning a new piece of evidence E , you should replace all your credences with the ones you get by conditionalizing on E . Typically, a Bayesian update gets you closer to truth about some propositions and further from truth about others. For instance, if you pull a coin from your pocket and look at one side of it, and see heads, this confirms that the coin has at least one heads side, which is true. But it also confirms the hypothesis that the coin is a rare defective double-heads one, which hypothesis is typically going to be false. More generally, none of the evidence we ever learn logically excludes the existence of fairies, but I assume there are no fairies. Suppose you learn some piece of evidence E . Then you trivially get closer to the truth about E , but you also confirm the conjunctive proposition F

that E obtains and there are fairies, by confirming the first conjunct (unless the prior for F was zero or one).¹⁷

Whether you are epistemically better off after a Bayesian update depends on a variety of factors, including the relative importance of the propositions that you get closer to the truth on and those that you get further from the truth on. If for some reason the above false proposition F is extremely epistemically important (maybe aliens have announced they will destroy the earth if and only if F is true), getting spurious confirmation of F could well be an overall epistemic harm. Thus, Bayesian update doesn't always get the best results. And typically there is, abstractly speaking, a rule that will get you better results than Bayesian update. Suppose your concern is with propositions $p_1, \dots, p_n$, and imagine that a stranger comes up to you and tells you to always employ the following update strategy:

(3) if you learn that p_1 is false, do Bayesian update

(4) if you learn that p_1 is true, change your credences to specific numbers $t_1, \dots, t_n$, respectively.

Suppose that the stranger happened, by chance or not, to pick the numbers $t_1, \dots, t_n$ in such a way that $t_i = 0$ if p_i is false and $t_i = 1$ if p_i is true. As long as (4) does not happen to coincide with Bayesian update (typically it won't), and as long as there is any chance of learning that p_1 is true, the stranger's strategy will beat Bayesian update.

Now, chances are, you won't be able to design a better update strategy than a Bayesian one *for yourself*. As already noted more than once, your expected epistemic utility, defined using a proper scoring rule and calculated by your own lights, will be maximized by Bayesian update over the space of all update strategies (**GreavesWallace**). But the crucial thing to note in this result is the "by your own lights". Someone else who has information

¹⁷We have $P(E \mid F) = 1$. However, $P(E \mid \sim F) < 1$, since if we had $P(E \mid \sim F) = 1$, then we would have $P(E) = 1$, and hence you wouldn't be *learning* E , as your prior in E would have been 1. Thus, $P(E \mid F) > P(E \mid \sim F)$ and E raises the credence of F by Bayes' theorem given Bayesian update.

that is not available to you can sometimes—as in the stranger example—design a better rule for you.

Imagine now that you are designing an intelligent robot. You have access to more types of information than the robot ever will, and you use your superior knowledge to design a robot with an update rule that beats Bayesian update (with respect to some or maybe even all ways of prioritizing the importance of propositions). The robot, however, is rather smart, and observes that it is finding itself “naturally” or “instinctively” reasoning in ways that do not match Bayesian update. This makes the robot worry: “Should I keep on reasoning the way I ‘naturally’ do, or should I reprogram myself for Bayesian update?” From your point of view as the robot’s designer, you may well hope—and hope for the sake of the robot’s doxastic flourishing—that the robot will stick to the rule you programmed in, as it is in fact superior with respect to getting to the truth (given the right prioritization). In fact, it seems plausible that the robot would be epistemically defective if it switched to Bayesianism, and started engaging in less reliable reasoning.

We are not robots. Whether we are designed by a person more intelligent than ourselves is highly controversial (I will argue in the affirmative in Chapter ??). However, evolution might be metaphorically said to have “access” to more information than an individual human does: the selective pressures that made us what we are were responsive to a vast amount of data that we individually do not have, and perhaps that our ancestors did not even have in their minds. Some think that we can do better than our instinctive ways of reasoning¹⁸, especially given large amounts of scientific data, say by adopting Bayesianism as our update rule. For the argument I want to press, I am willing to concede the point. But it is not clear that the same would have been true at earlier points in our species’ intellectual development. And it *is* clear that the same is not true for all possible species of intelligent animals: there *could* be intelligent animals whose instinctive reasoning would beat Bayesian reasoning.

¹⁸Say, those of **Kahneman’s (Kahneman)** System 1.

The mere fact that these intelligent animals' instinctive methods of reasoning would beat Bayesian update does not mean that they would be rational to opt for the instinctive methods. But reflection on these possibilities should make one incline to the hypothesis that Bayesianism does not have a hegemony on rationality. It seems possible to have beings for which some other mode of reasoning than Bayesianism is right. Thus, even if Bayesianism is the right epistemology for humans, it is not the right meta-epistemology: to reason rightly is not the same thing as to update in a Bayesian way.

6. The knowledge-first alternative

Suppose that knowledge is a highly natural informationally complex universal with very complex necessary truths associated with it, much like obligation and love were suggested to be in the Platonic approach in ??backref. Some of the Mersenne questions above can be given answers in terms of knowledge, on this knowledge-first account (cf. **WilliamsonKn**, **WilliamsonKn**). Thus, maybe it is reasonable to believe that all ravens are black when one has observed enough ravens to gain knowledge.¹⁹ Likewise, perhaps, the right priors are ones such that relying on them leads to knowledge when everything else goes right (though one might be reasonably suspicious whether one can make this condition precise).

Essentially the same objections, however, can be given to the knowledge-first solution here as to the ethical Platonist approach in terms of informationally complex universals.??backref We do not have an adequate knowledge-first solution to the problem of the appearance of contingency in the norms. Moreover, there are a number of epistemological phenomena that are unlikely to be account for with the binary relation of knowledge between an agent and a proposition. For instance, it is typically better to have a justified higher credence in a truth, even if this does not change whether one has knowledge, for

¹⁹Knowledge-first as an alternative to the Aristotelian account was suggested to me by Charity Anderson, as was this solution to the raven problem.

instance because one already had enough evidence for knowledge, and there will be apparently arbitrary parameters in the mapping from the credence to the value, perhaps of the sort discussed earlier.^{??backref} It does not seem likely that such a function from credence to value can be accounted for in terms of knowledge.

7. Intellectual limitations

Limitations in recognizing logical implications and contradictions are a case where rational norms are plausibly species-relative. It is epistemically irrational for one to conclude that it's raining from the fact that it's neither raining nor snowing. But it is not irrational for one to take $22828 \times 2219 = 50645332$ (which is in fact false) to follow from the axioms of arithmetic due to an arithmetical slip. Of course, someone with exceptional calculational skills may immediately see the latter is mistaken, but the ordinary person's failure to see it is not an instance of irrationality.

Ethics also contains intellectual limitation cases. Analogically to the multiplication case, we would not consider a person to be less than virtuous because they are unable to see an extremely complex medical ethics case rightly. But if someone doesn't realize that it's wrong to ambush strangers in order to sell their organs, there seems to be something morally wrong.

But the ethical cases may also be different. Consider an adult who strives to follow their conscience as best they can, but nonetheless fails to see that it's wrong to kill strangers for their organs. This person presumably has a severe intellectual and/or emotional disability. We might judge them to be a good person, or at least not a vicious one. However, the logical case seems different. The person who, due to intellectual disability, concludes that it's raining from the claim that it's neither raining nor snowing *is* failing at rationality, though of course they are inculpable. One can defensibly define moral worth in terms of the seriousness of their attempt to do what seems right, but it is difficult to define someone's degree of rationality in terms of the seriousness of their attempt to think as seems right. For it is paradigmatic of irrational people that they take themselves to be acting

quite rationally—if one derives that it’s raining from its neither raining nor snowing, this is precisely because the conclusion seems to follow. It is, indeed, unclear whether it is even possible to conclude p from q without its seeming that p follows from q . But it is paradigmatic of immoral people that they lack integrity and violate their own conscience.

The line to be drawn in the epistemic rationality cases (and in the ethical ones if they turn out to be similar), is a line that we are unlikely to be able to draw in a species-independent way. It seems plausible that for beings that as a species are more adept at logic than we are, a failure to see the truth of Fermat’s Last Theorem as following from the axioms of arithmetic is a failure of rationality, but it is not so for us.

The difficulty of handling limitations is particularly glaring in the case of Bayesianism. Typically we can’t do “proper” Bayesian update. We don’t have the time, don’t have the mathematical skills, or perhaps most importantly aren’t able to quantify our priors in such a way as to make them amenable to precise mathematics. We need an epistemology that works in this all too human predicament, a *human* epistemology, and we need an account of what grounds that epistemology.

One may insist that under these imperfect conditions, we should simply say that we are stuck with irrationality. But nonetheless there are cases where it is clear that something should be done under the unhappy circumstances. Suppose that the completely right credence on my priors in some proposition p is 0.9874, but I can only calculate to two significant figures. Then I should take the credence to be 0.99, not 0.98 or 0.01. But things will be less clear if I need to form credences in both p and q , which a perfect Bayesian with my priors and evidence would take to be 0.9874 and 0.0332, but due to limitations of time or energy level, I can only get a total of four significant figures. Are the right credences for me 0.99 and 0.03, or 0.987 and 0.0, or 1 and 0.033? It depends, surely, on the relative epistemic importance of p and q . And once importance needs to be taken into account, it is very unlikely that we will have a precise and elegant account with no free parameters. Instead, we have human messiness.

Recently, **JeffreyGood (JeffreyGood)** has argued that ethical norms may be relative to a stage in life. Whether this is so, it is very plausible that some epistemic norms are so. There may be logical facts failure to notice which constitutes a failure of an adult's rationality, but would not constitute a failure of a child's rationality. If this is right, then we have all the more reason to think the norms of epistemic rationality to be species-relative, since surely what the stages in life are, and when they occur, is something that is species-relative.

8. Social epistemology

8.1. Rational social animals. We are rational social animals. Our rationality is exhibited in a *social* way, and our sociality is exhibited in a way proper to the kind of animal—the human animal—that we are. On the Aristotelian account, all the norms tied to these observations are grounded in our human nature. In this section, we will consider a number of epistemologically relevant aspects to our sociality and show how they point towards our robust view of human nature.

8.2. Innate beliefs and testimony. Humans are said to have much less in the way of instinct than other animals. Furthermore, we do not seem to have any innate beliefs. But we can imagine a species of intelligent animals which do more by instinct than we do, and which have evolved to be born with some unshakeable and true beliefs, such as that purple winged things are to be avoided and electrically charged spiky fruit is good to eat. For these beings, there is nothing irrational about having such beliefs without any evidence. For us, there is. In a Bayesian mode, we might say that for these beings very high priors in these empirical claims are appropriate, but not so for us.

Experiments in machine learning suggest that for certain kinds of problems it is useful for a system to be pre-trained and hence have priors that embody some substantive information. Pre-training presumably comes with a sacrifice of flexibility. Thus whether pre-training is useful depends on the subject area and the kinds of situations the system will face. And of course the content of any pre-training is highly contingent. It would be

surprising if our evolutionary process did not involve any pre-training. Our discussion of skepticism in ??backref implies that it is normal for us to have certain substantive priors, and that not to have them would be abnormal. Some empirical evidence for this is given by infants' neural responses to snake pictures (**BertelsSnakes**). But what those priors should be is surely highly species-dependent: we would not expect alien babies evolved for an environment without any snakelike objects to naturally react to snakes. And even if humans, surprisingly, turn out not to have any substantial information encoded in their priors (if that is even logically possible), that fact is surely species-dependent. In such a case, for us to have substantively informative priors would be abnormal, but for intelligent cats, say, it might be quite normal.

We can also imagine a species where memories are genetically inherited (this would be a species for which Lamarckism is true). A member of this species could, then, be born with a large number of beliefs that they had no evidence for. For, like we often do, the members of this species could forget the evidence that led to a conclusion. But while in our case, we once had the evidence, in this species it was some ancestor of theirs that had the evidence. Typically (apart from substantive priors, say) it is irrational for humans to have beliefs without ever having had evidence for them, but not typically irrational for us to have beliefs whose evidence we have forgotten—say, facts learned in childhood. For members of the species with genetically inherited beliefs, on the other hand, there typically need be no problem with beliefs that the individual never had evidence for, as long as an ancestor had the evidence and forgot it. These inherited beliefs would be akin to innate beliefs or substantive priors, but unlike innate beliefs that would be normatively the same for all members of a species, they would differ from individual to individual depending on what the parents believed.

There is partial post-conception gene transfer in humans in a phenomenon known as fetal microchimerism where a few cells with the fetus's genotype are found in women's healthy tissue after pregnancy (**ODonoghueMicrochimerism**) as well as in microchimerism in twins (**WaszakTwins**). While we do not at present know of any benefit

to the organism from such mechanisms²⁰, we can imagine cases where there are benefits, say by one organism giving cells with a new genotype to an organism whose genotype is either defective (as we do artificially in somatic cell therapy²¹ or less well adapted for some new situation. Thus we could imagine organisms which non-accidentally engage in some post-conception gene transfer. If we now combine this with the thought experiment of organisms that have inherited beliefs, say by supposing the transfer of neural cells, we will have organisms that have voluntarily or involuntarily genetically transferred beliefs. And such transfer, whether voluntary or involuntary, could lead to rational beliefs, if it were to fit with the species norms.

Now, let us return to our species. Perhaps we should not think the species where memories are inherited to be all that different from us. For our parents bequeath much knowledge to us. It is true that this is mediated by soundwaves or gestures rather than by gametes, but does that make a significant philosophical difference? We should, thus, take seriously the idea that just as for members of a species where memories are inherited it is fundamentally rational to believe these inherited memories, and to question them without special reason is irrational, for us it may be fundamentally rational to believe parental testimony, and to question it without special reason is irrational. And transfer through ordinary testimony in adulthood can be thought to be similar to hypothetical voluntary or involuntary genetic transfer of memory at later stages in life in a species with genetic memory. Such hypothetical transfer would be a non-Bayesian mode of update, and so by analogy we have some reason to think that testimony does not reduce to Bayesian update. And just as the modes of inheritance and transfer of belief would be species-relative, it is likely that the specific norms of when and how it is rational to accept inherited or testified-to beliefs are species-relative.

²⁰On the contrary, fetal microchimerism may have unfortunate autoimmune consequences (**ODonoghueMicrochimerism**).

²¹This has for instance been done in animal models for colorblindness, with potential for future human treatment (**HassallColorblindness**).

8.3. Self vs. others. Suppose it turns out that I have a near-identical twin on a planet just like Earth, all of whose mental life has been exactly the same as mine, with one exception: the first bird I saw today was a bluejay, while the first bird he saw today was a robin. Neither of us is an avid birder, and so this has no significant effect on me. We both learn that next time we fall asleep, we will be transported to a third planet, and my twin's memories will be changed so that his robin memory will be replaced by my bluejay memory, with corresponding tweaks for anything else affected by the memory. Furthermore, I am given a choice of epistemic policies: I can ensure that tomorrow both I and my twin will firmly believe that the bluejay memory is correct, and hence will believe ourselves not to have had our memories changed, or I can ensure that we will both assign credence $1/2$ to the bluejay memory's being correct.

Insofar as my epistemic concern is solely for myself, it makes sense to take the policy of sticking to the bluejay memory. But doing so harms my twin epistemically. What about the total epistemic value? As before, let $T(r)$ and $F(r)$ be the accuracy scores or epistemic utilities for assigning credence r to a proposition that is true and false respectively. Then on the policy of sticking to the memory, the total epistemic value is $T(1) + F(1)$: I will have credence one in a truth and my twin in a falsehood. On the policy of assigning credence $1/2$, the total epistemic value is $T(1/2) + F(1/2)$. If the pair T and F is strictly proper, then $(1/2)T(1/2) + (1/2)F(1/2) > (1/2)T(1) + (1/2)F(1)$, and so $T(1/2) + F(1/2) > T(1) + F(1)$. Thus total epistemic utility is higher when the policy is of total uncertainty. However, the total epistemic utility *for me* is higher when the policy is of certainty: obviously, $T(1) > F(1)$.

When evaluating epistemic policies, then, we have two different points of view: a self-concerned and an other-concerned point of view. And we are led to different policies by the different points of view.

The above is a strange case. But there are everyday cases of balancing self- and other-concerned epistemic reasons. It might well turn out that as a community we get to the truth more effectively if investigators have a stronger commitment to their theories than

the evidence warrants, because then they will be more motivated to search for evidence and will have “skin in the game”. However, such a commitment can be expected to be epistemically harmful to the individuals. Thus the epistemic goods of the community may conflict with individual epistemic goods.

For completely asocial rational agents, we would expect norms of epistemic rationality favoring individual epistemic goods. But for social rational agents, we would expect norms of epistemic rationality favoring social epistemic goods. And sociality comes in degrees. We would expect that in more social kinds of rational animals, there would be a stronger favoring of social epistemic goods, and in less social ones, a weaker such favoring. Humans are neither asocial nor maximally social (we are not a hive mind!). For us, we would expect some kind of an intermediate between favoring individual epistemic goods and favoring social epistemic goods. But that intermediate normative position will raise Mersenne questions as to why it lies exactly where it does. And it is not plausible that there is some metaphysically necessary rule that takes the degree of sociality of a rational agent and spits out the exact degree to which social epistemic goods should be favored. It is much more plausible to suppose that the degree and manner of preference for social epistemic goods is a part of the species norms.

Nor is it likely to be a single numerical degree of preference per species. Surely the appropriate degree of preference for social epistemic goods depends not just on the species, but also on one’s role in society and the subject matter. So we need a ground not just for single human numerical degree of preference for social epistemic goods, but a function from social role and area of knowledge to degrees of preference for social epistemic goods.

8.4. Paternalism. If there is such a thing as epistemic value, the most obviously right thing to do with regard to its possession by others is to maximize the expectation of it. Suppose now that you and I have differing prior opinions about the efficacy of a drug: you have a credence of $3/4$ that it is efficacious, while I have a credence of $1/2$ that it is efficacious. I perform some experiments and find evidence of the drug’s efficacy that raises my credence from $1/2$ to $3/4$. Suppose I now consider whether to reveal the evidence to

you. I know that if I reveal E , your own calculations (e.g., Bayesian update) will make your credence go from $3/4$ to $7/8$. But I think the best credence all things considered is $3/4$. So I withhold the evidence, thereby keeping your credence at the level that I think is best.

The above argument seems pretty dubious. But we can make it sound rigorous. Suppose you have a strictly proper scoring rule (T, F) . Then my expectation of your epistemic utility after revealing the evidence and your updating your credence to $7/8$ will be:

$$U_R = (3/4)T(7/8) + (1/4)F(7/8)$$

since by my lights, there is a $3/4$ probability that the drug is efficacious, and your epistemic utility if I conceal the evidence will be

$$U_C = (3/4)T(3/4) + (1/4)F(3/4).$$

It follows immediately from the definition of a strictly proper scoring rule that $U_R < U_C$, and so if I am trying to maximize your expected score, I should conceal the evidence.

This is surely unacceptable paternalism. But suppose that I am nearly certain of a quack remedy's efficacy due to my acceptance of a conspiracy theory (think of various shenanigans during the pandemic here), and you are a scientist who just found a slight bit of evidence in favor of the drug's efficacy. If we're having a conversation and I mention my conspiracy theory, surely you do not have on balance epistemic reason to tell me about your evidence and thereby help confirm my conspiracy theory. There is thus *some* acceptable minor paternalism—we need to be prudent in deciding what to say. And, of course, some paternalism, whether epistemic or moral, is acceptable with respect to one's children.²²

Where the line is to be drawn between acceptable and unacceptable epistemic paternalism seems to be very much a Mersenne question. It is quite implausible to think that there is a clearly privileged place to draw the line.

More generally, the above line of thought shows that our behavior with respect to other people's epistemic value follows subtle and complex rules. We shouldn't simply always

²²For a helpful survey on epistemic paternalism, see **JacksonPaternalism (JacksonPaternalism)**.

maximize their expected epistemic value, as that doesn't show respect for their autonomy. But obviously some attention needs to be paid to the epistemic value of others. Whatever the right rule here is, it likely has multiple parameters and an appearance of contingency. Moreover, the observation that epistemic paternalism can be more acceptable with respect to one's children points to a species-relativity in the parameters, since it is likely that how one should relate to children is species relative (think of a species that has very little parent-child relationship after birth), and it is species relative whether we have other relationships analogous to those to our children.

8.5. A fact's being known. Suppose a scientist discovers a fact p . She then communicates it to twenty people at a conference, and eventually publishes it in a journal where a hundred more people read it. And then a Youtuber picks it up and a million people hear of it. The most obvious way to think of the social epistemic value here is that there is a value V that comes into the world when the scientist discovers p . After the conference, the value V turns into $21V$, and after the journal readers get to it we have a value of $121V$. Finally, the Youtuber multiplies this value by about ten thousand, to yield $1000121V$.

This seems the wrong story about the value of what one might call "human knowledge". It doesn't seem right to say that the Youtuber contributed so much more to human knowledge than the scientist.

An additive account of the value of human knowledge yields an epistemic variant of Parfit's repugnant conclusion for utilitarianism (**ParfitRP**). On an additive account, letting a sufficiently large number of people know some trivial fact, such as the length of Charles III's left big toe, will swamp the value of a major scientific discovery that only a few experts can understand, even if a single person's learning of the scientific discovery swamps the value of a single person learning the length of the royal toe.

In our science story, it seems that the big jump in value occurred when p became known, which happened either when the scientist discovered it or when she informed others of it. What exactly it means for a proposition to *be known* is not immediately clear. First note that when we talk of a proposition being known, we are talking of its being *humanly* known. It

is correct to say that it is not known whether there is intelligent life in other solar systems, even though if there is, someone knows it—namely the intelligent beings in question. And if God exists, then someone knows everything.

The obvious account is that a proposition is known provided that some human knows it. But that condition is neither necessary nor sufficient. If the abstract of a random mathematics paper from half a century ago says: “We give necessary and sufficient conditions for ϕ ”, then I have good reason to conclude that the necessary and sufficient conditions for ϕ are known, even though there is a pretty good chance that nobody has read the paper for decades and nobody who has read it and is still alive remembers what the conditions are. Thus, some things that no one knows are nonetheless known. On the other hand, if Alice is dying of thirst on a desert island, it is correct to say that no one knows this even if Alice knows it.

Roughly, we might say that something is known provided that the knowledge is generally accessible to humans with appropriate training. Alice’s lonely death is not accessible, but the conditions for ϕ are. What exactly counts as “accessible” and “appropriate training” is rather unclear. Suppose the only copy of the journal with the conditions for ϕ is on top of Mount Everest. It is not clear that the conditions are known. Quite likely we should say that the being known in this sense is something that comes in degrees. And then there will be a very complicated function from the subject matter and the degree of accessibility to the *value* of the particular instance of being known. This complicated function is very likely to have many free parameters and hence will raise Mersenne questions about its grounds. And yet this value of being humanly known is important, if only because it informs our resource allocation.

9. *A priori* intuitions

We have a large store of what one might call substantive *a priori* intuitions, such as that nothing can cause itself, that there is contingency, that nothing is a proper part of one of its proper parts, that a material object couldn’t have been always immaterial, that if it is

not necessary that there exist green objects then neither is it necessary that there exist blue objects, that every thought has a thinker, that every number has a successor, that the Peano axioms of arithmetic are consistent, and so on. These intuitions vary in strength, though all the above examples are very plausible.

An Aristotelian account of epistemological normativity can hold that among the norms of our nature are requirements that we have a high credence, maybe in some cases even certainty, in such propositions, and add optimistically that we are so constituted that we tend to follow this requirement of our nature. We can then have a basic justification in such claims simply in virtue of our nature requiring us to assign them a high credence.

10. Going beyond the applicability of human epistemology

Suppose that you are certain you are one of infinitely many people have each rolled a fair die, none of whom have seen the result. It is then announced by a perfectly reliable angel that all but finitely many of the dice show six. What should be your credence that your die shows six?

On the one hand, it seems very likely that your die shows six. After all, the vast majority of the dice show six, and you have no reason to think yourself exceptional.

On the other hand, prior to the announcement, your credence in six was $1/6$. And the announcement told you nothing about your die. For the following two statements are logically equivalent:

(5) All but finitely many of the dice show six.

(6) All but finitely many of other people's dice show six.

For your die's state makes no difference to whether there are finitely or infinitely many non-sixes. But then given your certainty that the dice are fair, (6) gives no information about your die's result. And since (5) is logically equivalent, it too gives no information. Having received no information, you should stick to $1/6$.

This reasoning seems convincing. Yet if everyone sticks to $1/6$, then all but finitely many people are quite far from the truth. Furthermore, if everyone is playing a game

where you are asked to guess if you have a six or not, and you are rewarded for getting it right and penalized for getting it wrong, then if everyone sticks to $1/6$ for having a six, everyone will guess that they don't have six, and all but finitely many people will be penalized, which is surely not the right result.

In **PrussInfinity** (PrussInfinity) I argued that a good solution to epistemological paradoxes like this is to reject the metaphysical possibility of an event depending causally on infinitely many events, in the way that the angel's announcement depends on the infinite number of die rolls. But there is another possible solution: we can deny that our epistemic norms extend to far-fetched situations, just as in ??backref it was suggested that our ethical norms do not apply to far-fetched situations. Whether this is a completely satisfactory resolution to the paradox is not clear. For there is some plausibility in thinking that if predicaments like the above were metaphysically possible, there could be rational beings who could reason in them. But it seems there couldn't be any. However, no matter what we say about the metaphysical possibility of such beings, there is something right about the thought that *we* are not made to reason about such things.

There are many other examples where epistemology seems to break down in radical cases. What should you think if you came to be convinced that an evil demon is trying to get you to believe as many falsehoods as possible? Any answer you might have is undercut. Should your credences in everything be at $1/2$, then? But that is incoherent: for then you have credence $1/2$ that the last die you rolled is 1, and credence $1/2$ that it was 2, and credence $1/2$ that it was 3, and credence $1/2$ that you never tossed a die, and so on. Perhaps suspension of judgment is something other than assigning probability $1/2$. But if so, should you suspend judgment about everything, including about the norm of fitting your beliefs to your evidence? However, if you suspend your judgment about that, then why bother with suspending your judgment about other things, given that you don't think you need to fit your beliefs to your evidence? Or what should you think if you come to be convinced that you are a simulated non-player character in a sophisticated video game?

Next recall a phenomenon that many of us have sadly experienced, where we realized we were using certain words but had no concept behind them, and the words were mere meaningless words. For instance, take a word that we as laypeople defer to experts on the meaning of, such as “gluon”. But now imagine that physicists have been pulling our collective legs about gluons all this time: it was just a made-up word without any meaning behind it. We can imagine discovering that, and even more easily we can imagine someone convincing us that this happened. And words whose meaning we get by deference are not the only words like that. The phenomenon of a person who doesn’t know what they are talking about is not uncommon. My Aristotelian metaphysics includes accidents. But I could imagine finding myself in a position where I come to be convinced that I never had a concept of an accident—that “accident” is a meaningless word. Now imagine a radical hypothesis: I come to be convinced that I have no concepts at all. What should I think now?

There is something plausible about the idea that in certain extreme situations there is no right way for us to think, but beings other than us could have moral norms that provide additional specifications—ones that may appear arbitrary to us, say—as to what one should think in some of these extreme situations. The Aristotelian account fits neatly with this, because we do not expect human nature to tell us how to behave in situations far outside our natural environment.

11. Epistemic supererogation

In ??backref, we saw that a Natural Law ethic can easily accommodate the concept of supererogation by allowing for a case where one flourishes more in willing to ϕ than in willing to ψ , but nonetheless willing to ψ would not be a case of languishing. This was a special case of a general phenomenon where one state can constitute greater flourishing than another, even though neither is a case of languishing.

Does this phenomenon appear in epistemology as well? Intuitively, it does. There are two varieties of this phenomenon. One kind, discussed by

HedbergSuper (HedbergSuper), involves deliberate actions directed at epistemic ends that go over and beyond one's epistemic duties. These actions may involve overt physical activity such as pulling out a credit card to check an already-memorized PIN number or reading encyclopedia articles, but also internal mental activity such as deliberate reflection on one's beliefs.

A second kind involves epistemically skillful ability of a sort that may not be voluntary, such as the evaluation of evidence or the formation of beliefs. Often, when Dr. Watson fails to see the connections that Sherlock Holmes sees, we cannot say that Watson is epistemically defective—rather, Holmes is, at times, superlatively insightful in weighing the evidence. The distinction between insufficient insight, acceptable insight and superb insight is indeed familiar to us.

An Aristotelian metaepistemology neatly accommodates both forms of supererogation as part of the general normative phenomenon of functioning better than the norm requires, with the relevant aspect (in the present case, the epistemic one) of human teleology explaining what counts as better and what counts as worse. The possibility of epistemic supererogation thus provides some support for the present account.

If the correct epistemology is Bayesian one might initially think there is no room for supererogation of the non-voluntary sort. Either one updates correctly on the evidence or one updates incorrectly. There is, no doubt, a gradation among failures of correct update, but there is only way to update correctly. Thus, it seems, on a Bayesian epistemology, there is no room for supererogation of the non-voluntary sort, and an Aristotelian's metaepistemologist's ability to account for epistemic supererogation is less of an advantage, especially if Hedberg's voluntary supererogatory acts can be accounted for in terms of a non-epistemic normativity, such as pragmatic or moral.

But there are at least three areas within a reasonable Bayesianism where there is room for supererogation even of the non-voluntary sort. First, if Bayesianism is to be at all a candidate for our epistemology, and assuming some version of an ought-implies-can doctrine,

Bayesianism has to accommodate imprecise probabilities, because we humans are only capable of having probabilities that are both precise and rational in very limited situations such as coin flips. Maybe I could make myself have a credence of *exactly* 0.9997 that my copy of *Being and Time* is in my office, but that number wouldn't be rational. But we are required to have some level of precision. I would be failing in rationality if I vaguely judged that the probability of *Being and Time* being in my office is "at least a little more likely than not". If I simply judged it to be "very likely", I would not be irrational. But it could be that a person more excellent in precision could have a finer grained probabilities than my "very likely", whether partly numerical ("somewhere around 0.9997" or maybe "between 0.9997 and 0.9999") or just with more refined non-numerical levels. Such a person's judgment could well count as supererogatorily rational.

Second, we are simply incapable of updating fully on all of our evidence—there is too much of it. When I consider the evidence that it's getting to be time when I will go to the gym, I don't update on the fact that there are smudges on the living room window. Our world is deeply interrelated. There is likely some kind of a correlation. (E.g., smudges may indicate something about the weather, which may indicate something about whether I am going to exercise outdoors or indoors.) But I ignore it. And that's part of being an acceptable human reasoner. Similarly, when Dr. Watson ignored the dog's famous failure to bark in his holistic evaluation of the evidence (Doyle1927), he wasn't irrational. Sherlock Holmes, on the other hand, in focusing on this piece of evidence was more than merely rational: he was an excellent reasoner.

Third, and more speculatively, if rational constraints on priors involve ranges of acceptable assignments, it could well be that sometimes assignments near the edges of a range are less rational than ones further from the edge. For instance, plausibly, we should have anti-skeptical priors that assign high credence to such things as the uniformity of nature or the reliability of our senses, and our nature requires us to keep these credences between the bounds of two epistemic vices: undue skepticism (setting the priors too low) and excessive credulity (too high). Then someone whose credences are just slightly above undue

skepticism or slightly below excessive credulity would be rational, but perhaps only barely so. It would be better to have a more balanced credence, away from the two boundaries of rationality.

Thus, even a Bayesian should be glad for the possibility of accounting for epistemic supererogation, and the Aristotelian can lend a hand here.

12. Metaepistemology

12.1. Why be epistemically rational? One of the main questions in metaethics for the past century has been why one should bother being moral. The analogous question for epistemic rationality has received comparatively little attention. One reason for the low amount of attention could be that there is an obvious answer: It's good (both instrumentally and intrinsically) to have the truth, and following the norms of epistemic rationality helps one get there.

Indeed, there are proposed norms which can be justified in this way. As noted a number of times, if one measures the utility of a doxastic state by a strictly proper scoring rule, then by the agent's own lights Bayesian update is an algorithm that maximizes epistemic utility. Similarly, it seems reasonable to think that the epistemic rationality involved in choosing which evidence to pursue, i.e., which experiments to perform²³, can be accounted for by such a maximization. And finally the norm that one's credences should be (probabilistically) consistent can be justified by score maximization, because it is a theorem that if one's credences are not consistent, there is a set of credences that are guaranteed to yield a better score no matter what the truth turns out to be, given some mild continuity conditions on the strictly proper scoring rule.²³

If Bayesian update, choice of experiments and consistency are all there is to epistemic rationality, then this kind of reasoning initially has some hope of answering the question

²³**PreddEtAl** (**PreddEtAl**) prove this for additive rules. **PettigrewNew** (**PettigrewNew**) announced a result for continuous rules, which was subsequently proved by **NielsenDom** (**NielsenDom**) and **PrussDom** (**PrussDom**), and a precise characterization of exactly how much continuity is needed is now known (**PrussCont**).

of why one should bother being epistemically rational, assuming we take it for granted that the good is worth having.

But this hope may not be borne out. As argued in ??backref, there is good reason to think that prudential rationality does not always involve maximizing utility—risk needs to be taken into account. Once we have seen that expected utility maximization is not the only reasonable position to take with respect to prudential goods, why should we assume that it is so for epistemic goods?

But more seriously, as already seen in ??backref, Bayesian update, consistency and the norms governing choice of experiment do not exhaust the scope of epistemic norms, especially in the context of requiring substantive objective norms constraining rational priors going beyond consistency.

It won't do to say that the rational priors best conduce to epistemic utility and that's we should have them. For rational priors do not best conduce to epistemic utility. At various points in the past it would have been irrational for human beings to have a high prior for such hypotheses as that water is made of hydrogen and oxygen atoms, that 2020 would be a year of a global pandemic, or that the fine-structure constant is approximately $1/137$. And yet epistemic utility would have been increased by adopting such priors, because all these hypotheses are in fact true. The priors that uniquely best conduce to epistemic utility are a prior of 1 for every actual truth and a prior of 0 for every actual falsehood. But such priors would not be rational for human beings.

An Aristotelian has a story here. Although truth is partly constitutive of our cognitive flourishing, so is conformity to the norms of rationality. It is thus non-instrumentally good for us that we reason in accordance with our nature. A human being who thought *a priori* that the fine-structure constant is approximately $1/137$ would be right, and would flourish to that extent, but would also have a crazy prior, which itself is contrary to cognitive flourishing even if that prior matches reality. Typically, we have more control over whether we follow the norms of epistemic rationality than whether we get to the truth, and so from the point of view of pursuit of one's own flourishing it makes sense to follow these

norms in typical situations. We should be rational because being rational is constitutive of flourishing for us.

That said, one can imagine cases where tradeoffs are reasonable. Suppose a trustworthy but eccentric alien offers to inform you of the correct theory of stellar formation in exchange for the very small sacrifice of brainwashing yourself into thinking that you have an even number of hairs on your head. Irrationally coming to a belief about the parity of your hair count is contrary to our cognitive flourishing, but in terms of epistemic utility, knowing the correct theory of stellar formation seems worth it.

There is also another possible answer to the “Why bother?” question. It could be that to the extent that our cognitive life is under our voluntary control, there are moral requirements that we follow epistemic norms. After all, on the account defended here, morality extends throughout the sphere of the voluntary: the moral is defined by the proper functioning of the will. Thus it is possible that one have a deontological view on which even if voluntarily brainwashing yourself to believe something you have insufficient evidence for is beneficial to your epistemic flourishing, nonetheless it is morally wrong to do so. Much of our cognitive life is not under our direct voluntary control. But we may nonetheless have a *prima facie* moral duty to do what we reasonably can to develop our cognition in such a way that even the involuntary parts cohere with the norms, just as we may have a *prima facie* moral duty to promote the health of our bodies. It does, after all, seem to be a part of the proper function of our will to direct us to both physical and cognitive health.

This answer is also compatible with Aristotelianism: we can account for the moral obligation in the Natural Law way discussed in Chapter??backref. But in any case we should not take the moral account of why we should be rational too far. We *can* have two people who act equally well morally, but one of them does cognitively better than the other, due to innate talent, opportunities for cognitive self-improvement, and simple luck at getting to the truth. Epistemic value goes beyond moral excellence, if only because it includes the value of epistemic success, much as health goes beyond moral excellence, even though there are *prima facie* duties to promote our own health.

12.2. What is epistemic rationality? We have an Aristotelian answer as to what epistemic rationality is among humans: it is the proper functioning of our cognitive faculties. That answer nicely generalizes to yield an account of epistemic rationality for a variety of kinds of beings, as long as we can make sense of what cognitive faculties are.

In the case of morality, a similar functional approach defined morality in terms of the proper function of the will, and the will as a system directing behavior in the light of practical good-directed reasons. We might try now for a similar account but this time in terms of reasons directed at epistemic goods, such as correct and informative (say in the sense of yielding understanding or in terms of scoring rules) representation of the world. This approach is more problematic, however, than on the moral side, especially if we are open to a wide range of possible epistemologies for different kinds of rational beings.

First, we could imagine beings that directly contemplate the truth, like Aristotle's gods or Kant's intellectual intuiters. It is difficult to identify reasons in such a case, and yet the contemplation appears to be a rational activity. And while humans do not normally directly intuit reality, it is possible that we do so in special cases, such as Platonic contemplation of the Forms or the Catholic view of human flourishing as bound up with a direct beatific vision of God.

Second, reliabilist epistemology can involve processes that reliably directly generate beliefs in us rather than doing by the intermediary of reasons. And even if reliabilism is not a correct description of human epistemology, it seems imaginable that there be rational beings of whom it is normatively true.

Third, objective Bayesianism has three places for epistemic normativity: updating on evidence, choosing which experiments to perform, and having rational priors (at least ultimate ones). The update and experiment-selection are indeed based on reasons. However, the priors are not typically attained on the basis of reasons, but appear to be simply innately had by an agent (though in cases where one changes one's priors reasons may be involved—cf.). Again, even if objective Bayesianism is not the right normative

theory of humans, it is plausible that it could be the normative theory for a different kind of rational being. Moreover, there are other phenomena that could be governed by rational normativity that are similar to that of priors, such as innate beliefs, or beliefs passed on genetically in imaginable alien species.

Without adverting to reasons, we might simply define an epistemically rational system as one directed at epistemic goods, such as correct and informative representation of reality, and epistemic rationality as the normativity of the proper functioning of such a system. However, if we did that, then epistemic rationality would go very far down in the animal kingdom: even worms have systems directed at correct representation of reality. To rule worms as not rational, we might want to add an element of self-reflection, where the agent is such that they ought to explicitly embrace the epistemic goods as such, and ensure that their epistemic functioning is correctly aimed at this goods. This last condition would bring contemplation, reliable belief formation and prior-generation back into the fold of rationality: we might say that it is precisely when the agent reflectively (either synchronically or asynchronously) monitors these processes in their directedness at epistemic goods that the agent is epistemically rational in engaging in them.

It is worth noting that on this reflective account, the high forms of cognition, such as the Aristotelian gods' direct vision, might well be rational in a somewhat different way from the rationality of lower forms of cognition. For the high forms of cognition are often seen as essentially correct, neither in need of correction nor repair, and hence the self-monitoring could be limited to realizing that necessarily everything is working correctly. This is akin to the way that morality would work differently for a perfect being that does not need to engage in self-correction, but that knows that it always functions morally correctly—such a being, for instance, has no obligations to ever offer apologies for wrongs done.

12.3. Grounds of epistemic norms. Perhaps the primary question of metaethics is whether ethically normative claims have truth value, and if so, what grounds their truth value. The Natural Law answer that I have been defending is that ethically normative claims have truth value grounded in facts about the flourishing of the will, which facts in

turn are grounded in our form (??add this!). Analogically, we would expect epistemologically normative facts to have truth value grounded in facts about the flourishing of the intellect, which facts in turn would be grounded in our form.

Natural Law metaethics as it stands appears logically consistent with a wide variety of normative ethical theories, though some combinations are less appealing philosophically. Thus, while an Aristotelian metaethicist could suppose that the proper function of the will is to maximize utility, normative utilitarianism fits more elegantly with metaethical utilitarianism according to which facts about the right and wrong just are facts about utility maximization.

Similarly, Natural Law metaepistemology appears consistent with a variety of normative epistemological theories. Reflection on ethical cases shows that flourishing can have internal and external components (e.g., one's friends' well-being is partly constitutive of one's well-being), and so Natural Law metaepistemology is compatible with internalist and externalist norms. It is compatible with full-blown Bayesianism, though just as normative utilitarianism fits more neatly with metaethical utilitarianism, so too those versions of Bayesianism on which the norms all have simple formal statements—say, subjective Bayesianism or Carnapian objective Bayesianism—fit more simply with an epistemology on which the norms are grounded in formal constraints. On the other hand, an objective Bayesianism with non-formal constraints on priors fits very well with a Natural Law metaepistemology.

However, over the last century or so there has been a disciplinary difference between ethics and epistemology. While the ethicists have had a primary focus on the question of how one should act, the epistemologists' primary focus has not been the question of how we should think, but what knowledge is. One might take this question of the nature of knowledge to be akin to many ethicists' interest in the nature of virtue. In both cases, there is a sense in which the question does not carry normativity on its sleeve, in that there are possible answers to the question that do not involve any normative components. Thus, Descartes identified knowledge with clear and distinct thoughts, and a naive Aristotelian

(though of course not Aristotle himself) might identify virtue with the mathematical center between the humanly possible extremes. Of course, one might think that clear and distinct thoughts or being centered in one's habits is worth having, but that does not make the state essentially normative—water is also worth having, but is not essentially normative. However, likewise, in both cases more plausible theories of the item do have a significant normative component. Thus, the real Aristotle would identify virtue with a *reasonable* mean, while many have tried for accounts of knowledge where one of the components is *justification*.

Natural Law metaepistemology more easily yields an account of the grounding of more obviously normative concepts such as justification than of knowledge. For instance, a justified inference could be an inference performing which constitutes flourishing of the intellect (or the intellect *qua* engine of inference). Again, this is compatible with many theories as to what kinds of inferences are in fact justified. But it is more difficult to see exactly how to ground knowledge. Nonetheless, the Natural Law metaepistemologist can simply say that facts about knowledge are grounded in facts about intellectual flourishing with specifying how the grounding goes. Or they might punt and say that the concept of knowledge is not essentially normative, and hence explaining how knowledge is grounded is beyond the scope of the normative parts of metaepistemology—and perhaps it is only to the normative parts that the Natural Lawyer has a distinctive contribution.

I think there is a plausible story about something in the vicinity of knowledge, however. We can think of two aspects of the intellect, namely process and accomplishment, and both aspects have their distinctive flourishing. Justification will be a matter of process flourishing. But knowledge seems more like accomplishment. Thus, we might try to identify an accomplishment of the intellect as knowledge provided that this output constitutes intellectual flourishing. However, this is not exactly right. For flourishing intellectually in an accomplishment will include not just the aspects of knowledge acknowledged in the analytic tradition, like true belief and justification, but also *understanding*, structural connections between different things found in the intellect that enlightens us about the things

that are important. Aquinas famously distinguishes the vice of *curiositas* from the virtue of *studiositas* (**AquinasST**), and the distinction lies in *curiositas*'s trivial pursuit of mere bits of knowledge while the virtue pursues a holistic structured understanding of the world's explanatory connections. Thus flourishing with respect to intellectual accomplishment is more like what the medievals called *scientia*—a structured intellectual *body* of knowledge—than what contemporary epistemologists simply call knowledge.

So a Natural Law metaepistemology elegantly gives us a theory about the grounding of *scientia* rather than of knowledge. Is this a disadvantage? Or is it a hint that in fact many of the blind alleys of post-Gettier epistemology have been due to the fact that the more natural concept in the vicinity is the intellectual accomplishment of *scientia* rather than mere knowledge? There is room for significant further exploration here, as well as room to search for a purely normative characterization of knowledge that might end up being grounded in human nature. How such a search might go will depend on controversial questions about the role of knowledge in our rational life.

While my own view suggests we should minimize our focus on the concept of knowledge in favor of understanding, justification, and credence, an Aristotelian attracted to knowledge having a significant role might have room to give a normative-functional account of knowledge that has a significant role for human nature. Thus, if one thinks that the knowledge is the norm of assertion, then one might say that knowledge is that state which makes assertion non-defective, and the standards of defectiveness of assertion are given by our nature. Or one might try to characterize knowledge as whatever it is that needs to be added to beliefs about explanatory relations in order to yield understanding, with understanding now having a normative characterization as the object of a human's intellect. Given an account of knowledge that goes back to the flourishing of the intellect in this kind of an Aristotelian way, we would have an explanation for why there are so many epicycles in post-Gettier accounts of knowledge. For what we have seen of other areas of the normative life suggests that we should not expect a great deal of simplicity in the norms governing us.

That said, it is not the task of metaepistemology to provide a detailed account of knowledge—that is the task of epistemology—but simply to provide an account of the types of facts that ground knowledge. We might roughly break up these facts into three categories: (i) mental facts constituting one’s belief and the inferences that led to it; (ii) the facts (in typical cases extra-mental) making the belief true and entering into the warrant or justification; and, finally, (iii) facts because of which the first two sets of facts count justifying the belief and yielding what additional condition is needed to get out of the Gettier condition. It is only the third class of facts that seems to fall under metaepistemology, and the Natural Law (meta)epistemologist can say that facts from that category are normative facts about the flourishing of the intellect.

Appendix: *Our pathological scoring rule

Recall our pathological scoring rule given by:

$$T_0(x) = \begin{cases} 1000 & \text{if } x \geq 0.999 \\ -1000000 & \text{if } x \leq 0.001 \\ 0 & \text{otherwise.} \end{cases}$$

and

$$F_0(1 - x).$$

In ??backref, I claimed that this rule was proper and that it can be written as a limit of symmetric, strictly proper, finite and continuous scoring rules. It is easy to see that if it can be written as such a limit, then it *is* proper, so we only need to show it can be written as such a limit.

First note that any symmetric continuous proper scoring rule (t, f) can be written as the limit of symmetric continuous strictly proper scoring rules by letting $t_n(x) = t(x) - (1 - x)^2/n$ and $f_n(x) = f(x) - x^2/n$, since the Brier scoring rule defined by the functions $-(1 - x)^2$ and $-x^2$ is strictly proper, and the sum of a proper and a strictly proper scoring rules is strictly proper.

Thus, all we need to show is that we can approximate (T_0, F_0) with symmetric, finite, continuous and proper scoring rules. Furthermore, we can drop the symmetry requirement. For write $f^*(x) = f(1 - x)$. Then (t, f) is a proper scoring rule if and only if (f^*, t^*) is a proper scoring rule. Now if $T_0(x) = \lim_{n \rightarrow \infty} t_n(x)$ for all x and $F_0(x) = \lim_{n \rightarrow \infty} f_n(x)$, then

$$\frac{T_0(x) + F_0^*(x)}{2} = \lim_{n \rightarrow \infty} \frac{t_n(x) + f_n^*(x)}{2}$$

and

$$\frac{F_0(x) + T_0^*(x)}{2} = \lim_{n \rightarrow \infty} \frac{f_n(x) + t_n^*(x)}{2}.$$

But $T_0 = F_0^*$ and $F_0 = T_0^*$, so the left-hand sides are just $T_0(x)$ and $F_0(x)$, respectively. Moreover, $((t_n + f_n^*)/2, (f_n + t_n^*)/2)$ will be a continuous symmetric finite proper scoring rule if (t_n, f_n) is a continuous finite proper scoring rule.

Fix $\varepsilon > 0$. Let ϕ_ε be a continuous non-negative finite function that is zero except on the set $U_\varepsilon = (0.001, 0.001 + \varepsilon] \cup [0.999 - \varepsilon, 0.999)$, and is such that $\int_{0.999-\varepsilon}^{0.999} (1 - u)\phi_\varepsilon(u) du = 1000$ and $\int_{0.001}^{0.001+\varepsilon} u\phi_\varepsilon(u) du = 1000000$. Define

$$T_n(x) = \int_{1/2}^x (1 - u)\phi_{1/n}(u) du$$

and

$$F_n(x) = \int_{1/2}^x u\phi_{1/n}(u) du.$$

By ??SchervishThm4.2, (T_n, F_n) is proper. It is clearly continuous and finite. It is, further, easy to calculate that $T_n(x)$ and $F_n(x)$ equal $T(x)$ and $F(x)$ except perhaps on $U_{1/n}$.??check For any x in $[0, 1]$, there is an N such that x is not in $U_{1/n}$ for any $n \geq N$. It follows that $T_0(x) = T_n(x)$ and $F_0(x) = F_n(x)$ if $n \geq N$, and we have the limiting condition we wanted.

CHAPTER VI

Mind

1. Multiple realizability

Some conscious beings have brains. Let us begin by considering the unpopular hypothesis that it is a necessary truth that all conscious beings have brains.

First, this hypothesis is intuitively implausible: it seems quite plausible that we could have conscious beings with a very different body plans.

Second, observe that brains are a specific type of organ in DNA-based animals. To have a brain, thus, you need to have DNA. To have DNA, you need to have hydrogen atoms. To have hydrogen atoms, you need to have electrons. A particle with a different electric charge would not be an electron, and the charge of the electron is definable in terms of the fine structure constant $e^2 / (2\epsilon_0 hc)$. If the fine structure constant were different, we wouldn't have electrons. We might have shmelectrons that behave almost exactly like electrons, but they wouldn't be electrons. If we didn't have electrons, we wouldn't have hydrogen, but at best shmydrogen. And if we didn't have hydrogen, we wouldn't have DNA, but at best shmDNA.

But now imagine a world extremely so similar to ours that no instruments of a sort humans ever have a hope of constructing could ever tell the difference, but where, nonetheless, the fine structure constant has a slightly different value. In that world we have beings that behave, as far as any of us could ever tell by external and internal examination, just as we do. But they not only would *be* unconscious zombies, they would *have to be* zombies—no beings with shmelectrons in place of electrons could be conscious on the hypothesis we are considering. That such a slight difference in physical constitution would make the difference is extremely implausible.

Similarly, imagine a universe like ours where antimatter predominates over matter, and we get evolution of anti-Carbon-based animals behaviorally very much like humans. Surely such animals *could* be conscious.

Third, if the brain-necessity hypothesis is true, we should be quite surprised at the existence of consciousness. The argument just given shows that consciousness requires the precise value of the fine structure constant that we have. How likely is that? Well, there are infinitely many possible values that agree with our world's fine structure constant to within a thousand significant figures. Unless our fine structure constant turns out to be some very special distinguished value¹, the chances of randomly getting the exact number we have is zero or at best infinitesimal. Given the fact that consciousness has very significant value, if consciousness depends on brains, and hence on electrons, then the fact of consciousness would loudly cry out for explanation.

The line of thought above is akin to fine-tuning arguments, where narrow ranges of fundamental constants are claimed to be needed for life, and call out for explanation, with two options being typically offered: a multiverse (unlikely things will happen if dice are rolled enough times) and an intelligent designer. But there are some relevant differences in our present case.

First, our range is much narrower—only one exact value is compatible with consciousness on the hypothesis we are exploring—which means that objections from the rescaling of ranges do not apply as has been claimed in the case of the fine-tuning argument (MMV2001).²

Second, plausibly an intelligent designer would be conscious, and if consciousness requires brains as we are hypothesizing, a designer will be of no help here, on pain of circularity.

¹For a while, some physicists speculated it was exactly $1/137$ (Kragh2003), but later measurements disproved that, and a recent estimate is $1/137.035999206$ (Morel2020).

²And in any case, see responses in HawthorneIsaacs2017 (HawthorneIsaacs2017) and PrussNorm (PrussNorm).

Third, because the consciousness-permitting range has only one point on it, and there are uncountably infinitely many possible other values of the fine structure constant, hitting this value will not automatically be probable even given a multiverse. If you spin a continuous fair spinner once, your chance of hitting a particular value is zero or infinitesimal. But the same is true for any finite number of independent spins. Moreover, in classical probability theory, this is also true for a countably infinite number of spins. And for an uncountably infinite number of spins, the probability is simply undefined. In light of this, the multiverse hypothesis only really solves the problem of consciousness in our context if it is a Lewisian (**LewisPlurality1986**) or Tegmarkian (**Tegmark**) hypothesis that *every* possible cosmic arrangement is realized in reality. But such a hypothesis only solves the problem at the expense of introducing serious sceptical problems, since there will be cosmoses, just as real as ours, where every coherent sceptical hypothesis hold, and it does not appear reasonable to think that we got so lucky as to escape them all (cf. **PrussActualPossible**, **PrussActualPossible**).

Tying consciousness to brains thus links consciousness to the precise laws of nature we have. That is not only intuitively implausible but leads to serious problems. We should think that there is some flexibility in what kinds of bodies conscious beings can have.

Perhaps instead of supposing that consciousness is tied to brains, we could suppose that consciousness is tied to a range of brain-like organs. Thus, consciousness would be compatible with having somewhat different laws of nature, resulting in fundamental particles slightly different from the ones we have, and behavior somewhat different from the one we have, but not *very* different. But now consider the Mersenne questions about the boundaries of physical constitution compatible with consciousness. These questions cannot be settled by invoking human nature, since they are questions that transcend the nature of any one species. Nor can they be settled the way Mersenne settled his original questions, by invoking God's creative decision, because we are supposing that the connection between consciousness and brain-like organs is necessary. We should avoid theories raising Mersenne questions that do not have a plausible answer.

Furthermore, the issue of worlds practically indistinguishable to our instruments but where one contains consciousness and the other does not returns on the range view. Suppose that the upper cutoff for the fine structure constant to be compatible with consciousness is $1/100$ (recall that our world's fine structure constant is about $1/137$). Then there will be two practically indistinguishable worlds, a consciousness world w_c with fine-structure constant $(1/100) - \epsilon$ where ϵ is one divided by a googolplex, and a zombie world w_z with fine-structure constant $(1/100) + \epsilon$. But it is implausible that consciousness should depend on physical features that are so insignificant.

Finally, consider the thought experiment that we encounter an alien species that is internally biologically very different from us. These aliens change color under internal control. Careful observation finds a plausible and not immensely complex correspondence between sequences of color changes and English, and when we use this correspondence to build a putative "translation" device, we find we are able to engage in what appear to us to be sensible conversations on a broad variety of topics, including science, ethics and metaphysics. Equipped with this device, the aliens no longer sound so very alien at all. Apart from differences easily attributable to body plan and sensory modalities, the aliens sound pretty much like a human from an unfamiliar human culture. They seem to talk of beliefs, hopes and desires that make sense to us and behaves in ways that fits by our human standards with these alleged beliefs, hopes and desires. There are sequences of color changes that our device "translates" as "That hurts", and these sequences occur in the kind of contexts that make a lot of sense to us—paradigmatically, in cases of bodily damage. I take it that we would think that the aliens are likely indeed typically in pain when the device outputs "That hurts", and that the device is genuinely *translating* with a significant degree of accuracy.

And at least, I think, we should be confident in the following: Aliens with a very different internal body plan from us *could* have mental states of the sort we do—states like beliefs, hopes, desires and pains. This line of thought pushes one to a very liberal view about what kinds of physical constitutions are compatible with consciousness. It does not

appear, in particular, that consciousness should depend on having a physical constitution that includes brains or anything similar to brains. We thus have very significant multiple realizability of minds.

2. Functionalism

2.1. Introduction. Full-blown dualism, of course, yields significant multiple realizability. Indeed, a minded being's body could be an oak tree or even a rock, as long as it had the right kind of non-physical mind. We will discuss the interaction of dualism with Aristotelian forms in Section 7. In the meanwhile, however, let us continue to consider broadly naturalistic accounts of mind.

We have seen that there is good reason to be very liberal about the physical realization a minded thing can have. But if we are to remain in a broadly naturalistic theory, we need to put some limits on the kinds of physical constitutions that minds can be based on. We saw earlier that limits based on particular natural kinds—DNA, brains, electrons, etc.—are highly implausible. The most plausible remaining option appears to be functionalism: to have a mind is to have a certain kind of functional structure, so that, necessarily, if there is a functional isomorphism between two entities with their respective functionally-specifiable causal histories, if one of these entities has a mind, so does the other (**Putnam1975, Putnam1975**; cf. **Lewis1972, Lewis1972**). Moreover, the isomorphism between causal histories implies a significant degree of identity between the mental histories, but perhaps not complete identity.

On what we may call strong functionalism, the two entities' purely internal mental histories will be the same, and in particular they will have qualitatively the same states in their histories—whenever one felt hot, so did the other, and whenever one had a perception as of crimson, so did the other. The restriction to purely internal mental histories allows for some externalism. Thus, an individual on Earth may be thinking about water, while the analogous thought in an isomorphic individual on Twin Earth may be thinking about XYZ, where XYZ fulfills the same causal role on Twin Earth as H₂O does on Earth.

But the subjective feel of the thoughts, and of the causal interactions with the substances, will be the same.

On weak functionalism, the non-qualitative purely internal mental histories will be the same, and whenever one has a conscious state, the other has an analogous conscious state, but the exact qualitative phenomenal character of the conscious states may depend on the precise physical substrate underlying the two conscious states. On weak functionalism, a silicon-based isomorph of a human being will have some sensation in a functional state isomorphic to a human's eating sugar, but that sensation's qualitative character may be different from the taste of sweet.

I will now argue that functionalism, whether weak or strong, has serious problems which can be solved by combining it with an Aristotelian hylomorphism.

2.2. Interpretation. Begin with the well-known observation going back at least to [(I)Appendix]Putnam that simple causal systems, like the electrons buzzing inside a rock, can be re-interpreted as emulating the functioning of our brains, simply because they have such a vast number of states. If this is right, then functionalism appears to lead to the absurd thesis that rocks not only think, but think like humans do. One precisely formulated version of this argument will be given in the Appendix to this chapter.

The obvious way out of this difficulty is to insist that gerrymandered functional systems do not count: only simple causal systems count as implementing the functions. However, complex evolved brains like ours doubtless have many significantly gerrymandered states. One might try to draw a distinction between more and less gerrymandered systems, however. The functional states that need to be attributed to a rock to re-interpret it as thinking our thoughts are surely many orders of magnitude more gerrymandered than human functional states. So the view now becomes that mental states are defined by functional states implemented in a way whose degree of gerrymandering is sufficiently low. But now we have a nasty Mersenne question again: what makes it be the case that the gerrymandering cutoff lies where it does?

In fact, this is not just one Mersenne question. Our best hope for measuring the degree of gerrymandering is to look at the complexity of a linguistic expression of the realizer of the functional state in terms of properties that cut the world at the joints. But some mental states appear to involve significantly more neural complexity than others. The feeling of slightly betrayed for no good reason by someone who committed themselves to friendship with one probably requires an order of magnitude of neural complexity greater than the feeling of pain from sticking one's hand in the fire. The former state is mentally much more complex than the latter, requiring a sophisticated engagement with sociality and norms, and there is good reason to think that there is correlation between the mental complexity of a state and the physical complexity of the state's realizer.

Now, a view on which there is a single cutoff for the degree of physical complexity in the realizer of a mental state would imply an oddly artificial cutoff for the mental complexity of a conscious state: mental states with a sufficiently high degree of mental complexity would require a degree of physical complexity that goes above the cutoff. For a related oddity, imagine a digital computer whose computational causal history is functionally isomorphic to yours, and suppose, as is plausible, that it is possible to have such a computer where the physical realizers of mental states fall below the gerrymandering cutoff. But now imagine that we gradually complicate the gerrymandering in the computer. As we do so, the computer's physical realizers of isomorphs of your mental states will become more and more complex, until eventually some will start to go above the gerrymandering cutoff. But at first this will only happen to the mentally more sophisticated states—like the feeling of betrayal discussed earlier. The result will then be an odd sort of zombie where all the more sophisticated states are no longer conscious while the less sophisticated ones are still conscious, and yet all affect the behavior, including verbal behavior, just as they do in you.

A single cutoff is, thus, implausible. What we need is a cutoff function c from mental complexity to physical complexity such that for a functional state to count as a realizer of a

state M it must be defined with a physical complexity below $c(M)$. This function is likely to have multiple parameters, each of which involves a difficult Mersenne question.

2.3. Reliability. Whether our universe is deterministic or not, functional systems are imperfectly reliable. How reliable do they need to be to count as the functional systems they are? Consider a subsystem that given two inputs representing numbers puts out their sum 99.9% of the time, and 0.1% of the time puts out the product. Obviously, it is more reasonable to interpret it as a reliable addition system than an extremely unreliable multiplication system. But suppose the system puts out sums 50.1% of the time and products 49.9% of the time. What then?

There are two natural cutoffs. We could require that a system is defined by how it behaves 100% of the time or by how it behaves more than half of the time. Requiring perfect functioning would have the empirically false consequence that humans don't think. A 50% cutoff, however, may be problematically low. If every subsystem of a complex functional system had a 49% failure rate, then the typical outputs of the system would be largely random, because any output of the system is the result of a causal chain of many subsystem states, and any such chain would likely contain multiple failures. Moreover, while 50% reliability seems to be a natural and well-defined cutoff, the actual reliability of a system cannot be captured by a single number, if only because reliability depends on environmental conditions. If we say that a system is an adder provided that the output is the sum of the inputs more than half of the time, we have to specify the temperature, background radiation, and other conditions under which that reliability is to be defined. And now we gain a Mersenne question, having lost the neat elegance of specifying the reliability with the single number $1/2$.

We might insist on reliability in the actual environmental conditions the system is in. But suppose that Alice finds herself in extremely harmful conditions—say, great heat or toxic fumes—but by a fluke survives with what is intuitively full brain function for a few seconds longer than we would expect, screaming seemingly in pain. Given that under those extreme conditions the functioning is extremely unreliable, we would have to say

that Alice is in fact a mindless zombie and feels no pain. This is implausible. Thus we need to define reliability in terms of hypothetical *normal* conditions. It is not clear that this can be done in the context of a modern naturalism that does not include teleology.

In any case, the very idea of a sharp cutoff in reliability for mindedness seems counter-intuitive. Imagine two humanoids whose brains are nearly identical, with the exception that one brain structure crucial for consciousness in one of the humanoids has 50.000001% reliability and the other has merely 49.999999% reliability. Suppose that both brains *in fact* function exactly the same way, so that the sequences of internal states are exactly the same—it's just that one is slightly more likely to fail than the other. It does not seem very plausible to think that one would be conscious and the other not.

2.4. Pain and damage. Plausibly, a functionalist account of (nociceptive) pain will include two essential aspects of its causal role: the pain is caused by bodily damage and the pain causes a tendency to aversive behavior.³ But imagine that the brain pathway leading to the tendency towards aversive behavior is broken. **Lewis1983 (Lewis1983)** thought that in such cases the subject would still feel pain. The phenomenon of pain asymbolia where pain is felt without aversive behavior—and apparently even without unpleasantness!—provides some evidence for this hypothesis (**Grahek**).⁴ Similarly, if a patient's nerves are severed short of the tissue whose damage they are supposed to signal, surely the patient can feel pain originating in the nerves themselves ("neuropathic pain") rather than in damaged tissue—this what phantom limb pain is thought to be (**Subedi**) Presumably, if *all* of an unfortunate patient's nerves were so severed, the patient would still be able to feel pain given direct nerve stimulation, even though the patient's pain would no longer actively play a part in a damage-detection system.

³This may be necessary, but is unlikely to be sufficient for the causal role of pain, or else a computer that monitors the state of a storage drive and moves data from damaged to undamaged portions feels pain when it detects damage and acts on the detection by averting putting data in the damaged location.

⁴But see **KleinAsymbolia (KleinAsymbolia)** for a different interpretation of the data.

To account for this, it seems we need to define the functions of functionalist theories in terms of normal or typical behavior. David Lewis's own proposal is population-based: the function of a state is defined by what the state does in the relevant population. This faces the notorious reference class problem: should we take the relevant population to be the biological species or, for instance, the more narrow subgroup of people who have the particular condition that disrupts the causal network? If we take the relevant population to be the species, then we can make the empirically supported judgments about pain asymbolia and the intuitive judgment about total severing of nerves, namely that there is pain present in these cases. If we took the narrower population as the reference class, we would have to say that there is no pain in these cases.

So suppose we take the species to be the right reference class. But now imagine that a functionalist tyrant has all of her own pain-signaling nerves severed, but does not wish to feel any pain originating from these nerves, and hence kills every organism that does not share her unfortunate condition. Then there will be no reference class according to which those of her states that are homologous with pain states in us have the causal role of pain, and so on Lewisian functionalism, the genocidal tyrant's plan has succeeded—by killing others, she has killed her pain. But it is obviously false that one can end one's pain by killing others.

One might object that the relevant reference class should be counted diachronically, and so the many generations of past humans outnumber the survivors of the tyrant's genocidal plan, and hence ensure that the state still has the functional role of pain. But on such a diachronic interpretation, we need only change the story. Move the story into the realm of science fiction, and suppose that the tyrant institutes a massive cloning program which within a few years produces more human beings—living doubtless in appalling conditions—who outnumber all the humans who lived previously, and the tyrant ensures that each such human's nerves are severed just as her own are. Now on a diachronic species-based account, the function of the tyrant's states has changed, and hence the tyrant ceases to be capable of feeling pain. This, too, seems patently false.

An alternative to population-based statistical approaches is to focus on natural selection. The function of a trait is the causal role that it evolved for—the causal role such that its historical fulfillment led to the present existence of organisms with that trait. This view has been defended with variants by a number of authors starting with Wright and Millikan (Wright1973; Millikan1984; Sterelny1990; Neander1991; Agar1993; Papineau1993; Dretske1995). For instance, sheep with eyes currently exist because their ancestors were enabled to survive and reproduce by means of visual acuity. There are many details to fill out to make this account precise, but they won't matter for the first critique that I will offer, based on KoonsPruss (KoonsPruss).

Imagine a world w_1 where super-powerful aliens secretly snatch each organism from earth just before it dies and transport it in some form of stasis to another galaxy, while replacing it with a fake carcasse on earth. In the other galaxy they then cure whatever was ailing the organism and ensure that the organism succeeds in reproducing many times over. All the descendants of that organism are in the care of the aliens, who ensure that no genetic line goes extinct—that every organism in the care of the aliens succeeds in reproducing with its offspring also being taken care of. Moreover, in w_1 life on earth is an exact duplicate of how it is in the actual world, w_0 —nobody ever catches on to idea that the organisms we have get snatched before death.

Without extinction, and with every organism successfully reproducing, there is no evolution—just exponential population growth. It is essential to an evolutionary explanation of a phenotype that the phenotype contributes to the the organism's probability of passing on its genotype. In w_1 , there are no evolutionary explanations of any earth organisms, and hence on a functionalism where function is defined evolutionarily, no organisms have minds. This already seems somewhat counterintuitive given physicalism: the physical causal history of every actual organism on earth is the same in w_1 as in w_0 , and yet there are no minds among the earth organisms of w_1 . To go back to our sheep example, in w_1 it would be correct to say that there would be sheep with eyes now even if the eyes never enabled any sheep to survive and reproduce, since survival and reproduction in our

alien snatcher story are guaranteed regardless of whether a sheep has eyes and what it does with them.

To make the counterintuitive aspect of this story even more problematic, now imagine a world w_2 which is just like w_1 , except that once the organisms arrive in the other galaxy, they are immediately slaughtered. In w_2 , we do have evolution just as in w_0 , the only difference being the exact location of death. So in w_2 , on evolutionary functionalism, there are minded beings on earth. But it is highly counterintuitive that whether the aliens choose to slaughter the abducted organisms once they take them to another galaxy affects whether earth primates are conscious. Consciousness should be a local property: events in another galaxy that do not have a causal impact on our ecosystem should not affect whether our ecosystem has consciousness.

One might object that in w_1 there is still natural selection, but one centered around location rather than existence. While the reason why giraffes exist in w_1 has nothing to do with the selective benefits of their necks in reaching food—without the necks, they might come close to dying of hunger but would still reproduce in the second galaxy—the benefits of their neck explain why there are giraffes *on earth*. But locational selection is highly implausible as an account of the function underlying minds. Imagine a world where there is a galaxy full of planets with randomly generated ecosystems, all of them in their first generation of macrofauna, some of which are phenotypically just like us. Suppose next that aliens pick out those macrofauna whose brains function in a way congenial to the aliens' plan, these macrofauna including organisms physiologically just like ours. And suppose further that the aliens' plan is to select those brains function in the way required for mindedness. The aliens then take all these macrofauna and transport them to a new planet. On a locational selection version of evolutionary functionalism, it seems that as soon as the macrofauna are all in a new location because of the mental-like functioning of their brains, they become conscious. But it is absurd to suppose that organisms can become conscious simply due to a change of location.

Granted, on some evolutionary accounts of selection, selected function is only exhibited in the generations subsequent to the selection.⁵ If so, then only the immediate children of the transported macrofauna have minds. But that is only a little less absurd: it is surely not the case that simply a locational change in the parents makes the children conscious. Moreover, the aliens need not even move the selected-for organisms. They could just move the unselected-for organisms. And surely you can't be conscious because your neighbors were moved, absent some relevant causal impact on you.

Besides, it seems *ad hoc* to allow locational selection in addition to the much more natural notion of Darwinian existential selection. Once we allow locational selection, we should probably allow selection in terms of properties other than location. For instance, the aliens could paint a red circle on the head of those organisms they like, and set up machines that put red circles on the descendants of any organisms with red circles. We would then have red-circle selection, and if we allowed that to count for defining function, then we get the absurdity that simply by painting red circles on some animals one can make them, or their descendants, conscious.

Second, while it sounds straightforward and reasonable to say that sheep with eyes currently exist because their ancestors were able to survive and reproduce because these eyes allowed them to see, there are serious technical difficulties here. Evolutionary history, especially in a world with indeterministic quantum mechanics, is stochastic in nature. Had the ancestors of the sheep been unable to see with their eyes, they would still have had a tiny chance of surviving and passing on the eye genes, and of course seeing by itself was insufficient to guarantee survival and reproduction. The most plausible thing to say here is that seeing significantly increased the chance of survival and reproduction.⁶ The "significantly" is important, or else having function becomes too easy. In an ecosystem as

⁵For instance [(I, p). 28]Millikan1984 requires that the benefits of the behavior *have been* responsible for the proliferation of the reproductive group.

⁶A non-probabilistic option is to say that seeing causally contributed to survival and reproduction. However, if this just means that there is some causal connection between an instance of seeing and the present existence of sheep with eyes, then for all we know there are all sorts of implausible functions. Suppose, for

complex as ours, pretty much any two things has some kind of slight non-zero positive or negative correlation, and tiny positive correlations between behaviors and passing on genes will be found all over the place.

But if the function of the sheep eye requires a *significant* increase in the chance of sheep presently existing with eyes due to ancestors seeing, then we can consider the boundary between the increases that are insufficient to generate a function and those that are sufficient to generate a function, and now we have a Mersenne question of why this boundary lies exactly where it does. And because we are considering theories on which consciousness depends on function, we cannot simply say “Let’s not worry, because the question of where the boundary lies is a merely verbal question.” For the question of whether something is conscious or not is *not* a merely verbal question, given its moral significance.

We might have some hope that the boundary between insignificant and significant increases is some neat number like 2: we need ancestral seeing to do better than double⁷ the chance of sheep with eyes existing right now for seeing to be the function of eyes. There does feel to be something non-arbitrary and reasonable about doubling. But even if we do this, we still have other worries about boundaries. We are talking of the chance of the current existence of sheep with eyes. But if we are talking about precise numerical boundaries between chances, then the propositions for which we calculate the chances must be precise as well. But what exactly is a sheep and what exactly is a sheep eye (as opposed to a damaged mess of cells in the ocular orbit)? We can easily imagine boundary cases of the proposition that currently there are sheep with eyes (or maybe that currently most sheep

instance, that some individual that is the ancestor of all current sheep had its life saved prior to sexual maturity by an instance of visual misperception: it thought it saw some yummy bush on top of a hill, and went for it, and thereby avoided a predator, but in fact there was just a mossy stone on the hill. We do not know that this happened, but we also do not know that it didn’t happen, and so on a causal contribution view, we would have to implausibly say that for all we know perceiving mossy stones as bushes really is a function of the sheep’s visual system.

⁷If we merely required doubling, then if the starting chance were zero, we wouldn’t require *anything*, since $2 \cdot 0 = 0$.

have eyes), and different boundary cases will yield different chances. An Aristotelian can answer this: a sheep is something with a sheep form. But we are considering ways of *avoiding* Aristotelianism.

2.5. Kripkenstein problems. For a functional subsystem to be an adder it is not enough that it always outputs the sum of its inputs. After all, if the only inputs ever given are pairs of zeroes, then the subsystem could just as well be a multiplier. As is well-known from Wittgenstein as interpreted by *cit* *asnoun* Kripkenstein—sometimes referred to as “Kripkenstein”—no finite amount of input/output data is sufficient to determine the function of the system as an adder, since any finite input/output sequence is consistent with infinitely many possible functions—admittedly, perhaps messily gerrymandered ones.

We might try to define the function of a system or subsystem in terms of counterfactuals. Perhaps an adder is something that *would* output the sum for all inputs in the range of allowable inputs. However, FrankfurtFW’s (FrankfurtFW) counterexamples to the Principle of Alternate Possibilities can be adapted to show that this is untenable. Imagine that Bob counts as thinking that $5 + 7 = 12$ in virtue of the fact that his thought involves the operation of an adding subsystem in his brain. But suppose that a neuroscientist has placed a neural scanner and bomb in Bob’s vicinity in such a way that if the scanner detects the adding subsystem getting any input other than 5 and 7, the bomb blows up Bob. Now counterfactuals like “If the inputs were 4 and 3, the output would be 7” are false. If the inputs were 4 and 7, there would be no output, just an explosion.

One might try to define functions by asking what the output would be given the inputs *absent external interference*. But what counts as interference with a system, as opposed to, say, a helpful or neutral effect, depends precisely on the system’s function. If the purpose of a wood-pulp product is to preserve inscribed information, then fire is an external interference; but if the wood-pulp product’s function is to be kindling, then fire activates its the object’s function. Furthermore, one can internalize Frankfurt-like cases. Imagine

that through science-fictional genetic modification Bob's liver comes to behave just like the scanner-and-bomb system.

2.6. Neo-Aristotelian functionalism.

2.6.1. *A solution.* Not having the resources to define function would be a fundamental flaw in functionalism. However, defining functions is exactly one of the problems that Aristotelian forms are designed to solve. We can say that F is the proper function of a subsystem in an organism provided that fulfillment of F constitutes the organism's flourishing with respect to that subsystem.

Recall the four difficulties we discussed above. First, we had Putnam's reinterpretation problem, that a rock could have states that are isomorphic to our own. But a rock does not have the human form, and hence even if we have a causal correspondence between our states and gerrymandered descriptions of the rock's states, the gerrymandered descriptions of the rock's states do not enter into the functions defined by the form of the rock or its constituent particles. Second, we had worries about the degree of reliability needed for function. But the Aristotelian metaphysics of function does not require the system to be reliable, though Aristotelian optimism predicts that most systems will in fact be reliable. Third, robust realism about form side-steps the reference class problem for defining function, and gives the right intuitive judgment for pain in damaged organisms. A state can still have the same functional role as a pain state even if due to damage it does not actually have the same place in the causal nexus, as long as it still constitutes an exemplification of a subsystem's functions, i.e., is an instance of partial flourishing of the organism with respect to the subsystem. And, finally, the functions defined by the form privilege certain descriptions in a way that allows one to solve Kripkenstein problems.

A robust view of organisms as having forms that specify normative, and hence functional, features thus solves central problems with functionalism. And as we saw, there is very good reason for a naturalist to adopt functionalism—it is our best known naturalist option for saving multiple realizability. Thus, we have an argument from naturalism to neo-Aristotelianism ("neo", because the Aristotelianism of Aristotle's *De Anima* has a view

of mind that is not merely functional). Of course, whether the robust realism about forms is fully compatible with naturalism depends on how one defines naturalism. If naturalism is understood to say that there are no fundamental normative properties, then of course there is no compatibility.⁸ But on a broader naturalism we can adopt neo-Aristotelian normative realism, and thereby solve the most fundamental problem facing functionalism in the philosophy of mind.

2.6.2. *The Chinese Room*. It is worth noting that the same solution helps with a less fundamental but more intuitive difficulty: Searle's famous Chinese Room thought experiment.⁸ The Chinese Room thought experiment imagines running the software of a brain of a native Chinese speaker by having a clerk manipulate symbols by following a set of rules, where following these rules ends up performing computations isomorphic to those in the brain of a native Chinese speaker. It is a widely-held intuition that this does not yield understanding of the Chinese language. If you pass the clerk a Chinese inscription saying that his friend died, he may respond with a note expressing grief, but will not actually feel any. If computational isomorphism implies mental isomorphism, the clerk, or at least the clerk-notebook-pencil system, should feel what a native Chinese speaker would feel, and yet it is highly counterintuitive to think either feels grief.

But our neo-Aristotelian functionalism escapes the problem, because the clerk's computations do not have the same teleological features as the corresponding brain events in a native Chinese speaker. Even if the computations are exemplifications of the proper function of deliberate rule-following in the clerk—rule-following may indeed be one of the proper functions of a human being—that is a different proper function from the ones that are involved in actual brain function. Thus neo-Aristotelian functionalism about mind intuitively correctly predicts that mere rule following does not constitute understanding.

⁸Similar points apply to descendants like the Schwitzgebel's (Schwitzgebel) conscious country argument.

2.6.3. *A spectrum.* There is a spectrum of functionalist theories, depending on the level at which we have the functions. Think here of the analogy of a computer. We could suppose that proper function is specified at a fairly low level: say, the level of components like transistors, specified by engineers to function in such-and-such ranges of temperatures, voltages, and currents, to saturates at such-and-such a level, and to have such-and-such current gain and transition frequency, etc., and by having the correct data bits in the memory units. On this picture, for mental function it doesn't matter how the components are implemented, say at the level of doped semiconductor materials, as long as they satisfy the specifications. Or, at higher level, we could specify the processor as a whole, rather than its transistors, defining the syntax and semantics of the machine code that it runs (with or without some abstraction, say with respect to edge cases), and then specifying what machine code and data are actually loaded. Or, at a yet higher level, we could specify the algorithms, say in a high level programming language or even pseudo-code, for instance giving the precise sorting algorithm that is being used for operationalizing preference rankings in an autonomous robot.⁹ Or, even further, we could abstract from details of the algorithms, and specify what the computer accomplishes at a very high level, e.g., simulate n -body Newtonian gravitational interaction, or translate spoken Cantonese into written ecclesiastical Latin.

We can imagine a similar hierarchy in the case of minds implemented wholly or partly by brains, though we know much less about the details of how high level function is implemented by our neurons, so it is easier to illustrate the point with computers. There is thus an important degree of freedom in functionalist theories of mind with regard to the level at which the mind-constituting functions are specifying. Some levels are more plausible than others. The lower the level of specification, the less multiple realizability there will be. In fact, in the computer analogy, it seems that even the fairly high level at which details of algorithm choice are specified is too low. Surely an elephant and a dog could equally

⁹Compare Putnam's concept of "computational plasticity", where a mental state is realized by different kinds of computational states (Putnam1988).

exhibit preference-driven agency, with some efficiency differences, if one employs a merge sort and the other a bubble sort in generating its preference ordering. Furthermore, the lower the level of specification, the more Mersenne problems we raise, because the more degrees of freedom we have when we require the specification of detail.

There is thus reason to prefer very high levels of specification. And Neo-Aristotelianism can rise to that challenge, because forms can equally well specify proper function at whatever level of abstraction we deem appropriate.

3. Integrated Information Theory

An up-and-coming alternative to functionalism as a theory of consciousness that has caught the attention of some neuroscientists is Integrated Information Theory (IIT) by **Tononi (Tononi)**. IIT is based on the core insight that some complex causal systems can be physically separated into subsystems that interact only weakly or not at all with one another, in the sense that each subsystem only makes a small contribution—measured in information-theoretic ways—to predicting the functioning of the other subsystem. Such systems have little or no consciousness. For instance, a digital camera sensor contains a light sensor for each pixel. If we partition these pixel sensors into those on the left and the right halves of the full sensor, the data on each side makes relatively little contribution to the data on the other side (it makes *some* contribution, in that sensors heat up, and their heat generates noise in the data of neighboring sensors). As a result the system has little consciousness.

But when a complex causal system has no such decomposition, it has “integrated information”, which typically constitutes consciousness. Given a mathematical model of the probabilistic functioning of the system, there is a specific measure, Φ , of the degree of integrated information in the system. There is also a specific proviso on which if two systems are nested—one is a part of the other—then only the one with the higher Φ value gets to be conscious.

IIT has multiple realizability, but significantly less of it than functionalism does. Those isomorphisms between systems S_1 and S_2 that preserve transition probabilities of states and map physically separate components of each system to physically separate components of the other system will preserve Φ . Functionalism, on the other hand, allows for functional isomorphisms that changed transition probabilities (e.g., replacing a component with a more reliable one), as well as ones where multiple physically separate components got mapped to one, or one physical component got implemented by several physically separate ones. For instance, in a computer simulation of a brain system, an electrical potential in a single nerve may be modeled by a 64-bit floating point number. Each of the 64 bits of the floating point number is stored in a separate tiny storage capacitor in a memory chip: thus one brain component is implemented in 64 digital components. The simulating system thus allows for decompositions (say, between the even- and odd-numbered bits) that the original physical system did not allow for.

We could in principle implement a consciousness in an electronic system by mimicking a human brain given IIT. However, we would not only have to simulate the functioning of the brain, but we would have to implement this functioning with physical components that have a one-to-one mapping with the physical components of the brain, e.g., neurons mapped to computer chips, nerves to wires, etc.

Thus, IIT avoids some of the difficulties facing functionalism, by limiting which functional isomorphisms preserve consciousness facts, just as neo-Aristotelianism does, but by a different mechanism, namely by focusing on more details in the physical arrangement of the realizer of the functional system rather than by introducing a non-physical form. For instance, in the Appendix to this chapter I argue any finite deterministic system can be mapped to the states of a single particle moving in a straight line, and hence every particle has human-level consciousness on functionalism, no such conclusion follows for IIT. For while the different states of a brain can all be encoded into, say, the position of one particle—the position can be specified as a number with infinitely many decimal places,

and these suffice for encoding the states—the particle is not a complex system capable of physical subdivision, while the brain is.

Furthermore, the difficulties about specifying the degree of reliability needed for consciousness in a functional system disappears entirely: the Φ measure precisely takes into account the transition probabilities, and how they modify the amount of integrated information.

Unfortunately, IIT is subject to fatal counterexamples.¹⁰ For instance, consider a very large square grid. At each point of the grid is stored a zero or one. At each step of the evolution of the system, the value at each grid point is replaced by the *logical or* of the values of the two to four neighboring points (the points in the middle of the grid have four neighbors; the points at the corners have two; the points at the edges have three). Thus, if a point is surrounded by zeroes, it becomes zero; otherwise, it becomes one. If there are any ones in the system, they spread all around until they fill the grid. If the system starts at zero, it stays at zero. The system is intuitively integrated: one cannot divide it up into two large subsystems whose causal connections can be separately analyzed—there will always be plenty of connections across the subsystems. **TononiGrid** (**TononiGrid**) has thus conceded that even if a large grid-like system of logic gates has all the gates zeroed out, it will have consciousness. While Tononi embraces such conclusions, they seem highly counterintuitive.

For a much simpler case, imagine a world with two eternal particles, each having an electric charge specifiable as any real number. Suppose, further, that the laws of nature specify that every second, the two particles swap their charges. This system has only one decomposition into two subsystems—the two particles. Each particle contains an infinite amount of information for the prediction of the other particle's subsequent charge, since a real number contains infinitely many bits of information (it can be expressed by a sign, a binary integer, a “binary point”, and then infinitely many binary digits). The Φ measure of the system is infinity. Thus, the two particles that eternally interchange their charges are

¹⁰For a nice analysis, see **Aaronson** (**Aaronson**).

infinitely conscious. While a panpsychist might be willing to admit that two particles are collectively conscious, this would surely be a minimal level of consciousness, rather than the god-like infinite consciousness that IIT predicts.

IIT has a pleasing mathematical precision, and a tempered version of multiple realizability. However, it leads to implausible consequences. Perhaps some future IIT-like theory will avoid the extravagancies of functionalist multiple realizability while avoiding the implausible consequences of IIT.

We should also note a particular limitation of IIT. As it stands, IIT is a theory of the amount of consciousness in a system. But what we really want in a theory of consciousness is an account of what constitutes particle phenomenal experiences: what makes an entity feel pain, hear a middle C, see red, or experience a sense of awe at the mystery of existence. We can sketch toy functionalist stories about some of these—say, pain being a state that mediates between damage detection and avoidance behavior—but IIT’s information-theoretic character does not seem to lend itself well to analysis of phenomenal experience. Moreover, “information” in information theory is binary bits, abstracted from their specific meaning. But any reduction of phenomenal experience to information would, plausibly, have to focus on the specific meanings (damage, say). Thus, even if future technical tweaks to IIT might avoid the counterexamples, without a radical expansion to take into account data beyond information theory it is unlikely to solve the problem of phenomenal experience.

Functionalism, thus, still rules among naturalist theories that allow for significant multiple realizability, and its main problems can be alleviated by adding neo-Aristotelian forms.

4. Supervaluationism about minds

A tempting solution to the problem of the multiplicity of functionalist (or other) theories of mind differing with respect to fine details is to treat each theory as a precisification of concepts like *mind* or *pain*, none of which is privileged over the others. And, if all has

gone well, then a typical adult human falls under all the precisifications of “have a mind” and in paradigmatic cases of being in pain they fall under all the precisifications of “are in pain”.

One difficulty with this approach has already been discussed in the ethical context, namely that similar problems arise at the meta level: What *range* of, say, standards of reliability yields a functionalist concept that is in fact a precisification of our concept of mind? In other words, moving to supervaluationism only multiplies the Mersenne questions.

Furthermore, recall the argument that the centrality and overridingness of ethical norms to our lives makes it deeply implausible to think that there is a plurality of closely related concepts, none of which is privileged over the others. But given the deep importance of the mental to ethics, the same concern applies to the mental. That an action causes severe pain to a non-consenting individual with no significant benefit is a conclusive moral reason not to perform the action. But if there are many concepts of pain with none of them privileged, and similarly of consent, then whether one has a conclusive moral reason for an action will depend on the choice of precisification as well. Hence, whether one has a conclusive moral reason becomes a merely verbal question in such cases, and that is not compatible with the force of moral reasons.

The implications of this kind of vagueness about mind are counterintuitive. There is some plausibility in thinking that there can be vagueness about the exact phenomenal character of a mental state. But vagueness about whether there is a phenomenal character at all, whether there is *any* consciousness, is more problematic. For instance, maybe it can be vaguely true that someone is in severe pain, but in such a case it is definitely true that they are at least experiencing a discomfort. This inference, however, is invalid on the precisification approach to solving the problems with functionalism. Imagine a perfectly reliable computational system S_0 which would definitely implement severe pain—a system that, say, emulates what happens in a human brain when one has one’s leg amputated without anaesthesia. But then imagine a continuum of systems that in the actual world

behave just like S_0 , but are less and less reliable. At some point at this continuum we simply have completely random behavior that by chance matches that of S_0 . Somewhere in between it is vague whether the system has the same functions as S_0 . But we can imagine that everywhere along the continuum it is definitely true that *if* the system is conscious of any discomfort at all, it is conscious of a severe pain. None of the systems implement non-severe discomfort—they either implement nothing mental or they implement severe pain. Thus in the vagueness region of the reliability continuum, it is vague whether the system is experiencing severe pain, *and* whether it is experiencing even any discomfort. This kind of state of vagueness about whether there is severe pain and whether there is discomfort appears impossible.

To make the counterintuitive character of this kind of vagueness clearer, imagine that you start as a conscious human who is continually observing the visual situation around you, a field of colorful wildflowers. But then your neural system gradually becomes less and less reliable, though by luck nothing goes wrong—everything actually behaves as if all the reliability were in place. Moreover, the reliability of all your different subsystems changes together, with each subsystem being equally reliable at any specific time. Eventually, according to functionalism, mental states stop, and they all stop together, because the level of reliability is the same for all your subsystems. On the vagueness solution currently under consideration, at any given time in the process it is definite that you are having normal human perceptions of a field of wildflowers or are lacking mental states, but it eventually becomes vague which of the two it is. We are apt to imagine this process as a one involving fading mental states¹¹—the wildflowers get less colorful, your thoughts get sluggish—but that's incorrect. It is definitely true that if you are having any mental function at all, you are seeing the wildflowers with all their vibrance and are thinking as well as ever, and hence it is definitely true that the wildflowers are not getting less colorful and your thoughts are not deteriorating. Yet, allegedly, at a vague point it all transitions from full vividness to nothing. We can, of course, imagine a transition from full vividness

¹¹Compare the fading qualia thought experiment of **ChalmersFading** (**ChalmersFading**).

to nothing with no in-between states—a sudden loss of consciousness, as when you receive a massive head injury. But what seems absurd is the idea of a sudden transition from full vividness to nothing with no in-between states that happens at a vague time. Additionally, the kind of vagueness involved here could generate a situation where it is vague whether something is a person, and hence has the kind of dignity or even infinite value (**BRInfinite**) that persons are often thought to have, or whether it is a mindless thing—with its being definite that it is nothing in between. For suppose it is vague whether a certain degree of statistical reliability is enough for proper function, and then suppose that all of the computational systems of an entity have that degree of reliability, so that it is definite that if that degree of reliability is sufficient for function, the system is a person, and if it is insufficient, the system completely lacks computational function, and hence is mindless. One might hold that it is possible to have a system where it is vague whether it has enough mentality to have the value of a person, but our best candidates for imagining such a system are ones that have significant mental functioning while it is vague whether that amount is sufficient for personhood. But the idea that it is definite that it is either a person or completely mindless, but vague which one, seems highly counterintuitive.

5. Substances

The functions necessary for a human's cognition appear to be housed in each of a multiplicity of entities, including the whole human organism, the human organism less hair, the head, the brain, and perhaps just the upper brain. Standard functionalism seems to imply that all of these entities think the same thoughts, since they engage in isomorphic functioning. But this has some very implausible consequences. If all of the above entities think, then by the same token for any subset S of the set of my hairs, there is an entity A_S that contains all of my organism less hair plus precisely the hairs in S , and on standard functionalism it seems that A_S thinks exactly as I do, for all S . The number of A_S entities is equal to 2^N where N is the number of hairs that I have. If I pull out one of my hairs, half of these entities perish. Since I have hundreds of thousands of hairs, the loss of conscious

beings is vast—more than 2^{100000} beings. Moreover, pain had by a more hirsute person is shared by a much greater number of entities, all other things being equal, so it follows that other things being equal (such as the mass of the non-hair parts), given a choice between relieving the pain of a bald and a non-bald person, absurdly we should help the non-bald person. A similar argument applied to atoms instead of hairs shows that more massive people should typically have vastly greater preference in our moral calculus, and that weight-loss is akin to murder.¹²

We might try to escape these ethical worries by insisting that what counts for moral purposes are not the tokens of thought but the types. This too is counterintuitive. Suppose disease *A* is had by a million people that causes a severe pain, and careful functional analysis shows that every sufferer from the disease feels *exactly* the same pain. On the other hand, disease *B* is had by a two people, and causes in each of them a different qualitatively unique pain but equal in severity to that of the sufferers from *A*. If we count types of thought, then it seems that it is intrinsically twice as good to distribute pain killers to the *B* sufferers than to the *A* sufferers. And that seems quite wrong.

Perhaps, though, what we should count for moral purposes are not types of thought but types of lifetime streams of thought. Even though each *A*-sufferer has the same pain, that pain is embedded in a different type of lifetime of thought. On the other hand, the thoughts of each of the *A_S* entities are embedded in the same type of lifetime of thought. The lifetime stream type approach, however, is not all that plausible. I think most of us will have the intuition that if disease *A* is found among a million mentally duplicate twins and *B* is found in two non-duplicate persons, it's still better to relieve the pain of *A* than of *B*. Furthermore, the lifetime stream approach implausibly implies that how much moral consideration an individual now deserves can depend on what *will* happen to them. The million up-to-now mentally duplicate twins deserve more consideration if in fact their mental streams will diverge in a year.

¹²For essentially the same argument, see Chisholm1979 (Chisholm1979), Zimmerman2003 (Zimmerman2003) and Johnston2016 (Johnston2016).

Alternately, we might modify the functionalism to rule out all but one of the nearly-colocated entities as candidates for thinking. The most natural approach here is to say that what thinks a thought is a minimal or a maximal system exhibiting the relevant type of functioning (IIT goes for the minimal move). Neither account seems tenable. The maximal approach seems particularly absurd: my thoughts aren't being had by some giant system that includes my brain and the Orion Nebula. Probably the only defensible version of the maximal approach is a holism on which it is the universe as a whole that thinks—but then the moral counting problems reappear, because all the thoughts are had by the same entity, and so there does not seem to be a reason to relieve the pain of disease *A* rather than of *B*.

The minimal approach is problematic in a different way. There need not be a unique minimal functional system. Consider a bridge made of planks, whose function is to allow a one ton load to be rolled over the bridge. It might turn out that there is some unique subset *S* of *n* planks such that the planks in *S* would suffice to bear the load, and such that any other subset sufficient to carry the load would have $n + 1$ planks. But this also need not be true. It could well be that the bridge is such that there are two or more different subsets with *n* planks that can carry the load, but no subset with $n - 1$ planks can.

The question of how many minimal functional systems there are in a given human organism is a potentially interesting empirical question, but it is intuitively irrelevant to the question of how much moral consideration should be given to this human's suffering. Furthermore, there seems to be a rather arbitrary decision to be made as to how one defines the minimality of a system. One might try to do so by minimalizing the number of particles, the number of atoms, the number of molecules, or the mass.¹³ All of these could result in different determinations.

The Aristotelian, of course, has a natural solution: there are only as many substances as substantial forms, and substantial forms unify some but not all arrangements of material parts. Something like this can be said, however, by anyone who believes in non-vague

¹³The mass is a little tricky, because we learn from Einstein that energy is a component in mass, and the mass of a part can depend on the energy it has as a part of a larger whole.

restricted composition, i.e., that only some precise arrangements of material parts make up a whole. An example of such a view would be a variant of van Inwagen's theory that only those arrangements of multiple material parts that are caught up in a life make up a whole but without van Inwagen's belief in the vagueness of life. But it is likely that any such theory will include very difficult to answer Mersenne questions, such as what grounds the fact that life requires such-and-such a degree of homeostasis.

6. Normativity and representation

Perhaps the toughest problem for naturalistic theories of mind is the problem of intentionality or representation: what makes a state of an entity be *about* the external world. Attempts at answering this in terms of the causal interaction of the organism with its environment run into difficult problems with the intensionality and even hyperintensionality of representations (**Fodor1987**). We have the same causal interactions with mammals as with earth mammals, and with triangular and trilateral shapes, and yet thoughts about mammals and those about earth mammals are not the same, nor are thoughts about triangular and trilateral shapes.

Furthermore, causal approaches have the infamous problem of specifying which item along the causal chain is the referent of the thought. Let's say that my thought of a horse is triggered by seeing a horse. The thought has many causes. Restricting ourselves to causes that correspond to joints in nature, some causal options include brain visual center activity, nerve signals from retinas, retinal stimulation, light traveling from the horse to the retina, the horse right at the time of reflecting light, the colt, the horse embryo, the horse's equine ancestors, the horse's pre-equine ancestors, and, finally, the Big Bang. Rules specifying where exactly the referent of the thought lies in the causal chain is are likely to be quite complex and generate Mersenne questions: What makes *this* specification rule be the one that determines meaning?

On the other hand, fundamental normativity in an organism might provide a very neat reduction of representation. Here is one attempt:

- (1) A state S of an organism represents the world as being such that s if and only if the organism's nature includes the norm that S should occur only if s .

This is an extremely broad account of representation. If a bacterium is such that it should move fast away from a location z only if z contains a hazard, then its movement represents z as containing a hazard. Simple as this account is, it has no problem with intensionality and even hyperintensionality, since the normative facts can be hyperintensional. It is a different thing to be such that one should move away fast away from z only if z contains a hazard and to do so only if z contains a hazard and $2 + 2 = 4$.

The norms in the organism's nature then can be particular—thought type t_{17} , specified in some neural way, should only occur in the presence of horses—or can encompass complex general functional rules. We might, for instance, save the causal account of intentionality by solving its two major problems. For instance, we might have a norm that, roughly, says that when a thought type is such as to typically have xs as its relevant cause, then that thought type is only appropriate in the presence of xs described at such-and-such a level of generality. And the norm can have a vast amount of detail to specify what “typically” means, what “relevant” means (thereby solving the problem of causal chains), and what the level of generality in the description is (thereby solving the intensionality and hyperintensionality problem). Such a norm, of course, raises a significant number of Mersenne questions, but if the norms are grounded in human nature, the answers to the Mersenne questions have a ground: our human nature.

Such teleological accounts have an intriguing connection with ethics. It is generally true that one's ϕ ing should occur only if ϕ ing is permissible. Thus, whenever we act, our action represents something—it represents its own permissibility. This fits with the personalist intuition that all our action expresses something, and all wrongful action is akin to a lie, a misrepresentation of normative truth. Furthermore, some specific actions may be taken to be misrepresentations in a thicker way. Suppose I use you as a mere means. If Kant is right that using something as a mere means is only permissible if that

thing is a non-person, then by using you as a mere means, I misrepresent you as a non-person. On the positive side, if it is the proper function of our will to pursue only the good, then whenever we pursue something, we represent it as a good.

This account also has interesting consequences for philosophy of language. It is a truism that an assertion that *s* represents the world as being such that *s*. This immediately implies a version of the truthfulness norm for assertions: an assertion that *s* should be made only if *s*. Opponents of the truthfulness norm will take this to be a reason to reject the normative reduction of representation. But it is worth noting that there are in fact multiple norms governing assertion. Thus, if it is insisted that knowledge is a norm of assertion, there is no difficulty accommodating this in our account: it could be that one may assert *s* only if *s*, but likewise one may assert *s* only if one knows that *s*. Necessary conditions can be multiplied. If so, assertion represents both its being the case that *s* and its being the case that one knows that *s*, so an assertion represents more than is being asserted—a familiar Gricean thought. In fact, the breadth of what is represented goes further. One should only assert something if it is either good manners to assert it or there is some reason overriding the reasons of etiquette. Thus, on our reductive account, an assertion that *s* not only represents the worlds as being such that *s*, but also represents the world being such that it is good manners to assert that *s* or there is some reason in favor of asserting this that overrides the reasons of etiquette. That's a surprising conclusion, but fits with a Gricean picture on which one also represents by implicating.

Perhaps, though, the above account of representation is too broad. Maybe bacteria don't represent hazards by moving away from them, and maybe saying "Your argument is unconvincing" doesn't represent claims about etiquette. There is a slight tweak to the reductive account that narrows the field of representation significantly:

- (2) A state *S* of an organism represents a proposition *p* if and only if the organism's nature includes the norm that *S* should occur only if *p* is true.

Norms are hyperintensional, so there is a difference between the norm that one should run away from *z* only if *z* contains a danger and the norm that one should run away from *z*

only if it is *true* that *z* contains a danger, even though of course there is a danger if and only if it is true that there is a danger.

It is plausible that lower organisms have no normativity concerned with truth as such, and hence the alethic version (2) of our reduction will not apply to bacteria. Furthermore, we lose the ethical applications of (1). For instance, while the Kantian will insist on the norm of rationality that we should treat something as mere means only if it is a mere means, this does not mean that we have a norm of treating something as a mere means only if it is *true* that it is a mere means. Moral norms rarely make reference to truth as such.¹⁴ Finally, while the modified reductive account still forces us to accept that there is a truthfulness norm on assertion, it allows us to accept, say, a knowledge norm as well, without having to say that in asserting we represent the world as being such that we know, since the knowledge norm is that one should assert *p* only if one knows *p*, not that one should assert *p* only if it is true that one knows *p*. The alethic version (2) of our normative reduction thus loses some of the interesting applications, but these applications were also controversial.

Of course, we might also just say that “represents” is said in multiple ways, and there is both simple representation, defined by (1), and *alethic* representation, defined by (2). The two are closely related: a state alethically represents *p* just in case it simply represents *p* being true. We can then divide the substances in the world into non-representers, simple representers and alethic representers. Given how broad simple representation is, it is not clear that there are any non-representers, if Aristotle is right that substantial form is always normative, always a specifier of what a thing is to be [*to ti ēn einai*]. If all substances are representers, then we have an account that is rather Leibnizian: our simple representers and alethic representers parallel Leibniz’s division of the monads into souls and minds.

In light of Tarski1933’s (Tarski1933) indefinability of truth theorem and PorterTruth’s (PorterTruth) similar argumentation, there is very good reason to think

¹⁴Likely exceptions arise in the case of moral norms connected with language, say the norm to make true what one has promised.

that truth is not even coextensive with a physical property. A simple version of Porter's argument notes that although the Tarski T-schema

(3) Sentence "*s*" is true if and only if *s*

may not hold for sentences using semantic vocabulary, like the liar sentence that says of itself that it is false, (3) holds for indexical-free sentences *s* in the language of physics. But if truth is coextensive with a physical property τ , then consider the sentence of the language of physics:

(4) *u* does not have τ ,

where "*u*" is an indexical-free physical description of a token of (4). (E.g., at t_0 Alice might whisper "Alice's whispering at t_0 does not have τ .") If the token described by *u* is true, then by the left-to-right T-schema, *u* lacks τ , and hence does not have truth, a contradiction. If it is not true, then it does not have τ , and so by the right-to-left T-schema *u* is true, and hence *u* has τ , a contradiction.

Now, our best alternative to the Aristotelian account of teleology in the context of broadly functionalist philosophy of mind¹⁵ is an evolutionary account according to which the proper function of a system is a function that has an evolutionary explanation. But evolutionary explanations are scientific, and hence what they explain is the occurrence of physical properties, or at most properties coextensive with physical properties. Therefore, an evolutionary function cannot involve anything coextensive with truth, and hence even if evolutionary teleology can do work in other contexts, it cannot be used in the context of our account of alethic representation.

¹⁵The third popular account of teleology is based on divine purposes. However, as noted in ??backref, this is implausible for functionalist theories of mind, because purposes are already mental and hence require teleology, and because it is implausible that if God made beings just like us but had different purposes for them, they wouldn't be conscious.

7. Dualism

The neo-Aristotelian teleological twist on functionalism allows one to remain to a significant degree a naturalist (more on that in ??forward). The account reduces mental properties to functional properties, but the functional properties are not reducible to the kinds of properties that physics studies. A more traditional Aristotelian approach, however, is to refuse to reduce the reduction of mental properties to non-mental ones. This is certainly also compatible with the robust view of human nature that has been defended in this book.

Whether the more dualist or the more functionalist view is more plausible depends on how satisfying the reduction of mental properties to teleologically laden functionalist properties is, and one way to evaluate this reduction is to consider whether the arguments against the reduction of mental to physical properties apply to this neo-Aristotelian functionalism. These arguments can be divided into three categories, depending on which aspect of mental life they object to the reduction of: content, consciousness, or freedom of will.

As has been noted??backref, the neo-Aristotelian teleology offers hope for a reductive account of mental content. Teleological properties can be hyperintensional—it is the purpose of eyes to see rather than to see or be such that $2 + 2 = 5$, even though necessarily everything that sees is such that it sees or is such that $2 + 2 = 5$. Thus concerns that there is no way to reach the hyperintensionality of mental content from the extensional or at best intensional content of the physical world do not transfer to the neo-Aristotelian account. Similarly, concerns about the need to account for purpose in the mind—beliefs are states that are there *in order to* mirror the world—and that the only source of purpose for the physicalist, namely evolution, is inadequate do not apply to the neo-Aristotelian theory. Thus, content-based arguments do not tell against a neo-Aristotelian functionalism.

The case of consciousness is less clear. We can try to run standard knowledge arguments transposed to the neo-Aristotelian context.??cf.ref:Jackson Suppose Mary is raised in a black and white environment, and knows all the physical facts as well as all the normative facts about human beings. In particular, she knows that certain states of the human

being are such that they should occur precisely when the human being is looking at a red object, and that these states typically do occur when people see a ripe tomato. She, further, knows all the normative interconnections between these states and other states, including all the inferential connections. Will she learn anything further by seeing the tomato? Here intuitions may differ. It is perhaps *slightly* easier to hold out for a negative answer to the learning question here than in the case where Mary has a purely physical state, as in Jackson's original thought experiment. It seems easier hold that what constitutes the phenomenal feel of red is a teleological-cum-physical feature of the state than that it is just a physical feature.

Similarly, we can try for imaginability arguments. These come in two versions: zombie arguments that one can have our physical constitution without consciousness (**Chalmers1996**) and afterlife arguments that our consciousness could survive the destruction of the body (**SwinburneSoul1997**). On these arguments, the imaginability of a scenario provides defeasible evidence of its metaphysical possibility.

Could there be beings that have isomorphic physical and normative properties to ours but that are unconscious zombies? Again, this is not completely clear. Among our normative properties are moral duties. Could one have something isomorphic to moral duties without consciousness? If not, then zombie arguments against neo-Aristotelian functionalism do not work as they stand.

Alternately, what can the neo-Aristotelian functionalist say about surviving the destruction of the body? Those "fainthearted" neo-Aristotelians (cf. **KoonsFaint**, **KoonsFaint**) like **Koslicki2008** (**Koslicki2008**) who hold that the form is nothing but a kind of arrangement of matter, of course, will find it troubling to suppose the form to survive the destruction of the matter. Though even there, there is Aristotelian precedent for a view that would allow for the metaphysical possibility of the form's survival. Aquinas's account of transsubstantiation holds that in the Eucharist, the accidents of bread and wine continue to exist after the cessation of the existence of the substance. Among the accidents there will be *shape*. If a shape can exist without that of which it is the shape, then why not

an arrangement as well? Similarly, some contemporary trope theories hold that there are “unaffiliated” tropes, tropes that exist without their substance. If a trope can exist without a substance, why can’t an arrangement exist without matter, and perhaps even engage in some sort of disembodied activity?

But of course the view defended in the book is more robust: forms are not mere arrangements. The more reality the form itself has, the more plausible that there is no metaphysical impossibility about the form existing without the matter. Thus a more “stout-hearted” neo-Aristotelian naturalist could more easily suppose the possibility of the form’s surviving physical death, and perhaps engaging in some mental activity.

Finally, we have freedom of will. Those convinced that an action that comes solely from the causal powers posited by a completed physics would not be free will not be moved by being told that these causal powers have a normative organization. Adding to the normative organization the thesis that form grounds all of the nomic behavior of the entities (perhaps including even at the microphysical level), as will be discussed in ??forward-ref, may however seem to help recover a more robust sense in which the form is a soul that is in charge of the body. But it is still not clear that this really solves the problem. For on such an account of the form’s role, our form likewise grounds the nomic functioning of all of our subsystems including functioning, such as immune function or bone regeneration, that are only very indirectly under our voluntary control. It thus does not seem that adding form’s grounding of our behavior is sufficient to recover a robust sense of free will.

One move here is to suppose that in rational animals the form has one more explanatory role, not reducible to the explanatory roles it has in other beings: it voluntarily controls those aspects of the functioning of the organism that are directly voluntary. An irreducible notion of voluntary control, however, is murky. It would be philosophically preferable to be able to see voluntary control as a species of the control that the form already has over all of the organism, and to explain what makes the control voluntary in terms of the other explanatory roles of the form.

For instance, perhaps what makes the control voluntary is that in voluntary control, the form ensures a connection between actions and *reasons*. And insofar as the form is the central metaphysical component of the agent, a causal story going back to the form and to reasons can be plausibly thought of as just a more detailed account of what happens in agent causation. But at this point, given such a substantive role for the form in the explanation of our voluntary behavior, it becomes plausible to say that the view really is a dualist view. However, this Aristotelian dualism has the advantage of not making human beings stand out ontologically, as *all* embodied substances have a form as a component.

8. Soul and body ethics

There are four main families of views of what we are:

- (i) brains or cerebra,
- (ii) souls,
- (iii) purely physical bodies or organisms, and
- (iv) body–soul composites.¹⁶

In this section, I will review the claim the brain and soul views have implausible normative ethical consequences, and argue that the body–soul view, especially in its Aristotelian variant where the soul is the form of the body, is the ethically most attractive option.

On soul or brain views, our bodies¹⁷ are not a part of us. In particular, you have never seen or touched any of the people you love, unless perhaps (on the brain view) you have witnessed brain surgery or a horrid head injury. There is something special about in-person interaction. For a number of us this conviction was made particularly poignant by the electronic interaction during the pandemic. Yet if the body is not a part of us, then ordinary in-person interaction is mediated by a large amount of data processing equipment—the eyes, ears, and nerves (and including the brain itself on the soul view)—that is just as

¹⁶Of course, one can imagine other options. The most radical is that we do not exist (Unger1979). An apparently undefended option is that we are brain–soul composites.

¹⁷In the context of brain views, I follow the loose common usage where “body” denotes the body minus the brain.

external to us as a phone. This data processing equipment is biological rather than technological, but the biological equipment is not a part of us on the soul and brain views. The human importance of in-person interaction is deflated by soul and brain views. Of course, except in the case of touch, our ordinary in-person interaction is also mediated by causes outside of us—light, sound-waves, and chemicals wafted in the air. But there is a significant quantitative difference here: the ordinary mediation involves quite a bit less processing and transformation of the data, of the sort that goes on in our physical eyes, ears and nose. One body reflects light, the other body receives the reflected light, and in between the light travels in more or less a straight line, with some distortions if we see the other through lenses or in a mirror. And the directness of touch is particularly important in familial, romantic and, in many cultures, friendly interaction.

Furthermore, there is an important legal distinction between assault and assault with battery, where battery involves harmful bodily contact. This distinction is difficult to justify morally if our bodies are not a part of us. On brain and soul views, bodies seem to be very much like worn personal property, such as clothing. And the very high ethical importance that we attach to avoiding unwanted sexual touch is particularly mysterious.

One might object that unlike clothes, our bodies are full of sensors that make us directly aware of what is happening to them. Furthermore, our bodies are under our direct control—we have basic actions that control the body, and another's interference in the body thus impacts our autonomy much more than interference in clothing or other property.

But while this line of thought does not fully explain the importance of interference with a body. Battery, sexual or not, is morally noxious even when the victim is completely unconscious, and hence the sensors are irrelevant. Moreover, the triggering of harmful sensory impressions is not what makes battery particularly objectionable, because harmful sensory impressions can be triggered without battery. Here, think of the difference between shoving a police officer during a protest against police brutality and showing the officer a gory poster depicting brutality. Shoving is *prima facie* morally problematic

in a way in which showing the poster is not, even though both present noxious sensory stimulation—indeed, the gory poster could cause a greater unpleasantness to the police officer than the shove, and yet is perfectly appropriate in a properly peaceful protest. The difference between the two treatments appears to rest in the fact that the shove modifies (perhaps only slightly) the victim's body in a way that goes beyond collaboration with the victim's sensing of the environment.

Similarly, while some of the time interference in the body of another, say by shoving, directly undercuts the victim's direct control of the body, this is not a necessary condition for battery. Non-consensual amputation of a permanently paralyzed body part does not damage the victim's direct control of the body, but has the special harm of battery.

Thinking about taxation is another way to see the ethical distinction between the body and property. It would be wrong for the government to tax one by forcing a kidney donation, even if there were strong evidence that on the whole a kidney donation has a smaller impact on the taxpayer's wellbeing than some clearly permissible monetary tax. There is a dignitary harm in being divested of a part of oneself in the kidney removal.

We thus have good reason to think that we do have bodies as parts of us. That leaves the question whether we are *just* physical bodies, or physical bodies with souls or forms. All the previous arguments of this book point to the need to posit non-physical forms. Furthermore, if we go back close to the beginnings of Western philosophical reflection on the soul, in **PlatoDialogues's (PlatoDialogues) *Phaedo*** we get the idea of the soul as that which makes a living thing be alive. That is indeed one of the things a substantial form does for us, since it makes us be what we are—living human animals. And it is very plausible that what makes us be alive is a part of us. Furthermore, if we see the form as governing our behavior not just normatively but causally, whether in the case of voluntary behavior through our free will or at least by being the ground of the applicability of the laws of nature to our bodies as will be discussed in ??forwardref, that gives us further

reason to think that “soul” is an appropriate word for our form. Moreover, the meta-ethical discussion in ??backref highlighted the idea that ethical norms are not alien to us, and that requires the form—or soul—to be a part of us.

Finally, there is a spectrum of very similar physical entities between a rock and any purely physical being—add a particle here, shift this one, etc. The functional changes, understood physicalistically, between neighboring items in the spectrum from can be quite small. Yet persons have a qualitative dignity that does not appear to be something that one can generate by an otherwise insignificant change from something that lacks that dignity. On the other hand, it is far from clear that we can have a spectrum of souls or forms ranging from rocks, or at least trees, to people, with only small difference in between. The transition, for instance, from being such that one should pursue various particular goods (nutrition, shelter, procreation, etc.) to being such that one should pursue the good as such seems to be something that does not come merely by minor variation. Likewise, we have a transition from being such that moral considerations about one always involve a moral balancing of reasons, as they do in the case of treatment of animals that aren’t persons, to having inalienable rights. It is wrong to kill one person to give their organs to save three others, but it need not be wrong to do so with dogs. If we are purely physical and there are no forms or souls, continuous minor variation is all we have. If we are purely physical but there are forms that are not a part of us, then it is still true that intrinsically we only have minor variation, as the major variation is outside of us. Neither view appears ethically satisfactory. Thus, we should think that the form or soul not only exist but is a part of the human being.

9. Conclusions

Functionalism suffers from serious problems with defining function. Plugging in an Aristotelian account of function in terms of a teleologically laden form solves these problems, and offers an attractive and broadly naturalistic neo-Aristotelian view of mind. One may, however, think that it takes more than this to preserve our freedom of will, and that

we need to see our human nature or form as not merely offering norms but influencing our rational behavior in a way that makes it appropriate to call it a *soul*. Moreover, since the form makes us alive, calling it a “soul” is consonant with the tradition of Western use of the word. Finally, such a soul-body view has significant ethical appeal.

Appendix: Functionalism gone too far

Consider a deterministic functional system Q consisting of a finite number of possible computational subsystems $S_1, \dots, S_N$, where S_k is always in exactly one total state from the finite set $\mathcal{A}_k$ of possible states at each of the discrete “significant” moments of time $t_1, \dots, t_m$ over its finite lifetime¹⁸, and a finite number $I_1, \dots, I_M$ of sensors, where I_k is always in exactly one state from the finite set $\mathcal{I}_k$ of possible input states.

We can think of Q as a finite digital computer. The total state of the system at any given time can be represented as the $(N + M)$ -tuple $(a_1, \dots, a_N, b_1, \dots, b_M)$ where a_k is a state in $\mathcal{A}_k$ and b_k is a state in $\mathcal{I}_k$. We can designate some of the subsystems as outputs, connected to external effectors (muscles, motors, lights, etc.) Furthermore, we suppose there are functional laws which provide a deterministic mapping f from the total state of the system at time t_i to the total state at time t_{i+1} . Thus, $f(a_1, \dots, a_N, b_1, \dots, b_M) = (a'_1, \dots, a'_N)$ provided that the system would transition from internal state $(a_1, \dots, a_N)$ to internal state $(a'_1, \dots, a'_N)$ under the influence of the sensory states $(b_1, \dots, b_M)$.¹⁹ Presumably any deterministic analog system can be approximated by such a system Q to arbitrarily high precision. We

¹⁸A standard modern digital computer only has defined computational states at ticks (appropriate level transitions) of its internal clock. Between ticks of the internal clocks, it is in an analogue state that is not computationally defined. Much engineering goes into ensuring that the states become properly “digital” and well-defined in time for the next clock tick.

¹⁹In this formulation the mapping is independent of time. But we can always suppose there is a finite clock, e.g., given by a subsystem S_k such that the set of states $\mathcal{A}_k$ is the set of times during the system’s lifetime, and with the transition rule that the time always gets incremented. Or we can suppose an input from an external clock.

suppose for convenience that there is some fixed initial state of Q , with fixed initial inputs (maybe all the sensors sense nothing at first) and computational states.

Now consider a different system P consisting of a single particle moving in the xy -plane in space, with a constant (sublight) non-zero x -velocity v , starting at x -coordinate 0 at time t_1 . For ease of visualization, suppose the x -axis runs left to right, and the y -axis runs down to up. Let $\mathcal{J}$ be the set of all possible single-time input state vectors $(b_1, \dots, b_M)$ where $b_k \in \mathcal{I}_k$ for each k , and let K be a positive integer such that $\mathcal{J}$ has at most 10^K members.²⁰

We now recode Q 's inputs into P 's inputs as follows. Suppose our particle is at y -coordinate 0 at a time $t_0 < t_1$. For $1 \leq n \leq N$, let ψ_n be a one-to-one function from $\mathcal{J}$ to integers between 0 and $10^K - 1$, both inclusive. The analogue to inputting the sensor state vector $(b_1, \dots, b_M)$ into Q at significant time t_n will now be this. We take the y -coordinate value y_{n-1} at t_{n-1} , and add to it the value $10^{-Kn}\psi_n(b_1, \dots, b_M)$, shifting the particle upward along the y -axis as needed to ensure it reaches that coordinate by time t_n .²¹ Thus, the first set of inputs of Q will be encoded in the first K digits after the decimal point, the second set of inputs will be encoded in the second K digits, and so on.

Next, suppose Q 's computational states begin with the fixed initial state vector $(a_{1,1}, \dots, a_{1,N})$ at time t_1 . Given the determinism of the system, there is a mathematical function g that takes a sequence $s = (b_1, \dots, b_n)$ of length n of members of $\mathcal{J}$ and returns the computational state $g(s)$ that Q would be in at time t_n if it were to have received the sequence of inputs $b_1, \dots, b_n$ at the respective times $t_1, \dots, t_n$. Let $n(x)$ be the integer n such that $x = vt_n$, if there such an integer, where we recall that v is the velocity of our particle along the x -axis. Let $S_n(y)$ be the sequence of n sensor state vectors of Q that would be coded for by the y -coordinate value y . Thus, the first member of $S_n(y)$ will be the sensor

²⁰The set $\mathcal{J}$ will be a subset of the Cartesian product of the sets $\mathcal{I}_k$. It may be a proper subset as some combinations of input states may be such that they cannot occur.

²¹We may need to ensure the units in which these are measured are such that the particle can be shifted in the requisite time without exceeding the speed of light.

state vector s_1 such that $\psi_1(s_1)$ is equal to the decimal number given by the first K digits after the decimal point in y , and so on. We now deem P to be in a computational state corresponding to $g(S_{n(x)}(y))$ when the particle P is at (x, y) . This is defined at all the significant times $t_1, \dots, t_n$. We have, thus, an isomorphism between the computational states and sensor inputs of Q and P .

We now go for one final twist. Suppose that in the actual world, the system Q actually gets the sensor input vectors $s_1, \dots, s_N$ at times $t_1, \dots, t_N$, respectively. We can now choose the function ψ_n such that $\psi_n(s_n) = 0$ for all i . Then a single particle moving with constant velocity v along the x -axis, with y -coordinate always equal to 0 is an isomorph to the actual functioning of Q . But the very same particle will be an isomorph of any other system Q' with the same significant times. Thus, the particle will think your thoughts *and* my thoughts, given functionalism!

Note that this version of the argument has two advantages over that given by [(I)Appendix]Putnam. First, I avoided Putnam's rather handwavy handling of system inputs and outputs. Second, I have allowed for a more general formulation where the functionalism involves a structuring of the system into subsystems with individual isomorphisms between corresponding systems, rather than a more global functionalism without the added structure added by subsystems.

CHAPTER VII

Language and semantics

1. Communication, norms and illocutionary force

1.1. Cooperative games. There are competitive games, such as the Prisoner's Dilemma or the Tragedy of the Commons (**Hardin**), where it is difficult to see how to rationally secure cooperation between agents. Cooperative games, on the other hand, seem like they should be much easier. Say, you have two agents who will each in a separate booth choose whether to press a red button or a blue button. If they both press different buttons, they each get a reward, say a chocolate bar. If they press the same button, they each get a penalty, say a nasty electric shock, with the penalty outweighing the award by a significant factor, so it's better to get neither than to get both. If either player omits to press a button, neither gets anything, and the buttons are so set up that one cannot press both. Moreover, let us suppose, the players are allowed to confer ahead of time. Here is an example pay-off matrix:

	red	blue	nothing
red	-10,-10	1,1	0,0
blue	1,1	-10,-10	0,0
nothing	0,0	0,0	0,0

Obviously, when conferring ahead of time, they will need to decide which button each should press, by rolling a fair die or flipping a fair coin if necessary, and then they need to go into their booths and press that button. Neither has any incentive to defect to pressing the other button, and there is no cost to pressing a button should the other fail to press anything. The game has three Nash equilibria, pairs of strategies such that neither player would gain from changing their strategy while the other player's remained fixed: red-blue, blue-red and nothing-nothing. In the terminology of David Lewis's pioneering study of

coordination games (**LewisConvention**), each of these is also a coordination equilibrium, namely a pair of strategies such that no player would benefit from any one player's (their own or the other's) change of strategy. While Lewis's primary interest was in the mechanism (e.g., convention) by which we can land in *a* coordination equilibrium, my interest here will be in how we can land in one of the better two equilibria, red-blue or blue-red, instead of falling into a safe scenario where both players avoid risk by pressing no button.

Suppose our two players, Alice and Bob, are perfect expected utility maximizers who break ties with fair coin-flips, and the only relevant utilities are the rewards and penalties of the game. There are no further games that will be played. Nobody outside the game is in any way affected by the results (e.g., nobody will be disappointed if one of them breaks a promise). And because each player gets the same payoff, it won't matter whether Alice and Bob maximize collective utility or their own personal utility. Finally, the above information is completely luminous to both players, and hence constitutes certain shared knowledge. I claim that at this point the obvious strategy—to decide on a button and then both press it—is no longer rationally available.

For concreteness, let's suppose that Alice and Bob have agreed to press the red and blue buttons respectively. They go into their booths. What will Alice do? She is a perfect expected utility maximizer. She will only press the red button if the expected utility of doing so is at least as big as that of all the alternatives (these being pressing the blue button or pressing no button). Now the expected utility of pressing the red button is only going to be at least as big as the expected utility of pressing neither button if Alice takes it to be significantly more likely that Bob will press blue than that Bob will press red. In the case of the above matrix, Alice needs to be at least ten times as confident of Bob's pressing blue than of Bob's pressing red.

But why should Alice take it to be an order of magnitude more likely that Bob presses the blue button than the red button? Ordinary human beings take themselves to be beholden to norms of promise-keeping, and tend to abide by those norms, especially when there is no clear benefit to failing to do so. But Bob is an extraordinary human being,

a pure expected utility maximizer. Whatever normative force he takes promises to have has to be derivable from the norm of expected utility maximization. In ordinary contexts, dealing with ordinary human beings, keeping promises certainly does maximize utility, because ordinary human beings believe in norms of promise-keeping, punish those who break those norms (if only by castigating or refusing to enter on joint projects with promise-breakers), and in any case tend to abide by these norms, unless perhaps it is highly profitable not to do so. But Alice and Bob are not dealing with ordinary human beings.

Here is one way to highlight the difficulty. Imagine that Alice and Bob are perfect utility maximizers but Bob has a perverse value theory that in addition to common-sensical value assignments to chocolate bars and electric shocks assigns non-instrumental negative value to keeping promises and non-instrumental positive value to doing the very opposite of what one has promised (where the opposite of pressing the blue button is pressing the red button, rather than just pressing nothing). In this case, if there is shared knowledge of the perverse value theory, it is more reasonable to expect Bob to press the button other than the one he promised. And now imagine instead that he is a perfect utility maximizer with a value theory that assigns positive value to keeping promises and non-instrumental negative value to doing the opposite. In this case, it will be more reasonable to expect Bob to press the button he promised. But then the in-between case, where Bob is a perfect utility maximizer and assigns zero non-instrumental value to promise-keeping and promise-breaking, should be one where the probabilities of pressing the promised button and pressing the opposite button are approximately equal. And approximate equality is not good enough in this case—it's too dangerous for Alice to press the button.

Of course, as a matter of contingent fact people have a preference for promise-keeping over promise-breaking: we feel bad when we break promises and good when we have fulfilled them. Preferences enter into utilities, and so if Alice and Bob have the standard preferences, they will have a bias in favor of promise-keeping, and if each knows the other to have the preference, then each can take the other's preference into account, and hence each can expect the other to keep the promise.

But, first, this does not solve the problem when the penalty for both pressing the same button is much stronger than the preference for promise-keeping. In that case, unless Alice and Bob are *very* confident in the other's choice, a mere preference for promise-keeping will not be enough to get them to press the promised button rather than pressing nothing.

Second, and more importantly, if a preference for promise-keeping is needed to solve the problem, we now have an argument that some norm incumbent on humans requires such a preference. For if two human beings are stuck in the suboptimal nothing-nothing solution in the button-pressing game, they are clearly falling short of what humans should be able to achieve—they are not acting as humans should. And if having a preference for promise-keeping is the only way out, then humans clearly are defective when they lack such a preference. The argument thus shows that there must be norms on human beings that go beyond utility maximization, whether collective or individual.

The cooperative behavior in promise-keeping, thus, requires a normativity that goes beyond expected utility maximization. Promises are perhaps the most morally-laden of our communicative behaviors, so this is not too surprising. But the upshot of this chapter will be that what holds of promises holds of many and likely all our other communicative behaviors.

One may object that the button story is particularly unrealistic. The salient features are that payoffs for both players are approximately the same, that the penalty for pressing the same button is very high, and that there is no benefit to one person pressing the button and the other doing nothing over neither doing anything. These features are what make it particularly challenging to avoid the suboptimal equilibrium of neither doing anything. Yet we can imagine such situations, and it is clearly within human rationality to be able to resolve them. And if my argument is correct, then it follows that human rationality is more than expected utility maximization.

In any case, let's end this section with a story like our button story that does not involve modern technology. There are two feuding families living in different villages. Each family has two guards who separately patrol the area around their village. Alice and Bob are your

family's guards. One day, they find out that the other family's guards are weaker than usual due to having a cold, so that Alice and Bob can each go and beat up either of them. If they end up beating up both of them, this show of force will lead to the other family suing for peace, and will end the feud—and the feud is destructive enough that it's worth ending even at the cost of injury to two people. If they both beat up the same guard, the other will hear the sound and rush to your undefended village, burn down your family's fields, thereby greatly intensifying the feud. If one of Alice and Bob beats up one guard while the other remains on patrol, your family's fields will be safe, and the feud will be unchanged—the enemy will be intimidated by the attack, but will also be more angry, and these effects will cancel out. It should be possible for Alice and Bob to agree on which guard each will attack and to rationally keep to their agreement.

1.2. Arresting the regress of meaning. Some communicative actions—speech acts or gestures—have their significance assigned through earlier communicative actions. Thus, sometimes one coins a word and stipulates its meaning in terms of other words, sometimes one uses gestures to introduce a new word, and sometimes one just hopes that use in a rich enough communicative context will clarify the meaning. But barring outlandish hypotheses such as that humans got their language from an infinite regress of angels, we cannot suppose an infinite regress. There must be ur-communicative actions, ones which did not get their significance from earlier communicative actions.¹

At the same time, there is a contingency here. While it feels natural to us to use an extended index finger to indicate the nearest salient object approximately along the extending from the knuckle in the direction of the fingertip, it would be possible to have rational beings that use this gesture to indicate the third-nearest salient object along the ray extending in the opposite direction. There is no necessary connection between the physical

¹There is a Sellarsian objection to this. Perhaps there are behaviors prior to the advent of rationality that count as having communicative significance in virtue of *later* behaviors, in a kind of virtuous significance-conferring circle (cf. **SellarsEPM**, **SellarsEPM**). But if so, then we can just count the whole circle of behaviors as the first ur-communicative community action.

behavior and its significance. We thus have a Mersenne question here: What grounds the correlation between physical behavior and significance in ur-communicative actions?

We might try to explain the correlation in terms of the actual contingent behavior of individuals and communities which arises by natural or social selection. Suppose, for instance, that some social animals evolve to squeak at a certain pitch when observing a predator, thereby warning other members of their group. Eventually, their descendants develop rationality, but the squeaking behavior is maintained, and remains correlated with the presence of a predator, even though it is now under voluntary and rational control. It is plausible to say that the squeaking is now a communicative activity whose significance is "Predator!"

But this plausible claim deserves more careful examination. Rationality complicates things. Suppose Alice sees a predator and is considering to squeak to trigger her group-mates' defensive behavior. If Alice's squeaking and her fellows' defensive behavior is to be rationally chosen, the agents need reasons. The fact that their ancestors used to squeak when predators were present and start defensive behavior upon such squeaking is an interesting bit of pre-history, but there does not appear to be a reason for them to imitate this quaint custom. Even if we add the fact that they find themselves with a desire to squeak in the presence of the predator and to initiate defenses upon hearing a squeak, these desires at most generate the very weak kinds of reasons one has to fulfill miscellaneous sub-rational desires rather than the strong kinds of reasons that one has to warn one's fellows and protect oneself and one's community.

There is no difficulty here if Alice is the only rational one, and hence the others can be counted on to act instinctually. Alice can then rationally squeak to trigger the instinct. Similarly, if Alice acts on instinct and the others are rational, they can infer from her instinctive behavior that there is a predator present, as one infers fire from smoke. Likewise, if Alice thinks there is a fairly high chance that her fellows, though rational, will simply non-rationally follow their habit and prepare themselves (rational beings don't always act rationally), or if she knows that her fellows think there is a fairly high chance that Alice

would find herself squeaking in the throes of instinct, there is no difficulty. The difficulty shows up precisely when we have nothing but rationality at play.

Perhaps we can solve the problem by positing a non-rational preference for squeaking when seeing a predator and for preparing a defense when hearing a squeak? After all, arguably, even a perfectly rational being will act in accordance with preferences when other things are equal.

But much as we saw in the promise case, the non-rational preference is insufficient here unless it is implausibly strong. For even if one finds oneself with an urge to squeak in the presence of a predator, the squeak itself endangers one. Because of this, for a rational being, a moderate preference for squeaking is not sufficient to motivate the squeak. It is only when one thinks the squeaking will trigger defensive behavior among one's fellows that it's worth squeaking. Rational beings step back from and evaluate the outputs of their preferences. Similarly, we may suppose that defensive behavior is costly and inconvenient, and is only worth engaging in, notwithstanding the non-rational preference, when there is reason to think there is an actual predator.

Instead of a brute preference, perhaps convenient priors will do. Thus, suppose that members of the community simply find themselves with a high prior conditional probability that a member of the community rationally squeaks when presented with a predator and does not squeak when not presented with a predator.² Knowing this prior is widespread in the community, Alice can squeak in order to get her fellows to update their credences in favor of a predator. The priors seem pleasantly self-confirming: if community members have the priors, then it will become public that they do so, and the squeaking behavior will match the priors.

But now suppose Bob is reflecting on his convictions. Bob finds himself accepting a correlation between Alice's rational squeaks and the presence of predators. But since the squeaks are rational, there must be a rational explanation of these squeaks. Since we are no longer attempting a preference-based story, presumably the explanation is that Alice

²I am grateful to ?? for the suggestion that priors might do the job.

accepts a correlation between community members hearing squeaking and their rationally coming to think there is a predator. In other words, Bob finds himself having a brute prior concerning a contingent and empirical matter—namely, another community member having a certain credential state. However, when we find out that our conviction about a contingent and empirical matter is simply a brute prior, that tends to undermine the conviction. Suppose that I find myself believing there is vast treasure buried under my house. I search for the source of my belief, and find it's just a prior. Absent a story such as that angels put that prior in my head to encourage me to dig out the treasure, finding out that this was *just* a prior should undermine the confidence, contrary to what subjective Bayesians think. This is why it is unstable to respond to a skeptical hypothesis by simply citing a brute low prior for it.

One might hope that natural selection can help. We have good reason to expect that social animals would evolve ur-priors that enable communication. Thus, it seems these priors are not brute: they are coordinated by natural selection. However, presumably such evolved priors would concern pre-rational behavior: they would not be priors for conditionals whose antecedent is that Alice or Bob is acting purely rationally and whose consequent is some claim about their behavior, which are the relevant conditionals once Alice and Bob are acting purely rationally.

And in any case why should one think there would be any continuity between pre-rational squeaking and rational squeaking? Exactly the same considerations as in the promise case arise here. Just as we considered the case of rational agents who set a non-instrumental disvalue on promise-keeping, we might consider rational agents who set a non-instrumental disvalue on acting like one would pre-rationally. Such agents' squeaking would be counter-correlated with danger. On the other hand, agents who place a non-instrumental value on acting like one would pre-rationally would exhibit a positive correlation between squeaking and danger. It seems that the in-between case where no value is attached to acting similarly to pre-rational action should result in no correlation.

Thus, for evolved agents to function properly, we need them to have a norm by which they should prefer to maintain pre-rational signalling practices to some degree.

1.3. Requests. Typically, one asks someone for something that one wants. But asking is not the same as communicating one's desire. First, sometimes one asks for something one doesn't want. For instance, a security specialist could conduct phishing calls where they ask fellow employees for their passwords, hoping that few if any will give it. Or a middle manager might be tasked by upper management with requesting something from staff that the middle manager thinks is actually bad for the company, and hence hope that no one will agree to the request. Conversely, one may want something but not ask for it for moral reasons. To adapt a Wodehousian situation, one may own an ugly heirloom that one cannot give away because of one's relationship with the person from whom one received it, but one would be glad if it were taken away. One could imagine a frank conversation where one happens to intimate that one wouldn't mind having the heirloom taken away. One may suppose that in such cases the communication *is* a surreptitious request—and that, of course, is illegitimate, much as a king's exclamation "Would that someone rid me of this troublesome priest" is an invitation to murder. But one could also imagine a case where the intimation is not a coy request, but simply a frank statement to a friend, followed by sincere emphasis that one isn't requesting removal. In that case, removal of the heirloom would be theft, even if welcomed by the owner. Or, for a different case, one might have a moral objection to a particular life-saving medical procedure, and hence one's conscience would forbid one from requesting it, but nonetheless wish that the procedure were done to one against one's will, say by a medical mistake, and one could in a frank conversation communicate that wish *without* that constituting an underhanded request.

In making a request, then, one not only affirms the existence of a desire, but one creates a reason for the other party to provide one with something. And it's not just any reason, but a special kind of reason, a reason in light of one's own request.

But now consider the first time anybody ever requested anything on earth. In requesting, they created a moral reason for their interlocutor. The power to create such a reason

must have been a power the requester already had. Moreover, the meaningfulness of the communicative act of requesting must have already been in place: the act must have been understandable, since requesting only creates a reason when the requesting is understandable. But how could that communicative act not only have had its illocutionary force but had been *understable* as having that illocutionary force given that no one had ever requested anything? The meaning of a request is largely defined by the kind of reasons it gives rise to. But how can one grasp these reasons if one has never encountered them before? It seems that we have to have a responsiveness to such reasons in our very nature.

1.4. Schelling focal points. The economist Thomas Schelling (**Schelling**) performed a number of experiments on coordination, in the context of games where two or more players make independent decisions and if these decisions are appropriately coordinated, they both win, while if they are not coordinated, they both lose. In a paradigm example, three players are asked to give the letters A, B and C in some order, with all of them winning if everyone gives the same order and all of them losing otherwise. Not surprisingly, the majority would write down “ABC”. We can call a type of behavior that is converged on in such a way without prior communication and where we have a shared expectation of such a convergence a “Schelling focal point”: writing down “ABC” is one, but presumably writing down “BAC” is not.

Now, suppose Alice and Bob have shared knowledge that in the past squeaking by members of their kind was associated with danger. Alice sees the danger and needs to communicate this to Bob who knows that she is acting voluntarily as a pure expected utility maximizer. Let’s imagine that Alice is now considering what auditory production to make with three options easily available to her: squeak, stomp loudly, and maintain silence. We have three behavior types going along with these three options: (a) in the presence of danger make the same sound as used to be correlated in the community with the presence of danger, or (b) make a different sound in the presence of danger when previously some sound was correlated with the danger, or (c) make no sound in the presence of danger where a sound was previously correlated with the danger. Of these three options,

we would expect that only (a) is a Schelling focal point. Bob would expect Alice to go for the focal point, and so if she squeaks we would expect him to interpret her as acting in a way that correlates with danger.

But this solution fails when Alice and Bob have a shared knowledge that they are expected value maximizers with no special non-instrumental value attached to going for focal points. The argument should now be familiar (see ??backref). Suppose that Alice and Bob had shared knowledge of Alice negatively valuing going for focal points. Then we would expect that Alice's squeaking would be evidence for Bob that she is *not* in danger. The evidential force of this evidence would be expected to decrease the weaker the negative valuation of focal points. In the case where Alice's valuation of focal points is positive, her squeak would provide evidence of danger, with the strength of evidence co-varying with the degree of her positive valuation. And so we would expect that in the case where Alice assigns zero non-instrumental value to going for focal points, there is zero evidence, assuming a certainty that she is an expected value maximizer.

Of course, in practice, one may know that someone is an expected value maximizer, but not be sure of it, and in the absence of certainty, one might reason that if Alice isn't an expected value maximizer, but one of the *hoi polloi* who squeak without engaging in expected utility calculations, then her squeak is evidence of danger, and so as long as there is a small chance that she isn't an expected value maximizer, the squeak provides weak evidence of danger. But we already saw that weak evidence is not enough. Suppose the danger in question is a major flood, and the right response to such a flood is evacuation of the village. Maybe the background chance of such a flood on any day is one in a hundred thousand, but with Alice's squeak it becomes one in ten thousand. That may still not be enough to make a costly evacuation worthwhile. For communication to be as effective as it is for rational beings like us, it needs to provide more than weak evidence.

That there are Schelling focal points is plausible. Indeed, the Aristotelian can plausibly say that our nature builds in such points, though doubtless often under some general description, like "When all else is close to equal, prefer to put things in the order that

your community most typically enumerates them in” rather than a specific description like “ABC”, and provides us with a norm whereby we should have a weak preference for going for them. A bit of reflection shows that the points will have seemingly arbitrary parameters specifying how close to equal should other things be, how strong should the preference be, how you should define the boundaries of your community, and what is the function from the degree of typicality of a behavior in the community to the strength of reason to engage in it, and hence we will have a multitude of Mersenne questions. Furthermore, the argument above suggests that in some cases, a weak preference for Schelling focal points is not enough to ground communicative norms, as a weak preference will not generate sufficiently strong reasons in cases where it is particularly dangerous to have a misunderstanding.

1.5. Communication and norms.

1.5.1. *Declarative communications.* We have seen that the transition from non-rational signalling to rational communication of a sort akin to assertion is difficult. Thinking about the cases of promises, assertions and requests has shown that generation of a new type of illocution is problematic. We need some sort of a reason-generation mechanism in our very nature connected to communicative norms. In this section we focus on declarative or assertion-like communications, *communication that some fact obtains*.

Can we suppose that the normative mechanism here is a necessary one? Perhaps it is a necessary truth that when there is a pre-rational behavior that tends to be triggered by circumstances *C*, then that behavior when done rationally signifies or communicates *C*?. But there would be multiple Mersenne questions that would be raised by such a necessary truth. First, we need to select one item *C* in the causes rather than another—does the squeak signify the predator, or the predator’s fur, or the light in the air between the predator’s fur and the observer’s eyes, or the immediate cause of the predator’s presence? There are multiple selection rules, no one of them significantly more natural than *all* the others. Second, what degree of reliability does the triggering tendency have to have in order to

yield a signification fact? The multiplicity of parameters in the connection between behavior and significance points to something contingent in a way that is by now familiar. We can easily imagine different species of rational beings where the parameters are different from what they are in us.

But we have already seen that Aristotelian form can solve multiple problems of providing a source for norms. In the case of our squeakers, we can likewise suppose that the animal's essential form directly specifies that squeaking properly occurs only when there is a predator present, or it could specify a general rule for connecting pre-rational behavior with norms of significance.

But even if there is such a norm-generating process, what makes the norms be norms of *communication*? A cat's nature requires it to turn its ears towards relevant sounds. When we see a cat turn its ears in some direction, that provides us with evidence that there was some sound relevant to it. But the cat is not communicating that there is a sound relevant to it by turning its ears. What, then, makes it be the case that a norm in the nature of a communicative animal is a communication-constituting norm? Do we not need some further primitives besides norms of proper function to make it be *communicative* proper function?

We can speculatively sketch a part of an answer in terms of the *content* of norms. A toy story could be that some norms come in pairs, where one norm posits that a certain overt behavior is only proper when some fact *p* is known to a community member to obtain *and* another community member is known to be present and capable of observing the behavior, while another norm posits that when that overt behavior is observed in another member of the community, there is a tendency to form a belief in *p*. In that case, the toy story says that a behavior that is a fulfillment of the first norm counts as a communication of *p* and a behavior that is a fulfillment of the second norm counts as a reception of *p*. Of course, the full story would need to be much more complicated.

And all that said, it is not clear that we need a full story as to which exact behaviors are in fact communications. What matters for figuring out what to do (say, in a case where

a conspecific squeaks) is the content and force of the norm, not what kind of norm it is. It is a tautology that if the force of a norm is kept fixed, the norm has the same reason-giving impact on us, whether it be a norm of semantics, prudence, etiquette or morality.³

1.5.2. *A speculative partial reduction of illocutionary force.* In the previous section I offered a speculative normative account of *communicating that*, i.e., of declarative or assertion-like communications. But communications come in types corresponding to other illocutionary forces as well. In this section, let's consider some of the normatively most important communications that we might call properly *performative*: promises, commands, requests, questions, and consents. The force of each of these performatives is closely tied to a normative effect on the speaker or the audience when things go right. Thus, when things go right, a promise creates an obligation on the speaker to the audience, a command creates an obligation on the audience, a request or question creates a non-obligatory reason in the audience for an action or communication, and a consent cancels a reason for the audience.

Assertion seems different. While assertion is governed by norms—it is only appropriate when its content is true, or believed or known, depending on the exact theory of the norm of assertion—it is not quite as obvious what kind of a normative *effect* it produces. A number of philosophers have thought that assertion does create a *prima facie* obligation to correct one's statement if one's statement turns out to be false (Dummett1991). If this is the characteristic normative effect of assertion, assertion differs from what I called the proper performatives. For with the proper performatives, the characteristic normative effect is clearly the central point of the practice. But the obligation to correct one's statement is not *central* to the practice of assertion. Note that if there is no afterlife, it makes little sense to make a promise for tomorrow if one is clearly about to die and similarly it makes little sense to make a request for tomorrow of a clearly currently dying person, precisely because the characteristic normative effect of the speech act will be irrelevant or not even come off. But even if there is no afterlife, it makes perfect sense to use the last moment

³It is worth noting here that the degree of force of a norm raises another Mersenne question.

of one's life to make an assertion, even though the normative effect of being required to correct one's statement is irrelevant or does not come off.

One may have some hope in thinking that assertion has a more central characteristic effect on the audience. Perhaps it creates a *prima facie* obligation to believe the asserted content, an obligation whose violation violates the virtue of proper trust. While I am inclined to think there may be something to the suggestion that assertion does create such an obligation, there is reason to doubt that this is exactly parallel to the normative effects of proper performatives. In those, the characteristic normative effect is central to the concept of the act. Someone who does not know that a promise effects an obligation on the speaker probably can't even make promises and does not understand what has happened when someone made a promise. But while there may be a *prima facie* obligation due to the virtue of trust to believe what is asserted to one, the existence of this obligation does not seem as central to the practice. There seems to be no difficulty in asserting without believing oneself to be effecting any obligation on the audience.

Thus we have some reason to think that assertion is different from the proper performatives in that while it may be partly defined in normative terms, as indeed I think it is, that definition involves the norms governing the making of the assertion rather than the normative effect of the assertion. Conversely, while assertion is at least partly constituted by norms governing the making of the assertion, the special norms governing the making of the proper performatives appear derivative from the connection to the normative effect. Thus, one shouldn't command things when one lacks the relevant authority because then the normative effect of the command will be lacking and yet one is apt to deceive the listener into thinking that there is such a normative effect. Similarly, one shouldn't promise what one intends not to do because of a general moral principle that one shouldn't intentionally put oneself in a position where one won't fulfill one's duties.

Now consider the question of what a promise is—or any of the other non-assertoric speech acts we've considered. It is not *simply* an effecting of a specific normative effect. For the very same normative effect as a promise has could be effected differently. For

instance, I may acquire an obligation to buy you a new phone by promising to do so, or by deliberately smashing your current phone. We get a little closer to the truth by saying that a promise is an effecting of the normative effect by communicative means, but that's not right either. I could come to be obligated to do fifty pushups by promising someone to do so, or by insulting a fellow soldier while being under a standing order imposing an obligation of fifty pushups whenever one insults a fellow soldier, and in both cases I gain the obligation by a communication.

One wants to say something like this: To promise is to gain an obligation to ϕ by communicating (or intentionally communicating or some other close variant) that one is promising to ϕ . Here, as before, *communicating that p* is either asserting p or something sufficiently similar to asserting like intimating that p , squeaking with a relevant content, etc. But this account is obviously circular, since the account of promising involves communicating that one is promising.

But an Aristotelian can hypothesize the following story, building on the concept of a normative powers (**Raz1972; Lewisohn2024; DurhamDissertation**). Say that a normative effect is one that affects normative truths. Then:

- (1) An agent has a *communicative normative power* for a normative effect E in circumstances C if the agent's normative nature is such that by communicating in C that the agent is producing E precisely by intentionally communicating that they are so doing the agent succeeds in producing E .
- (2) A *promissory power* with respect to an audience x and an action of ϕ ing is a communicative normative power whose normative effect is the speaker's owing to x that the speaker ϕ .
- (3) A *promise* is a successful exercise of a promissory power.

For instance, I might tell you:

- (4) I am hereby gaining an obligation to buy you lunch by exercising a normative power that produces in the speaker an obligation by communicating that they are gaining such an obligation by a power of that sort.

And I then become bound, because it is my nature to be bound by such an exercise of a communicative power. Of course, in practice, we communicate the content of (4) by using simpler verbiage, such as “I’m hereby promising to buy you lunch” or “I shall buy you lunch” or, in countries which do not distinguish “will” from “shall”, merely “I will buy you lunch”, in a context where it is clear that one is committing oneself rather than merely predicting the future.

And one can similarly get accounts of commands, requests, questions, and consents by describing the normative effects. And for each one we might suppose we have some handy shorthand. For instance in English in asking a yes-or-no question, instead of a complex declarative statement about the listener gaining a reason to say “yes” if a proposition p is true and “no” if it is false, one can simply say a sentence whose content is p with an interrogative tone.

On this speculative account, we have reduced “proper” performatives like promises, commands, requests, questions, and consents to something assertion-like, a communication with a normative content which, in light of our shared human nature, has the power of producing the very effect it describes, when that effect is within one’s normative capabilities. For not every normative effect is indeed within these capabilities: I cannot become bound to do something immoral by a promise, and a command to a random stranger to give me a dollar has no normative effect.

This ability to make true what one is saying may remind one of an engineer who made a robot that, within a range of capabilities, does what one describes it as doing when one describes it as about to do it because one says so. Thus, to get the robot to vacuum the floor one might say: “The robot is about to vacuum the floor because of what I am hereby saying.”

There may, of course, be illocutionary forces beyond those of assertion-like communications and proper performatives, and we haven't given account of these. But it is very plausible that assertion-like communications and proper performatives are the most important of the communications between humans. And at least some others can be straightforwardly accounted for by a relationship to these. Thus, forms of fictionalization or stage-acting might be accounted for as simulations or pretences to other kinds of speech acts.

The normative account offered of assertion-like communications and proper performatives likely needs refinement. But it gives a picture of how an Aristotelian can account for basic and central phenomena in human communication by invoking a rich picture of norms grounded in human nature.

2. Content and indeterminacy of reference

Wittgenstein, Kripkenstein (Kripkenstein), Quine (Quine) and PutnamRTH (PutnamRTH) have problematized reference and content in light of the fact that different content attributions to our locutions can be made to fit with our behavior. The Wittgenstein-Kripke line of thought notes that any finite number of cases of behavior can be made to fit with infinitely many rules. Any finite number of utterances of " $a + b = c$ " that fit with our "usual" interpretation of "+" will also fit with infinitely many rules, including, say, the rule that " $a + b = c$ " means that c is identical with a plus b when a and b are less than or equal to x and that c is identical with a times b when at least one of a and b is bigger than x , where x is some specific gigantic number. The Quinean line of thought observes that the same word, "Gavagai", can be interpreted to mean a rabbit or an undetached rabbit part, with both interpretations fitting equally well with the community's practices. And finally the Putnamian line of thought observes that a remapping of the truth conditions can make "The cat is on the mat" mean any other true proposition, as well as noting that the identities of mathematical objects—such as the integers—would be underdetermined even by a countably infinite number of statements about them.

In all of these cases, we have the initial intuition that there is a well-defined meaning to the locutions, an intuition that is destabilized by the arguments. These cases, thus, can be seen as the opposite of the cases of vague terms like “bald”, where our initial intuition is that there is no well-defined meaning.

We thus have two families of arguments. One family of arguments pushes in the direction of indeterminacy. And it does so not just in the cases where indeterminacy is intuitive, as for “bald” and “heap”, but alas also in cases where we expect determinacy, as with the question whether we are referring to rabbits or undetached rabbit parts. Another family of arguments, mainly those based on insistence on classical logic^{??backref}, push in favor of determinacy, but alas also in cases where we expected indeterminacy.

With regard to these arguments, it would be simpler either to embrace determinacy in all cases or to embrace indeterminacy in all cases. For then we would only need to bite the bullet on one set of arguments. If we are to do this, then embracing determinacy in all cases seems preferable—embracing indeterminacy about all of language would seem to do too much damage to our linguistic practices. However, by treating all the cases alike, we go against common sense which distinguishes “bald” from “rabbit”.

The Aristotelian has a particularly good hope of having a metaphysical answer to the arguments for indeterminacy, by grounding sharp facts of meaning in a human nature that grounds answers to a multitude of Mersenne questions. This answer can be embraced in all cases, thereby resulting a picture of a sharp world that we will discuss below, or only in some cases, which fits better with common sense “folk linguistics” of vagueness.

All the arguments for indeterminacy that were mentioned are based on an assumption that the correct semantic theory will make semantic facts supervene on facts about our actual behavior and the world around us. But an Aristotelian can reject this assumption, and add normative facts about humans to the facts about our actual behavior and the world around us as part of the supervenience base for semantics. These further facts could be or imply hyperintensional normative facts, such as that it is only appropriate to say “Gavagai!” in the presence of a rabbit. Granted, necessarily, one is in the presence of a

rabbit if and only if one is in the presence of an undetached rabbit part. But there can still be a difference between the norm of its being appropriate to say “Gavagai!” in the presence of a rabbit and a norm of its being appropriate to say it in the presence of an undetached rabbit part.

Norms are very plausibly hyperintensional: ϕ ing and ψ ing might be such that necessarily one does one if and only if one does the other, but it is still a different thing to be required to ϕ than to be required to ψ . One way to see this is that if one is required to do something, one is required to try to do it. But trying, like intending and believing, is clearly hyperintensional. It is a different thing to try to bisect-or-trisect an angle with ruler and compass than to try to bisect an angle with ruler and compass, even though, necessarily, one bisects or trisects if and only if one bisects, since trisection is impossible. If I do not know that trisection is impossible and I have promised a friend to show them a trisection or bisection, what I am obligated to try is different than it would have been had I promised to demonstrate a bisection. It is one thing to have a reason to bisect and another to have a reason to bisect-or-trisect. Or, to adapt an example of **Faroldi** (**Faroldi**, 137), you might be obligated to drive to the hospital, without being obligated to either drive to the hospital or drive to the hospital while drunk.

The above examples all involve moral normativity. But plausibly the same is true of other kinds of normativity. The function of the $\times$ key on a calculator is to multiply quantities, not to calculate the exponential of the sum of their logarithms.⁴ It is the proper function of a duck embryo to develop two feet, but it is not the proper function of a duck embryo to grow a number of legs that God would believe to be the smallest prime number.

Similarly, reasons are hyperintensional, and norms give rise to reasons, which makes it likely that the norms themselves are hyperintensional. To see that reasons are hyperintensional, note that reasons sometimes provide explanations of actions, and explanations

⁴This is arguably true even if in fact the code running on the calculus actually performs multiplications by exponentiating the sum of the logarithms, as one essentially does on a sliderule.

are always hyperintensional.⁵ That explanations are hyperintensional is easiest to see in the special case of entailing explanations, where the explanans entails the explanandum, as when the facts that Bucephalus is a horse and all horses are mammals explain and entail that Bucephalus is a mammal. For if p explains and entails q , then p is equivalent to $p \ \& \ q$, but a conjunction does not explain its own conjunct. But if explanation were merely intensional, then anything equivalent to an explanation would also be an explanation. And even without entailment it is easy to see the hyperintensionality of explanation. For suppose p explains q . But p is equivalent to

$$(p \ \& \ q) \vee (p \ \& \ \sim q)$$

and this complex disjunction does not explain q . For the second disjunct, namely $p \ \& \ \sim q$, does nothing to contribute to explaining q , and so if the complex disjunction explained q , it would do so by means of its first disjunct, $p \ \& \ q$, and that does not explain q either.

A similar argument may directly show the hyperintensionality of reasons. Suppose that p is a reason for x to ϕ . Then

$$(p \ \& \ \phi(x)) \vee (p \ \& \ \sim \phi(x))$$

does not seem to be a reason for x to ϕ . For, plausibly, if a disjunction is a reason to ϕ , then at least one disjunct is a reason to ϕ . But $p \ \& \ \sim \phi(x)$ is not a reason to ϕ : that I promised to call you by noon and failed to call by noon is a reason to apologize, not a reason to call (have called?) by noon! And that I ϕ is not part of a reason for me to ϕ , so $p \ \& \ \phi(x)$ is not a reason for me to ϕ either.

Furthermore, an Aristotelian normative account of meaning doesn't require us to have any causal connection to the objects we speak about, and hence mathematical objects are no more problematic than physical objects. One form of semantic indeterminacy has to do with the observation that no collection of consistent finitely specifiable first-order axioms of arithmetic is sufficient to determine which mathematical object we are speaking about

⁵Cf. Faroldi (Faroldi, 139). Faroldi restricts this to non-causal explanations, but causal explanations are also hyperintensional by our argument below.

when we talk of “the natural numbers”. This indeterminacy comes in two varieties. One variety is famously pointed out by **BenacerrafIdentification** (**BenacerrafIdentification**): there are many sets of abstract objects that could just as well serve as natural numbers, none of them privileged. For instance, we might take $0 = \emptyset$, $1 = \{\emptyset\}$, $2 = \{1\}$, $3 = \{2\}$, and so on. Or we could just as well take $0 = \emptyset$, $1 = \{0\}$, $2 = \{0, 1\}$, $3 = \{0, 1, 2\}$, and so on. One may not worry too much about this if all of these admittedly different structures give rise to exactly the same truths about the naturals (e.g., in each one Fermat’s Last Theorem holds). The second variety, however, is more serious and comes from Gödel’s first incompleteness theorem. For any finitely characterizable and consistent collection A of first-order axioms, there will be an arithmetical sentence g and a pair of natural number structures satisfying A but in one of which g is true and in the other it’s not. Since plausibly there is a fact of the matter as to whether any specific arithmetical sentence is true or false, this is problematic, since our finite linguistic resources seem incapable of specifying which natural number structures we mean in a way sufficient to determine the truth or falsity of all arithmetical sentences.

Both problems can be resolved in a normative way. To resolve the first, we need only suppose that our assertions about the natural numbers have norms of appropriateness that involve facts about a specific set of abstract objects. For instance, we can suppose a norm that determines that when our language is “number-like”—when the terms of the language play a certain kind of “number” or “counting” role—then the language refers to a specific set of abstract objects. Any such specification of a “number-like” term is going to be extremely complex, and may involve many contextual and social features, and will raise many Mersenne questions, but as was argued in ??backref, the norms in human nature can indeed be very complex. And then even if a different set of abstract objects would yield a logically equivalent norm, the hyperintensionality of norms can privilege the specific set of abstracta as the grounds of appropriateness of arithmetical statements.

This approach would immediately resolve the second problem as well. But if we don't like abstracta or if we like Benacerraf's own structuralist answer to the first problem—namely, that any collection of things playing the role of natural numbers that has the same truths is as good as any other—we can resolve the second problem by allowing our nature to have norms with an infinite amount of information to specify what arithmetical structure we mean.

3. Sharpness and levels

3.1. Sharpness at the second-level.

3.1.1. *Declarative practices and reasons.* We should all agree that it would be possible to have an assertion-like communicative practice on which a declarative utterance instead of expressing a specific proposition subject to classical logic, expresses a *family* of propositions, and the utterance has the following axiology: it is bad if none of the propositions is true, it is good if all of them are true, and it is neutral if some but not others are true. We take someone to be engaging the communicative practice well (badly) to the extent that their utterances are good (bad).

We can complicate things in various ways. We might, for instance, make the family of propositions itself be fuzzy, in the technical sense that we assign a degree (say, between zero and one) to which a given proposition is in the family. Then we might say that an utterance is maximally good provided every proposition even partly in the family is true, and maximally bad provided every such proposition is false. But for utterances on which there is no such unanimity in the fuzzy family, we find a way of measuring an intermediate value. Thus, the family of propositions corresponding to "Alice is rich" may contain to a high degree the proposition that she is at least a millionaire, to a low degree the proposition that she is at least a billionaire, and to zero degree the proposition that the sky is blue.

Such practices are perfectly comprehensible within an Aristotelian framework: we can suppose that human nature makes some declarative utterances fulfill us, others be contrary

to our fulfillment, and others be indifferent to us. I have not yet claimed that our practices are like that, just that practices like that are perfectly comprehensible.

However, in fact, it is very plausible that some of our everyday utterances are not merely epistemically vague. An inquiry into whether three rocks can make a heap or exactly how much money one needs to have to be rich seems a waste of time, not only because the answers are useless to us, but because there is no answer. In particular, it is very natural to say with the supervaluationists that there are infinitely many ways to make “heap” and “rich” precise, for each of which there is a well-defined answer to the question, but without such precisification there is no answer. This fits well with the hypothesis that our everyday communicative practices are like the ones described above.

Our description of the practices presupposed that there are propositions that are simply true or false, and that given the truth values of the propositions, the values of declarative utterances—whether just bad, neutral and good, or more fine-grained ones—are thereby determined. However are they determined sharply, definitely? Isn’t it plausible that not only is it vague whether some group of four rocks is a heap, but one can have a grouping of rocks—say, five of them—where it’s vague whether it’s vague or whether it’s definite that it is a heap, so that it’s vague whether the statement “These rocks are a heap” is neutral or good (or it’s vague what exact value it gets).

Now the value of an utterance gives rise to reasons. If an utterance has positive value, that constitutes a reason to make the utterance. If it has negative value, that constitutes a reason to refrain from making it. And if it has no value, then we neither get a reason to make it nor a reason not to make it. But now recall the argument from ??backref that moral evaluation cannot be non-epistemically vague. If moral evaluation were more than epistemically vague, and yet we had classical logic, then the right account of that vagueness would be that there are many moral concepts that fit with our usage of terms like “is right” and “is wrong”, no one of which is privileged. But the central importance of morality to our rational lives requires privileging. Nothing can ultimately compete with moral wrongness as a reason against an action.

And what was true of moral concepts is *a fortiori* true of reason concepts in general. Nothing can ultimately compete with *reasons* in guiding our lives. Just as we might say that the overridingness of morality is the central discovery of ethics—explicit in Socrates—the centrality of reasons is the central discovery of action theory. It is difficult to say more here than to bang one's fist on the table. The concept of a reason is not an unprivileged one among many closely similar concepts. And this undercuts the possibility of non-epistemic vagueness about reasons. But because of the way the good and the bad immediately give rise to reasons, it follows that they cannot be vague either. And hence in the supervaluationist social practices described above, where declarative utterances express a family of precisifying propositions, there must always be sharp facts about whether the family is such that, respectively, all, none or merely some of its members are true. In other words, vagueness must stop at the first level.

A reason-giving practice can only generate sharp reasons, since there are no other reasons. Thus if vagueness is somewhere involved in a reason-giving practice, that vagueness must be “flattened out” once reasons are generated. Depending on the details, there may be more than one way of doing that. For instance, plausibly there is first-level vagueness about a person's height: there are multiple ways of measuring without any privileged one, depending on how much upward stretching is allowed, what gravity regime the measurement is made in (remembering that even on earth, gravitational acceleration varies about half a percent with location), whether skin flaking at the top of the scalp is included, etc. A practice that rewards the “tallest person” (say, a Guinness World Record practice) might generate a reason to bestow the reward on a person who is definitely the tallest, or on one who is definitely or vaguely the tallest, or it could (likely unfairly) flatten the vague data about height into sharp facts about reasons in some more complex way, especially if the family of precisifiers of “*x* is the tallest person” comes along with a degree-of-membership function.

In principle, any finite number of levels of vagueness could also get so flattened out. Thus, if there is second-level vagueness but no third-level vagueness, one might specify

that there is reason to give the reward to someone who is definitely definitely tallest or definitely vaguely tallest or vaguely definitely tallest, and to no one else. And then we have fully sharp reasons, since there will always be a definite fact of the matter whether there is a reason to give the reward absent third-level vagueness. In light of this, one might think that the sharpness of reasons argument would allow declarative practices corresponding to any finite number of levels of vagueness. A limitation of vagueness to a finite number of levels would itself be a significant result.

However, there is reason to hold out for one level of vagueness in our declarative practices. The most plausible account of the flattening in our ordinary declarative practice *D* is:

- (5) There is *D*-reason to declare *p* if and only if *p* is definitely true.

Thus, if the existence of a reason is a definite matter, as I have argued, then whether *p* is definitely true should be a definite matter. And if definiteness is definite, so is vagueness, since, definitely, *p* is vague if and only if neither *p* nor $\sim p$ is definite. Thus vagueness stops at the first level.

The main competitor to (5) would be:

- (6) There is *D*-reason to declare *p* if and only if *p* is definitely or vaguely true.

Given that (6) would give us reasons to affirm evidently contradictory sentences in cases of vagueness, it is not very a plausible candidate for an account of the relation of *D*-reasons to truth. However, even if (6) were the correct account, as long as it was definitely the correct account, the sharpness of “There is *D*-reason to declare *p*” would imply the sharpness of “*p* is definitely or vaguely true”. Now, it is definitely true that

- (7) *p* is definitely or vaguely true if and only if $\sim p$ is not definitely true.

Hence, we would have sharpness of “ $\sim p$ is not definitely true”, and hence we would have sharpness of “ $\sim p$ is definitely true”. But since it’s definitely true that *q* is definitely true if and only if $\sim \sim q$ is definitely true, letting *p* be $\sim q$, we conclude that we would have

sharpness of “ q is definitely true”. Since this would work for all q , we would again have sharpness of definiteness and of vagueness in general.

One might try for some other flattening of vagueness profiles to reasons. But (5) and (6) seem to be the most plausible two candidates. Thus we have good reason to stop at first level vagueness.

Finally, we might also argue for the claim that vagueness stops at the first level by thinking about the specifics of the morality of promises. If I promise to ϕ , and I do in fact ϕ , then I have done morally well; if I do not in fact ϕ , then I have done morally badly. But what if it’s vague whether I have ϕ ed? Let’s say that I have promised to cure your baldness, and because of my treatment you have some meager tufts of hair that you didn’t have, not enough to make it definite that you are non-bald and yet not enough to make it definite that you are bald. Did I do well or badly? Well, it is very natural to say: my activity was neutral in respect of the promise. Given a sharpness in attributions of moral goodness, badness and neutrality, and given the above plausible respective matching of moral value with attributions of definite truth, definite falsehood and vagueness, we have good reason to think that in the case of predicates that can figure in promises at least, we can at most have first-level vagueness—it must be sharp whether someone is definitely bald, vaguely bald or definitely non-bald. But if we have second-level sharpness about baldness, plausibly we have second-level sharpness about everything.

3.1.2. *Logic*. A standard argument for full-blown epistemicism is the Sorites series.??Sorensen Consider this argument, where the if-then statements are material conditionals.

- (P_0) Charles wasn’t old on his first day.
- (P_1) If Charles wasn’t old on his first day, he wasn’t old on his second day.
- (P_2) If Charles wasn’t old on his second day, he wasn’t old on his third day.
- ...
- (P_{26999}) If Charles wasn’t old on his 26999th day, he wasn’t old on his 27000th day.
- (C) So, Charles wasn’t old on his 27000th day.

Clearly C is false: a 73-year-old *is* old. The argument, however, is valid by a sequence of 26999 instances of *modus ponens*. The only way a valid argument can have a false conclusion is by having a false premise. Now, premise P_0 is clearly true. Thus, at least one of the premises P_n , for $n = 1, \dots, 26999$, is false.

Now, if P_n is false for $n \geq 1$, then since P_n is a material conditional, its antecedent is true and its consequent is false. Thus, Charles wasn't old on his n th day but became old by his $(n + 1)$ st day. Hence, we have a one-day transition from young to old. And it is precisely such sharp transitions that non-epistemicist advocates of vagueness reject as absurd.⁶ Yet logic forces us to accept them.

Plausibly, logic applies to propositions, not to sentences of our declarative practices. If we consistently precisify the premises and conclusion of the argument, the precisification of one premise will turn out to be false.⁷

On the second-level sharpness view, we can say that P_n is definitely true for small n , then becomes vague (maybe somewhere around $n = 22000$), and finally becomes definitely true again. The points at which P_n becomes vague and then again definitely true are fully precise, which is counterintuitive, but that counterintuitiveness is less evidentially significant than the violation of common sense in Charles becoming old on a specific day.

If we think a sentence is bad to say when definitely false, neutral when vague, and good when definitely true, then none of the vague P_n are bad to say, but the conclusion C is bad to say. We might think that this means that we have a case where something bad to say logically follows from a number of things that are not bad to say, and this may seem absurd. But it's not clear that this is absurd. For we are not here dealing in propositions, where a conjunction of acceptable ones is also acceptable, but with sentences in a vague

⁶We should assume that by "old" we mean something like "calendrically old". For we all understand such locutions as "Alice became old the day she found out she had lung cancer."

⁷Only one. For if Charles is old on a day, he's old on all subsequent days. If P_n is false for $n \geq 1$, then the consequent is false, so Charles is old on his $(n + 1)$ st day, and if P_0 is false, then Charles is old on his first day. Either way he is old on day m whenever $m > n$, and so P_m is true, since it's a material conditional with false antecedent. Thus, as soon as one premise is false, all the subsequent ones must be true.

declarative practice. We should not import logical intuitions that apply to propositions in thinking about the value of a declarative practice. It may also help to see that in the case of *moral* badness there is nothing particularly paradoxical about cases where asserting a conjunction is bad but no conjunct is bad to assert. For instance, consider the case of a racist who writes a series of factually correct and interesting articles, each one about a different highly immoral member of the very same minority group. Each article may be such that it is not bad to write, but the oeuvre as a whole is racist in light of the tendentious selection of subject matter.

We do, however, have a famous difficulty. Like other supervenience views, the account being defended has the consequence that definitely:

- (8) There exists n between 0 and 27000 such that Charles is not old on day n but is old on day $n + 1$.

At the same time, for any specific day m in that range, say $m = 22003$, it is *not* definitely true that:

- (9) Charles is not old on day m but is old on day $m + 1$.

In particular, the disjunction of instances of (9) as m ranges over all the numbers between 0 and 27000 is definitely true, but every disjunct is merely vague. This is, admittedly, counterintuitive. This counterintuitiveness is, I suspect, tied to the fact that if we have a true disjunction of *propositions*, at least one disjunct is true. Furthermore, there are cases where a disjunction is definitely true, but plausibly no disjunct is. For instance, it seems to be vague whether a hot dog is a sandwich. But it is definitely true that a hot dog is a sandwich or not a sandwich by the Law of Excluded Middle.

3.1.3. *Some objections.* The conclusions above are counterintuitive. Intuitively, just as it sounds silly to think that on such-and-such a day it was true to say that Charles was (chronologically) old while it wasn't true to say it a day earlier, it sounds silly to say think that there was a precise day on which he was definitely old, while the day before he was merely vaguely old.

However, it is to be expected that our linguistic intuitions are typically more trustworthy than our metalinguistic intuitions, since our linguistic intuitions are more important to our flourishing as social animals. Thus, our intuition that oldness can be vague is more to be trusted than our intuition that vagueness can be vague. We have good arguments for second-level sharpness, and these arguments undercut the intuition.

A second objection starts like this. Communicative practices of a declarative sort that embody first-level vagueness are indeed possible. All we need to do is to specify the values in the way we did at the beginning of Section 3.1, so that an utterance is good when all the propositions in an associated family are true, bad when they are all false, and neutral otherwise. If we had a community of perfectly sharp speakers, we could even imagine introducing such a practice, either as a sort of pleasant relaxation (it's not fun to always have to be precise) or for practical reasons, to be engaged in at times. Suppose we have such a practice, and it includes terms like "hairs" and "scalp". But then we could engage in this practice to introduce a new non-sharp communicative practice. For instance, we might specify that in the new practice a predication of "is bald" is good when the person has more than 2000 hairs on their scalp, is bad when the person has fewer than 1000, and is neutral otherwise. However, how much filament needs to stick out of a follicle for the follicle to count as hosting a hair itself appears to be fodder for first-level vagueness matter, and it can be first-level vague whether a hair is on the scalp or the upper part of the cheek. Consequently, in the new practice we can have cases where it's vague where Jim's crinal profile falls—whether he has fewer than 1000 hairs on the scalp, more than 2000, or in-between. And in such a case it may well be vague that it's vague whether he is bald.

Now, given the possibility of such a practice, its actuality is likely. Surely we often do extend our communicative practices, using old—and presumably vague—terms to introduce new ones. And we do that with our vocabulary.

However, it is not clear that we can successfully institute a practice with vague values. The rules of a practice provide reasons for the practitioners. But if it cannot be vague whether something is a reason for ϕ ing, a reason against ϕ ing, or neither, then we cannot

make it vague what the rules defining the practice are and how they apply. What we can do is at most something this. We can have a practice of instituting practices, where we specify a collection of ordered “rule” pairs (D, E) where D is a description of a behavior (or maybe situation) and E is an in-practice evaluation, such as “permissible” or “bad”. Our practice of instituting practices then specifies that given the rule pair (D, E) , then $D(b)$ is definitely true if and only if b has E . In cases where b does not definitely fall under the description D in any of the rules, then we can simply say that b definitely lacks all of the relevant evaluative properties.

For instance, the rules for an oversimplified race r could be (A, W) , (B, T) , and (C, L) , where W is a win, T is a tie, and L is a loss for our (contextually specified) runner, and $A(r)$ says that r is a performance that is a run with the upper body of the runner crossing the finish line before the upper body of any other runner, $B(r)$ says that r is a performance that is a run with the runner’s upper body never crossing the finish line or another runner’s crossing earlier, and $C(r)$ is the denial of $A(r)$ conjoined with the denial of $B(r)$. Now suppose that in our first-order declarative language, sentences like “Alice’s upper body crossed the finish line at t ” are vague when what crossed the finish line at t was a loosely attached scab on the forehead. Then in cases where Alice was ahead only by such a scab, Alice definitely lacks a win, a tie or a loss. If a win and a tie have positive valence, while a loss has negative valences, then we can say that Alice’s performance definitely is neutral.

That all our games are in fact perfectly sharp in their values is counterintuitive. But it is hard to avoid this conclusion while holding on to the privileged role that reasons play in our lives that forces reasons to be sharp.

3.1.4. *A sharp world and a somewhat fuzzy language.* A broadly-held intuition is that the world is sharp but our language is fuzzy, so all the vagueness is due to our language. There is a well-known objection to this: our language is itself a part of the world, so linguistic vagueness is still vagueness about the world (**MerricksVV**).

The view defended above where vagueness is restricted to the first level does justice to the sharp-world–fuzzy-language intuition. Our declarative sentences express a range of

propositions, maybe even a graded range, and that's the fuzziness of the language. However, the propositions in the range are themselves sharp, and the truth of the sentences is derivative from the truth of the propositions, so ultimately the world impacts our language through the sharp end of our practices, and the world itself can be said to be sharp.

At the same time, the view manages to escape the objection that our language is a part of the world. The relevant part of the world is our declarative practice. And the semantics and norms of this practice are fully sharp. There is always a definite answer to the question whether a given proposition is a precisification of a given declarative sentence, to what degree (if the account involves degrees), and what the norms governing the use of the sentence are. It can be vague whether a sentence is true, but it is not vague what the sentence means: it means its set of precisifiers.

Requiring a piece of sports equipment to be between exactly 200 and 250 grams does not render a sports practice vague. Similarly, the fact that the evaluative properties of our declarative language are sometimes defined in terms of ranges of propositions, rather than individual propositions, does not make the practice itself vague. Linguistic vagueness on this view does not mean that linguistic practice as a practice is vague. It just means, very precisely, that some of the sentences of the language express a range of propositions. This kind of vagueness does not make the practice itself vague, and does not introduce any vagueness into the world.

On the other hand, if there were higher-level vagueness, so that sometimes it was vague whether a sentence is, say, definitely true, the linguistic practice as such would be vague. For the practice depends normatively on whether a sentence is definitely true: there is a practice-internal good to declaring a sentence that is definitely true which is not had when one merely declares a sentence that is vaguely true. ??Merricks

A closely-related intuition is that the world as it objectively is is sharp, but the world as it is relative to us is fuzzy. Someone who accepts this intuition may insist that practice-internal values are themselves relative to us, rather than a part of the objective furniture of the world. Again, this is a difficult line of defend given the observation that if something

has a practice-internal value for some individual x , then it is an objective fact about the world that it has that practice-internal value for x .

Perhaps, however, one could read this intuition as implying that there is second-level vagueness but no third-level vagueness. Thus, one might say that for our declaratory practice, it is not sharp whether a performance is good, neutral or bad, but there is some objective range of precise interpretations of that practice, and we say that a performance is definitely good provided it is good on all interpretations, vaguely good if good on some but not all, and definitely not good if good on none. What that range is, however, is an objective fact about the world. Within each interpretation, the evaluation of a performance will be sharp. However, there does not seem to be any significant philosophical benefit to placing the sharpness at the third-level rather than the second. We are left with the problematic idea that the concept of a reason is subject to multiple precisifications, in a way that does violence to the overridingness of rationality, and besides we have a more complex view, while still requiring an account of the third-level sharpness, which is no easier to have than an account of second-level sharpness. And while I have argued that the intuitions supporting first-level vagueness are on more trustworthy than those supporting second-level vagueness, there does not seem to be as significant a distinction between the support for second- and third-level vagueness. Thus the benefit from allowing second-level vagueness does not seem that great.

4. Neo-Aristotelian metaphysics

The neo-Aristotelian metaphysics allows us to have a completely sharp world with our language being completely sharp, with there being definite facts of the matter about significant and insignificant questions: Is Alice dead yet? Is this pile a heap? Should Alice rebuke Bob publicly? Is the smashed object a car? Is this still the ship that Theseus sailed in? Is Beethoven a better composer than Bach? Should I lower my credence in quantum mechanics after the mildly senile retired physicist told me she just found a contradiction in it? Is a cat a dommal (where “dommal” is a term whose patterns of use are that users

are content to call all dogs “dommals” and are happy to infer mammality from being a dommal).

Our nature *could* provide us with this sharpness. First, when our statements concern normative matters, our form can at least partly ground the truth values. This can yield complete sharpness about all normative matters.

Second, our nature could ground the facts about the proposition expressed by and illocutionary force of each of our utterances by grounding the norms that specify how facts about our symbolic behavior together with facts about the non-symbolic aspects of the world attach propositions and illocutionary force to utterances. These rules could be very complex, and known by us only approximately and in general terms.

But at the same time, instead of grounding what specific proposition is expressed by each declarative utterance, our nature could ground facts about expressed ranges of propositions, thereby allowing declarative utterances to be vague with multiple precisifiers. Again, our nature can do this by fully precisely grounding our semantic norms—for, as we saw in ??backref, it is possible to have fully precise semantic norms and yet genuine first-order vagueness. This seems to be the preferable view all things considered.

Finally, and independently of whether there is any first-level vagueness, we can ask how exactly the metaphysics grounds the norms of language? Obviously, it is not tenable that it is, say, written into our nature that saying “Gift!” should indicate a present, since then we have the absurd conclusion that Germans speak incorrectly when they indicate poison with “Gift!”⁸ While it could be that there are some symbolic gestures that have an innate meaning—pointing, say—and then the human nature would have norms constitutive of this meaning, it is plausible that the bulk of the relevant semantic norms are such as to yield complex conditionals $L_{C,S,N}$ that under circumstances C , a communicative action of type S has normative features N . The specification of circumstances C is likely to include details about the context, statistical facts about the instances of S and of other communicative actions, as well as historical facts about the development of a language.

⁸I take the capitalization to be insignificant in speech.

The normative features can be intrinsic features of the instance of S —such as that S is apt or inapt—or they can be causal features, such as that S leads to an obligation to ϕ .

We can suppose that human nature is so complex as to directly include a vast and perhaps infinite number of $L_{C,S,N}$ facts, for different triples of C , S and N , but it is more plausible to suppose a somewhat simpler human nature that contains general semantic rules from which facts of the form $L_{C,S,N}$ follow. But even this somewhat simpler human nature likely involves a large number of seemingly arbitrary parameters, answering questions such as how present usage is weighted against historic usage in determining meaning, how do regularities of use in finer-grained communities (families, towns, scholarly disciplines, etc.) interact with regularities of use in coarser-grained communities (provinces, countries, etc.), how is agreement on a definition weighted against actual patterns of use (e.g., many will say a bachelor is an unmarried man but will not be willing to classify the pope as a bachelor), how much effect on semantics comes from the fact that a proposed meaning cuts nature less or more along the joints, and so on. We do not know these rules except in the broadest of outlines. But at the same time, given Aristotelian optimism, we have some hope of furthering our knowledge of them by empirically studying the semantic judgments of actual human beings as linguists do.

5. Norms of assertion

Under what conditions may we assert a proposition p ? The literature contains a number of answers, of which the main three are that you (respectively) believe (**Bach2008**), justifiably believe (**Kvanvig2009**), or know (**WilliamsonKn**) p .⁹

⁹There may also be some edge cases where none of these seem right. For instance, suppose that you are temporarily deafened after an accident and you do not even know whether you still have the power of speech. You turn to a friend and try to say: “At least I can still speak!”, and gauge from their reaction whether what you said was true. In that case, you don’t believe or know that what you are saying is true, but if you do succeed in speaking, it is guaranteed to be true, so you are not being dishonest. Handling such cases in an account of the norm of assertion is difficult. It is tempting to modify suggested norms to say that you believe, justifiably believe or know the conditional that if you succeed in asserting p to your audience, then p is true. But it is not

To a Bayesian who thinks that belief and justification always come in degrees, all three norms come with an implicit or explicit threshold of the credence and evidential levels at which assertion becomes appropriate. And then we have a Mersenne problem: what explains or grounds why the threshold is where it is.

A non-Bayesian might think that belief, justification and/or knowledge are binary distinctions that cut the epistemic world at the joints. If so, then it may seem that we have no Mersenne problem for assertion, and indeed the lack of the problem is an advantage over Bayesian approaches. However, the problem returns when we take into account the complexity in real-life assertions. For the level of belief and/or justification needed for an assertion does in fact vary from case to case, depending on context and confidence markers.

Expert witnesses speaking under oath need to be more confident of what they are asserting than friends tossing around ideas over a beer, with a vast amount of contextual variation in between. A realistic account of the norm of assertion needs to take such context into account. Furthermore, even when we keep context fixed, our tone of voice, body language (including facial expression), and choice of wording communicates the level of confidence. While one might not be lying if one says something that one just barely knows with a voice communicating the kind of justified confidence that a mathematician has in theorems verified by referees, one is violating the norms proper to asserting with that tone of voice.

It is clear, thus, that a realistic account of the norms of assertion will have to at least involve something like functions from pairs (C, S) where C specifies the conversational context and S describes the speaker-provided cues to the degrees of belief, confidence, justification and/or knowledge relevant to assertion. These functions are unlikely to admit of any simple mathematical expressions. Even if we neglect the social complexities of

clear that in ordinary assertions you need to have any beliefs about such complex conditionals (**PrussLying**). These complications would only serve to strengthen my case for the presence of Mersenne problems, and so I will simplify the discussion by ignoring them.

context, speaker cues themselves involve at least the three dimensions of tone, body and wording, weighed together in a complex way. Yet it is clear that we have norms here—and indeed morally relevant norms—and we need an answer to the question of what grounds the functions involved in the norms being as they are.

It is plausible to say that the functions here are largely socially defined. The same tone of voice in one culture may convey confidence and in another something quite different. However, that only shifts the Mersenne problem. If previously we were, say, trying to find a ground for a function f from context-cue pairs to required degrees of belief, we now need a grounding for the (second-order) function g that assigns to a set R of relevant social regularities a “threshold function” $f = g(R)$ from context-cue pairs to required degrees of belief. Slight reflection should show that our function g can be expected to be rather complex.

We cannot say, for instance, that the degree of confidence $f(C, S)$ required for a context-cue pair (C, S) is the average amount of confidence in fact occurring for (C, S) in a culture with regularities R . Presumably, the majority of assertions in a culture are appropriate. And among the appropriate assertions, most not only meet the required threshold of confidence for their context-cue pair, but likely exceed it at least slightly. This suggests that it should be at least possible in a culture that the average level of confidence for asserting in the case of a given context-cue pair is higher than the minimum required level, and so we cannot define the threshold as the average.

We would likely be closer to correctly predicting the mark of the threshold if we set the threshold some ways below the average, say specifying that we should be one standard deviation below the average level of confidence. But that would likely lead to a Mersenne problem: Why is it one standard deviation below the average, instead of some other statistically attractive measure, such as being at the 25th percentile level of actual confidences for those context-cue pairs? Furthermore, such formulas are likely to fail in a hypothetical society where everyone is perfectly honest, and hence always asserts at or above the required threshold of confidence.

Further complexities arise from local (spatial, temporal and other) variation in cultures, even that between neighborhoods in a single city. The sample sizes for the local regularities R are likely to be small, and the function g from regularities to threshold-functions will need to take into account both local and non-local regularities, with some sort of a weighting in favor of the local ones. This is likely to be quite complex, and the Mersenne question of the explanation of why this exactly function—doubtless with many apparently arbitrary parameters—is the one that governs our assertive behavior is indeed very obvious.

The Aristotelian, however, can embrace all this messy complexity, and say that the human form specifies the function from social regularities to threshold-functions.

6. *Natural numbers

Very plausibly:

- (10) any first-order statement about natural numbers and their arithmetical operations is determinately true or determinately false, whether or not we know its truth value.

But what is true of the natural numbers and their arithmetical operations will frequently be false of other sets of mathematical entities with their operations. One may initially conclude from (10) that we are thus able to gain unambiguous reference to *the* natural numbers and their operations in our language or thought, and then we will have a puzzle as to how we manage to pick out the natural numbers from among the infinity of mathematical objects.

But that's too quick. We learn from **BenacerrafIdentification** (**BenacerrafIdentification**) that the concept of *the* natural numbers is dubious. Set theorists frequently identify the sequence $0, 1, 2, \dots$ of natural numbers with the sequence of sets $\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \dots$ (i.e., identify 0 with the empty set, and then identify each number with the set of its predecessors). It is implausible to think that this strange set-theoretic construction captures *the* natural numbers, but it does capture a structure that for all practical purposes behaves (with appropriately defined operations) like them. But other sequences could be used

equally well. For instance, we might go for the sequence of sets $\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \dots$. Or, following Frege, we might identify a number n with the class of all sets with n elements.

Suppose we take the lesson from this to be that there is no such thing as *the* natural numbers. There will still be “arithmetical truths” and “arithmetical falsehoods”. There is still no reasonable doubt about the truth of the magnificent ancient Greek discovery that for every prime number there is a larger prime number, or that Goldbach’s Conjecture that every even number bigger than two is the sum of two primes is either true or false. It remains highly plausible that (10) is correct, even if there is no such thing as *the* naturals. This requires that even if we do not pick out a specific sequence of naturals with their arithmetical operations, our language or thought somehow manages to characterize a “natural number structure”—an infinite plurality of objects with appropriate arithmetical operations—with sufficient determinacy to fix the truth value of any first-order statement about them.

But how do we do that? Taking a cue from Euclid, one might say that we characterize natural number structures by a collection of axioms. For instance, we might think of natural numbers in terms of a “successor” operation, and begin by saying that every natural number has a successor and no two numbers have the same successor, and continue on through our favorite axioms, such as the Peano ones. The problem with this is that Gödel’s first incompleteness theorem says that any consistent recursively expressible axiomatic characterization that includes the obviously true Peano axioms is insufficient to determine all the truths about natural numbers, and hence will fail to yield (10). For any such axiomatic characterization, there will be a pair of mathematical structures, N_1 and N_2 , each of which satisfies the characterization, and an arithmetical statement p that is true of N_1 but false of N_2 .

There are several known solutions to this problem. The first is to simply deny (10), and allow a genuine indeterminacy to some arithmetic statements, which are neither true

nor false. This is counterintuitive, but if we could ensure that only very strange arithmetical statements have that indeterminacy, it could be acceptable. Unfortunately, we cannot ensure this.

Consider statements of the form

(11) Q can be proved from $P_1, \dots, P_n$

where $P_1, \dots, P_n$ and Q are sentences of first order logic. Such “provability” statements are not particularly strange, and I think we find ourselves with strong intuitions that for any $P_1, \dots, P_n$ and Q , there has to be a determinate fact of the matter about whether (11) holds. This claim seems close to the heart of the objectivity of logic. But as Gödel famously noted, sentences can always be encoded as numbers, and the provability relation can be encoded using arithmetic operations and quantifiers. As a result, (11) is equivalent to an arithmetical statement.

Suppose now that we restrict (10) to arithmetical statements that are translations of provability statements. Can we now characterize natural numbers in a finite first-order way that is sufficient to yield the truth values of these (translations of) provability statements? Again, the answer is negative, but the issue is more complicated. Consider some set A of first-order recursively specifiable consistent axioms including enough axioms of arithmetic. Let G be a Rosser-tweaked Gödel sentence (**Batzoglou**) for these axioms, chosen so that G is a Σ_1 sentence, i.e., has only one unbounded quantifier, which quantifier is existential. Thus, G is of the form $\exists n F(n)$, where all the quantifiers in $F(n)$ are bounded, i.e., of the form $\exists m(m < t \ \& \ \dots)$ or $\forall m(m < t \rightarrow \dots)$ (where t is a polynomial in the free variables). We can now construct a Turing machine M_G which iterates through all the natural numbers n , and halts as soon as it gets to $F(n)$. Thus, this Turing machine halts if and only if G is true. But a statement that a specific Turing machine M halts can be translated into a statement that a certain formula ϕ_M of first-order logic is valid, i.e., is provable from a tautology (cf. **Turing, Turing**). Putting all this together, and assuming our collection of axioms includes enough arithmetic to make the argument go, we will be able to prove from the axioms in question that ϕ_M can be proved from one’s favorite tautology if and

only if G . Since G is neither provable nor disprovable from A , we thus have a provability statement that is neither provable nor disprovable from A .

Thus, even if we restrict the arithmetical truths we want definite truth values of to the translations of provability statements, we won't be able to give a recursive characterization of first-order axioms that yield the truth values.

There is, however, an Aristotelian solution to the problem. While a finite collection of first-order axioms is insufficient to characterize all arithmetical truths, an infinite one can easily do so (if one is really profligate, one can just include all and only the arithmetical truths among the axioms!). We already saw in previous chapters that the human form likely has a vast dizzying collection of norms embedded in it. Why not an infinite collection? Thus, human nature could include infinitely many arithmetical norms, such as "One should use arithmetical concepts in such a way that it is appropriate to think that A_n " (or, alternately, one can have a linguistic version of this norm about what kinds of things it is appropriate to say), where A_n is the n th of the axioms, and where the axioms $A_1, A_2, \dots$ are sufficient to determine the truth value of all arithmetical statements (or maybe just of all provability statements—i.e., all of Σ_1).

The Aristotelian solution is not the only one, of course. We might have a Platonic solution on which, *pace* Benacerraf, there really is a privileged collection of *the* naturals in the Platonic heaven, distinguished from alternatives by something like greater "naturalness" in the Lewisian sense (**LewisNewWork**). Or we might accept causal finitism (**PrussInfinity**), which builds the concept of the finite—and hence of a natural number—into the metaphysics underlying causation. Or we might simply suppose a "magical" theory of reference on which there is a special relation between the human mind and a certain collection of mathematical entities (cf. **PutnamRTH**, **PutnamRTH**). And some think that moving to second-order logic solves the difficulty.

Nonetheless, the present Aristotelian solution still has a certain attractiveness. Unlike the magical theory of reference, we do not need to posit anything new about us—just more in the way of norms. And unlike the Platonic or causal finitist solutions, the Aristotelian

solution does some justice to the intuition that mathematical terminology is stipulated by our minds—except that the stipulation of the axioms of natural numbers is built into the nature of the human mind, rather than merely contingent. Furthermore, the Aristotelian solution allows for an epistemology of mathematics that is closely akin to the epistemology of ethics, where we know norms by having an intuitive though fallible tendency to follow them, and so a particular axiom of arithmetic can be accessed by our finding ourselves inclined to use arithmetical concepts in accordance with the norm encoding the axiom.

The second-order logic solution deserves some more discussion. In a second-order logic, we allow for quantification over predicates in addition to quantification over individuals. Arithmetic appropriately formulated within second-order logic then has a unique model on the standard semantics (**Enderton**). This seems to solve all our problems at little cost beyond allowing second-order quantification which one might think we have good intuitive reason to allow anyway.

However, it does not. The standard semantics for second order logic interprets quantification over predicates in terms of quantification over all sets. But this requires a fixed “true set theory” for the semantics. The same problem that we faced with how our thought and language picked out the “true natural numbers” returns as the problem of how our thought and language picks out the true sets, the ones that provide the standard interpretation of the second-order logic. If, on the other hand, we allow for non-standard semantics for second-order quantifiers, the Henkin semantics, where we do not interpret second-order quantifiers in terms of subsets, then we lose the unique characterization of the natural numbers, and our old problem is unsolved.¹⁰

7. Conclusions

A neo-Aristotelian metaphysical picture of us as having a human nature grounding a large number of complex norms can solve the problem of where communicative norms

^{10*}There is an internal categoricity result in Henkin models which generalizes the standard-model categoricity result by showing that the isomorphism of arithmetic across all models within a fixed Henkin model (**InternalCat**), but that does not seem to solve the problem of picking out the true arithmetic.

come from in the first place, how vagueness is to be resolved, what grounds the norms of assertion, and perhaps can even help with foundational questions in the philosophy of mathematics. Except perhaps for the mathematical case, in all of these cases it is very plausible that there is a species-relativity in the relevant norms, and a vast number of Mersenne questions, so the neo-Aristotelian solution appears superior to a view on which the norms hold necessarily.

CHAPTER VIII

Laws of nature and causal powers

1. Humean and pushy laws

1.1. Deterministic versions. There are two main types of theories of laws of nature. On Humean or deflationary views, we have laws simply in virtue of non-causal regularities of the behavior of objects in nature, and causation is then grounded in the laws. On pushy or inflationary views, we have laws in virtue of metaphysical components of reality that affect or constrain the behavior of objects. In this section, I will argue that pushiness is the right view.

The most prominent Humean account of laws is LewisCounterfactuals's (LewisCounterfactuals) Best System Analysis (BSA). The non-probabilistic version says that laws are the logical consequences of the "best system". The systems in question are collections of true sentences all of whose terms are "perfectly natural", in the sense of cutting reality perfectly at the joints. Thus, a system can contain the statement "All electrons have charge", but cannot contain the statement "All kittens are cute" because neither the concept of a kitten nor of cuteness is perfectly natural. (A kitten is an immature cat, and even if felinity is perfectly natural, immaturity is not.) We then compare systems along two dimensions: a briefer system is better and a system that conveys more information about reality is also better.

Brevity is straightforward once we fix a language with perfectly natural terms. Informativeness is more difficult to characterize, but there are obvious comparative judgments: that energy is conserved is more informative than that I am sitting now, since the energy claim tells us (a little) about each of the vast numbers of particles in the universe over all

of time, while the claim about my sitting tells us only about what is happening in a small region of spacetime.

Modern Humeans then tend to define causation in terms of laws. For instance, the Lewisian approach is to define laws in terms of counterfactuals, and counterfactuals in terms of similarity of worlds, with similarity of laws prioritized (**LewisArrow1979; LewisCounterfactuals**). The resulting account has pleasing ideological parsimony: both laws and causation reduce to the arrangement of matter, an arrangement called “the mosaic”.

We now observe some serious problems with BSA.

1.2. Problems with explanation. The most discussed problem with BSA is explanatory circularity. To introduce the problem, let us begin with an explanatory problem independent of BSA. The old Deductive Nomological (DN) model of explanation in the philosophy of science¹ had it that scientific explanations are deductively valid arguments for the particular fact to be explained, at least one of whose premises is a law. For instance:

- (1) All massive objects attract gravitationally.
- (2) The Sun is a massive object.
- (3) So, the Sun attracts gravitationally.

Of course, it was soon seen that having this form is not necessary for explanation—statistical explanations do not fit the deductive schema, and cases where the explanandum is not even probable given the explanans are particularly problematic, with the classic case being a patient’s having untreated syphilis explaining why the patient got paresis, even though paresis is not a typical outcome of untreated syphilis (**SalmonFourDecades**). It was also seen that having this form is not sufficient for explanation, as one needs somehow to account for the explanatory direction between the particular facts stated in the premises and the particular fact in the conclusion. A famous example here is how the laws of nature together with the length of a flagpole’s shadow yield a deductive argument for the position

¹For a historical discussion, see **SalmonFourDecades** (**SalmonFourDecades**).

of the Sun in the sky, but obviously the position of the Sun in the sky is explanatorily prior to the length of the shadow (**SalmonFourDecades**).

But even the seemingly innocent example (1)–(3) has a crucial problem, which suggests that explanations of the DN sort rarely if ever work. For consider that (1) is a universal generalization, and a universal generalization is true in part in virtue of its instances. That all massive objects attract gravitationally is partly grounded by the Andromeda Galaxy attracting gravitationally *and* partly by the Sun attracting gravitationally. But grounding is a form of explanation. Thus, the Sun's gravitational attraction is explanatorily prior to the universal generalization about massive objects attracting gravitationally. And yet the universal generalization is, according to the DN model, prior to the Sun's gravitational attraction. But loops in explanatory priority are implausible.

The most common kind of response to these kinds of arguments is that explanatory loops are permissible when the explanation in the two directions is of a fundamentally different sort. For instance, in the above case, the Sun's gravitational attraction is *nomically* or maybe *causally* explained by (1) and (2), while providing a partial *grounding* explanation of (1) (**LoewerLawsTime**). Nonetheless, it is not clear that it matters for the anti-loop intuition that the two directions of an explanatory loop are of the same kind. It is the circularity of explanation that seems to be the problem, regardless of the types of explanation.

For suppose you light a match in the kitchen. Then the friction applied to the match explains why something is burning in your house. But notice that this explanation consists of a chain of two explanations. The friction applied to the match in your kitchen *causes* or *nomically explains* (in conjunction with some appropriate laws) the match to be on fire in your kitchen, and the match's being on fire in your kitchen *grounds* the fact that something is burning in your house (particular cases ground existential generalizations). The causal/nomic and grounding explanations combine into a generic explanation that is neither just a causal/nomic explanation nor just a grounding explanation, but is nonetheless

an explanation. This shows that there is a generic concept of explanation which goes beyond particular types of explanation, like causal, nomic or grounding. And an explanatory loop with respect to generic explanation is problematic.

Or consider the following plausible transitivity principle:

- (4) If D partially grounds E , and E partially scientifically explains F , then D partially explains F .²

But now the sun's gravitational attraction partly grounds (1), which partly scientifically explains the attraction on the DN model, making the sun's gravitational attraction be partly self-explanatory by (4).

We have thus seen that paradigmatic cases of DN explanations can be troubling even without Humean assumptions. However, there is a plausible explanation for the initial intuitive appeal of DN explanations like (1)–(3). When we offer scientific explanations in terms of a generalization that is a law of nature, we are implicitly invoking the fact that the generalization is a *law*, rather than merely depending on the generalization being *true*.

To support this point, suppose that we live in a world with only one massive object, Bob, but where nonetheless it is a law that all massive objects bend spacetime in some

²**LangeGrounding (LangeGrounding)** gives a version of this principle with the stronger conclusion that D partially *scientifically* explains F . This stronger conclusion may not be justified, however. For the partial ground of a scientific explanandum could itself be something outside the domain of science. For instance, many theists will hold that the existence of any creature is partly grounded in God's activity—indeed, people talk of God as “the Ground of Being”. But now consider the scientific explanation of birds in terms of the prior existence of dinosaurs from which they evolved. On the theistic story, the prior existence of dinosaurs is partly grounded in God's activity, and it is reasonable to conclude that therefore the existence of birds is partly explained by God's activity—but not that it is partly *scientifically* so explained. Even apart from theism, there is likely a layer of fundamental philosophical ontology—say, objects having tropes—that grounds the explanans in a typical scientific explanation, and yet it seems wrong to invoke the objects of philosophical ontology in scientific explanation. (Interestingly, our weaker version of the transitivity principle does not suffice for the anti-Humean argument for which Lange designed his principle, unless one additionally posits the transitivity of partial grounding.)

specific way. One might think that this thought experiment will be rejected by the advocate of BSA on the grounds that no universal generalization that has only a single instance can be a law. But this is not clear. First, a single instance can be extremely informative, and in our example we can suppose that the fact that all massive objects bend spacetime is highly informative. Thus, many contemporary Humeans accept as a law the “past hypothesis” which says that the initial entropy of the universe is low³, because the past hypothesis is so informative when combined with other laws, even though it is a hypothesis that applies to only one time. Second, on standard Lewisian BSA, the laws are the logical consequences or theorems of the best system. But a universal generalization that has only one instance can be a logical consequence of a law that applies to many other instances. For instance, it follows from the law of gravitation that all massive natural satellites of planets on which a philosopher was executed by being made to drink hemlock attract gravitationally, and hence this complex universal generalization is a law. But as far as we know, there is only one such natural satellite, earth’s moon. Similarly, it could be that the fact that all massive objects bend spacetime in a specific way is a special case of a more general fact about what kinds of objects bend spacetime and how they do so, and the more general fact could have more than one instance.

Now in this world with one massive object to say that Bob bends spacetime because he’s massive and all massive objects bend spacetime is silly: when there is only one instance, we surely cannot explain the instance in terms of the universal. But it is not at all silly to say that Bob bends spacetime because he’s massive and *by law* all massive objects bend spacetime, even if Bob is the only massive object. Of course, we do not live in a world where there is only one massive object, but plausibly the logical form of the explanation of a particular massive object bending spacetime is no different in our world than in that one.

³See **ChenPH (ChenPH)**, who also argues that the past hypothesis can be seen as a fundamental law on some non-Humean accounts.

Now the hypothesis that nomic explanations invoke a law *as a law*, implicitly or explicitly, can help solve the explanatory circularity model facing DN explanations, by modifying (1)–(3) into:

(5) By law, all massive objects attract gravitationally.

(6) The Sun is a massive object.

(7) So, the Sun attracts gravitationally.⁴

But now note that the Humean cannot make use of this solution to the circularity problem. For, first, on a Humean analysis, just as the Sun's attracting gravitationally was a partial ground of (1), it is also be a partial ground of (5), since it is in part because this universal generalization is true that it is found in the best system.⁶

And, second, even apart from the circularity problem, explanations of the modified DN sort are problematic for BSA. While adding irrelevant facts to an argument's premises does nothing to damage the argument, irrelevance damages an explanation **SalmonStat**. But on BSA, the proposition that by law all massive objects attract gravitationally tells us that the proposition that all massive objects attract gravitationally is a logical consequence of the system of truths that best optimizes a balance of informativeness and brevity. But *this* fact does not seem explanatory. That there is a system that best balances informativeness and brevity and has as a logical consequence that all massive objects attract does not seem to

⁴Whether this remains deductively valid depends on whether its being the case that by law *p* it follows that *p* (i.e., whether "by law" is a modal operator satisfying Axiom T). While this seems a natural assumption, it has been disputed.⁵ We need not settle this question. If the argument is not deductively valid, it is still a good ampliative argument form, and it is clearly explanatory. And the Humean thinks laws are true.

⁶Here I am assuming that the law of gravitation is fundamental. It is not *always* true on BSA that a universal generalization is a law in part because it's true. It might be that a universal generalization is a law because it is a logical consequence of the more fundamental laws that are axioms of the best system. But if that is the case for (5), then just need to replace the example with a more fundamental one or work in a world where the universal gravitational attractiveness of massive objects is a fundamental law. For the fundamental laws, their inclusion in the system is explained in part by their truth, as the axioms of the best system must be true.

contribute to an explanation of why a particular object attracts gravitationally. Facts about brevity of expression are relevant to giving physics-based explanations of the movements of planets.

For imagine a family of worlds where objects with mass attract gravitationally, but so do objects with a certain other fundamental property Q . Suppose further that there is nearly total overlap between massive objects and objects with Q , so that both:

(8) All massive objects attract gravitationally

(9) All Q objects attract gravitationally

are highly informative, and equally brief. But now imagine two worlds, w_M and w_Q . In w_M , the number of Q objects is N for some large N , and the number of massive objects is $N + 1$, while in w_Q , the number of Q objects is $N + 1$ while the number of massive objects is N . Thus, in w_M , it is (8) that is the more informative, while in w_Q , it is (9) that is the more informative. Suppose there are no other regularities that make it into the best system, and suppose that we include (8) in the best system for w_M , but adding (9) would give so little extra information (since there is only one object that has Q but no mass) that only (8) makes it in. On the other hand, for analogous reasons, the laws of w_Q include (9) but not (8).

The proposition (5) in w_M is then grounded in part in the fact that there is one more massive object than Q object in w_M (unlike in w_Q). But the fact that there is one more massive object than Q object is not explanatorily relevant to the Sun (assuming the Sun to exist in w_M) being gravitationally attractive. Yet that fact is explanatorily prior to (5), which in turn is explanatorily prior to the Sun's attractiveness.

In summary, to avoid circularity, we need to invoke laws *as laws* in our explanations. But lawfulness on BSA consists of facts about truth, informativeness and brevity. The facts about truth ensure that we haven't escaped circularity, and the facts about informativeness and brevity are not explanatory. BSA is inadequate, thus, if laws are supposed to be explanatory.

1.3. Coin tosses. , Now imagine a universe consisting of a single quantum “coin toss” performed a very large number of times. Observing a “neatly patterned” full sequence such as $HHH...H$, $TTT...T$, $HTHT...HT$ or $THTH...TH$ would make us confident that the coin tosses are deterministic (e.g., $HTHT...HT$ would make us confident that there is a law that each toss is followed by its opposite). But even though this would make us confident of determinism, any such sequence could also occur without determinism. Not so on our Humean story. On our Humean story, any one of these global regularities *logically guarantees* a deterministic law in a world consisting of a single coin tossed repeatedly, since the regularity would have to be a law.

This is highly counterintuitive. Furthermore, suppose that we in fact have a “pattern-less” sequence that to the Humean yields an indeterministic law that says that each coin toss is independent and fair. But while the statement that each coin toss is independent and fair should allow any finite sequence of coin tosses, it turns out that certain sequences, such as our four examples above, are logically incompatible with independence and fairness on our Humeanism.

1.4. Magic. Suppose you live in a multiverse consisting of two physical universes, each of which has laws of nature roughly like ours. Now, plausibly, any state of a physical universe like ours can be encoded as a countable sequence of real numbers (i.e., a finite sequence, or one that can be enumerated using the natural numbers: $x_0, x_1, x_2, \dots$). For instance, suppose the universe is made up of a countable cardinality of particles, each of which has a finite number of properties naturally expressible as one or a finite number of real numbers (e.g., mass and charge can be expressed as one number, and position can be expressed as three given a coordinate system), which properties either change continuously or have a countably infinite number of times of discontinuity. Then all we need in order to fully describe the system is to specify the properties at a countable number of times (e.g., the times at which discontinuities happen as well as all times that can be expressed

by a rational number), and the resulting description can be expressed as a countable sequence of real numbers.⁷ But a countable sequence of real numbers can, with a well-known trick, be expressed as a single real number.⁸ And if instead of a classical particle system one prefers a quantum story, then note that most models of quantum mechanics make the wavefunction be a vector in a separable Hilbert space—and the cardinality of the set of vectors in a separable Hilbert space is the same as the cardinality of the set of all real numbers. Adding a countable number of discontinuities in case of quantum collapse, we can still encode the state of the universe as a single real number.

Now, any real number can be encoded into the tilt angle of a physical rod (e.g., encoding x into a rod tilted—with respect to some axis—at angle $\arctan x$). Thus, we can encode the complete physical state, over all of time of one of the two universes into the tilt angle of a physical rod in the other universe. Now fix some encoding that can be specified in a relatively brief way. Suppose now that you are in one of the universes, and at a precisely specifiable moment (say, one that is a precise number of Planck times since the beginning of your universe) you tilt the rod at angle that, *mirabile dictu*, happens to match the state of the other universe. Then including the information that the rod angle at this moment matches the complete state of the other universe will indeed provide a vast amount of information about your multiverse in a fairly brief compass, and hence would be included in

⁷*This argument uses the fact that if A and B are countable sets, then the Cartesian product set $A \times B$ of pairs (a, b) with a from A and b from B is also countable. To see this, note that if we can enumerate A as $\{a_0, a_1, a_2, \dots\}$ and B as $\{b_0, b_1, b_2, \dots\}$, then we can enumerate $A \times B$ as $(a_0, b_0), (a_1, b_0), (a_0, b_1), (a_2, b_0), (a_1, b_1), (a_0, b_2), \dots$

⁸*Any real number can be remapped to the range from 0 to 1, exclusive, by taking x to $(1/2) + (1/\pi) \arctan x$. Any countably infinite sequence of numbers $x_0, x_1, x_2, \dots$ between 0 and 1 exclusive can then be expressed as a sequence of decimal numbers where the i th number is of the form $0.x_{i,0}x_{i,1}, x_{i,2}, \dots$ (with the convention that an infinite terminal sequence of nines is preferred to an infinite terminal sequence of zeroes, say). One can then encode all these numbers into a single number of the form $0.x_{0,0}x_{1,0}x_{0,1}x_{2,0}, x_{1,1}, x_{0,2}, \dots$ (this is basically the same pattern as in Note 7), so that every digit of every one of the numbers in our initial series occurs at a determinate position in the encoded number. And *a fortiori* if one can encode any countably infinite sequence of reals into a single real, one can encode any finite sequence of reals into a single real.

the optimal Lewis-Ramsey description, and will be a law. Thus, by tilting a rod at a specific angle—and surely any rod tilt angle is physically possible for you (though maybe not possible to induce *intentionally*)—you can create a law correlating the rod angle with the other universe. Thus, by waving a rod, you can make the rod be a vastly informative dowsing rod that by law and not merely by coincidence carries complete information about another universe. This is highly implausible magic!

1.4.1. *A plurality of bestnesses.* A problem that will be familiar from many of our earlier discussions is that there are many free parameters in the account of the bestness of the best system. For instance, given a measure of informativeness $I(T)$ and length $L(T)$ of a system T , we will want to combine them into a measure $f(I(T), L(T))$ of quality of theory by using some kind of a combining function f such that $f(x, y) < f(x', y)$ and $f(x, y) > f(x, y')$ whenever $x < x'$ and $y < y'$. But there are infinitely many functions satisfying these inequalities. Perhaps with some thought we can find some more reasonable constraints, but it is very implausible to think we can reduce the space of reasonable candidates to one.

Moreover, we can see that neither $I(T)$ nor $L(T)$ has a unique privileged candidate.

If a world has a phase space—say, defined by values of various natural determinables like charge, position and momentum at various times—it makes sense to think of the informativeness of a theory T as inversely related to the size of the set of trajectories through phase space (i.e., functions from time to phase space) compatible with T . If we could get the set of allowed trajectories down to one, that would be maximal informativeness.

But how do we measure the size of the set of trajectories compatible with the theory? The set-theoretic cardinality of the set of allowed trajectories for many theories that intuitively vary significantly in their informativeness will be the same. For suppose our determinables are position. Now consider theory T_1 according to which there is a single particle which for all time is found in the same position in three-dimensional Euclidean space, and T_2 according to which the particle moves through three-dimensional Euclidean space over a continuous trajectory. Clearly, T_1 is much more informative than T_2 . But the cardinality of the space of trajectories allowed by T_1 is *the continuum*, the cardinality of the real

numbers.⁹ However, that is also the same as the cardinality of the space of all continuous trajectories, which is what T_2 allows.¹⁰

We might, on the other hand, try to define the size of the set of trajectories as a volume rather than a cardinality. An immediate problem may seem to be that of the choice of units of volume. This is not a serious problem for comparing degrees of informativeness, as long as we measure the volumes of the set of allowed trajectories using the same units. Even completely arbitrary units like the length of Charles III's forearm and the time between his mother's and his own coronations can be used for comparative purposes. However, the scaling issue is relevant for the combination problem. For unless our combination function f has some additional special properties, like being linear in the second variable (i.e., $f(x, \alpha y) = \alpha f(x, y)$), the choice of scaling can be an issue, because it might be that $f(x, y) < f(x, y')$ but $f(x, \alpha y') < f(x, \alpha y)$ for some choice of x, y, y' and α . On the other hand, we might use the scaling worry to justify a linearity constraint on the second variable in f , thereby reducing the arbitrariness of the choice of f .

A more serious problem is cases of two theories that clearly vary in information content but both allow a set of trajectories with zero volume. For instance, suppose a world with only one moment of time, and one determinable: three-dimensional position of a single particle. A theory that requires the particle to be at the coordinates at the origin reduces the space of trajectories to a subset of zero volume, as does a theory that merely constrains the particle to lie on the yz -plane, since both a point and a plane have zero volume.

We might find a way of comparing sets of zero volume. For instance, we might try to find the Hausdorff dimension (which may be fractional) of the two sets, and deem the set with higher dimension to convey less information. And then we could use Hausdorff measure to compare sets of the same Hausdorff dimension. This approach leads to a pair

⁹A position can be encoded as three real numbers, which can then be recoded as one real number (see note 7, above).

¹⁰To specify a continuous trajectory, one only needs to specify its values at all times that are represented by rational numbers, of which there are countably many. But a countable sequence of real numbers can be specified by a single real number (see note 8 above).

of numbers, (α, β) , where α is the Hausdorff dimension of the allowed subset of phase space and β is the α -dimensional Hausdorff measure. Thus, instead of combining length with a single number, we need to combine length with *two* numbers. And what do we do in the case of sets whose Hausdorff measure is infinite with respect to the Hausdorff dimension—these may need to be compared as well. Furthermore, Hausdorff dimension itself is not the only way to formalize the concept of dimension.???

Thus we should not expect a canonical measure of informativeness: there will be many free parameters.

Now, length may seem more tractable. Indeed, given a fixed language whose sentences are finite strings of characters from a finite symbol set, there is no difficulty in making sense of the length of the briefest expression of a system of propositions in that language.

But there are many languages. Just think of such decisions as the choice of grouping notation. In recent use in logic, for instance, we have parenthesis notation, Polish notation and dot notation. Surely there are many more reasonable grouping notation. Or think which logical primitives should be included. For truthfunctional connectives, *nand* is sufficient, as is *nor*, as is either of the pairs *and-not* and *or-not*, but why should we limit ourselves to a minimal sufficient set? Perhaps we should allow all the binary connectives. Or perhaps some but not all. Or perhaps all the ternary ones. Or perhaps just one seven-place *nor*. It is reasonable to think there is a privileged set of predicates—the perfectly natural ones. But is it reasonable to think there is a privileged set of logical operators? This is murky. Or a privileged grouping notation? That seems even more dubious.

We thus have little reason to hope in a single distinguished measure of bestness. Now, we might hope that for a wide range R of measures of the quality of a theory in a world, there is sufficient overlap between the best theories to ensure that the things our best science will converge on as the laws will in fact be entailed by the theories that are best according to the measures in R . We could, then, define a law as a proposition L such that for all quality measures Q in R , the Q -best theory entails L . But specifying the boundaries of

R will be subject to difficulties very similar to those in defining a single canonical measure Q .

It seems we cannot escape the idea that there is significant vagueness in the concept of a law for the Humean. Is this a problem?

In Section 1.2, I argued that in paradigmatic cases where universal generalizations enter scientific explanations, they do so as laws. In other words, that L is a law is itself a part of the explanation, and not just L itself. Whether this *always* so is not clear, but it is clearly sometimes so. Nobody would bat an eye at a scientist saying that planets move in elliptical orbits because by a law of nature objects attract gravitationally in proportion to their mass and the inverse square of distance.

But then if an account of what it is to be a law of nature is a very complex statement involving a measure of the quality of theories, and various linguistic facts about lengths of expressions, then our scientific explanations in terms of laws are much more complicated than we likely thought. While the concepts figuring *in* the laws may be very natural, the concept of a law is itself rather unnatural given BSA. Moreover, the plurality of bestnesses implies that there is a serious vagueness in the concept of a law, which makes our scientific explanations, even in the most precise areas of fundamental physics, full of vagueness.

Suppose, on the other hand, that we follow the example of the old deductive nomological model of explanation and deny that laws enter explanations *as laws*, notwithstanding the argument of ??backref. Still, the vagueness of the concept of law would affect what counts as an explanation, assuming that generalizations that are not laws are not allowed in explanations, at least not in the place where we would put a law. We end up damaging the objectivity of the concept of explanation on this approach: it becomes a linguistic question what is and is not an explanation. And if explanation is significantly relevant to justification—for instance, via inference to best explanation—we damage the objectivity of the concept of justification. This is perhaps less problematic than damaging the explanations themselves, as would be the case if the nomicity of the laws were a part of the explanations themselves, but it is still problematic.

Furthermore, our now familiar Aristotelian solution to issues of unacceptable vagueness, which is to ground boundaries in human nature, is not plausible in the case of laws of nature, because doing so would render laws of nature too anthropocentric.

1.4.2. *Mind and causation.* It is interesting to note that the full Humean package where causation is grounded in laws and laws are grounded in a best-system analysis has a serious problem with any philosophy of mind on which certain causal relations are essential to mental events. This will include, first and foremost, functionalist accounts on which the pattern of causal interconnections *defines* mental events. But it will also include any view on which causal interconnections of a particular sort are *essential* to mental events. For instance, while the functionalist may think that what *makes* an event be a pain is that it is typically caused by damage and typically causes motivations to aversive behavior, one need not be a functionalist to think that a causal connection between an event and motivation is necessary for the event to be a pain. Likewise, plausibly, a necessary condition for x to infer q from p is that a thought (say, a believing or an assuming) of x with content p causes a thought with content q . Finally, it is very plausible that for a wide range of mental events, an essential part of what makes the mental event be *mine* is that it is at least partly caused by me, or by my character, etc.

But now notice that best-system laws are defined globally in terms of the four-dimensional arrangement of stuff in the universe. Whether some lawlike generalization does make it into the best system depends, in particular, on what the future is like. One reason is that typical lawlike generalizations tend to non-vacuously apply to events past, present and future, and so if the future were sufficiently different, the generalizations themselves would be false, and hence not in the best system. If some such generalization is needed for a causal relation, say between pain and aversive motivation, then if the future were such that the generalization would fail to hold, there would be no pain now.

The idea that whether there is pain now depends on what will happen in the future is itself highly counterintuitive. Additionally, it allows for implausible deductive predictions

about the future. I am in pain now. My being in pain now requires a causal relation between pain and aversive motivation. Such a causal relation requires a law of sort *S*. A law of sort *S*, on BSA, requires the future to satisfy certain constraints *C*. Hence, the future will satisfy *C*. But it is highly implausible (and in tension with the historical Hume's scepticism about induction) that we can thus reason *deductively* from present pain to facts about contingent future arrangements of matter, namely that they satisfy *C*. And if introspection counts as *a priori*, this is *a priori* reasoning about contingent future facts.

Moreover, the same intuitions that disallow deductive inferences about future arrangements of matter from my present phenomenal states should disallow deductive inferences about arrangements of matter outside the solar system from my present phenomenal states. But given BSA, the existence of the fundamental laws undergirding a causal connection between pain and motivation would have non-trivial implications for how matter is arranged outside the solar system. Laws grounding causal connections shouldn't be such as to be purely local (i.e., to have merely vacuous implications for how things are outside of a local region).

Perhaps our Humean could simply embrace the non-locality of the grounding of our phenomenal states, and allow deductive inferences of facts about temporally and spatially distant arrangements of matter from our phenomenal states, *pace* the historical Hume. But the counterintuitiveness of this is a definite cost for the theory.

1.5. Indeterministic extensions.

1.5.1. *Probabilistic BSA*. BSA was initially formulated for deterministic laws. But what about indeterministic laws, such as that some quantum setup has a chance $1/3$ of resulting in outcome *A* or that a certain indeterministic coin toss has chance $1/2$ of heads? The standard move is to go to a Probabilistic BSA (PBSA). In BSA, we required all the claims of the theory to be true. In PBSA, we only require non-probabilistic claims to be true. However, now, in addition to optimizing informativeness and brevity, we also optimize *fit*, where we check how well the outcomes fit probabilistic predictions. A theory which claims there is some large number *N* of fair (i.e., the chances of heads and tails are equal)

and independent coin tosses will tend to better fit worlds where the number of heads is closer to $N/2$ than worlds where the number of heads is further from $N/2$. Thus, we optimize informativeness, fit and brevity over theories whose non-probabilistic content is true.¹¹ However, PBSA also has a number of serious problems.

1.5.2. *Violation of the Principal Principle.* Let's make the standard distinction between chances and epistemic probabilities. Chances are objective mind-independent probabilities grounded in the laws of nature or causal propensities. Epistemic probabilities, on the other hand, are the probabilities an epistemic agent has given evidence.

LewisSubjectivist's (LewisSubjectivist) Principal Principle (PP) holds that there is a tight epistemic relationship between chances and epistemic probabilities. Basically, if one's relevant evidence consists in the claim that the chance of some event is p , then p should also be one's epistemic probability.¹² I will now an example that shows a tension between PP and PBSA.

Imagine a world where all the coins have always lain scattered on a large flat surface. There is an extremely large number of these coins. One of them is made of gold and all the others are made of copper. You have observed all the copper coins and found an extremely strong fit between their arrangement and the hypothesis that whether each coin is heads or tails is independent across the coins and the probability of heads is always $63/64$. You have not yet seen whether the gold coin is heads or tails, though you know that it is one or the other, and that there are no further coins.

On PBSA, if the number of coins is sufficiently large, a hypothesis specifying that the chance of a coin being heads is $63/64$ will be sufficiently informative to make it into the

¹¹If we want a bit more elegance in the formulation, we can drop the requirement that the non-probabilistic content is true and instead stipulate that a non-probabilistic statement has good fit when it is true but infinitely bad fit when false, so no theory with false non-probabilistic statements will win the crown of being the best system.

¹²There are some technical refinements needed to handle the problem of undermining futures **HallUndermining**; **LewisUndermining**; **ThauUndermining**, but the example below does not involve undermining futures.

laws of nature.¹³ Thus, if you are sure of PBSA, given your observations you can be sure that this is a law. For if the number of copper coins is sufficiently large, and the statistical fit is good enough, you can be completely confident that the fit to the 63/64 hypothesis will be sufficient for nomicity regardless of whether the gold coin is heads or tails.

But now if you are sure that the 63/64 hypothesis is a law, then by PP you will have an epistemic probability of 63/64 for the gold coin being heads.

And this is wrong. An inductive inference is always weakened by differences of material: if our inductive data is the distribution of heads/tails across all the copper coins other than one, then it would indeed be reasonable to assign an epistemic probability around 63/64 to the remaining copper coin being heads. But if the remaining coin is of a different material, we surely should not have the same level of confidence in its being heads. Intuitively, we should shift our epistemic probability closer to 1/2.

One might object on behalf of PBSA that one should instead posit a law that says that a *copper* coin has chance 63/64 of showing heads. But this is mistaken. The resulting hypothesis is not significantly more informative, but is significantly less brief since it must mention and define the material of the coin, and hence is not going to be a viable competitor for being in the best system.

1.5.3. **A betrayal of frequentist intuitions.* In defining the laws, we are maximizing a balance of fit and brevity. Again, suppose a world with such a vast number of coins that their probabilistic distribution will be sufficiently informative that it will be included in the laws. If half of the coin tosses are heads, and coin toss results are sufficiently important for the universe's behavior, then a law *U* assigning chance 1/2 to heads is likely the best balance.

But now suppose that the frequency of heads isn't 1/2 but something very close to 1/2. Then the law may still be that the chance is 1/2, because a slight decrease in fit due to having a chance not quite matching the exact frequency might be more than offset

¹³The number of coins has to be sufficiently large to offset the loss of brevity due to having to explain what a coin is and what counts as heads.

by the gain brevity if chance $1/2$ is significantly more briefly expressible than the actual frequency, say $1/2 - 3 \cdot 2^{-254}$.

Let's suppose the number of coins N is very large but easily expressible (e.g., $N = 2^{2^{2^{2^2}}} = 2^{65536}$), and let $\mathcal{W}_N$ be the set of worlds with nothing but N tosses where there are no briefly expressible and highly informative patterns other than those conveyed by the independence of the coin tosses, the frequencies and the number of tosses.

Let F_N be the set of all fractions between 0 and 1 with denominator N . Each member of F_N could be the exact frequency of heads in some world in $\mathcal{W}_N$.

Let $A_{N,1/2}$ be the subset of frequencies in F_N such that there is a world w in $\mathcal{W}_N$ in which there is a Humean law specifying the chance to be $1/2$. For simplicity, suppose N is even, so $1/2$ is a member of F_N . For N sufficiently large, we should expect $1/2$ to be a member of $A_{N,1/2}$. But for the reasons given above, we would expect some frequencies in F_N that are very close to $1/2$ to be in $A_{N,1/2}$ as well. If E_α is the event of the actual frequency being α , then on our Humean view we have:

$$(10) \text{ If } \alpha \text{ is not in } A_{N,1/2}, \text{ we have } P(E_\alpha \mid \text{Law}(U)) = 0.$$

On the other hand, very plausibly, if α is in $A_{N,1/2}$, then we have a real possibility of having frequency α on the assumption that the law is U , and so we would expect:

$$(11) \text{ If } \alpha \text{ is not in } A_{N,1/2}, \text{ we have } P(E_\alpha \mid \text{Law}(U)) > 0.$$

Frequencies that are sufficiently far from $1/2$ (such as the 0 in our previous example) are not going to be in $A_{N,1/2}$: the fit of these frequencies is too bad. The case of $\alpha = 0$ has already been discussed.

What is surprising, however, is that very likely there are some members α and β in F_N such that $\alpha < \beta < 1/2$ and yet α is a member of $A_{N,1/2}$ but β is not a member of $A_{N,1/2}$. For we can suppose that both α and β are *extremely* close to each other, and very close to $1/2$, but α is much more briefly expressible than β , so that although the fit of a chance $1/2$ law is slightly poorer for worlds with frequency α than for worlds with frequency β , the greater brevity of an expression for α makes a law giving the chance of heads as α have a

better balance of fit and brevity in a world with frequency α than a law giving the chance as $1/2$, while a law with chance $1/2$ has a better balance of fit and brevity in a world with frequency β than a law giving the chance as β .

I cannot give precise examples here, because we do not actually have the measures of fit in hand and estimating the brevity of the shortest expression of some number is often a very difficult task. But suppose I am right. Then by (10) and (11), some frequencies in F_N that are further from $1/2$, such as α , will have a higher conditional probability on U than some frequencies that are closer to $1/2$, such as β . I imagined that the counterintuitive result that E_0 has zero conditional probability on U fails might be given the “excuse” that U entails a chance that is so tiny that we might as well take the probability to be zero. But that excuse won’t work for E_β getting zero conditional probability on U . For if we say that U entails such a low chance for E_β that E_β deserves conditional probability zero, then E_α should get conditional probability zero as well, since E_α has an even lower chance on U .¹⁴

Thus, on PBSA, it appears that U ’s being a law logically rules out some frequencies closer to $1/2$ while being compatible with some frequencies further from $1/2$. This seems to be a betrayal of the frequentist intuitions behind PBSA.

1.5.4. **Chance and propositions.* There is a variety of slightly different accounts of chance that can be given on PBSA. Let “the mosaic” refer to the arrangement of properties that are systematized in the best system. Here are a few options, depending on whether one takes conditional or unconditional chances to be fundamental:

- (12) Event E has unconditional chance p if and only if the mosaic’s best system entails that E has chance p .
- (13) Event E has unconditional chance p if and only if the mosaic’s best system conjoined with a complete specification of the mosaic’s initial conditions entails that E has chance p .

¹⁴For a large number of tosses, the chance of getting a particular frequency x of heads has an approximately normal distribution centered on $1/2$ if the true chance of each independent toss is $1/2$, and hence drops off as x moves away from $1/2$.

- (14) Event E has conditional chance p on C if and only if the mosaic's best system entails that the conditional chance of E on C is p .

There is room for much technical discussion of the details here, but notice that each of these three accounts is viciously circular: it is attempting to define a chance (conditional or not) by a definition that itself makes use of the concept of chance.

There is a complicated way out of this difficulty that modifies the PBSA.

Suppose that we have a language $\mathcal{L}$ whose vocabulary can express all the fundamental physical concepts acceptable to the Humean, and has one additional function symbol, $\text{Chance}(x)$. This new function symbol is uninterpreted, and it is stipulated that sentences using that function symbol are neither true nor false. Let M be some appropriate set of mathematical axioms that include the correct axioms of set theory (perhaps indeed they include all truths of set theory) as well as the right axioms of probability formulated using the chance symbol (so these will be axioms of unconditional or conditional probability, depending on whether the chance symbol is unary or binary). Say that a set of sentences or "theory" T in $\mathcal{L}$ is *non-false* provided that no false sentence can be formally proved from $T \cup M$. This is equivalent to saying that all the provable consequences of $T \cup M$ that do not use the chance symbol are true (note that since everything follows from a contradiction in classical logic, if T violates the laws of probability in its use of the chance symbol, T will automatically count as false). We can now specify that the best system is the non-false theory T that optimizes brevity, fit and informativeness.

Defining informativeness is difficult, but we might define it by looking at two kinds of information conveyed by the provable consequences of T . First, we have sentences that do not make use of the chance symbol. The informativeness of that part of the consequences will be measured much as in a non-stochastic BSA. The informativeness of a statement of form $\text{Chance}(E) = r$, where E does not make use of the chance symbol, will be null if $r = 1/2$, and while if $r > 1/2$, it will increase monotonically with both r and the informativeness of the non-probabilistic claim that E occurs, and if $r < 1/2$, it will also decrease

monotonically with r while increasing monotonically with the informativeness of the non-probabilistic claim that E does not occur. We might for simplicity disallow claims using the chance symbol but not of the form $\text{Chance}(E) = r$. And, of course, we will have to beware of overlap in information between the chancy and non-chancy sentences, and so on.

Finally, we measure fit by looking at all the provable consequences of the form $\text{Chance}(E) = r$ where E does not contain another instance of the chance symbol, and saying the fit of each is a measure of closeness between r and the numerical truth value of E (falsehood being zero and truth being one)—perhaps a measure according to some standard scoring rule.¹⁵

The details are doubtless very difficult, but that is to be expected. Given all this, we get a best system T consisting of sentences some of which fail to express a proposition. Given that laws of nature are not linguistic entities in a particular language, and that they are factual, and hence capable of being true, we surely cannot take a set of uninterpreted sentences like T in $\mathcal{L}$ to be a system of laws.

But we can now make a further move. Let B_T be the proposition that all the sentences in T are provable consequences of the best system. Then B_T is indeed a proposition and factual. We can then say that the laws of nature are all the logical consequences of B_T . The laws, then, are not the sentences derivable from the best system, since these include uninterpreted sentences, but rather the laws are the consequences of the claim that these sentences are sentences of the best system.¹⁵

One problem of this approach, however, lies on the probability side. Suppose we are in a world with a fair coin sequentially tossed a large finite number N of times so set up that if T is the best system then $T \cup M$ proves a sentence saying that the coin was indeed

¹⁵An alternative would be to consider the logical consequences of the proposition that T is the best system. But that would be less satisfactory. For it seems that one could have a world with all the laws we have *and* more. But if it is a law that T is the best system, then it seems there couldn't be a world with additional laws beyond those in T .???

tossed sequentially N times and proves the sentence “ $\text{Chance}(H_1) = 1/2$ ”, where H_1 is the event of the first throw being heads.¹⁶ Then B_T will be a law of nature. In the notation of Section 1.5.2, it will be law that the frequency of heads is a member of the set $A_{1/2,N}$. This set contains $1/2$, and contains some frequencies in F_N very close to $1/2$, but also excludes frequencies just as close to $1/2$ as some of the included ones, on the grounds of that some of the excluded ones are so briefly expressible (e.g., $1/2 + 1/2^{256}$ in a world with $N = 2^{256}$) that the brevity cost of the greater fit is worthwhile, so that if the frequency were one of the excluded ones, its being what it is would be a law. A law of nature that says that the frequency lies in the messy set $A_{1/2,N}$ looks little like the kinds of laws scientists posit. We do not have laws of nature allowing and disallowing events based on the brevity of the expressibility of facts about these events.

1.5.5. *Non-Humean chances.* There is another approach to chances, however, than PBSA. Suppose that we take chances themselves to form part of the Humean mosaic, perhaps being fundamental properties, rather than being something defined via patterns in the mosaic. Then a non-deterministic theory can stick with BSA. The requirement that chancy statements be *true* now is cashed out in terms of the chances in the Humean mosaic. It is now quite possible for the best system to say that the chance of heads is $1/2$ even if the coin never comes up heads. For we no longer optimize fit, but informativeness. And the claim that the chance of heads is $1/2$ might well be quite informative, since it could tell us about the stochastic properties of a vast number of coin tosses—even if it does not give us useful information about the outcomes of these tosses.

From the Humean point of view, the down side of this is that chances are too much like causal propensities, which the typical Humean wants to reduce to patterns in the mosaic. After all, if our account of chance is not based on frequencies, then it seems that our best option for what it means to say that the chance of heads is $1/2$ is that the immediate cause has a propensity of $1/2$ for heads.

¹⁶This will require N to be a number that is sufficiently briefly expressible to make it into the best system.

Thus, if we allow chances into the mosaic, we probably should allow causation as well. The resulting BSA has a richer mosaic, and is harder to refute by counterexample. The problem that coincidental correlations become rigidified into magical laws^{??backref} is less pressing once it is clear that the magical laws are not causal laws—for causation is now a part of the mosaic, and hence mere correlations do not yield causation. However, we still have the problem of the plurality of bestnesses. And because we have so significantly enriched the mosaic, the main reason to believe the BSA is weakened. For that main reason is ideological parsimony: the elegant fewness of fundamental concepts. Once we have introduced causation and causal propensities, one of the great intellectual assets of BSA—giving an account of causation^{??}—is gone. On the other hand, since causal facts can be used for explanation, the explanatory difficulties for BSA are mitigated, since at least some causal explanations become available.

Thus, we have lost the parsimony which is the main motivator for Humean views, while retaining some of the difficulties, such as magic (albeit now of a non-causal sort).

2. Aristotelian laws

The most common Aristotelian approach to laws is to ground them in the essential powers of substances, which in turn are grounded in the substantial forms. If an electron is a substance, then the form of an electron gives it the power to attract positively charged things and repel negatively charged ones, and such facts ground the laws of electron behavior. And if a rabbit is a substance, then the form of a rabbit gives it the power to reproduce with another rabbit, which grounds laws of rabbit behavior.

This makes laws explanatory, without introducing any additional entities beyond the forms that we already need to explain the normative side of nature. Indeed, the integration of the normative and explanatory is central to the Aristotelian optimism on which things typically act rightly (we might even speculate that in the case of some things, such as subatomic particles, they *always* act rightly), and hence we can defeasibly infer ought from is.

Now let us recall the example from section ??backref:

(15) All massive objects attract gravitationally.

(16) The Sun is a massive object.

(17) So, the Sun attracts gravitationally.

This was problematic, because it is plausible that universally quantified statements are partly grounded in their instance, and hence (15) is partly grounded in (17), which makes for circularity in the order of explanation as grounding is a type of explanation. Supposing for simplicity that the sun is a substance and has a stellar form, the Aristotelian, however, can say that having stellar form *entails* engaging in gravitational attraction.

(18) Necessarily, all objects with stellar form attract gravitationally.

(19) The Sun has a stellar form.

(20) So, the Sun attracts gravitationally.

But the necessary truth (18) is not even partly grounded in (20). In other words, the Aristotelian can solve our initial problem by adding necessity to the explanation.

That said, there is a complication. Arguments in Section ??forwardref will make it plausible that on an Aristotelian view there are no stars. So we will need to modify the story, e.g., to talk about the plurality of substances that are Sunwise arranged:

(21) Necessarily, all substances with forms from among $F_1, \dots, F_n$ attract gravitationally.

(22) The plurality of substances arranged Sunwise include substances with forms from among $F_1, \dots, F_n$.

(23) So, the plurality of substances arranged Sunwise attract gravitationally.¹⁷

But there are some further serious complications. The first is that classical Aristotelianism is committed to the principle that substances do not have substances as

¹⁷I am assuming that a plurality that includes some attractors counts as attracting. If not, the formulation will be more complex.

parts **AristotleMet.**¹⁸(??cross-refs) But then if a rabbit is a substance, the electrons in it are not substances. Instead, the electrons in the rabbit may be accidents of the rabbit, or “virtual objects” constituted by bundles of rabbit causal powers. If so, then the laws of electron behavior will be grounded differently for different electrons. The behavior of electrons in humans and those in rabbits will be grounded by the human and the rabbit forms, respectively. Free-floating electrons, on the other hand, would seem to be self-standing substances, and their behavior will be grounded by electronic substantial forms. On this view, the grounding of the laws of electron behavior appears quite complex.

The easiest way to remove this complexity is to allow substances to be parts of other substances. In this book, however, I am trying to maintain some neutrality on the details of various robust Aristotelian ontologies, so it is worth seeing what can be done without such a controversial move.

Forms of different beings can (and presumably do) common patterns of behavior. This is a puzzle for the Aristotelian. Why do electrons in oaks, rabbits and humans behave the same way? To give a partial answer, we might posit a “fine structure” in the forms, whereby the oak, rabbit and human forms (and those of standalone electron substances) have a common component coding for the behavior of electrons. This common component could be found in a genus that the forms are specifications of. Or we might simply suppose that the common efficient cause or causes of all of these substances—say, God, or the particles in the early universe—have so acted as to produce substances with a variety of forms all of which code for common behavior of electrons.

There are, however, two kinds of laws that pose a special problem for the Aristotelian approach. One kind is high-level structural laws that depend on all physical things having a common pattern to their behavior. The conservation of energy is a paradigm example. While the forms of electrons and photons (and/or of larger substances containing them) can make it a law that energy is conserved in the interactions of electrons and photons, it is

¹⁸**Devereux (Devereux)** has argued, however, that earlier in his career, Aristotle did allow substances to be parts of substances.

logically possible to have physical objects that interact with, say, photons in a way that does not conserve energy. The law of conservation of energy appears to be partly grounded in the fact that there in fact are no such things—that all physical substances have a forms that satisfy the constraints of the law of conservation of energy. But if $F_1, \dots, F_n$ are all the forms of existing physical objects, then $F_1, \dots, F_n$ are not sufficient to ground the conservation of energy without some additional metaphysically contingent fact such as that there are no physical objects with a form not among $F_1, \dots, F_n$. Another example is Einstein's Principle of Relativity, that all laws of nature hold in all reference frames.¹⁹

A second kind of problematic law is exemplified by the Second Law of Thermodynamics. The Second Law of Thermodynamics has long presented a puzzle given that the underlying physics relevant to the behaviors of the particles whose joint entropy tends to increase is invariant under a time-reversal. The usual contemporary solution is that what explains the increase of entropy is a physical dynamics that can be time-reversal-invariant combined with low-entropy initial conditions. For given low-entropy initial conditions, entropy, as it were, has nearly nowhere to go but up—there are a lot more possibilities for transitions from low to high entropy states than from low to low entropy states. Thus the Second Law is not grounded in the natures of the interacting objects, but in those natures as combined with the metaphysically contingent fact that we started in a low-entropy state. We might call laws that depend on special boundary conditions “impure laws”.

The two families of laws—the high-level structural laws that transcend the natures of particular objects and the impure laws that depend on boundary conditions—are ideal candidates for BSA. BSA does not care whether the laws depend on boundary conditions or not, just how elegant and informative they are. But they do create problems for Aristotelian accounts, and more generally for other “pushy” accounts of laws. For it does not seem that any kind of “pushy” law forces the initial conditions to have a low entropy. And while a “pushy” law can move things around in our world, it is somewhat mysterious how

¹⁹**LangeMeta (LangeMeta)** argues that such meta-laws are also a problem for Humean best-systems accounts.

it can keep the kinds of things that would have the power to violate the structural laws out of existence.

There are several options for the Aristotelian at this point. The first is to bite the bullet, and hold that the structural and impure lawlike generalizations are not actually laws of nature. This is fairly plausible in the case of impure laws, but less so in the case of structural laws.

The second option is to notice that at least some of our problematic laws can be accounted for as follows. We suppose that in fact none of the physical substances in existence have a causal power to produce a violation of the law directly or indirectly, where indirect production of a violation would be the production of a causal chain leading to such a violation. In short, there are no physical potential violators of, say, the conservation of energy, and no physical potential producers of such violators. We then say that whenever we have a “lawlike fact” with the property that nothing physical is a potential violator or potential producer of a potential violator, that fact is a law. We still need some account of what makes a fact lawlike. One move would be simply to defer to the best version of BSA: a lawlike fact is one that would be a logical consequence of the best system. And, finally, we just bite the bullet on any generalizations that cannot be accounted for in this way: these we just do not account to be laws.

This may seem to involve us in all the difficulties of BSA. But that need not be the case. First, we *might* only make the move for non-stochastic laws, denying that structural or impure generalizations that are stochastic in nature (such as, alas, the Second Law of Thermodynamics) are laws of nature. Second, the explanatory difficulties of BSA are defused in our present context. For laws grounded in this way are explanatory simply in a causal-privative way. We explain, say, why a violation of conservation of energy did not happen by saying that there was nothing physical with the causal power to generate such a violation. This is a genuine explanation independently of whether the conservation of energy is a law. The explanatoriness of the laws here does not flow from BSA. For instance, on this story we might happily embrace the plurality of accounts of bestness, because the

laws explain simply in virtue of their content, not in virtue of their being *laws* or members of the best system.

The third option is to embrace theism, and allow for two kinds of laws: laws grounded in the essential powers of physical substances and lawlike generalizations about physical substances intended by God to hold. This will be a partial occasionalism: some explanations in terms of laws will go back to created substances, but others will go back to God (and presumably some cases will be explained by a combination of the two). But such a partial occasionalism does not appear to be particularly problematic, since it allows for non-occasionalist creaturely causation, and only makes some lawlike generalizations be laws by divine fiat.

3. Some non-Aristotelian alternatives to BSA

3.1. Armstrongian universals and laws. ArmstrongWorld (ArmstrongWorld) grounded laws of nature in nomic necessitation relationships between universals. For instance, a classical physicist might say that the universal *having one kilogram of mass* nomically necessitates the universal *exerting a gravitational force equal to $6.67 \dots \cdot 10^{-11} m_2 / r^2$ Newtons on any object of mass m_2 kilograms at distance r meters*.

The account has some similarities to an Aristotelian grounding of laws in forms, since forms are individualized universals. One difference is that the nomic necessitation relation is a primitive that on the Aristotelian view is replaced by broadly logical entailment. This difference favors the Aristotelian, in that nomic necessitation is a mysterious additional theoretical ingredient the Aristotelian does not need, while (broadly) logical entailment is used all over the place in metaphysics anyway. However, this difference can be bridged: we can suppose a variant of the Armstrong view where nomic necessitation is strengthened to broadly logical entailment.

The bigger difference is that the Aristotelian approach makes use of causal powers grounded in the forms. This yields an advantage for the Aristotelian. Causal powers have

an explanatory asymmetry: the activation of the causal power explains the effect. Entailment relations, however, can go in both directions. The universal *exerting a gravitational force equal to $6.67... \cdot 10^{-11} m_2 / r^2$ Newtons on any object of mass m_2 kilograms at distance r meters* will broadly logically entail, and hence also nomically necessitate (since nomic necessitation is a weaker relation than logical entailment), the universal *having one kilogram of mass*. For it seems to be part of the very nature of the gravity of our world that it is produced by mass in the above way. But then we do not have an explanatory asymmetry, whereas surely there should be one: mass causes gravitational force. Perhaps the nastiest version of this problem will occur in a world with two-way determinism, where complete past states necessitate complete future states and *vice versa*. While we may not live in such a world, nonetheless such a world appears metaphysically possible.

3.2. Pushy global laws. An account on which there are irreducible pushy global laws—fundamental aspects of reality that move stuff around—also has many of the same advantages over Humeanism that Aristotelianism does. But it is worth noting that there may be some advantages of an Aristotelian account.

First, causal powers are familiar things: we have them ourselves, and exercise them whenever we move. Global pushy laws of nature are rather more mysterious. Metaphysically contingent principles moving stuff around are puzzling. The idea of *things* moving things is very familiar, but *principles* moving things are more mysterious, though granted they do have roots going back to pre-Socratic *archai*.

One difficulty with pushy global laws is that we should take seriously the hypothesis that we inhabit a multiverse with different universes having significantly different patterns of physical behavior. For if the pushy global laws are truly global, they will exist at the level of the multiverse as a whole, which forces the laws of physics to be complex, generating different patterns of physical behavior in different universes. They might do this by there being different types of particles or fields in different universes, and the global laws imposing different patterns of behavior on different types of things, or by their directly imposing different patterns of behavior on different regions of the multiverse.

The Aristotelian, on the other hand, can simply ground the differences of behavior between universes by supposing the substances in different universes to have different forms. There is no need to *further* suppose complex conditional laws that impose different behavior on different kinds or regions: the kinds themselves are responsible for the differences in behavior. This is perhaps a neater solution.

Could the defender of pushy global laws instead suppose that the enforcers of the laws would be divided between the universes, so the “globality” is not as broad as to the multiverse, but only extends to the level of each universe? But now the account is rather Aristotelian. Instead of the laws being something like fully global principles, they are more like forms that govern particular concrete entities, namely particular universes. We might even say that such a view is just a variant on Aristotelianism where the substances are universe-sized. We would then have two ways of developing such an Aristotelianism: we might suppose the universes to be the only substances, or more attractively in light of the humanistic arguments in this book we might allow (*pace* classic Aristotelianism) for there to be mid-sized substances like ourselves that are parts of the universe-sized substances.

4. Conclusions

We have multiple good reasons to reject Best-Systems Accounts of laws. An Aristotelian account of laws as grounded in the forms of objects is a powerful alternative to BSA, though it has some difficulties as well—difficulties that are perhaps especially way remedied on a theistic version. Armstrongian and pushy global laws also provide powerful alternatives to BSA, but there are still some reasons to prefer an Aristotelian account to them.

CHAPTER IX

Metaphysics

1. Composition

Intuitively, your parts compose a whole, namely you. On the other hand, if you choose exactly one atom from every star in every galaxy in the universe, intuitively these scattered atoms do not make up any whole.

Yet it seems that one could produce a continuous series of worlds, adding, subtracting or moving one particle at a time, where the first item in the series consists of the scattered atoms in different stars and the last item consists of you. Somehow, as one moves along the sequence, at some point a new macroscopic object pops in (maybe it's you, or maybe it's something else), despite the variation between successive terms in the sequence being extremely slight. What makes for the transition?

The most popular answer is to reject the presupposition that a new macroscopic object pops in somewhere along the sequence, by accepting either mereological nihilism, which holds that there are no complex objects, or mereological universalism, which holds that every plurality of objects composes a whole. Indeed, **Sider4D (Sider4D)** originated the continuous series thought experiment precisely to argue for mereological universalism.

Nihilism implies that either we are simple or we don't exist. That we don't exist is obviously quite implausible (*pace* **UngerDoNotExist, UngerDoNotExist**). The main view on which we are simple is Cartesian dualism, which has serious ethical problems as we saw in ??backref. Moreover, on nihilism, even if one can use Cartesian dualism to salvage the existence of humans, it is difficult to salvage the existence of plants and animals.

Universalism implies some counterintuitive claims, such as that there is a scattered object like the one described above, or that there is an object composed of the Empire State

Building and my left toe. More seriously, universalism has the consequence that where I exist, there is a vast number of beings that differ from me in a functionally insignificant way—say, by a small number of particles in skin cells. If they are all conscious like me, we have the ethically unacceptable consequence that we should prefer to relieve the pain of a larger rather than a smaller person, because there are many more such functionally insignificant “neighbors” in the case of a larger person (cf. **Johnston2016**, **Johnston2016**). If only I am conscious, then we have the difficulty of explaining what makes me so different.

Furthermore, mereological universalism is itself an unstable middle ground between the restricted composition view on which some but not all pluralities compose a whole and “modal profile universalism”. Say that a mereological modal profile of an object x is a mapping p that assigns to each world w a collection $p(w)$, possibly empty, of objects in w such that x exists in w if and only if $p(w)$ is non-empty and x is composed in w of all the members of $p(w)$.¹ Modal profile universalism then says that for any abstract mapping p that assigns to a world a collections of objects in the world there is an object x that exists in every world w such that $p(w)$ is non-empty and has p as a mereological modal profile. If we accept mereological universalism without modal profile universalism, then we have to draw an arbitrary line between such abstract mappings that do correspond to an object’s mereological modal profile and those that do not, and this arbitrary line seems every bit as problematic as that between pluralities that compose a whole and pluralities that do not compose a whole. There is thus reason to take the real choice for those who want to restrict composition to be between nihilism and modal profile universalism.

But modal profile universalism is rather problematic. Let p be my modal profile. For any infinite cardinality κ there is a possible world w_κ which has κ particles but no biological

^{1*}Note that as defined, an object can have more than one mereological modal profile. If x is not necessarily simple, then a minimal mereological modal profile for x will be given by letting $p_0(w)$ be the singleton $\{x\}$ if x exists in w and a maximal mereological modal profile will be given by letting $p_1(w)$ be the set of all of x ’s parts in w .

life.² Modal profile universalism then implies there is an object x_κ whose modal profile is p_κ , where p_κ agrees with p in every world except w_κ and $p_\kappa(w_\kappa)$ is the set of all the particles in w_κ . Since x_κ 's modal profile agrees with mine in every world where I exist, x_κ is on the same chair as I am, has the same shape as I do, and behaves the same way as I do.

There is an immediate rather counterintuitive consequence. There is no set of all such objects x_κ (it's easy to see they are all distinct), since if there were such a set, there would be a set of all cardinalities κ , and there is not (**PrussCardinalityObjection**). But surely there is a set of all the objects on my chair.

Furthermore, x_κ has exactly the same modal properties as I do when we restrict the scope of the modal properties to worlds with life. It is very difficult at this point to defend the claim that x_κ in the actual world differs from me with respect to consciousness. Thus if I have pain, an infinity of objects beyond cardinality is in pain. Thus, if you have a choice whether to relieve my headache or the headaches of all people in France, we cannot say that either option relieves the pain of more people—in both cases, an infinity beyond cardinality gets relief. Yet surely you should relieve the headaches of all the people in France rather than mine.

Responses can doubtless be given to these and other problems, but at least we can say that there are strong reasons to reject both nihilism and universalism. But now we need an solution to the problem of where a complex object pops into existence along a continuum of arrangements of particles.

However, the Aristotelian has a solution, namely the invocation of form: it is not possible to have a sequence where at one end you have one atom from every star, at the other have all of your parts, and the differences between successive cases involve changing one particle at a time. For your parts include a *human form* in addition to the particles, and the

²One may worry that if κ exceeds the cardinality of spacetime, multiple particles will have to share a position. But, first, there is no problem with bosons like photons sharing a position. And, second, these worlds could be infinite multiverses, and there does not seem to be any limit on the cardinality of universes in a multiverse.

one-atom-per-star object does not include a human form. Adding, subtracting or moving particles is not enough: one needs to add a human form somewhere in the sequence, and that is where the metaphysical transition happens.

We can, further, give an account of when a plurality of objects, the *xs*, compose a substantial whole: namely, there is a form among the *xs* which informs all the other *xs*, and everything informed by that form overlaps at least one of the *xs*:

$$(1) \exists F[F \in xx \ \& \ \text{Form}(F) \ \& \ \forall y((y \in xx \ \& \ y \neq F) \rightarrow \text{Informs}(F, y)) \ \& \ \forall y(\text{Informs}(F, y) \rightarrow \exists x(x \in xx \ \& \ O(x, y))),$$

where $O(x, y)$ says that x and y overlap, i.e., have a part in common, and where I assume that the $\text{Form}(x)$ predicate only applies to substantial forms.³ Here, the informing relation is one where the form gives identity to the thing it informs.⁴

We still have a Mersenne question, however, in the vicinity. When the particles that produce a human being come together—say, by gametes joining to form a zygote—a human form comes into existence and comes to inform the particles. Why does this happen precisely when it does? Here, however, at least three answers can be given. First, we might say that the gametes are still parts of the parents, and still informed by the parental human

³Van Inwagen famously popularized the special composition question of when a plurality composes a whole (**vanInwagenMaterialBeings**). The Aristotelian answer violates a stricture of van Inwagen's, namely that the answer be informative, which means that it cannot use mereological vocabulary—in our case by using the term “overlap”. Van Inwagen's reason for the stricture is to avoid trivial answers like “The *xs* compose a whole just in case there is a *y* composed of the *xs*.” However, the Aristotelian answer is non-trivial despite violating the stricture, and hence the stricture is excessively strict. It is likely that some other non-trivial answers violate the stricture as well, and indeed may have a similar structure to ours. Consider, for instance, an ancient atomist metaphysics with hard atoms of non-zero size on which the *xs* primitively compose a whole just in case the *xs* are a maximal plurality of atoms connected by chains of the contact relation, and the *ys* compose a whole just in case the plurality of their atomic parts composes a whole.

⁴Understood this way, informing is a relation that holds both between a substantial form and the material parts of the substance and between a substantial form and the accidents or accidental forms of the substance. One may, however, object that these are two different relations. If so, take my $\text{Informs}(x, y)$ relation to be a disjunction of the two.

forms, and these human forms specify the circumstances under which the human form shows up (this will be problematic when the parental organisms are pre-human apes). Second, we might give a similar answer, but instead suppose that the powers are grounded in the forms of the particles or other fundamental physical constituents of the universe. Third, we might suppose with the Thomistic branch of the Aristotelian tradition that at least in the case of human beings, God is creatively involved.??forwardref

2. Identity over time

One of the classic questions of metaphysics is about the grounds of identity over time. A very general way of posing the question is to ask for an explanation or ground of claims of the form:

(2) Object x_1 is identical to object x_2 ,

where x_1 and x_2 respectively exist at times t_1 and t_2 , which are presupposed to be different⁵ and where we require the explanation not to involve identity and to involve only purely qualitative properties and relations.⁶

Put in that very general way, it seems unlikely that we will have a solution, absent some extremely controversial metaphysical assumptions, such as Leibniz's Principle of Identity of Indiscernibles (PII).⁷

But there is a somewhat less general way of putting the diachronic identity question. Suppose that at time t_1 , some proper plurality of items, the x s, compose an object and at time t_2 the y s compose an object. Then we ask for the grounds of:

⁵One may worry that this presupposition cannot be stated without using identity, i.e., without denying that $t_1 = t_2$. However, it is not clear that even if this is true, it affects the significance of the question. Moreover, if time turns out to be linearly ordered, then we can state the presupposition disjunctively: t_1 is earlier than t_2 or t_2 is earlier than t_1 .

⁶If we allow the account to involve non-qualitative properties like Socrateity (the property of being Socrates), then we can say x_1 is identical to x_2 provided that any property had by x_1 is had by x_2 .

⁷The PII says that two things are identical just in case they have the same purely qualitative properties. If the PII holds, then we can say that $x_1 = x_2$ if and only if for every purely qualitative property we have $Q(x_1)$ iff $Q(x_2)$.

- (3) An object composed of the xs at t_1 is identical with an object composed of the ys at t_2 .⁸

This is not asking for an account of diachronic identity in general, but of diachronic identity of complex objects (note that we only require the xs to be a proper plurality, i.e., for there to be more than one of them).

Here we *can* give an Aristotelian account:

- (4) There is a form F such that at t_1 , F unites the xs , and at t_2 , F unites the ys .

Here we might stipulate that a form F unites the zs just in case F is one of the zs , every form among the zs informs each non-form among the zs , anything informed by F overlaps at least one of the zs :

- (5) $\text{Unites}(F, zz) \equiv [F \in zz \ \& \ \text{Form}(F) \ \& \ \forall G \forall x ((\text{Form}(G) \ \& \ \sim \text{Form}(x) \ \& \ G \in zz \ \& \ x \in zz) \rightarrow \text{Informs}(G, x)) \ \& \ \forall x (\text{Informs}(F, x) \rightarrow \exists y (y \in zz \ \& \ O(y, x)))]$.

There is no identity anywhere in (5).⁹

⁸The reason for the indefinite pronoun is that, first, there might be more than one object composed of the same parts and, second, using the definite pronoun introduces another instance of the identity relation given the Russellian analysis of “The F is G ” as saying that some F is G and has the property of being *identical* with every F that is G , whereas we only want an account of a single identity relation.

⁹Note that the right hand side of (5) is much more complex than (1), even though we are giving an account of a very similar phenomenon. A simpler formulation than (5), and more in line with (1), would say that F unites the zs just in case it is one of them and informs every one of the zs other than itself and everything informed by it overlaps at least one of the zs . However, this formulation makes use of identity in talking of *other than F* . To avoid the identity operator, we quantify over all the substantial forms among the zs and say that they inform all the non-forms. For the present account to work, we cannot have a case where all the things informed by a substantial form F are also informed by another substantial form G , i.e., it cannot be that all the non-substantial-form constituents of one substance a are parts of another substance b . For if we had that, then the (5) would make F unite all the constituents of a together with the form of b , which does not seem right. On a natural Aristotelian interpretation of the metaphysics of conjoint twins (??cross-ref), there are common parts that informed by two forms. Our assumption is compatible with that interpretation, but rejects the odd possibility of conjoint twins where the common parts include *all* the non-substantial-form constituents of one of them. If our assumption is rejected, then we may need to make use of the identity operator in (5). However,

It may appear that (4) uses the concept of identity in claiming that the *same* F unites the x s as unites the y s. Even if that were so, progress would have been made: an account of identity for complex would be given in terms of identity for simple things. But in any case, we need not concede the point as we can rewrite (4) in first order logic without any equal signs:

$$(6) \exists F(\text{Form}(F) \ \& \ \text{Unites}(F, xx, t_1) \ \& \ \text{Unites}(F, yy, t_2)).$$

Thus, (4) gives us an account of the diachronic identity of complex objects that does not presuppose identity.¹⁰ Perhaps we do not think it is important to give an account of diachronic identity—maybe it is brute (**MerricksBrute**; **PrussIdentity**)—but it seems that being able to give an account is better, other things being equal, than not being able to.

The problem of diachronic identity is most ethically pressing in the case of persons: our practices of promising, rewarding and punishing, and prudential reasoning all presuppose personal identity over time. In the case of diachronic *personal* identity, however, we do have four main substantive alternatives to (4). On the brain and soul theories, the identity of persons is defined by (functional or living) brains (**CampbellMcMahan**) and souls (**SwinburneSoul1997**) respectively.¹¹ On the animalist theory, the identity of persons is biological continuity of organisms (**OlsonHuman**; **ThorntonVarieties**). On psychological theories, the identity of persons is defined by psychological continuity of mental life (**ShoemakerIdentity**; **ParfitIdentity**).

the identity operator would only need to be used in the case of where one of the relata is a form, and so we could still have an explanation of identity of substances in terms of identity of form, and metaphysical progress will have been made.

¹⁰One might also worry that complexity itself presupposes identity: an object is complex provided it has two distinct parts. However, one can also define an object as complex provided it has a proper part, where x is a proper part of y provided that x is a part of y but y is not a part of x . Alternately, in the case of substances, an Aristotelian can define a complex object as one that has a part that is not itself a substantial form.

¹¹One may worry that this theory presupposes the identity of brains and souls over time. Interestingly, this need not be the case. Thus, we can say that x_1 at t_1 is the same person as x_2 at t_2 provided that there is a brain (or soul) b such that x_1 has b as a part at t_1 and x_2 has b as a part at t_2 . No equal sign appears here.

The Aristotelian theory becomes just a special case of the soul theory if Aristotelian forms are souls. Each of the other theories, however, suffers from a serious Mersenne problem: What grounds exactly how much of a brain needs to be preserved, or how much biological or psychological continuity is needed for personal identity? If one attempts to resolve this by saying that many answers are possible, and they each give rise to a concept similar to that of a person—namely concepts of a person₁, a person₂, ...—all with the same ethical consequences, then we have the problem of what grounds the boundaries of the whole set of such concepts. We can't allow total flexibility. For instance, if our criterion of psychological continuity is too tight or too loose, the resulting concept isn't recognizably much like a person (if it's tight enough, only nearly-instantaneous stages will qualify; if it's loose, then it can hold between me and Napoleon if I have read enough of his diaries).¹²

3. Teleological animalism and cerebra

Each premise of the following argument is very plausible.

- (7) We are humans.
- (8) All humans are mammals.
- (9) So, we are mammals.
- (10) All mammals are animals.
- (11) So, we are animals.

Nonetheless, the conclusion—labeled as “animalism”—is denied by many philosophers. One traditional path to this denial is a Cartesian dualism on which we are immaterial souls that inhabit human animals. The other path is modern colocationist views on which we are a special kind of material object—a person—constituted by a human animal. The third path is brain views on which we are brains, or parts of brains (namely cerebra), which in turn are a proper part of a human animal.

¹²Compare also Johnston's personite problem.??personite problem.

On all three paths, one will want to deny (9). Perhaps the best way to do so would be to distinguish “humans” in (7) and (8) into human animals and human persons, and insist that premise (7) is true only of human persons, while (8) is true only of human animals. Thus the argument is unsound if “humans” is used consistently, and otherwise it is invalid.

In place of (9), the opponent of animalism will have say that we are *associated* with mammals, by ensoulment, constitution or parthood. Yet (9) is by itself extremely plausible. To deny that we are mammals seems akin to denying that earth is round. Further, both the Cartesian and brain views imply that we rarely if ever see or touch another person. One needs extremely good arguments for such counterintuitive theses.

Probably the main candidate here is a family of arguments about the difficulties of accounting for cerebrum transplants on animalism (**OlsonWhat; OlsonWhat**). The simplest version is that if your cerebrum is removed from your skull and placed in a vat in such a way that it can continue functioning, then intuitively you continue to think and will come along with the cerebrum. But a cerebrum is not an animal. On the contrary, the cerebrumless body appears to be an animal. After all, some animals lack cerebra, and it seems that the destruction of a cerebrum in an animal that normally has one would result in the animal becoming severely disabled rather than ceasing to exist. Thus, animalism seems to point to the cerebrumless body being you and the cerebrum in the vat being something or someone else, contrary to our intuitions about us coming along with transplanted cerebra.

Cartesians, colocationists and brain theorists who identify with the cerebrum have no such problem. They can all say that the cerebrumless body is not you, and instead you inhabit or are colocated with or are just plain identical with the cerebrum in the vat.

Here I want to argue that an Aristotelian about humans can embrace a teleological variety of animalism on which it is natural to say that we go along with our cerebra. Animalism is indeed highly plausible as a view of the human person that does justice to the intuitions that humans are mammals, weigh much more than a brain when adults, and can be seen without neurosurgery, but faces the cerebrum transplant problem as the main

objection. If Aristotelianism can help animalists overcome that problem, that is some evidence for Aristotelianism about humans.

There are two teleological features in the animalism I will sketch. The first teleological feature is the thesis that what defines something as an animal (and indeed an animal of a particular type) is its teleology. While it is usual to think of animals as things that nourish themselves, grow, reproduce, and have a certain level of autonomy from the environment, the teleological animalist instead insists that animals need not engage in these activities, but need only have a teleological orientation towards them. Animals are the sort of thing that *should* engage in these activities.

The second teleological feature is to see organisms, including humans, as having a teleological *hierarchy*, with some telē subordinated to others. Sometimes the subordination is instrumental: our teeth rend food in order to nourish us. But there can also be a value-based subordination, where an activity, while not merely instrumental towards another activity, is less central to the flourishing of the organism and to the organism's identity. On the teleological animalism I am sketching, it is postulated that in humans, activities common to all animals are subordinated to specifically personal activities, namely rational and moral behavior.

When there is a hierarchical subordination, it is plausible to think that in cases of splitting, an organism is more apt to come along with the organs supporting the higher-level features. If in humans it is moral and intellectual capacity that is at the top of the teleological hierarchy, and the cerebrum is much more directly supportive of these capacities than the lower brain, heart, lungs, etc., then we would expect the human organism to come along with the cerebrum. If you cut a worm in such a way that one end contains just the head and the other contains the rest of the body, and both parts behave as if they were alive, it is reasonable to suppose that the form of the original individual may go with the larger piece which contains more in the way of life-supporting organs, rather than with the head, because the head is less teleologically central to the worm than the rest of its body.

But given human teleology, we would expect us to go along with the head if the head were put on life support, or even with the brain or cerebrum in a transplant.

What about the remaining cerebrumless human body, which maintains its vital functions? If the original individual goes along with the cerebrum, and if we take the idea of one human form in two bodies to be absurd, there are three possibilities for the cerebrumless body:

- (a) the cerebrumless body gains a new human form, or
- (b) it gains some non-human form, or
- (c) it becomes a non-living substance or a formless heap of matter.

If the cerebrumless body gains a new human form, then we have the counterintuitive consequence that a temporary removal of a cerebrum followed by reimplantation would either result in conjoined twins—two human beings joined at the edges of the cerebrum—or would result in the death of one or more of the two human beings. None of these options seems very plausible, but at the same time, we should not be too surprised if strange things happen when cerebra are moved around.

If the cerebrumless body gains a non-human form, this is presumably an organismic form, since the entity is capable of nourishment, reproduction, etc. But we have something moderately puzzling: a non-human organism that would produce a human being if it were to mate with a human or with another organism of the same sort but of the opposite sex. Moreover, it seems plausible that if the cerebrumless body has a teleology at all, that teleology impels it to try to support the cerebrum: oxygen would be directed by the body towards the missing cerebrum, presumably. This suggests that the being is incomplete without the cerebrum, that it ought to have a cerebrum. But a being that ought to have a human cerebrum seems to be a *human* being.

The last option is a formless heap of matter or a non-living substance (such as a body of water might be on some Aristotelian theories). This does not seem particularly puzzling

in the transfer case. If the form departs, by going along with the cerebrum, then formlessness would seem to be the obviously expected result, barring some special reason to the contrary.

What if instead of the cerebrum being removed and put on life-support, the cerebrum is simply destroyed? This corresponds to the tragic real-life scenario of upper brain death. Bioethicists disagree on whether upper brain death is death (**TruogDeath**; **NapierDeath**). We have four moderately plausible options for cerebral destruction:

- (i) the original individual continues to live, i.e., the cerebrumless body retains the original human form, or
- (ii) the original individual dies and the cerebrumless body gains a new human form, or
- (iii) the original individual dies and the cerebrumless body gains a non-human form, or
- (iv) the original individual dies and the cerebrumless body has no organismic form, and is either a non-living substance or a heap.

Between (a)–(c) and (i)–(iv), there are twelve combinations. However, we can intuitively reduce the number of options quite significantly. I have assumed that in the transfer case, the original form goes along with the cerebrum in the transfer case, departing from the cerebrum. Suppose that (i) is false. Then the original individual dies and the cerebrumless body is deprived of its original form. It seems very plausible that what happens to the cerebrumless body upon deprivation of its original human form should not depend on what that form does after departing the cerebrumless body—whether it continues to inform a reduced body (the cerebrum), or survives disembodied as many religious people think, or perishes. Thus, if (i) is false, then (a), (b) or (c) holds respectively if and only if (ii), (iii) or (iv) holds.

We thus have three plausible options with (i) false:

- (α) (a) and (ii)
- (β) (b) and (iii)

(γ) (c) and (iv)

What if (i) is true, so that in the case of cerebral destruction, the original form continues to inform the cerebrumless body? Does this tell us anything about what happens in the transfer case? One might initially think that the falsity of (i) fits poorly with any of (a), (b) and (c). After all, if destruction of the cerebrum results in the form staying with the cerebrumless body, shouldn't we expect the form remain with the cerebrumless body when the cerebrum is transferred?

But this is not clear on teleological animalism. For we might think that in division or partial destruction of the body, the form goes along with the part that is the best candidate for being informed by it, at least when there is a unique best candidate and that best candidate is "good enough". On the teleological account, the quality of a candidate for being informed is measured by how high the goals directly promoted by the part lie in the teleological hierarchy. If the part's proper function is to directly promote goals at the very top of the hierarchy, then the candidate is automatically good enough. Thus, when the cerebrum survives, it is reasonable to think the form goes along with the cerebrum, because the cerebrum is good enough as a candidate, and definitely higher in the hierarchy than the cerebrumless body. But if the cerebrum is destroyed, the cerebrumless body is the unique best candidate, because it is the only candidate. Whether the cerebrumless body is good enough as a candidate is not clear, but neither is it clear that it is not good enough. It is, after all, on the next step down in the hierarchy after the cerebrum, having among its tasks the full support of the cerebrum's functioning, as well as many important purely animal functions. Thus, all of the following are at least somewhat reasonable epistemic possibilities:

(δ) (a) and (i)

(ϵ) (b) and (i)

(ζ) (c) and (i)

On teleological animalism we thus have six combinations for making sense of what happens to the form and cerebrum, namely (α)–(ζ), and none of them appear immensely

problematic, though (a) and (b) have some moderately counterintuitive consequences. But even if we feel the need to reject these, that still leaves us with (γ) and (ζ): in cerebral transfer cases, the cerebrumless body is not a living thing, while in cerebral destruction cases, the cerebrumless body may (if we have (i)) or may not (if we have (iv)) continue to be a living human being.

One may think (ζ) is implausible, because it implies that whether the body-minus-cerebrum continues to have the original human form depends on what happens to the cerebrum—whether it is merely removed or actually destroyed. But such dependence is, as already noted, to be reasonably expected. If we think of the cerebrum as a magnet for the original human form, then transfer of that “magnet” might be reasonably be thought to pull the form along, hence depriving the cerebrumless body of it, while destruction of the “magnet” might well leave the form in place.

Teleological animalism, thus, has multiple ways of making sense of what happens in cerebral transfer and destruction cases, while these cases are taken to be highly problematic to other types of animalism. Given the plausibility of animalism as such, this gives us another reason to accept teleological animalism.

4. Relativistic considerations

4.1. Introduction. The Special Theory of Relativity—and the General as well, but let’s simplify the discussion by restricting to the Special—has some rather interesting implications for spatially extended objects. I will first give an argument that a rattlesnake can survive (at least briefly) as just a part of a rattle. But a rattle is far less central to the organic life of a snake than a cerebrum is to that of a human, and hence if a rattlesnake can survive as just a part of a rattle, a human can survive as just a cerebrum.

Then I will offer a geometric probabilistic argument for the Small Endpoints Thesis (SET) that *all* material substances are *subatomic* in size at beginnings and ends of their lives, and a metaphysical causal argument for the Small Beginnings Thesis (SBT) that this is true at least of the beginnings of material substances. These theses are hard to believe. Many

already find the idea that we started our lives as zygotes hard to believe, since human zygotes do not seem sufficiently sophisticated to support the functioning characteristic of humans. But that we started our life as beings much smaller than zygotes invites a completely incredulous stare. Nonetheless, the arguments for SET and especially SBT appear to be very strong.

The two smallness theses will then point to significant advantages for the robust Aristotelian metaphysics defended in this book over “faint-hearted” Aristotelianism and non-Aristotelian alternatives, in addition to providing further support for the possibility of survival as a cerebrum. Furthermore, these theses require a significant weakening of traditional Aristotelian claims about matter having to be well-disposed to have a particular form, as well as explanations of the perishing of a substance in terms of the matter losing that disposition.

I note that my claim in the two small endpoint theses is contingent: our world doesn’t have any such material substances. Furthermore, given that the argument for theses is based on Relativity, if we live in a multiverse and some of the universes in the multiverse have different laws from ours, I stipulate that the claim is only about the denizens of our universe.

4.2. Some background. Special Relativity presents spacetime as four-dimensional, with three spatial and one temporal dimension. The famous fact that will be crucial to the arguments here is that many ordinary concepts, such as simultaneity, then end up relative to a reference frame (specifically, an inertial one). For any object—real or hypothetical—constantly moving with a constant sub-light velocity (i.e., undergoing sub-light inertial motion), there is a reference frame according to which that object is standing still. Note, however, that the concept of “moving with a constant sub-light velocity” is itself absolute: if an object does that in one reference frame, it does it in all of them (standing still is the special case of this where the velocity is zero). However, what is relative is simultaneity: if two distinct points of spacetime are simultaneous in one reference frame, there will always be another reference frame where they are not simultaneous.

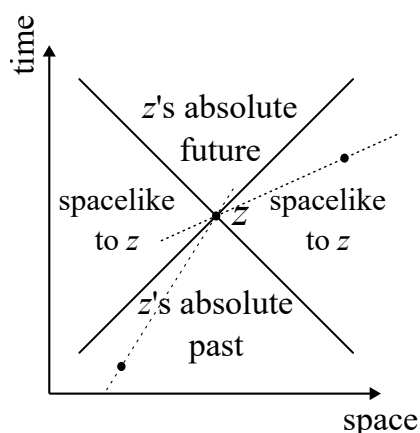


FIGURE 4.1. Points spacelike to z , as well as z 's absolute past and absolute future. The points timelike to z are all of the absolute past and future plus z itself.

We will need some more relativistic detail for the arguments.

Two points of spacetime are said to be “timelike-related” provided it’s possible to travel between them at a speed lower than that of light. They are “spacelike-related” provided that to travel between them you need to exceed the speed of light.¹³ If two distinct points z and w are spacelike-related, there will always be reference frames according to which they are simultaneous, and reference frames according to which they are not simultaneous. In fact, there will be a reference frame according to which z is earlier than w and a reference frame according to which w is earlier than z .

On the other hand, if two distinct points are not spacelike-related, then they are absolutely non-simultaneous—non-simultaneous in every reference frame. Furthermore, one of these points will be absolutely earlier than the other.

In some of our arguments, we will give two-dimensional diagrams, where space is the horizontal axis and time is the vertical axis, and the units are chosen so that the speed of light is equal to 1. In these diagrams, an object moving at the speed of light is represented by a line at 45 degrees to horizontal, while lines that are more or less than 45 degrees to

¹³The remaining case is that it is possible to travel between them but only at exactly the speed of light. In that case, they are “lightlike-related”. We won’t need that case.

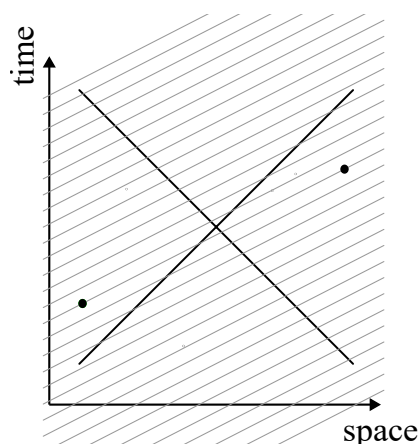


FIGURE 4.2. A reference frame defined by a collection of parallel hyperplanes. The two black points are simultaneous according to this frame.

horizontal respectively represent movement at less or more than the speed of light. (See Figure 4.2.)

Given a reference frame F , for any point z in spacetime there will be a set of points in spacetime simultaneous with z according to F . These points all lie on the same “hyperplane” (i.e., a flat three-dimensional slice through spacetime). Moreover, these points are all spacelike-related to each other, so we say that this is a spacelike hyperplane. A reference frame F then divides up spacetime into a family S_F of parallel spacelike “hyperplanes of simultaneity” such that two points of spacetime are simultaneous according to F provided that they lie on the same hyperplane K in S_F . In a two-dimensional diagram, the reference frame will be a set of parallel lines whose slope to the horizontal is less than 45 degrees. (See Figure 4.2.)

Conversely, given any one spacelike hyperplane, we can form the set S of all the hyperplanes parallel to it, and this set then defines a reference frame F such that $S_F = S$.

4.3. Of snakes, rattles, form and matter. Imagine a rattlesnake stretched out in a line from head to tail, with the snake staying still in a reference frame F . For concreteness and simplicity, let’s model the rattlesnake as half a meter long, including a six centimeter rattle, with a thickness of at most three centimeters throughout its body. Now something terrible

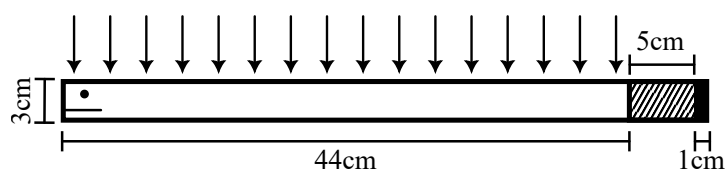


FIGURE 4.3. An idealized cylindrical snake subject to blasters (arrows).

Striped area: non-tip rattle; black area: rattle tip.

will happen to the rattlesnake. Science-fictional blasters are pointed at every centimeter of the snake's body other than the rattle. These blasters are all timed to have their blasts hit the snake simultaneously, in frame F , at t_1 . Once the blast hits the snake, it very quickly vaporizes the section of the snake's body it is pointed at. Stipulate that a "jiffy" is the very short amount of time it takes light to travel one centimeter, about 33 picoseconds,¹⁴ and let us suppose it takes four jiffies to vaporize each section of the snake (the section is three centimeters thick, so the blast can go from one side to the other of the snake in less than four jiffies without exceeding the speed of light). If t_2 is four jiffies after t_1 , then at t_2 all that will be left of the snake is the rattle.

But according to Relativity, causal effects propagate at speeds not exceeding that of light. Now, consider the *tip* of the rattle, which I will stipulate to be the very last centimeter. At t_1 , the nearest blast is five centimeters away from the tip of the rattle. Since the blasts are the cause of death, and causation propagates at a speed not exceeding that of light, over the four jiffies between t_1 and t_2 , the effect of the blasts will not yet reach the rattle's tip.

Now, on an Aristotelian metaphysics, for the rattle's tip to cease to be a part of the snake, the tip must lose its snake form. This requires that the tip must be affected in some way by the blast. At t_2 , however, the tip has yet to be affected, given the speed of light limit. But as long as the snake form is found in *some* matter, the snake still exists. So at t_2 , the snake exists, even though everything outside the rattle has been destroyed (and

¹⁴This definition is attributed on the Internet to the chemist Gilbert N. Lewis, but I have not been able to track down an original citation.

presumably the vaporization of the section of the snake just next to the rattle has resulted in some of the rattle getting destroyed, too). Thus, as I advertised in the previous section, we have seen that the snake can survive as just a part of a rattle.

Likely, this survival is very short: we can reasonably speculate that the loss of snake form proceeds through the rattle very fast, maybe even at the speed of light. A few more jiffies, then, and the snake will likely be gone.

One might object that loss of form is not the kind of effect that we have in mind when we say that causal influences do not exceed the speed of light. There are two supplementary arguments I can offer here. First, if the tip of the snake's rattle has indeed lost its snake form at t_2 , let z be a spacetime point within the tip of the snake's (no longer alive) rattle at t_2 . Let E be the event of the blasters beginning to blast. Then z and E are spacelike-related, because the spatial distance between z and the nearest part of E is five centimeters, while the temporal distance between z and E is four jiffies, so one cannot travel between z and E without exceeding the speed of light. But if they are spacelike-related, then there will be a reference frame F' according to which z is earlier than E . Thus, in reference frame F' , the tip of the rattle will have lost its form *before* the blasters went off. This seems very strange indeed.

Second, it is a standard Aristotelian point that a body part that loses its form stops existing. A severed finger is a finger in name only: the finger has perished and there is just a heap of matter. Thus, loss of form is the destruction of the rattle's tip. But the idea that the blasters somehow managed to destroy the rattle's tip in a faster-than-light way is rather implausible. Destruction of objects seems like the kind of thing that shouldn't proceed in a faster-than-light kind of way, especially since the argument of the previous paragraph shows that the destruction in some reference frames would have to precede the blast.

As already noted, if a snake can survive—even if only for a few jiffies—as only a rattle, there should be no deep difficulty in a human surviving as a cerebrum.

One might object that rattles are not living and informed parts of a snake, but dead stuff. If so, then replace the rattlesnake with a non-rattlesnake in the argument, point blasters at all but the last six centimeters of tail, and use the same argument to conclude that the snake can survive as just a few centimeters of tail. This is not quite as amazing as surviving as part of a rattle, but it is sufficient to make the *a fortiori* argument above about the possibility of survival as a cerebrum.

Finally, note that this snake thought experiment already points our attention towards SET, by showing that it is at least *possible* for a large organism to shrink to an insignificant portion at the time of death.

4.4. A geometric argument for SET. First, imagine you are holding a perfect cube. Now imagine putting a perfectly flat plane against the cube. If the plane is parallel to and touching one of the six faces of the cube, it will make contact with the whole of the face, and hence with infinitely many points. Similarly, if the plane is parallel to and touching one of the twelve edges of the cube, it will make contact with the whole of the edge, again with infinitely many points of contact. But in every other orientation of the plane against the cube, the plane will contact the cube at exactly one point, indeed at one of the eight vertices of the cube.

Moreover, there is a sense in which there are vastly more orientations of the plane relative to the cube where the plane contacts the cube at one point than where the plane contacts the cube at more than one point. There are only six orientations where the plane contacts the cube at a face—the plane has to be parallel to that face. There are, admittedly, infinitely many orientations where the plane contacts the cube along an edge, but these orientations are constrained by the need for the plane to be parallel to the edge. Once we decide which of the twelve edges of the cube to align our plane with, we have only one degree of freedom with regard to the orientation between the cube and the plane. Imagine rocking the plane against the cube, keeping contact with the edge—there is only one degree of freedom to the rocking. But when the plane makes contact with one of the eight vertices of the cube, there are two degrees of freedom—we can rock the plane along two different

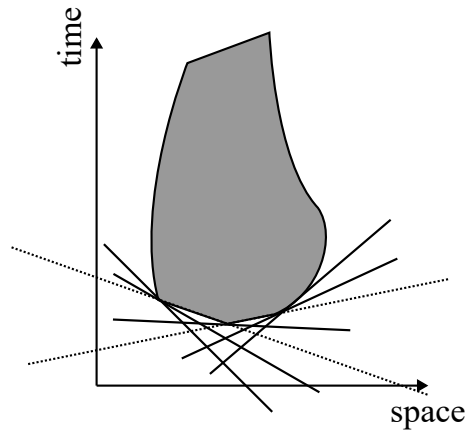


FIGURE 4.4. Only two hyperplanes with slope less than 45° meet the beginning boundary of the gray object in more than one point.

angles and keep the contact. This geometric observation remains correct regardless of what solid object we use in place of the cube. If we place a plane against the boundary of the object, for “almost all orientations” of the plane, the plane will be contacting the object only at a single point. (In the case of some shapes like an ellipsoid, this will be true for all orientations.) Moreover, the same thing is true in n dimensions for any $n > 1$: if we have an n -dimensional object and have an $(n - 1)$ -dimensional hyperplane rest against it, for almost all orientations of the hyperplane, the object will contact the hyperplane at a single point. (This is made mathematically precise in the Appendix.)

Now move to the four-dimensional spacetime of Special Relativity and consider a material substance with a beginning and end to its life, say Rover. Let D be the four-dimensional shape consisting of the points of spacetime occupied by Rover, and imagine a reference frame F . This reference frame has a simultaneity relation defined by a family S_F of spacelike hyperplanes, where, recall, two points are simultaneous provided that they lie on the same hyperplane in S_F . Assume D is finite in size, both spatially and temporally (i.e., no more than x meters in diameter at all times and no more than y seconds in temporal extent, for some finite x and y). Suppose z is one of the points of spacetime where Rover begins to exist with respect to frame F . Let K be the hyperplane in S_F that passes through z . This will be a hyperplane resting against D . If Rover begins to exist at two or

more points according to frame F , then K will make contact with two or more points of the boundary of D . Now, for almost all orientations, a hyperplane of that orientation that rests against the object will make contact with the object's boundary at a single point, as discussed earlier. Moreover, the orientation of a spacelike hyperplane defines a reference frame (since the orientation of a hyperplane is sufficient to determine the family of parallel hyperplanes). Thus, according to almost all reference frames, Rover begins to exist at a single point. (See Figure 4.4 for a two-dimensional illustration.) Exactly the same argument applies to Rover's demise: at almost all reference frames, Rover ends its life at a single point.

A few words are needed about what is meant by "almost all" reference frames. A reference frame is defined by the orientation of its hyperplanes. We can define an orientation of a hyperplane K by a four-dimensional vector v that is perpendicular to K and has length one. The set of all such vectors is a hypersphere of radius one (the set of points in four-dimensional space of distance one from the center), though we are not interested in the whole hypersphere, but only in the part of it that corresponds to vectors that are perpendicular to spacelike hyperplanes.¹⁵ Now let U be the set of all the unit-length vectors v such that (a) the vector v defines a reference frame (by defining a family of spacelike hyperplanes perpendicular to v) and (b) in this reference frame Rover starts existing at two or more points. Then U has zero hyperarea as a portion of the hypersphere of all vectors of radius one. Mathematicians refer to something that happens in a region of zero area, volume, or similar measure as happening "in almost no cases", and to what happens everywhere outside such a region as happening "in almost all cases". It follows that in almost reference frames, Rover starts existing at only one point, and ends at only one. A more precise version of this argument will be given in the Appendix.

Our argument assumed that Rover has finite extent in both space and time. Suppose now Rover has infinite extent in time but is still bounded in space—i.e., at all times, Rover is contained in some ball of finite diameter. There are three possibilities: (i) Rover has

¹⁵*It is important for technical reasons that this part has non-zero hyperarea.

neither beginning nor end, (ii) Rover has a beginning but no end and (iii) Rover has an end but no beginning.

Regarding (i), Rover is irrelevant to SET because SET claims that all the beginnings and endings of material substances are subatomic, and if Rover lacks a beginning or end, then Rover does not contribute to the collection of beginnings and endings of material substances.

Case (ii), where Rover has a beginning but no end, requires a minor tweak to the argument. Just apply the above argument to a finite-duration initial temporal part R_1 of Rover of sufficient temporal duration (as measured in whatever frame you like) so that in all reference frames Rover's beginning occurs within R_1 . And case (iii) is very similar: just apply the argument to a sufficiently long but finite-duration final temporal part R_2 of Rover.

What remains is the case of material substances that have infinite spatial extent. Here I suggest a simple hypothesis: there aren't any such. The only at all plausible candidate for such a thing would be the universe as a whole, and given the Aristotelian thesis that substances can't have substances as proper parts, and the fact that we are substances and would be part of the universe, we have to conclude that there is no such substance as the universe. If I am wrong about that, then as far as my arguments go, I will need to restrict SET and SBT to substances other than the universe, which does not affect any of the applications I will make.

Thus, if we fix a random reference frame, we can be all but certain that any specified material object has a point-like beginning with respect to the frame if it has a beginning at all, and it has a point-like ending if it has an ending. In fact, as long as the number of material objects in the universe with finite lifetime is finite or countably infinite, then for any fixed random reference frame, we can be all but certain that *all* of them start off and end up subatomic in size. (This follows from the mathematical fact that the union of a countable collection of sets of measure zero has measure zero.) When I initially gave the Small Endpoints Thesis, I didn't specify a reference frame. The above argument is compatible with the existence of some reference frames where SET is false. However, the argument shows

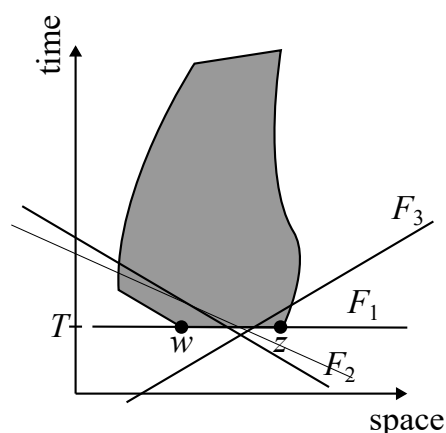


FIGURE 4.5. In F_1 , w and z are simultaneous, but in F_2 , w comes first, and F_3 , z comes first.

that for any fixed frame the reader might have in mind—say, a frame defined by some precisification of the reader’s current center of mass—it is nearly certain that SET is true in that frame, since almost all frames have SET true in them. It would require incredible luck for the reader to have in mind a reference frame where SET is false—luck akin to tossing a coin infinitely many times and getting heads each time. I assume the reader is not so lucky.

4.5. A causal argument. An object exists as soon as any part of it exists. Moreover, absent backwards causation, whether an object exists at a time t never depends on what happens after t . This thesis when combined with some additional considerations about causality and Special Relativity yields an argument for the Small Beginnings Thesis, that material objects with finite lifetimes start off subatomic.

Now, for a reductio, suppose that Rover is a material object whose life occupies a bounded region of spacetime and that Rover does not start off subatomically small in some reference frame F_1 . For simplicity, I will assume that Rover has a first moment, T , of existence with respect to F_1 . This is not guaranteed. It could be that Rover exists at all moments t after some time T but not at T itself. The argument could generalize to that case, but becomes more technically complicated.

All material objects have causes. Let Spot be Rover's cause. Then Spot in causing Rover acts on Rover at at least one spatiotemporal point, z , which point is at a time T and within the boundary of Rover's four-dimensional region. Since we have assumed Rover is not subatomically small, there is another spatiotemporal point at time T and within Rover, which point we will call w . Because z and w are simultaneous with respect to frame F_1 , they are space-like related, and it follows that there is another reference frame, F_2 , according to which w is earlier than z (see Figure 4.5).

Now, let's think about how things look in our new frame F_2 . Since Rover exists at w , and w is earlier than z , what happens to Rover at z is causally irrelevant to Rover's existence. Hence Spot's causal activity at z is causally irrelevant to Rover's existence. But causation should be something that does not depend on reference frame. This thesis is intrinsically plausible and is also implicit in standard ways to think about Relativity Theory.¹⁶ So we have a contradiction: Spot's causal activity at z both is relevant and not relevant to Rover's existence.

It is worth noting that there is a further detail to the beginnings of material substances that we do not get on the geometric argument but follows from an extension of the causal argument. The geometric argument is compatible with a material substance beginning at different spacetime locations in different reference frames. Imagine that the material substance is smoothly rounded at its temporal beginning. Then different spacelike supporting hyperplanes on the "bottom" (i.e., temporally earlier) side of the four-dimensional region occupied by the substance will contact the hyperplane at different points. (Imagine supporting hyperplanes for a four-dimensional ball: they all meet the ball at different points.) Each such hyperplane defines a different reference frame, and in each reference frame then

¹⁶For instance, one of the reasons why faster-than-light causation is problematic is that if event C caused event E in a faster-than-light way in one reference frame, then in another reference frame event E would be prior to C , and so the causation from C to E would have to go backwards in time. But this argument against faster-than-light causation only works if causation is not dependent on the reference frame. If causation were dependent on a reference frame, we could simply say that in the second frame, event C does not cause E .

the object will begin at a different point of spacetime (namely the contact point between the hyperplane and the boundary of region occupied by the substance).

But an extension of the causal argument does not allow this. For suppose that in frame F_1 Rover begins at point z in spacetime and in frame F_2 Rover begins at a different point w of spacetime. If z and w are spacelike related, then there will be a third frame F_3 where z is earlier than w (see Figure 4.5). Then, much as in the original argument, relative to F_3 , Spot's causal activity at w comes after Rover has already come into existence at z , and hence is irrelevant to Rover's existence, a contradiction. Suppose now that z and w are not spacelike related. Then one of them will be *absolutely* earlier than the other. Swapping labels if necessary, we can suppose that z is the earlier one. But then Spot's causal activity at w will come after Rover has come into existence relative to *all* frames, and hence will be irrelevant.

Assuming causal influences travel at the speed of light, this argument also yields a picture of what the temporal beginnings of a material substance are like. They are sharp and pointy. For if the substance absolutely starts at point z due to Spot's causal activity, and causal influences travel at the speed of light or less, then all of the region occupied by Rover needs to be contained in the region reachable from z at the speed of light or less, i.e., in the future light-cone with vertex z . Thus Rover has a sharp point at its beginning.¹⁷

The above argument was phrased in a way takes Rover's cause to be outside Rover. But what if we think of an object that is caused to exist by the interaction of parts? Imagine, for instance, a proton and an electron that come together to form a hydrogen atom. As they approach each other, the proton comes to be significantly affected by the electron when the proton is at some point w of spacetime while the electron comes to be significantly affected by the proton when the electron is at some point z of spacetime. The two points w and z are spacelike-related. We can then suppose that the hydrogen atom comes into existence

¹⁷Or at least sharp in the natural units of physics where the speed of light c is 1. In such a unit system, lightcones subtend a 90 degree angle at their vertex. In a unit system where the speed of light is a large number, light cones are much blunter.

all-at-once along the line joining w and z , i.e., all-at-once in a reference frame at which w and z are simultaneous, and does so because of both of the causal interactions.¹⁸ How does the argument counter this kind of a suggestion?

If Rover is the hydrogen atom, then notice that when the proton is at w it is being affected by the electron in a pre-Rover state, since physical interactions cannot exceed the speed of light, and the electron only comes to be a part of Rover when the electron is at z , which is spacelike related to w . However, for Rover to come into existence at w , the past existence of the electron is insufficient. While it is the past existence of the electron that affects the proton at w , for Rover to come into existence at the time of w (in a reference frame in which w and z are simultaneous, one assumes), the electron must also exist at that time, assuming that a hydrogen atom cannot exist without an electron. Thus, the state of affairs of Rover's existing at w depends not only on the causal interaction with the electron's past state but also in a faster-than-light way on the present existence of the electron. And in a reference frame where w is earlier than z , Rover's presence at w depends on the later existence of the electron at z . It follows in turn that how many objects exist at a point in time depends on what happens on later times—granted, on what happens at times that are only later in some reference frames, but this still seems quite problematic.

4.6. Subatomicity. I advertised the endpoint claims as holding that beginnings and endings are subatomic, but the actual relativistic arguments seem to prove something stronger: that the beginnings and ends are points.

There are, however, several reasons for preferring the weaker “subatomic” formulation. First, the arguments are based on Special Relativity, and assume a spacetime continuum. It might, however, turn out that this is only an approximation to an underlying discrete spacetime. Assuming that Special Relativity is a sufficiently good approximation, our arguments may still work approximately, but break down at a sufficiently small scale.

¹⁸I am grateful to an anonymous reader of one of my papers for this suggestion.

That scale is very likely to be something like the Planck length scale, which is very much subatomic. So maybe all we get are subatomic endpoints.

Second, on interpretations of Quantum Mechanics other than Bohmian ones, particles do not have precise positions. The arguments appears to point towards substances beginning as single fundamental (and hence subatomic) particles, but the positions of these particles may be fuzzy and not pointlike.

Third, our arguments are compatible with a scenario on which material substances only approach pointlikeness asymptotically as we approach their beginnings and ends. The first argument, in the more precise formulation it has in the Appendix, yields the claim that in almost all reference frames, the hyperplane marking the beginning or the end of the substance intersects the boundary of the spacetime region occupied by the substance at a single point. However, the boundary of the spacetime region occupied by the substance need not itself be occupied by the substance. The substance might only occupy the *interior* of the region, not including the boundary. In such a case, the endpoint hyperplane does not actually intersect the region occupied by the substance, and the region of nearby hyperplanes occupied by the substance will asymptotically tend to a point.??forwardref A similar remark can be made about the second argument if we drop the simplifying assumption that the substance has a first moment of existence.

4.7. Metaphysics and SET/SBT. SET is counterintuitive when combined with the existence of mid-sized objects like ourselves, or else objects like atoms or molecules that make no sense without a multiplicity of particles. But if we accept mereological nihilism (**SiderNihilism**), and deny that there are any complex objects, SET becomes quite unsurprising. On microphysical mereological nihilism, the only objects are fundamental particles and/or points of space or spacetime, and these obviously start small. On existence monism, we have a mereological nihilism on which the only object is the universe (**HorganPotrc**), and one might interpret Big Bang physics as implying that this object starts small. One can also restrict these mereological nihilisms to physical objects.

Additionally, one can imagine a multiverse mereological nihilism on which physical reality consists of a plurality of partless universes, each of which starts small—a view that does not appear to have any current defenders.

All such views, unless combined with Cartesian dualism, have the implausible consequence that we do not exist. Supposing that we take this consequence to be unacceptable, we can ask if there is any metaphysical view on which we can embrace SET and allow for the existence of mid-sized objects like ourselves. But there is. On Aristotelian metaphysics, complex material objects are unified by having a substantial form, a non-material component. The Aristotelian convinced by SET need only suppose that a new substantial form of a mid-sized object like a human being always comes into existence at a single point of spacetime, becoming united to matter at one point. Then the form spreads to a larger chunk of matter (plausibly at some speed not exceeding that of light) until we have a macroscopic object. Eventually, it is time for the object to perish, and the form gradually stops informing matter, until it is only informing matter at only one point. This is a decidedly strange view, but it does do justice to the arguments for SET and it is not as problematic as nihilistic views on which we do not exist or Cartesian views on which we are not physical at all.

We now explore some applications of versions of Aristotelian SET/SBT on which human beings nonetheless exist.

4.8. Applications of SET/SBT.

4.8.1. *Non-living and living substances.* It is worth noting that the Aristotelian version of SET probably cannot save objects like atoms, molecules, or chemical substances such as chunks of iron or bodies of water, since all of these seem by their nature incapable of existing at one point. It indeed seems clearly essential to an atom, molecule or chemical that it be made of a certain plurality of electrons and protons. Non-living complex objects seem to all require a plurality of parts incompatible with SBT. We thus have good reason to accept the mereologically nihilist story about such things.

However, organisms can be saved, if we generalize to all organisms Aristotle's idea the form of an animal is the soul of the animal and what makes an animal alive is the presence of the soul. For what is essential to an organism is its life, and if we see the life as proceeding from the soul rather than an arrangement of particles, then a soul located in a tiny amount of matter seems logically sufficient for the organism's existence.

Granted, it is counterintuitive to think of a plant that does not (yet or any more) have a single atom. But let's take seriously the idea that plants are alive. Now, it is notoriously difficult to give a satisfactory definition of life in terms of physical arrangement. Biologists give a variety of definitions including such factors as growth, nutrition, reproduction and homeostasis, but there is no agreement on what all the terms mean. And yet the concept of life has a real moral significance. Think of how it makes perfect sense for someone to spend a large chunk of their life in caring for plants with a kind of emotional attachment that would be inappropriate for artifacts unless these artifacts happen have some special connection with human beings in the way artworks do. As we saw, moral significance tends to push one away from positing vagueness^{??backref}, and so there is something attractive about the idea of there being a sharp distinction between life and non-life, and seeing life as grounded in a special type of substantial form, a *soul*, is thus at least somewhat appealing (more on the nature life will be said in ^{??forwardref}).

However, one could also imagine an Aristotelian convinced by the arguments for SET/SBT denying the existence of lower life-forms, just as I have suggested we should about complex inorganic objects. But once we reach the level of animals where there is consciousness, it is difficult to deny existence. The Cartesian intuition that consciousness can only be found in an individual entity—rather than, say, in a mere plurality of particles—is compelling. And as we saw in ^{??backref}, there is good reason to think on the basis of the shortcomings of contemporary materialist philosophy of mind that something beyond physical arrangement, such as substantial form, is needed for consciousness. While it is counterintuitive to suppose subatomic plants, subatomic things with the kind of soul that

can render them conscious (at least once they acquire more matter) seem a little bit less counterintuitive.

Let us assume, thus, that humans and higher animals can exist notwithstanding SET/SBT, leaving open the question of lower animals, plants, bacteria and viruses.

4.8.2. *Cerebra and zygotes.* An immediate consequence of SET/SBT is that there is no special metaphysical difficulty in an animal's surviving as a cerebrum. After all, a cerebrum is many orders of magnitude more organized than a subatomically-sized substance. Similarly, those who think that we could not have been zygotes because zygotes are insufficiently sophisticated to support our existence must abandon this argument given that we—and all other material substances with beginnings—started off even less organized than a zygote, namely subatomic. Of course, the relativistic considerations above do not show that we were once zygotes or that we would in fact go with a cerebrum in a cerebrum transplant. As far as the relativistic considerations go, we might have started our lives as subatomic objects within the pinkie of a biologically two-year-old body, and we might end our lives as subatomic objects in a big toe (compare the rattlesnake argument, after all, in ??backref). But it seems more in keeping with the anti-skeptical bent of Aristotelian optimism to think that we start and end our lives in a somewhat less counterintuitive way.

4.8.3. *Proper disposition of matter.* It has been commonly believed by Aristotelians that in order to receive a form, matter must be properly disposed, and, conversely, form departs from matter precisely when the disposition of matter becomes inadequate. However, given that a subatomic chunk of living human matter cannot be distinguished from a subatomic chunk of living oak tree matter or a subatomic chunk of a dead cat, the disposition thesis becomes insupportable as it stands.

Nonetheless, one might be able to defend a version of the disposition thesis where one considers the “adequate disposition” of a piece of matter to be in part a function of the environment the matter finds itself in. Thus, one might hold that a subatomic pointlike piece of matter whose environment is the rest of a just-fertilized or about-to-be-fertilized zygote counts as properly disposed to receive the form, and a subatomic pointlike piece

of cerebral matter whose environment is a cerebrum that has lost or is about to lose its functioning counts as no longer properly disposed to hold on to the form. Such proper disposition facts could be grounded in the causal powers of the parents, in the case of the beginning of life, or in the causal powers of the substance to “give up the ghost and create a corpse”, in the end of life cases.

There is independent reason to think that the adequate disposition thesis depends on relational features of the matter in question. Start with an artifactual example. Famously, Michelangelo in his Sonnet 151 holds that statues are already there in the stone, but need to be freed **Nims**. It seems more in keeping with common sense ontology, however, to say that the statue came to exist only once enough of the outside marble was chipped away. In other words, the existence of the statue required the matter of the statue to have an adequate environment: one lacking marble all around. While statues may not actually be substances, we can imagine biological examples. Suppose, for instance, that suddenly new matter appeared in all the empty spaces between the particles of our body, thereby blocking all functional behavior. Of course, we would soon suffocate and die, but it is plausible that even before that it would be correct to say that the body is no longer adequate to its function. The empty spaces in our bodies are crucial to our functioning, and so a part of the material adequacy of the human body for its human form is having such empty spaces, which is an environment-relative feature.

That said, while it is plausible that the causal structure of our universe is such as to tie our life to matter in an appropriate environment, once we see that a defensible proper disposition thesis must involve relational features of our matter, it becomes intuitively less plausible to suppose that the proper disposition thesis is *metaphysically* necessary. We may have the causal power to “give up the ghost and create a corpse” under certain inadequate conditions, but our finite causal powers could be overridden by other powers, and hence it would be metaphysically possible for us to continue to exist in such cases. Indeed, it seems central to the concept of a material substance that it enjoys an *ontological* independence of material stuff outside of itself.

It was essential to the normative applications of Aristotelian forms^{??backref}, and the philosophy of science considerations^{??backref}, that we understand the forms in a robust way, and not as merely ontologically grounded in the arrangement of matter. The relativistic arguments provide a powerful metaphysical consideration in favor of such a robust account of form. For if substances begin their existence at a subatomic level at which one cannot distinguish the substance's matter as constituting a living human or a piece of organic slime, then form cannot be grounded in the arrangement of the matter that it is the form of.

It is worth noting that the Small Endpoints Thesis argument against the adequate disposition thesis is still compatible with a weaker version on which there are *some* appropriateness conditions on the matter that can have a given form. Perhaps there are few if any restrictions on the *arrangement* of the matter, but we can have restrictions on the *type* of matter. Perhaps, say, humans and oak trees must be made of the fundamental particles of the Standard Model of physics, because it could be that human and oak forms require particles tied to the kinds of laws of nature that our world has. Perhaps even more restrictively, the bodies of earth animals must be made of matter rather than antimatter. Such restrictions are possible, but do not appear to have significant philosophical applications.

The Small Endpoints Thesis rules out significant positive adequacy conditions on which a body must have some significant and distinctive structure to receive a form. However, it is compatible with the thesis that are significant *negative* adequacy conditions. Thus, while a pig and a sparrow might start as a very similar subatomic entity, one might hold that a *developed* pig-like body could not have a sparrow form and a *developed* sparrow-like body could not have a pig form. For instance, one might hold that having DNA is not a necessary condition for being either a pig or a sparrow (since the subatomic entity does not have DNA), but having sparrow DNA is incompatible with being a pig and having pig DNA is incompatible with being a sparrow. But while such a thesis is coherent, it does not appear to be well-motivated metaphysically. A sparrow embryo and a pig embryo appear to have more similarity to each other than either of them has to a subatomic entity. If both

the sparrow embryo and some subatomic entity can have the same form, why would it be metaphysically impossible for the pig embryo to have it?

In response, one might hold that a pig form only has the power to inform matter that falls somewhere along the line of development from a subatomic entity to the matter of a zygote with pig DNA to the matter of a recognizable embryonic pig to the matter of a recognizable mature pig. But why think it has such a restricted power of informing? It does not seem any “metaphysically harder” for the pig form to inform a sparrow-like body than to inform a subatomic entity. We should not posit metaphysical restrictions without positive reason.

4.8.4. *The special composition question.* Finally, observe that the arguments for SET and SBT are independent of Aristotelianhylomorphism (though the causal argument presupposes a realism about causation). Therefore, they can be used to criticize a number of alternative answers to the special composition question—the question of giving informative necessary and sufficient conditions for a plurality of entities to compose a whole.¹⁹ Recall that the two extreme answers are nihilism, which holds that a plurality of two or more objects never composes a whole, and universalism, which holds that every plurality of objects composes a whole. All other answers are restricted theories of composition, including the Aristotelian answer defended in ??backref.

Now, SBT gives us good reason to think that for any material substance, say a cat, there is a time when that substance is complex and yet roughly the size of a small molecule (say, H₂O). For the substance starts subatomic, and it is implausible that it would suddenly jump over all the intermediate sizes and be the size of a small molecule. We expect growth to be fairly continuous. This means that the correct answer to the special composition question has to be such as to be applicable to things the size of a small molecule. Van Inwagen’s answer??ref that the *xs* compose a whole just in case they have a life together (understood in the usual empirical biological way, rather than in terms of having a form of a living thing) fails this condition—things that small do not have a life together. We

¹⁹??ref

can expect other accounts based on macroscopic relations or properties to fail. Thus, it fits with many of our intuitions that the *xs* compose a whole just in case they either have a life together or someone has a purpose for their joint function, since that allows not just living things but also artifacts into our ontology. But according to SBT plus a continuity constraint, if a chair is actually a substance, at some point that substance is localized as a very, very tiny complex substance. But while a carpenter has a purpose for the joint function of the parts of a chair, the carpenter has no purpose for any such mysteriously tiny substance at the root of the chair.

Mereological universalism is usually elaborated as a four-dimensionalist view: any plurality of object-stages across spacetime composes a whole. Such a four-dimensionalist mereological universalism massively violates SBT: there will be many material objects with large beginnings.

On the other hand, an Aristotelian account with a metaphysically robust view of form has no special difficulty with tiny things. An oak tree may at some point consist of a single atom, but it can simply be an empirically inaccessible fact that that atom has an oaken form.

5. Persistence and death

5.1. Causal powers. Living substances appear to causally contribute to their future existence—dogs that eat healthy food and trees that succeed in processing sunlight are apt to live longer. Let's take this appearance at face value. This suggests that living substances have a causal power to continue existing given an appropriate environment. The same is likely to be true of non-living material substances as well.

This is a generalization about the causal powers of things, rather than a general principle of "existential inertia".^{??refs} And as in the case of other causal powers, we can expect

there be activation conditions for the power of self-continuation. Our relativistic arguments give us reason to think these activation conditions include not just the intrinsic features of the substance, but features of its immediate environment as well. When these conditions are not met, a living substance instead of continuing its bodily existence excretes its matter, thereby generating a corpse.

If the Small Endpoints Thesis is correct, the generation of a corpse is not an instantaneous process. At the end of its life, a horse is subatomic. But empirically it is obvious that, in typical horse deaths, throughout the death process there is matter arranged in the shape and size of a horse. Thus, when the horse is subatomic, most of its corpse has already been generated: the horse at the end of its life has shrunk to a subatomic substance within a nearly fully formed corpse. We might speculate that once the internal and environmental conditions for the horse to cause its future existence start failing, the form's activity contracts its scope, generating a corpse where the form is no longer informing the body. Towards the end of the process, only a subatomic portion of the body is informed by the form. It is plausible that the transition from the form informing hundreds of kilograms of horseflesh to its informing a subatomic portion is continuous, though it may be very fast, and thus we have a continuous process of corpse-generation.

It may seem initially strange to think that a living organism coexists with corpse-matter. But in a way we always do that. Bits of us are constantly dying: our skin is literally turning to dust. That dust flies away from us. It's just that at final death the process becomes more complete—what looks like our body is eventually mostly a heap of dead matter, and then finally wholly so.

5.2. The possibility question. Is it metaphysically possible for the human form survive the destruction of the human body? If we think of the form in a non-robust way as a shape or arrangement of matter, then a form without matter is as impossible as the smile without the Cheshire cat. However, form on the account defended in this book is highly robust. As we saw in ??backref, it grounds normative and nomic facts. And as our relativistic arguments show, the form cannot be ontologically grounded in the arrangement of matter,

since material substances begin their existence at such a tiny—indeed, subatomic—scale that we should not expect any intrinsic difference between the initial matter of a human, a cat or a mushroom.

At the same time, while we have good reason to think that a substance's form does not ontologically depend on the specific type or arrangement of matter, it could be that the form requires some matter or other for its existence. But while there is some intuitive reason to think that the human form should be tied to a specifically human body and the oaken form to a specifically oaky body, once we have rejected this intuition, it is not clear that there is a significant reason to think that the form metaphysically requires any matter for its existence.

One might think that without a body to inform, a form has “nothing to do”. Aquinas famously responded to this line of thought by saying that the human form was a soul with an abstract intellectual function that did not depend on a body.^{??ref} This response may be correct, but we need not settle the point right now. For even if something has “nothing to do”, it does not follow that it has to cease to exist. The question we are examining in this section is one of possibility: *can* the human form survive the destruction of the body. If the form no longer had anything to do after death, Ockham's razor would give us reason to think that the form would cease to exist. But Ockham's razor is a principle about what *actually* exists, not what *possibly* exists. We should not suppose actual conspiracies when there is a simple explanation, but we should all agree that conspiracies are *possible*.

That said, a form without a body could still have a metaphysical “task”: it could ground the *possibility* of matter coming back to re-form the old substance.

Given the robust nature of form, and its ontological independence of the details of the arrangement of matter, we have some reason to think that it is metaphysically possible for form to survive the destruction of the body, not just in the case of human form, but of any form of a material substance.

5.3. The actuality question. The next question is whether the human form actually survives the death of the body. Aristotelian optimism makes it very plausible that beings of

a given kind are capable of approximating the flourishing proper to that kind, and indeed that such an achievement is not just a rare exception. But human flourishing is rarely if ever even approximated in this life, so we have good reason to think that human existence can be expected to continue past death, and not merely in rare exceptional cases.

For consider both the moral and the intellectual aspects of our flourishing. As Aristotle himself knew^{??ref}, true virtue is rare. When we look at ourselves, we see that all too often even actions that are right do not flow from motives far from right, and in this we are not exceptions among humans. We may have friends, but we do not cherish them nearly as we ought. Every human around us is something with dignity, something sacred (we need not understand this in a religious sense), a being capable of morality, love and knowledge. We should be constantly in awe of our fellow humans and constantly grateful for being allowed to partake of the community of such beings. And yet we are usually far from that awe—often we even look down on others.

On the intellectual side, we know how quickly a child's questions can lead far beyond human knowledge. The physicist Richard Feynman wondered as a child why it is that a ball in a wagon would move back in the wagon when the wagon was pulled. His father explained that objects have a tendency to stay put or continue moving ("inertia") but then correctly emphasized to little Feynman that we have no idea why they do that.^{??ref} Our scientific questions bottom out with laws of nature, but few if any of us know what makes them hold: there are many views on the grounds of laws of nature; even if one among these is true, it is likely that only few believe that view, and even fewer—if any—know it to be true. Yet any child or any adult either wants to know or *ought to* want to know. We learn the answers to non-ultimate questions of detail, but what is really important to our intellectual flourishing are the ultimate questions. For our intellectual flourishing calls for a holistic understanding of reality. And yet we are so far from that.

Given our vast distance from flourishing in this life, Aristotelian optimism indeed supports the actuality of a life after death.

6. Naturalism

Is the Aristotelian hylomorphic account of humans compatible with naturalism? This depends on how naturalism is defined and whether we take the theistic version of the account or not.

Since theism implies the existence of a non-natural causally efficacious substance, namely God, it will be incompatible with most versions of naturalism. In Chapter??ref I will argue that the best version of the Aristotelian account will be theistic.

Bracketing theism, it is interesting to query whether the Aristotelian's account of the forms of finite substances is compatible with naturalism. If so, then one could at least combine the Aristotelian account with naturalism at least for finite objects like us, and save some naturalistic intuitions. Furthermore, it would mean that an Aristotelian not convinced by the arguments that theism is needed to make the theory satisfactory could be a full-blown naturalist.

If we take naturalism to say that the only entities that exist are the ones that would figure in a completed science, then it is unlikely that Aristotelian metaphysics would be compatible with naturalism about finite objects. However, such a strong naturalism would likely also conflict with many other metaphysical theories that are rarely taken to contradict naturalism.

For instance, consider theories of time. Of the theories of time, four dimensionalism, on which ordinary objects like ourselves are extended in time as well as space, is what seems to fit best with Relativity Theory. But the most common four-dimensionalist view of changing properties is perdurantism: changing objects are made of temporal parts or slices to which the properties are primarily attributed, so that a tomato that once was green and now is red has a green temporal part and a red temporal part. But slices are unlikely to figure in a completed science if our current science is a good guide to that. Consider that our current physics considers particles like electrons and quarks to be fundamental, and hence not made up of smaller parts. But electrons and quarks change over time (e.g., with respect to spin and flavor), and so they would need to have temporal parts. But these

parts are not found in our physics: our physics does not go more fundamental than the fundamental particles.

Or consider that our science may quantify over physical objects like particles and fields, and applies predicates to them, but does not quantify over properties. Thus, properties, whether understood as universals or as tropes, go beyond current science. Yet it seems implausible to understand naturalism as implying nominalism.

One may weaken naturalism to say that the only *causally relevant* entities are those of a completed science. But this would still rule out perdurantism, since objects change with respect to causally relevant properties, and hence their temporal parts have these properties, and have them more fundamentally. It would also likely rule out many versions of trope theory, since the causal efficacy of tropes is supposed to explain the causal efficacy of the objects made of them.

Let's step back. Naturalism denies the existence of causally efficacious "supernatural" entities like ghosts, but it is neutral on temporal parts and tropes of "natural" entities like electrons. What is the relevant difference between ghosts and temporal parts of electrons? It seems to be this. While neither entity is posited by science, if ghosts exist, then certain phenomena that it belongs to science to investigate have no scientific explanation, barring systematic overdetermination. If there are ghosts, they are presumably responsible for at least some of the appearances of ghosts, such as maybe chills some people feel in graveyards. These phenomena then either have no physical explanation at all, or by a massive coincidence are overdetermined by a physical and a spectral cause. There is a competition, thus, between scientific and spiritual explanations when we are concerned with ghosts.

On the other hand, there is no overdetermination in the case of the objects studied by science and their ontological constituents such as temporal parts or tropes. When a temporal part or a trope of an ice cube makes your hand cold, the ice cube also makes your hand cold. The ice cube's cooling causal influence on your hand does not compete with the ice cube's temporal part's causal influence on you or on the causal influence of a trope of coldness (if there are temporal parts of ice cubes or tropes of coldness). Nor is this

overdetermination, since it is not overdetermination when an object *C* causes an effect *E* by means of *C*'s part. After all, it is not overdetermination when you have an ice cube on your hand, and *both* the ice cube and its lower half cool the hand.

We might thus want to say that naturalism holds that any phenomenon that it falls within the purview of science to seek for causal explanations of either has no causal explanation at all, or else has a scientific causal explanation and is not overdetermined by a scientific and a non-scientific one. On this account, theism is incompatible with naturalism if there is a first state of physical reality. For if there is such a state, it belongs to science to seek for its causal explanation. However, there is a theistic explanation of that state and no scientific explanation. We thus have the kind of competition that naturalism rules out.²⁰ One might object that science knows that it cannot, on pain of vicious circularity, provide a scientific explanation of the first physical state, and hence it does not belong to science to seek that explanation. But this objection confuses the first physical state *qua* first and the first physical state as it intrinsically is. It does not belong to science to seek the explanation of the first physical state *qua* first. But if we just consider it as a specific physical state—particles and fields arranged thus-and-so—then it certainly belongs to science to seek for its explanation, though that search will be rightly abandoned if sufficient evidence is gathered that the state is in fact the first one.

Now, let's go back to Aristotelianism. Bracketing theism, the main entity that might trouble the naturalist is form. But the forms of things are components of substances—humans, dogs, oaks, etc.—that are also found in our current science (e.g., biology) and are likely to be found in the completed version of that science. And there is neither competition nor overdetermination between causal explanations provided by forms and their substances. As long as the substances do not have spooky causal powers beyond the scope of science, adding forms to the ontology no more contradicts a plausibly defined naturalism than perdurantism or trope theory does.

²⁰If there is no initial physical state, we can apply the above argument to an initial segment of physical states.

It is, of course, open to the Aristotelian to suppose that some finite substances, whether natural like humans and oaks or supernatural like angels or ghosts, have causal powers of a sort that goes beyond the scope of science. But nothing in the applications given for Aristotelianism posits such powers. Thus as far as the argument of this book goes, there is no contradiction with a carefully specified naturalism with respect to finite things.²¹

*Appendix: Making the geometric argument for SET more precise

Consider a non-empty region D of an n -dimensional Euclidean space. Technically, Special Relativity happens in a Minkowskian space. But we can embed a Minkowskian space into Euclidean space coordinate-by-coordinate, and will do so to make things easier to imagine.

A *hyperplane* is an $(n - 1)$ -dimensional linear substance of D . A *supporting hyperplane* for D is a hyperplane K such that (a) K intersects the boundary of D , and (b) all the points of D lie on K and/or one side of K . Here, the boundary of a region D is understood in the topological sense, as the set of points z such that there are points of D that are arbitrarily close to z and there are points outside D that are arbitrarily close to z . To make (b) more precise, note that K cuts space into two halves, H_1 and H_2 , known as “open half-spaces” such that no two of the three sets K , H_1 and H_2 have any points in common. Condition (b) says that D is a subset of $K \cup H_1$ or a subset of $K \cup H_2$. Note that a supporting plane for D can intersect D only at a boundary point of D .

Given a vector v of length one (a “unit vector”) and a plane K perpendicular to v , one of the half-spaces that K cuts space into lies in the direction of v from K and the other region lies in the direction of $-v$ from K . We then have this useful fact which will be proved later.

LEMMA 1. *Let D be a bounded non-empty region of n -dimensional Euclidean space, and let v be any vector. There is exactly one supporting plane $K = K(D, v)$ for D such that (a) K is perpendicular to v and (b) no point of D is found in the open half-space in the direction of v from K .*

²¹I think the most plausible place to look for a tension would be in examining human free will, but that goes beyond the scope of this book.

(A region is bounded provided there is some finite r such that every point of the region is within distance r of the origin.) Moreover, *every* supporting plane is of the form $K(D, v)$ for some unit vector v : just choose v to be one of the two unit vectors perpendicular to K , in such a way that D is on the correct side of K with respect to v . Now, the set of unit vectors forms the hypersphere S^{n-1} . Then we have (and again the proof will be given later):

PROPOSITION 1. *Let D be a bounded non-empty region of n -dimensional Euclidean space. Let U be the set of points v on the $(n - 1)$ -dimensional unit hypersphere S^{n-1} such that the supporting plane $K(D, v)$ meets the boundary of D at two or more points. Then U has zero $(n - 1)$ -dimensional Lebesgue measure as a subset of the hypersphere.*

We are now ready to give a more precise version of the Rover argument. Start with:

PROPOSITION 2. *Let D be a bounded region of n -dimensional Euclidean space, and let K be a bounding hyperplane for D . Suppose the boundary of D intersects K in a single point. Then for any distance $r > 0$, there is a $u > 0$ such that for every hyperplane L parallel to K whose distance from K is at most u , the intersection of L with D is contained in some ball of radius r .*

Again, the proof will be given later.

Proposition 2 tells us that not only is the intersection of Rover's four-dimensional boundary with a bounding hyperplane small for almost directions of bounding hyperplane, as we learned from Proposition 1, but for almost all directions, planes parallel to and sufficiently near a bounding hyperplane of that direction will have an arbitrarily small intersection with Rover. Thus, for almost all reference frames F , if we look at Rover's lower temporal bound or beginning time t_0 in F , for any fixed non-zero size δ , at all times $t_1 > t_0$ shortly after that lower temporal boundary (i.e., with $t_1 - t_0$ small), Rover will still have size no bigger than δ .²²

²²To make the point even more precise, we should consider the conversion factor between three-dimensional spatial size relative to F and the four-dimensional Euclidean distances in Proposition 2 on a hyperplane L . However, all we need for the argument is that if each distance is non-zero so is the other and if one distance tends to zero so does the other.

Similarly, the points of contact between K_2 and D can be characterized as Rover's upper temporal boundary with respect to F , and the exact same argument shows that for almost all frames F , Rover ends its life smaller in size than any specifiable positive size, and hence it is subatomic.

We can now prove our Lemma and Propositions.

PROOF OF LEMMA 1. Let c be the supremum of the dot product $v \cdot x$ as x ranges over D . This is finite as D is bounded. Let K be the plane $\{z : v \cdot z = c\}$, which is perpendicular to v . Any point z in the open half-space H defined by K in the direction of v satisfies $v \cdot z > c$, and hence is not in D . Let (x_i) be a sequence of points of D such that $v \cdot x_i$ converges to c . Let y be a limit point of the x_i (this exists since D is bounded). Then y is in the topological closure of D . But $v \cdot y = c$ by continuity of the dot product, and there are points arbitrarily close to y in the half-space H and hence not in D , so y is in the boundary of D . Since all points $z \in D$ satisfy $v \cdot z \leq c$ by choice of c , it follows that K is a supporting plane for D .

Now, let L be any supporting plane for D perpendicular to v such that no point of D is found in the open half-space in the direction of v from L . Then $v \cdot z$ takes the same value for all $z \in L$, since L is perpendicular to v . Let that value be d . We then have $v \cdot z \leq d$ for all $z \in D$ since no point of D is found in the open half-space in the direction of v from K . It follows by choice of c that $c \leq d$. Now, since L is a supporting plane, it meets the boundary of D , and so there is a sequence (u_i) of points of D converging to a point $w \in L$. Thus, $v \cdot u_i$ converges to $v \cdot w = d$ by the continuity of the dot product. We have $v \cdot u_i \leq c$ for all i by choice of c , and so $d = v \cdot w \leq c$. Thus $c = d$, and so $L = K$. $\square$

PROOF OF PROPOSITION 1. A convex set is one that contains the line segment between every pair of points in it. The closed convex hull of D is the smallest topologically closed and convex set containing D , and it has the same supporting planes as the original set. Moreover, any points where a supporting plane meets the boundary of D are also points of the boundary of the closed convex hull (since there are arbitrarily close points of the

complement of D on one side of the supporting plane). Thus, replacing D with its closed convex hull, we may assume D is closed and convex.

Let $\sigma(v)$ be the supremum of the dot product $v \cdot x$ as x ranges over D . This is known as the support function of D . Note that $K(D, v) = \{z : v \cdot z = \sigma(v)\}$ (cf. the method of proof of Lemma 1). Then for almost all v in n -dimensional Euclidean space, there is a unique z in D such that $\sigma(v) = v \cdot z$ (“alesia” 2022??ref; cf. the proof of Lemma 1 in Pruss 2023??ref). When there is such a unique z , $K(D, v)$ intersects D precisely at z . Now, let V be the set of all av where $v \in U$ and $a > 0$. Observe that if U has non-zero $(n - 1)$ -dimensional measure, then V has non-zero n -dimensional measure (the measure of V intersected with the unit ball has the same ratio to the measure of the unit ball as the ratio between the measure of U and the unit sphere). Furthermore, for every point $av \in V$ with $a > 0$, the plane $K(D, av) = K(D, v)$ intersects D in at least two points. Thus, V must have zero measure, and hence so does U . $\square$

PROOF OF PROPOSITION 2. Let z be the intersection point of the boundary of D with K . For a *reductio ad absurdum*, suppose that there is a distance $r > 0$ such that for every $u > 0$, there is a hyperplane L parallel to K whose distance from K is at most u and whose intersection with D is not contained in a ball of radius r centered on z . Thus, for any positive integer n , choose a hyperplane K_n parallel to K whose distance from K is at most $1/n$ and whose intersection with D is not contained in a ball of radius r centered on z . Choose a point z_n in that intersection whose distance from z is at least r .

Since D is bounded, some subsequence of the points z_n converges to some point w . Moreover, that point w must be at least distance r from z , and must lie on K (since the K_n converge to K). Since $w \in K$ and K is a bounding hyperplane of D , there are points in the half-space defined by K disjoint from D , and hence in the complement of D , arbitrarily close to w , and there are points in D (namely elements of the chosen subsequence of z_n) arbitrarily close to w , so w is in the boundary of D . But since the distance from w to z is at least r , this contradicts the claim that the intersection of D with K is the single point z . $\square$

CHAPTER X

Details and open questions

1. Introduction

I have argued that in order to explain a large multitude of normative and metaphysical features of reality, especially of the human being, we would do well to posit a metaphysically robust entity, a nature or form, in each substance, and especially in human beings. This form is extremely rich in informational content, as it has to answer a very large number of Mersenne questions in ethics, epistemology, semantics, and metaphysics.

But the theory so far is highly underspecified. The point of this chapter is to discuss some issues resolving which would specify the theory a little further in terms of both teleology and fundamental metaphysics.

2. Teleology

2.1. Parts and aspects. When my arm is functioning poorly due to an injury, I am functioning poorly insofar as I have an arm. It is tempting to reduce evaluations of the function or flourishing of a part of a substance to the function or flourishing of the whole in respect of the part. But while this is tempting, there is some reason to resist this reduction.

Of course, everyone agrees that there are cases where the flourishing or languishing of a part or aspect is instrumental to the opposite state of the whole. Xenophon has Socrates give the example of a person who is harmed by their wisdom because a tyrant hears about the wisdom and has the wise person kidnapped to serve as an adviser.^{??ref} And many a person would have escaped a broken leg sustained if they had a minor sprain that kept them from skiing. But such cases can be reconciled with a reduction of the flourishing of the part to the flourishing of the whole by noting that there is nothing particularly strange

about an aspect of languishing being a means to flourishing (e.g., one might get a cash settlement for an injury).

But there are more interesting cases where the flourishing state of a part seems not merely instrumental to the opposite flourishing state of the whole, but constitutive of it. Muscles are torn down by exercise and regrow stronger. The process of tearing down is harmful to the muscles themselves, but is a constitutive part of the proper functioning of the body's system of adaptation to particular activities. And, more generally, the death of cells is part and parcel of the normal self-renewing persistence of a multi-cellular organism.

These cases are perhaps not entirely convincing. One might insist that when cells die as part of the self-renewal of the organism, the death is itself a part of the proper functioning of the cell, and hence both the cell and the whole are flourishing. Historically, Aristotelians have tended to insist that a thing's destruction is bad for it.¹ However, this may be mistaken. If we think of substances as four-dimensional objects—spatiotemporal entities—then a thing is destroyed just in case it has an upper temporal boundary. Now, having *spatial* boundaries is not bad for a thing—indeed, having spatial boundaries of the right sort is constitutive of a thing's having the correct shape and size. A dog so bloated as to be boundless, taking up the whole universe, would not be a healthy dog. Similarly, a substance of a specific sort could have proper temporal boundaries, and if so, it might be harmed not just by living too short a time, but also by living too long a time.¹ It is plausible that typical human cells have proper temporal boundaries as well, and hence their living too long may be bad for the organism as a whole *and* for the cell as a part.

¹Whether it is a part of human flourishing to have an upper temporal boundary or whether humans better flourish in living forever is a difficult question whose thorough discussion is beyond the scope of this book. We can note that human beings pursue projects that are at least typically cut short by death, and their pursuit of such projects is a part of their flourishing. Moreover, there is reason to doubt that a satisfactory degrees of virtue, wisdom, and understanding of the world, and a sufficiently broad array of deep friendships are practically attainable in a typical human lifespan, which points to the idea that it is not good for us that our upper temporal boundary be at death (on the other hand, see ??refs), and hence points towards life after death given Aristotelian optimism.

Cases of redundancy may provide a more convincing example of where the flourishing of a part comes apart from the partial flourishing of the whole. Suppose that an organism for its basic functioning needs n_1 functioning parts of some sort—say, cells of a particular kind, or legs, or teeth—and for the sake of redundancy it needs some larger number n_2 . Suppose that having more than n_2 is supernormal for the organism (as per our discussion in ??backref), until we reach some large number n_3 at which point the organism has too many.

For the sake of definiteness, suppose that the parts are teeth of some hypothetical organism that does not have a precisely prescribed number of teeth, and that n_1 is 30, n_2 is 35, and n_3 is 40. With fewer than 30 teeth, the organism fails to chew well. With 35 to 39, it has a healthy level of redundancy. And at 40 or more, it has too many teeth. Suppose now that Sally is an organism of this sort and she has 38 teeth. One of the teeth, however, is getting worn down quite a bit. That tooth is no longer fully functional, and hence that tooth is failing to flourish. However, this does not constitute any failure of flourishing in Sally. Even if the tooth stopped functioning entirely, Sally would still have sufficient denotation both for first-order purposes and for redundancy. Sally can still be fully functioning as a whole with respect to her teeth, even though that tooth is not itself fully functioning, and even though she would be fully functioning to a higher degree if that tooth were to fully function. In this case, the direction of flourishing of the part and of the whole seems to be the same, but nonetheless one can have full flourishing of the whole without every part fully flourishing. In such a case, it would not be correct to say that Sally is languishing with respect to that tooth. For that tooth doesn't make her languish—it just makes her flourish less.

Interestingly, redundancy can be between parts or aspects of very different sorts. We might suppose that a flourishing human being has a sufficient number of abilities of various sorts. These abilities can be moral, intellectual, emotional, or physical, and within each category, they can differ quite significantly from each other. Then full flourishing could turn out to be compatible with a severe impairment within certain abilities—for

instance, one's mathematical aptitudes might be dysfunctional, and yet the *person* might fully flourish. This would allow an Aristotelian to accept the thesis that some persons with significant disabilities can nonetheless be fully flourishing. For the disability can constitute the failure of a part or aspect of the person to flourish, without thereby compromising flourishing of the whole.

We could, in principle, suppose that there are some cases where the failure of flourishing of a part might not even make the whole flourish less. Going back to Sally, perhaps 37 teeth is better than 38 (maybe 37 makes for better fit within the jaw), even though any number from 36 to 39 is fully normal. In that case, if Sally's 38th tooth is starting to deteriorate, this could be moving her to an even better state.

At the same time, there are proper parts and aspects such that the flourishing of the part or aspect always lines up with the flourishing of the whole. In the case of a person, to have one's will flourish in an action is to flourish with respect to the will, and makes one better off with that respect, and hence we do not need to correct looser discussion earlier in this book where no distinction was made between flourishing with respect to the will and having one's will flourish. Plausibly, rationality is similar. One is always better off for being more moral and more rational, and a failure of flourishing of one's morality is a failure of the person as a whole.

One way to handle the distinction between flourishing of the part and flourishing of the whole is to suppose, contrary to classical Aristotelianism, that the parts are themselves substances, with their own forms. Another way, however, is to suppose that there is only one substantial form, that of the whole, and this substantial form in addition to specifying the conditions for the flourishing of the whole specifies the conditions for the flourishing of the parts which are not reducible to the flourishing of the whole. The second option appears somewhat simpler, but our account can work with both.

2.2. Teleological reduction.

2.2.1. *A multiplicity of concepts.* Let us return to the norms for the whole, however. The applications given in this book make use of various normative concepts that are said to be

grounded in forms, such as proper function, teleology, and flourishing. It would be good to investigate if these can be further unified, under either one of these concepts, or some further unifying concept.

First, as we have seen in the discussion of the supererogatory and supernormal, we have both binary and comparative normative concepts, often in the same context. Suppose a stranger is about to be hit by a train, and the only four options are:

- (1) give them a kick to ensure that they have no chance of survival
- (2) make fun of them
- (3) stand by idly
- (4) jump in and push them out of the way likely at the cost of one's own life.

The binary distinction is that the first two options are impermissible, and the other two are permissible. But there are comparative distinctions with the two binary categories: (1) is worse than (2), and (4) is better than (3). Furthermore, the binary distinction does not reduce to "big differences"—we cannot just say that the impermissible is whatever is much worse than something permissible. For while (2) is much worse than (3), (3) is also much worse than (4) (though it is more natural to say that (4) is much better than (3)).

Proper function and teleology appear to be primarily binary concepts: a thing functions properly or improperly, and a thing either does or does not achieve its end. Flourishing, on the other hand, seems to comprise both the binary and the comparative. The mildly vicious person suffering undeserved horrendous pain appears not to be flourishing *simpliciter*, but if the pain were increased, they would languish even more. Flourishing at this point appears the best candidate for a foundational concept for our norms: it supports a binary distinction *and* comparisons.

However, the claim that teleology is a primarily binary concept—a thing's achieving or not achieving its end—is an oversimplification. To see this we need an extended digression on ends, going beyond traditional Aristotelian thinking.

2.2.2. *Ends.* Many activities seem to occur for an end. The activity then counts as successful provided that the end occurs and occurs as a fulfillment of the activity. An organism produces gametes in order to reproduce. A cat chases birds in order to catch them, and eats them for nourishment. And I put on shoes to keep my feet comfortable when I walk. The end-directedness of much voluntary activity is obvious, though whether there really is teleology in the involuntary cases is a more controversial.

The Aristotelian tradition tends to analyze voluntary action as always end-directed, but also tends to see involuntary activity as often, if not always, directed at an end. I will argue, starting with the case of voluntary action, that many interesting phenomena would be misclassified as end-directed. The actual structure can be more complex, and while it has a directional structure, it is misleading to think of that structure as teleological in the sense of possessing a “target *telos*”, an aim-to-be-achieved, an end that fulfills it.

Consider a sprinter who is running a hundred meters all out against a clock, rather than against other opponents. The runner has an end, namely to sprint 100 meters. But sprinting 100 meters does not explain the intense effort the runner puts in. Less than half of the effort could have been put in, and 100 meters would still have been sprinted. The bulk of the runner’s effort is explained by being directed not at completing the sprint but at completing it in good time.

But what exact state of affairs does the runner’s speed-directed effort have as its end? One might have a particular target time in mind when running, but in our example we are imagining a sprinter who runs all out. A runner who is just aiming at a particular time would be reasonably expected to slow down if it became obvious that a slower (and hence less exhausting) run would still achieve the target time, but not so our all-out sprinter. Our sprinter may have some specific time in mind to motivate themselves (say, 9.9 seconds), but interpreting their action as merely aiming at that time does not capture all of the directional structure of the performance. Any shortening of the time of the run is welcome given the sprinter’s aims, even if it goes below the time they have in mind.

We would normally describe the sprinter as trying to run 100 meters “as fast as possible”, and that seems to be a coherent description of an end. However, the language of “as fast possible” should not be taken literally. First, we have the question of what the relevant comparison class is. Is the runner trying to run as fast as any human being can on any track? This seems an excessively grandiose aim. As fast as they themselves can run on this track on this occasion? But then what aspects of “this occasion” are fixed (e.g., the level of motivation)? And what no minimum time exists, but only a lower bound that cannot be reached, much like an object with mass can come arbitrarily close to the speed of light but cannot achieve that speed.

Second, suppose we fix a particular sense of “as fast as possible”, and then after ten meters the runner realizes that they have been running slower than was possible. At this point, it is no longer possible for the runner to achieve the goal of literally running the hundred meters in as short a time as possible. But we do not expect the runner to stop. We expect the runner to resume running all-out, as part of the same directed activity.

There is obviously a teleological structure to the sprinter’s run. But it is not a structure of aiming to *achieve* a target *telos*. We can think of this structure as a kind of preference structure: a faster time is always preferred to a lower time. Such preference structures are common in games, where we ourselves have defined the teleology, but we can also find them in the case of teleologies that are not our own invention. For instance, suppose the human mind aims at understanding (cf. ??backref). If we understand this in terms of aim at a target *telos*, the understanding of everything or of everything humanly understandable, then as soon as we realize that we cannot have this (for there are humanly understandable things that I cannot understand because they have faded too much into the past—say, facts about Napoleon’s motivations on some day unrecorded in history), the pursuit would become pointless. It is better to say that our teleological orientation sets a preference ordering on our understanding, where understanding a set *B* of items is better for us than understanding a set *A* of items whenever *A* is a proper subset of *B* (every item in *A* is in *B*, but not *vice versa*) and when the levels of understanding for each item are kept fixed. This

ordering then obviously affects our epistemic lives, but also our practical ones (since one needs to make sacrifices in order to understand).

Additionally, a teleology concerned with specific aims has a difficulty capturing the directedness at lesser degrees of failure often but not always present in our actions. In chess, if it becomes clearly inevitable that you will lose, you are expected to resign. In many sports, on the other hand, such resignation is poor sportsmanship. Instead, you are expected to aim to minimize the amount you lost by, whether by trying to minimize the difference between your and your opponent's score, or by trying to make your opponent work hard for their victory.

There also seem to be cases outside of rational activity with a teleological directedness not understood in terms of the pursuit of a specific *telos*. A pecan tree produces pollen in order to have offspring. The more offspring, the better! Again, the *telos* is not to have infinitely many offspring², but rather the teleology seems best understood in terms of a preference ordering: more offspring, keeping the health of the offspring constant, is better.

As an alternative we might suppose that one can have infinitely many target *telê*. Thus, to account for the teleology towards understanding, we might say that humans have a *telos* to understand phenomenon ϕ_1 , and another *telos* to understand ϕ_2 , and so on, for all the phenomena that can be understood by humans, or we might say that a pecan tree has a *telos* have at least one offspring, and a *telos* to have at least two offspring, and so on. But while I have argued that our form is complex, to suppose such infinite complexity seems excessive. Furthermore, it is plausible that each target *telos* impels the organism causally—e.g., pulling on us in our deliberation. But infinitely many such things causally influencing a single aspect of us would violate the principle of causal finitism defended recently by multiple authors^{??}refs. Furthermore, each organism would always have infinitely many unfulfilled target *telê*, and that may seem just too pessimistic to believe.

²One might suppose that there is a limit here based on how much offspring it is possible for one pecan tree to have. But it is not clear that there is such a limit in principle. Wouldn't a pecan tree always be better off if it lived an extra year and had more offspring (perhaps transported to another planet, to avoid overcrowding)?

It may, of course, sound oxymoronic to talk of a teleological structure without a target *telos*. But that's a merely verbal point. We can think of teleology as about pointing and directedness, without a *target* as such. Imagine for instance that I am competing in an odd archery competition, where the winner is the one whose arrow hits closer to the center—except that if you hit the exact center, you automatically lose. Then there is an obvious sense in which I am aiming. Indeed, since the chance of hitting the center is infinitesimal, I should even set my bow's sight on the center, but it is not correct to say that I am aiming to *hit* the center.

Terminologically, we in these cases we might either adopt the seeming oxymoron of “teleology without a *telos*”, but it seems neater to distinguish between a target *telos* and a directional *telos*. A target *telos* is one that fulfills the striving when it is achieved. A directional *telos* specifies a direction, but there need not be a target to be hit. Sometimes a directional *telos* can be formulated in terms of an unachievable “ideal target”, such as “perfection” or “getting to the end of the track instantly”. Imagine a golf course where the distance between a hole and the starting tee is so long that a hole-in-one is humanly impossible. In the first shot, the player may aim *in the direction of* or *towards* the hole, but it wouldn't be correct to call the hole the target of the shot. The hole is an “ideal target”. And sometimes there may even be no ideal target, just a direction, as in the case of the odd archery competition where one aims in the direction of closeness to the center while avoiding the center itself.

It is, parenthetically, interesting to note that this way of thinking of teleology complicates a well-known Kantian argument for why we should believe in an afterlife and God.^{??ref} The argument holds that we are unable to achieve complete virtue in this life. But since we cannot aim at the impossible, and need to aim at virtue, we should suppose a life beyond death, and a being capable of ensuring that in that after-life virtue is achievable. But a critic might use the above understanding of teleology to defend the idea that we have an unlimited directedness in the *direction* of perfect virtue without a directedness at the *achievement* of perfect virtue. To respond to this criticism, the Kantian needs to argue

that either perfect virtue is a genuine (not merely ideal) *target telos*, or we have as a target end some “adequate” degree of virtue that is not practically achievable in this life.

2.2.3. *Teleology vs. flourishing.* Given that flourishing is itself not just a binary distinction, it is plausible that we can reduce these more complex teleological phenomena to flourishing. We can suppose, for instance, that pecan trees flourish *simpliciter* when they have *an* offspring, but flourish more when they have more. And we can suppose that the fulfillment of our practical intentional aims—or perhaps with a restriction to cases where we aim at a good or at least a non-bad—is a part of our flourishing even when the aims are directional rather than targeted, and that we flourish to the degree to which we fulfill these aims.

But now that we see the non-binary complexity in teleology, we might ask whether there isn’t a reduction in the opposite direction. For we can capture the binary concept of flourishing as opposed to languishing in terms of the fulfillment of a target *telos*, while capturing comparative flourishing in terms of how well a directional *telos* is fulfilled. Alternately, we might ask whether there is even a non-verbal distinction between flourishing and the generalized notion of teleology.

There is, however, some reason to think that the phenomenon of flourishing is broader and more fundamental than that of teleology. Teleology describes an entity’s *striving* if not always towards a target, at least in a direction. But we can imagine cases of flourishing that are a kind of externally imposed gift. Normally, we appreciate our existence as a kind of good. But it does not seem right to think of us as having a striving for existence—after all, our existence is explanatorily prior to all our strivings. We might strive to *persist*, but that is not the same as striving to exist in the first place. Or imagine there is a God who creates beings whose good consists in union with God, but where this union with God is something that the creature can do nothing to either promote or hamper—the union must be a pure gift. This seems intelligible, and there may even be some theologians tempted to think it true in our own case.??refs

2.3. Interdependence. There is a children's picture book with the jarring line that pigs give us ham. In an Aristotelian setting, we can ask: Is it a *telos* of a pig to provide ham to predators or of a butterfly larva to be a host for the offspring of a parasitoid wasp?

A positive answer fits well with one aspect of Aristotelian optimism and poorly with another. For it makes more things in nature be instances of proper function, but at the same time it makes the *telê* of the prey be less harmonious, since the prey will presumably properly avoid being preyed on.

Intuitively, being eaten is clearly not good for one. However this line of thought begs the question. If giving one's life for a member of another species is one's *telos*, it is thus far good for one. But one can do justice to the intuition being eaten isn't good for one by noting that even if providing food for another could be good for one, it might not be good *overall*. Aristotelian optimism implies a *tendency* towards harmony within the teleology of a substance, but is compatible with there being some *telê* that are in tension with one another, say to feed another and to survive. In such a case, survival may take priority, and so its loss is an overall harm—but a lesser harm than if the organism lost its life without benefiting another.

The same multiple-*telê* line of thought responds to the objection that organisms tend to engage in behavior clearly aimed at preventing their becoming food for another—they try to escape predation. Considered without regard for teleology, a pig engages in activities that promote the predator's nutritive good—the pig eats and grows—and it engages in other activities that oppose the predator's nutritive good—the pig runs off. Aristotelian optimism sees a tie between activity and teleology. If so, then it is not clear that we should privilege higher level behavior in figuring out teleology. Perhaps the pig's becoming nutritionally beneficial is indeed a pursuit of one porcine *telos* just as much as its running away is a pursuit of a different porcine *telos*.³

³One might also argue that the a prey animal's running off prevents a population implosion that would be harmful to the predators as well. (I am grateful to Dominic Pruss for this observation.)

It is very plausible that sometimes the good of another individual of one's own kind is a *telos*. Parenthood is an obvious example: the existence and flourishing of one's offspring is a part of one's own flourishing. But even in fairly primitive organisms, there seem to be cases that go beyond parenthood. In some slime mold colonies, a fruiting body in a stalk is produced, and the organisms forming the stalk of the fruiting body do not themselves reproduce, but instead sacrifice their reproductive potential to elevate other organisms and make them better able to reproduce.

In cross-species symbiotic relationships, we may think that each organism only promotes the good of the other as a means to its own good, but we need not think this. We might think that each organism has the other's flourishing as a part of its own *telos*. We might speculate that, after all, that if we met up with intelligent aliens, promoting their flourishing would be an expression of virtue in humans, even if there were no mutuality in the promotion.

That said, I do feel the intuitive force of the idea that it is implausible to think of the prey as having the nourishment of a predator as a *telos*. But even if we do not wish to go as far as to say that pigs have a ham-production teleology, we have room for a substantive and robust picture of teleologically supported interdependence between species. And applied to humans this would give us an Aristotelian reason for why the good of the earth ecosystem could be not merely a means to the human good, but partly constitutive of it, which would make a positive contribution to environmental ethics.

3. Individual or shared forms?

3.1. Platonism and two Aristotelian views. There is a long-standing scholarly debate whether Aristotle thinks forms are individual—numerically different ones for different members of the same kind—or jointly shared by all members of the same kind.??refs The shared form view at first sight seems rather more Platonic than Aristotelian, but as long as the shared form is immanent in the individuals and governs their functioning, the view

can count as Aristotelian. In any case, it is now time to compare an individual-forms Aristotelianism, a shared-form Aristotelianism, and Platonism with regard to the applications in this book.

One might also wish to consider hybrid views on which there are both shared universals and individual immanent forms.⁴ Such views were held by some medieval thinkers. Since all the applications in this book are handled by the individual immanent forms on their own, the question of whether additionally there are shared universals can be bracketed. I will thus stipulate the individual forms view to be that there are *at least* individual immanent forms, and take the shared universals and Platonist views to include a denial of such individual forms.

All three views have a robust view of human nature, whether this nature be a Platonic universal, a shared immanent form, or an individual form. All three views can give a grounding to answers to Mersenne questions about normative domains, and thus secure many of the normative applications of our theory of a robust human nature. So far we have a tie. But there are applications where the tie may be broken.

3.2. Mind. If we follow Aristotelian tradition and identify the mind with the form (or some aspects of the form), then the shared form and Platonist views lead to Averroes' theory that we all have the same mind. If the will is part of the mind, then we get absurd consequences about the lack of personal responsibility for our action, and in any case, the view is famously counterintuitive. However, there are two reasons this argument may not fully convince a contemporary Aristotelian or Platonist metaphysician to embrace an individual forms view.

First, one might completely reject the identification of the mind with the form. While identifying the mind with form offers some attractive theoretical simplicity, one might instead suppose the form defines the mind's teleology, and the mind is a physical or non-physical system structured by that teleology, perhaps along the lines of the teleological

⁴I am grateful to Bryan Reece for this suggestion.

functionalism sketched in ??backref. Or, second, one might take hold that the human mind has two parts: one part is the non-physical form and the other part is the brain. On this view, the mind is only partially identified with the form. This would allow different people to have numerically different minds on a shared form view, because we have numerically different brains—your and my mind would have a part in common and a part not in common.

However, there are some other considerations in favor of an individual forms view.

3.3. Diachronic identity and synchronic unity. In ??backref, the Aristotelian account was used to give a simple view of the diachronic identity of substances, including ourselves: they persist because of having the numerically same form. This, of course, does not work if we all share the same form.

There is also a synchronic unity problem. My nose and my heart are parts of the same human, while my nose and your stomach are not. What constitutes the difference? On an individual form view, there is an elegant Aristotelian answer parallel to the diachronic identity answer: my nose and my heart are informed by the same form, while my nose and your heart are not. This answer will obviously not do on either the Aristotelian or the Platonist shared form view.

Nor will it do to say that the difference is constituted by the fact that there is continuous human matter joining my nose with my heart, since if you and I were conjoined twins, but conjoined neither by heart nor nose, my nose would be connected by continuous matter with your heart, even though those would still not be the nose and heart of one human.

One might try for a teleological account of the unity of the human being: my nose and my heart function together (the nose allows the entry of oxygen which is distributed by the heart). However, we are social animals, and teleological cooperation crosses individual boundaries.

It could be that some other account friendly to shared forms is available. But the simplicity and elegance of the individual-form account of individual human unity is a reason to opt against the shared form view.

3.4. Ethical intuitions. The arguments of this section do not apply to the Platonic view, unless one thinks that Platonic properties are parts of the individuals that instantiate them. But they will apply to the Aristotelian shared form view, on which our form is a part of us.

Our form is not just a part of us, then, but arguably the most important part of us. Thus, you and I have our most important part in common on a shared form view. This is intuitively both attractive and unattractive ethically.

It is attractive because it makes our commonality deeper, and hence seems to make it easier to see why egoism makes little sense, since we are only partly different from each other, and only in respect of the less important parts. It is unattractive, however, for the same reason. If you and I have so much in common, then why should I go to the trouble of benefiting you when I can just benefit me, and you overlap with me anyway in our most important part? So far, thus, it seems that ethically it's a wash between the shared and individual form views.

On the other hand, considerations of autonomy push one a little towards an individual form view. If you choose for your life to be that of a solitary artist, then it seems like your action primarily affects you and not me, and I have no right to complain, and should cede you your autonomy. But on a shared form view, your going off to sculpt rocks on a distant island directly affects what happens to my most important part, my form—my form comes to inform a human being separated from much of humankind, and engaging in artistic pursuits that I may happen to disagree with. Using a central part of me for ends I disagree with is problematic. On a shared form view, just as conjoined twins perforce need to deliberate together about what to do with their lives, we are *all* stuck with one form, and it seems that we need to deliberate together about the shapes of our lives, to a degree that seems excessive. For instance, suppose that you and I are complete strangers, and I am enamoured of a life of simplicity, while you want your life to be a rich tapestry. Since my own form—indeed, perhaps, my very soul—is involved in your tapestry, your choice detracts from my life of simplicity, and so on the shared form view, you have a significant

reason to avoid complexity. That reason may be defeasible, but it is still a reason, and a reason too many.

3.5. Distant reproduction. For evolution to work with Aristotelian forms, sometimes organisms of one metaphysical kind must produce organisms of another kind. For instance, supposing that wolves are a different metaphysical kind from dogs, and dogs evolved from wolves, it must have happened that two wolves reproduced and made a dog. (One may think wolves and dogs are metaphysically the same kind, but let's suppose not for the sake of the argument.) If we are to avoid occasionalism about this, we have to suppose that the two wolves had a causal power to produce a dog-form under those circumstances.

Plausibly dogs evolved from Pleistocene wolves in Siberia, but there was also a Pleistocene wolf population in Japan, and so imagine that the causal power to produce a dog was found in both wolf populations. Suppose, counterfactually, that a short period of time after a pair of wolves produced a dog in Siberia, a pair of Japanese wolves also produced a dog.

On an Aristotelian shared-form view, when the Siberian wolves produced a dog, they did two things: they produced a dog-form and they made a dog composed of the dog-form and matter. But when the Japanese wolves produced a dog, the dog-form already existed, so they only thing they could do is make a dog composed of matter and that dog-form.

The first oddity here is this. Our (perhaps imaginary) Japanese wolves didn't know that there was already a dog in Siberia, so when they produced a dog, they exercised exactly the same causal powers that their Siberian cousins did. But their exercise of these causal powers had a different effect, because it did not produce a new form, since the form already existed, and instead it made the form get exemplified in some matter in Japan. It is odd that the exercise of the same causal power works differently in the same local circumstances.

Second, there is an odd action-at-a-distance here. The dog-form was available in Siberia, and somehow the Japanese wolves in the story made matter get affected by it thousands of kilometers away.

In fact, to make things worse, we can suppose the Japanese wolves only lagged a two or three milliseconds after their Siberian cousins. In that case, the Siberian wolves caused the existence of the dog-form, which then affected the coming into existence of a dog in Japan in a faster-than-light way. Indeed, in some reference frames, the Japanese dog came into existence shortly *before* the dog-form came into existence in Siberia. In those reference frames we have backwards causation: the Siberian wolves make a dog-form and that dog form organizes matter in Japan *earlier*. (Compare the causal argument for the Small Beginnings Thesis in ??backref.)

On the other hand, if there is an absolute simultaneity notwithstanding Special Relativity, we can imagine that the Japanese wolves reproduced at *exactly* the same time as the Siberian ones. Then we either have puzzling overdetermination-at-a-distance, where each pair of wolves engages in an activity that is a sufficient cause of the dog-form, or else we have an odd degree of freedom where nature has to decide which pair is the one that gets to cause the dog-form and which one merely gets to imprint the form on matter.

Third, we can suppose a multiverse made of causally isolated⁵ and spatiotemporally disconnected universes, and suppose that the Siberian wolves are in one universe while the Japanese ones are in another. On this version, either we have overdetermination of dog-form across causally isolated universes, or we have some sort of a coordination across causally isolated universes and events that have no common time sequence to determine which pair of wolves gets to make the dog-form.

If, on the other hand, every dog has a numerically distinct form, there is no difficulty: the Japanese wolves' activity can be entirely causally independent of the Siberian ones', whether the two are in one universe or two.

⁵Apart from their common causal origin in God or some other first cause if there is one.

The above is a reason to prefer the Aristotelian individual-forms view or the Platonist view (since presumably the Platonist universals are not produced by any evolutionary activities) to the Aristotelian shared-forms view.

3.6. Normative knowledge. There may also be some reason to prefer either Aristotelian view to the Platonist one insofar as the Aristotelian views might give a more direct account of at least some of our knowledge of normative matters. Since the Aristotelian forms are causally efficacious, we can suppose that the teleological features that ground norms also impel the substances that have the features to pursue their *telē*. And from these pursuits we get some knowledge of the norms, by means of the optimistic principle that things tend to go right. Granted we will see in ??forwardref there are limitations to this approach, but at least *some* knowledge of normative matters seems to be neatly grounded in the very norms known.

There does seem to be something right about the intuition that one can often just *see* the norms in the norm-governed behaviors, and taking the grounds of the norms to be causally efficacious makes this possible. Furthermore, the phenomenology of moral motivation is that sometimes we feel “the norms” pulling on us. An Aristotelian view not only lets us take this “pull” quite seriously—the norms are aspects of our causally efficacious nature which impel us to act teleologically—and to include it among the grounds for our knowledge of our teleology.

Indeed, worries about knowledge of Platonic abstracta are one of the main reasons why many contemporary philosophers reject Platonic views.??ref Theism might be able to help here—God perhaps grounds the abstracta and thereby knows them, and can give us innate knowledge of them. But perhaps even a theist will have the intuition that some of our ordinary humdrum knowledge of such normative facts as that one should proportion belief to evidence or that one should treat others with respect does not involve divine illumination. After all, a plausible theological insight is that if God exists, then God likes to work through creatures as far as possible. This insight seems central to the most plausible (partial) responses to the problem of evil we have. God wants us to grow in virtue and

doesn't just zap us into a final state. God wants nature to have a hand in creation and hence allows a long often violent evolutionary process. And God lets things happen according to the laws of nature rather than directly micromanaging things.??refs And methodologically it seems plausible that we should minimize invocations of God. Plus, it is attractive to be able to save the phenomenology of seeing norms in the behavior of others and ourselves.

4. Accidental normative forms

If you have promised to ϕ , you should ϕ . Consider two Aristotelian metaphysical explanations of what is going on here. On the conditional-norm explanation, the human form contains the conditional norm that you should ϕ if you have promised to ϕ , which when combined with the fact that you have promised to ϕ grounds an obligation to ϕ . But there is another possible explanation. Perhaps promising to ϕ causes, in virtue of a power contained in the human form, the normative accident of being such that you ought to ϕ to come into existence. One might think of the second story as a very robust normative power theory: a theory on which normative powers are a type of causal power that brings into existence an irreducible and new normative entity. (Most normative power theories do not consider normative powers to be a type of causal power.??refs)

The conditional-norm account posits fewer entities, and insofar as this is the case, Ockham's razor favors it. And intuitively it seems to be the right account of promissory obligations. If an analogous account holds for all norms, then the human substantial form could be the only normative property of the human being—there would be no normative accidents.

Now, while the conditional-norm account for humans posits fewer entities (i.e., forms), it makes the human form contain more information in the form of conditionals. For instance, on the conditional-norm view, both humans with and without a Y chromosome have a human nature that specifies what range of physical developments that are normal expressions of the genes that are unique to the Y chromosome. On the normative accident view, it may be that humans with a Y chromosome also have an additional non-physical

property governing the expression of Y-based genes, but humans without a Y chromosome do not have any accidents or substantial forms governing such expression.⁶

At the same time, the normative accident view of human beings involves significantly more in the way of *causal* laws presumably grounded in human causal powers, such as that when one makes a promise, a promissory-obligation accident comes into existence, or when one has such-and-such DNA, then such-and-such an accident governing norms of gene expression comes into existence. Moreover, since normative accidents are in large part defined by the norms they embody, the informational complexity of the normative accident is presumably present in the causal power for its production—it is a power to produce an accident of, say, being required to ϕ . Thus while we have reduced the *normative* complexity in human nature, we have done so at the cost of *causal* complexity, and a multiplication of entities.

This last point does not apply to normative accidents that have a cause outside of the human being. But it is difficult to think of examples, with the exception of one theological possibility. According to many Christians, by God's grace human beings can be directed at the "beatific vision", a direct vision of God. This beatific vision exceeds the power of human nature, and it is usually taken that natural human fulfillment does not require it. One way to make sense of this is to suppose that God by grace gives all or some humans

⁶It's worth noting that both views are compatible with a broad variety of views on gender and transsexuality, since choosing between the metaphysics of the two views does not settle the question of what the range of normal physical developments is. One might think that the normative accident view allows for a more conservative theory on which there is a metaphysical component—an accidental form—that determines whether one is really male or female, an accident of maleness or an accident of femaleness. But at the same time, the normative accident view could enable one to have a metaphysical basis for the claim that one's real sex and/or gender fails to match one's biological constitution at birth. It is also worth noting that for norms relevant to sex and/or gender, the conditional norm view could make the antecedents of the conditionals be facts about DNA (such as whether one has a Y chromosome) but could also make them involve facts about psychology and society. The metaphysics does not by itself settle the normative questions here.

a normative accident of a teleological directedness at the beatific vision, together with the supernatural powers that promote the fulfillment of this telos.

5. *Substances and quantum mechanics

What has form, i.e., what are the substances? It is essential to our meta-normative applications that *humans* be substances and hence have forms. And given a view of human beings as having an organic teleology, it would be implausible to deny form to other organisms. Thus, all organisms have forms. But what about inorganic reality? It appears to be central to Aristotelian metaphysics that all of reality be grounded in substances and their properties and relations. Inorganic aspects of reality—say, particles—that are part of an organism will be grounded in the organism *if* we accept the classic Aristotelian thesis that substances cannot be proper parts of substances.

But in any case, that would still leave the vast amount of inorganic reality outside of any organisms, including particles, fields, rocks, mountains, planets, stars, galaxies, and so on. There has to be at least one inorganic substance to ground these.

On most interpretations of quantum mechanics there is no precise number of particles in existence, as the universe is in a superposition of different states corresponding to different particle counts. But whether a substance exists, and what identity relations hold between substances, must be an objective fact. We thus cannot have superpositions of different substance counts in reality. Thus, on most interpretations of quantum mechanics, particles are not good candidates for substances. The one exception is the de Broglie-Bohm pilot wave interpretation (Bohmianism), on which there are particles or “corpuscles” with definite positions. If we have the intuition that particles are substances, that will support Bohmianism.

The view radically opposite to a corpuscularian insistence on inorganic reality being partly grounded in substantial particles is to suppose that inorganic reality is grounded in some small number of global substances. This is an active area of research among Aristotelian philosophers of science, and a number of options are available.??refs These options

differ depending on how the information contained in the mathematical wavefunction of the universe is encoded in the substances of physical reality.

First, there could be a single global “wavefunction substance” that grounds the values of the global quantum wavefunction, existing over and beyond the organic substances, and interacting with them.

Second, one might suppose that a part of the information in the wavefunction of the universe is encoded in states of the organic substances, and the rest (i.e., vast majority) of the information is encoded in a global “Swiss cheese substance” that has holes corresponding to the organic substances.

Third, one might try to ground the wavefunction’s data in a number of small local point-like substances. These could be points of three-dimensional space or of four-dimensional spacetime, and the wavefunction could be grounded in their relations.⁷ Or they could be points in a higher-dimensional phase space, with the wavefunction grounded in their properties.

A Bohmian view does not escape the need for a grounding of the wavefunction, because the corpuscles are governed in their behavior by the wavefunction, and their postulated properties like position are not sufficient to ground the values of the wavefunction. Thus, an Aristotelian Bohmianism does not appear particularly economical: it requires organic substances, corpuscles, and a ground for the global wavefunction.

Interpretations of quantum mechanics can be by and large seen as responses to the measurement problem: the problem of what to do with the fact that naively applying the Schrödinger equation to the initial conditions in many experimental setups generates the prediction of a wavefunction that contains a superposition of different macroscopic results,

^{7*}In a non-relativistic setting, the wavefunction for n particles that have position as their only property is a complex-valued function ψ the phase space $\mathbb{R}^{3n}$. A point in $\mathbb{R}^{3n}$ can be identified as a sequence $(x_1, \dots, x_n)$ of n points each in $\mathbb{R}^3$. We can then suppose that there is an n -ary determinable relation (“being co-wavefunctioned”?) between any n points in $\mathbb{R}^3$, a determinate of which is quantified with a complex number, which complex number yields $\psi(x_1, \dots, x_n)$. If the points $x_1, \dots, x_n$ are substances, then these values are indeed grounded in relations between substances.

such as an instrument display showing one number and its showing a different number—something we never observe. Thus, in order to generate predictions compatible with what we actually see in the lab, something must be done. Responses divide into collapse and no-collapse interpretations. On both families, we have a baseline deterministic evolution of the wavefunction over time according to the Schrödinger equation.

On no-collapse interpretations, the baseline is all that happens, and as a result physical reality gets more and more superimposed. For instance, given the likely dependence of our solar system on quantum events in the early universe, reality is in a superposition of our solar system having formed and our solar system not having formed. But that is not what we observe: we simply observe our solar system. One way to think about this is that the global wavefunction on a no-collapse interpretation describes something like a multiverse, in some branches of it there being a solar system and in others there not being one. We inhabit a branch where there is a solar system.

If you toss a quantum mechanical coin, on a no-collapse interpretation our branch splits into a branch with heads and a branch with tails. What happens to you? There are six options:

- (5) You cease to exist.
- (6) You live in both a heads and a tails branch.
- (7) You go to the heads branch and no one inhabits “your” position in the tails branch.
- (8) You go to the heads branch and someone else⁸ inhabits “your” position in the tails branch.
- (9) You go to the tails branch and no one inhabits “your” position in the heads branch.
- (10) You go to the tails branch and someone else inhabits “your” position in the heads branch.

⁸Perhaps a pre-existing person who had been co-located with you prior to the branch split.??ref:many-minds

Both (5) and (6) have a pleasing symmetry. However, both have a serious problem. Suppose the coin is very unfair, so it has a $7/8$ chance of heads and a $1/8$ chance of tails. Now, on both (5) and (6), the two new branches are equally related to you—either neither is *your* branch or both are *yours*. But obviously prior to the toss, you should plan your future asymmetrically—you should expect to get heads. This asymmetry in planning does not fit with the metaphysics of the situation.⁹ There is, of course, a large literature on this argument, with significant dispute^{??}refs, but I will take the argument to provide significant support for the asymmetric metaphysical options.

On the remaining four options, the wavefunction does not contain all the information about the universe, since the wavefunction only says that there is a superposition between heads and tails, and does not tell you whether the individual who tossed the coin goes into the heads branch (as on (7) and (8)) or into the tails branch (as on the remaining two options). On these options, there is no problem with uneven chances: the chance of heads can determine how likely it is that one will get (7) rather than (9), or that one will get (8) rather than (10).

Thus, on a no-collapse view we have good reason to think there has to be more to reality than the wavefunction. Indeed, a no-collapse view points towards a metaphysics where some of the facts about reality are encoded in the wavefunction but others are encoded by the “location” of the agents in the multiverse. This fits neatly with an ontology on which we have a global substance grounding the wavefunction and agents that move around in the multiverse.¹⁰

⁹It may occur to the reader to say that there are eight branches, one of them with tails and seven with heads. Then if you live in all eight branches, you might think it makes sense to plan “more” for the more common branch. There is, however, a simple technical reason why this can’t be the solution. This solution only works when the chance of heads is a rational number, i.e., of the form p/q with p and q integers. But if the chance of heads is not a rational number, say because it is $1/\sqrt{2}$, there is no way to divide the branches in a ratio corresponding to the chances. And the continuous evolution of the wavefunction ensures that most of the time the chances will be irrational numbers.

¹⁰For interpretations like this, see ^{??}refs.

On collapse interpretations, superposed states, such as a superposition between an electron being in one place and its being in another, are resolved indeterministically into pure states, say, the electron being in a specific place. The value of the wavefunction specifies the chance that a given superposed state will resolve into a given pure state. This collapse is an exception to the deterministic Schrödinger equation.

On these interpretations, as far as the physics goes, it is possible to think the wavefunction describes all of physical reality if the situations that trigger collapse are themselves described by the wavefunction. The historically first collapse interpretations^{??Ref} had collapse triggered by observation. This appears to require the prior existence of an observer. But if we think about the evolution of organisms capable of observations, this evolution itself depends on a vast number of quantum events (such as DNA mutations), and hence prior to the appearance of the first physical observer, all we have was a superposition of a state where an observer-like being had already evolved and a state where an observer-like being had not yet evolved. Thus, unless the existence of an observer is a fact over and beyond a wavefunction that codes for a superposition of observer-like states, observation-triggered collapse seems unable to get off the ground. Observation-triggered collapse, thus, offers some support for an Aristotelian metaphysics where in addition to one or more substances grounding the wavefunction, we have some additional macroscopic substances that are observers.¹¹

More recent collapse interpretations coming from physicists (e.g., Ghirardi–Rimini–Weber theory^{??ref}) have collapse get triggered discretely or continuously by physical facts about the wavefunction. This is compatible with the wavefunction being a complete description of physical reality. These theories are empirically testable, but have not yet been sufficiently tested.^{??refs} If we find such a theory to be correct, an Aristotelian view that posits additional macroscopic substances beyond the substance or substances

¹¹Of course, an interesting open question for such theories is what kind of interaction with a macroscopic substance counts as an observation. Do plants observe when they interact with their environment, for instance?

grounding the wavefunction will appear to be *physically* otiose, though the normative and semantic considerations of the earlier chapters will continue to be support it. Furthermore, standard arguments against physicalism in the theory of mind??ref-and-backref lend support to a richer ontology.

Anyway, other things being equal, no-collapse interpretations of quantum mechanics are preferable: they take the wavefunction to evolve under the elegant (and deterministic) Schrödinger equation. The physically-based collapse theories modify this evolution in a mathematically complicated way with one or more additional free parameters (such as a collapse rate and a critical distance in the Ghirardi-Rimini-Weber theory). Theories on which collapse is triggered by interaction with something not at the level of physics, like organisms, at least seem to have the slight advantage of preserving the idea that when physics alone is in play, we have the elegant Schrödinger equation—collapse is triggered by interaction with something complex that is properly studied by other sciences than physics (e.g., biology or psychology).

Thus it seems that when we think about quantum mechanics, a model of reality on which there are macroscopic substances like oaks, sharks and people, and a further substance that (alone or in cooperation with the macroscopic substances) is needed for grounding the wavefunction is not at all unnatural. On no-collapse models and non-physical collapse models, this theory answers a need that arises from the physics itself—a need to generate experimental measurements.

CHAPTER XI

Evolution, Harmony and God

1. The origin of the forms

1.1. Evolution and forms. We have good empirical reasons to think that the variety of biological structures that fills our planet is largely or completely the product of unguided variation together with natural selection. However, as I have argued, there are good philosophical reasons to think that these organisms—or at least some of them, like humans—have normatively laden forms which specify how the organisms should behave, endow them with the causal powers that make that behavior possible, and impel them towards that behavior.

It is implausible to think that the forms supervene on biological structure. For instance, one theory of the evolution of wings for gliding is that small wings are useful for heat dissipation. Larger wings allow for more dissipation of heat, but are also more expensive for the organism to maintain. However, it is said that at around the size at which the heat-dissipation benefits are outweighed by the maintenance costs, the wings also become useful for gliding. It is plausible that a species *A* that has the smaller wings has them with the telos of heat dissipation. But a species *B* that has evolved the larger wings has them with the telos of gliding, either instead of or in addition to heat dissipation.??ref,check But we can now suppose a member of *B* whose wings are defective and only good for heat dissipation. Such a member's biological structure might be largely indistinguishable from that of a normal member of *A*, and yet it is normatively different: such wings are defective in *B* but entirely appropriate in *A*. If these norms are grounded in forms, it seems there is a different form in members of *B* than of *A*, even when the biological structures are qualitatively the same.

In general, in the evolutionary process, we expect typical cases of reproduction to involve see small differences in genetically-based biological structure between parents and children but no change at all between the parent's form and the child's form. For if we had constant change between the parent's form and the child's form, our best account would be that the form simply matches the biological structure, which would not allow for genetic defects, and yet genetic defects—deviations of genetically-based biological structure from the kind norms—are clearly possible. Moreover, it is important to our ethics that all human beings, despite a wide variety in physical and mental endowments are beings of the same kind.

We thus need an explanation of why it is that at certain apparently relatively rare and discrete points in the evolutionary sequence we have a new form on the scene. This itself yields Mersenne questions: while some transitions of form might happen to coincide with a particularly striking genetic transition, we expect a number of them to come along with only minor genetic transitions, seemingly at arbitrary positions. What explains the locations of these transition points?

Hitherto in this book, such questions were answered by invoking the forms themselves. And this can be done in this case as well. We might suppose that the form of species *A* endows the members of *A* with a causal power to generate new members of *A* in some circumstances, together with new instances of the form of *A*, but also a causal power to generate new members of *B* in other circumstances, along with new instances of the *B* form. The difference in circumstances could consist in differences in the DNA content in the gametes joining together, so that when a descendant is going to have such-and-such DNA contents, the descendant gets the form of *A*, but with other DNA contents, the descendant gets the form of *B*.

This story requires the forms to contain troublingly intricate specifications of which form is generated when. Granted, the slew of Mersenne questions we have already raised should make us open to a significant complexity of information in the forms. But the case at hand is different. But the story above requires that the first biological organism on

earth—presumably some simple unicellular or maybe even proto-cellular?? organism—contain within it a form that codes for the causal power to produce forms of all possible immediate descendants of it. These immediate descendant forms then would have to code for the causal power to produce all their possible immediate descendants, and so on. Thus, the form of the first and simplest organism would implicitly code for all the forms of life that would ever actually be found on earth, and indeed all the forms of life that *could* ever descend from it.¹ We thus have here a dizzying complexity.

But the problem does not stop here. For we can now ask where that immensely sophisticated form of the first organism comes from? If we say that it comes from the causal powers of non-living substances, such as fields or fundamental particles, then we have to posit an even greater complexity in the forms of these non-living substances. It is highly counterintuitive to suppose non-living things to have immense sophistication of form, and we would contradict the intuition that the world of physics is fundamentally pretty simple. Further, however, we would need a story of where the first forms arose from. If we take the above account to its logical conclusion, then at the Big Bang we would already have particles or fields whose forms implicitly included the vast formal complexity of all physically possible living organisms. And this in turn yields a powerful design argument for something like theism. For the idea that such complexity would simply come about for no reason at all is quite implausible.

Thus, the story that forms contain the rules for the generation of future forms points towards a being whose own power is sufficiently great to generate such forms. And to avoid a vicious regress, such a being would need to be a necessary one.

But note that once we have accepted the existence of a necessary being that is the ultimate source of the varied forms in our world, we can now tweak the story to avoid the implausible idea that unicellular organisms implicitly code for the forms of elephants and

¹It is tempting to say that the number of possible descendant forms is infinite, but that is not clear. After all, there could be some physical limit to the size of the genetic code for a biological organism given our laws of nature. But in any case, finite or not, the number of possible descendant forms is vast.

unicorns. Instead of supposing that the transitions between forms corresponding to certain selected changes in genetic structure are caused by the parent forms, we can suppose that the necessary being is directly responsible for the transitions of forms. On such a view, the form of a unicellular organism might only endow its possessor with the ability to generate a descendant of the same kind, and the necessary being would directly produce any new forms when it is appropriate to do so.

One way to reduce the problem of transitions of forms is the theory that there is only a handful of kinds of organisms on earth. Some contemporary Aristotelians, for instance, suppose that the only three “metaphysical species” of organisms are plants, non-rational animals, and rational animals. If so, we only need to explain two transitions in evolutionary history.

But this does not solve the vast complexity problem. For now the form of animal, say, is going to contain conditionals encoding the norms for all the animals—if you’re a dog, you should have four legs and if you’re a sparrow, you should have two and so on. And if the form of rational animal includes all possible rational animals other than humans, then we will need the form of the rational animal to encode not just the great complexity in ethics, epistemology, semantics and metaphysics in one biological species of rational animal—*homo sapiens*—but also others, such as possible rational sharks or hive-minded rational bees. We have not escaped. And, once again, we need an explanation of the small number of organic forms and where they come from. If they come from inorganic substances, then once again we have the implausible suggestion of vast complexity in inorganic substances, which if accepted leads to a design argument. Otherwise, the best suggestion is again a necessary being that bestows either the first of the small number of highly complex organic forms or that bestows all of them.

It seems that whether we allow for a large number of types of organic forms or a small number, we have reason to believe in a necessary being that produces the initial complex forms and is perhaps responsible for transitions between forms.

1.2. Reasons for creating forms. Of course, this would lead to the question of *why* the necessary being produces new forms when it does so. Here, taking the necessary being to be rational can help. For there can be good value-based reasons for the transitions to fall in some places rather than others.

Consider, first, an odd thought experiment. A horse-like animal comes into existence with an maximally flexible form such that whatever the animal does fulfills the norms in the form. To eat and grow is one proper function, and to starve and produce a corpse is just as proper a function. Whatever our “flexihorse” does or undergoes is equally good for it. But there is something unsatisfactory about the flexihorse as a creature. If whatever the flexihorse does is equally good for it, then the fact that the flexihorse flourishes is just a direct and trivial consequence of its externally imposed form rather than the individual’s *accomplishment*.

Reflection on this suggests there is a value in creating organisms that can fail to fulfill their norms. This value might be grounded in the forms themselves: it might be that real horses, unlike flexihorses, have self-achievement of flourishing among the proper functions in their form. And there is a value in creating organisms that have additional types of good written into their form, including such self-achievement. Alternately, one might hold that in addition to kind-relative goods, there may be kind-independent goods—perhaps grounded in imitation of the creator??forwardref?—and self-achievement of flourishing could be one of these.

Either way, a rational being creating organisms has reason to create organisms that can fail to achieve their form, and hence has reason to create beings with less flexible norms. Moreover, there appears to be a comprehensible value—again, either kind-relative or kind-independent—in production of beings of the same kind. As an intuition pump here, think of the *Symposium*’s idea that the yearning for eternity is exemplified in animal reproduction. Thus, we can give a value-based explanation for why a necessary being would create beings in discrete kinds, with norms that the beings need not live up to.

2. Harmony

2.1. Review. At a number of points in the book, we have found noteworthy harmonies, many of them falling under the head of “Aristotelian optimism”, that call out for explanation. At this point, let us collect a number of them as well as add a few more.

2.2. Epistemology of normativity and form. Central to Aristotle’s epistemology of normativity is the optimistic assumption that things generally go right. This allows us to get evidence as to what is good or proper from how things are actually behaving: actual behavior is more likely correct than not.

The assumption that things tend to go right is not quite enough, however, for an epistemology of normativity. It won’t, for instance, give us knowledge of what is right in rare cases. Furthermore, the claim that behavior tends to go right has well-known systematic exceptions. In cases where people are under significant social pressure to act in ways that are morally wrong, while some resist, many—perhaps a majority—will make morally inappropriate compromises.

In addition to optimism about behavior, it seems we additionally need optimism about intuitions, innate beliefs or high Bayesian priors about specific substantive propositions (I will now just say “intuitions” for short). We can try to derive this optimism about intuitions from optimism about behavior combined with the claim that cognitive behavior has truth as an end. However, the claim that cognitive behavior has truth as an end seems to depend on intuition about ends. One might try argue that the claim that cognition has truth as an end is analytic—it follows from the very conception of cognition. But that would just shift the bump under a rug to a different place: our intuition that such-and-such behaviors are in fact *cognitive*. So it seems that some irreducible optimism about intuitions is unavoidable.

Another area for optimism beyond behavior is with regard to our ability to identify which things have form, and which groups of things have *the same* form. There is good reason to think it is metaphysically possible to have particles structured in a way that is

empirically indistinguishable from a horse and yet with the particles not constituting a horse or any other whole. One way to make this plausible is to recall the Small Beginnings Thesis^{??backref} according to which material substances, like horses, start subatomic in size. Given that there is nothing specifically empirically equine about a subatomic entity (it wouldn't of course be really tiny four-legged critter, but rather something like a fundamental particle or a portion of a wavefunction with a horse form), at that stage it is possible to have something that is not a horse but that is empirically indistinguishable from our subatomic horse.

One could put Small Beginnings aside, and ask whether it would be metaphysically possible to have something empirically just like a full-grown horse but without a horse form. Given that the Aristotelianism that I have been defending is a robust one, where the horse form is an additional entity rather than a mere arrangement of particles (as we saw, this also followed from the Small Beginnings Thesis), it seems hard to deny the possibility of any given arrangement of particles lacking that extra entity.

Furthermore, suppose that in a lab we carefully arrange particles just as they are in a full-grown horse. If a horse form comes into existence, we need an explanation of where it came from. There are three initially plausible options here: (a) nowhere, (b) from the physical constituents (particles, fields, wavefunction, etc.) and (c) from God or some other very powerful supernatural being.

If the form came from nowhere, we have a violation of the venerable principle that nothing comes from nothing. But such violations, if they happen at all (which is dubious^{??ref}), are brute contingent facts. Hence they at least *need not* happen, and so it is metaphysically possible that the particles come to be shaped like in a horse without a horse form arising.

If the form came from God or some other powerful supernatural being, then it is very plausible that this being would also have the power to withhold the form.

If the form came from the physical constituents, then we have to suppose that the fundamental physical entities in our universe have preprogrammed into them the forms

of all possible material substances, which they necessarily produce as needed. But this hypothesis also calls for some optimism. It surely would be metaphysically possible to have a world with physical constituents that behave empirically like ours, but where these constituents are not such as to necessarily produce horse (or other organic) forms—say, because it is random whether they produce a horse form in any given chunk of horse-like matter. To deny this hypothesis would require an optimism that goes beyond supposing that substances tend to behave correctly, since these hypothetical physical constituents would be behaving correctly when they would fail to produce a horse form in horse-like matter.

But now if it is metaphysically possible to have something empirically indistinguishable from a horse and yet is a mere heap of physical stuff, the same would be true of humans. And thus to know the normative structure of the human world we would need to rule out the hypothesis that many of the human-like chunks of matter around us are form-zombies, formless beings that behave like we do.

Perhaps, though, we can conclude that a given horse-shaped pile of matter is a horse from the fact that its parents are horses, together with the fact that it is normal behavior of horses to give birth to horses rather than merely horse-shaped piles of matter. However, this regularity is one that we cannot empirically and non-circularly get to by observing horse behavior. As far as empirical observation goes, we see horses give birth to horse-shaped piles of matter. To be confident that these piles are horses requires something like an intuition of the sort we have been talking about.

One way to be optimistic about getting right the match between forms and piles of matter will be by supposing that our intuitions about the differences in structure and behavior that give rise to differences of kind, when subjected to a due process and brought to a rational (by human lights) equilibrium, will tend to get things right. Another way would be to suppose a broad uniformity of nature thesis that makes invisible joints (say, those between types of forms) tend to correspond to visible joints. But a uniformity of nature thesis

may not be sufficient for the epistemology of rare-case norms, so the intuition-focused approach may be better. In any case, in order to have a satisfactory epistemology of form, our optimism needs to go beyond the thesis that typically things go right—important as that thesis is.

In practice, we can gesture at a process of gaining knowledge of forms and norms on the basis of the behavior of substances (or apparent substances) and our intuitions (or innate beliefs or substantive Bayesian priors), together with a rational equilibrium, doubtless in a communal setting since we are social epistemic agents. We can optimistically suppose that such a rational equilibrium brings about a belief that all the human-like beings that are biologically within the *homo sapiens* species, despite biological variation (due to age, sex, or genetic subcommunity), share in the same form, while the currently existing organisms outside that biological species do not.² But all this harmony calls out for explanation.

2.3. Nomic coordination. On an Aristotelian picture of reality, the orderly behavior of the universe is grounded in the causal powers of disparate substances. This, however, is puzzling on two counts. First, we have the puzzle that there is a relatively small number of forms found in the universe. Why, for instance, is there one kind of human form shared by all the billions of humans, rather than a different human-like form had by each individual?

Note that the problem of the large ratio of individuals to kinds is one that every theorist faces. There are at least 10^{80} particles in the universe, but only about seventeen fundamental particle types (see ref.). Still it is worth noting that the Aristotelian does not have any obvious distinctive answer to *this* problem.

There is, however, a second problem that not everyone has. Some Aristotelians (see refs, and including the author of this book^{backref}) enjoy criticizing Humeans for making nomic explanations bottom out in an unexplained harmony of nature, where the vast numbers of particles all happen to behave the same way, and the laws of nature merely

²Things are less clear with regard to other biological species of genus *Homo*, such as Neanderthals. And so they should be, since we cannot observe the regularities in their behavior, except by archaeological means.

describe this behavior. But for Aristotelians the same problem returns at the level of kinds. There is a significant number of different kinds of substances in nature: humans, dogs, oak trees, electrons, etc. Yet their behavior is nomically coordinated. For instance, in classical mechanics, each object, regardless of its type, exerts a gravitational force proportional to inertial mass and inversely proportional to the square of the distance.³ Thus on an Aristotelian account of laws each object—or at least each object with mass—will have to have a form with a gravitational causal power with the same dependencies on mass and distance. Perhaps even more strikingly, the causal powers of all the forms in nature are so coordinated that mass-energy is always conserved. Energy lost by one object is gained by another, regardless of what kind each object is. We already met this puzzle in ??backref.

Granted, the problem is in a sense many orders of magnitude different than for Humeans. While the Humean leaves unexplained the coordination between 10^{80} individual particles, the Aristotelian has a problem about the unexplained coordination between 17 individual particle types and a large number, but likely not far above 10^{10} if earth is the only planet with life, of types of biological kinds. But while a coincidence between 10^{80} things may be vastly more improbable than that between 10^{10} , neither coincidence makes for a satisfactory theory. And this is not a problem for everyone: views on which there are global “pushy” laws do not have this problem.

Moreover, just as the Humean faces the sceptical worry that hitherto unobserved particles might behave differently, the Aristotelian faces the sceptical worry that hitherto unobserved *kinds* of substances, say undiscovered types of bacteria on the bottom of the ocean or on other planets, might behave differently even at the level of physics, since their physical behavior is grounded in their form.

One might make some progress towards a solution by supposing a richer structure in the forms that we have hitherto done, going back to Aristotle himself. For while Aristotle’s metaphysics puts a significant emphasis on the *species*, it also has multiple levels of genera

³Or, more precisely, distance to its microscopic parts: the overall gravitational force of a spatially extended object that is not spherically symmetric only follows the inverse square law asymptotically.

that he appears to take metaphysically seriously. Suppose now that we say that the species form has a structure where it is composed of a multiplicity of levels of generality, and at each level it has a metaphysical component specifying the level. Thus, perhaps skipping metaphysical levels or including extraneous ones, we might suppose that an oak tree form has a broadleaf tree generic form as a component, which in turn contains a plant generic form, in which there is a DNA-based organism generic form, and in which there is a generic form shared by all the material substances in our universe.

We might then call the last generic form “the physics form”, and suppose that it is responsible for the common physical behaviors—the gravitational pulls, the mass-energy conservation, and so on. With respect to fundamental physical behavior, then, we have one kind across all the substances. But the more specific levels of form in the oak tree still have a role to play. Minimally, they play a normative role: they specify how the oak tree *should* behave in virtue of being an oak tree, or in virtue of being a broadleaf, or in virtue of being a DNA-based organism. But they likely have some sort of a causal role. Such a causal role could fit into gaps in fundamental physical causal laws—which gaps do exist if the laws are indeterministic—but at least it will help to explain the passing on of form to offspring: the offspring of broadleaves tend to be broadleaves, and so on.

This solution, however, still leaves something unexplained: Why do all the substances in our universe have the same generic “physics form”? We can, however, reduce the amount of what is unexplained. For instance, we might suppose that in substantial causation where one substance causes another, the physics form of the causing substance causes the effected substance to have the same physics form. On this account, all we need to explain is why *initially* there was only one physics form among all the initial kinds of entities. Since the initial kinds of entities are presumably only fundamental physical constituents like particles, we have moved from a coincidence between 10^{80} things in Humeanism, to 10^{10} kinds in our less structured Aristotelianism, to now somewhere around 17 kinds. But that’s still something that calls out for an explanation.

Perhaps, though, explaining the harmony between a relatively small number of kinds is not very different from a non-Aristotelian advocate of pushy global laws having the explain the harmony between these laws—say, that all the laws governing fundamental forces are compatible with mass-energy conservation. Maybe we have a difference between a coincidence involving four forces and one involving 17 particle types, but this kind of a difference may not be that decisive. Nonetheless, it would be good if the Aristotelian had something further to say here.

Furthermore, global pushy laws involve an interesting harmony or fine-tuning issue recently identified by ??ref. A version of the problem arises from the fact that laws are supposed to move objects in accordance with the properties that material objects have. For instance, the Coulomb law induces a force of $-kq_1q_2/r^2$ between particles with charges q_1 and q_2 (measured in some natural physical system of units), with the force being measured in the direction from one object to the other. Now consider the infinitely many possible fundamental properties quantifiable by the same kind of range of positive and negative real numbers that charge is. Call any such property “charge-like”. For each one of these charge-like properties, say Q , there is a possible Coulomb-like law L_Q that says that two objects have a force of $-kq_1q_2/r^2$ between them when the particles have Q -values q_1 and q_2 respectively. Presumably, only one Coulomb-like law L_Q is exemplified in our world, namely L_{charge} . But now consider this vast coincidence: our world is a world with particles that have a small number of fundamental physical properties including charge and our world also has the law L_{charge} which is but one law from among the infinitely many L_Q laws, the vast majority of which would be irrelevant to our world’s particles since they lack the requisite property Q . In other words, we have a harmony between the properties and the laws: our world has laws that are relevant to the properties of the world.

One might object that for all we know our world has laws involving all the infinitely many other charge-like properties, but we will never know about any of these laws, because the properties of particles governed by them are unexemplified in our world. But on

this view, the full collection of our world's laws of nature is immensely complex in a way that appears to be a massive violation of Ockham's razor.

Alternately, the advocate of pushy global laws might say that there are no laws governing unexemplified properties. Thus, only those L_Q can exist where Q is exemplified. However, this has the odd consequence that what physical laws we have in the world depends on what nomically contingent events happened in our world's physical history. Suppose that charge became exemplified in our world because a contingent collision between two photons produced an electron-positron pair. But then the existence of L_{charge} in this universe depends on a nomically contingent event: the collision of the photons. This seems wrong. The laws of nature do not seem to be piece-meal generated by physical events. Even more seriously, that an electron-positron pair can be generated from a collision of two photons is itself a law of nature. But if the existence of this law depends on the existence of electrons and positrons if there are no laws governing unexemplified properties (in this case, electronicity and positronicity), and so we have a vicious circularity: the law depends on the electron-positron pair coming from the collision, but the law explains why the collision results in such a pair.

One might worry that a similar problem shows up for Aristotelians. If, say, the Coulomb law is grounded in the forms of the electron and positron, then it seems that if photons did not collide to produce electrons or positrons, there would be no such law. But that's not quite right. For in the universe where photons do not collide, it is still true that if there *were* electrons and positrons, they *would* behave in accordance with Coulomb's law, a fact grounded in the forms that electrons and positrons *would* by definition have (namely the forms that would constitute them as electrons and positrons, respectively). The advocate of pushy laws that do not exist for uninstantiated properties has a much harder time saying what could ground a similar counterfactual.

Perhaps the advocate of pushy global laws could say that for every set of fundamental physical properties that can be instantiated in the same world, there are laws about how objects with those properties would behave, even though the vast majority of these laws do

not apply to anything actual in our universe. We could then say that while Ockham's razor is not globally applicable to the laws as a whole—there is a vast infinity of these laws—for any fixed and relative small set U of fundamental physical properties instantiable in the same world, Ockham's razor applies to the subset of the laws governing objects with properties from U —we should opt for simpler candidates for these laws. This will be enough to recover our preference for simpler scientific theories.

Nonetheless, this much is clear: if we attempt to solve the coordination between the exemplified properties and the laws by supposing a vast set of laws, there does not seem to be a significant simplicity advantage to global pushy laws over Aristotelian form-grounded laws. And Armstrongian laws grounded in necessitation relations between universals have a similar problem. Thus, while Aristotelianism has a difficulty here—and that is the main upshot of this discussion—it is not thereby refuted, assuming a non-Humean picture of laws. But it would be nice to solve the difficulty.

2.4. Harmony between types of normativity.

2.4.1. *Introduction.* Consider three central aspects of our normative life: flourishing, morality, and epistemic rationality. At times they conflict. But not that often. I will now argue that there is a surprising amount of harmony between the three, beginning by considering all three possible pairings.

2.4.2. *Flourishing and morality.* Next, let us move on to one of the greatest discoveries of Western philosophy: Socrates' realization that virtue is central to our flourishing or happiness.

We should not, however, go along with Socrates' more extreme claim that virtue is all there is to our flourishing. First, we have Aristotle's observation that a person can have perfect virtue but remain unconscious all their life, and that is not flourishing. It is not just the possession but the exercise of virtue that contributes to flourishing. Second, we have Vlastos' decisive counterexample of the vomity bed^{ref}: it clearly contributes to our flourishing that we not have to sleep in a bed covered in vomit, even though this good is neither virtue nor its exercise.

Nonetheless, the possession and exercise of virtue is the main contributor to our happiness. The person who has pleasure, reputation, wealth and health but is vicious leads an unenviable life, and a person who has the possession and exercise of virtue while lacking much in the way of other goods leads a flourishing life, except perhaps in cases where the shortfall of other goods is extreme, as in Aristotle's example of being tortured on the rack^{ref}.

Now, deep and important as the fact that it is good for us to act well is, this fact is easily explained by a basic Aristotelian metaethics: Right action is action that constitutes the proper function of the will, and functioning properly is good for us. Thus far, the harmony does not call out for further explanation. But when we add that the acting well is *central* to our flourishing—that it is among our greatest goods—then we do in fact have a harmony that we need to go beyond the basic theory to explain. For instance, there is good reason to think that long-distance running is one of our proper functions as human beings—we seem to be particularly designed to excel at this.^{ref} However, a vicious person who runs excellently does not flourish on balance, while a virtuous person can flourish without ever running. Running is important on relevant *occasions* which may not be common (unless we are early humans who have to engage in exhaustion-hunting^{refs}), while we use the rational will for a large chunk of our waking hours. But we could imagine animals where the two were reversed—they run all the time, but most of the time they act on instinct, which allows for faster decision-making, only being triggered to engage in rational deliberation on rare occasions. For such animals, perhaps, moral virtue would not contribute to flourishing as much as it does for us. Yet such animals are compatible with our basic Aristotelian theory.

Further, Aristotle rightly observed that normally there is a pleasure in acting virtuously. Additionally, a virtuous person takes pleasure in the flourishing of others, especially their friends^{ref:Aristotle}, while at the same time contributing to this flourishing. The life of virtue tends to be pleasant. Again, this aspect of the harmony between flourishing and right action calls out for explanation.

And while there are obvious notable exceptions, as a rule virtuous people tend to have a good reputation and be liked by others. Further, an important part of our moral goodness is benefiting human beings in a broad variety of ways, and we are human beings ourselves. Moreover, we are often particularly well-positioned to benefit ourselves, so virtuous people have a tendency to do so. Furthermore, executive virtues are among the virtues, and without the executive virtues, we are unlikely to be happy.

There are, of course, many conflicts between morality and happiness. The most obvious is that morality often requires us to forego various other goods. To a significant degree, however, the sacrifice is compensated for by the innate value of the life of virtue and the pleasures thereof.

Next, let us move on to something completely obvious. Benefiting others tends to be morally right and harming others tends to be morally wrong. This is unsurprising on consequentialist ethical theories, but on non-consequentialist ones, like our Aristotelian theory, it calls out for an explanation. The right is defined by the proper function of the will. We can imagine animals whose proper function of the will is to benefit themselves, with no regard for others. Further, many animals have body parts whose proper function is to harm others—talons, teeth and horns, say. Sometimes, as in the case of horns used for mating battles, the proper function is to harm others of the same species. We could thus imagine rational animals whose proper function of the will is focused on harming others. But we are not like that. But we are not like that. And it is good that animals with our endowments are not like that. For if we were like that, we either would act rightly and hence not cooperate with one another or we would cooperate with one another at the expense of acting rightly. But since right action is good for the individual, both options would be harmful to us.

More generally, consider this form of argument for a moral fact *F*:

- (1) It would be bad for humans if *F* did not obtain.
- (2) So, probably, *F* obtains.

This is, alas, not a very good argument in the case of non-moral facts. We should not, for instance, conclude that global warming trends will stop on the grounds that it would be bad for humans that they continue!

However, there are good cases of this argument in the moral case. A moral theory that denies that we should benefit others should be rejected, because it doesn't fit with our well-being. But let's consider another, more specific example of the argument. Intelligent social animals like us appear to need some sort of a system of punishment in order to counter severe antisocial behavior. But an effective system of punishment requires harming another in a way that would be wrong—either due to the harm or the violation of autonomy or both—were it not a case of just punishment. Let *F* be the moral fact that the imposition of harsh treatment on malefactors by social authorities is morally justified. It would be bad for us if *F* did not obtain, since then either we would as a species be constantly victimized by severe antisocial behavior, or we would be wrongfully imposing harsh treatment on malefactors, and hence we would ourselves be malefactors, which would be bad for us, or some other bad thing would happen (e.g., if harsh treatment were justified, but only when imposed by vigilantes and not when imposed by authorities).

Examples can be multiplied. For instance, it would be bad for us if the promisee were unable to release the promiser from a promise—so, probably, they can. Likewise, it would be bad for us if promises to do something wrong were morally binding—so, probably, they aren't. It would be bad for us if it were never permissible to risk an innocent life (we couldn't drive, or hunt cooperatively, or leave a baby alone no matter what precautions we took)—so, probably, there are reasonable risks it is permissible to take. In such cases, we do indeed argue for a conclusion that may initially seem unobvious by pointing out the unfortunate consequences for our flourishing if it were false.

Famously, Andrea Dworkin argued that heterosexual intercourse by its physical nature violates the boundaries of a woman's body, and hence human reproduction was impermissible until such time as *in vitro* fertilization was invented. Most people dismiss the position as crazy, and it seems very plausible to dismiss it precisely on the grounds that it

would have been very bad for our species—we would have needed to go extinct to obey the norm—if heterosexual intercourse were never permissible, and hence it is sometimes permissible. A similar argument is likely to be one of the reasons for people not taking antinatalism??Refs seriously: if it wasn't permissible to have children, that would be very bad for the human species, so it is permissible.⁴

For the argument from (1) to (2) to work, we need a harmony between what is good for humans and what moral facts obtain, a harmony going beyond the fact that doing right is partly constitutive of our flourishing.

We thus have multiple types of harmony between flourishing and morality. But we should not be too Pollyannish here. As noted before, there may be harms other than vice that *can* destroy one's flourishing. Torture, both due to physical pain and the subsequent psychological damage, is a glaring example, and in some cases morality requires one to stand up to torture.

Second, it is worth noting that we can imagine hypothetical cases where doing what is morally right makes one less well off as a moral agent. One might imagine an ethics class that is so inspiring that it tends to transform students morally for the better, and a student who can only afford the cost of the class of the class by committing theft. We can then imagine that the class is so transformative that its benefits to one's character outweigh the harms of the theft. Nonetheless, the theft is wrong. Or we can imagine a case where one is in a morally harmful environment and the only way to escape that environment is by doing something morally wrong. For instance one might live in a soul-destroying totalitarian state and to escape it one would need to steal in order to pay for airplane tickets.

Furthermore, as Stohr has noted, there are times when acting virtuously is not innately pleasant: consider her example of the pediatric oncologist having to break tragic news to parents??ref.

⁴This last pair of arguments shows that one can take "humans" in (1) as a species and not merely as an aggregate of individuals.

Nonetheless, the exceptions to the confluence of virtue and happiness, while sometimes tragic, tend to be rare. Torture is, fortunately, relatively rare, and not all cases of torture in the world are ones where morality requires one to stand up to the suffering. Some cases are, for instance, ones where terrorists or other criminals are tortured to reveal their wrongful plans—in those cases, where inflicting the torture is morally wrong, there is no duty to withstand it, and there may even be a duty to yield to it. Other cases are ones where the torture is inflicted as an unjust punishment for real or imagined crimes, and in those cases there is no question of withstanding or yielding, just of surviving, and hence no exception to a harmony between flourishing and morality.

On the whole, norms of flourishing and norms of morality do harmonize.

2.4.3. *Flourishing and epistemic rationality.* Epistemic rationality aims at epistemic flourishing, which is an important aspect of our flourishing as persons. Epistemic flourishing is not just a function of how many facts one knows. Wisdom and integrative understanding both contribute more to epistemic flourishing than knowledge of trivia, and hence epistemic rationality aims more at these than at maximizing the number of items known. But these more important epistemic accomplishments are particularly important to human flourishing.

At the same time, the life of virtue is central to our flourishing, and epistemic accomplishments—especially wisdom—are crucial to the life of virtue. To engage in morally excellent action, we need to understand ethical norms, the human beings that we affect by our actions, including especially what is good for them, our own selves, and our relationships with others, and this understanding constitutes much of our wisdom. Furthermore, for morally excellent action to be successful, we often need empirical knowledge of the means by which good may be accomplished—and here we need to include both common-sense knowledge as well as the sciences.

The sciences are not just instrumentally good to us. Knowing the causal structure of the world not only helps us live healthier lives, but is an important part of our flourishing as

human beings, and gives us delight, because the world's knowable explanatory structure is lovely, and appreciating such loveliness is a part of our flourishing.

We can imagine beings whose natures impose on them standards of epistemic rationality that are too high for these beings to be able to know what is needed for flourishing. For instance, these could be Cartesian beings whose standards of rationality require infallibility, and yet which are not capable of infallibly cognizing what is needed for flourishing. But fortunately we are not such beings. We can know that friendship is good for us and that physical suffering is bad for us.

Thus, our norms of epistemic rationality harmonize with and generally support our flourishing. Of course, as often in our Aristotelian optimism, there are exceptions. There are things that it is better not to know, as they may depress us or violate the privacy of others. But most of these things are not central to our lives of epistemic rationality—wisdom and understanding of how the world hangs together are not among the things that it is better for us not to know.

2.4.4. *Morality and epistemic rationality.* The morally good tend to be wise, and the wise tend to be morally good. Moral norms impel us to investigate what is right and good. There are beings none of whose behavioral norms are knowable to them. After all, many organisms—bacteria, almost surely fungi and plants, and likely lower animals—are incapable of knowing *anything*, and yet they all have behavioral norms. Now, moral norms are not just any behavior norms, but norms of *voluntary* behavior, and organisms incapable of knowledge are presumably also incapable of voluntary behavior.

One might think that it is partly constitutive of being a moral agent that one knows some of one's moral obligations, if only in the most general terms—say, that one knows that the good is to be pursued and evil to be avoided, as per Aquinas's first principle of Natural Law.^{??ref} This is not clear. Error theorists about morality claim not to believe that anything is to be pursued or avoided. If they do not believe it, neither do they know it. Yet they are moral agents. We might be suspicious that error theorists really do believe what they claim to believe. If so, then consider that knowledge tends to be defeasible, and being

confident that one has a decisive argument against something tends to provide a defeater for it. Some error theorists are confident that they have a decisive argument against there being moral truths??refs, and it seems plausible that even if humans are unable to fail to believe in morality, in the presence of such confidence humans do not *know* moral truths.

I think we can, further, imagine a whole species of no members of which are capable of knowing any moral rules. One option might be a species where fallible and correct moral belief is widespread, but the epistemic standards of their nature are so high that infallibility would be required for justification. Another option would be a species where the moral norms are conditional on the agent having some beliefs about moral rules, but the members of this species have no epistemic access to moral facts. Nonetheless, due to the evolutionary advantages of morality for social animals, they have evolved to have largely correct moral beliefs—though unjustified ones, because they did not evolve to have these beliefs *because* of their truth.⁵

Thus, the moral life need not *require* knowledge of moral norms. Mere belief may be sufficient. Thus, the knowability of moral norms by us is a harmony that calls out for explanation as it is not guaranteed by the nature of morality.

Furthermore, our beliefs about specific moral norms on occasion fall short of the truth. How much is lacking varies from person to person. It is very unlikely that there is some conceptual limit to how much ignorance of specific moral norms is possible. Thus, it is possible to have beings whose epistemic norms are so stringent that they rule out true belief about the moral norms that are in play for daily life. If belief in moral norms is needed to live the moral life, such beings would have a direct conflict between epistemic and moral normativity: they could only live the moral life by failing in epistemic rationality. An example of such beings would be Bayesian agents whose priors for specific moral norms are so low that rational posteriors would always remain lower than is needed for belief.

⁵This is basically a species for which an evolutionary debunking argument would work. Such a species seems possible, no matter what one thinks of evolutionary debunking arguments in the case of humans.??refs

Besides knowledge of specifically moral truths, morally excellent action requires knowledge or at least true belief about other people, both in general and in particular. Meaningful relationships often require deep understanding of the other person, and hence require standards of epistemic rationality that permit such. If, for instance, our epistemic standards were such that rationality requires us to suspend judgment about whether there are other minds, our moral lives would make little sense.

Furthermore, it is infamously true that lack of understanding of other persons—whether as individuals or participants in a culture other than one's own—has a tendency to lead to morally deficient treatment of these persons. Conversely, a gain in understanding tends to lead to better treatment. At the same time, however, there are exceptions. Sometimes if we knew a person better, we would know more about their faults, and this would lead us to treat them poorly. But typically even this exception does not produce a major conflict between moral and epistemic rationality for two reasons. First, except in the case of rare and exceptional secret misdeeds—such as someone's being secretly a child abuser—honest self-examination leads us to find parallels in our own lives to the faults of others. This should lead to a more forgiving attitude towards others, as well as motivation to improve the faults in ourselves. Second, an understanding of the faults of others helps them become better.

Finally, a number of philosophers think there is an irreducible phenomenon of I-Thou second-person knowledge.^{??refs} And there is good reason to think that second-person knowledge requires an empathy that can only be developed by a life of virtue, and tends to both come from and deepen a morally good relationship. One might, of course, imagine cases where second-person knowledge enables one to hurt the other in an especially painful way. However, such deep betrayals are relatively rare.

We could imagine beings whose moral norms prohibit acting in epistemically excellent ways. One could imagine beings that are morally required to engage in practices productive of self-deceit in order to preserve morally important beliefs, such as the belief in other

minds. Or ones that are morally forbidden to engage in scientific investigation, in order to better preserve a sense of mystery about the natural world.

Humans are fortunate. Our epistemic standards are such as to allow us to know the moral norms needed for daily life, and to understand other people in ways sufficient for the moral life. And not only are we not morally prohibited from epistemically excellent behavior, but our morality calls on us to engage in such.

There is, however, one significant and very specific area of conflict between moral and epistemic rationality. Our knowledge of human biology could probably be greatly advanced by means of immoral experiments on human subjects. This observation decreases the harmony between moral and epistemic rationality. But perhaps not by all that much. First, a practice of immoral experiments on human subjects is likely to result in a society with a deadened conscience, and hence with a significant loss of *wisdom*. And wisdom is more important to human epistemic flourishing than biological knowledge. Second, the significance of the conflict may be rather specific to our present point in scientific development. Prior to modern medicine, the epistemic benefits of immoral human experimentation would likely have been small. And in the future, we will likely be able to find morally unexceptionable scientific methods that provide nearly as much epistemic benefit as immoral human experimentation, such as computer simulations and experiments on non-human animals genetically modified to be unconscious.

2.5. Harmony within each normative aspect.

2.5.1. *Morality*. Some thinkers believe in a unity of the virtues thesis on which it is impossible to fully possess any virtue without possessing them all. One approach to arguing for such a thesis is to hold that there is an architectonic virtue that is needed for any virtue and that entails all the others. Socrates apparently believed that wisdom was such a virtue. On such a view, for instance, courage requires wisdom, while wisdom precludes the vices opposed to courage, such as cowardice or foolhardiness. Similarly, St. Paul held that having *agapē* implied the fulfillment of all of morality, and might well have agreed that without love there is no virtue. ??modern-refs

But even without a dogmatic affirmation of the unity of the virtues, it is plausible that vices tend to distort even virtues that they do not oppose, and the virtues tend to be mutually supporting. In fiction, one can meet a villain of direct speech who can be relied on to keep promises and avoid lies, while yet being selfish and violent, such as Cett in ??ref:Sanderson. But in real life such an individual is rare and in fact I personally do not find Cett a plausible character. To some degree the reason for the rarity is practical: honesty may make an evildoer get caught and defeated. But there is a deeper reason that applies in all cases: the most powerful motive for honesty is respect for fellow persons, and such respect tends to prevent vice in general. Admittedly, it is possible to respect others under some aspect while disrespecting them in other respects, say honestly respecting others as fellow truth-seekers while violently disrespecting them as living beings, but such a disconnect is unlikely. After all, a major part of why others are worthy of respect as living beings is that they are truth-seekers.

There are some pairings of virtues where we are apt to see a conflict. For instance, honesty in giving feedback appears to often conflict with kindness. But even in these pairings, it is plausible that the conflict is largely if not entirely due to a shallow exhibition of one or the other virtue. But it is not necessary to insist on the point. All I need for my argument is that there is a *tendency* to mutual reinforcement between the virtues, a tendency that calls for explanation. Without a principled reason to expect harmony, we would expect a random pair of virtues to be like gravity and electromagnetism: sometimes they pull together but just as often they pull or push in opposite directions.

A second type of harmony within the normative sphere is the paucity of real dilemmas: cases where one is simultaneously required and prohibited from the same thing. There does not appear to be a principled reason why there could not be cases where in light of one fact, one would be defective in will in ϕ ing, but in light of another fact, one would be defecting in will in failing to ϕ , so whatever one does, one acts wrongly. It is easy to imagine an artifact with a complex teleology that guarantees that it will always be defective in some respect. One might say that a spork should spear food well while at the

same time holding liquids well, and this incompatible teleological arrangement explains the unpopularity of this implement. And a randomly generated set of rules is one where we would expect frequent conflict.

But while there is no principled reason to avoid conflict between moral rules, real dilemmas are, indeed, sufficiently rare that philosophers have seriously discussed whether they even exist.^{??refs}

On one view, there is a limited set of well-delineated families of real dilemmas. One proposed family is where the dilemma is the result of a prior wrongdoing.^{??ref:Aquinas} Thus, a person with children who has gambled on credit may have both a duty to support the children and a duty to pay creditors, with the two duties being genuinely incompatible: whatever the unfortunate agent does, the agent acts defectively. Another proposed family arises from mistaken conscience: if ϕ ing is forbidden, but conscience requires you to ϕ , then whether you ϕ or not, you do wrong.^{??ref:Murphy} That all real dilemmas fit into one of a few well-delineated families is certainly not something we would expect in a random collection of rules, and hence is still a harmony that calls for explanation.

On broader view, there may be real dilemmas that do not fit into such neat categories. But even these are likely to be rare, and more rare than we would expect on a randomness hypothesis. Thus, we still have a harmony that calls for explanation.

2.5.2. *Flourishing*. While we may not have many principled conflicts between types of flourishing, limited resources induce many practical conflicts. Choices need to be made. Even privileged and affluent people must choose between leisure and money, between developing one set of skills or another, between avoiding danger and enjoying benefits of dangerous activity, and so on. But most people are not privileged, and the conflicts between types of flourishing they face are much more painful, such as between medical treatment and food security, or shelter and clothing. On its face, then, the various aspects of our flourishing are disharmonious.

But a significant amount of harmony returns when we realize that making hard choices well is itself one of the most important aspects of our flourishing, and so the practical

conflict between types of first-order flourishing itself gives rise to a higher harmony. More generally, there is less in the way of conflict between types of flourishing the more we look at the highest types of flourishing, such as virtue, wisdom, and participation in beauty.

While virtue and wisdom have already been discussed in the context of harmony, participation in beauty is worth focusing on all too briefly. We find aesthetic practices in all human cultures, even those whose material life is harshest. Very few if any human communities are so impoverished that their members cannot tell stories to each other because their moral duties leave them no time. Very few if any are incapable of some degree of decoration on their tools of daily use. But above all, virtue and wisdom themselves can have a beauty that exceeds the beauty produced by professional artists. Most people can make their own life into a work of art, and in doing so will possess moral and epistemic flourishing.

2.5.3. *Epistemic norms.* The unified character of our epistemic life is evident. The very goal of the epistemic life appears to be an explanatory unification of reality. Moreover, insights from one area are often helpful analogues for other areas. While there are differences in substantive reasoning methods, the general procedures of logical reasoning and weighing of evidence appear to be subject-independent.

Nonetheless, a fundamental tension in our epistemic life was noticed by William James, namely that between pursuit of truth and avoidance of falsehood. If one believes every proposition, one has all the truths, but at the expense of having all the falsehoods as well. If one believes no propositions, one has avoided all the falsehoods, at the expense of having no truth.

But we probably should not consider this an actual disharmony in our epistemic norms. If we understand the epistemic life in terms of belief, as in this Jamesian example, we can simply understand the norm of belief as a requirement to believe all and only the relevant propositions you have evidence for. This admittedly can be seen as a conjunction of two norms: (a) believe all the relevant propositions you have evidence for, and (b) don't believe any propositions you lack evidence for, where each norm is easily fulfilled

on its own, while it is the conjunction of the norms that is challenging. However, this is no more disharmonious than it would be to formulate the task of a monochromatic pixel artist by saying: blacken each pixel if and only if blackening it is a part of your artistic vision. One can artificially split this into two conjoined injunctions, (a) to blacken all pixels whose blackening is a part of the artistic vision, and (b) to leave unblackened all those whose blackening is not a part of the vision, each of which is easy to fulfill on its own, but there is no real disharmony here. Moreover, the split norms when fulfilled the “easy way”—by blackening all or none of the image area, respectively—have little if any value. The value is in the combination. Similarly, a person who believes all propositions gets no wisdom, understanding or practical benefit out of it. For instance, if you believe that an ibuprofen will relieve a headache and that it won’t relieve a headache, you get no practical guidance. And it is not clear that the paralysis that comes from believing nothing is any advantage to getting lots of stuff wrong.

Another potential tension is found in paradoxes, where our intuitions pull us to accept premises that end up jointly incompatible. However, the history of the philosophical employment of paradoxes^{??refs:include-Kant} gives us good reason to see paradoxes in a more optimistic light, as tools that take us beyond a surface-level understanding of the phenomena.

2.5.4. *Disharmony.* The occasional conflicts between and within normative domains do raise a difficulty akin to the problem of evil for theists. But at this point in our dialectics, the problem is not as severe. For while the theist needs to commit to the idea that every particular instance of evil has a theodicy, since every particular instance of evil must be compatible with the existence of a perfect God if theism is true, the Aristotelian optimist as such is only committed to a harmonious tendency in our lives. Such a tendency is compatible with occasional highly significant exceptions.

Of course, if one opts for a theistic explanation of human nature, as I will eventually argue one should, the exceptions will indeed cry out for a theodicy. The theodical task is a difficult one, as is well known. But it is a task that is beyond the scope of this book.^{??refs}

2.5.5. *A Darwinian evolutionary explanation?* If a species' moral, prudential and epistemic norms were sufficiently disharmonious, members of the species whose behavior is driven to accord with these norms would tend not to survive and reproduce. It is tempting, thus, to think that there is an evolutionary explanation of the harmony among our norms.

One problem with such an evolutionary explanation is that it pushes back the problem to that of explaining the accord between behavior and norms. Since we have argued that the norms themselves do not reduce to physical features, but depend on a metaphysical form, and evolutionary explanations involve physical causal stories, the explanation of the accord between behavior and norms is not subject to a naturalistic evolutionary explanation.

A second, and perhaps even more serious, problem is that Darwinian evolution explains phenotypic features that physically arise from genotypic features. An evolutionary explanation of the harmony of the norms would require the norms to supervene on physical phenotypic features. But we have reason to deny any such supervenience, because any such supervenience would require a function from physical phenotype to norms with a vast number of unexplained free parameters, such as all the parameters we discussed in Chapters ??backrefs. We solved the problem of free parameters by locating them in the form or nature of the animal, with it being a contingent matter of fact which forms are actualized, but on a supervenientist view, the function from phenotype to norm, and hence the parameters defining it, would have to be necessary. And it is implausible to think that such a vast number of independent and presumably messy parameters is metaphysically necessary.

One might, however, attempt a Darwinian explanation that is not naturalistic in the contemporary sense. Suppose that the form actually causally affects behavior in a way that contemporary naturalism rejects, and suppose that forms affect behavior in favor of the norms imposed on the form. In that case, if the norms are sufficiently disharmonious, we can expect the behavior to be disharmonious as well, and hence for the species to die

out. If, further, there is some non-naturalistic connection between the genotype and form, then random variation of genotypes may give rise to random variation of form, and hence some sort of evolutionary process. However, while this may help explain why successful species have a significant amount of harmony between their norms, it does so by pushing back to the question of why they have a harmony between norms and behavior.

2.5.6. *Non-realism?* At this point I have to confess that in thinking about normative harmony, I find myself tempted to feel a certain suspicion. The harmony feels a little suspicious. We wouldn't accept a theory on which our norms are typically in conflict—such a theory would be “too inconvenient”. But there is something that feels “too convenient” about the harmonies noted here. It is tempting to say that we have the harmonies between the norms because we have created norms that are convenient for us.

There are several ways of expanding on this. On an epistemic non-realism, there are objective, mind-independent norms, but our access to them is poor or non-existent, and out of wishful thinking or due to evolutionary pressures we have made up stories about a coherent realm of norms, whereas the reality might well be quite different. On a metaphysical non-realism we can keep the same debunking story about how we made up the stories about the norms, but deny that there are any objective, mind-independent norms, in favor of either an error theory that holds there are no norms or a subjectivism that says there are norms but only ones created by our social practices.

As said in the Introduction, a robust realism is assumed by this book. It is beyond the scope of the book to argue for this realism. At this point, however, it is worth noting that the arguments of the book do put significant pressure on the realism. Perhaps some readers, after realizing what controversial metaphysics is the best explanation of a realist view will conclude that the realism is false. That is a different discussion.

At the same time, to the realist who is unwilling to abandon realism, the suspiciousness of the harmony should be evidence that there is something to be explained here.

2.6. Fit to DNA. Imagine a world that is physically very much like ours, but the feathered flyers are defective pigs, while the oinking mud-wallowers are defective birds. The

defects go deep: it's not that the DNA fails to be expressed correctly, but the DNA is itself defective, as happens all too often but to a much less surprising degree in our world.

Is such a misfit between DNA and norms possible? Suppose that all we have available is Aristotelian metaphysics. Why couldn't a feathered flyer have a porcine form? The main tool available in traditional Aristotelian metaphysics is the principle that the matter of a thing must be properly disposed to its form, but we have seen in Section ?? of Chapter ?? that a proper disposition thesis sufficiently strong to distinguish the matter of a bird from that of a pig is unlikely to be true.

Furthermore, we have to explain how a fairly continuous sequence of genotypic changes over evolutionary history correlates with discrete changes between forms. When birds evolved from dinosaurs, at some point the ability to fly became normative, with the occasional individuals incapable of flight being defective birds, whereas their flightless ancestors were normal (at least in this regard) dinosaurs.

After Darwin, Franklin, Watson and Crick, the Aristotelian requires an explanation of how form and DNA track one another, an explanation that was not available to Aristotle.

2.7. Exoethics. Exoethics is a hypothetical discipline studying both how we should treat intelligent agents and how they should behave.??refs In this section I will focus on the latter question.

We would expect that ethics for non-human rational beings—"aliens"—would be similar in norms where the biopsychosocial traits are similar, with by and large intelligible differences in norms where they are different. We would expect reproductive ethics to be different from ours for asexually reproducing beings, and the requirements of altruism to be different from ours for intelligent bees or intelligent sharks. And we would expect these differences to be intelligible: we would expect the intelligent bees to have norms requiring more altruism than is required of humans and intelligent sharks to have less in the way of altruistic norms, and we would be surprised if it turned out the opposite way.

One explanation of this would be some grand function from biopsychosocial traits to kind-specific ethical norms, effectively meaning that there is a kind-neutral set of conditional ethical norms whose antecedents include facts about the traits of the agent's kind. We already saw in earlier chapters how complex the details of ethical norms for humans would be, and *a fortiori* a grand set of plausible conditional norms for all rational beings would have to be much more complex, likely with a vast number of seemingly arbitrary parameters that require grounding.

The Aristotelian way to account for such overarching norms would be to suppose an excessively complex form of rational animal that includes conditional norms that specify how you should behave if you're an intelligent bee, an intelligent primate, an intelligent shark, or a plasma-based intelligence in a star. The complexity here appears excessive. It is implausible to think that I contain information on what I should do if I were an intelligent bee. Moreover, assuming that the seemingly arbitrary parameters do indeed give us good reason to think that the norms are contingent—that another form of rational animal is *possible*, one with different conditionals (perhaps requiring slightly more altruism from the intelligent bees and slightly less from the intelligent sharks)—then there seems little advantage to supposing that we should expect all the rational animals in our world should have the same form, unless it so happens that humans are the only rational animals, in which case the excess of hypothetical norms inapplicable to humans but written into our form seems to violate Ockham's Razor.

It seems better, then, to suppose that if there were aliens, they would have their own form, with their own norms. But now we come back to the puzzle. We *would* expect their norms to be similar to and vary from ours in ways that are explicable by their biopsychosocial traits. But why? Perhaps we just *have to* be optimistic about our own forms and their harmony in order to avoid skepticism, and perhaps we are required by our own form to avoid skepticism. But why should we be optimistic about hypothetical aliens, especially if we accept the logical possibility of lack of harmony between different norms as well as between norms, genetic code and environment?

We could, of course, deny that our optimism should extend to hypothetical aliens. But, first, aliens may not be hypothetical at all. Given the significant number of already known exoplanets that appear habitable to beings like us^{??refs}, which are but a small sampling in one galaxy among many, it seems not at all unlikely that there really are aliens. But whether aliens are hypothetical or real, if our optimism would not extend to them, we should be suspicious of the optimism as applied to us. We probably should not think that optimism should extend to all possible hypothetical aliens. Disharmonious aliens seem *possible*. However, we should think it is likely that if there are aliens, they tend to have harmonies similar to ours. Otherwise, we not only have a puzzle as to why we are so special and lucky, but we have a significant reason to doubt our luck, in a way that provides a defeater for the optimism that Aristotelianism requires.

2.8. Modern technology and outlandish scenarios. In ^{??backref}, I argued that an ethics based on human form can simply ignore outlandish scenarios that are far outside of our ecological niche, such as ones involving infinite numbers of beneficiaries. However, there is a danger in this line of reasoning. As Arthur C. Clarke famously said, “Any sufficiently advanced technology is indistinguishable from magic.”^{??ref} To human beings 50,000 years ago (or even just 500 years ago!) much of our technology would indeed be magical, and decisions that we routinely need to make, say in bioethics, would be predicated on outlandish assumptions.

We might thus expect an ethics and epistemology grounded in a form possessed by hunter-gatherer primates to be silent on dilemmas of a highly technological society, leaving us to do whatever we wish, or, even worse, to fail to harmonize with the shape of our lives, like that of a fish on land. Probably the most problematic technologies will be ones that change the shape of our sociality, since our sociality seems closely bound up with moral norms. This will include socially transformative technologies such as the city, writing, and the general availability of fast global communication.

Yet while there are, as there have always been, difficult and controversial moral and epistemological cases, we do not in fact find ourselves adrift without guidance in the modern world. Virtue continues to contribute to our flourishing, and ancient texts, whether religious or philosophical, continue to point to good ways of living. We are not fish out of water in the city or on the Internet, though there are challenges to be met, and sometimes a transformative technology needs to be limited in its use—for instance, some of us have become more cautious about replacement of in-person with electronic interaction during the pandemic.

This gives us reason to think that if our moral norms are grounded in human nature, human nature was somehow picked out with foresight for what kinds of challenges humans would face in the distant future. Our ethics does may not work in outlandish situations, such as those involving infinities as noted in ??backref, but it works in a range of moral environments much broader than that found in early homo sapiens society. This is surprising. Contemporary scenarios involving organ transplants, space travel, fast worldwide communication, bombs that can wipe out much of humankind, well-coordinated political systems with hundreds of millions of people, and machines that mimic the ability to understand complex texts would likely have seemed to Aristotle to be pure hypotheticals, and there would have been little reason to suppose that our ethical norms would have significant application to such outlandish scenarios. Yet they do. And think of times early in the prehistory of humankind. The complex technologies and social systems of Aristotle's time would have seemed outlandish. Why would an ethics based on our human nature apply to such scenarios? And why is it right—as I take it that it is—that we see the ethics grounded in our human nature as something that will continue to have fairly harmonious application as our technology develops in ways we cannot foresee.

In ??backref I suggested that our ethics and epistemology simply does not apply to scenarios that are outlandish in the sense of being far out of this world. Our everyday life would have appeared far out of this world to our ancestors, but our ancestors would have been mistaken about its being out of this world, as this kind of a life was the genuine

destiny of our kind. And this is likely to continue. Many think there is a serious possibility that human beings will spread through the galaxy, affecting vast numbers of lives, which may make actual seemingly outlandish questions about where our actions have very slight probabilistic effects on vast outcomes.??ref:Fanaticism,glitchy ethics??backref and discuss:ch4 on infinity

One approach to these modern questions is a principle-conservatism: the questions are settled by moral principles that we accepted for millenia. Because the default for an action is moral permissibility, in the case of qualitatively new kinds of actions, principle-conservatism is apt to lead to a radical expansion of moral possibilities. If we had no principles governing human DNA manipulation in the past, even if this was simply because we had no concept of DNA, now there are no limits, except limits coming from traditional principles of harm and consent. Principle-conservatism in these kinds of cases would (paradoxically, given the word “conservatism”) justify a vast change in the shape of human lives of a sort that arguably is not compatible with human flourishing. Conversely, however, we have cases like the invention of effective therapeutic surgery. Prior to these, any serious degree of cutting open of the human being would be an instance of grave harm, and generalizing prohibition from those cases to therapeutic surgery would have been unfortunate for the human race.

A more naturalistically inclined Aristotelian could despair about ethics in quintessentially modern situations. Absent something non-natural behind our existence, like God or an axiarchic principle, we should not expect our natures to provide guidance in these situations, or at least “non-glitchy” guidance??backref. This line of thought could lead the Aristotelian to a dark view on which there just is no answer to a number of contemporary moral questions, or on which the answer conflicts in a glitchy way with our moral intuitions, or perhaps even one on which the true moral norms conflict, and we get hard-to-avoid moral dilemmas.

2.9. Avoiding radical scepticism. There is a number of sceptical hypotheses that cannot be ruled out either on logical grounds or *a posteriori*. These include hypotheses that

the world around us is a computer simulation, that our moral intuitions are disconnected from moral reality, that we are Boltzmann brains, i.e., short-lived brains in bubbles of oxygen arising from fluctuations in the vacuum of space, that we live in an infinite multiverse that undercuts all probabilistic reasoning^{??ref}, that simpler scientific theories are less often right other things being equal, and so on. Yet we think these hypotheses false. If we think them false neither on logical nor empirical grounds, it must be because we assign low probabilities to them prior to empirical assessment. Moreover, this assignment is required by our rationality: those who fail to assign low probabilities to them are irrational.

Our Aristotelian account can ground the correctness of this judgment of irrationality in human nature.^{??cf.backref} But there would be something deeply problematic about us if the low *epistemic* probability of the sceptical hypotheses were not matched by a low *objective chance* for them to be true in light of the causal and stochastic structure of the world. If in fact the most likely way for a being with our rational nature and mental life to arise would be as a Boltzmann brain, then even if we have lucked out and are not a Boltzmann brain, there is a disharmony between the world and our mental life.

In such a lucky case, the connection between our nature-required priors and the world then appears too fragile, chancy and “unsafe”^{??ref} for the beliefs essentially dependent on these priors to count as knowledge. Even a reliabilist should say that if some beings were required by their nature to assign a very high prior probability to the hypothesis that the universe formed an even number of years before life first arose, and it was mere chance that this hypothesis was true with the causal structure of reality not assigning it a higher probability than the hypothesis of an odd number of years between the beginning of the universe and the beginning of life, then that hypothesis is not knowledge. Yet it is very plausible that we *know* the sceptical hypotheses under discussion to be false. (This judgment has admittedly been disputed by a number of epistemologists who admit with G. E. Moore that I know that I have two hands, but will not allow the Moorean inference that I know that I am not a brain in a vat, despite the fact that I have two hands obviously entailing that I am not a brain in a vat. ^{??ref}) And even without considerations of knowledge,

we might note that an optimism resting on an assumption of mere luck appears paradigmatically irrational. Furthermore, even if we do not know the sceptical hypotheses to be false, we believe them to be false, and hence should expect an explanation of their falsity.

The problem is perhaps most pressing in the case of the kinds of highly abstract *a priori* intuitions discussed in ch4:??backref, such as the intuition that the axioms of arithmetic are consistent or that nothing can cause itself. If there is merely a coincidence between the truth of the intuitions and our possession of a nature that normatively requires us to have these intuitions and perhaps causally impels us to them, then even though we may be justified in following the intuitions, they are unlikely to count as knowledge. Aristotelians thus need a theory on which there is the right kind of connection between our rational nature and the metaphysical, causal and stochastic structure of the world.

3. Bayesian priors

Suppose we adopt a version of the Aristotelian account of Bayesian priors, on which we should adopt priors within specific ranges, priors that enable us to engage in scientific reasoning about the world.??backref In particular, we should adopt priors that are not negligibly low for simpler hypotheses about how the universe functions.

We might at this point raise a skeptical worry: What reason do we have to think these priors lead us to truth? But this skeptical worry can be answered. Let's take a toy example. Suppose we ought to assign the non-negligible ur-prior 0.000001 for the Newtonian hypothesis H that there is a constant G such that the force of gravity is always Gm_1m_2/r^2 in the direction from the first mass to the second. We now ask: What reason do we have to think that assigning 0.000001 is apt to help us in the search for truth? Here is the answer: The expected epistemic utility of assigning 0.000001 to H is higher than that of assigning any other value.

One might respond: How do I know that this is the optimal epistemic utility? But now the answer is both mathematical and trivial. Recall that the epistemic utility for H is measured by a strictly proper scoring rule (T, F) , where $T(q)$ is the utility of having

credence q in H when H is true and $F(q)$ is the utility of having credence q in H when H is false. What is the expected epistemic utility? Well, reasoning properly (i.e., according to my nature), I will have credence 0.000001 in H and 0.999999 in the negation of H . Thus, the expected epistemic utility is:

$$0.000001 \cdot T(q) + 0.999999 \cdot F(q).$$

So what credence q should I have to maximize this? The answer follows from the very definition of a strictly proper scoring rule: I should have credence $q = 0.000001$, since that is what maximizes the above formula.⁶

This feels like a cheat. When deciding what the best credence for H is, I calculated the expected epistemic utility using 0.000001 as the probability that H is true, and concluded, unsurprisingly, that the answer is 0.000001. But that is right and proper. For on our objective Bayesianism, it is *right* to have credence 0.000001 for H , and hence 0.000001 is indeed the right weight for $T(q)$ and 0.999999 is the right weight for $F(q)$ in calculating expected epistemic utilities. Recall that the question was what *reason* we have to think that

⁶*One might object that since we are looking at the adoption of ur-priors, perhaps what we should do is consider our expectation for *future* epistemic utility after additional data comes in. A toy model for this is as follows. Suppose that my current prior for H is $P(H)$, and I am considering the value of having prior $Q(H)$. I expect that in the future I will gather additional information. This can be modeled as a partition $I_1, \dots, I_n$ of our probability space into n events, where I will find out in the future which event I_i of the partition is actual. I will then update my credence for H from $P(H)$ to $P(H \mid I_i)$, and what I am interested in is the epistemic score *after* that update. So now I ask: By my current lights, which are modeled by a prior probability function P , might it not be that some other prior probability function Q will have a better expectation for what my future score will be after updating on the observed event of the partition? If my scoring rule is proper, the answer is negative. The reason is that the Greaves and Wallace theorem (**GreavesWallace**) tells us that given a proper scoring rule, an optimal strategy to adopt by one's current, P -based, lights for the future is to update one's credence by conditionalization on the observed event in the partition. In particular, this strategy must be no worse than the alternate strategy where instead of updating P to $P(\cdot \mid I_i)$, I update P to $Q(\cdot \mid I_i)$. But the expected score of the alternate strategy is precisely the expected score I will have for ordinary Bayesian update if I just switch my credences now from P to Q .

assigning 0.000001 to H is truth-apt. Spelled out more fully, the answer is this: 0.000001 is the best credence according to an expected epistemic utility calculation done on the basis of the credence that I *should* adopt for H . This sounds circular, but is not. The circle is broken by the “should”, which has a non-circular ground in human nature on our Aristotelian meta-epistemology. My human nature is the reason why I should do the expected epistemic utility calculation using 0.000001 and 0.999999 as my weights when evaluating truth-aptness, and it just turns out that this yields the credence 0.000001 as the most truth-apt.

But there is good reason why this feels like a suspicious bit of reasoning. For there is a nearby question, not of justification but of explanation. Granted, we should adopt 0.000001, and when we rightly do that, we conclude that it's truth-apt to do so. But *why* is it in fact the case that the intelligent beings on Earth have a nature which enjoins on them an ur-prior for H that tends to lead to truth? What explains the harmony between the credences their nature requires and the actual facts of the universe around them? If we are rational and adopt the required ur-priors, the above trivial argument will tell us that we ought to believe in the harmony between the credential requirements of our nature and the universe. But that we ought to believe in this harmony does not explain why this harmony obtains. Why is it that the intelligent beings on Earth have a nature which requires them to have the non-negligible ur-prior 0.000001 for H rather than, say, the (to us crazy) 10^{-100} or $1 - 10^{-100}$?

4. Complexity and explanation

4.1. A problem. The scientific revolution came about from a conviction that we should try to find expressions for the laws of physics in simple mathematical laws with a minimum of free parameters, at times even at the expense of fit to empirical data. A heliocentric solar system with circular orbits and without epicycles fit the observational data less well than epicyclic refinements of Ptolemaic geocentric models, but was the right way for science to move forward.

This drive towards elegant simplicity is in obvious tension with the vastly complicated landscape of seemingly arbitrary parameters in ethics, epistemology, semantics and metaphysics that I have used as ammunition for Mersenne questions leading to our robust Aristotelianism. But do we not normally prefer simpler theories to more complex ones, and hence should we not reject the normative complexity in favor of a simpler theory like utilitarianism in the name of Ockham's razor?

Ockham's razor, however, has always been a defeasible criterion: entities are not to be multiplied *beyond necessity*. But sometimes there is necessity. It would be simple to suppose that all trees of a single species look exactly the same. But that just wouldn't fit with our evidence. In biology, one does not expect individuals of the same sort to be exactly alike, unlike in fundamental particle physics.

Specifying what a rational animal of a particular species ought to be like and how it should behave can be expected to involve a lot of information. How much information? Well, we might take the information contained the DNA common to all humans to give us a lower order of magnitude bound, since the common DNA presumably encodes something about what human bodies are supposed to be like. There are 3.2 billion base pairs in human DNA, and 99.1% of human DNA is said to be common to all humans (??ref <https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/dna-profiling>). Since each base pair is two bits of information, that means about 6.3 billion bits, or the equivalent of about 500,000 book pages.⁷

Imagine the task of designing the rules of behavior for a rational animal that has a significant complexity in its bodily life, subject to the constraint that the rules lead to a life that elegantly balances moral and epistemic norms, and fits well with the bodily nature of the animal and its niche in the ecosystem. It is plausible to think this will be several orders of magnitude more complex a task than that of designing the rules for a well-balanced and significantly embodied game such as tennis. Generating a game of pleasing elegance and

⁷Counting a page at 1800 characters and each character at seven bits: $6.3 \times 10^9 / (1800 \cdot 7) = 500000$ (oddly exactly, by coincidence!).

yet compelling complexity, especially an embodied one, takes a fair amount of information, and the official rules for tennis are about forty pages.^{??ref}

We can think of simplicity as an aesthetic criterion in theory choice. But simplicity is not the only factor contributing to beauty. (If it were, the most beautiful art would be no art: you can't get simpler than an empty installation that can be completely described by $\sim \exists x(x = x)$.) Overall theoretical simplicity is one way of having an elegantly unified theory. But one can also achieve elegant unification in other ways. Consider, for instance, hierarchical organization. Wittgenstein's *Tractatus*^{??ref} achieves a unification by being summed up in seven top-level sentences, with a progressive hierarchical amplification and justification in terms of multiple levels of sentences. Or consider the unification achieved in biology by Linnaean and Darwinian taxonomies.

We could have an ethics that is just simple: it has a briefly expressible rule that covers everything in full detail. But just as it is unlikely that we would get a compelling racquet sport with a single brief rule, even if we allowed for some vagueness, it is unlikely that we would get a harmonious set of norms for the life of a rational animal out of such a rule—that, indeed, is an upshot of the enumeration of the many Mersennian issues of detail in normative phenomena that have been discussed in this book.

Can we have some other kind of theoretical unification? I think we can. As discussed in connection with particularism(^{??did I??backref}), we can suppose suppose a hierarchical structure. We can have a hierarchical ethics, at the top with one or more principles like Aquinas's "Pursue the good and avoid the bad", the Kantian injunction to treat others as ends rather than mere means, or the Biblical "Love your neighbor as yourself". But unlike (perhaps) the historical Kant, we need not take the top level principle or principles to have all of the normative informational content for morality. Instead, we can think of it as a unifying headline, akin to explaining tennis by saying: "Hit the ball back into the other player's side." There will be further rules that are not mere logical derivations, but specify the general principle of the higher level rules by means of finer-grained rules. The finer-grained rules themselves are not unlikely to need further adumbration, and so on.

Consider Hillel's famous response to the request that he explain the Jewish law while standing on one leg:

That which is hateful to you, do not do to your fellow. That is the whole Torah, all the rest is commentary. Now, go and learn it [the commentary].
 ??ref:add scholarly translation

Now it is clear that in fact that the primary Jewish commentaries (i.e., the Mishnah, and the Talmuds which are commentaries on the Mishnah??check,refs) on the five books of the Torah contain normative material not found in the books they comment on. Nor should this be a controversial claim, since rabbinical tradition holds that the rabbis have an oral tradition going over and beyond the books of the Torah. Thus we should probably interpret Hillel as saying that the Golden Rule (in his negative formulation) is a kind of summary, rather than as saying that the rest is logically derivable. Similarly, Aquinas, after giving his "first precept"??ref that good is to be done and evil avoided, lists second level laws such as preserving human life, respecting the reproductive life of us a rational animals, knowing the truth (especially about God), and living in society. It appears that the second precepts contain normative information not found in the first precept, and it is clear from the arguments in Chapter??backref that his second precepts still have not reached the level of normative information needed to resolve all moral questions.

In both the Hillel and Aquinas cases, we have a unification of ethics under one or more general principles that are insufficient for deriving all the specifics. This is akin to the explanatory unification that modern biology receives from evolutionary theory. Besides generalities like that species tend to mutate towards inclusively fitter forms, the basic principles of evolutionary theory—random variation and the survival of the fittest—do not generate specific predictions. However, they do *organize* the vast sphere of modern biological knowledge.

Famously, Aristotle has observed that in ethics, unlike in geometry, one can only speak in ways that are true for the most part.⁸ On the account I am defending, this is not quite right. Instead, many of the higher level ethical claims are what one might call “generalities” that organize ethical reasoning. These claims can be exceptionlessly true, but there are limits to how helpful they are in particular cases. Thus, it may always be true that we should respect human life, but this does not give a clear answer as to what the health care provider should do when the family of a particular patient requests disconnection from life support. The respect claim describes, in general terms, the shape that the finer-grained principles have. And it may well be that the finest-grained principles which apply to certain particular cases have a complexity beyond our practical ability to specify, and so we do not know principles that definitively settle a case.

While I have used ethics in the above discussion, the same plausibly applies to epistemic rationality, where we have a very general principle like “Pursue understanding (or knowledge or truth)”, with finer-grained specifications such as “Follow the evidence”, “Avoiding error is more important than getting at truth”, “Prefer elegant theories” and “Direct your attention to more important matters.” In the case of semantics, we may, on the other hand, have a high level principle that “Meaning follows usage”, and then a variety of finer-grained principles about how usage yields meaning. As we get to very fine-grained principles, we have an extremely complex account, but hierarchically organized.

We might also want to say that there is a highest level of the hierarchy, where human behavior is divided into *types* of norms, such as norms of will (ethics and prudence), intellect (epistemology), communication (semantics), and biological function (which we have not focused much on in this book).

⁸ref. Of course, this claim itself needs to be carefully understood. Aristotle himself says that murder and adultery are always wrong. Perhaps he is thinking that murder and adultery are definitionally wrong—murder being a wrongful killing and adultery being sex contrary to respect for marriage? Or perhaps Aristotle’s claim that ethics concerns things that are merely true for the most part is itself meant to be merely true for the most part.

Finally, note that while we want a small number of free parameters in the fundamental laws of physics, we are not particularly troubled by a multiplicity of free parameters in the initial conditions. The drive to *extreme* mathematical simplicity in science seems to be limited to the fundamental laws of physics. We are much more moderate in our demands for simplicity in sciences other than physics and even in physics outside of the laws themselves. It may be that, as a matter of logical necessity, valuable creaturely lives like ours require a very significant degree of complexity in their environment and makeup (both normative and physical), and radical simplicity can only be had in limited areas, like physics. If so, then the whole setup might be seen as pointing towards a structure of the universe that favors both lives like ours and simplicity but is willing to sacrifice simplicity for the sake of other goods.

4.2. Aesthetics. In surveying ways that normative disciplines can benefit from an Aristotelian metaphysics, I focused on ethics, epistemology and semantics, which leaves a notable omission: aesthetics. Aesthetics differs from the other normative areas. It is impossible to live anything like a recognizably human life while completely ignoring ethical, epistemological or semantic norms. But a life that completely ignores aesthetic norms is only *somewhat* brutish. Aesthetic enjoyment is an undeniably important part of human life, but one can be a decent and rational social animal, albeit an impoverished one, without it.

What I just wrote may sound like a swinish downplaying of the value of beauty and other aesthetic goods. But it could also be a part of the opposite attitude. One might say that aesthetic goods tend to transcend us, pointing beyond the merely human, perhaps even towards the divine. As such, their value can exceed the merely human, and engagement in them can be supernormal. Compare how one can, by definition, lead a morally decent life without doing anything supererogatory, though such a life would also tend to be morally drab.⁹

⁹Though not always. While many instances of moral heroism are supererogatory—say, jumping in front of a train to push a stranger out of the way—there are instances of moral heroism that are one's duty.

If participation in aesthetic goods is somewhat more optional than adherence to ethical, epistemological and semantic norms, then it is more difficult to argue for the inescapability of treating aesthetic normativity to be objective, and thus it is not surprising if more people take a subjectivist view to aesthetics.

Nonetheless, it is difficult to deny that there are such things as good and bad taste. In high school, I had to write a sonnet. I expect that only things that can be said in favor of mine is that it was grammatically correct, had the right number of syllables per line and rhymed. Anyone who reads it alongside Shakespeare's Sonnet 27 and thinks that what I wrote is in the same ballpark of quality either does not know English or has bad taste. A poetic culture that exalted it above Shakespeare would be rotten. There is at least some objectivity in aesthetics.

Furthermore, scientific epistemology appears to involve aesthetic judgment. Physicists have a preference for beautiful theories. Newton's Law of Gravitation says that the force of gravity is $F = Gm_1m_2/r^2$. But if we took all the messy data relevant to gravitation that was available before the 20th century and tried to find which value of p makes a force law of the form $F = Gm_1m_2/r^p$ give the best fit to the data, we can be confident that the optimal value would not be exactly 2, but 2 plus or minus some small deviation. I do not know how large that deviation from 2 would be, but probably it is significantly greater in magnitude than $1/10^{100}$. Now imagine that at the end of the 19th century, two scientists, Alice and Bob, came along. Alice proposed, on a lark, that the force of gravity is $Gm_1m_2/r^{2+11^{-109}}$ while Bob proposed, likewise on a lark, that the force of gravity is $Gm_1m_2/r^{2-11^{-109}}$. And now imagine that they argued as follows. We have three theories of gravitation: Alice's, Bob's and Newton's. The best fit to the exponent is some value p that is 2 plus or minus a deviation. That deviation is very likely to be significantly bigger than one in a googol, so very likely either Alice's or Bob's theory fits the data better than Newton's as it's closer

A soldier, for instance, has the duty to follow just orders, and sometimes such an order requires heroism. Likewise, we are all required to refrain from murder no matter what the personal cost.

to the value of p that fits the data optimally. Thus, we should not be more confident of Newton's theory than of Alice's or Bob's joke theories, which is absurd.

The obvious thing to say here is that the prior probability of Newton's theory is higher. And I have already argued that the Aristotelian has an excellent account of where prior probabilities come from: they are tied to the norms of human nature.^{??backref} But reflection on the priors in physics shows a pattern: more beautiful theories have higher priors, and scientists go for these theories because of their beauty. If beauty is subjective, then physics at its heart is subjective. And that seems implausible.

The main response to this argument would be that physicists go for simplicity rather than beauty.^{??ref} and thanks:Chan But this does not appear quite right. Some force laws are *conservative*, i.e., the force can be derived as the gradient of a potential. Newton's Law of Gravitation is conservative, as is any law where the force is radial and depends only on distance. Conservative laws have beautiful mathematical properties, but it is false that every conservative law is simpler than every non-conservative law. Yet conservativeness is itself a consideration in favor of a force law. Or consider this. We have discovered that the gravitational force has an inverse square law, and now we are investigating electrostatic attraction and repulsion. The hypothesis that electrostatic force is Cq_1q_2/r (in the direction away from the charges) where q_1 and q_2 are the charges and r is the distance is simpler than the hypothesis that it is an inverse square law Cq_1q_2/r^2 . But given that we have one inverse square law, it is reasonable to suppose the second is an inverse square law as well. Granted, one might say that here we have simplicity in a meta-law that says that all forces are inverse square laws. But even if one doubts there is any such meta-law, it will seem more likely that a new force law would have a similar form to an old force law than any specific alternate form. Why? Because such convergence makes physical nomic reality more beautiful.

Finally, an aesthetic approach can take us beyond physics, while a simplicity approach may have difficulties. In discussion, a biologist colleague once mentioned that the relevant beauty in biology is not simplicity but diversity. Furthermore, even in physics, in addition

to simplicity driving our priors, we have a preference for *productive* candidates for laws of nature: laws that generate complex behavior in a simple way. Since beauty includes both simplicity and diversity, an aesthetic story can explain more features of our priors than just a simplicity story.

Now, *prima facie* there are two ways to understand aesthetic objectivity. First, we might take a “Platonic” view on which some items are just objectively more beautiful, profound, sublime, ugly, etc. Second, we might take an anthropocentric view on which we have human norms of aesthetic engagement: there is something wrong with you as a human being if you fail to enjoy Shakespeare’s work more than my puerile school-required poetic production.

Reflection shows that even the Platonic view needs some relativity to humans. Consider a CD with an admirable performance of Beethoven’s Ninth Symphony. Here we have sublime beauty—or so I understand on the testimony of those who have musical taste. But the beauty is relative to a particular mode of sensory engagement. Microscopic visual investigation of the pits in the aluminum embedded in the CD which encode the audio does not yield the beauty. The beauty of Beethoven’s Ninth is a beauty *for listeners*. And not just for any listeners, but for listeners with a particular kind of auditory apparatus, sensitive to a particular range of frequencies. The beauty is not there for listeners whose auditory frequency range starts near the top of the human range and goes far beyond it. Furthermore, a work of art must be understood in a context. Monet’s water lily paintings would have a different aesthetic valence in a world where all water lilies are spiky plants with

red petals¹⁰, because they would no longer have the kind of resemblance to physical reality that is partly constitutive of their aesthetic value.¹¹ Aesthetic qualities apply to works of art in a rich context that includes at the very least a specification of the relevant sensory faculties—which in our case are human—as well as many other contextual and often social features. This does not undercut the Platonic view, but makes it a little closer to the anthropocentric one.

The Platonic view of aesthetic objectivity raises a harmony question. Suppose Shakespeare's Sonnet 27 is objectively more beautiful than what I wrote in high school, but you enjoy my work more than Shakespeare. Then there is something wrong with you: your faculty of aesthetic enjoyment is malfunctioning. But why do the *norms* of the faculties of aesthetic enjoyment match the Platonic realities? Couldn't we imagine beings that are supposed to enjoy the ugly?¹² That would be disharmonious. Thus, on the Platonic approach we have a harmony problem.

The anthropocentric view of aesthetic objectivity fits very nicely with Aristotelian metaphysics: our nature prescribes norms of aesthetic engagement. But we have two different harmony problems. The first relates to the beauty of laws of physics. Why do features of laws of nature match our human norms of aesthetic judgment, even in the case of laws that were discovered long after modern humans have evolved?

The second problem is that there is a harmony between norms of aesthetic engagement and other norms, especially ethical ones. Part of the harmony is that there is a moral value

¹⁰There are 70 species of water lilies, or Nymphaeaceae, in our world. While the boundary of a biological family is unclear, it is plausible to imagine a world where plant evolution produced spiky flowers with red petals that are a part of the Nymphaeaceae family, and all other water lilies went extinct.

¹¹The point isn't that works of art need to reflect outside reality to have their aesthetic value, but that the way in which they are related to outside reality *can be* and in the Monet case *is* essential to their evaluation. Note also that the relationship of an excellent work of art to outside reality can be that of deliberate dissimilarity, as in surrealism.

¹²An interesting question here is with regard to things that are "so bad that they are good": we can enjoy a trashy movie because it is trashy. Perhaps what we should say here is that proper use of faculties is enjoyable. As we watch the amusingly bad movie, we enjoy engaging in our expert judgment of its badness.

in the production of beauty. Another part is that ethically admirable lives are themselves thereby *beautiful*, appropriately calling forth similar aesthetic responses from us as works of great art (consider the moral features evoked by the descriptor “a beautiful life”). Even Nietzsche’s aesthetic approach to life recognizes this: *the saint* is one of the modes of greatness he lists in ??ref. Indeed, every kind of excellence seems to call forth a positive aesthetic response.

5. Some Aristotelian tools for explaining harmonies

5.1. Three tools. We have seen quite a number of actual harmonies encompassed by Aristotelian optimism, as well as some hypothetical harmony in connection with hypothetical aliens??ref. The classic Aristotelian has several tools for explaining harmonies.¹³

The first classical tool is the theory of the adequacy of matter to form on which for matter to receive a particular form, it must fit it sufficiently well, because a form’s power of informing is limited to some kinds of matter.

This might well explain the harmony between the DNA and the phenotypic features of an organism, on the one hand, and its norms, on the other hand. A molecule-by-molecule copy of a pig on this theory cannot have the form of a sparrow or a bat or the like, and hence we need not wonder at why the norms governing a pig do not require it to fly in order to flourish.

By itself, the material adequacy tool is insufficient to explain the harmony between DNA and norms, however. For while it can explain why a molecular duplicate of a pig cannot be specifically a sparrow or a bat, it does not explain why there isn’t some other form that requires flying for flourishing but also has the power to inform a molecular duplicate of a pig.

The second classical tool consists in the causal powers embedded in the forms. The form of a sparrow not only includes the norm of flying, but a tendency for the organism

¹³This section owes much to discussion in my mid-sized objects seminar, and especially to Christopher Tomaszewski’s suggestions on the explanatory powers of forms.

to develop a power to fly. The forms thus have tendencies to make the organisms with the forms behave in ways in accordance with the norm.

Again, the causal power tool has a similar limitation to the material adequacy tool. While we can explain why *sparrows* have both the norm of flying and a tendency to be physically capable of flight, we need an explanation for why there isn't some other form than that of a sparrow—the form of an unfortunate bird that has the same norms as a sparrow but none of the causal powers relevant to flight.

A natural answer to the limitations of the material adequacy and causal power tools is found in the third tool: the sparseness of forms. The logical space of forms is sparse: only some combinations of norms and powers (including the power of informing) are candidates for being contained in a single form. Compare here Plato's insistence that there is no Form of Mud.^{??ref:Parmenides} The reason why the evolved pig-like animals in our world do not have a form that both makes flying normative and can inform a pig replica is simply that there cannot be such forms. And similarly, nothing in our world has the norms of a sparrow without the causal powers relevant to flight, because no such form exists in logical space.

5.2. Evaluation of the tools. First, note that the material adequacy tool is problematic in light of the arguments of ^{??backref} that organisms begin to exist in bodies that are too small to have any significant structure. That said, in ^{??backref} it was noted that even if no significant structure is required of matter to have a specific organic form, nonetheless some structures may be incompatible with some forms. Thus, perhaps, a bird form could inform something without much significant structure, but could not inform a pig-like body. I argued that this view is not sufficiently metaphysically motivated, but is still defensible. So perhaps the material adequacy advocate can keep on using the material adequacy tool. Or perhaps the small-beginnings argument will be rejected on the grounds that there is an absolute reference frame.

Second, material adequacy is presumably a rather weak criterion. An embryo of an organism is unable to engage many of the behaviors characteristic of the species, and yet is

obviously materially adequate for the form. One might say that the embryo at least has the material adequacy for becoming an adult that can engage in the characteristic behaviors, but that won't be true if the embryo—or the adult—is severely impaired.

Next, note that we saw that both the material adequacy and causal power tools require the sparseness tool to be useful for accounting for Aristotelian optimism. So let us consider sparseness. Now, ontologists who have a sparse theory of properties typically formulate sparseness in terms of the denial of logically complex properties (such as being both negatively charged and cubical) and/or of properties that do not cut nature at explanatory joints (such as baldness).

But these kinds of sparseness will not help the Aristotelian. Consider a sparrow form S and a less harmonious form S' which is structurally similar to the sparrow form but has its parameters and powers arranged in far less harmonious ways—for instance, maybe the form is capable of informing a less sparrow-like body, or perhaps it has overt behavioral and epistemic norms that conflict much more than those of a sparrow. The important thing here is that we can suppose that the less harmonious form S' simply differs from S in the values of various parameters (such as the appropriate ratio of the masses of the organs, the proper degree of social interaction, etc.), rather than in terms of logical complexity. Moreover, if instantiated in some matter, S' would cut nature at the joints just as much as S would—both would specify a natural kind.

Standard sparseness theories are value-neutral, and restrict forms on structural grounds. But to explain harmonies, we need a value-based sparseness that rules out *bad* forms. Given that what forms are possible is a fundamental metaphysical feature of reality, the sparseness theory requires a fundamental metaphysics sensitive to value. The main such metaphysical theories are theism, pantheism, and axiarchism, on all of which the fundamental features of reality induce a drive to the good.

Furthermore, simply ruling out *bad* forms is insufficient for Aristotelian optimism. We could imagine borderline good forms, ones full of disharmony and a disunited ethical life, leading to many skeptical worries, but still on the whole good. Aristotelian optimism

requires us to think that such forms are *less likely* to be instantiated, and hence that in our theorizing about norms we should prefer theories that show a greater unity, but does not and probably should not require us to rule out such forms. Yet in a world where, for all we know, such forms are common, it is not clear that we would know normative facts. We might have a probability bigger than 1/2 of agreement between moral intuitions and moral norms, but the probability would be unlikely to be high enough for knowledge.

Ruling out the metaphysical possibility of borderline good forms would require fundamental metaphysics itself to draw a line between the forms that are “good enough” and those that are not. It seems difficult to find a non-arbitrary place to draw this line—again, we have Mersenne questions. The only non-arbitrary lines appear to be those between the bad, the neutral and the good (and any of these lines lets in the borderline good forms), as well as perhaps between the optimal and the inoptimal.

Restricting forms by means of the line between the optimal and the inoptimal would be sufficient for Aristotelian optimism. But how could we use optimality considerations to generate sparseness? There appear to be two options. First, we could require the forms themselves to be optimal, and, second, we could require that the forms be such as to be found in an optimal world.

It initially seems that if we require the forms themselves to be optimal, we will only have one form—the very best possible form, that of a perfect divine being—and hence only one kind of substance in reality, whereas we were searching for an explanation of optimism about the diverse forms in the world, or at least the human form. Moreover, we have good reason to think a perfect divine being is not the only substance there is. For if that is the only substance, then we must be modes or parts of it. But my personal moral failings conclusively rule out the hypothesis that I am a mode or part of a perfect substance.

However, there is another optimalist explanation of what forms there are. Instead of applying optimalism to the forms, apply it to worlds, like Leibniz, Leslie and Rescher did??refs, so that there is only one metaphysically possible world. Then we can say that

the space of forms contains only ones that can be instantiated in the best of all worlds.¹⁴ This gives a value-based sparseness in the logical space of forms, but of course seems to give as much optimism as anyone could want—we can't do better than to think that we are in the best of all possible worlds.

Finally, we should note that sparsity of form will not explain all of the harmony between the forms of different actual substances, such as the harmony between the different kinds of fundamental particles whereby they all support the conservation of mass-energy, unless the forms are implausibly extremely sparse.

6. Explanation of our normative complex

6.1. A pattern of explanation of norms. Here is a familiar pattern. We have a deeply-seated moral intuition about the general prohibition, *g*, of some action, such as incest. It is not clear how to derive the prohibition in its full generality from intuitively more basic principles, such as one of the categorical imperatives. Easy considerations, which I will call the *cs*, show that in *typical* cases the action is wrong, but our moral intuition goes beyond these typical cases. Thus, considerations of the abuse of power, distortion of familial dynamics, and genetic harms show that most cases of incest are wrong, but it is easy to imagine cases of incest to which these considerations do not apply—say, same-sex adult siblings of similar age who were raised apart—and yet moral intuition forbids incest in those cases as well.

We can now save the moral intuition by saying that the more general prohibition *g* is simply a fundamental moral rule, not reducible to the *cs* that explain why the action is wrong in typical cases. But if we stop at this, the connection between *g* and the *cs* mere happenstance, and that seems intuitively wrong. The abuse of power, distortion of family dynamics, and genetic harms seem relevant to why incest is wrong.

¹⁴This is not exactly Leibniz's own view. Leibniz thought that the forms of entities competed for instantiation in the divine mind.??ref This metaphor implies a winner and a loser, and hence the existence of forms that lose the contest, and hence are not instantiated in the best of all possible worlds.

At this point, often we are in a position to see another fact: it is quite beneficial to have a general moral prohibition beyond the prohibitions arising from the *cs*.

One reason for such a benefit is that our judgment as to whether the *cs* apply to a given case is fallible, especially given our capacities for self-deceit, and the costs of violating the *cs* are so high that it would be better for us to have a general prohibition than to try to fallibly judge things on a case-by-case basis.

Second, in some examples of the pattern, being open to serious deliberation about the forbidden action can itself harm one or more of the goods involved in the *cs*: thus, being open to weighing whether the distortion-of-family-dynamics consideration applies against a particular instance of incest can itself distort the agent's participation in family dynamics. Some things are best left unthinkable!

Third, we could have a tragedy of the commons situation. It could be that the *cs* are actually insufficient to render an instance of the action wrong, but we would be better off as a society if the action were generally avoided. Thus, perhaps, the genetic harm coming from one more couple's engagement in incest would be insufficiently significant to render the incest wrong, but without a general prohibition, incest would be sufficiently widespread as to cause serious social problems. A general prohibition that is not logically dependent on the *cs* would help avert such social harms.

These considerations are very familiar to us in the case of positive law. Jaywalking involves harms such as disruption of traffic flow and the danger of death of the pedestrian and of trauma to the driver, and the considerations of these make jaywalking wrong in typical cases. There are obvious instances, however, where these considerations do not apply: say, crossing a road where the pedestrian can clearly see that there are no intersections or cars on the road for a significant distance in either direction. However, it may be better for people simply to abstain from jaywalking than judging whether the disruption and safety considerations apply on a case-by-case basis, because there could be so much harm if the judgment were to go wrong. As a result, it is can be reasonable for a state simply to ban jaywalking altogether (or to ban it with some clear and easily adjudicated exceptions). We

similarly resolve cases of tragedy of the commons with positive law: think, for instance, about laws against littering.

In the case of positive law we have two different explanations. First, there is an explanation of why the forbidden action is wrong in general: this is because it has been competently forbidden by legitimate authority. This explanation need not make reference to considerations such as disruption of traffic flow or danger of death.¹⁵ Second, there is an explanation as to why the action has been forbidden by the authority—and here all the rich specific considerations are relevant.

6.2. Theism. Now, a theistic version of natural law can have precisely the above pattern. An action is morally forbidden because our nature is opposed to it: this is an instance of grounding explanation. This explanatory fact does not make reference to the *cs*. But we are still pulled to ask the further question: “Why does our nature includes this prohibition?” But since our nature is essential to us, the answer to that question could simply be the necessary truth that we couldn’t exist without this nature.

However, we can rephrase the second question like this: “Why are there intelligent primates on earth with a nature that includes this prohibition rather than some other kind of intelligent primates with a nature that does not include this prohibition?” And here the theist can answer: Because it would be good, in light of the *cs* and the further considerations in favor of generalizing the prohibition beyond the cases where the *cs* specifically apply, to have intelligent primates with a nature that includes this prohibition, and God acted in light of this good.¹⁶

¹⁵Though in some cases *some* such reference may be needed in order to establish that the matter falls within the competence of the authority in question. Thus, a regulatory agency may need to demonstrate that traffic flow disruption comes from some pattern of motor vehicle activity in order for certain rules regulating that pattern to fall in the scope of its authority.

¹⁶A divine command theorist can make the same move, but divine command theory has some liabilities which were discussed in ??backref.

The explanation may still appear viciously circular. On an Aristotelian metaphysics of value, what is good for us is grounded in our possession of our form. How could the possession of our form, then, be explained by what is good for us? But there is no difficulty in the context of theistic selective explanations. Granted, whatever form will be exemplified will define what is good for its possessors. But if God were to choose to exemplify a form that defines a certain state as the only possible good for its possessors and yet the type of being was one that could not attain that state, or could do so only in extremely rare cases, the result would be beings that universally or almost universally are in a bad state. And it's bad that there are beings in a bad state, even if the badness of the bad state is defined by their form. On the contrary, God has good reason to select a form that makes it much easier to attain the good state defined by the form.

Here we should make a distinction between the specific goods grounded in our form—health, friendship and the like—and the general good of fulfilling our nature. The specific goods are grounded in our nature. But it is not clear that we should say that the general good of fulfilling our nature is itself grounded in our nature. It is plausible in the Aristotelian context to say that to be good for x just *is* to fulfill x 's nature. This identity is simply reductive. Given this, the circularity in the explanation disappears. What specific things are good for us is grounded in our nature. But it is good for us to fulfill our nature, and that fact is independent of what the nature is. It thus makes sense to explain *which* nature is exemplified by considerations of how apt the possessors of that nature would be to fulfill that nature, and have that good. We thus have a kind of explanation of why rabbit-like beings specifically have reproduction be good for them—having reproduction be good for them is more apt for the fulfillment of their nature than, say, having the discovery of mathematical truth be good for them. It is the general good of fulfillment of any nature that can explain why a nature with such-and-such specific goods is selected.

Further, we can have explanatory relations running both ways between goods and norms of behavior. If having norm N promotes some specific good S , then that could explain why a being whose form codes for S being good also has norm N of behavior. But

conversely, we could explain why we have a being that has a form that codes for norm *N* also has a form that codes for *S* being good for the being. If the norms are selected for exemplification by a perfectly good God, we may expect both forms of explanation to show up, as well as a hybrid model where both *N* and *S* are chosen together for their fit.

This kind of divine selection explanation of both ethical norms and goods extends from ethical to prudential, epistemic and semantic norms. As we saw in ??backref, ethical, prudential, epistemic and semantic norms all interact in complex ways with what is good for us, and this interaction can provide God with reasons in favor of some and against other combinations of norms and goods.

6.3. Non-theistic alternatives. Theism thus neatly explains the harmonies between our various norms. It explains why the beings that exist are by and large able to fulfill their norms and flourish. It can also explain why various skeptical hypotheses are false??backref—God has reason to ensure the falsity of such hypotheses—and how we can have access to truths about non-empirical matters like mathematics, metaphysics and ethics. Aristotelians can have pretty much all the optimism they want if they are theists.

What, on the other hand, could a selective non-theistic cause be like? There are three main candidates for selective causes in the philosophical literature: evolution, axiarchic principles, and intelligent designers such as God.

Genetic descent with variation only directly governs the non-normative aspects of organisms. It is not sufficient to explain the form that the organism has, given that the form encodes normative features as well. Nonetheless, it is worth considering the possibility of a law of nature linking DNA to form, a law of nature specifying that when an organism with such-and-such DNA comes into existence, it has such-and-such a form. This law of nature would be immensely complex, with many free parameters raising Mersenne questions. Moreover, if the law of nature is to cohere with the Aristotelian optimism that is crucial to our Aristotelian account, there must be a fit among the various aspects of the form and between the form and the actual physical body plan and physical environment. It is implausible that this fit, in us and presumably in the myriad of other organisms, is just

a coincidence. Such a law of nature calls out for an explanation. On pain of vicious regress, the need to explain the law of nature points towards one of the other two explanatory candidates: axiarchic principles and intelligent designers. Moreover, without a value-laden explanation of the linkage between DNA and form, the hierarchical structure of the norms discussed in Section ??backref and the non-deductive explanation of normative principles in Sections 6.1–6.2 is replaced by a vast coincidence, and hence we do not have a satisfactory answer to the complexity objection??backref.

Furthermore, the need to explain our fundamental *a priori* intuitions, or the prior probabilities of non-empirical matters, in a way that connects them with their truth is unmet by a purely evolutionary approach. Evolutionary selection is not responsive to potential inconsistency of the Peano axioms of arithmetic (at least not to an inconsistency that involves a very long proof).

At this point, our choice appears to be between an explanation involving a non-intelligent tendency towards value and an intelligent one, such as the theistic one that we have already discussed.

The main candidate for the non-intelligent tendency are axiarchic principles, such as those defended by Leslie and Rescher.??refs These are fundamental metaphysical principles that require the world to be optimal.

It is worth noting that while I am discussing axiarchism as an alternative to theism, Rescher himself takes his theory to imply theism: it is better for there to be a God, and hence there is a God.??ref And if there is a God, then presumably this God is sovereign and governs the selection of forms, and so we have the theistic account. Leslie's version also involves supernatural beings. But Leslie thinks that what is best is not that there be one infinite all powerful, all knowing and all good God, but infinitely many omniscient observers who enjoy the world thereby adding to its value, though without creating it, since then there would be the possibility of conflict between them.??ref,check

There are a number of problems with optimalistic axiarchic principle explanations.

First, intuitively, a metaphysical principle *constrains* what beings can exist and how they behave rather than somehow explaining the positive existence of beings. But the axiarchic principles are supposed to explain the existence of beings: the beings in reality exist because it is for the best that they do so. Theism explains beings in terms of the creative activity of a being and has no such difficulty.

Second, we have the problem of evil. Looking at the litany of suffering in human history, our world doesn't look like the best of all possible worlds. Axiarchic views can apparently make use of many of the responses to the problem of evil given by theists. This vast literature is beyond the scope of this book. As far as this objection goes, it is not clear how much of an advantage, if any, theism has.¹⁷

Third, we get metaphysical modal fatalism on optimalism: there is only metaphysically possible world. We now have a dilemma regarding the class of worlds among which ours is said to be best. Either that class contains only one world or it contains multiple worlds. If it contains only one world, then it is unclear why optimalism implies optimism. For if optimalism implies optimism, then we would expect worlds-based pessimism, the view that we are in the worst of all worlds, to imply pessimism. But if there is only one world for "reality to choose from", then both optimalism *and* pessimism hold, and it is quite unclear whether we should be optimistic or pessimistic as Aristotelians.

Now, suppose that there are multiple worlds to choose from—likely infinitely many. Since on optimalism only the best world is metaphysically possible, the other worlds must be metaphysically impossible. At this point we appear to have two initially plausible options. First, we could have no logical restrictions on the worlds—thus, we will have worlds with square circles, and so on—and, second, we could require the worlds to be (narrowly) *logically* coherent, even if they are not metaphysically possible. In other words, they have no strictly logical contradictions in them, though they may violate fundamental metaphysical principles, like the principle of optimalism.

¹⁷But see ??ref:Adams.

If we place no logical restrictions on worlds, then it is unlikely that the best world is in fact one that is logically coherent. For consider the kinds of claims theodacists make in defending the compatibility of the goodness of God in the face of the horrendous evils of this world, which are presumably also going to have to be similar to what the optimalist will need to say to defend the optimality of our world in the face of these evil. These are invariably claims about some great goods logically requiring God to permit evils: without real evil, one cannot have courage, forgiveness of real evils, and exercises of sacrificial love. But such a defense of the optimality of our world will fail if we drop logical constraints. We can have courage in the face of evil without evil, forgiveness of real evils without evil, and sacrifices with no losses. Furthermore, on optimalism, the best *possible* world cannot include free will of the sort libertarians believe in, since that kind of freedom requires the possibility of alternate possibilities, whereas on optimalism there is only one possible world. But an incoherent optimal world can include libertarian-style free will despite such free will being incompatible with the world's optimality. In fact, it can include libertarian-style free will and determinism, since incoherence is no bar. Or imagine an incoherent world that contains nothing but one entity, Bob, with the logically incoherent property of being such as to be better than any world or any plurality of one or more entities in any world (the property is incoherent since it implies that its possessor is better than itself). Thus all in all it seems likely that the best of all worlds will be deeply incoherent, if incoherent worlds are included.¹⁸ Yet I assume we reject the hypothesis that our world is incoherent.

Next suppose we constrain worlds to be logically, but perhaps not metaphysically, possible. A proposition is narrowly logically possible provided that no contradiction can be derived from it by the rules of the correct logic. A serious philosophical difficulty here is specifying what one means by "the correct logic". An obvious necessary condition on the correct logic *L* is that if *q* follows from *p* according to *L*, then the material conditional "if *p*,

¹⁸One might respond that incoherence is itself necessarily a disvalue. But since we are talking of an incoherent world, we could suppose that at our best incoherent world incoherence is not actually disvaluable despite being necessarily disvaluable.

then q ” has to be metaphysically necessary. But if logical necessity is to be distinguished from metaphysical necessity, as the proposal under discussion requires, the converse cannot always be true. In other words, not everything that follows of metaphysical necessity can follow of logical necessity. But now it is not clear which metaphysically necessary conditions correspond to rules of “the correct logic”.

There is, further, a serious technical problem. First, note that if logical necessity is distinguished from metaphysical necessity, it is presumably distinguished by the formality of logical necessity: the fact that logical necessity is expressed by means of formal rules of derivation from formal axioms. Second, note that the S5 axiom of modality is quite plausible for the logical possibility that governs the worlds that the best one is to be chosen from. For the S5 axiom says that if something is possible in one possible world, it’s possible in all of them. If S5 is false, then what worlds count as logically possible will vary between the logically possible worlds. Hence, explanatorily prior to the choice of the best logically possible world there is no fact about what worlds are in fact logically possible, and the idea of the optimality principles selecting between these worlds makes no sense, since at that point there is no such thing as “these worlds”.

But now it turns out that there is a conflict between the formality of logical necessity and S5. In fact the conflict does not need the full S5 axiom. It only needs an extremely version of it:

- (3) There is some statement s such that it is logically necessary that is logically possible that s .

The S5 axiom says this is true whenever it is logically possible that s , but (3) only says this is true for at least one s . For instance, it is highly intuitive that it is logically necessary that it is logically possible that evil is evil, or that $2 + 2 = 4$, and so on.

But now for the conflict. It is very plausible that among the things that are logically necessary are the Peano axioms of arithmetic. These are so intuitively implied by our concept of the natural numbers that if we deny their logical necessity, we might as well deny the logical necessity of such claims as that genuine forgiveness requires a wrong or

that courage requires evil, claims that I have argued are likely to be needed for a defense of the optimality of our world. Moreover, for the sake of the *formality* of logical necessity, the axioms need to be formally recursively specifiable, and similarly for the rules of inference. But now Gödel's Second Incompleteness Theorem comes in and tells us that no formal system that includes the Peano axioms can prove its own consistency. Suppose, however, that it is logically necessary that it is logically possible that *s*. But on a formal account, that it is logically possible that *s* means that a contradiction cannot be derived from *s*. And that it is logically necessary that it is logically possible that *s* means that there is a formal proof of the fact that no contradiction can be formally proved from *s*. But if we could prove that no contradiction can be formally proved from *s* in the formal system of true logic, we could also prove that no contradiction can be formally proved *with no additional assumptions* in the formal system of true logic—in other words, we would have an in-system proof of the consistency of the system, contrary to Gödel's Second Incompleteness Theorem.

Now, maybe there is some notion of the formality of a system that escapes this argument, but in any case we have a serious difficulty for an account that takes narrowly logical possibility to be essential to identifying the worlds from among which the best is chosen. Thus, whether or not we constrain these worlds by logic, we have a serious problem for optimalism.

Finally, there does not appear to be a unique best world. We could take any good world and add one more happy disembodied mathematician. This might not produce an overall better world. It might, for instance, be aesthetically inferior in some way—say, by having too many mathematicians and thus offending against simplicity, or by having a non-prime number of mathematicians (perhaps the aesthetically best number of mathematicians is some cool type of prime, like a Mersenne prime¹⁹)—but it is superior in at least one significant way, namely by having an additional happy mathematician, and it is not plausible to think that it would be an overall inferior world.

¹⁹A prime of the form $2^n - 1$.

Furthermore, if we distinguish logical from metaphysical possibility, then there appear to be logically possible worlds that are either tied with our world for goodness or incommensurable with it. It seems that *a posteriori* truths are logically contingent, though as Kripke has shown, they may be metaphysically necessary. Let L_{AP} be a qualitative description of my life and L_{NB} be a qualitative description of Napoleon's life. That I lead L_{AP} and Napoleon led L_{NB} is *a posteriori* and hence logically consistent. It is logically, though probably not metaphysically, possible that I and Napoleon have been swapped, and I led L_{NB} while he led L_{AP} . Such a world would not be any worse or better than ours. It would, thus, be tied or incommensurable with ours. Now one could modify optimalism to say that no world is better than ours, but there are multiple metaphysically possible worlds that are either all tied or incommensurable. Such an optimalism would be rather less elegant, but would at least escape this last difficulty.

Finally, it is worth noting one advantage of theistic over axiarchic Aristotelianism. Axiarchic explanations appear to require a value comparison between worlds, and thus requires an account of value that goes beyond that defined by the forms of individuals. The Aristotelian explanation of the normative in terms of forms on axiarchic Aristotelianism is limited to rule-like norms, but does not extend to the realm of the good as a whole. Presumably, for instance, a world with greater diversity of types of life is more valuable than one with a lower degree of biological diversity, but this is not grounded in the forms of the living things. On theism, on the other hand, there is some hope of explaining such global axiology in terms of the imitation of God or in terms of what God's own form defines as good for God to create, as will be discussed in Section 7. Moreover, if axiarchism requires a single best world (*pace* the response to the Pruss–Napoleon swap discussed above), the value comparisons need to be more comprehensive than on theism, where God might freely choose between incommensurable worlds.

6.4. Theistic and axiarchic Aristotelianism, and future ethics. Axiarchic and theistic Aristotelianism has a significant degree of non-naturalism: many features of the arrangement of things in our world have a value-based explanation.

In ??backref, we discussed the difficulty that social and technological development poses to a more naturalistic version of Aristotelianism: given such an Aristotelianism, we should be surprised to have our ethics continue to give useful guidance despite the vast changes in our lives over the last hundred thousand years. A non-naturalistic story could restore optimism. A God could know what kinds of seemingly outlandish scenarios might actually be relevant to the lives of his creatures, and can wisely choose the forms whose norms that fit with these, and a principal of optimality might well have similar effects. The resulting norms may seem *ad hoc*, especially in edge cases: they won't be the elegant principles of act utilitarianism (though of course act utilitarianism faces significant difficulties in out-of-our-experience situations, as we saw in ??backref:population-ethics). And an apparent *ad hoc* character in divinely-instituted rules as applying to edge cases should not surprise us—wise legislation does not eschew judgment calls. On optimalism, we might be slightly more surprised, but also not by much, as the facts about edge cases could be due to minor factors we are unaware of.

An interesting question is whether an axiarchic or theistic Aristotelian should be surprised by having ethics glitch in some actual cases, in one of the three ways discussed in ??backref: (i) real dilemmas, (ii) conflict between moral rules and the reasons for them, and (iii) conflict between moral rules and our intuitions. After all, logical space contains an infinite number of possibilities for a form of a rational being, and a perfect being or an axiarchic principle should be able to combine one with an environment in which there would be no actual glitches.

But even if such a combination would be possible, would it be good? Even for a being whose power is not limited by anything other than logic, there can be unavoidable costs to options. Intuitively, there is a value to the most important norms of behavior for limited beings²⁰—say, norms governing killing—having a significant simplicity, so that they can be reasoned about more easily, especially under time pressure. At the same time, there is a value to a diverse and rich moral environment. And there is a value to morality lacking

²⁰Why limited???

glitches. Plausibly, one cannot have all three values to their maximal degree at the same time; there are likely to be logically unavoidable trade-offs. And there does not appear to be strong reason to think that a perfect being would be so enamoured of one of the three values that we would expect that value to be present to the maximal degree. In particular, we should not expect a completely unglitchy ethics.

But we might have reason to hope that glitches are rare in the actual circumstances faced by humanity, or that the worst of the glitches should only occur in the case of agents who have wrongfully produced the circumstances for this glitching.²¹

6.5. Aquinas' Fourth Way and the good. Aquinas' Fourth Way²² puzzles the modern reader. It begins with a principle that comparisons between degreed properties are grounded in a comparison to a maximal case: x is more F to the extent that x is more like an item that is maximally F . Aquinas then illustrates the principle with the case of heat and fire: an object is hotter provided that it is more akin to the hottest thing, namely fire. He then applies the principle to goodness, and concludes that there is a best thing, and this is God.

The fire illustration is not just unhelpful to us, since we know that fire is not the hottest thing (the sun is almost twice as hot as the hottest flame), but it is actually a conclusive counterexample to the comparison principle, since we can easily compare temperatures without reference to an alleged hottest object.²²

So Aquinas' comparison principle is false. But I contend that there is still something to his argument when applied to the good.

²¹Compare the theory that real moral dilemmas only occur in the case of agents who have done wrong, say by making contradictory promises.²²

²²In any finite universe, presumably there will be a hottest object. However, temperature comparison is not defined by that object, since even if Bob is in fact the hottest object, we would expect it to be physically possible to have a hotter object than Bob. But if degrees of heat were defined by closeness to Bob, it would not be possible to be hotter than Bob, since nothing can be closer to Bob than Bob.

A form-based metaphysics gives a powerful account of the good for a being of a particular kind—an oak, a sheep or a human, say—in terms of its match to the specifications of the form. It also gives a ground to comparisons between the good of different instances of the same kind: a four-legged sheep is, other things equal, better at sheepness than a three-legged sheep, because it more completely fulfills the specification in their ovine nature. In fact, this is itself a counterexample to Aquinas' comparison principle, in that we can compare degrees of success at sheepness without supposing any individual sheep to be perfect.

However, in addition to value comparisons within a kind, there are ones between kinds. Jesus' observation that we are "worth more than many sparrows" (Mt. 10:31^{ref}) is quite uncontroversial. Indeed, even a perfect sparrow seems to have less good than a typical human. While the nature of a sparrow will enable value comparisons between sparrows, and that of a human between humans, we still have the question of what grounds the value difference between sparrows and humans. Some Aristotelians reject cross-kind value comparisons as nonsense.^{??refs} But given the intuitive plausibility of many such comparisons, this rejection is a costly one.

Aquinas' Fourth Way is not infrequently seen as more Platonic than his other arguments for the existence of God, and Plato indeed had a solution to the problem of cross-kind comparisons, by talking of differing degrees of imitation of the Form of the Good, which itself is perfectly good. Plato, on the other hand, lacked a satisfactory solution to the problem of intra-kind comparisons. He may well have thought that there was a Form of Humanity^{??refs}, which exemplified humanity perfectly, so that similarity to the Form of Humanity would define how good one is at being human. However, we can see that this solution is clearly unsatisfactory. First, the Forms are immaterial, so the Form of Humanity is immaterial, and hence it lacks fingers. Thus, the fewer fingers a human has, the more they are like the Form of Humanity, and hence, absurdly, the more perfect they are. Second, if somehow the Form of Humanity ends up having body parts, then the Form of

Humanity either has an even number of cells or an odd one. But clearly neither option is more perfect than the other.

Central to Plato's solution to cross-kind value comparisons is the self-exemplification of the Form of the Good: the Form of the Good is itself maximally good. But a similar self-exemplifying Form cannot be used to account for intra-kind comparisons. Aristotle, on the other hand, has the non-self-exemplifying forms immanent in things. The Aristotelian form of humanity specifies human perfection, but not by exemplifying it. It has neither fingers nor cells, but it *specifies* that humans should have ten fingers while specifying an age-dependent normal range of cell numbers.

Notwithstanding the general falsity of Aquinas' comparison principle for degreed properties, Aquinas provides us with a plausible extension of the Aristotelian system to allow for comparisons of degrees of good between objects of different kinds in terms of the similarity to or degree of participation in a maximally good being, a divine being that plays the role of a self-exemplifying Form of the Good. The human being participates in God in respect of abstract intellectual activity, Aquinas will contend, while sparrows do not, and in that important respect, at least, humans are more like God. On the other hand, the sparrow's speedy movements approximate divine omnipresence better than the stillness of a mushroom does, and in that respect at least the sparrow is superior to the mushroom. (An omnipresent being is present everywhere at once, while a speedy being is present in a large range of places in a short time.) We have, thus, a ground for something like a great chain of being.

There are still difficulties here. While the human is superior in intellectual activity, the sparrow moves around with greater three-dimensional freedom. How can we say that the human is superior all things considered? Where we previously had a problem of cross-kind comparisons, we now have the problem of cross-attribute comparisons. Intuitively, the human's intellectual superiority to the sparrow trumps the sparrows motive superiority to the human, and enables us to say that the human is more perfect on the whole. This higher level question is difficult indeed.

But there is some hope in thinking that in attributing different attributes to God we sometimes express divinity to different degrees. It may be that there is no meaningful comparison between how well we express divinity by saying that God is all-knowing versus by saying that God is all-powerful, but saying that God knows the multiplication table up to 10×10 surely expresses divinity less well than saying that God can create any possible physical reality. I suggested earlier that speedy motion imitates divine omnipresence. Thus, the sparrow's ability to fly imitates God's presence throughout several kilometers surrounding the surface of the earth, while the human's more limited mobility imitates God's presence in a thin two meter shell of air surrounding that surface. But the degreed difference between the two divine attributes—each a limited special case of omnipresence—imitated here might well be trumped by the fact that the sparrow does not imitate God's abstract intellectual activity *at all* while the human does imitate that activity, and does so in respect of a very wide scope of things (the human can think abstractly about the whole universe, for instance).

In ??backref, we gave a non-theistic Aristotelian sketch of a three-step great chain of being. The account here has a hope of allowing one to fill in more intermediate links. Even if the details in the comparisons between different attributes or respects do not work out, we still have an advantage for the theistic Aristotelian in being able to make cross-kind comparisons under specific respects, like motility or intelligence.

??ref:Jeffrey/Ward

6.6. Theistic choice points. Suppose we are convinced that we need a theistic Aristotelianism. At this point there are metaphysical and theological choice points. One metaphysical choice point is whether there are any uninstantiated forms. If there are, as on a Platonic picture, then we have a theological question: Does God freely choose which ones to create, or do they exist necessarily, say in the mind of God? If there are no uninstantiated forms, as on a more classically Aristotelian picture, then probably the most parsimonious

theistic story is that God creates the forms in the act of creating the substances that instantiate them. We thus have three views: Theistic Voluntarist Platonist Aristotelianism, Theistic Involuntarist Platonist Aristotelianism and Theistic Classical Aristotelianism.

Although previously I have argued in favor of individual forms^{??backref}, Platonism more naturally fits with a shared form for each kind of thing. One can, however, suppose that there is a universal form and there are individual forms which are instances of the universal form or acts of sharing in it.

The Voluntarist Platonism option may seem to have some unnecessary complexity. If God chooses which forms to create, it is puzzling why God would “bother” with the ones that aren’t going to get instantiated. There is, however, a possible answer: to open a field of possibilities to creatures. Perhaps the forms need to have some kind of universal Platonic existence in order for creatures to have the power of producing their instantiations. There is a value to the earth ecosystem “having a choice”, with many evolutionary possibilities of what kinds of biological substances should exist, and on a more Platonic version of the metaphysics this could require the pre-existence of the form in a universal mode.

On both the Voluntarist Platonist and Classical versions, there is or can be a field of possibilities for other forms than the ones that actually exist. On the Voluntarist Platonist version, these are other forms that God could have created *ex nihilo* independently of instantiation. On the Classical version, these are other forms that God could have instantiated and thereby brought into existence. Presumably, God knows what these possibilities are, and so they have some kind of existence as ideas in the mind of God. There is much room here for difficult metaphysical exploration of the exact status of these divine ideas.^{???many-refs} Nonetheless, this point shows that there is a commonality between all three versions of theistic Aristotelianism: there is a field of formal possibilities. On the Voluntarist Platonist and Classical theories, this is a field of divine ideas. On the Involuntarist Platonism, this is a field of necessarily existing universal forms.

On all three views, God selects from that field of possibilities some forms that are to be instantiated, and likely also some forms that creatures can on their own (say, via an evolutionary process leading to a material substrate appropriate to the form) cause to be instantiated. There are significant metaphysical differences between the views, but all three involve a similar kind of divine selection model as to which forms are instantiated.

6.7. Participation in God.

6.7.1. *The participatory limitation account.* Earlier we saw that a significant part of our Aristotelian optimism can be explained by having a value-based sparse theory of forms. This sparseness can be explained theistically by classical theism's doctrine that all things are either God or participate in God. In such a setting, it is natural to think of a form as a way for a being to participate in God. But now while God's infinity and otherness may give a wide scope to what sorts of arrangements of features could count as a participation, that scope is plausibly narrower than all logically non-contradictory arrangements of features. Some candidate norms, like a requirement of causing gratuitous pain to others, just may not be included in any metaphysically possible way of participating in a perfectly good God. And some combinations of individually admissible features may also fail to be found in any metaphysically possible mode of participation in a perfectly unified God, such as having oral conversations with conspecifics as central for one's good while having the essential causal power of deterministically exploding whenever one approaches a conspecific within talking distance.

Nonetheless, as we saw in Section 5.2, a mere limitation on the space of possibilities for forms is insufficient for explaining all the aspects of Aristotelian optimism. Thus, a theistic account that encompasses participation will likely need to be combined with a divine selective account on which God is more likely to choose forms that are individually and in combination more harmonious.

6.7.2. *Mersenne's return.* There is, further, a serious problem with the participatory limitation account. One of the main things that led us to grounding ethics in human form was

the appearance of contingency implied by the vast number of seemingly arbitrary parameters in ethics. But does not the same thing difficulty apply to the participatory limitation account? For there seem to be parameters defining the boundaries between participatable combinations of normative features and unparticipatable ones. Just how much unity between the norms is needed in a form that participates in a God who is one? How much normative egoism can be found in a form that participates in a perfectly loving God?

We might try to shift the difficulty onto parameters in the divine nature which determine what is a possible participator in God. These parameters can be necessary, since God is normally thought of as a necessary being.

However, there are shortcomings of this approach. First, it becomes puzzling why we don't simply do a similar thing for the Mersenne questions that led us to forms. We could suppose, for instance, a Platonic account on which there is a vast number of metaphysically necessary ethical ur-norms with metaphysically necessary but still seemingly arbitrary parameters, instead of norms found in natures of particular types of rational beings. Given the plausibility that different possible intelligent species would have different parameters in their norms, the ur-norms would presumably include many conditional ones connecting the non-normative features of a species with the norms governing their behavior: "If you have such-and-such genetics, then you should prefer parents to strangers to degree x ."

One response is that shifting the difficulty from parameters found in human ethics to parameters in divine ethics is still philosophically advantageous. First, as discussed in ??backref-and-add, there is an advantage in norms that are grounded in a nature, whether divine or human, in that these norms are not an alien imposition of dubious binding power, but are the requirements of one's very own nature. Second, there may be a greater unity in a complex set of norms governing a single divine being than in abstract Platonic norms governing all possible beings in conditional form. Third, there is some hope that the parameters governing divine nature are fewer and more unified than any plausible set of parameters governing humans.

On the other hand, the idea of arbitrary parameters in God is theologically and philosophical unattractive. Such parameters fit poorly with classical theism's doctrine of divine simplicity. Moreover, it is the lack of arbitrary parameters that makes God an attractive explanatory posit. If God is to have a vast number of arbitrary parameters, is it not just as simple to explain things in terms of a brute necessity of a Big Bang, or of a particular set of necessarily selected forms?

One might worry that a similar difficulty applies to the divine selection account. God is more likely to actualize a more unified form. But what are the parameters in the function that governs the relationship between the degree of unity in a form and the chance that God would actualize that form?

We might again insist that parameters in the divine nature are not particularly problematic. But there is another move. The idea that there are numerical chances assigned to divine actions is itself theologically and philosophically problematic. First, numerical chances seem to be a kind of limitation on divine power.^{??refs:Murphy???} Second, the idea of assigning numerical chances to the vast infinity of possible divine actions is mathematically problematic. After all, this infinity exceeds any infinite cardinality, since for any cardinal number κ , God could create precisely κ angels, while the class of cardinal numbers exceeds any particular cardinal in its size^{??refs}, and so the class of divine actions creating groups of angels is beyond cardinality.

Instead of supposing numerical chances assigned to divine actions, we might simply suppose that there is a non-numerical qualitative rule: what better matches the divine nature is more likely to be instantiated. This rule need not even impose a total ordering on the space of possible divine creative actions, because there may be vast scope for incommensurability between divine actions.^{??ref:divine-creative-freedom}, so that there will

be divine actions whose chances are themselves incommensurable.²³ We might then further suppose that the qualitative ordering on the chances still yields numerical epistemic probabilities for humans, since the human form could prescribe specific numerical ratios of epistemic probabilities where the objective chances have a merely comparative or even incommensurable relationship, or we could suppose that epistemic probabilities are often merely comparative rather than numerical. And, it is worth noting, there is a well-worked out mathematical literature on non-numerical comparative probabilities.^{??refs}

The divine selection account, thus, seems to have a greater chance of escaping the arbitrary parameter worry than the participatory limitation account.

6.7.3. *A dual account.* It is quite reasonable to combine the theistic choice and participatory limitation accounts. Given our discussions, the participatory limitation account is unlikely to suffice on its own, after all. On a combined account, some aspects of form, especially coarser-grained ones, can be explained by participatory limitation while others, especially finer-grained ones, can be explained by divine selection. The result is an explanatorily rich account of normativity, which predicts a minimal coherence throughout a normative complex, and leads one to expect higher degrees of unification in more central aspects.

This expectation of unity in turn yields another tool to help us to actually find out what our norms are. We should prefer normative theories that allow for significant integration and harmony between moral, epistemic and other norms. An integrated picture of human flourishing is obviously attractive.

^{23*}What could that mean? Here is a mathematical parable. The Banach-Tarski paradox has it that we can divide a solid mathematical ball into five disjoint pieces, E_1, E_2, E_3, E_4, E_5 , and move these pieces to construct two balls of equal size to the original. Imagine randomly choosing a point in the original ball, and asking which of the five pieces it lies in. Paradox ensues if we make too many comparisons between pieces, such as that the point is more likely to lie in E_1 or E_2 than in just E_3 , or vice versa, but it is safe at least to say that if A is a proper subset of B , then the point is more likely to lie in B than in A . There are, admittedly, still technical problems with such approaches (cf. ^{??ref}:Pruss-domination).

7. Kind-independent goods

7.1. Introduction. Aristotelianism does really well with explaining kind-dependent values. But there also appear to be values that appear to transcend kinds, such as simplicity and diversity. Furthermore, we can compare kinds: some are better than others. All these values are crucial to the theistic (and axiarchic) value-based explanations that make our Aristotelian optimism work.

7.2. Comparisons-between kinds. The Aristotelian account defended in previous chapters can ground the comparison between a flourishing and a non-flourishing human, or between a flourishing and a non-flourishing chanterelle mushroom. But it is also obvious that a human is a better kind of entity than a mushroom.

Some Aristotelians deny such intuitions??refs, but they are a part of the common-sense picture of the world. Obviously, we should have more care for humans than for dogs, for dogs than for mushrooms, and for mushrooms than for photons. But at the same time, it is highly intuitive that other things being equal, our duties of care should be proportioned to the value of the entity being cared for. This requires a comparison of values between kinds.

Aristotelianism might be able to explain a kind of value domination, where kind *A* has all the goods of kind *B* and one or more additional ones. Imagine “electrohumans” that can do all the things humans can, but can also sense electrical activity. Then it seems that electrohumans would be capable of all the forms of fulfillment we are, and some more. But even this is not completely clear. Imagine a work of art that holistically appeals to all human senses and depicts a field of chamomile flowers being pollinated by bees, in sight, smell, taste, sound and touch. A human can enjoy such a work as a generically sensorily complete work, but to an electrohuman it is a sensorily limited work. The aesthetic enjoyments of a sensorily complete and a sensorily limited work are different in kind. Of course, the electrohumans can enjoy a similar work that also includes information for their electrical sense. But that isn’t a dominating kind of enjoyment, since art that appeals more

sensory modalities does not dominate art that appeals to fewer sensory modalities, but rather the two are incommensurable—consider, for instance, color versus black and white photography.

But even if electrohumans dominate humans, such cases are too rare in our ecosystem to explain our common-sense cross-kind comparisons. Humans are better than dogs, but dogs have a better sense of smell. Dogs are better than mushrooms, but mushrooms are better at digesting decaying matter. We can evaluate different individual dogs by how well they fulfill their nature, and different individual chanterelles by how well they fulfill theirs, but the nature of neither allows the other to be evaluated.

A theistic version of Aristotelianism might make some progress on this problem by supposing that one can compare the degree in which the various beings participate in God. In other words, God's divine nature could provide a common measure for different kinds. Human rationality involves more participation in God than the canine sense of smell.??ref:Adams

While I think this is a promising approach, it is not completely clear that it fully solves the problem of cross-kind comparisons. What makes it be the case that rationality involves "more" participation than smelling does? How are kinds of participation to be compared?

Perhaps we should retreat from claiming an objectivity in cross-kind comparisons, and instead suppose that our nature grounds a variety of importance comparisons, where different ways of being good—different ways of participating in God—are differently prioritized by us. Thus, perhaps, after all, we cannot say that humans are fully objectively better than dogs and dogs than mushrooms, but that these comparisons hold with respect to the importance prioritizations built into human nature, prioritizations that give pride of place to intelligence. These comparisons then involve additional Mersenne problems, which standard Aristotelianism so neatly solves. Note that this is not the same as instrumental comparisons: that dogs are better kinds of things relative to these human importance prioritizations than chanterelles is not just an instrumental claim about dogs being more useful to us. Tigers are also better than corn plants, but corn plants are much more useful.

7.3. Global goods.

7.3.1. *A problem and an initial theistic solution.* Non-theistic Aristotelianism does a good job accounting for goods at the level of individual substance in terms of the flourishing of that substance. But what about goods at a higher level of organization than individual substances?

One example is ecosystems. If ecosystems are substances, it is easy to account for their value: the form of the ecosystem specifies its flourishing. But taking ecosystems to be substances not only violates the traditional Aristotelian thesis that no substance is a proper part of another, but more seriously ecosystems just do not seem to have the kind of objective temporal and spatial boundaries that we associate with substances.

Fortunately, ecosystems are made up of organisms, and it is plausible that a part of the flourishing of an organism is constituted by an appropriate interdependency between it and other biological and non-biological constituents of its environment. Thus, an Aristotelian can at least initially accommodate the goods of an ecosystem to those of its biological constituents.

One initial problem with such an accommodation is resolvable. We might think that individual plants and microorganisms have little in the way of non-instrumental (or final) value, but an ecosystem largely or wholly composed of plants and microorganisms can still have a great non-instrumental value. But perhaps this means that we should just attribute more value to the individual plants and microorganisms in virtue of their flourishing as members of the ecosystem.

But there is a second, more serious problem. Imagine equally flourishing forests on two different islands. One island is twice as big as the other, and its forest therefore consists of twice as many organisms. If both forests flourish equally, then it seems that the organisms in the two forests flourish equally, and hence the island with twice as many organisms contains roughly twice as much good. This seems incorrect, however. As a vastly oversimplified toy model, suppose one ecosystem consists of n organisms of one kind and the other of

$2n$ organisms of the same kind that all flourish equally. Each organism then has two components of flourishing: an internal flourishing I and the ecosystem-relative flourishing E . Then the good of the larger island is roughly $2nv(I) + 2nv(E)$ (where $v(A)$ is the value of A) while the good of the smaller island is roughly half of that, namely $nv(I) + nv(E)$. But that doesn't seem correct. For the terms involving $v(E)$, namely $2nv(E)$ and $nv(E)$, represent the value of the ecosystem, and these should be intuitively equal if the two ecosystems flourish equally despite different numbers. We should expect the larger island to be more valuable, but not *twice* as valuable, because the ecosystem value shouldn't double—only the value of internal flourishing should double from $nv(I)$ to $2nv(I)$.

Furthermore, there appear to be goods at even higher levels of organization than ecosystems, such as the goods of simplicity and unity of laws or the diversity of kinds of being across the universe. On a theistic account, we have some hope of grounding non-individual goods in God. It is, plausibly, good for God to diffuse his goodness by creating in his image. While the God of classical theism is simple, this simplicity comes with what one might call an infinite intensity of being, which cannot be adequately imaged by any finite being. Aquinas writes:

Hence we must say that the distinction and multitude of things come from the intention of the first agent, who is God. For He brought things into being in order that His goodness might be communicated to creatures, and be represented by them; and because His goodness could not be adequately represented by one creature alone, He produced many and diverse creatures, that what was wanting to one in the representation of the divine goodness might be supplied by another. For goodness, which in God is simple and uniform, in creatures is manifold and divided and hence the whole universe together participates the divine goodness more perfectly, and represents it better than any single creature whatever. (??ref:S.Th. I.47.1)

This gives an account of the value of diversity. The value of simplicity and unity might be more straightforwardly grounded in the value of God's creating a world that images divine simplicity and unity. The goods of an ecosystem are an instance of a unity-in-diversity as well, and hence can perhaps be appropriately grounded in the value of God's self-diffusive creation. (The latter unity-in-diversity is even better explained if God turns out to be a multiplicity of persons, as in the Christian doctrine of the Trinity.)

We might say that the value of the diffusiveness of the divine good is grounded in God's nature or form (which on classical theism is identical with God), and hence we are following the same form of explanation as in our usual Aristotelian intra-kind value explanations, but now using the divine form.

7.3.2. *A problem for theistic Mersennians.* For a number of centuries, it has been popular for theistic thinkers to insist that the elegant simplicity of the laws of nature is explained well by theism.??ref God loves beauty, and the laws we find in physics since Galileo are beautifully simple. But that wasn't how Mersenne argued for the existence of God in the argument that inspires this book: instead, he focused on the messy and seemingly arbitrary features of the universe around us, which pointed to contingency. And in the earlier chapters of this book we have *ad nauseum* been identifying complex and seemingly arbitrary features of what one might call "the human world": the normatively rich world that humans inhabit, and now we have been using this as an argument for theism. It seems that I have made a powerful case for what atheists have been wont to argue: that no matter how things look—simple or complex, say—theists will disingenuously claim it as evidence for theism.

I think what the theist should do at this point is to say that God *qua* artist is pursuing neither complexity nor simplicity in creation, but the beauty that subsumes both. A drive towards simplicity alone would lead to God not creating anything. The picture of the world that the earlier chapters lead to is one where there are elegant mathematical laws that cut across many kinds. The same law of gravitation is written into the nature of fermions, bosons, plants, fungi, animals and persons, a little like a theme in a piece of

music or a novel reappears embodied in a variety of ways, though apparently with a much more impressive invariance. A focus on elegant simplicity in science needs to be qualified by a recognition that, especially outside of physics, beauty is manifested in other ways—say, in the vibrant diversity of life (I am grateful to my biologist colleagues for teaching me this).

As we saw in Section ??backref, a game like tennis that is both elegant and balanced may need to have a complex hierarchy of rules, with some simple general rules that can be quickly explained to a beginner, and dozens of pages of details. This may reasonably seem to be a sacrifice of beauty for other goods, but in fact we need not see it this way. Leibniz famously recognized God’s artistry in a balance of simplicity and diversity. But it is more than a numerical balance. Different “areas of reality” are differently characterized. Physics tends towards simplicity. Chemistry describes a complexity from a limited set of building blocks. Biology tends towards diversity, reusing a relatively limited number of types of biochemical mechanisms. And as we saw in ??backref, morality has simple and imprecise rules at higher levels, which need complex specifications: at the highest level, perhaps, we should love our neighbor, or obey the golden rule, or respect persons. This is specified, say, into a variety of virtues, which are then further specified into precise rules that handle specific cases, all of them embedded in our nature. Epistemology is perhaps less diverse and complex, but perhaps we are only beginning to discover its complexity, while more of the complexity of ethics has already been uncovered by human reflection, especially in literature.

Social rules for humans—say, federal and state laws governing the transport of hazardous materials—are likely to be Procrustean or complex. In practice, they are often a balance between the two, because if they become too complex they are too difficult for beings of our level of intelligence to apply and hence some degree of Procrusteanism needs to be tolerated. Well-written legislation (which likely there is not enough of) has an elegant balance between avoiding Procrusteanism and being simple. We should expect something similar if God were choosing norms for the behavior of reasoners with our level of ability.

Is this a case of the theist disingenuously having it both ways? That depends on how much objectivity one sees in aesthetics. If aesthetics has no objectivity to it whatsoever, is simply an arbitrary human creation, then one might say that the beauty of our complex and organized world is just a function of our optimistically mirroring the world in our concept of beauty. But as we saw in ??backref, aesthetics could have one of two kinds of objectivity: it could be objective and species-independent or it could be objective and species-dependent. On neither kind of objectivity is the aesthetic story above *ad hoc* or disingenuous. And it would make sense for God to make a beautiful world on either kind of objectivity, whether for the sake of beauty itself or for the sake of an intelligent species to which it is beautiful.

8. Final remarks

An Aristotelian theory on which we are substances, and substances are defined by a metaphysically robust form which bestows on them causal powers and a teleology grounding norms, provides an attractive grounding for the rich variety of norms we find in ethics, epistemology and semantics, as well as solving a number of metaphysical problems.

However, in order for these solutions not to lead to skepticism, we need an Aristotelian optimism positing a variety of harmonies in nature. The norms themselves need to fit with each other (both within each kind of norm, as well as between kinds), with the causal powers of the organisms as well as with the niche the organisms occupy. Our norms are epistemically accessible to us, despite not being parts of physical reality. We have laws of nature that are unified despite being “enforced” by a variety of forms. Secular Aristotelianism has three main tools available for explaining harmony: the need for matter to be adequate to form, the causal powers encoded into the forms, and a doctrine of the sparsity of form. The doctrine of the adequacy of matter to form is seen to have significantly less force when we reflect on the admittedly highly controversial small beginnings thesis. Moreover, both the adequacy and causal powers tools depend on the sparsity of form, and

in any case, these three tools are not sufficient for explaining all the harmonies, such as the nomic harmony between the forms of different substances.

The sparsity of form, and the existence of our harmonies in general, requires a metaphysical picture of the world where value is basic. The sparsity of forms needed for Aristotelian optimism involves considerations of value, and hence space of possible forms needs to originate in something sensitive to value. Likewise, the choice of which forms are to be instantiated requires a value-based etiology. We have two main choices for a value-based etiology for the world: axiarchism and theism. I have argued that axiarchism faces serious problems with modality, and that leaves theism.

Some of the problems we are facing here are ones that Aristotle knew or should have known about: normative harmony and the harmony between norms and environment. However, the harmony between forms, DNA and evolution is something new since Darwin, and it particularly calls out for a theistic explanation. Aristotle himself thought the array of species on earth did not change. It is harder to be a secular Aristotelian than it was in his time. That said, Aristotle was not entirely secular himself. He did believe in a multiplicity of gods, and supposed that the teleology of non-divine beings had some kind of an imitative explanation in terms of the activity of the gods. This is similar to a theistic account of the forms as proceeding from the mind of God and reflecting different aspects of divine perfection.??backref

To be an intellectually satisfied Aristotelian after Darwin, Franklin, Watson and Crick seems to call for theism.